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Domains with Pseudoconvex Neighborhood Systems

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§1. Introduction

Let $\Omega=\{z\in\mathbb{C}^n: r(z)<0\}$ be a bounded domain in $\mathbb{C}^n$ with $r\in C^2(\mathbb{C}^n)$, $dr(z)\neq 0$ for $z\in\partial\Omega$. The set Ω is pseudoconvex if the Levi form of the boundary $\partial\Omega$ is positive semidefinite everywhere on $\partial\Omega$. Here we discuss the problem of determining whether $\bar{\Omega}$ has a neighborhood system of pseudoconvex sets. It is well known that if r is strictly plurisubharmonic in a neighborhood of $\partial\Omega$, then the sets $\Omega_\varepsilon=\{z\in\mathbb{C}^n: r(z)<\varepsilon\}$ form a system of pseudoconvex neighborhoods of Ω. If $\partial\Omega$ fails to be strongly pseudoconvex at some set $E\subset\partial\Omega$, the situation is more complicated. The existence of pseudoconvex neighborhoods was shown by Diederich and Fornæss [6] under the assumption that $E=M_1\cup\cdots\cup M_n$ is the union of submanifolds with a nondegeneracy condition. If E contains a complex submanifold, this nondegeneracy condition is not fulfilled, and in fact a pseudoconvex neighborhood system need not exist in general, as was shown by Diederich and Fornæss [7].

The present paper looks at the situation where E contains a complex submanifold of codimension 1. First we discuss the case where $E=\overline{M}$, and M is a locally closed complex submanifold of codimension 1 in $\mathbb{C}^n$. The approach is to try to find a holomorphic vector field v, in a neighborhood of $\overline{M}$, that is transverse to $\partial\Omega$. (A vector field $v=(f_1, g_1, \ldots, f_n, g_n)$ on $\mathbb{R}^{2n}=\mathbb{C}^n$ is *holomorphic* if the functions f_j+ig_j are holomorphic.) If such a v can be constructed, then by standard techniques $\partial\Omega$ can be pushed out in a neighborhood of $\overline{M}$ to obtain a pseudoconvex set $\tilde{\Omega}\supset\Omega\cup\overline{M}$. We note that if M is simply connected, then v exists regardless of the codimension of M (Theorem 3.4).

The obstruction to finding v is given by a certain cohomology class $c(M,\partial\Omega)\in H^1(M,\mathbb{R})$ which measures the turning of $\partial\Omega$ around M. It is shown (Theorem 4.6) that v exists if and only if $\mu<\pi/2$, where μ solves the extremal problem

$$\mu=\inf_{u\in\mathscr{F}}\sup_{z\in M}|u(z)|,$$

$$\mathscr{F}=\{u\in C^2(\overline{M}),\ u \text{ pluriharmonic},$$

$$\text{and}\quad \int_\gamma d^c u=\int_\gamma c(M,\partial\Omega)\quad \text{for all } \gamma\in H_1(M,\mathbb{R})\}. \tag{1.1}$$

The minimization (1.1) is related to certain seminorms $N\{\gamma\}$ defined by Chern, Levine and Nirenberg [3]. In order to use this seminorm, however, we need to prove a bounded approximation theorem (Theorem 4.3) which shows that the norm varies continuously as the domain is perturbed. If $H_1(M,\mathbb{R})$ is generated by γ, the existence of v is determined by $N\{\gamma\}$, which is intrinsic to M, and by $\int_\gamma c(M,\partial\Omega)$,

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which measures the amount that $\partial\Omega$ winds around M. (See Section 3 for a precise statement.)

Theorem 1.1. *Let Ω be a domain with C^4 boundary which is strongly pseudoconvex at every point of $\partial\Omega \setminus \overline{M}$. Assume that $M \subset\subset \tilde{M}$ is a strongly pseudoconvex domain in a Stein manifold $\tilde{M}$, and that ∂M is C^4. If $H_1(M, \mathbb{R})$ has at most one generator γ, then $\bar{\Omega}$ has a neighborhood system of pseudoconvex sets if for some $\varepsilon > 0$*

$$(\pi - \varepsilon) N\{\gamma\} \geqq \left|\int_\gamma c(M, \partial\Omega)\right|.$$

(This inequality is vacuous if $H_1(M, \mathbb{R}) = 0$.)

The transverse vector field can be used to obtain function-theoretic results about Ω. For instance, there is the following result of Fornæss and Nagel [10].

Theorem. *Let $\Omega \subset \mathbb{C}^n$ be a domain of holomorphy with C^3 boundary, and let E be the set of points where $\partial\Omega$ fails to be strongly pseudoconvex. If there is a holomorphic vector field v that is transverse to $\partial\Omega$ in a neighborhood of E, then $\mathcal{O}(\bar{\Omega})$, the set of functions holomorphic in a neighborhood of $\bar{\Omega}$, is uniformly dense in $A(\Omega) = \mathcal{O}(\Omega) \cap C(\bar{\Omega})$, the functions holomorphic on Ω and continuous on $\bar{\Omega}$.*

Here we prove an additional result.

Theorem 1.2. *Let E and Ω be as in the preceding theorem. Then Ω is holomorphically convex with respect to the algebra $\mathcal{O}(\bar{\Omega})$. That is, if $K \subset \Omega$ is compact, then*

$$\hat{K} = \{z \in \bar{\Omega}: |f(z)| \leqq \sup_K |f| \text{ for all } f \in \mathcal{O}(\bar{\Omega})\}$$

is a compact subset of Ω.

To look at the converse to Theorem 1.1, we restrict our attention to the special case where $\tilde{M} = \mathbb{C}$, i.e. M is a schlicht Riemann surface with C^2 boundary. Instead of looking for a holomorphic transverse vector, we try to fit an appropriate Hartogs domain inside $\bar{\Omega}$. For this another problem in potential theory, related to mapping multiply connected domains onto canonical regions, is used.

Let us suppose that $M \subset \mathbb{C}$ is a connected open set with C^2 boundary components $\gamma_0, \gamma_1, \ldots, \gamma_g$ and that γ_0 lies in the noncompact component of $\mathbb{C} \setminus M$. Given numbers $c_1, \ldots, c_g \in \mathbb{R}$ there is a unique solution to the problem

$$\begin{aligned} &w \text{ is harmonic on } M, \ w \in C^1(\overline{M}) \\ &\int_{\gamma_j} d^c w = c_j \quad j = 1, 2, \ldots, g \\ &w(z) = a_0 = 0 \quad \text{for } z \in \gamma_0 \\ &\text{there exists a constant } a_j \text{ such that} \\ &w(z) = a_j \quad \text{for } z \in \gamma_j \quad j = 1, 2, \ldots, g. \end{aligned} \tag{1.2}$$

(see Ahlfors-Sario [1]).

Theorem 1.3. *Let $\Omega \subset \mathbb{C}^2$ be a domain with C^4 boundary which is strongly pseudoconvex at all points of $\partial\Omega \setminus \overline{M}$.*

Let M be conformally equivalent to a finitely connected domain in $\mathbb{C}$ with boundary components $\gamma_0, \ldots, \gamma_g$ and let the numbers $c_1, \ldots, c_g$ be given by $c_j = \int_{\gamma_j} c(M, \partial\Omega)$. If $a_1, \ldots, a_g$ are the numbers determined by the problem (1.2) *and if*

$$\max_{0 \leqq j,k \leqq g} |a_j - a_k| > \pi g,$$

then there is an open set $\tilde{\Omega}$, $\Omega \subsetneqq \tilde{\Omega} \subset \mathbb{C}^2$ with the property that if $\Omega_2 \subset \mathbb{C}^2$ is pseudoconvex and $\bar{\Omega} \subset \Omega_2$, then $\tilde{\Omega} \subset \Omega_2$.

On the other hand, if $\max_{0 \leqq j,k \leqq g} |a_j - a_k| < \pi$ then $\bar{\Omega}$ has a neighborhood basis of pseudoconvex open sets.

Now let us consider the situation where the Levi form degenerates on a larger set. If there is an open set $N \subset \partial\Omega$ where the Levi form vanishes identically, then N is foliated by complex submanifolds of codimension 1. If we assume that $E = \bar{N}$, i.e. Ω is strongly pseudoconvex on $\partial\Omega \setminus \bar{N}$, then it is reasonable to look for a holomorphic vector v that is transverse to $\partial\Omega$ on $\bar{N}$. This seems to be significantly different from the case already discussed. For instance, if each leaf M of the foliation is locally closed, then by Proposition 6.1 $c(M, \partial\Omega) = 0$.

The main result is that $\bar{\Omega}$ has a system of Stein neighborhoods if $\bar{N}$ does and if the leaves of the foliation of N are locally closed (the precise hypotheses are that N satisfy (6.1) and (6.2)). In particular this applies to certain cases where N is CR-equivalent to a domain in

$$\mathbb{C}^{n-1} \times \mathbb{R} = \{z \in \mathbb{C}^n : \operatorname{Im} z_n = 0\}.$$

The paper is organized as follows. Section 2 contains assorted elementary results and observations concerning pseudoconvex neighborhood systems. In Section 3, the class $c(M, \partial\Omega)$ is defined. A bounded approximation theorem related to intrinsic seminorms is proved in Section 4, and this is then used to prove Theorem 1.1. In Section 5, Theorem 1.3 is proved. Foliations in boundaries are the subject of Section 6; several examples are given. In Section 7, a special system of strongly pseudoconvex domains associated with Ω is produced to prove Theorem 1.2.

§ 2. Preliminary Remarks

Let Ω be a relatively compact pseudoconvex set with C^4 boundary in the complex manifold Ω_1, and let $M \subset \partial\Omega$ be a locally closed complex submanifold of Ω_1. We will compile a series of elementary remarks that should motivate the hypotheses of Theorems 1.1 and 1.3. We use the notation

$$d = \partial + \bar{\partial}, \quad d^c = i(\bar{\partial} - \partial)$$
$$dd^c = 2i\partial\bar{\partial}$$

and

$$f_j = \frac{\partial f}{\partial z_j}, \quad f_{\bar{j}} = \frac{\partial f}{\partial \bar{z}_j}, \quad f_{j\bar{k}} = \frac{\partial^2 f}{\partial z_j \partial \bar{z}_k}.$$

Let us suppose that there is a connected domain $W \subset \mathbb{C}^{n-1}$ and a holomorphic imbedding Φ: $\overline{W} \to \Omega_1$ such that $\Phi(W) \supset M$ and $\Phi^{-1}(\partial\Omega) = \Phi^{-1}(M)$. We will call $\Phi(W) = \hat{M}$.

Remark 2.1. It follows that $\hat{M} \cap \Omega = \emptyset$.

Otherwise $\Phi^{-1}(\Omega)$ is non empty. Since W is connected it follows that there is a point w_0 in the boundary of both $\Phi^{-1}(\Omega)$ and $\Phi^{-1}(M)$. If $w \in \partial\Phi^{-1}(\Omega)$, then $w \in \Phi^{-1}(\partial\Omega)$ and thus $w \in \Phi^{-1}(\overline{M})$. Thus a neighborhood of w_0 lies in $\Phi^{-1}(\partial\bar{\Omega})$. But now we can find a disk containing w_0 and intersecting both $\Phi^{-1}(\Omega)$ and $\Phi^{-1}(\partial\Omega)$. By the sliding disk theorem and the smoothness of $\partial\Omega$ this contradicts the pseudoconvexity of Ω.

Remark 2.2. M (i.e. $\Phi^{-1}(M)$) is pseudoconvex.

Let $\{D(t)\}$ be a 1-parameter family of disks in $\hat{M}$, (i.e. there is a continuous function $F(t, z) = \{|z| < 1\} \to \hat{M}$ which is holomorphic in z, such that $D(t) = F(t, \{|z| < 1\})$. We will show that if $D(0) \cup \bigcup_{0 < t \leqq 1} \partial D(t) \subset M$, then $\bigcup_{0 < t \leqq 1} D(t) \subset M$. Without loss of generality, we may assume the diameter of $\bigcup D(t)$ is less than a fixed $\varepsilon > 0$. Let ε be chosen so that if $v(z_0)$ is the outward normal to $\partial\Omega$ at z_0, then $v(z_0)$ is transverse to $\partial\Omega$ at z if $|z - z_0| < 2\varepsilon$. It follows that for all $\delta > 0$

$$D(0) \cup \bigcup_{0 < t \leqq 1} \partial D(t) - \delta v(z_0) \subset \Omega,$$

where $z_0 \in D(1)$. Since Ω is pseudoconvex, it follows that $\bigcup_{0 \leqq t \leqq 1} D(t) - \delta v(z_0) \subset \Omega$. Noting that this holds for all $\delta > 0$, we conclude that $\bigcup_{0 \leqq t \leqq 1} D(t) \subset \bar{\Omega}$. Since $\Phi^{-1}(\partial\Omega) = \Phi^{-1}(\overline{M})$ and $\hat{M} \cap \Omega = \emptyset$, it follows that $\bigcup_{0 \leqq t \leqq 1} D(t) \subset \overline{M}$.

Remark 2.3. If $\bar{\Omega}$ has a Stein neighborhood basis in Ω_1, then $\overline{M}$ has a Stein neighborhood basis inside $\hat{M}$.

Proof. Let $\{\Omega_\delta\colon 0 < \delta\}$ be pseudoconvex open sets in Ω_1, $\cap \Omega_\delta = \bar{\Omega}$ with $\Omega_{\delta_1} \subseteq \Omega_{\delta_2}$ for $\delta_1 < \delta_2$. By Remark 2.1 it follows that $\hat{M} \cap \bar{\Omega} = \overline{M}$. Thus for $\delta > 0$ sufficiently small, $\Omega_\delta \cap M = M_\delta$ is a Stein manifold which is relatively compact in $\hat{M}$, and so $\{M_\delta\}$ is the desired Stein neighborhood basis.

The fact that $M \subset \partial\Omega$ sometimes forces a certain amount of smoothness on $\overline{M}$. Suppose r, the defining function for $\partial\Omega$ is of class C^k. The Levi form of $\partial\Omega$ has exactly p nonzero eigenvalues if and only if

$$dr \wedge d^c r \wedge (dd^c r)^{p+1} = 0,$$
$$dr \wedge d^c r \wedge (dd^c r)^p \neq 0.$$

Remark 2.4. Suppose the Levi form of Ω has p nonzero eigenvalues at each point of $\overline{M} \subseteq \partial\Omega$, and codim $M = p + 1$. If $\partial\Omega$ is of class C^k, $k \geqq 2$, then the tangent bundle TM is a C^{k-2}-subbundle of $T\Omega_1|_{\overline{M}}$.

This is immediate since $dr \wedge d^c r \wedge (dd^c r)^p$ defines the conormal bundle to M and is of class C^{k-2}.

The following remark shows that if M is a Riemann surface and Ω has complex dimension 2, then it is natural to assume that M lies in a larger Riemann surface.

Remark 2.5. If $\dim_{\mathbb{C}} M = 1$, and $\overline{M}$ is a topological manifold with boundary, then there is a Riemann surface $\hat{M}$ and a conformal mapping $\Phi: M \to \hat{M}$ such that $\Phi(\overline{M})$ is a compact subset of $\hat{M}$ and Φ is a C^k diffeomorphism between $\overline{M}$ and $\Phi(\overline{M})$.

Proof. Since the conormal bundle to M, generated by $dr \wedge d^c r$, is C^{k-1} on $\overline{M}$, there is a C^k, 2 (real)-dimensional surface $S \subset \Omega_1$ containing $\overline{M}$. If we give Ω_1 a smooth Hermitian metric, then S inherits a Riemannian metric, and this metric determines a conformal class. Let $\hat{M}$ be any Riemann surface in this class, and let $\Phi: S \to \hat{M}$ be a conformal equivalence. Since M is a complex submanifold of Ω_1, the analytic and conformal structures on M are the same, and thus $\Phi|_M$ is analytic. Finally, since S is of class C^k, Φ and Φ^{-1} are of class C^k.

Let us give an example to show why one might expect M always to be finitely connected. The following example shows that if $M \subset \partial\Omega$, then the "holes" of M cannot be too small if $\partial\Omega$ is smooth.

Example. Let $D = \{z \in \mathbb{C}: |z| < 1\}$, and let $\Phi: D \to \mathbb{C}^n$ be a holomorphic imbedding which is continuous on $\bar{D}$. Let $K_1, K_2, \ldots$ be disjoint compact subsets of D such that each K_j is the closure of its interior. Suppose that $\lim_{j \to \infty} \operatorname{diam}(K_j) = 0$. If there exists an open set $\Omega \subset \mathbb{C}^n$ with C^1 boundary $\partial\Omega$ such that

$$\Phi(\overline{D \setminus \bigcup_j K_j}) = \partial\Omega \cap \Phi(D)$$

then Ω is not pseudoconvex.

Proof. Let $z_0 \in \bar{D}$ be an accumulation point of $\bigcup_j K_j$. Since $\partial\Omega$ is C^1, the outward normal $v(\Phi(z_0))$ is transverse to $\partial\Omega$ at $z \in \partial\Omega$ if $|z - z_0| < \varepsilon$. Choosing $\varepsilon > 0$ even smaller if necessary, there exists $\eta > 0$ such that $z - \eta\, v(\Phi(z_0)) \in \Omega$ if $z \in \mathbb{C}^n$, $|z - z_0| < \varepsilon$. We may choose j sufficiently large that $|z - z_0| < \varepsilon$ for $z \in \Phi(K_j)$, because $\lim_{j \to \infty} \operatorname{diam} \Phi(K_j) = 0$. Thus for any $t > 0$, we define

$$\omega(t) = \Phi(K_j) - t\, v(\Phi(z_0)),$$

and so $\omega(\eta) \subset \Omega$. By transversality,

$$\partial\omega(t) = \Phi(\partial K_j) - t\, v(\Phi(z_0)) \subset \Omega.$$

for all $0 < t < \eta$. Now we see that if Ω were pseudoconvex, then $\Phi(K_j) \subset \partial\Omega$, which is contrary to assumption.

Proposition 2.6. *Let $\Omega \subset \mathbb{C}^n$ be a domain with C^3 boundary, and let $M \subset \partial\Omega$ be a complex manifold of codimension 1. If there is a Stein manifold $\tilde{M}$ such that $M \subset\subset \tilde{M}$ is a strongly pseudoconvex domain with C^4 boundary, then there exists a function f of class C^3 in a neighborhood of $\overline{M}$ such that*

$$\overline{M} \subset \{z \in \mathbb{C}^n: f(z) = 0\},$$
$$df \neq 0 \quad \text{on } \overline{M} \text{ and}$$
$$|\bar{\partial} f(z)| = 0(\operatorname{dist}(z, \overline{M})^2). \tag{2.1}$$

Proof. Let $\mathcal{N}(M)$ denote the normal bundle of M. Since ∂r is a section of the conormal bundle, this bundle is toplogically trivial and smooth of class C^2. Thus the problem of finding a global section of $\mathcal{N}(M)$ (a line bundle) reduces to solving a Cousin I problem where the data are C^1 on $\overline{M}$. This may be solved using bounds on the $\bar{\partial}$-operator (see Kerzman and Nagel [16]) to obtain a global section of $\mathcal{N}(M)$ which is continuous on $\overline{M}$. Thus there is a vector field $v=(v_1, \ldots, v_n)$ with $v_j \in A(M)$ which is nowhere tangent to $\overline{M}$. Let $\xi=(\xi_1, \ldots. \xi_n)$ be a vector field which is holomorphic in a neighborhood of $\overline{M}$ in $\tilde{M}$ and transverse to $\overline{M}$ (see the remarks preceding Lemma 4.5). For $\varepsilon>0$ small, the map $F: \overline{M} \times \{|\zeta|<\varepsilon\} \to \mathbb{C}^n$ given by $F(z, \zeta)=z+\zeta v(z)$ is nonsingular and of class C^3. If we define f as any smooth extension of the function ζ to a neighborhood of $F(\overline{M} \times \{|\zeta|<\varepsilon\})$, then f satisfies (2.1).

§ 3. The Mapping to Cohomology

In the first part of this section we construct the cohomology class $c(M, \partial\Omega)$ and discuss the geometric nature of a complex submanifold in a pseudoconvex boundary. At the end of this section we give examples to show which cohomology classes do in fact arise in this manner.

We let $\Omega=\{z \in \mathbb{C}^n: r(z)<0\}$ be a pseudoconvex set with $dr \neq 0$ on $\partial\Omega$, $r \in C^4(\mathbb{C}^n)$. If f satisfies (2.1), there exists a nonzero function $\alpha=e^{u+iv}$ with v defined modulo 2π such that

$$\partial f(z)=\alpha \partial r(z) \tag{3.1}$$

for $z \in \overline{M}$. Thus if $r_j=\dfrac{\partial r}{\partial z_j} \not\equiv 0$, then r_j/r_k is meromorphic on M, since $\bar{\partial} f$ vanishes to second order on M. We would like a defining function r such that α is holomorphic. This is possible, locally, since by Proposition 3.1 v is pluriharmonic on M. If v^* is the harmonic conjugate of v, then the function $r^*=e^{(v^*+u)}r$ satisfies

$$\partial f=e^{-v^*+iv}\partial r^* \tag{3.2}$$

on M.

Proposition 3.1. *If r, f, M, Ω and v are as above, then v is pluriharmonic on M.*

Proof. We may choose coordinates $(z_1, \ldots, z_n)$ near z_0 such that $z_0=(0, \ldots, 0)$, $f=z_1+O(z_1^2)$, and $r_j(0)=0$ for $2 \leqq j \leqq n$. We must show that $v_{j\bar{k}}(0)=0$ for $2 \leqq j, k \leqq n$. The function $\rho=re^u$ is a defining function for $\partial\Omega$, and by (3.1) we see that $\rho=\mathrm{Re}(z_1 e^{-iv})+O(|z_1|^2)$. Since Ω is pseudoconvex near M, the Levi form is non negative. We compute

$$\begin{aligned}
1 \leqq j \leqq n \quad & \rho_j=\tfrac{1}{2}(-z_1 e^{iv} i v_j+\bar{z}_1 e^{iv} i v_j) \\
& =v_j \,\mathrm{Re}(i\bar{z}_1 e^{iv}), \\
2 \leqq j, k \leqq n; \quad & \rho_{j\bar{k}}=v_{j\bar{k}}\,\mathrm{Re}(i\bar{z}_1 e^{iv}) \\
& \quad +v_j v_{\bar{k}}\,\mathrm{Re}(-\bar{z}_1 e^{iv}) \\
& =v_{j\bar{k}}\,\mathrm{Im}(\bar{z}_1 e^{iv})-v_j v_{\bar{k}}\,\mathrm{Re}(\bar{z}_1 e^{iv}).
\end{aligned} \tag{3.3}$$

Thus each entry of the complex hessian of ρ is dominated by $O(|z_1|)$. Near M, we may choose tangent vectors to $\partial\Omega$

$$T_j=\frac{\partial}{\partial z_j}+\sum_{j\neq k} a_j^k\frac{\partial}{\partial z_k} \qquad 2\leqq j\leqq n$$

where $a_j^k=O(|z_1|)$. Thus if we apply the Levi form to the vector field $T=\sum\xi_j T_j$ we obtain

$$\langle\partial\bar\partial\rho, T\wedge\bar T\rangle=\sum_{n\geqq j,k\geqq 2}\rho_{j\bar k}\xi_j\bar\xi_k+O(|z_1|^2). \tag{3.4}$$

Now (3.4) is non-negative for all $z\in\partial\Omega$ near M and all choices of $\xi_2,\ldots,\xi_n$. By (3.3), we see that

$$0\leqq\sum\xi_j\bar\xi_k(v_{j\bar k}\operatorname{Im}(\bar z_1 e^{iv})-v_j v_{\bar k}\operatorname{Re}(\bar z_1 e^{iv}))$$

for z_1 sufficiently close to 0. Since for $z\in\partial\Omega$, z near M we have $\operatorname{Re}(\bar z_1 e^{iv})=O(|z_1|^2)$, it follows that

$$0\leqq\sum\xi_j\bar\xi_k v_{j\bar k}\operatorname{Im}(\bar z_1 e^{iv}). \tag{3.5}$$

But we observe that $\operatorname{Im}(\bar z_1 e^{iv})$ can be both positive and negative for $z\in\partial\Omega$ near M, and so it follows from (3.5) that $v_{j\bar k}=0$.

Since M is a complex submanifold of $\mathbb{C}^n$, the pullbacks ∂_M and $\bar\partial_M$ of ∂ and $\bar\partial$ to M are intrinsically defined on M. Since v is defined on $\overline{M}$ modulo 2π, the form

$$c(M,\partial\Omega)=d_M^c v \tag{3.6}$$

is well defined. By Proposition 3.1, $d_M^c v$ defines a class in $H^1(M,\mathbb{R})$.

A slightly different derivation of $c(M,\partial\Omega)$ is as follows. Locally there is a defining function r^* of the domain Ω with the property that $\partial r^*=\sum_{j=1}^{n}\frac{\partial r^*}{\partial z_j}dz_j$ is a holomorphic 1-form on M, i.e. $\bar\partial_M(r_j)=0, j=1,\ldots,n$. This defines r^* uniquely up to constant multiple on M, and we may define

$$c(M,\partial\Omega)=d_M^c\log|dr^*|,$$

which is independent of the choice of r^*.

Proposition 3.2. *The class $c(M,\partial\Omega)$ is the obstruction to finding a defining function r such that*

$$\partial f=\alpha\partial r$$

where $\alpha\in\mathcal{O}(M)$.

Proof. If $[d^c v]=0$ in $H^1(M,\mathbb{R})$, then there is a function $w\in C^1(M)$ such that $d_M w=d_M^c v$. Since $\bar\partial_M=\frac{1}{2}(d_M-\mathrm{id}_M^c)$, w is the harmonic conjugate of v. Thus we set $w=v^*$ in (3.2) and have the desired r^*. Conversely, if $\alpha\in\mathcal{O}(M)$ then u is already the harmonic conjugate of v and solves the equation $d_M u=d_M^c v$.

Now we wish to construct a large family of pseudoconvex surfaces that exhibit the behavior discussed earlier in this section. Let Ω_1 and $f\in\mathcal{O}(\Omega_1)$ be given, and

suppose that $df \neq 0$ on $M_1 = \{z \in \Omega_1 : f(z) = 0\}$. Let $\sigma \in C^2(M_1)$ be a strictly plurisubharmonic function such that $\overline{M} = \{z \in M_1 : \sigma(z) \leqq 0\}$ is compact and $d\sigma(z) \neq 0$ on $\{z : \sigma(z) = 0\}$.

Proposition 3.3. *Let Ω_1, M_1, and M be as above. If $u : M_1 \to \mathbb{R}/2\pi\mathbb{Z}$ is pluriharmonic on $\overline{M}$, then there exists a domain $\Omega \subset \Omega_1$ with smooth boundary $\partial\Omega$ containing M. Furthermore $c(M, \partial\Omega) = [d_M^c u] \in H^1(M, \mathbb{R})$.*

Proof. There exists a Stein neighborhood $U \supset \overline{M}$ and a holomorphic retraction π: $U \to M_1$ (see Docquier and Grauert [9]). Thus $u = u(\pi)$ and $\sigma = \sigma(\pi)$ are defined in a neighborhood of $\overline{M}$. One can show that there are a large constant $K > 0$ and a suitable convex function λ, $\lambda(t) = 0$ for $t \leqq 0$, $\lambda(t) > 0$ for $t > 0$, such that $\Omega = \{z \in \Omega_1 : r(z) = \mathrm{Re}(f e^{iu}) + (K + \sigma)|f|^2 + \lambda(\sigma) < 0\}$ has the desired properties. Observe that at points where $\lambda(\sigma) = 0$, one may compute

$$-\partial r \wedge \bar{\partial} r \wedge \partial\bar{\partial} r = -\tfrac{1}{4}|f|^2 \partial f \wedge \overline{\partial f} \wedge [(K + \sigma)\,\partial u \wedge \bar{\partial} u$$
$$-8\,\mathrm{Im}(\partial u \wedge \overline{\partial \sigma}) + \partial\bar{\partial}\sigma] + O(|f|^3),$$

which will be a positive (2,2)-form near M if K is chosen sufficiently large. The addition of $\lambda(\sigma)$ serves to "cap off" the domain near ∂M. By the construction it is clear that $c(M, \partial\Omega) = [d_M^c u]$.

Theorem 3.4. *Assume that Ω, M are as in Theorem 1.1 except that $H_1(M, \mathbb{R}) = 0$, and M is allowed to have arbitrary dimension. Then there is a vector field v which is C^1 and transverse to $\partial\Omega$ in a neighborhood of $\overline{M}$ and which is holomorphic on M.*

By the arguments in §4 it will be possible to make v holomorphic in a neighborhood of $\overline{M}$.

Proof. Let r be a defining function for $\partial\Omega$. By the proof of Proposition 3.1, $v_j = \mathrm{Im} \log r_j$ is pluriharmonic on M, where $r_j \neq 0$. For $z_0 \in \overline{M}$, there is some j such that $r_j \neq 0$ in a neighborhood U of z_0 with $U \cap \overline{M}$ simply connected. Thus there exists v_j^* such that $v_j^* + iv_j \in C^2(\overline{M} \cap U) \cap \mathcal{O}(M \cap U)$. We replace r by the defining function $\lambda_j r = e^{v_j^*} r$ so that the coefficients of $\partial(\lambda_j r)$ are holomorphic on M. Since λ_j is unique up to a constant real multiple and $H_1(M, \mathbb{R}) = 0$, it follows that $\lambda \in C^2(\overline{M})$ exists with the same property. By Lemma 4.5 there exist $g_j \in \mathcal{O}(M) \cap C^1(\overline{M})$ such that $\sum g_j (\lambda r)_j = 1$. If we set $v = (g_1, \ldots, g_n)$, then $\mathrm{Re}\langle v, \bar{\partial} r\rangle > 0$, and so v is transversal.

§ 4. Existence of Holomorphic Transverse Vectors

We will use a seminorm defined by Chern, Levine, and Nirenberg [3] for homology classes on a complex manifold. For $\gamma \in H_1(M, \mathbb{C})$ (we are concerned here only with the first homology group) define

$$N\{\gamma\} = \sup_{\gamma} \int d^c u,$$

with the supremum being taken over the class $F = \{u$ pluriharmonic on M, $0 < u < 1\}$. For the purposes at hand, it is more useful to have a slightly different

norm:

$$N_a\{\gamma\} = \sup_{\gamma} \int d^c u$$

with supremum taken over the smaller class $F_a = \{u \text{ pluriharmonic on } \overline{M}, 0 < u < 1\}$. It is obvious that $N_a\{\gamma\} \leqq N\{\gamma\}$. We will show that in certain cases these two norms are equal.

For a positive integer k, let $L^p_k(M)$ denote the space of functions f such that $\int |D^\alpha f|^p \, dV < \infty$ for all α such that $|\alpha| \leqq k$, and where dV is any smooth volume element on $\overline{M}$. Then by interpolation one may define $L^p_k(M)$ with norm $\| \ \|_{k,p}$ for any real number k (see for instance Taylor [21]). We will use the following result which seems to be well known, but for which we could not find an exact reference.

Theorem 4.1. *Let M be a compact manifold with boundary which is smooth of class C^k. Let ω be a d-closed j-form with coefficients in $L^p_s(M)$, $|s| < k-1$, $1 < p < \infty$. If ω represents the trivial cohomology class in $H^j(M, \mathbb{C})$, then there exists a $(j-1)$-form η with coefficients in $L^p_{s+1}(M)$ such that $d\eta = \omega$, and $\|\eta\|_{L^p_{s+1}} \leqq C \|\omega\|_{L^p_s}$ for a constant C independent of ω. Furthermore the constant C may be chosen to work for all small C^k perturbations of M.*

Proof. The method of proof for this result is to observe that the d-Neumann problem satisfies the Lopatinski condition and is a coercive elliptic boundary value problem. The parametrix for this problem is a pseudodifferential operator E of order -1. The L^p-theory for this problem has been studied by several authors and it is known that E: $L^p_s \to L^p_{s+1}$ (see [21]).

In the following theorem we consider domains of the form $M(\varepsilon) = \{z \in M: r(z) < \varepsilon\}$, where $r \in C^4(M)$ and $dr \neq 0$ on $\partial M(0)$. Thus for $|\varepsilon| < \varepsilon_0$, $M(\varepsilon)$ has smooth boundary and is diffeomorphic to $M(0)$. For a homology class $\gamma \in H_1(M(0), \mathbb{C})$, we will denote by $N\{\gamma, M(\varepsilon)\}$ its norm inside $M(\varepsilon)$. It is immediate that if $\varepsilon_1 < \varepsilon_2$, then

$$N\{\gamma, M(\varepsilon_1)\} \geqq N_a\{\gamma, M(\varepsilon_1)\} \geqq N\{\gamma, M(\varepsilon_2)\}. \tag{4.1}$$

Thus right continuity (in ε) of N will imply that $N = N_a$.

Theorem 4.2. *Let $\tilde{M}$ be a Stein manifold, and let $M = \{r < 0\}$, $r \in C^4$, $dr \neq 0$ on ∂M be a strongly pseudoconvex, relatively compact domain in $\tilde{M}$. Then $N(\varepsilon) = N\{\gamma, M(\varepsilon)\}$ is continuous,* i.e. $N(0) = \lim_{\varepsilon \to 0} N(\varepsilon)$, *and as a consequence it follows that $N_a\{\gamma, M(\varepsilon)\} = N\{\gamma, M(\varepsilon)\}$.*

Proof. By inequality (4.1) we see that $\lim_{\varepsilon \to 0^-} N(\varepsilon) \geqq N(0) \geqq \lim_{\varepsilon \to 0^+} N(\varepsilon)$. Since $F(M(\varepsilon))$ is a normal family, there exists $u(\varepsilon) \in F(M(\varepsilon))$ such that $N(\varepsilon) = \int_\gamma d^c u(\varepsilon)$. Furthermore, since $\{u(\varepsilon): \varepsilon < 0\}$ is bounded, there exists $u \in F(M(0))$ such that $u(\varepsilon)$ converges uniformly to u on compact subsets of $M(0)$. It follows that $d^c u(\varepsilon)$ converges uniformly on γ, which is compactly supported in $M(0)$, and so $N(\varepsilon)$ is left continuous

$$\lim_{\varepsilon \to 0^-} N(\varepsilon) = \lim_{\varepsilon \to 0^-} \int_\gamma d^c u(\varepsilon) = \int_\gamma d^c u \leqq N(0)$$

For right continuity we need to use the hypotheses of the theorem. If $u(0)$ satisfies $\int_\gamma d^c u(0) = N(0)$, then by Theorem 4.3, there exist $u(\varepsilon) \in F(M(\varepsilon))$ such that the $u(\varepsilon)$ converge to u uniformly on compact subsets of M. Thus the continuity of $N(\varepsilon)$ follows from the inequality

$$N(0) = \int_\gamma d^c u(0) = \lim_{\varepsilon \to 0^+} \int_\gamma d^c u \leqq \lim_{\varepsilon \to 0^+} N(\varepsilon).$$

To complete the proof of Theorem 4.2 we need the following result on bounded approximation of pluriharmonic functions. The proof could also be applied to give a similar result for analytic functions, which was obtained by Cole and Range [5].

Theorem 4.3. *Let M and $\tilde{M}$ be as in the hypotheses of Theorem 4.2. For each pluriharmonic $u \in L^\infty(M)$, there exists a sequence $\{u_\varepsilon\}$ of pluriharmonic functions on $\overline{M}$ such that*

$$\|u_\varepsilon\|_{L^\infty(M)} \leqq \|u\|_{L^\infty(M)}$$

and the u_ε converge to u uniformly on compact subsets of M.

Proof. Let us begin by showing that without loss of generality, we may assume that $\tilde{M} = \mathbb{C}^n$. We may assume that $\tilde{M}$ is properly and holomorphically imbedded in $\mathbb{C}^N$ for some large N. By a theorem of Rossi [18], there is a strongly pseudoconvex domain $W \subset \mathbb{C}^N$ with C^4 boundary such that $W \cap \tilde{M} = M$, and there is a holomorphic projection $\pi\colon W \to M$. It is sufficient to prove the bounded approximation theorem for $\tilde{u} = u \cdot \pi$ on W.

The approximating functions u_ε are obtained by the method of using a partition of unity to divide the support of u into small sets on ∂M, locally translating each piece out, and then solving the $\partial\bar{\partial}$-equation. Let $\{\chi_j\}$ be a partition of unity such that $\operatorname{supp} \chi_0 \subset M$ and $\operatorname{supp} \chi_j \cap \partial M \neq \emptyset$ if $j = 1, 2, \ldots, J$, and $\operatorname{diam}(\operatorname{supp} \chi_j) \leqq c$. Such partitions exist for $c > 0$ arbitrarily small. Let ξ_j be a vector that is transverse to ∂M and outward-pointing on $\operatorname{supp} \chi_j \cap \partial M$. If c is sufficiently small, vectors $\{\xi_j\}$ may be chosen such that $\bigcup (\operatorname{supp} \chi_j \cap \partial M) + \varepsilon \xi_j$ is disjoint from M_ε for $0 < \varepsilon < \varepsilon_0$. If we set

$$F(\varepsilon, u)(z) = \sum_{j=1}^{J} \chi_j(z)\, u(z - \varepsilon \xi_j) + \chi_0 u$$

then $F(\varepsilon, u)$ is defined on M_ε. Since u is pluriharmonic,

$$\partial\bar{\partial} F(\varepsilon, u) = \partial\bar{\partial}(F(\varepsilon, u) - u) = \sum_{j=1}^{J} \partial\bar{\partial}\chi_j (u(z - \varepsilon\xi_j) - u(z)) + 2\,\mathrm{Re}(\partial\chi_j \wedge (\bar{\partial} u(z - \varepsilon\xi_j) - \bar{\partial} u(z))).$$

We will use the notation

$$\omega_\varepsilon = \partial\bar{\partial} F(\varepsilon, u) = \frac{1}{2i}\, dd^c F(\varepsilon, u)$$

and $k(\varepsilon)=\|\omega_\varepsilon\|_{L^p_{1}(M)}$. Thus $\lim_{\varepsilon\to 0} k(\varepsilon)=0$ for all $p<\infty$. For fixed $p<\infty$, there exists $\varepsilon_0>\delta=\delta(\varepsilon)>0$ such that

$$\|\omega_\varepsilon\|_{L^p_{1}(M(\delta))}\leqq 2\,k(\varepsilon).$$

By Theorem 4.1, there exists a 1-form η_ε such that $d\eta_\varepsilon=\omega_\varepsilon$ on $M(\delta)$ and

$$\|\eta_\varepsilon\|_{L^p(M(\delta))}\leqq 2\,C k(\varepsilon).$$

Now we split $\eta_\varepsilon=\eta'_\varepsilon+\eta''_\varepsilon$ into parts of type (1,0) and (0,1). Thus

$$\partial\eta'_\varepsilon=\bar\partial\eta''_\varepsilon=0$$

and $\bar\partial\eta'_\varepsilon+\partial\eta''_\varepsilon=\omega_\varepsilon$. We claim that there exist functions $h'_\varepsilon, h''_\varepsilon\in L^\infty(M(\delta))$ such that $\partial h'_\varepsilon=\eta'_\varepsilon$, $\bar\partial h''_\varepsilon=\eta''_\varepsilon$ and

$$\sup_{z\in M(\delta)}(|h'_\varepsilon(z)|+|h''_\varepsilon(z)|)\leqq 2\,CC'k(\varepsilon). \tag{4.2}$$

For this we fix $p>2n$ and solve the $\bar\partial$ (and ∂) operators on $M(\delta)$ with the kernel of Henkin and Romanov [13]. Denoting this kernel by $K_\delta(z,\zeta)$, we have

$$h''_\varepsilon(z)=\int_{M(\delta)}\eta''_\varepsilon(\zeta)\wedge K_\delta(\zeta,z)\,dV(\zeta).$$

where dV is Lebesgue measure. A simple check shows that the coefficients of $K_\delta(\zeta,z)$ satisfy

$$\int_{M(\delta)}|K_\delta(\zeta,z)|^q\,dV(\zeta)\leqq C''$$

if $q<\dfrac{2n}{2n-1}$ and if $\partial M(\delta)$ is of class C^3. Furthermore, since C'' can be chosen to hold for $|\delta|\leqq\delta_0$ sufficiently small, we obtain (4.2) by Hölder's inequality with $C'=C''$. Thus u is approximated by $u_\varepsilon=F(\varepsilon,u)-h'_\varepsilon-h''_\varepsilon$.

Proposition 4.4. *Let Ω and M satisfy the hypotheses of Proposition* 2.6, *and let $c(M,\partial\Omega)$ be the associated cohomology class. Assuming that $H_1(M,\mathbb{C})$ has at most one generator γ, it follows that there is a vector field transverse to $\partial\Omega$ which is holomorphic on M and $C^2(\overline{M})$ if for some $\varepsilon>0$,*

$$\left|\int_\gamma c(M,\partial\Omega)\right|\leqq(\pi+\varepsilon)\,N\{\gamma\}.$$

Proof. By Theorem 4.3 there exists a function $0<w<1$, which is pluriharmonic on a neighborhood of $\overline{M}$ in $\tilde M$ such that

$$\frac{1}{\pi}\left|\int_\gamma c(M,\partial\Omega)\right|<\int_\gamma d^c w.$$

By changing sign if necessary, we may find a constant c, $0<c<1$ such that

$$\int_\gamma c\pi d^c w-d^c v=0 \tag{4.3}$$

where v is the function defining $c(M, \partial\Omega) = [d^c v]$. Now we write $v = c\pi w + (v - c\pi w)$, and so by (4.3), $v - c\pi w$ has a harmonic conjugate $(v - c\pi w)^*$. Thus

$$h = (v - c\pi w)^* + i(v - c\pi w) \in \mathcal{O}(M) \cap C^3(\overline{M})$$

from which it follows that

$$\begin{aligned} \partial f &= e^{u+iv}\,\partial r \\ \partial f &= e^{u-(v-c\pi w)^*} e^h e^{ic\pi w}\,\partial r. \end{aligned} \tag{4.4}$$

Now if we extend e^h to a function H in a neighborhood of $\overline{M}$ so that $\bar\partial H(z) = O(\mathrm{dist}(z, \overline{M}))$, then the new defining function $F = f/H$ for $\overline{M}$ will satisfy (2.1). This may be done by extending e^h to be constant in the direction df. That is, e^h is first extended to a set of codimension 1 (containing $\overline{M}$) by making it constant along integral curves of $\nabla \mathrm{Ref}$. Then it is extended to a neighborhood of $\overline{M}$ to be constant in the direction $\nabla \mathrm{Imf}$.

Now let $v_1, \ldots, v_n \in \mathcal{O}(M) \cap C^2(\overline{M})$ be analytic functions such that

$$\sum_{j=1}^{n} v_j \frac{\partial F}{\partial z_j} = 1.$$

(The existence of the v_j's will be established in Lemma 4.5.) We claim that $v = (v_1, \ldots, v_n)$ is transverse to $\partial\Omega$. It suffices to show that

$$\mathrm{Re}\langle v, \bar\partial r\rangle > 0,$$

and so we compute

$$\begin{aligned} &\mathrm{Re}\left(\sum_{j=1}^{n} v_j \frac{\partial r}{\partial z_j}\right) \\ &= e^{(v-c\pi w)^*}\,\mathrm{Re}\left(\sum_{j=1}^{n} v_j \frac{\partial F}{\partial z_j} e^{-ic\pi w}\right) \\ &= e^{(v-c\pi w)^*}\,\mathrm{Re}(ie^{-ic\pi w}) > 0 \end{aligned}$$

since $0 < c\pi w < \pi$.

For the following results we will use an approximation theorem: $\mathcal{O}(\overline{\Omega})$ is uniformly dense in $A(\Omega)$ if Ω is a strongly pseudoconvex domain with C^4 boundary. This result is due to Henkin [12], Kerzman [15], and Lieb [17]. The work of Rossi [19] allows us to conclude that if $f_1, \ldots, f_j \in A(\Omega)$ have no common zeros then the ideal they generate is all of $A(\Omega)$.

Lemma 4.5. *Let $f_1, \ldots, f_n \in C^2(\overline{M}) \cap \mathcal{O}(M)$ be given, where M is a strongly pseudoconvex domain in a Stein manifold $\tilde{M}$. Suppose M is relatively compact and has a C^4 boundary. If $|f_1| + \cdots + |f_n| > 0$ on $\overline{M}$ there exist functions $h_1, \ldots, h_n \in \mathcal{O}(M) \cap C^2(\overline{M})$ such that*

$$\sum_{j=1}^{n} f_j h_j = 1.$$

Proof. Let us choose $\tilde{h}_1, \ldots, \tilde{h}_n \in C(\overline{M}) \cap \mathcal{O}(M)$ such that $\sum f_j \tilde{h}_j = 1$. If $H_1, \ldots, H_n \in C^2(\overline{M}) \cap \mathcal{O}(M)$ are chosen such that $\|\tilde{h}_j - H_j\|_{L^\infty}$ are sufficiently small then $h_j = H_j(\sum f_j H_j)^{-1}$ will have the desired properties.

Proof of Theorem 1.1. By Proposition 4.4 there exists a vector field $v \in \mathcal{O}(M) \cap C(\overline{M})$ which is transverse to $\partial\Omega$. If we show that a transverse vector field $\tilde{v}$ exists which is holomorphic in a neighborhood of $\overline{M}$, then $\partial\Omega$ can be pushed out by Lemma 7.2. Since v is transverse, the mapping

$$\Phi\colon M \times \{|w| < \varepsilon\} \to \Omega_1$$

given by $\Phi(z, w) = z + w v(z)$ is a nonsingular holomorphic map for $\varepsilon > 0$ small. Now since M is a strongly pseudoconvex Stein domain in $\tilde{M}$, and ∂M is smooth, it is possible to find a strongly pseudoconvex Stein neighborhood ω of $M \times 0$ in $M \times \{|w| < \varepsilon\}$, and we may make $\partial\omega$ of class C^2. Since Φ is of class C^2, $\Phi(\omega)$ will be a strongly pseudoconvex domain with C^2-boundary. If we extend v to $M \times \{|w| < \varepsilon\}$ by making it constant in w, then we may use this to define v on $\Phi(\omega)$. Thus v is a vector field in $\mathcal{O}(\Phi(\omega)) \cap C^2(\overline{\Phi(\omega)})$. Now we approximate v uniformly on $\Phi(\omega)$ by a vector ξ holomorphic on $\overline{\Phi(\omega)}$. Since ξ may be taken to be transverse to $\partial\Omega$ in a neighborhood of $\overline{M}$, the existence of Stein neighborhoods follows by Lemma 7.3.

Remark. For a converse to Theorem 1.1 it is possible to ask whether an annulus $A(\lambda) = \{z \in \mathbb{C}\colon 1 < |z| < \lambda\}$ can be mapped into M, $\Phi\colon A(\lambda) \to M$ such that $\Phi_*(|z| = 1) = \gamma$. The norm of γ in the annulus is $N_A\{|z| = 1\} = \dfrac{2\pi}{\log \lambda}$, and it follows by the norm-decreasing property of holomorphic maps that $N_M\{\gamma\} \leqq \dfrac{2\pi}{\log \lambda}$. If it is possible to imbed an annulus $A(\lambda)$ such that

$$(\log \lambda) \int_\gamma c(M, \partial\Omega) > \pi,$$

then Stein neighborhoods do not exist, as is seen in the following section.

Theorem 4.6. *Let Ω be a domain which is strongly pseudoconvex at every point of $\partial\Omega \setminus \overline{M}$, let M satisfy* (2.1), *and let μ be the number defined by* (1.1). *If $\mu < \dfrac{\pi}{2}$, then $\overline{\Omega}$ has a pseudoconvex neighborhood system. If $\mu \geqq \dfrac{\pi}{2}$, then there is no vector field transverse to $\partial\Omega$ which is holomorphic in a neighborhood of $\overline{M}$.*

Proof. If $\mu < \dfrac{\pi}{2}$, then a holomorphic vector field on M can be constructed as in Proposition 4.4. By the proof of Theorem 1.1, this vector field may be taken to be holomorphic in a neighborhood of $\overline{M}$. The existence of pseudoconvex neighborhoods follows from Lemma 7.3.

If $\mu \geqq \dfrac{\pi}{2}$, we show that no holomorphic vector field can exist. If v is a holomorphic vector field transverse to $\partial\Omega$, then

$$\operatorname{Re}\langle v, \bar{\partial} r\rangle = \operatorname{Re} \sum_{j=1}^{n} v_j \frac{\partial r}{\partial z_j} > 0. \tag{4.5}$$

By Proposition 3.1, $g = \log i\sum v_j \frac{\partial r}{\partial z_j}$ is pluriharmonic on M, and by (4.5) it follows that $0 < \min_{\bar{M}} g \leqq \max_{\bar{M}} g < \pi$. By the definition of $c(M, \partial\Omega)$, we see that this cohomology class is given by $[d^c g]$. It follows from (1.2), then, that $\mu < \frac{\pi}{2}$, because we may choose

$$u = g - \tfrac{1}{2}(\max_{\bar{M}} g - \min_{\bar{M}} g).$$

§ 5. A Riemann Surface in the Boundary

Here we will give the proof of Theorem 1.2 and give some examples of what happens for non-schlicht Riemann surfaces. Let M be a bounded connected domain in $\mathbb{C}$ with C^2 boundary curves $\gamma_0, \ldots, \gamma_g$, and let γ_0 lie in the unbounded component of $\mathbb{C} \setminus M$. If $\varphi_j\colon M \to \mathbb{C}$ is a conformal mapping of M onto a circular slit domain such that φ_j maps γ_0 to the outer (unit) circle, γ_j to the inner circle, and each other γ_k is taken to a circular arc (everything is centered at the origin). Then the solution to (1.2) is given by

$$w(z) = -\sum_{j=1}^{g} \frac{c_j}{2\pi} \log|\varphi_j(z)|.$$

Since μ is determined by all harmonic functions on M which have prescribed periods (see (1.1)),

$$\tfrac{1}{2} \operatorname{osc} w = \tfrac{1}{2}(\sup_{z \in M} w(z) - \inf_{z \in M} w(z)) \geqq \mu.$$

Since w is harmonic, it follows from the maximum principle that

$$\operatorname{osc} w = \max_{0 \leqq j,k \leqq g} |a_j - a_k|.$$

If we define the matrix $B = (b_{jk})$ by

$$b_{jk} = \log|\varphi_j(z)|_{z \in \gamma_k}$$

then the vector $a = (a_1, \ldots, a_g)$ of (1.2) is given by $a_k = -\sum_{j=1}^{g} \frac{c_j}{2\pi} b_{jk}$. Thus the correspondence between a and c in (1.2) is given by a linear mapping B which is determined by the conformal structure of M.

Let us observe that if M is an annulus then

$$\operatorname{osc} w = |a_1 - a_0| = 2\mu.$$

To see this, let M be a circular annulus so that $w = c_1 \log|z|$ solves (1.2). Any other harmonic function v with $\int_{\gamma_1} d^c w = \int_{\gamma_1} d^c v = c_1$ is of the form

$$v = c_1 \log|z| + \operatorname{Re} F$$

with F analytic on M. If $F(z)=\sum_{m=-\infty} \alpha_m z^m$ is the Laurent series for F, then the constant term α_0 does not change the variation of v.

It follows that

$$\operatorname{osc}(c_1 \log|z| + \operatorname{Re} F) > \operatorname{osc}(c_1 \log|z|)$$

so the infimum in (1.1) is achieved by $u=-\frac{c_1}{2\pi}\log|z|+\frac{a_1}{2}$.

On the other hand, if M is triply connected, then the solutions of (1.1) and (1.2) have different variation. Let w solve (1.2), and let $w=a_j$ on γ_j with $a_2<a_1<a_0$. Then there is a conformal map $F\colon \overline{M}\to\mathbb{C}$ with the properties

$$\begin{aligned} &\operatorname{Re} F(z)>0 \quad && z\in\gamma_0,\\ &\operatorname{Re} F(z)<0 && z\in\gamma_2,\\ &|F(z)|=1 && z\in\gamma_1. \end{aligned}$$

It follows that $\int_{\gamma_j} d^c(w-\varepsilon \operatorname{Re} F)=c_j$ $j=1,2$ but for $\varepsilon>0$ sufficiently small

$$\operatorname*{osc}_M (w-\varepsilon \operatorname{Re} F) < \operatorname*{osc}_M w,$$

since w is decreased on γ_0 and increased on γ_2.

This shows that for a triply connected domain, the solutions of (1.1) and (1.2) differ by a constant if $c_1=0$ or $c_2=0$ but they do not differ by a constant if $c_1, c_2\neq 0$. Thus the solution $u(c_1, c_2)$ of (1.1) is nonlinear in (c_1, c_2).

The following observation is a corollary of Theorem 1.1.

Theorem 5.1. *Let the C^4 pseudoconvex domain $\Omega\subseteq\mathbb{C}^2$ be given, and let $\partial\Omega$ be strongly pseudoconvex at each point of $\partial\Omega\setminus\overline{M}$. Let M be a locally closed Riemann surface in $\partial\Omega$ with C^2 boundary ∂M, and let M be conformally equivalent to a finitely connected domain in $\mathbb{C}$. Let $c(M,\Omega)\in H^1(M,\mathbb{R})$ be the associated cohomology class, and let w be the solution of* (1.2) *with* $c_j=\int_{\gamma_j} c(M,\Omega)$. *If* $\max_{0\leq j,k\leq g}|a_j-a_k|<\pi$, *then $\overline{\Omega}$ has a neighborhood system of pseudoconvex domains.*

Proof. If $\max|a_j-a_k|<\pi$, then by the maximum principle $\operatorname*{osc}_M w<\pi$ and thus the solution μ of (1.1) is less then π. The existence of pseudoconvex neighborhoods follows by Theorem 4.4 and Lemma 7.3.

Theorem 5.2. *Assume that M is an annulus and that M, Ω satisfy the hypotheses of Theorem* 5.1. *If $|a_0-a_1|>\pi$, then there exists an open set $\tilde{\Omega}\supsetneq\Omega$ such that every pseudoconvex neighborhood of $\overline{\Omega}$ must contain $\tilde{\Omega}$.*

Proof. We will construct a Hartogs domain inside $\overline{\Omega}$. We may assume without loss of generality that $a_1=\frac{\pi}{2}+\varepsilon$ and $a_0=-\frac{\pi}{2}-\varepsilon$. Let f be the defining function for M satisfying (2.1), let u be the solution of (1.2), and let α be defined by (3.1). Thus $\operatorname{Arg}\alpha - u$ has a harmonic conjugate u^*, and so $h=\exp(u^*-i(u-\operatorname{Arg}\alpha))$ is holomorphic on M. We may extend h to be holomorphic in a neighborhood of M. Replacing f by

$F=f/h$, we obtain a new defining function for M satisfying (1.1) (on perhaps a slightly smaller domain), and

$$\partial F(z)=\alpha'(z)\,\partial r(z) \qquad z\in M$$

with $\operatorname{Arg}\alpha'(z)=u(z)$.

Let us define the vector $v(z)=\left(\frac{\partial F(z)}{\partial z_1}, \frac{\partial F(z)}{\partial z_2}\right)$. By the choice of a_1 and a_2 it follows that $v(z)$ points into Ω for $z\in\gamma_1\cup\gamma_2$. Let us set, for $\eta\in\mathbb{R}$,

$$M(\eta)=\{z+\eta v(z): z\in M\}.$$

Then $\eta_0>0$ may be chosen such that

$$\partial M(\eta)=\{z+\eta v(z): z\in\gamma_1\cup\gamma_2\}\subset\bar{\Omega}$$

for $0\leqq\eta\leqq\eta_0$. Since $M(0)\subset\bar{\Omega}$, we conclude that any pseudoconvex open set $\tilde{\Omega}\supset\bar{\Omega}$ must contain $M(\eta)$ for $0\leqq\eta\leqq\eta_0$.

We expect that the conclusion of Theorem 5.2 is still valid if $|a_0-a_1|=\pi$. However we have verified this only in the following special case:

Let $\Omega\subset\mathbb{C}^2$ be the smoothly bounded pseudoconvex set $\{\rho<0\}$ where

$$\rho(z,w)=|w+e^{i\ln z\bar{z}}|^2-1+\lambda\left(\frac{1}{z\bar{z}}-1\right)+\lambda(z\bar{z}-e^{\pi})$$

where $\lambda(x)$ is a sufficiently convex smooth function with $\lambda(x)=0$ for $x\leqq 0$ and $\lambda(x)>0$ when $x>0$. It has been shown in [7] that $M=\{(z,0);\ 1\leqq|z|\leqq e^{\pi/2}\}$ and that Ω is strongly pseudoconvex on $\partial\Omega-M$.

One sees easily that $\partial\rho=e^{i\ln z\bar{z}}\,dw$ on M and hence $c(M,\partial\Omega)=[d^c\ln z\bar{z}]=[2d\operatorname{Arg}z]$ on M. Hence the solution to the problem (1.2) is $w(z)=\ln z\bar{z}-\ln e^{\pi}$. Therefore in this case $|a_0-a_1|=|-\ln e^{\pi}-0|=\pi$.

Fix a sufficiently small number $0<\theta_0\ll 1$. For any $0<\theta<\theta_0$ let $p(\theta)$ denote any complex number $|p(\theta)|<1$, close to the unit circle such that $\ln|p(\theta)|^2=-\theta$. Similarly let $q(\theta)$ denote any complex number with $|q(\theta)|\gtrapprox e^{\pi/2}$ such that $\ln|q(\theta)|^2=\pi+\theta$. Since λ vanishes to infinite order at the origin it is easy to see that

$$\Omega_\theta=\{w\in\mathbb{C};\ (p(\theta),w)\in\Omega \text{ and } (q(\theta),w)\in\Omega\}$$

is a nonempty connected open set.

In fact Ω_θ is an intersection of two discs of radius slightly less than one. Notice that Ω_θ intersects the imaginary axis in an open interval I_θ which contains the point $(0,\sin\theta)$. Hence any point on the positive imaginary axis close to zero is in some I_θ, although the origin is in none of them. Moreover one easily checks that $b\Omega_\theta$ changes continuously with θ.

We define $\tilde{\Omega}=\Omega\cup\bigcup_{0<\theta<\theta_0} V_\theta$ where

$$V_\theta=\{(z,w);\ w\in\Omega_\theta \text{ and } |p(\theta)|<|z|<|q(\theta)|\}.$$

Then $\tilde{\Omega}$ is an open set and it is easy to see that $\tilde{\Omega}\supsetneq\Omega$. We will show that $\tilde{\Omega}$ is contained in the nebenhuelle of $\bar{\Omega}$. So assume f is a holomorphic function on some

small tubular neighborhood of $\overline{\Omega}$. Then already there exists a $0<\theta_1<\theta_0$ such that f is defined on

$$\tilde{\Omega}_{\theta_1}=\Omega\cup\bigcup_{0<\theta<\theta_1}V_\theta.$$

This is because the tubular neighborhood of $\overline{\Omega}$ must contain such V_θ. Of course θ_1 depends on the size of this tubular neigborhood. Next let $\theta_1\leqq\theta_2\leqq\theta_0$ be the largest θ_2 such that f has an analytic continuation to $\tilde{\Omega}_{\theta_2}$. We will show that $\theta_2=\theta_0$ which will complete the proof.

Assume to the contrary that $\theta_2<\theta_0$. The annulus

$$\{(z,w);\ w=i\sin\theta_2 \text{ and } |p(\theta_2)|\leqq|z|\leqq|q(\theta_2)|\}$$

is in the interior of $\tilde{\Omega}_{\theta_2}$. It follows therefore by using Hartogs domains that f has a holomorphic extension to $\tilde{\Omega}_{\theta'}$ for some $\theta_2<\theta'$, which is a contradiction.

So we have shown that $\overline{\Omega}$ has a nontrivial nebenhuelle.

Theorem 5.3. *Let M, Ω satisfy the hypotheses of Theorem* 5.1. *Let M be a triply connected domain with $a_0\leqq a_1\leqq a_2$ and $a_2-a_0>\pi$. If there is a holomorphic mapping $\varphi\colon M\to\mathbb{C}$ such that*

$$\operatorname{Re} e^{-ia_j}\varphi(z)>0 \quad \text{for } z\in\gamma_j,\quad j=0,1,2, \tag{5.1}$$

then Ω has a nontrivial nebenhuelle.

Proof. As was shown in the proof of Theorem 5.2, a defining function F may be chosen to satisfy (2.1) and to have the property $\partial F(z)=\alpha(z)\partial r(z)$

$$\operatorname{Arg}\alpha(z)=u(z)\qquad z\in M$$

where u solves (1.2). Again, we set

$$v(z)=\left(\frac{\partial F(z)}{\partial z_1},\frac{\partial F(z)}{\partial z_2}\right).$$

Thus if $\eta\geqq0$ is chosen sufficiently small

$$z-\eta\beta e^{-ia_j}v(z)\in\overline{\Omega}$$

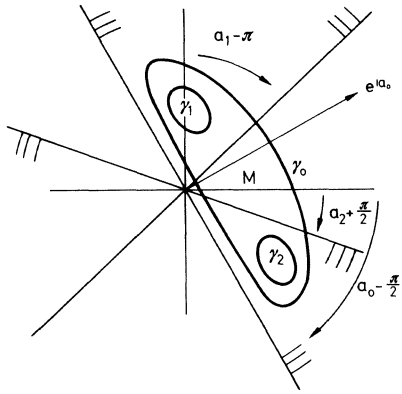
when $z\in\gamma_j$ and $\operatorname{Re}\beta>0$. If we set

$$M(\eta)=\{z-\eta\varphi(z)\,v(z)\colon z\in M\}$$

then the condition on φ assures that $\partial M(\eta)\subset\overline{\Omega}$ for $0\leqq\eta\leqq\eta_0$. Since $M(0)\subseteq\overline{\Omega}$, every pseudoconvex domain containing $\overline{\Omega}$ must also contain $M(\eta)$ for $0\leqq\eta\leqq\eta_0$.

Remark. There are three possibilities for the mapping $\varphi\colon M\to\mathbb{C}$, one of which is sketched here; the other two are obtained by inversions.

The fact that this mapping is not always possible corresponds to the fact that it is possible that $\operatorname{osc}_M w<\pi$ where w solves (1.1) (and thus $\overline{\Omega}$ has Stein neighborhoods) but that $\operatorname{osc}_M u>\pi$.



Theorem 5.4. *Let M and Ω satisfy the hypotheses of Theorem* 5.1. *Let M be an arbitrary multiply connected domain, and let $a_0 \leqq a_1 \leqq \cdots \leqq a_g$. If there exists j such that $a_{j+1} - a_j > \pi$, then Ω has a nontrivial Nebenhuelle. In particular, this occurs if*

$$a_g - a_0 = \operatorname{osc} u > g\pi.$$

Proof. This is proved by repeating the proof of Theorem 5.2, using the set $M' = \{z \in M: a_j < u(z) < a_{j+1}\}$. M' may be treated as an annulus, because $\partial M'$ consists of the two components $\gamma'_j = \{z \in \overline{M}: u(z) = a_j\}$ and $\gamma'_{j+1} = \{z \in \overline{M}: u(z) = a_{j+1}\}$.

Corollary 5.5. *Let M be a multiply connected domain, and let real numbers $c_1, \ldots, c_g$ be given. If w solves* (1.1), *and u solves* (1.2), *then*

$$\operatorname*{osc}_M w \leqq \operatorname*{osc}_M u \leqq g \operatorname*{osc}_M w.$$

Proof. The first inequality has already been observed. To establish the second inequality we observe that if $c_1, \ldots, c_g$ are multiplied by $\lambda > 0$, then $\operatorname*{osc}_M w$ is also multiplied by λ. Thus we may assume that $\operatorname*{osc}_M w < \pi$. By Proposition 3.3, there are Ω and M generated by w. By Theorem 4.4, $\overline{\Omega}$ has a Stein neighborhood system, and so by Theorem 5.4, $\operatorname*{osc}_M u < g\pi$.

Let us construct some examples to show what may happen if M is not schlicht. Let $\tilde{M}$ be a compact Riemann surface, and let M be a domain in $\tilde{M}$ with finitely many smooth boundary components. Then there is a neighborhood M_1, $\overline{M} \subset M_1 \subset \tilde{M}$ which may be imbedded in $\mathbb{C}^2$. Let $u: M \to \mathbb{R}/\mathbb{Z}$ be a harmonic function which is C^2 on $\overline{M}$. By Proposition 3.3 the function u may be used to construct a pseudoconvex domain $\Omega \subset \mathbb{C}^2$ with $\overline{M} \subset \partial\Omega$, which is strictly pseudoconvex on $\partial\Omega \setminus \overline{M}$.

Let us suppose that $M \setminus u^{-1}(0)$ has a relatively compact connected component Σ, and that there is a lifting $\tilde{u}: \bar{\Sigma} \to \mathbb{R}$ of $u|_\Sigma: \Sigma \to \mathbb{R}/\mathbb{Z}$. It follows that any pseudoconvex set $\tilde{\Omega}$ containing $\Omega \cup \bar{\Sigma}$ must contain a larger set. Let $\sigma_1, \ldots, \sigma_q$ be the boundary components of Σ. Since $\sigma_j \subset u^{-1}(0)$, it follows that $\tilde{u}|_{\sigma_j}$ is an integer n_j, and not all the n_j's are the same. By taking a subset of Σ if necessary, we may suppose that $\tilde{u}|_{\sigma_j} = 0$ if $1 \leqq j \leqq r$ and $\tilde{u}|_{\sigma_j} = 1$ if $r+1 \leqq j \leqq q$. Thus $\Sigma = \{0 < \tilde{u} < 1\}$ may be treated

as an annulus, and the existence of a nontrivial nebenhuelle for $\Omega \cup \bar{\Sigma}$ may be derived by repeating, in essence, the proof of Theorem 5.2.

We consider the case where $\tilde{M}$ is a torus; for simplicity we take $\tilde{M} = \mathbb{C}/\mathbb{Z} + i\mathbb{Z}$. The harmonic functions

$$u\colon \tilde{M} \to \mathbb{R}/\mathbb{Z}$$

are of the form $u(z) = mx + ny$, $m, n \in \mathbb{Z}$: Let us set

$$\Sigma = \{z \in \tilde{M}\colon\ u(z) \in (2k, 2k+1) \text{ for some } k \in \mathbb{Z}\}.$$

If $M \supset \bar{\Sigma}$, then the domain Ω formed from M and u will not have a Stein neighborhood basis.

On the other hand many examples can be constructed where there is a Stein neighborhood basis. Let $\Gamma \subset \tilde{M} \setminus \{p_1, \ldots, p_r\}$ be a union of smooth closed curves, and let $u\colon M_1 \to \mathbb{R}/\mathbb{Z}$ be harmonic on some neighborhood M_1 of Γ. It is clear that $[d^c u]$ defines the "same" cohomology class on all neighborhoods $\Gamma \subset M \subset M_1$, and by making M sufficiently small, we may make the μ of (1.1) as small as we wish (since Γ is totally real). It follows by Theorem 4.6 that for sufficiently small $M \supset \Gamma$ $\left(\text{i.e. such that } \mu < \frac{\pi}{2}\right)$ the domain $\bar{\Omega}(M, u)$ will have a Stein neighborhood basis.

§ 6. A Foliation in the Boundary

If Ω is a pseudoconvex domain and an open set $N \subset \partial\Omega$ is foliated by complex submanifolds, then the problem of finding a transverse holomorphic vector on $\bar{N}$ is different from the case of a single manifold in the boundary. In this section we will assume that Ω is strongly pseudoconvex at all points of $\partial\Omega \setminus \bar{N}$ and that N has a foliation $\mathcal{M}$ by complex submanifolds of codimension 1. If the leaves of $\mathcal{M}$ are locally closed, then there is always a transverse vector field on each leaf of $\mathcal{M}$ (Proposition 6.1). We show that a pseudoconvex neighborhood basis can be found if $\mathcal{M}$ takes the especially simple form (6.1) and $\bar{N}$ satisfies (6.2). But in general, a leaf is not locally closed, and we can no longer write it as the zero set of a holomorphic function. For this case we can only give examples.

We begin with an observation on the behavior of N in a neighborhood of a closed leaf. We assume N is real analytic. So for $z_0 \in N$, there exists an open set U, $z_0 \in U$, and a function $F\colon (-\varepsilon, \varepsilon) \times U \to \mathbb{C}$ with the properties

$$N \cap U = \bigcup_{-\varepsilon < t < \varepsilon} M(t), \quad \text{where } M(t) = \{z \in U\colon F(t, z) = 0\}$$

$$F \in C((-\varepsilon, \varepsilon) \times \bar{U}),\ d_t F \neq 0 \quad \text{and} \quad F \text{ is holomorphic in } z. \tag{6.1}$$

Proposition 6.1. *Let $\mathcal{M}$ be a foliation of N and let M be a leaf of $\mathcal{M}$. If there is an open subset U of N containing M and a function F such that* (6.1) *holds, then $c(M, \partial\Omega) = 0$. If M is a strongly pseudoconvex domain with C^2 boundary in some Stein manifold $\tilde{M}$, then there is a transverse holomorphic vector field in a neighborhood of $\bar{M}$.*

Proof. If $c(M, \partial\Omega) \neq 0$ then there is a relatively compact open subset $M' \subset M$ such that $c(M', \partial\Omega) \neq 0$. We will extend the foliation $\mathcal{M}$ to a neighborhood of M'. Let us set $\zeta = t + ir$ and

$$F(\zeta, z) = F(t, z) + irF_t(0, z) - \tfrac{1}{2}r^2 F_{tt}(0, z).$$

Thus $\dfrac{\partial F}{\partial \bar{\zeta}}$ vanishes to second order on M. It is easily seen that there is an open set $\tilde{U}$ containing $\overline{M}'$ and a small $\varepsilon' > 0$ such that the leaves

$$M(\zeta) = \{z \in \tilde{U} : F(\zeta, z) = 0\} \quad |\zeta| < \varepsilon'$$

form a foliation $\tilde{\mathcal{M}}$ of $\tilde{U}$, (compare with Theorem B of [2]). We define the function $\varphi(z) = \zeta$ if $z \in M(\zeta)$, and $r = \operatorname{Im} \varphi$. It follows that φ satisfies (1.1), and $\partial\varphi = 2i\partial r$ which shows that $c(M', \partial\Omega) = 0$.

If M is strongly pseudoconvex with smooth boundary inside $\tilde{M}$, it follows from Theorem 4.6 that there is a holomorphic vector field v which is transverse to $\partial\Omega$ in a neighborhood of $\overline{M}$.

Proposition 6.2. *Let $N \subset \mathbb{C}^n$ be a Levi-flat hypersurface with $N \subset \{r = 0\}$, $r \in C^{4n+1}$, $dr \neq 0$ on $\bar{N}$. Assume that $\bar{N}$ has a neighborhood basis*

$\{\omega_j\}$ *of Stein open sets such that for some constant c*

$$\left\{z : \operatorname{dist}(z, \bar{N}) < \frac{1}{cj}\right\} \subset \omega_j \subset \left\{z : \operatorname{dist}(z, \bar{N}) < \frac{c}{j}\right\}. \tag{6.2}$$

If v is a CR-vector field of class C^{4n+1} which is transverse to $\partial\Omega$ on $\bar{N}$, then there is a transverse vector field holomorphic in a neighborhood of $\bar{N}$.

Proof. Following the method of Hörmander [14], we may extend each coordinate of v to a neighborhood of $\bar{N}$ in such a manner that $\bar{\partial} v$ vanishes to $(2n+1)$-st order on N. Since v is transverse on $\bar{N}$, there is an $\varepsilon > 0$ such that any vector field ξ with $|\xi - v| < \varepsilon$ is also transverse. Now there is a constant k_1 such that

$$\Big(\int_{\omega_j} |\bar{\partial} v|^2\Big)^{1/2} \leqq k_1 j^{-1/2} \sup_{\omega_j} |\bar{\partial} v| \leqq c^{2n} k_1 j^{-2n-1/2}.$$

We will use the fact that the $\bar{\partial}$-operator may be solved in $L^2(\omega)$ with the bound k_2 on the solution depending only on the diameter of ω. Thus there exists h_j with $\bar{\partial} h_j = \bar{\partial} v$ on ω_j and

$$\Big(\int_{\omega_j} |h_j|^2\Big)^{1/2} \leqq c^{2n} k_1 k_2 j^{-2n-1/2}.$$

We recall that there is a constant k_3 such that for any function u on $B(\delta) = \{z \in \mathbb{C}^n : |z| < \delta\}$

$$|u(0)| \leqq k_3 \Big(\varepsilon^{-2n} \Big(\int_{B(\delta)} u^2\Big)^{1/2} + \varepsilon \sup_{B(\delta)} |\bar{\partial} u|\Big).$$

It follows, then, by (6.2) that for some constant C

$$\sup_{z\in\bar{N}}|h_j(z)| \leqq C(j^{2n}j^{-2n-1/2}+j^{-1}j^{-2n-1/2}).$$

Thus if j is chosen such that $2Cj^{-1/2}<\varepsilon$ then $v-h_j$ is holomorphic on ω_j and transverse on $\bar{N}$.

Theorem 6.3. *Let Ω be a pseudoconvex domain with C^{4n+1} boundary. Let the Levi form vanish on $\bar{N}\subset\partial\Omega$, and let $\partial\Omega\setminus\bar{N}$ be strongly pseudoconvex. Let N satisfy* (6.1) *and* (6.2), *and let each leaf M of $\mathcal{M}$ be a strongly pseudoconvex domain with C^4-boundary in some Stein manifold $\hat{M}$. If $\bar{N}\setminus\bigcup_{\mathcal{M}}\overline{M}$ is totally real, then $\bar{\Omega}$ has a neighborhood system of strongly pseudoconvex sets.*

Proof. We show that there is a holomorphic vector v transverse to $\partial\Omega$ in a neighborhood of $\bar{N}$. The neighborhoods of $\bar{\Omega}$ are obtained by Lemma 7.2. By Proposition 6.1 there is a holomorphic vector field in a neighborhood of $\overline{M}$ for each leaf $M\in\mathcal{M}$. Since $\Gamma=\bar{N}\setminus\bigcup_{M\in\mathcal{M}}\overline{M}$ is totally real, we may approximate the normal vector Vr by a vector field holomorphic in a neighborhood of Γ. In a neighborhood of the closure of any leaf $\overline{M}$ we may obtain (by (6.1)) cutoff functions $\chi=\chi(t)$. Thus by a partition of unity, we may obtain a transverse vector field $v=\Sigma\chi_j v_j$ of class $C^{4n+1}(\bar{N})$ whose components are CR-functions on N. It follows by Proposition 6.2 that the desired holomorphic vector field exists.

The following corollary is a special case of Theorem 6.3.

Corollary 6.4. *Let Ω be a smoothly bounded pseudoconvex domain in $\mathbb{C}^{n+1}$, let $N\subset\partial\Omega$ be an open subset of the boundary, and let Ω be strongly pseudoconvex at all points of $\partial\Omega\setminus\bar{N}$. Let there be an open subset $\Sigma\subset\mathbb{C}^n\times\mathbb{R}$ such that each slice $\bar{\Sigma}(t)=\{z\in\mathbb{C}^n: (z,t)\in\bar{\Sigma}\}$ is the closure of a strongly pseudoconvex domain in $\mathbb{C}^n$ or else is totally real and such that there is a CR-equivalence $F:\Sigma\to N$ which is smooth on the closure. It follows that $\bar{\Omega}$ has a neighborhood basis of pseudoconvex sets.*

Now we look at an example. Let us consider

$$S_R=\{(z,w)\in\mathbb{C}^2: 1\leqq|z|\leqq R,\ |w|\leqq 1 \text{ and } \mathrm{Re}(we^{i\log z\bar{z}})=0\},$$

which is a Levi flat hypersurface in $\mathbb{C}^2$. We will show that there is a pseudoconvex domain $\Omega\subset\mathbb{C}^2$ such that $S_R\subset\partial\Omega$ and $\partial\Omega\setminus S_R$ is strongly pseudoconvex. S_R is foliated by leaves of the form

$$M(c)=\{w=cz^{-2i}: 1<|z|<R,\ |w|<1 \text{ and } \mathrm{Re}\,c=0\}$$

which are solution curves to the differential equation $dw=-2icwz^{-1}dz$. The only locally closed leaf of S_R is the annulus $M(0)$. If we apply the proof of Theorem 5.2 to $M(0)$, we see that S_R has a nontrivial Nebenhuelle if $R\geqq e^{\pi}$. Thus $N=S_R$ satisfies all the hypotheses of Theorem 6.3 except (6.1).

To show how S_R may be "capped off" to lie in the boundary of a pseudoconvex set, we introduce smooth functions $\lambda(t)$ and $\mu(t)$ such that.

$\lambda(t)=0$ if $1\leqq t\leqq R^2$, $\lambda''(t)+t\lambda'(t)>0$

if $t\notin[1,R]$, and $\lim_{t\to 1/2}\lambda(t)=\infty$

$\mu(t)=0$ for $t\leqq 1$

$$2t\mu'(t)>\mu(t), \quad t>1 \tag{6.3}$$

$$\mu''(t)>(16+t)\mu'(t), \quad t>1 \tag{6.4}$$

If we let λ and μ be any such functions (which clearly exist) and we set

$$\Omega=\{(z,w)\in\mathbb{C}^2\colon \lambda(z\bar{z})+\mu(w\bar{w})+\mathrm{Re}(we^{i\log z\bar{z}})<0\}$$

then $S_R\subset\partial\Omega$, and (6.3) guarantees that $\partial\Omega$ is smooth. By condition (6.4), Ω may be shown to be strongly pseudoconvex at all points of $\partial\Omega\setminus S_R$.

To exhibit some boundaries with more complicated foliations, we use a construction suggested to us by W. Thurston. We consider the torus $T^n\subset\mathbb{C}^n$ embedded as the distinguished boundary of the unit polydisk

$$T^n=\{z\in\mathbb{C}^n\colon |z_1|=\cdots=|z_n|=1\}.$$

On T^n we use coordinates $(\theta_1,\ldots,\theta_n)$ corresponding to $(e^{i\theta_1},\ldots,e^{i\theta_n})$, and thus $d\theta_j=-ie^{-i\theta_j}dz_j$. Let $\mathscr{F}$ be a real analytic foliation on T^n generated by the real analytic one-form $\omega=\sum_{j=1}^n f_j(\theta)\,d\theta_j$, and let $\tilde{\omega}=\sum_{j=1}^n F_j(z)\dfrac{dz_j}{z_j}$, where $F_j(z)$ is the unique holomorphic function extending $f_j(\theta)$ from T^n. Now we define the surface S by complexifying the leaves of $\mathscr{F}$, i.e. S is the union of the leaves of $\tilde{\mathscr{F}}$ that intersect T^n. Clearly, S is smooth near T^n, and we may define S by $S\cap U=\{z\in U\colon r(z)=0\}$ $dr\neq 0$ on S for some neighborhood U of T^n. Since $\partial r\wedge\bar{\partial}r=\lambda\tilde{\omega}\wedge\bar{\tilde{\omega}}$ for some scalar λ, and since $\dfrac{\partial r(z)}{\partial\theta_j}=0$ for $z\in T^n$, we may assume $\lambda=1$ and it follows that $\partial r=\tilde{\omega}$ on U. If U is sufficiently small, we may find holomorphic functions $v_1(z),\ldots,v_n(z)$ on U such that

$$\sum_{j=1}^n \frac{F_j(z)}{z_j}v_j(z)=1$$

and thus $v=(v_1,\ldots,v_n)$ is a holomorphic transverse vector field to S on U.

For example, we may take

$$\omega=\sum_{j=1}^n a_j d\theta_j=\sum_{j=1}^n \frac{a_j}{z_j}dz_j$$

to be a constant 1-form, with a_j real. The leaves of the induced foliation $\tilde{\mathscr{F}}$ are of the form

$$M(c)=\{z_1^{a_1}\ldots z_n^{a_n}=c\}$$

and $S=\{|z_1|^{a_1}\ldots|z_n|^{a_n}=1\}$ is foliated by $M(e^{i\theta})$. The vector field

$$v(z)=(a_1z_1,\ldots,a_nz_n)$$

is transverse to S.

It is possible to construct a domain Ω which is pseudoconvex and contains a portion of S in its boundary. Let $\varepsilon>0$ be chosen such that

$$\{z\in\mathbb{C}^n\colon 1-\varepsilon<|z_j|<1+\varepsilon,\ j=1,2,\ldots,n\}\subset U,$$

and let $\lambda\in C^\infty(1-\varepsilon,1+\varepsilon)$ have the properties:

$$\lambda''(t)+t\lambda'(t)>0,\qquad \lambda\geqq 0$$
$$\lim_{t\to 1\pm\varepsilon}\lambda(t)=\infty$$
$$\lambda(t)=0\qquad t\in(1-\varepsilon/2,1+\varepsilon/2).$$

It may be shown that for suitable $k>0$

$$\Omega(k)=\left\{z\in\mathbb{C}^n\colon r(z)+k\sum_{j=1}^{n}\lambda(z_j\bar{z}_j)<0\right\}$$

is a smooth pseudoconvex domain such that

$$S(\varepsilon/2)=\{z\in\mathbb{C}^n\colon 1-\varepsilon/2\leqq|z_j|\leqq 1+\varepsilon/2\}\cap S$$

is contained in $\partial\Omega$, and $\partial\Omega\setminus S(\varepsilon/2)$ is strongly pseudoconvex.

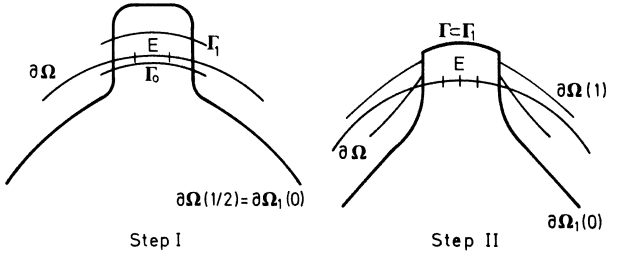
§ 7. Applications to Holomorphic Convexity and Approximation

This section contains a proof of Theorem 1.2. This uses Lemma 7.1 which gives the existence of an increasing 1-parameter family of strongly pseudoconvex domains which start inside Ω and ultimately contain $\bar{\Omega}$. These methods also yield some approximation theorems.

Lemma 7.1. *Let $\Omega\subset\mathbb{C}^n$ be a pseudoconvex domain with C^2 boundary, $E\subset\partial\Omega$, and let Ω be strongly pseudoconvex at $\partial\Omega\setminus E$. Let v be a holomorphic vector field defined in a neighborhood of E which is transverse to $\partial\Omega$. Given a compact set $K\subset\Omega$, there are domains $\Omega(t)$ $0\leqq t\leqq 1$ with the properties*

(i) $K\subset\Omega(0)\subset\Omega\subset\bar{\Omega}\subset\Omega(1)$,
(ii) $\partial\Omega(t)$ is piecewise smooth and strongly pseudoconvex,
(iii) $\Omega(t_1)\subset\Omega(t_2)$ if $t_1\leqq t_2$,
(iv) $\Omega(t)=\bigcup_{s<t}\Omega(s)=\operatorname{int}\bigcap_{s>t}\Omega(s)$.

Proof. We will proceed in two steps; the first is to push a strictly pseudoconvex surface out of Ω in a neighborhood of E. The second step is to make the surfaces $\Omega(t)$ engulf $\partial\Omega\setminus E$.



Let $E \subset V_1 \Subset V_2 \Subset V_3$, where V_1, V_2, V_3 are open sets such that v is defined on $\bar{V}_3$. Let us assume that for some $c>0$ the map

$$\Phi\colon (\bar{V}_3 \cap \partial\Omega) \times [-c, c] \to \mathbb{C}^n$$

given by $\Phi(z, s) = z + sv(z)$ is one-to-one and that

$$\Phi(\bar{V}_3 \cap \partial\Omega, s) \cap \bar{V}_3 = \emptyset$$

for $|s| \geqq c$. Let $r \in C^2(\mathbb{C}^n)$ be a defining function for $\Omega = \{r<0\}$ with $|\nabla r| \neq 0$ on $\partial\Omega$. Then there exists $\varepsilon' > 0$ such that for every function $h(z)$ with $\|h\|_{C^2} < \varepsilon'$ the surface $S = \{z \in \mathbb{C}^n\colon r(z) = h(z)\}$ is strongly pseudoconvex at all points of $S \setminus V_1$.

Let $\chi_1(z)$ be a smooth function such that $\chi_1 \equiv 1$ on V_1 and $\chi_1 \equiv -1$ on $\mathbb{C}^n \setminus V_2$, $-1 \leqq \chi \leqq 1$. Let us define

$$\Omega_1(\varepsilon) = \{z \in \mathbb{C}^n\colon r(z) < \varepsilon\chi_1(z)\}.$$

By a theorem of Diederich and Fornæss [8], there is a smooth defining function $\rho(z)$ for Ω such that $-(-\rho)^\alpha$ is strongly pseudoconvex for some $\alpha > 0$. For $\delta > 0$ small,

$$\Gamma(\delta) = \{z \in V_3\colon \rho(z) = -\delta\}$$

is strongly pseudoconvex, and v is transverse to $\Gamma(\delta)$. We define

$$\Omega_2(\delta, \eta) = \{z + sv(z)\colon z \in \Gamma(\delta), \eta \geqq s > -c\} \cap \bar{V}_2$$
$$\Gamma(\delta, \eta) = \{z + \eta v(z)\colon z \in \Gamma(\delta)\} \cap \bar{V}_2.$$

Observe that for each $\varepsilon > 0$

$$V_1 \cap \partial\Omega \subset \Omega_1(\varepsilon) \quad \text{and} \quad (V_3 \setminus V_2 \cap \bar{\Omega}_1(\varepsilon)) \subset \Omega. \tag{7.1}$$

By the first inclusion there exists $\eta_0 > 0$ such that if $-c \leqq \eta < \eta_0$, then

$$\Phi(V_3 \cap \partial\Omega, \eta) \cap V_1 \subset \Omega_1(\varepsilon),$$

and by the second inclusion η_0 may be chosen so that if $-\eta_0 < \eta < \eta_0$ then

$$(V_3 \setminus V_2) \cap \bar{\Omega}_1(\varepsilon) \cap \Phi(V_3 \cap \partial\Omega, \eta) = \emptyset.$$

It follows that $\delta = \delta(\eta_0)$ can be chosen small enough that

$$\Gamma(\delta, \eta) \cap V_1 \subset \Omega_1(\varepsilon) \quad \text{and} \quad (V_3 \setminus V_2) \cap \bar{\Omega}_1(\varepsilon) \cap \Gamma(\delta, \eta) = \emptyset. \tag{7.2}$$

Thus for $|\eta| \leqq \eta_0$ and $0 < \delta < \min(\delta_0, \delta(\eta_0))$ we see that

$$\Omega_3(\varepsilon, \delta, \eta) = [\Omega_2(\delta, \eta) \cap \Omega_1(\varepsilon)] \cup (\mathbb{C}^n \setminus V_2) \cap \Omega_1(\varepsilon)$$

is a pseudoconvex domain. In fact the boundary of $\Omega_3(\varepsilon, \delta, \eta)$ consists of portions of $\Gamma(\delta, \eta)$ and $\partial\Omega_1(\varepsilon) \setminus V_1$ and is thus piecewise smooth and strongly pseudoconvex. Finally we observe that for δ sufficiently small $E \subset \Omega_3(\varepsilon, \delta, \eta_0)$. To complete the first step of the proof, we take $\Omega(t) = \Omega_3(\varepsilon, \delta, (2t-1)\eta_0)$ $0 \leqq t \leqq 1/2$. Observe that by taking $c > 0$ small enough $\Omega(0)$ contains K.

The second step of the proof is similar to the first. Set $\tilde{\Gamma} = \Gamma(\delta, \eta_0) \cap \partial\Omega(1/2)$. By (7.2), $\tilde{\Gamma} \subset V_2 \cap (\mathbb{C}^n \setminus \bar{\Omega}) \cap \Omega_1(\varepsilon)$, and it follows that $\chi_1(z) > 0$ for $z \in \tilde{\Gamma}$. Let W_1, W_2 be open sets such that

$$V_1 \cup \tilde{\Gamma} \subset W_1 \Subset W_2 \Subset \{z \in \mathbb{C}^n : \chi_1(z) > 0\} \Subset V_2,$$

and let $\chi_2(z)$ be a smooth function such that $\chi_2 \equiv 0$ in a neighborhood of $\overline{W}_1$, $0 \leqq \chi_2 \leqq 1$, and $\chi_2 \equiv 1$ on W_2. We define the domains, for $1/2 \leqq t \leqq 1$:

$$\Omega(t) = [W_1 \cap \Omega(1/2)] \cup \{z \in \mathbb{C}^n \setminus W_1 : r(z) < \varepsilon\chi_1(z) + 4\varepsilon(t - 1/2)\chi_2(z)\}.$$

By the definition of χ_1 and χ_2 and by (7.1) it follows that $\Omega(1) \supset \bar{\Omega}$. To check pseudoconvexity, we observe that only the boundary of the second set changes, but the change occurs away from the closure of the first set. The part of $\partial\Omega(t)$ that is in the closure of the second set is

$$\{z \in \mathbb{C}^n \setminus W_1 : r(z) - \varepsilon\chi_1(z) - 4(t - 1/2)\chi_2(z) = 0\}.$$

Thus if we select ε so that $\|\varepsilon\chi_1 + 2\varepsilon\chi_2\|_{C^2} \leqq \varepsilon'$, then this boundary is strongly pseudoconvex. By the construction, properties (iii) and (iv) are immediate.

Theorem 1.2 is a consequence of the following.

Theorem 7.2. *Let Ω and E be as in Lemma* 7.1. *If K is a compact subset of Ω, then*

$$\{z \in \bar{\Omega} : |f(z)| \leqq \max_K |f| \quad \text{for all } f \in \mathcal{O}(\bar{\Omega})\}$$
$$= \{z \in \Omega : |f(z)| \leqq \max_K |f| \quad \text{for all } f \in \mathcal{O}(\Omega)\}.$$

Proof. For each $p \in \partial\Omega$ we will produce a function $f = f_p \in \mathcal{O}(\bar{\Omega})$ such that $|f(p)| > \sup_K |f|$. If $p \in \partial\Omega \setminus E$, f_p is known to exist (see Gunning and Rossi [11]), so we consider $p \in E$. By Lemma 7.1 there is a family $\{\Omega(t)\}$ of domains satisfying (i), ..., (iv). Let t_0 be the smallest value of t such that $p \in \bar{\Omega}(t_0)$. By the construction of $\Omega(t)$, p lies in the portion of $\partial\Omega(t_0)$ which is contained in $\Gamma(\delta, \eta)$. Since $\Gamma(\delta, \eta)$ is strongly pseudoconvex, there is a function $h_p \in \mathcal{O}(\overline{\Omega(t_0)})$ which peaks at p. Now the domains $\{\Omega(t)\}$ satisfy the hypotheses of a theorem of Docquier and Grauert [9], and so $\mathcal{O}(\Omega(1))$ is uniformly dense in $\mathcal{O}(\overline{\Omega(t_0)})$. Since h_p peaks at p and $K \subset \Omega(t_0)$, we may take $f_p \in \mathcal{O}(\Omega(1))$ such that

$$2\sup_{\Omega(t_0)} |f_p - h_p| < 1 - \sup_K |h_p|.$$

This shows that the first hull above does not intersect $\partial\Omega$. To see that the two hulls are equal, suppose $p \in \Omega$ and there exist $f \in \mathcal{O}(\Omega)$ with $|f(p)| > \sup_K |f|$. We may assume that $\{p\} \cup K$ is a compact subset of $\Omega(0)$. Since $\Omega(0)$ is Runge in $\Omega(1)$, we may let $h \in \mathcal{O}(\Omega(1))$ approximate f uniformly on $\{p\} \cup K$. Thus there exists $h \in \mathcal{O}(\bar{\Omega})$ such that $|h(p)| \geqq \sup_K |h|$.

By reworking part of the proof of Lemma 7.1, we obtain special pseudoconvex neighborhood systems (see Fornæss and Nagel [10]). Any of these special neighborhoods can be bumped out to give a neighborhood of $\bar{\Omega}$.

Lemma 7.3. *Let Ω and E be as in Lemma 7.1. Then $\Omega \cup E$ has a fundamental neighborhood system $\{\Omega_j\}$ of strongly pseudoconvex domains with smooth boundaries. Furthermore, given a neighborhood U of E, the $\{\Omega_j\}$ may be chosen so that $(\mathbb{C}^n \setminus U) \cap \partial\Omega_j = (\mathbb{C}^n \setminus U) \cap \partial\Omega$.*

Proof. We redo the first part of Step I, but this time we set

$$\Omega_1(\varepsilon_1, \varepsilon_2) = \{z \in \mathbb{C}^n : r(z) = \varepsilon_1(\chi_1(z) + 1) + \varepsilon_2\}.$$

For $\eta > 0$, we set $\Gamma(\eta) = \Phi(\bar{V}_3 \cap \partial\Omega, \eta)$ and $\Omega_2(\eta) = \Phi(\bar{V}_3 \cap \partial\Omega, (-c, \eta)) \cap V_2$. Choose $\varepsilon_1, \eta > 0$ small enough that $\|\varepsilon_1 \chi_1\|_{C^2} < \varepsilon'/2$ and $\Gamma(\eta) \cap V_1 \subset \Omega_1(\varepsilon_1, 0)$. Then since $\Gamma(\eta) \cap (V_3 \setminus V_2) \cap \bar{\Omega} = \emptyset$, $\varepsilon'/2 > \varepsilon_2 > 0$ may be chosen small enough that $\Gamma(\eta) \cap (V_3 \setminus V_2) \cap \bar{\Omega}_1(\varepsilon_1, \varepsilon_2) = \emptyset$. It follows, as before, that

$$\Omega_3 = [\Omega_1(\varepsilon_1, \varepsilon_2) \cap (\mathbb{C}^n \setminus V_2)] \cup [\Omega_2(\eta) \cap \Omega_1(\varepsilon_1, \varepsilon_2) \cap V_2]$$

is piecewise strongly pseudoconvex, i.e., is of the form $\Omega_3 = \{\max(r_1, r_2) < 0\}$ where r_j is strictly plurisubharmonic in a neighborhood of $\{z \in \partial\Omega_3 : r_j = 0\}$.

Let $\delta > 0$ be chosen such that

$$\{z \in \Omega_3 : |r_j(z)| < \delta, j = 1, 2\} \subset U.$$

Let χ be a smooth nonnegative function such that $\chi(t) = 0$ for $t \leqq -\delta$, $\chi(0) = 1$, and χ is strictly convex for $t > -\delta$. The set

$$\tilde{\Omega}_3 = \{z \in \Omega_3 : \chi(r_1(z)) + \chi(r_2(z)) < 1\}$$

is just Ω_3 with the corner "rounded off" and has the desired properties.

For $0 \leqq j \leqq \infty$, we define $A^j(\Omega) = \mathcal{O}(\Omega) \cap C^j(\bar{\Omega})$ and give it the topology of a closed subspace of $C^j(\bar{\Omega})$.

Theorem 7.4. *Let Ω and E be as in Lemma 7.1. Let $0 \leqq j \leqq \infty$ and let $\partial\Omega$ be of class C^{j+4}. Then $\mathcal{O}(\bar{\Omega})$ is dense in $A^j(\Omega)$. If $\partial\Omega$ is of class C^4, then $\mathcal{O}(\bar{\Omega})$ is boundedly dense in $H^\infty(\Omega)$, i.e., if $f \in H^\infty(\Omega)$, then there is a sequence $\{f_j\}$ in $\mathcal{O}(\bar{\Omega})$ which converges to f uniformly on compact subsets and such that $\sup_\Omega |f| \geqq \sup_\Omega |f_j|$.*

Proof. Let $\chi \in C_0^\infty(\mathbb{C}^n)$ be equal to 1 in a neighborhood of E, and let v be holomorphic in a neighborhood of supp χ. By Lemma 7.3 there is a strictly pseudoconvex domain $\tilde{\Omega} \supset \Omega$ with smooth boundary such that

$$\{z \in \partial\tilde{\Omega} : \chi(z) < 1\} = \{z \in \partial\Omega : \chi(z) < 1\}$$

and $\tilde{\Omega}$ may be taken to be of class C^{j+4}. Now we proceed as in Theorem 4.3 and let $\{\chi_j\}$ be functions such that $\{\chi, \chi_j\}$ is a partition of unity for $\tilde{\Omega}$ and ξ_j is a transverse vector to $\partial\Omega$ on $\partial\Omega \cap \text{supp}\, \chi_j$. The function

$$T_\varepsilon f = \chi(z) f(z - \varepsilon v(z)) + \sum \chi_j(z) f(z - \varepsilon \xi_j)$$

is defined in a neighborhood of $\bar{\Omega}$, and if $f \in A^j(\Omega)$, then $\bar{\partial} T_\varepsilon f \to 0$ in $C^j(\bar{\Omega})$ as $\varepsilon \to 0$. Furthermore $\bar{\partial} T_\varepsilon f = 0$ on the interior of $\{z \in \Omega : \chi(z) = 1\}$ so we may extend $\bar{\partial} T_\varepsilon f$ by zero to $\tilde{\Omega}$. If $r \in C^{j+4}(\mathbb{C}^n)$ is a defining function for $\tilde{\Omega}$, then $\bar{\partial} T_\varepsilon f$ is defined on $\tilde{\Omega}(\delta) = \{r(z) < \delta\}$ for $0 < \delta$ sufficiently small (depending on ε). Let $\delta(\varepsilon) = \delta$ be chosen so

that

$$\|\bar{\partial} T_\varepsilon f\|_{C^j(\tilde{\Omega}(\delta))} \leqq 2\|\bar{\partial} T_\varepsilon f\|_{C^j(\tilde{\Omega})}.$$

By Siu [20], there exists g_ε such that $\bar{\partial} g_\varepsilon = \bar{\partial} T_\varepsilon f$ on $\tilde{\Omega}(\delta)$ and

$$\|g_\varepsilon\|_{C^j(\tilde{\Omega}(\delta))} \leqq C\|\bar{\partial} T_\varepsilon f\|_{C^j(\tilde{\Omega}(\delta))}.$$

Thus $T_\varepsilon f - g_\varepsilon \in \mathcal{O}(\overline{\Omega})$ and converges to f in $C^j(\overline{\Omega})$.

The bounded approximation result follows similarly, using the kernel $K(z, \zeta)$ as in Theorem 4.3.

Under the hypotheses of Theorem 7.4, it follows from Čirka [4] that $\mathcal{O}(\overline{\Omega})$ is dense in $A^j(\Omega)$ in the $A^{j-n-1}(\Omega)$-topology.

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On Exponential Sums in Finite Fields, II

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I. Introduction

We follow in this introduction Serre's report [S]. Let p be a prime and let k denote a finite field with $q=p^a$ elements. Let als $\bar{k}$ be an algebraic closure of k and for $n\geqq 1$ let k_n be the intermediate field, $k\subseteq k_n\subseteq\bar{k}$, of degree n over k. Let $\mathrm{Tr}_n(x)$ be the trace

$$\mathrm{Tr}_n(x)=x+x^p+\cdots+x^{p^{an-1}} \tag{1.1}$$

from k_n to the prime field $\mathbf{F}_p$ and let $\psi_{k_n}(x)$ be the additive character

$$\psi_{k_n}(x)=e^{2\pi i\,\mathrm{Tr}_n(x)/p}; \tag{1.2}$$

every additive character of k_n is of type $\psi_{k_n}(cx)$ with $c\in k_n$.

Let V be an algebraic variety over k and let f be a regular function on V, again defined over k. Denoting by $V(k)$ the set of k-rational points on V, we define

$$S=S(V,f)=\sum_{x\in V(k)}\psi_k(f(x)).$$

By extension of scalars, we may replace V by $V\otimes_k k_n$ and define more generally $S_n=S_n(V,f)$ by

$$S_n=S(V\otimes_k k_n,f)=\sum_{x\in V(k_n)}\psi_{k_n}(f(x)). \tag{1.3}$$

The $\mathbf{Z}/(p)$-covering $z^p-z=f(x)$ of V, with group action $g(z,x)=(z+g,x)$ and the character $\psi(g)=e^{2\pi ig/p}$ determine an Artin L-function $L(t)=L(V,f;t)$, explicitly given by

$$L(V,f;t)=\prod_P(1-\psi_{k(P)}(f(P))\,t^{\deg P})^{-1} \tag{1.4}$$

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where P runs over all closed points of V and where $k(P)$ is the residue finite field and $\deg P = [k(P):k]$. From (1.4) one deduces

$$L(V, f; t) = \exp(\sum_n S_n t^n/n). \tag{1.5}$$

It is known that $L(t)$ is a rational function of t, hence with finitely many zeros β_j^{-1} and finitely many poles α_i^{-1}; in view of (1.5), this implies the formula [1]

$$S_n = \sum_i (\alpha_i)^n - \sum_j (\beta_j)^n. \tag{1.6}$$

We shall refer to the α_i, β_j as the characteristic roots of S_n. By (1.6), the fundamental problem of estimating the exponential sums S_n reduces to:

(A) finding upper bounds for the absolute values of the characteristic roots;

(B) finding upper bounds for the number of characteristic roots.

Part (A) of this program has recently found a very satisfactory solution in the work of Deligne [D1] and [D2]. The object of this paper is to deal with problem (B), in the most general case.

We shall prove the following result.

Theorem 1. *Let $\mathbf{A}^N$ be the N-dimensional affine space over k and let $f \in k[x_1, \ldots, x_N]$ be a polynomial over k. Then the total number of characteristic roots α_i, β_j associated to $S_n(\mathbf{A}^N, f)$ does not exceed $(4 \deg f + 5)^N$.*

At the end of Section IV of this paper, we shall also show how Theorem 1 implies a similar result for the most general sum $S_n(V, f)$, by means of an elementary reduction process. We shall prove:

Theorem 2. *Let V be a closed affine subvariety over k of $\mathbf{A}^N$, of degree $d = \deg V$. Let f be a regular function on V and let P be a polynomial in $k[\mathbf{A}^N]$ inducing f on V and of degree $\delta = \deg P$. Then the number of characteristic roots α_i, β_j associated to $S_n(V, f)$ does not exceed $(4D+5)^{2N+1}$ where $D = \max(d+1, \delta)$.*

As pointed out to me by Dwork, an alternative approach to the result of Theorem 1 can be obtained using the methods developed in [DW], in particular Lemma 14.1, p. 489, and Equation (21.5), p. 518. These results imply immediately good upper bounds for the degrees of the numerator and denominator of the Zeta function of a hypersurface. It is however a simple matter to extend Dwork's arguments to the case of exponential sums considered in Theorem 1 and calculations by Dwork show indeed that the number of characteristic roots of the sum $S_n(\mathbf{A}^N, f)$ does not exceed

$$3((3+2\sqrt{2}) \deg f + 1)^N < 3(6 \deg f + 1)^N.$$

I want to thank here J.-P. Serre, B. Dwork and P. Deligne for very useful discussions and comments about this paper, which led to better proofs and sharper results. In particular, (3.16) is due to Deligne and the computation (3.19) to Dwork.

[1] Zeros and poles are counted with multiplicity

II. Deligne's Results

We quote here from [S] and Deligne's SGA $4\frac{1}{2}$ the basic theorems proved by Deligne. Let $\bar{V}=V\otimes_k \bar{k}$ and let F be the Frobenius map. For every prime l, $l\neq p$, there is a certain locally free l-adic sheaf $\mathscr{F}_l(f)$ on $\bar{V}$ such that

$$S_n=\sum_i (-1)^i \operatorname{Tr}((F^*)^n \mid H_c^i(\bar{V};\mathscr{F}_l(f))) \tag{2.1}$$

where H_c^* denotes cohomology with proper supports. We have $H_c^i=(0)$ if $i<0$ or if $i>2\dim \bar{V}$ and in any case the groups H_c^i are finite dimensional vector spaces over a finite extension K_l of $\mathbf{Q}_l$ containing the p-th roots of unity. Now let $B_i(\mathscr{F}_l(f))=\dim_{K_l} H_c^i(\bar{V};\mathscr{F}_l(f))$ and let ω_{ij}, $j=1,\ldots,B_i$ be the eigenvalues of the endomorphism F^* of the K_l-vector space H_c^i. Then Deligne proves:

ω_{ij} is an algebraic integer and all its conjugates over $\mathbf{Q}$ *have the same absolute value $q^{r(i,j)/2}$, where $r(i,j)$ is an integer called the weight of ω_{ij};* (2.2)

$$0\leqq r(i,j)\leqq i; \tag{2.3}$$

$q^{\dim \bar{V}}/\omega_{ij}$ *is also an algebraic integer.* (2.4)

Clearly (2.1) implies

$$S_n=\sum_{ij} (-1)^i(\omega_{ij})^n \tag{2.5}$$

for all n, whence the characteristic roots associated to S_n in (1.6) are among the ω_{ij} and in particular satisfy (2.2), (2.3) and (2.4). We also get

$$|S_n|\leqq \sum_i B_i(\mathscr{F}_l(f))\, q^{ni/2}, \tag{2.6}$$

but as yet no general method has been suggested for computing the Betti numbers B_i and even their independence from the auxiliary prime l is not known in general. The calculations by Deligne in SGA $4\frac{1}{2}$, p. 219–229 for the case of generalized Kloosterman sums show that this problem may be quite deep. It should be noted that our Theorem 2 does not yet give an upper bound for the Betti numbers B_i, because of possible cancellation of the roots ω_{ij} in (2.5). In the proof of Theorems 1 and 2 we use only the integrality statement (2.4); a proof can be found in SGA 7 II, XXI Cor. 5.5.3 (iii), p. 394.

At the request of Serre, we end this section by making the following corrections to [S]: page 115, instead of $i>\dim(X)$, read $i>2\dim(X)$; p. 122, formula (6.9): the stated formula holds only if one replaces affine space X by the space $x_1 x_2 \ldots x_r\neq 0$.

III. p-Adic Theory

Let Ω be the completion of an algebraic closure of the p-adic field $\mathbf{Q}_p$; we shall denote by ord the additive valuation in Ω normalized so that $\operatorname{ord}(q)=1$.

Let V^N be the variety $V^N \subset \mathbf{A}^N$ given by

$$V^N = \{x_1 \dots x_N \neq 0\} \tag{3.1}$$

and consider the sum

$$S_n(V^N, f) = \sum_{\mathbf{x} \in V^N(k_n)} \psi_{k_n}(f(\mathbf{x})) \tag{3.2}$$

where $f \in k[\mathbf{x}]$ is a polynomial of degree $\deg f = d$.

The L-function $L(t) = L(V^N, f; t)$ associated with the sums (3.2) may be written as

$$L(t) = \exp\left(\sum_{n=1}^{\infty} S_n t^n/n\right) = \left\{\frac{\prod(1-\omega t)}{\prod(1-\eta t)}\right\}^{(-1)^{N-1}} \tag{3.3}$$

so that (1.6) becomes[2]

$$S_n(V^N, f) = (-1)^N \{\sum \omega^n - \sum \eta^n\}. \tag{3.4}$$

Our goal is to estimate the total degree of $L(V^N, f; t)$, that is the total number of characteristic roots ω, η.

The function $L(t)$ has been studied as a meromorphic function of $t \in \Omega$ by means of Dwork's p-adic methods, in [B]. The main result of [B] is that there exists a p-adically entire function $D(t)$, obtained as a Fredholm resolvent of a certain endomorphism of a p-adic Banach space, with the following properties. Let us define Dwork's δ operator by means of

$$f(t)^\delta = \frac{f(t)}{f(qt)};$$

then we have

$$L(t)^{(-1)^{N-1}} = D(t)^{\delta^N}, \tag{3.5}$$

the Newton polygon of $D(t)$ lies above the polygon with vertices

$$\left(\binom{N+m}{N}, \quad \frac{mN}{d(N+1)}\binom{N+m}{N}\right)$$

where $m \geqq -1$. (3.6)

We recall that, if

$$D(t) = \sum_0^\infty d_m t^m = \prod(1-\alpha t),$$

the Newton polygon of $D(t)$ is the polygon with vertices at the points

$$\left(\sum_{\text{ord}(\alpha) \leqq y} 1, \sum_{\text{ord}(\alpha) \leqq y} \text{ord}(\alpha)\right) \quad \text{for } y \in \mathbf{R};$$

[2] For simplicity, we do not write the ω, η with subscripts ω_i, η_j as in (1.6). Again, it is understood that $\omega \neq \eta$, ω, $\eta \neq 0$ and that the ω, η are repeated according to multiplicity

it is understood that if $D(t)$ is a polynomial then the Newton polygon of $D(t)$ has a vertical half-line as its last side.

For the proof of (3.5) and (3.6) above we note that (3.5) is in [B], Equation (34), p. 77 while (3.6) is in fact [B], Lemma 2, Equation (37), p. 78. It is proved there that the Newton polygon of $D(t)$ lies above the polygon with vertices $\left(\sum_0^m c(j), \frac{1}{d}\sum_0^m jc(j)\right)$ where $c(j)=\binom{N-1+j}{N-1}$. Since $\sum_0^\infty c(j)\,x^j=(1-x)^{-N}$, the sum $\sum_0^m c(j)$ is the coefficient of x^m in $(1-x)^{-1}\cdot(1-x)^{-N}=(1-x)^{-N-1}$, that is $\binom{N+m}{N}$ while similarly $\sum_0^m jc(j)$ is the coefficient of x^m in $(1-x)^{-1}\left(x\frac{d}{dx}\right)(1-x)^{-N}=Nx(1-x)^{-N-2}$, that is $N\binom{N+m}{N+1}=\frac{mN}{N+1}\binom{N+m}{N}$; this proves (3.6).

Following a suggestion of Deligne, we reformulate (3.5) and (3.6) as follows. Let Ω_t^* be the subset of Ω^* consisting of elements of order $\mathrm{ord}(\alpha)\leqq t$ and let $\mathrm{Div}(\Omega^*)$ be the set of integral divisors on Ω^* which are locally finite in the sense that $\mathrm{supp}(D)\cap\Omega_t^*$ is finite for every t. We shall write $D=\sum_{\alpha\in\Omega^*} n(\alpha)\,\delta_\alpha$ where $n(\alpha)\in\mathbf{Z}$ for such divisors. The composition law $\delta_\alpha\,\delta_\beta=\delta_{\alpha\beta}$ gives a ring structure in $\mathrm{Div}(\Omega^*)$.

If $D=\sum_{\alpha\in\Omega^*} n(\alpha)\,\delta_\alpha$ is a divisor and $n(\alpha)\geqq 0$ we shall write $D\geqq 0$. In this case, the Newton polygon of D is the polygon with vertices

$$\left(\sum_{\alpha\in\Omega_t^*} n(\alpha),\ \sum_{\alpha\in\Omega_t^*} n(\alpha)\,\mathrm{ord}(\alpha)\right), \tag{3.7}$$

with $t\in\mathbf{R}$.

If f is a p-adically meromorphic function in Ω and if $f(0)=1$ we define its divisor by

$$\mathrm{div}(f)=\sum_{\alpha\in\Omega^*} v_{\alpha^{-1}}(f)\,\delta_\alpha \tag{3.8}$$

where $v_{\alpha^{-1}}(f)$ is the order of zero of f at α^{-1}. We introduce the divisors

$$\Lambda_+=\sum\delta_\omega,\qquad \Lambda_-=\sum\delta_\eta \tag{3.9}$$

$$D=\mathrm{div}\,D(t), \tag{3.10}$$

and proceed to restate (3.5), (3.6) as follows:

$$\Lambda_+-\Lambda_-=(-1)^{N-1}\,\mathrm{div}\,L(t)=D(1-\delta_q)^N; \tag{3.5}'$$

we have $D\geqq 0$ and the Newton polygon of D lies above the polygon with vertices $\left(\binom{N+m}{N}, \frac{mN}{d(N+1)}\binom{N+m}{N}\right)$, $m\geqq -1$, *that is the Newton polygon of* $(1-\delta_{q^{1/d}})^{-N}$. (3.6)′

Since $D(t)$ is entire, we have $D \geqq 0$; now (3.6)′ is equivalent to (3.6). Finally (3.5)′ is obvious from (3.5) and

$$\operatorname{div}\left(\frac{f(t)}{f(qt)}\right) = (\operatorname{div} f)(1-\delta_q). \tag{3.11}$$

It is immediate from $D \geqq 0$ and (3.5)′ that we have

$$\Lambda_+ + \Lambda_- \leqq D(1+\delta_q)^N. \tag{3.12}$$

Theorem 1A. *The sum $S_n(V^N, f)$ has at most $(4 \deg f + 4)^N$ characteristic roots.*

Proof. By Deligne's result (2.4), the characteristic roots ω, η satisfy

$$\operatorname{ord}\omega, \quad \operatorname{ord}\eta \leqq N \tag{3.13}$$

therefore in the right-hand side of inequality (3.12) we need only retain the terms of order at most N. To do this we introduce the cutoff operator γ_t on divisors $\Delta = \sum n(\alpha)\delta_\alpha$, by means of

$$\gamma_t(\Delta) = \sum_{\alpha \in \Omega_t^*} n(\alpha)\,\delta_\alpha \tag{3.14}$$

and now it is clear that

$$\begin{aligned}\Lambda_+ + \Lambda_- &\leqq \gamma_N(D(1+\delta_q)^N) \\ &= \sum_{j=0}^{N} \binom{N}{j} \gamma_{N-j}(D)\,\delta_{q^j};\end{aligned}$$

hence

$$\deg(\Lambda_+ + \Lambda_-) \leqq \sum_{j=0}^{N} \binom{N}{j} \deg \gamma_{N-j}(D). \tag{3.15}$$

In order to estimate $\deg \gamma_{N-j}(D)$ we use the information on the Newton polygon of D provided by (3.6)′. In fact, $\deg \gamma_{N-j}(D)$ is the projection on the X-axis of the sides of the Newton polygon of D with slopes at most $N-j$. By (3.6)′, if we intersect the Newton polygon of $(1-\delta_{q^{1/d}})^{-N}$ with a line $Y = tX$, $t \geqq N-j$, the projection on the X-axis of the intercept will give a majorant of $\deg \gamma_{N-j}(D)$. If we take $m_j = d(N-j+1)$, we have

$$t_j = \frac{m_j N}{d(N+1)} \geqq N-j,$$

for $j = 0, 1, \ldots, N$ and by the explicit estimate (3.6)′ the projection on the X-axis of the intercept is $\binom{N+m_j}{N}$. We have proved

$$\deg \gamma_{N-j}(D) \leqq \binom{N+d(N-j+1)}{N}$$

and combining this with (3.15) we find

$$\deg(\Lambda_+ + \Lambda_-) \leqq \sum_{j=0}^{N} \binom{N}{j} \binom{N+d(N-j+1)}{N}. \tag{3.16}$$

In order to estimate the right hand side of (3.16), we note that

$$\sum_{j=0}^{N} \binom{N}{j} \binom{N+d(N-j+1)}{N}$$

is the coefficient of $x^{d(N+1)}$ in the Taylor expansion of $(1+x^d)^N(1-x)^{-N-1}$, therefore it equals

$$\mathrm{Res}_0 \{x^{-d}(1+x^{-d})^N(1-x)^{-N}\omega\} \tag{3.17}$$

where ω is the differential form $\omega = \dfrac{dx}{x(1-x)}$ and where Res_0 is the residue at the origin. Following a suggestion of Dwork, we now make the substitution $x \to \dfrac{z}{1+z}$ which carries ω into $\dfrac{dz}{z}$. By the invariance of resisdues, (3.17) becomes $\mathrm{Res}_0 \left(G(z)\dfrac{dz}{z}\right)$, where

$$G(z) = (1+z)^N \left(1+\frac{1}{z}\right)^d \left(1+\left(1+\frac{1}{z}\right)^d\right)^N, \tag{3.18}$$

hence (3.17) is the constant term in the Laurent expansion of $G(z)$. Since the coefficients of the Laurent expansion are non-negative, $G(z)$ itself can be used to give a bound:

$$\deg(\Lambda_+ + \Lambda_-) < G(z), \quad \textit{all } z > 0. \tag{3.19}$$

A very simple choice is $z = d$, using $\left(1+\dfrac{1}{d}\right)^d < e$, which yields

$$\deg(\Lambda_+ + \Lambda_-) < e(e+1)^N(d+1)^N < (4d+4)^N$$

if $N \geqq 4$. The same bound holds if $N \leqq 3$ by direct computation in (3.16). This completes the proof of Theorem 1A. A better choice for z would give

$$\deg(\Lambda_+ + \Lambda_-) < \frac{(d+\lambda)^N}{(\lambda-1)^{N+1}}$$

where $\lambda = 1.27846454\ldots$ is the solution of $\lambda = 1 + e^{-\lambda}$.

We conclude this section by mentioning that Dwork's alternative method yields the bound (for $d \geqq 1$):

$$\deg(\Lambda_+ + \Lambda_-) \leqq d^N \sum_{j=0}^{N} \binom{N}{j} \binom{2N+1-j}{N} < (\sqrt{2}+1)((3+2\sqrt{2})\,d)^N,$$

which is slightly weaker than our bound (3.16). However, it should be noted that in Dwork's approach the p-adic estimate (3.13) is replaced by the much simpler archimedean estimate $|\omega|, |\eta| \leqq q^N$.

IV. Proof of Theorems

The proof of Theorem 1 is obtained as an easy consequence of Theorem 1A. In fact, denoting by A a subset of $\{1, 2, \ldots, N\}$ and denoting by f_A the polynomial obtained from f by setting $x_i = 0$ if $i \in A$, one sees that

$$S_n(\mathbf{A}^N, f) = \sum_A S_n(V^{N-|A|}, f_A);$$

here A runs over the 2^N subsets of $\{1, 2, \ldots, N\}$, including the empty set. By Theorem 1A, the number of characteristic roots of $S_n(\mathbf{A}^N, f)$ will not exceed *a fortiori*

$$\sum_A (4 \deg f + 4)^{N-|A|} = (4 \deg f + 5)^N.$$

We shall obtain Theorem 2 as a consequence of a simple reduction to sums of the type considered in Theorem 1. Let $V \subset \mathbf{A}^N$ as a closed affine subvariety and let f be a regular function on V. There is a polynomial $P \in k[\mathbf{A}^N]$ inducing f on V and we can also find a finite set of polynomials g_i, $i = 1, \ldots, M$ defining V in the set theoretic sense, i.e. $V(k_n)$ is the subset of points $\mathbf{x} \in \mathbf{A}^N(k_n)$ such that $g_i(\mathbf{x}) = 0$ for $i = 1, \ldots, M$. We introduce additional variables $\mathbf{y} = (y_1, \ldots, y_M)$ and consider the new polynomial

$$h(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{M} y_i g_i(\mathbf{x}) + P(\mathbf{x}). \tag{4.1}$$

Proposition. *We have*

$$S_n(\mathbf{A}^{N+M}, h) = (q^M)^n S_n(V, f). \tag{4.2}$$

Proof. By orthogonality of characters, summation in $S_n(\mathbf{A}^{N+M}, h)$ over y_i yields 0 unless $g_i(\mathbf{x}) = 0$, $i = 1, \ldots, M$, that is $\mathbf{x} \in V(k_n)$, and in that case $P(\mathbf{x}) = f(\mathbf{x})$, while if $\mathbf{x} \in V(k_n)$ summation over y_i yields multiplication by q^{Mn}.

The above Proposition reduces the study of $S_n(V, f)$ to that of sums considered in Theorem 1. In order to get a bound for the number of characteristic roots of $S_n(V, f)$ we have then to control M, the number of equations defining V, and the degrees of the polynomials $g_i(\mathbf{x})$. If we replace V by $\bar{V} = V \otimes_k \bar{k}$ then we can take $M = N + 1$ and g_i, the defining equations of V, "generic" cones over V, of dimension $N - 1$ and degree $d = \deg V$. Although this construction may not succeed over k, it will certainly work over k_n for *all* large n. Let α_i, β_j be the characteristic roots of $S(V, f)$; for every n such that $\alpha_i^n \neq \beta_j^n$ for all i, j, the characteristic roots of $S(V \otimes_k k_n, f)$ are exactly the n-th powers of the characteristic roots of $S(V, f)$, and there are arbitrarily large such n. In view of these remarks and (4.2) and Theorem 1, we obtain Theorem 2.

Finally, a remark about the most general case in which V need not be an affine variety. Every variety V can be covered by finitely many affine subsets V_h, with h in a finite set of indices H. For example, if V is quasi-projective in $\mathbf{P}^N$, one can find an affine covering of V with at most $r+1$ components, $r=\dim V \leqq N$. Now for every subset A of H, define

$$V_A = \bigcap_{a \in A} V_a \quad \text{if } A \text{ is non-empty;} \tag{4.3}$$

we may also define $V_\emptyset = V$. Note that if $A \neq \emptyset$ then each V_A is an affine variety. By a simple combinatorial principle we have

$$\sum_A (-1)^{|A|} S_n(V_A, f) = 0,$$

hence

$$S_n(V, f) = -\sum_{A \neq \emptyset} (-1)^{|A|} S_n(V_A, f) \tag{4.4}$$

which can be used for a reduction to the case in which V is affine.

V. An Example

Information about the characteristic roots of $S_n(V, f)$ is essential if we want to obtain bounds for S_n. To some extent, this may be reversed and the knowledge of bounds for S_n can give partial information about the characteristic roots themselves. We have:

Theorem 3.

(i) *if* $|S_n| = O(q^{n\vartheta})$ *for some* $\vartheta \geqq 0$, *then we also have*

$$|S_n| = O(q^{nr/2}),$$

where r *is the integral part of* 2ϑ;

(ii) *if* r *is a positive integer and if*

$$|S_n| \leqq C q^{nr/2}$$

for every large n, *then the characteristic roots of* S_n *have weight at most* r *and moreover there are at most* C^2 *roots of weight* r;

(iii) *if* r *is the highest weight of roots of* S_n *and if there are* B *roots of weight* r, *then for every* $\varepsilon > 0$ *and infinitely many* n *we have the lower bound*

$$|S_n| > (B^{1/2} - \varepsilon) q^{nr/2}.$$

Proof. By the fundamental result (2.2) of Deligne, (iii) implies (i); also it is obvious that (iii) implies (ii). Finally, (iii) is implied by the following elementary result: let $z_1, \ldots, z_m$ be *distinct* complex numbers of absolute value $|z_h| = 1$ and let $b_1, \ldots, b_m$ be arbitrary complex numbers; then we have

$$\limsup_{n \to \infty} |\sum_h b_h z_h^n| \geqq (\sum_h |b_h|^2)^{1/2}. \tag{5.1}$$

In fact,

$$\sum_{n=0}^{N} |\sum b_h z_h^n|^2 = (N+1)\sum |b_h|^2 + O\left(\sum_{h\neq j} |b_h b_j| \left|\sum_{n=0}^{N} (z_h \bar{z}_j)^n\right|\right)$$
$$= (N+1)\sum |b_h|^2 + O(1),$$

because $\sum_{n=0}^{N} (z_h \bar{z}_j)^n = O(1)$, uniformly in N. Letting $N \to \infty$ in the above inequality, we obtain *a fortiori* inequality (5.1).

Now we shall illustrate with the example of generalized Kloosterman sums how the various theorems of this paper and Deligne's result (2.2) can be combined with elementary estimates to obtain sharper results.

Let V be the variety in $\mathbf{A}^r$ defined over k by the single equation

$$x_1 \dots x_r = a, \tag{5.2}$$

where $a \in k^*$ and let $f(\mathbf{x})$ be the regular function on V defined by

$$f(\mathbf{x}) = x_1 + \cdots + x_r. \tag{5.3}$$

We want to estimate the Kloosterman sum

$$K_n(a) = S_n(V, f). \tag{5.4}$$

It has been proved by Deligne in SGA $4\frac{1}{2}$ that $K_n(a)$ has exactly r characteristic roots, all of the same weight $r-1$, hence

$$|K_n(a)| \leqq r q^{n(r-1)/2}. \tag{5.5}$$

We shall deduce from our results a weaker inequality, namely: *if* $p-1$ *does not divide* r, the roots of $K_n(a)$ have weight at most $r-1$, hence

$$K_n(a) = O(q^{n(r-1)/2}) \tag{5.5}'$$

where the constant implied in the symbol $O(\dots)$ depends only on r.

Lemma. *We have*

$$\sum_{a \in k^*} |K_1(a)|^2 = q^r - q^{r-1} - \cdots - q - 1. \tag{5.6}$$

Proof. Let χ be a character of k^*. We have

$$\sum_{a \in k^*} \chi(a) K_1(a) = \sum_{a \in k^*} \sum_{x_1 \dots x_r = a} \chi(x_1 \dots x_r)\, \psi_k(x_1 + \cdots + x_r)$$
$$= \sum_{x_1, \dots, x_r \in k^*} \chi(x_1 \dots x_r)\, \psi_k(x_1 + \cdots + x_r)$$
$$= (-\tau(\chi, \psi_k))^r,$$

where $\tau(\chi, \psi) = -\sum_{x \in k^*} \chi(x)\psi(x)$ is the Gauss sum. Now (5.6) follows from Plancherel's theorem applied to the function $a \to K_1(a)$ on the finite group k^*,

using the elementary estimations $|\tau(\chi, \psi_k)| = q^{1/2}$ if χ is non-trivial, and $=1$ if χ is trivial. This lemma (for the field k_n) and Theorem 3, (ii) imply that $K_n(a)$ has at most one root of weight r and all other roots have weight at most $r-1$. If there is no root of weight r then Theorem 2 yields (5.5)′, with an explicit (and large) constant. Suppose that there is a_0 such that $K_n(a_0)$ has one root ω of weight r. Then clearly

$$|K_n(a_0)|^2 \sim q^{nr} = |\omega|^{2n}$$

as $n \to \infty$, and again (5.6) for the field k_n and Theorem 3, (ii) show that $K_n(a)$ has no root of weight r if $a \neq a_0$. Now $K_n(a_0)$ is in the cyclotomic field $\mathbf{Q}(\mu_p)$ and its conjugates over $\mathbf{Q}$ are the sums $S_n(V, uf)$ with $u \in \mathbf{F}_p^*$. It follows that, applying the transformation $x_i \to x_i/u$, the conjugates of $K_n(a_0)$ over $\mathbf{Q}$ are simply the sums

$$K_n(a_0 u^r), \quad u \in \mathbf{F}_p^*.$$

The characteristic roots of $K_n(a_0 u^r)$ are thus conjugates over $\mathbf{Q}$ of the characteristic roots of $K_n(a_0)$, therefore by Deligne's result (2.2) we deduce that each sum $K_n(a_0 u^r)$ has one root of weight r. In view of our previous remark that this could happen only for the sum $K_n(a_0)$, we see that we must have

$$u^r = 1 \quad \text{for every } u \in \mathbf{F}_p^*,$$

hence $p-1$ divides r. This completes the proof of (5.5)′.

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Induced Modules and Extensions of Representations*

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Let G be a connected affine algebraic group over an algebraically closed field k, and let B be a Borel subgroup of G. If V is a rational B-module we can ask for conditions under which V extends to G, that is, under which the action of B extends to a rational action of G on V. Since G/B is a complete variety, we have $M|_B{}^G \cong M$ for any rational G-module M, cf. (1.3); so if any extension of V exists, it must be unique.

The main result of this paper is that V is extendible to G if and only if it is extendible to every minimal parabolic subgroup $P \supseteq B$. Here "minimal" means that P properly contains B and that no other closed subgroup of P does so; equivalently, P has semisimple rank one. We also show that if P contains a Levi complement L, then V extends to P iff $V|_{L \cap B}$ extends to L; so, for reductive G, the extendibility of V reduces to an SL_2 question.

We obtain the main result as a corollary to a general theorem on induced modules. For any parabolic subgroup $P \supseteq B$, and any rational B-module V, let $V|^P$ denote the corresponding module induced from B to P. If $P_1, \ldots, P_n$ is any sequence of such P's, we let $V|^{P_1,\ldots,P_n}$ denote the P_n-module $V|^{P_1}|_B|^{P_2}\cdots|^{P_n}$ obtained by successively inducing to P_{i-1}, restricting to B, and inducing to P_i. We prove the B-module structure of $V|^{P_1,\ldots,P_n}$ depends only on the *set-theoretic* product $P_1 \ldots P_n$, and that $V|^{P_1,\ldots,P_n} \cong V|^G$ as P_n-modules in case this product is G. At the same time we prove a "factorization" theorem for the affine coodinate ring $R(G)$ of G. Let $P_1, \ldots, P_n$ be parabolic subgroups containing B such that $G = P_1 \ldots P_n$. Then we have $R(G) \cong R(P_n) \otimes_B R(P_{n-1}) \otimes_B \cdots \otimes_B R(P_1)$, where in general for a right B-module M and a left B-module N, we denote by $M \otimes_B N$ the set of fixed points of B in $M \otimes N$ relative to the action $b \cdot (m \otimes n) = mb^{-1} \otimes bn$.

The above results are in turn derived from simple considerations involving the Demazure desingularization of Schubert varieties [9], and are inspired by Kempf [12] and by Andersen's efforts [1] toward a reorganization of Kempf's work. A simplified version of Andersen's main reduction is given in the appendix.

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Finally, we mention that the main result is false in the case of finite Chevalley groups. In § 4 we give a counterexample in the case $G = SL_3(2)$.

1. Preliminaries

Notation. Throughout k is an algebraically closed field of arbitrary characteristic which is to serve as the domain of definition for the varieties and algebraic groups under consideration. Varieties are taken in the sense of "classical" algebraic geometry, are identified with their k-rational points, and are not necessarily irreducible. Given a variety X, we set $R(X) = \Gamma(X, \mathcal{O}_X)$, the k-algebra of everywhere defined rational functions on X, where $\mathcal{O}_X$ denotes the structure sheaf on X. If $f\colon X \to Y$ is a morphism of varieties, we will denote by f^*: $R(Y) \to R(X)$ the comorphism of f defined by $f^*(a)(x) = a(f(x))$, $a \in R(X)$, $x \in X$.

All algebraic groups are affine. If G is an algebraic group, a G-module V is *rational* if it is a union of finite dimensional submodules W rational in the usual sense (that is, $G \times W \to W$ is a morphism of varieties). For example, the coordinate ring $R(G)$ is a rational left G-module with respect to left translation: $(g \cdot f)(x) = f(xg)$, $f \in R(G)$, $x, g \in G$; and a rational right G-module with respect to right translation: $(f \cdot g)(x) = f(gx)$.

The subspace of fixed points in a G-module V is denoted V^G. Also, if W is a right G-module, then $W \otimes_G V$ denotes the fixed point space for the action of G defined by $g \cdot (w \otimes v) = wg^{-1} \otimes gv$ on $W \otimes V$.

1.1. Locally Trivial Bundles. Let H be an algebraic group, and let E be an algebraic variety equipped with a right action of H which is *rational*, in the sense that the associated map $E \times H \to E$ is a morphism of varieties.

Suppose in addition we have a surjective morphism $\pi\colon E \to Y$, constant on the orbits of H, such that Y has an open cover $\{Y_\alpha\}$ with $\pi^{-1}(Y_\alpha) \cong Y_\alpha \times H$, compatible with the action of H and the projection π. We say that E, together with π and the base space Y, is a *locally trivial principal H-bundle* (over Y). Obviously the Y_α's can be taken to be affine, in which case the structure sheaf of Y is determined by that of E by the isomorphisms $R(Y_\alpha) \cong R(\pi^{-1}(Y_\alpha))^H$. In particular, Y is the quotient X/H.

Given one locally trivial principal H-bundle E, we can frequently construct others. Suppose $X \to Y$ is a morphism, with Y as above. Then the fiber product $F = X \times_Y E$ together with projection $F \to X$ is also a locally trivial principal H-bundle. The diagram

$$\begin{array}{ccc} F & \longrightarrow & E \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

is of course commutative, and moreover we have $F \cong X \times_Y E$ for any locally trivial principal H-bundle F which fits into such a commutative diagram (with $F \to E$ equivariant) [13; (3.1)]. The proof is entirely analogous to that for the corresponding result in algebraic topology [11].

Next, following [13], suppose in addition to the principal bundle E we have a variety K equipped with a rational action of H on the left. Define a rational H-action on $E \times K$ by $(e, k)h = (eh, h^{-1}k)$, and let $Z = E \times^H K$ denote the set of orbits of H under this action. Let $E_\alpha = \pi^{-1}(Y_\alpha)$ where $\{Y_\alpha\}$ is an open affine cover of Y with $\pi^{-1}(Y_\alpha) \cong Y_\alpha \times H$ as above, and give Z the quotient topology, so that the images Z_α in Z of the E_α form an open cover. Each Z_α has a natural variety structure isomorphic to $Y_\alpha \times K$, and as such is the quotient of $E_\alpha \times K$ under the action of H. This gives Z a natural variety structure[1], and Z is the quotient of $E \times K$ under the action of H. (In particular, there is a natural inclusion $R(K) \otimes_H R(E) \subseteq R(Z)$.)

Let $\eta: Z \to Y$ be the morphism obtained from $E \times K \to E \to Y$. We have $\eta^{-1}(Y_\alpha) = Y_\alpha \times K$, compatible with the projection η, for each member Y_α of our open affine cover. Accordingly, we say that Z, together with η and Y, is a *locally trivial fiber bundle* over Y (or locally trivial K-bundle). Also, Z is the K-bundle *associated with* the principal bundle E.

Finally, suppose that K above also has a right H-action, commuting with the left action, such that there exists a locally trivial principal bundle $K \to K/H$ for this action. Then H acts on Z, and gives a locally trivial principal bundle $Z \to E \times^H (K/H)$, as is immediate from the isomorphisms $Z_\alpha \cong Y_\alpha \times K$, which are compatible with the right action of H. (In particular, the quotient $(E \times^H K)/H$ exists and is isomorphic to $E \times^H (K/H)$.) This procedure will be used repeatedly in § 2.

1.2. Proper Morphisms. Recall a morphism $f: X \to Y$ is *proper* if it universally closed (i.e., $f \times 1_Z: X \times Z \to Y \times Z$ is a closed mapping for every variety Z). A proper morphism remains not only closed, but proper, upon taking products, fiber products, or base change. A composition of proper morphisms is proper, and the property of being proper is local (i.e., f is proper iff for every open U in Y the induced morphism $f^{-1}(U) \to U$ is proper).

To say a variety X is *complete* means precisely that the canonical morphism $X \to \operatorname{Spec} k$ is proper. *In this paper, we will always assume a complete variety X is irreducible, and thus $R(X) = k$, the constant functions.*

The following result is well-known. We include a proof for completeness.

Lemma. ([10; III, 4.3.12]). *Let $f: X \to Y$ be a surjective proper morphism, and suppose further that f is a birational isomorphism and that Y is a normal variety. Then $f^*: R(Y) \to R(X)$ is an isomorphism.*

Proof. It suffices to prove that f^* is surjective. Let $b \in R(X)$ and let $E \subseteq X \times k$ be the graph of b. Since f is proper and E is closed in $X \times k$, we see $\bar{E} = (f \times 1)(E)$ is closed in $Y \times k$. Since the irreducible components of the fibers of f are complete, it follows easily that the projection $pr_1: \bar{E} \to Y$ is a surjective birational isomorphism with finite fibers. Hence, this projection is an isomorphism by Zariski's Main Theorem [15; p. 178]. If we set $\bar{b} = pr_2 \circ pr_1^{-1} \in R(Y)$, we finally obtain $f^*(\bar{b}) = b$.

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[1] It is obvious that Z is at least a prevariety. To show that the diagonal Δ of $Z \times Z$ is closed, observe that $(Z_\alpha \cap Z_\beta) \times_Y (Z_\alpha \cap Z_\beta) = Z_\alpha \times_Y Z_\beta$ is closed in $Z_\alpha \times Z_\beta$, and contains $\Delta \cap (Z_\alpha \times Z_\beta)$ as a closed subvariety

Remark. Actually, one gets that the direct image $f_*(\mathcal{O}_X)$ of the structure sheaf of X is the structure sheaf $\mathcal{O}_Y$ of Y.

1.3. Induced Modules. Let G be an algebraic group, and let H be a closed subgroup. Each rational G-module V defines by restriction a rational H-module $V|_H$ (frequently denoted V when it is clear from context that V is being considered as an H-module). Conversely, each rational H-module W determines a rational G-module $W|^G$, the *induced* module, which can be defined as the set of H-fixed points in $R(G)\otimes W$ relative to the action $h\cdot(r\otimes w)=f\cdot h^{-1}\otimes hw$ ($h\in H$, $f\in R(G)$, $w\in W$). The G-module structure on $W|^G$ is obtained from the G-action on $R(G)\otimes W$ defined by $g(f\otimes w)=g\cdot f\otimes w$ ($g\in G$). The module $W|^G$ is equipped with an H-module homomorphism $\mathscr{ev}$: $W|^G\to W$, called evaluation, defined by $\mathscr{ev}(\sum f_i\otimes w_i)=\sum f_i(1)w_i$. The induced module is then characterized by the following universal property: given any rational G-module V and H-module homomorphism ϕ: $V\to W$, there exists a unique G-module homomorphism $\tilde{\phi}$: $V\to W|^G$ satisfying $\mathscr{ev}\circ\tilde{\phi}=\phi$. This fact is most easily seen by observing that $R(G)\otimes W$ can be identified with the set of morphisms $G\to M$, where M ranges over the finite dimensional subspaces of W. Many further properties of induced modules are given in the papers [5], [6], and [7]. We will require none of these in the present paper, except for the fact, obvious from the universal property, that $W|^K|^G\cong W|^G$ when K is a closed subgroup with $H\leqq K\leqq G$. Also, we will use the following property of induction for semidirect products.

Proposition. *Suppose an algebraic group P is a semidirect product $U\cdot L$, $U\lhd P$, of closed subgroups. Let B be a closed subgroup of P containing U and let W be a rational B-module. Then the canonical map $W|^P|_L\to W|_{B\cap L}|^L$ is an isomorphism.*

Proof. The map $W|^P|_L\to W|_{B\cap L}|^L$ is defined via universal mapping by regarding the evaluation map $W|^P\to W$ as a $B\cap L$-homomorphism. Hence it is enough to produce an isomorphism $W|^P|_L\to W|_{B\cap L}|^L$ commuting with the appropriate evaluation maps. Define a left action of L on $R(U)$ by $(x\cdot f)(u)=f(x^{-1}ux)$ for $x\in L$, $u\in U$, $f\in R(U)$. This gives a left (diagonal) action of $B\cap L$ on $R(U)\otimes_U W$ as a submodule of $R(U)\otimes W$. Since $B=U(B\cap L)$ we obtain from $R(P)\cong R(L)\otimes R(U)$ an isomorphism $R(P)\otimes_B W\cong R(L)\otimes_{B\cap L}(R(U)\otimes_U W)$ of L-modules. Also, one checks that the evaluation map $R(U)\otimes_U W\to W$ is an isomorphism of $B\cap L$-modules. This gives the desired isomorphism $W|^P|_L\to W|_{B\cap L}|^L$.

Note if G/H is a complete variety and V is a rational G-module, then $V|_H|^G\cong V$. This is clear from the fact that the H-fixed points coincide with the G-fixed points in any rational G-module (such as $R(G)\otimes V$) since $R(G)^H=R(G/H)\cong k$ (constant functions) when G/H is complete.

Corollary. *Let P be a parabolic subgroup of G which has a Levi decomposition $P=U\cdot L$ with U the unipotent radical of P. Let $B\leqq P$ be a Borel subgroup and let V be a rational B-module. Then V extends to a rational P-module iff $V|_{B\cap L}$ extends to a rational L-module.*

Proof. If $V|_{B\cap L}$ extends to a rational L-module, then $V|_{B\cap L}|^L\cong V|_{B\cap L}$ by evaluation. Hence the evaluation map $V|^P\to V$ is also an isomorphism by the proposition above. Q.E.D.

Finally we note that if W is a rational H-algebra, in the sense that W is a k-algebra on which H acts rationally as a group of algebra automorphisms, then $W|^G$ has a natural structure as a G-algebra. (In particular, the ring $R(G/H) \cong R(G)^H$ may be regarded as the G-algebra induced by the trivial one-dimensional H-algebra k. Also $R(G) \cong R(G/1) \cong k|^G \cong k|^H|^G \cong R(H)|^G$.) Moreover, if V is a rational G-algebra and $V \to W$ is an H-algebra homomorphism, then the induced map $V \to W|^G$ is a G-algebra homomorphism. The details are left to the reader.

2. The Demazure Desingularization

2.1. Let G be a connected algebraic group with fixed Borel subgroup $B = T \cdot U$ where T is a maximal torus and U is the unipotent radical of B. Let $\bar{G} = G/G_u$ be the reductive group formed from G by factoring out its unipotent radical G_u, and let W be the Weyl group of the image $\bar{T}$ of T in $\bar{G}$. Let $w_1 \dots w_N$ be a reduced expression for the long word in W, where the w_i are fundamental reflections relative to $\bar{B}$. We set $P_i = B \cup B w_i B$, the "minimal" parabolic subgroup containing B associated to the reflection w_i. Then $G = P_1 \dots P_N$ and moreover each element in the open cell $B w_1 \dots w_N B$ can be written as a product $x_1 \dots x_N$ with $x_i \in B w_i B$ which is unique up to insertion of factors $b_i b_i^{-1}$ between x_i and x_{i+1}, $1 \leqq i < N$ [3; (3.4)].

We now define inductively, for $i = 1, \dots, N$, a locally trivial principal B-bundle X_i with base space Y_i, projection π_i. Let $X_1 = P_1$, $Y_1 = P_1/B$ and take for π_1 the quotient morphism. Having defined X_i, we set $X_{i+1} = X_i \times^B P_{i+1}$ and $Y_{i+1} = X_{i+1}/B \cong X_i \times^B (P_{i+1}/B)$.

Note as in (1.1), Y_{i+1} is a locally trivial P_{i+1}/B-bundle over Y_i. Since the morphism $Y_{i+1} \to Y_i$ is locally a projection $U \times P_{i+1}/B \to U$, it is a proper morphism (because P_{i+1}/B is complete). Further, since Y_1 is complete, an induction argument implies that all the Y_i are complete.

We also obtain by induction that multiplication $P_1 \times \cdots \times P_i \to P_1 \dots P_i$ factors through X_i to give a birational isomorphism $f_i \colon X_i \to P_1 \dots P_i$. (Note that $B w_1 \dots w_i B \times^B B w_{i+1} B \cong B w_1 \dots w_{i+1} B$ by [3; (3.4)].) This gives a commutative diagram of locally trivial principal B-bundles

$$\begin{array}{ccc} X_N & \xrightarrow{f_N} & G \\ \downarrow & & \downarrow \\ Y_N & \longrightarrow & G/B \end{array}$$

which is a pull-back diagram by (1.1). Since Y_N is complete, $Y_N \to G/B$ is proper, so f_N is proper. Thus, by Lemma (1.2) we have that $f_N^* \colon R(G) \to R(X_N)$ is an isomorphism. On the other hand, we clearly have $f_N^*(R(G)) \subseteq R(P_N) \otimes_B \dots \otimes_B R(P_1) \subseteq R(X_N)$[2]. Thus,

[2] One can also show $R(X_N) = R(P_N) \otimes_B \cdots \otimes_B R(P_1)$ by observing X_N is the quotient of $P_1 \times \cdots \times P_N$ by an action of $B \times \cdots \times B$ ($N-1$ factors).

2.2. Lemma. *The map* $R(G) \to R(P_N) \otimes_B \dots \otimes_B R(P_1)$ *obtained from multiplication* $P_1 \times \cdots \times P_N \to G$ *is an isomorphism of k-algebras.*

In the next section we obtain a version of this result without the assumption that the product $P_1 \dots P_N = G$ corresponds to a *reduced* expression for the long word in the Weyl group. (There is still an isomorphism, but the map is not in general obtained just from multiplication.)

3. The Main Results

Again let G be a connected algebraic group with fixed Borel subgroup B. If $P_1, \dots, P_n$ is any sequence of parabolic subgroups containing B, and V is a rational B-module, we let $V|^{P_1,\dots,P_n}$ abbreviate $V|^{P_1}|_B|^{P_2}\dots|^{P_n}$.

3.1. Theorem. *The B-module structure of* $V|^{P_1,\dots,P_n}$ *is determined* (up to a natural isomorphism) *by the set* $P_1 \dots P_n$. *If this product is G, then* $V|^{P_1,\dots,P_n} \cong V|^G$ *as* P_n*-modules.*

Proof. It clearly suffices to prove the theorem when the P_i's are "minimal" parabolic subgroups.

Suppose first that $P_1 \dots P_n = G$ and that the sequence of P_i's correspond to a reduced expression for the long word in the Weyl group, as in § 2. By (2.2) we have an isomorphism $R(G) \cong R(P_n) \otimes_B \cdots \otimes_B R(P_1)$ by a map which is equivariant with respect to both the left P_n and the right P_1 actions. Hence,

$$V|^G \cong R(G) \otimes_B V = R(P_n) \otimes_B \cdots \otimes_B R(P_1) \otimes_B V \cong V|^{P_1,\dots,P_n}$$

as P_n-modules.

It remains to prove in general that $V|^{P_1,\dots,P_n}$ depends only on the set $P_1 \dots P_n$ when the P_i's are minimal parabolics. Write $P_i = B \cup B w_i B$ where w_i is a fundamental reflection in the Weyl group W. If the expression $w_1 \dots w_n$ is reduced, then, by a theorem of Steinberg [14; Lemma 83], we can get to any other reduced expression $w_1' \dots w_n'$ for the same word by repeatedly replacing some consecutive product of w_i's which gives a reduced expression for the long word in a rank two parabolic subgroup by another reduced expression for this long word. The corresponding consecutive product of the P_i's will be the rank two subgroup itself. Applying the case of the theorem already proved to this subgroup, we obtain $V|^{P_1,\dots,P_n} \cong V|^{P_1',\dots,P_n'}$ where $P_i' = B \cup B w_i' B$. Finally, suppose $w_1 \dots w_n$ is not reduced. By the exchange condition [4; IV; § 1, 1.6], there are indices $k < l \leqq n$, such that $w_k \dots w_l$ is reduced and equal to $w_{k+1} \dots w_{l+1}$. Correspondingly, we have $P_k \dots P_l = P_{k+1} \dots P_{l+1}$, since $P_k \dots P_l$ here is the closure of $B w_k \dots w_l B$ [3; (3.13)]. Hence applying the theorem in the reduced case, we can assume $P_l = P_{l+1}$ for some $l < n$. However, $M|_B|^{P_l} \cong M$ when M is a rational P_l-module, as remarked in (1.3), since P_l/B is complete. Thus, $V|^{P_1,\dots,P_n} \cong V|^{P_1,\dots,P_{l-1},P_{l+1},\dots,P_n}$ and the theorem follows in general from the reduced case by induction on n. Q.E.D.

We remark that when V is a k-algebra, all the isomorphisms in the proof above are *algebra* isomorphisms: cf. the remark at the end of (1.3).

3.2. Corollary. *If $P_1, \ldots, P_n$ are parabolic subgroups containing B with $P_1 \ldots P_n = G$, then there is a k-algebra isomorphism $R(G) \cong R(P_n) \otimes_B \cdots \otimes_B R(P_1)$, equivariant with respect to the obvious right action of P_1 and left action of P_n.*

Proof. We have

$$R(G) = R(P_1)|^G \cong R(P_1)|^{P_1, \ldots, P_n} \cong R(P_1)|^{P_2, \ldots, P_n} = R(P_n) \otimes_B \cdots \otimes_B R(P_1).$$

These maps are P_n-equivariant k-algebra homomorphisms by the theorem and remark above, and are P_1-equivariant by naturality. Q.E.D.

The isomorphism above is not always obtained from the product map $P_1 \times \cdots \times P_n \to G$, though this is true whenever a minimum length factorization of each P_i into a product of minimal parabolic subgroups containing B gives a minimum length factorization of G; cf. Lemma (2.2).

We now give the main result of this paper.

3.3. Corollary. *Suppose V is a rational B-module which extends to a rational module for every minimal parabolic subgroup P containing B. Then V extends uniquely to a rational G-module.*

Proof. If V extends to P we have $V|^P \cong V$ as B-modules, since P/B is complete. Hence, $V|^{P_1, \ldots, P_n} \cong V$ for any sequence of minimum parabolic subgroups containing B. Taking $P_1 \ldots P_n = G$, we have by Theorem (3.1) that V extends to a unique rational G-module, namely $V|^G$, cf. (1.3). Q.E.D.

Here is another interesting way to state this result in the finite dimensional case.

3.4. Corollary. *Let G, H be algebraic groups with G connected and B a Borel subgroup of G. Suppose $B \to H$ is a homomorphism of algebraic groups, which extends to an algebraic group homomorphism $P \to H$ for each minimal parabolic subgroup P containing B. Then $B \to H$ extends uniquely to an algebraic group homomorphism $G \to H$.*

For reductive groups G both results (3.3) and (3.4) above may be refined using the Levi decomposition of P and Corollary (1.3). We note also that the proof of (3.4) in the reductive case can be reduced to rank 2 groups using known results on generators and relations. This does not seem to be the case for (3.3) which applies to infinite dimensional rational modules.

A result similar to (3.4) holds for Lie algebras in characteristic zero. We thank the referee for pointing this out, and for supplying the argument. Let $\mathfrak{g}$ be a split semisimple Lie algebra over a field k of characteristic zero. Fix a split Cartan subalgebra $\mathfrak{h}$, and Borel subalgebra $\mathfrak{b} \supseteq \mathfrak{h}$. For each fundamental root α, let $\mathfrak{p}_\alpha \supseteq \mathfrak{b}$ denote the corresponding standard minimal parabolic subalgebra. Using the notation of [18; VIII, §4] let $\{X_\alpha, X_{-\alpha}, H_\alpha\}$ be the standard generators for $\mathfrak{g}$. Let $\mathfrak{l}_\alpha = \mathfrak{h} + kX_\alpha + kX_{-\alpha}$ denote the standard Levi complement of $\mathfrak{p}_\alpha$. Then we have

3.5. Proposition. *Let $\mathfrak{a}$ be a finite dimensional Lie algebra over k, and let $f: \mathfrak{b} \to \mathfrak{a}$ be a Lie algebra homomorphism. Assume for each α that $f|_{\mathfrak{b} \cap \mathfrak{l}_\alpha}$ extends to a Lie algebra homomorphism $f_\alpha: \mathfrak{l}_\alpha \to \mathfrak{a}$. Then the f_α's extend uniquely to a Lie algebra homomorphism $\tilde{f}: \mathfrak{g} \to \mathfrak{a}$ (which extends f).*

Proof. The uniqueness of $\tilde{f}$ is clear since we know the images x_α, $x_{-\alpha}$, h_α of X_α, $X_{-\alpha}$, H_α. We have to check the relations between these, in order to apply [18; Thm. 1, p. 100]. These are labelled (16), ..., (22) on [*op. cit*; p. 99]. Of these (16), (17), (18), (19), and (21) obviously hold, since they take place in the image of $\mathscr{E}$ or an $\mathscr{E}_\alpha$. To prove (20), notice that $\mathfrak{sl}_2$ acts on $\mathfrak{o}$ by means of $ad\, x_\alpha$, $ad\, x_{-\alpha}$, and $ad\, h_\alpha$. For that action, x_β is of weight $n = n(\beta, \alpha)$ by (17), and $(ad\, x_\alpha)^{1-n}(x_\beta) = 0$ by (21). By well-known properties of finite dimensional $\mathfrak{sl}_2$-modules, this implies that x_β is killed by $ad\, x_{-\alpha}$, i.e., that $[x_{-\alpha}, x_\beta] = 0$ ($\alpha \neq \beta$ of course). This gives (20). To prove (22), apply the same argument: by (20) and (18), $x_{-\beta}$ is of weight $-n$, and is killed by $ad\, x_\alpha$; hence $(ad\, x_{-\alpha})^{1-n}(x_{-\beta}) = 0$, which is (22). Q.E.D.

4. A Counterexample

Let G be the Chevalley group of type A_2 over the field F of two elements. Thus G is a finite group of order 168. Let B be a Borel subgroup of G, and let P_1, P_2 be the two proper parabolic subgroups of G containing B. The group B is dihedral of order 8, and has three distinct subgroups D_1, D_2, E of order 4, with $D_1 \lhd P_1$, $D_2 \lhd P_2$, and E cyclic. Choose subgroups T_1, T_2 of D_1, D_2, respectively, which have order 2 and are not contained in E.

Let $\overline{M}$ be the 10-dimensional FB-module $F|_E^B \oplus F|_{T_1}^B \oplus F|_{T_2}^B$. Then $\overline{M}$ extends to an FP_1-module and also to an FP_2-module. But $\overline{M}$ does not extend to an FG-module.

It is easy to show $\overline{M}$ extends to P_1 and to P_2: Both P_1 and P_2 are isomorphic to the symmetric group on four letters, hence contain copies S_1, S_2 of the symmetric group on three letters. Using the Mackey decomposition theorem, one readily computes

$$F|_E^{P_1}|_B \approx F|_E^B \oplus F|_{T_1}^B,$$

$$F|_{S_1}^{P_1}|_B \approx F|_{T_2}^B.$$

The sum is $\overline{M}$, so $\overline{M}$ extends to an FP_1-module. Similarly $\overline{M}$ extends to P_2.

To prove $\overline{M}$ does not extend to G, we will make use of the modular representation theory of finite groups, and in particular the theory of vertices and sources [16, § 5].

Let R denote the complete ring of 2-adic integers. Suppose $\overline{N}$ is an FG-module with $\overline{N}|_B \approx \overline{M}$. Since $\overline{N}$ is a direct summand of $\overline{N}|_B|^G$, which is a permutation module, there is an R-free RG-module N with $N/2N \approx \overline{N}$ [17, Prop. 1]. Moreover,

$$N|_B \approx M = R|_E^B \oplus R|_{T_1}^B \oplus R|_{T_2}^B.$$

Let N_1 be an indecomposable component of N such that the indecomposable module $R|_E^B$ is a direct summand of $N_1|_B$. If $T \leqq B$ is a subgroup of order at most two, then $R|_E^B$ is certainly not a direct summand of the indecomposable module $R|_T^B$, and so N_1 does not have vertex T. Since N_1 is a direct summand of $N_1|_B|^G$, it

must be that N_1 is a direct summand of $(R|_E^B)|^G \cong R|_E^G$, and that N_1 has vertex E and source R.

Next, in preparation for an application of the Green correspondence, we will show that there is only one R-free RP_1-module which is indecomposable with vertex E and source R, namely $R|_E^{P_1}$. It is enough to show this module is indecomposable. One finds readily that its character is $1+\chi_2+\chi_3$ where χ_2, χ_3 are irreducible of degree 2, 3, respectively. Also $R|_E^{P_1}|_B \approx R|_E^B \oplus R|_{T_1}^B$. Since E acts trivially on $R|_E^B$, the latter cannot have character $\chi_2|_B$, and so $R|_E^{P_1}$ must be indecomposable.

Thus $R|_E^{P_1}$ must be the Green correspondent on P_1 of N_1. (The Green correspondence is applicable, since $P_1 \geqq B = N_G(E)$.) In particular N_1 must be self-dual, since $R|_E^{P_1}$ is self-dual. Also $N_1|_{P_1} \approx R|_E^{P_1} \oplus Q$ for some RP_1-module Q.

Suppose $Q=0$. The ordinary irreducible characters of G have degrees 1, 3, 3, 6, 7, 8. The two characters of degree 3 are complex conjugates, and so the character of N_1 cannot be composed entirely of characters of degree 3 (recall that the character of $R|_E^{P_1}$ is $1+\chi_2+\chi_3$). Nor can the character of N_1 be irreducible of degree 6, since one finds that the latter vanishes on an element of order 4. It follows that $Q \neq 0$.

Since M has only three indecomposable components, two of which are components of $R|_E^{P_1}|_B$, we must have $N_1 = N$.

The character of N contains 1 at most once, since N_1 was shown to be a summand of $R|_E^G$. Also the character is real, since N_1 was shown to be self-dual. A moment's examination of the degrees above shows that G has no such character of degree 10, a contradiction.

5. Appendix

In this section G is a simply connected semisimple algebraic group with fixed Borel subgroup B. Given a rational character λ on B we also denote by λ the corresponding one-dimensional B-module.

In [12], Kempf proved a vanishing theorem which may be stated as follows in the language of rational cohomology; cf. [6; (1.2)].

5.1. Theorem. *Let λ be a dominant character on B. Then $H^n(B, R(G) \otimes -\lambda) = 0$ for all positive integers n.*

This result appears to lie very deep. In an effort to reorganize and simplify Kempf's proof, Andersen [1] proved a reduction theorem very similar to the following result.

5.2. Lemma. *Suppose $P_1, \ldots, P_n$ are "minimal" parabolic subgroups containing B with $P_1 \ldots P_n = G$. Suppose*

$$H^1(B, R(P_j) \otimes -\lambda|^{P_1, \ldots, P_{j-1}}) = 0$$

for all $j \leqq n$ (read $j=1$ as $R(P_1) \otimes -\lambda$) *and all dominant characters λ on B. Then* (5.1) *holds.*

Proof. We will assume the field k has positive characteristic p; in characteristic zero, (5.1) is a special case of the Borel-Weil-Bott theorem, cf. [9].

We will also assume (5.1) for groups of type A_1, where it is easily proved (using the Čech cohomology of the projective line $\mathbb{P}^1$, for example). This makes available for the groups P_i the results of [6] whose proofs used (5.1). (The fact that P_i is not semisimple causes no difficulty.)

Given a rational B-module V, $H^n(B, R(P_j) \otimes V)$ is the Čech cohomology of a certain $P_j/B \approx \mathbb{P}^1$ vector bundle [6; (1.2)], hence vanishes for $n>1$ automatically. Now let V be a rational G-module. We assert the hypothesis of (5.2) implies $H^n(B, V \otimes -\lambda) \cong H^n(B, V \otimes -\lambda|^G)$ for all $n \geqq 0$ and all dominant characters λ on B. In fact, using [6; (2.1)] and [6; (3.1)], we find

$$H^n(B, V \otimes -\lambda|^{P_1,\ldots,P_{j-1}})$$

$$\cong H^n(P_j, V \otimes -\lambda|^{P_1,\ldots,P_j}) \cong H^n(B, V \otimes -\lambda|^{P_1,\ldots,P_j})$$

and iteration together with (3.1) proves our claim.

We now assert that $H^n(B, R(G))=0$ for $n>0$. For this it is enough to prove $H^n(B, I)=0$, $n>0$, for any rational G-module I which is the rational injective envelope of an irreducible module [5; (1.8)]. Since $R(U)$ is a rationally injective B-module, we find easily that I is a direct summand of $(R(U) \otimes -\mu)|^G$ for some *dominant* character μ. From [7; (5.5)], $R(U)$ is a directed union of B-submodules of the form $St(q) \otimes -(q-1)\delta$, where δ is the sum of the fundamental dominant weights, q is a power of p, and $St(q)$ denotes the Steinberg module of high weight $(q-1)\delta$. Since cohomology commutes with direct limits, we find from [5; (1.5)], the previous paragraph, and [6; (2.2)] applied to the composition factors of $St(q) \otimes -(q-1)\delta - \mu$, that $H^n(B, (R(U) \otimes -\mu)|^G)=0$ for $n>0$ and μ dominant. Thus, $H^n(B, I)=0$ for all $n>0$, as desired.

Now for any rational G-module V we have $R(G) \otimes V$ isomorphic as a G-module to a direct sum of copies of $R(G)$ [5; (1.6)]. Thus, by above, we finally obtain for $n>0$,

$$H^n(B, R(G) \otimes -\lambda) \cong H^n(B, R(G) \otimes -\lambda|^G)=0. \quad \text{Q.E.D.}$$

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On the Signature of a Link

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0. Introduction

In [4], Goeritz described how a quadratic form could be obtained from a regular projection of a knot, and showed that some of the algebraic invariants of this form are invariants of the knot. The signature, however, is not. Later, in [20], Trotter introduced a different notion of quadratic form, defined in terms of an orientable surface spanning the knot, and showed that the signature of this form (now known as the signature of the knot) is a knot invariant. Now Goeritz's form has a natural interpretation in terms of a spanning surface obtained from the given knot projection (see [17] and §2), and the non-invariance of its signature can be regarded as a consequence of the fact that this surface need not be orientable. In the present paper, we show how to define a quadratic form using any spanning surface, which simultaneously generalizes the forms of Goeritz and Trotter, and are able to relate the signature of this form to the signature of the knot. In particular, we obtain a simple algorithm for calculating the signature $\sigma(k)$ of a knot k from any regular projection of k, which expresses $\sigma(k)$ as the signature of the corresponding Goeritz matrix plus a certain 'correction term'.[1] Since the Goeritz matrix is often considerably smaller than any Seifert matrix, (for instance, any torus knot of type $(2, 2n+1)$ has a 1×1 Goeritz matrix), this should be of interest to the practical knot theorist, so for convenience we summarize the algorithm now. First colour the projection of k in chequerboard fashion. To each double point D assign an incidence number $\eta(D) = \pm 1$ as shown in Figure 1.

Let G be the associated symmetric integral matrix of Goeritz (see [4] or §1 below). Divide the double points into two types as shown in Figure 2. (For this we must orient k, but the result is independent of the choice of orientation. Also, it does not matter which strand passes over.)

Define $\mu = \sum \eta(D)$, summed over all double points D of type II. (Note that if the shaded surface is orientable, then there are no double points of type II). The result is then

[1] For another algorithmic approach to the computation of the signature, see Conway's article [3].

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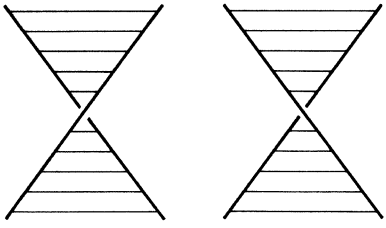

Fig. 1

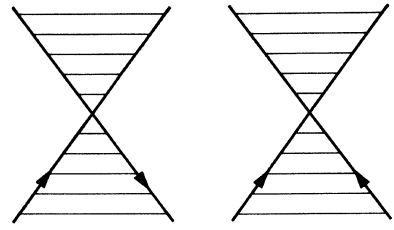

Fig. 2

Theorem 6[2]. $\sigma(k) = \operatorname{sign} G - \mu$.

We remark that, having somehow arrived at this function $\operatorname{sign} G - \mu$, the fact that it is an invariant of k can be verified directly by showing that it is unchanged by any of the Reidemeister moves on a knot projection [1], [15]. Indeed, the hard part of this was done by Goeritz, viz. describing the effect of these moves on G; keeping track of μ is comparatively simple. It is of some historical interest that, in this sense, the signature of a knot could have made its appearance thirty years before it did.

As the title suggests, everything we do works equally well for links as for knots, but, hoping to clarify the exposition, we treat knots first, in §§1–4, and indicate later, in §5, the modifications necessary for the general case. §1 contains a review of the Goeritz matrix. In §2, we defined our quadratic form $\mathscr{G}_V$ for any surface V spanning the knot or link. In §3, we define an 'Euler number' $e(V)$ for the spanning surface V. Then, by interpreting $\mathscr{G}_V$ as the intersection form of a certain 2-fold branched covering of the 4-ball (this was done for orientable V by Kauffman and Taylor [8]), and using the G-signature theorem, we show that $\operatorname{sign} \mathscr{G}_V + \frac{1}{2} e(V)$ is independent of the choice of V. By taking V to be orientable, we can relate this invariant to the signature of the knot or link. The algorithm described above is derived in §4. (See §5, Theorems 6′ and 6″, for the case of links.) In §6 we outline an alternative approach, avoiding the use of covering spaces, which involves a generalization to arbitrary spanning surfaces of the notion of S-equivalence. As a final illustration of the usefulness of non-orientable spanning surfaces, we give, in §7, a quick proof of a result of Shinohara [18] on the signature of knots contained in knotted solid tori.

Note on Conventions. We have reversed the convention of Goeritz regarding which regions of a knot projection are white and which black; this means that the spanning surface relevant to the discussion is now indicated by the *black* regions. (This convention agrees with Seifert [17].) Also, we have not assigned incidence number zero to double points at which a white region is incident with itself; to compensate for this we have adopted a definition of G which renders the incidence numbers of such points irrelevant. The advantage of this is that it eliminates separate consideration of these points when interpreting μ.

We work throughout in the smooth category.

[2] We are informed that an equivalent formula is known to A. Marin

To avoid repetition, we have used the word 'surface' (except where it appears in the phrase 'closed surface') to mean 'compact surface without closed components'.

Knots, links, and surfaces are unoriented unless otherwise specified, but the 3-sphere, the 4-ball, and, in general, all manifolds of dimensions 3 and 4, will be oriented.

1. The Quadratic Form of a Knot (after Goeritz)

For the convenience of the reader we shall give here the definition of Goeritz's matrix [4]. Let $k \subset S^3$ be a knot, and K (the image of) a regular projection of k onto the plane $R^2 \subset R^3 = S^3 - \{\infty\}$. Colour the regions of $R^2 - K$ alternately black and white. Denote the white regions by $X_0, X_1, \ldots, X_n$. Assign an incidence number $\eta(D) = \pm 1$ to each double point D as in §0. For $0 \leqq i, j \leqq n$ define

$$g_{ij} = \begin{cases} -\sum \eta(D) \text{ summed over double points } D \text{ incident to } X_i \text{ and } X_j, \text{ if } i \neq j \\ -\sum_{k=0,\ldots,n,\, k \neq i} g_{ik} \quad \text{if } i = j. \end{cases}$$

Let $G' = (g_{ij})$, $i, j = 0, \ldots, n$. Then the Goeritz matrix $G = G(K)$ associated to K is the $n \times n$ symmetric integer matrix obtained from G' by deleting the 0-th row and column; i.e. $G(K) = (g_{ij})$, $i, j = 1, \ldots, n$. (We shall use the notation $G(K)$ despite the fact that $G(K)$ depends on more than just K, namely, the ordering of the white regions, and, in particular, the choice of X_0. As regards the latter, the convention of Goeritz is to take X_0 to be the unbounded region; any other choice then corresponds simply to changing the projection by re-siting the point at infinity.)

It follows from [4] that matrices $G(K_1)$ and $G(K_2)$ obtained from any two projections K_1 and K_2 of k are related by a finite sequence of certain moves. The list of moves was subsequently reduced by Kneser and Puppe [10] to the following two.

$$G \mapsto RGR^T, \quad R \text{ integral and unimodular} \tag{1}$$

$$G \mapsto \left[\begin{array}{c|c} G & 0 \\ \hline 0 & \pm 1 \end{array}\right]. \tag{2}$$

In particular, the absolute value of the determinant of G is an invariant of the knot k. Goeritz also showed that the Minkowski units C_p of G at odd primes p are invariants of k. However, the signature sign G is clearly not invariant under move (2), and it is this shortcoming that we seek to rectify. (The Minkowski unit C_2, which is determined by the C_p for odd p together with the signature and the determinant, is also not an invariant of k [4].)

2. Spanning Surfaces

Let $V \subset S^3$ be a surface (orientable or not, but in any case unoriented) with ∂V the knot k. We define a bilinear form $\mathcal{G}_V$: $H_1(V) \times H_1(V) \to \mathbb{Z}$ as follows. Suppose $\alpha, \beta \in H_1(V)$ are represented by 1-cycles a, b. Then $2b$ can be pushed off V into $S^3 - V$, obtaining τb, say. (If b preserves orientation, we push off one copy of b to each side of V.) Define $\mathcal{G}_V(\alpha, \beta)$ to be the linking number of a and τb. More formally, let N be a closed tubular neighbourhood of V; N is the total space of an I-bundle over V with projection π: $N \to V$, say. Let $\tilde{V}$ be the corresponding ∂I-bundle. Then $\pi|\tilde{V}$: $\tilde{V} \to V$ is the orientation double cover if V is non-orientable, or the trivial double cover otherwise. Let τ: $H_1(V) \to H_1(\tilde{V})$ be the transfer map. Since V and $\tilde{V}$ are disjoint subsets of S^3, linking number defines a pairing

$$Lk\colon\ H_1(V) \times H_1(\tilde{V}) \to \mathbb{Z}.$$

We define

$$\mathcal{G}_V(\alpha, \beta) = Lk(\alpha, \tau\beta) \qquad (\alpha, \beta \in H_1(V)).$$

Remarks. (1) If V is orientable, this coincides with the quadratic form defined by Trotter in [20]. For in this case, let i_+, i_-: $V \to \tilde{V}$ be the two sections of $\tilde{V} \to V$; then $\tau\beta = i_{+*}\beta + i_{-*}\beta$, so

$$\begin{aligned}\mathcal{G}_V(\alpha, \beta) &= Lk(\alpha, i_{+*}\beta) + Lk(\alpha, i_{-*}\beta)\\ &= Lk(\alpha, i_{+*}\beta) + Lk(\beta, i_{+*}\alpha).\end{aligned}$$

(2) It will follow from Theorem 3 below that $\mathcal{G}_V$ is symmetric: a direct proof will be given in §6.

We have remarked above that $\mathcal{G}_V$ generalizes Trotter's quadratic form; we now show that it also generalizes Goeritz's. Let K be a regular projection of the knot k, and adopt the notation of §1. Associated to K is a spanning surface $V = V(K)$ of k, namely that indicated by the black regions. A little more precisely, we may assume that k coincides with K except in a neighbourhood of each double point. Then $V(K)$ is built up out of discs and bands. Each disc lies in $S^2 = R^2 \cup \{\infty\}$ and is a closed black region less a small neighbourhood of each adjacent double point. Each double point gives a small, half-twisted band (see Fig. 3).

We can embed a graph Γ in $V \cap S^2$ as a deformation retract of V; Γ has one node in each disc of V, and one edge running across each band (see Fig. 4). Let the complementary regions of Γ in S^2 be $Y_0, \ldots, Y_n$; we number these so that $X_i \subset Y_i$. Each Y_i inherits an orientation from S^2; the homology classes $\alpha_i = [\partial Y_i]$ generate $H_1(\Gamma) \cong H_1(V)$ subject to the single relation $\sum_{i=0}^{n} \alpha_i = 0$.

Theorem 1. *With the above notation,*

$$\mathcal{G}_{V(K)}(\alpha_i, \alpha_j) = g_{ij}.$$

Consequently, $\mathcal{G}_{V(K)}$ has matrix $G(K)$ with respect to the basis $\{\alpha_i | 1 \leqq i \leqq n\}$ of $H_1(V)$.

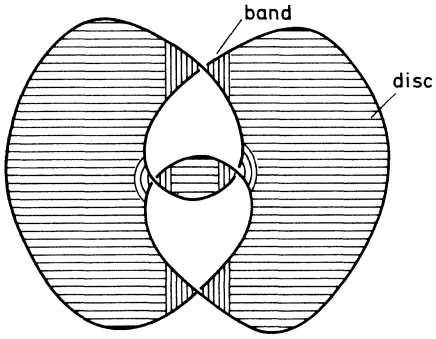

Fig. 3

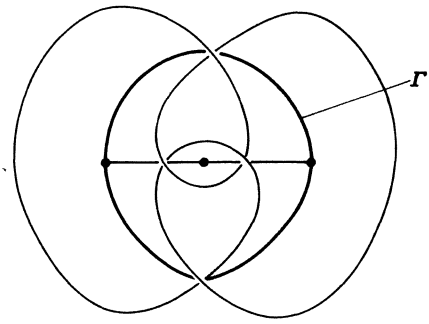

Fig. 4

Proof. Because $\sum_{i=0}^{n} \alpha_i = 0$ and $\sum_{j=0}^{n} g_{ij} = 0$, it suffices to prove the asserted equality for $i \neq j$.

We have

$$\mathcal{G}_{V(K)}(\alpha_i, \alpha_j) = Lk(\alpha_i, \tau\alpha_j) = Y_i \cdot \tau\alpha_j.$$

Now, over the discs of V, $\tilde{V}$ is contained in two parallel copies of S^2, so $\tau\alpha_j$ only intersects S^2 close to double points; in particular it only intersects Y_i close to double points incident to both X_j and X_i. Each such double point contributes one point of $Y_i \cdot \tau\alpha_j$, with sign $+\eta(D)$. (See Fig. 5.)

Hence, $Y_i \cdot \tau\alpha_j = g_{ij}$, as required. □

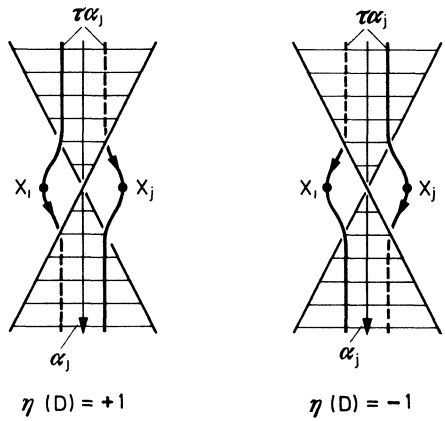

Fig. 5

3. Branched Covering Spaces

Suppose F is a (possibly non-orientable) surface properly embedded in D^4 with ∂F a knot k. Then, by a standard duality argument, $H_1(D^4 - F; \mathbb{Z}_2) \cong \mathbb{Z}_2$, generated by a meridian of F. Hence we can form the 2-fold branched cover of

D^4 with branch set F; we will denote this by M_F. Since F has a 1-dimensional spine, its normal 1-sphere bundle in D^4 has a section F'. (Note: we shall not distinguish between a section and its image.) Let $k'=\partial F'$, and $e(F)=-Lk(k,k')$. (k' is to be oriented similarly to k; $e(F)$ is then independent of the choice of orientation of k.) We might call $e(F)$ the *Euler number* of F; it is the normal Euler number of a closed surface obtained by capping off k with an orientable surface in S^3 and pushing into int D^4. Using σ to denote the signature of a 4-manifold, we have

Theorem 2. *Suppose F_1 and F_2 are properly embedded surfaces in D^4 with $\partial F_1 = \partial F_2$ a knot. Then*

$$\sigma(M_{F_1})+\tfrac{1}{2}e(F_1)=\sigma(M_{F_2})+\tfrac{1}{2}e(F_2).$$

Proof. Consider the pair $(S^4,E)=(D^4,F_1)\cup_\partial(-D^4,F_2)$. This has a 2-fold branched cover $M_E=M_{F_1}\cup_\partial(-M_{F_2})$. By Novikov additivity,

$$\sigma(M_E)=\sigma(M_{F_1})-\sigma(M_{F_2}),$$

while by the G-signature theorem ([2]; for an elementary proof in the case of an involution see [6])

$$\sigma(M_E)=2\sigma(S^4)-\tfrac{1}{2}e(E)=-\tfrac{1}{2}e(E)$$

where $e(E)$ denotes the normal Euler number of the closed surface $E\subset S^4$ (see [21]).

Now, $e(E)$ may be interpreted as follows. Take a section E' of the normal bundle of E which is transverse to E. At each point of $E\cap E'$ choose a local orientation of E. This determines a local orientation of E', and so an incidence number ± 1 for the intersection point; this is independent of the orientation choice. Then $e(E)$ is the sum of these incidence numbers over all points of $E\cap E'$.

It follows easily that $e(E)=e(F_1)-e(F_2)$, which completes the proof. □

Now suppose V is a spanning surface in S^3 for the knot k. Push int V into int D^4 in the obvious way to obtain a properly embedded surface $\hat{V}\subset D^4$ with $\partial\hat{V}=k$. The following result is already well-known in the case that V is orientable. (See, for example, [5], [8].)

Theorem 3. *With the above notation,*

$$(H_2(M_{\hat{V}}),\cdot)\cong(H_1(V),\mathscr{G}_V),$$

where $\cdot$ denotes the intersection form on $H_2(M_{\hat{V}})$.

Proof. We can construct $M_{\hat{V}}$ by gluing together two copies of the manifold obtained by cutting open D^4 along the trace of the isotopy which pushed int V into int D^4. But this manifold is homeomorphic to D^4; the part exposed by the cut corresponds to a tubular neighbourhood N of V in S^3. Thus $M_{\hat{V}}$ may be described as follows. Let $\iota\colon N\to N$ be the involution given by reflection of the fibre. Take two copies D_1^4 and D_2^4 of D^4, and let

$$M_{\hat{V}}=D_1^4\cup D_2^4/x\in N\subset D_1^4\sim\iota(x)\in N\subset D_2^4.$$

Now consider the Mayer-Vietoris sequence for $M_{\hat{V}}$:

$$0=H_2(D_1^4)\oplus H_2(D_2^4)\to H_2(M_{\hat{V}})\xrightarrow{\varphi} H_1(N)\to H_1(D_1^4)\oplus H_1(D_2^4)=0.$$

The inverse S of the isomorphism φ may be described as follows. If a is a 1-cycle in N, let

$$Sa=(\text{cone on } a \text{ in } D_1^4)-(\text{cone on } \iota(a) \text{ in } D_2^4).$$

Then $S([a])=[Sa]$. Hence if a, b are disjoint 1-cycles in N,

$$S([a])\cdot S([b])=Lk(a,b)+Lk(\iota(a),\iota(b)).$$

Finally, let $i_*\colon H_1(V)\to H_1(N)$ be the isomorphism induced by inclusion, and let α, $\beta\in H_1(V)$. Represent α, β by 1-cycles a and b in V. Then a and τb are disjoint 1-cycles in N, and τb is homologous to $2b$ in N. Moreover, $\iota(a)=a$, $\iota(\tau b)=\tau b$. Hence

$$\begin{aligned} Si_*\alpha\cdot Si_*\beta &= \tfrac{1}{2}S([a])\cdot S([\tau b]) \\ &= \tfrac{1}{2}(Lk(a,\tau b)+Lk(\iota(a),\iota(\tau b))) \\ &= Lk(a,\tau b) \\ &= \mathcal{G}_V(\alpha,\beta). \end{aligned}$$

Thus Si_* is the required isomorphism

$$(H_1(V),\mathcal{G}_V)\xrightarrow{\cong}(H_2(M_{\hat{V}}),\cdot). \quad \square$$

Now let k^V be a parallel copy of k missing V, and define $e(V)=-Lk(k,k^V)$. One easily proves

Lemma 4. $e(V)=e(\hat{V})$. $\square$

Corollary 5. *If V is a spanning surface for the knot k, then*

$$\sigma(k)=\operatorname{sign}\mathcal{G}_V+\tfrac{1}{2}e(V).$$

Proof. By Theorems 2 and 3 and Lemma 4, the right hand side of the asserted equation is independent of the spanning surface V. But if V is orientable, $\operatorname{sign}\mathcal{G}_V$ is by definition $\sigma(k)$, while $e(V)=0$. $\square$

We digress to note the following consequence of Theorem 3. Let M be a compact 4-manifold with $H_1(M)=0$, and suppose the intersection form on $H_2(M)$ is given, with respect to some basis, by the integral matrix A. By Lefschetz duality,

$$H_2(M,\partial M)\cong H^2(M)\cong \operatorname{Hom}(H_2(M),\mathbb{Z})=H_2(M)^*,$$

say, and the exact sequence of the pair $(M,\partial M)$ gives an exact sequence

$$H_2(M)\xrightarrow{A}H_2(M)^*\to H_1(\partial M)\to 0,$$

where the left-hand map is represented by the matrix A with respect to our basis for $H_2(M)$ and the corresponding dual basis for $H_2(M)^*$. It follows that A is a relation matrix for $H_1(\partial M)$.

Now suppose $H_1(\partial M;\mathbb{Q})=0$, and consider the linking form λ: $H_1(\partial M) \times H_1(\partial M)\to\mathbb{Q}/\mathbb{Z}$. Suppose α, $\beta\in H_1(\partial M)$ are represented by 1-cycles a, b. Let $\Delta\in\mathbb{Z}-\{0\}$ be an annihilator of $H_1(\partial M)$; for example, we could take $\Delta=\det A$. Then $\Delta a=\partial u$, $\Delta b=\partial v$, for some 2-chains u, v in ∂M, and by definition

$$\lambda(\alpha,\beta)=\frac{u\cap b}{\Delta} \quad (\mathrm{mod}\,1),$$

where $\cap$ denotes intersection number. We also have $a=\partial x$, $b=\partial y$ for some 2-chains x, y in M. Then $\Delta x-u$ and $\Delta y-v$ represent classes X, Y, say, $\in H_2(M)$. It is easy to see geometrically that

$$X\cdot Y=\Delta x\cap\Delta y-u\cap\Delta b,$$

and hence

$$\lambda(\alpha,\beta)=-\frac{X\cdot Y}{\Delta^2} \quad (\mathrm{mod}\,1).$$

Note that the map $H_2(M)\to H_2(M,\partial M)$ takes $X\mapsto[\Delta x]$, $Y\mapsto[\Delta y]$. Let $[x]$, $[y]$ correspond, under the isomorphism $H_2(M,\partial M)\cong H_2(M)^*$, to $\hat{X}$, $\hat{Y}\in H_2(M)^*$. Then (regarding X, Y, $\hat{X}$, $\hat{Y}$ as column vectors with respect to our bases of $H_2(M)$ and $H_2(M)^*$), we have $AX=\Delta\hat{X}$, $AY=\Delta\hat{Y}$. Therefore, recalling that A is invertible over $\mathbb{Q}$,

$$\frac{X\cdot Y}{\Delta^2}=\frac{X^tAY}{\Delta^2}=\frac{\Delta^2\hat{X}^tA^{-1}AA^{-1}\hat{Y}}{\Delta^2}=\hat{X}^tA^{-1}\hat{Y}.$$

Thus $\lambda(\alpha,\beta)=-\hat{X}^tA^{-1}\hat{Y}\,(\mathrm{mod}\,1)$. Also, the map $H_2(M)^*\to H_1(\partial M)$ takes $\hat{X}\mapsto\alpha$, $\hat{Y}\mapsto\beta$. In other words, λ is represented by the matrix $-A^{-1}$ with respect to the images in $H_1(\partial M)$ of our basis for $H_2(M)^*$.

Hence we recover the results of Seifert [17], that if k is a knot with 2-fold branched cover N, and G is any Goeritz matrix for k, then G is a relation matrix for $H_1(N)$, and the linking form on $H_1(N)$ is given by $\pm G^{-1}$ (the sign depending on the choice of orientation of N).

4. The Algorithm

Let K be a regular projection of the knot k, and recall from §0 the definition of $\mu=\mu(K)$: a crossing point is of type I or II according as the string orientations are or are not consistent with respect to the black regions. (See Fig. 2.) Then $\mu(K)$ is the sum of the incidence numbers of double points of type II.

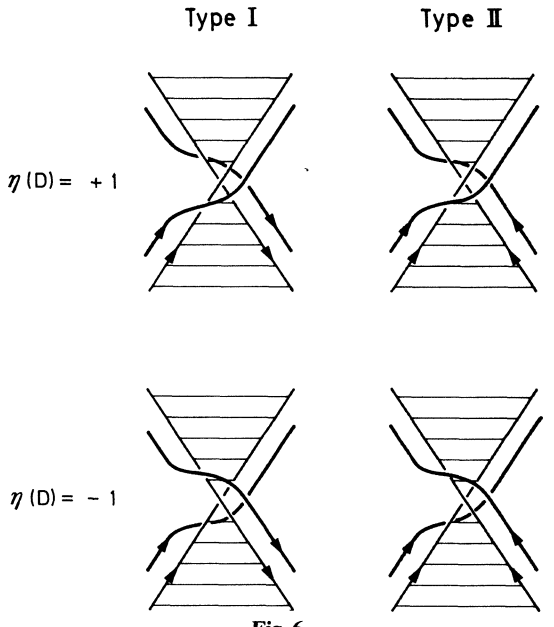

Fig. 6

Theorem 6. *For any regular projection K of the knot k, with associated Goeritz matrix G,*

$$\sigma(k) = \operatorname{sign} G - \mu(K).$$

The theorem is immediate from Corollary 5, Theorem 1, and

Lemma 7. $\mu(K) = -\frac{1}{2}e(V(K))$.

Proof. Let $V = V(K)$. We can take the push-off k^V of k which misses V to lie in the plane R^2 except in a neighbourhood of the double points. Figure 6 shows k^V close to a double point D in each of the four possible cases for D. If we calculate $Lk(k, k^V)$ by counting the number of times k^V passes under k from right to left, we see that D contributes $2\eta(D)$ if it is of type II, 0 if of type I. Hence $Lk/k, k^V) = 2\mu(K)$; i.e.

$$-e(V) = 2\mu(K). \quad \square$$

5. Links

We now describe how the theory has to be modified to take care of links.

Everything in §1 remains valid, except that the following additional move on the Goeritz matrix must be permitted. (See [11]; also §6.)

$$G \mapsto \left[\begin{array}{c|c} G & 0 \\ \hline 0 & 0 \end{array}\right]. \tag{3}$$

§2 carries over exactly as stated.

Regarding §3, the following comments are in order. Let F be a surface properly embedded in D^4, with ∂F the link $l=k_1\cup\cdots\cup k_m$, say. Then $H_1(D^4-F;\mathbb{Z}_2)\cong\mathbb{Z}_2^{\beta_0(F)}$, generated by meridians of F, and we can form the 2-fold branched cover M_F of D^4 with branch set (the whole of) F. Since F has no closed components, its normal 1-sphere bundle has a section F', with $\partial F'=l'=k_1'\cup\cdots\cup k_m'$, say. We then define

$$e(F)=-\sum_{i=1}^{m} Lk(k_i,k_i'),$$

where each pair k_i, k_i' is oriented compatibly but otherwise arbitrarily. Again, this can be interpreted as a normal Euler number, as follows. Attach 2-handles to D^4 along tubular neighbourhoods of the components of l, using the 0-framing for each component, giving a 4-manifold W, say. The union of F with the cores of the 2-handles is a closed surface E, and the normal Euler number of E in W is just $e(F)$. In particular, we see that $e(F)$ is independent of the choice of section F'. The definition of $e(V)$ for a surface V in S^3 extends analogously to the case where V has several boundary components:

$$e(V)=-\sum_{i=1}^{m} Lk(k_i,k_i^V).$$

Theorems 2 and 3 and Lemma 4 carry over exactly as stated, and as a consequence we have

Corollary 8. *If V is a spanning surface for the link l, then*

$$\operatorname{sign}\mathscr{G}_V+\tfrac{1}{2}e(V)$$

depends only on l. □

From our present point of view, in which non-orientable spanning surfaces are just as good as orientable ones, and (hence) links are unoriented, the invariant of Corollary 8 is the natural generalization to links of the signature of a knot. To relate this invariant to the classical signature of a link, we must now orient our links. More precisely, we consider a link together with an orientation of each of its components, modulo simultaneous reversal of these orientations. Let $\bar{l}$ be such a *semi-oriented* link, with underlying (unoriented) link l. One then defines the signature $\sigma(\bar{l})$ of $\bar{l}$ to be $\operatorname{sign}\mathscr{G}_V$ for any orientable spanning surface V of l which has a semi-orientation inducing the specified semi-orientation on $\bar{l}$. (Actually the original definition, due to Murasugi [13], was described in terms of a regular projection, in a purely combinatorial fashion, but this can be interpreted in terms of a certain orientable spanning surface associated with the projection (see [7, 19]).)

Now let V be any surface in S^3 whose boundary is the semi-oriented link $\bar{l}=\bar{k}_1\cup\cdots\cup\bar{k}_m$. ($V$ need not be orientable, and even if it is, we do not assume that it is compatible with the semi-orientation of $\bar{l}$.) Define

$$\bar{e}(V)=-\sum_{i,j=1}^{m} Lk(\bar{k}_i,\bar{k}_j^V).$$

Then

$$\bar{e}(V) = -\sum_{i=1}^{m} Lk(\bar{k}_i, \bar{k}_i^V) - 2 \sum_{1 \leqq i < j \leqq m} Lk(\bar{k}_i, \bar{k}_j) = e(V) - 2\lambda(\bar{l})$$

where $\lambda(\bar{l})$ is the *total linking number* of $\bar{l}$ (see [13]).

We then have the following analogue of Corollary 5, which identifies the invariant of Corollary 8 with the invariant $\xi(l) = \sigma(\bar{l}) + \lambda(\bar{l})$ of Murasugi [13]. (Note that this gives another proof of Theorem 1 of [13], that $\sigma(\bar{l}) + \lambda(\bar{l})$ is independent of the orientation; this proof is closely related to that of Kauffman and Taylor [8].)

Corollary 5′. *If V is any spanning surface for the link l, then $\xi(l) = \operatorname{sign} \mathscr{G}_V + \frac{1}{2} e(V)$.*

Proof. The right hand side is independent of V by Corollary 8. So choose V orientable; then a semi-orientation of V will detrmine a semi-oriented link $\bar{l}$, say. Since V is orientable, $\bar{e}(V) = 0$, hence $\frac{1}{2} e(V) = \lambda(\bar{l})$, while, by definition, $\operatorname{sign} \mathscr{G}_V = \sigma(\bar{l})$. □

Corollary 5′ can be equivalently stated in terms of the signature:

Corollary 5″. *If V is any spanning surface for the link l, then*

$$\sigma(\bar{l}) = \operatorname{sign} \mathscr{G}_V + \tfrac{1}{2} \bar{e}(V)$$

for any semi-oriented link $\bar{l}$ with underlying link l. □

We can now obtain two different (but of course closely related) generalizations of Theorem 6, to give algorithms for computing $\xi(l)$ and $\sigma(\bar{l})$ respectively from a link projection. (Since $\lambda(\bar{l})$ is easy to compute, any one of $\xi(l)$, $\sigma(\bar{l})$ can be readily obtained from the other. We discuss them both, however, since each has some claim to being the more natural generalization of the signature of a knot. Thus, on the one hand, $\xi(l)$ requires no mention of orientations, while on the other, the algorithm for $\sigma(\bar{l})$ involves the most natural generalization of the knot projection function μ.)

First consider $\xi(l)$. We need to be able to calculate $e(V(L))$ for the shaded surface $V(L)$ corresponding to a projection L of l. To this end, attach a number $\zeta(D)$ to each double point D of L as follows: if a single component of l crosses itself at D, let $\zeta(D) = 0$ or 1 according as D is of type I or II; if two distinct components cross at D, let $\zeta(D) = \frac{1}{2}$. Define $\nu(L) = \sum \zeta(D)\, \eta(D)$ summed over all double points D of L. Note that $\nu(L)$ is an invariant of the unoriented diagram L, and coincides with $\mu(L)$ when l is a knot.

One proves easily

Lemma 7′. $\nu(L) = -\frac{1}{2} e(V(L))$. □

And hence,

Theorem 6′. *For any regular projection L of the link l, with associated Goeritz matrix G,*

$$\xi(l) = \operatorname{sign} G - \nu(L).$$ □

Now let $\bar{l}$ be a semi-oriented link, and let $\bar{L}$ be some (correspondingly semi-oriented) regular projection of $\bar{l}$. Define $\mu(\bar{L})=\sum\eta(D)$, summed over all double points D of type II of $\bar{L}$. (Note that a semi-orientation is necessary to make the distinction between points of type I and points of type II meaningful.)

Then

Lemma 7''. $\mu(\bar{L})=-\frac{1}{2}\bar{e}(V(\bar{L}))$. □

And hence

Theorem 6''. *For any (semi-oriented) regular projection $\bar{L}$ of the semi-oriented link $\bar{l}$, with associated Goeritz matrix G,*

$$\sigma(\bar{l})=\operatorname{sign} G-\mu(\bar{L}).$$ □

6. A Down-to-Earth Approach

We have developed the properties of our quadratic form by paralleling the approach of Kauffman and Taylor [8] in the case of an orientable surface. A development along more traditional lines, avoiding the use of covering spaces, can also be given; we now outline this.

The following result is, as previously remarked, a consequence of Theorem 3; we give here a direct proof.

Proposition 9. *Let V be a surface in S^3. Then*

$$\mathscr{G}_V\colon H_1(V)\times H_1(V)\to\mathbb{Z}$$

is symmetric.

Proof. We adopt the notation of §2.

Orient $\tilde{V}$ so that a positive normal points out of N, and let $i_+, i_-\colon \tilde{V}\to S^3-\tilde{V}$ be given by translation in the positive and negative normal direction, respectively. Let $\alpha, \beta\in H_1(V)$. Then

$$\begin{aligned}\mathscr{G}_V(\alpha,\beta)&=Lk(\alpha,\tau\beta)\\&=Lk(\alpha,i_{+*}\tau\beta)\\&=\tfrac{1}{2}Lk(\tau\alpha,i_{+*}\tau\beta)\end{aligned}$$

so $\mathscr{G}_V(\alpha,\beta)-\mathscr{G}_V(\beta,\alpha)=\frac{1}{2}Lk(\tau\alpha,i_{+*}\tau\beta-i_{-*}\tau\beta)=\frac{1}{2}\tau\alpha\cdot\tau\beta$. But each point of $\alpha\cap\beta$ gives rise to two oppositely signed points of $\tau\alpha\cap\tau\beta$, so $\tau\alpha\cdot\tau\beta=0$. □

Suppose V_1, V_2 are two surfaces in S^3 with $\partial V_1=\partial V_2$. Suppose further there is a 3-ball $B^3=B^1\times B^2\subset S^3-\partial V_i$ such that

$$V_1\cap B^3=\partial B^1\times B^2$$

$$V_2\cap B^3=B^1\times\partial B^2$$

and

$$\operatorname{cl}(V_1-B^3)=\operatorname{cl}(V_2-B^3).$$

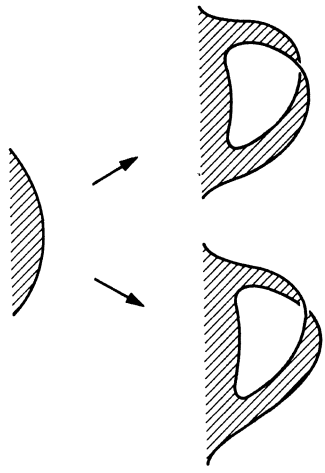

Fig. 7

In this situation we say that V_2 *comes from* V_1 *by a* 1*-handle move.*

(Here, and in the proof of Theorem 11 below, we leave corner-smoothing to the reader.)

We say that two surfaces in S^3 are S^**-equivalent* if they are related by a finite sequence of the following moves and their inverses

(A) Ambient isotopy

(B) A 1-handle move

(C) Addition of a small, $\frac{1}{2}$-twisted handle (see Fig. 7).

The boundaries of two S^*-equivalent surfaces are necessarily of the same link type.

(The terminology arises as follows. The relation of S-equivalence of Seifert matrices ([20, 14]) has a natural geometric analogue, namely the equivalence relation on orientable surfaces generated by the moves (A), (B) and $(B)^{-1}$ (through orientable surfaces only). It seems reasonable to call this S-equivalence also, whence S^*-equivalence.)

Proposition 10. *If* V_0, V_1 *are* S^**-equivalent surfaces in* S^3, *then*

$$\operatorname{sign} \mathscr{G}_{V_0} + \tfrac{1}{2} e(V_0) = \operatorname{sign} \mathscr{G}_{V_1} + \tfrac{1}{2} e(V_1).$$

Proof. That $\operatorname{sign} \mathscr{G}_V$ is invariant under (B) is proved just as in the orientable case ([14, 20]); clearly $e(V)$ is also invariant.

The effect of (C) on $\mathscr{G}_V$ is to take orthogonal sum with the form $[\varepsilon]$, where $\varepsilon = \pm 1$ depending on the sense of the twist in the band; since 2ε is subtracted from $e(V)$, the result follows. □

(In this way we see that (the matrices of) the quadratic forms of S^*-equivalent surfaces are related by the moves (1) and (2) of §1, provided the boundary is a knot. For a link, we must include the move

$$G \mapsto \left[\begin{array}{c|c} G & 0 \\ \hline 0 & 0 \end{array}\right] \tag{3}$$

corresponding to a 1-handle move connecting two components of a disconnected surface. This is also necessary for the Goeritz matrix of a link, as was observed by Kyle [11].)

Proposition 10 and the following theorem now imply that for any spanning surface V of a link l, $\operatorname{sign} \mathscr{G}_V + \frac{1}{2} e(V)$ depends only on l.

Theorem 11. *If V_0, V_1 are surfaces in S^3 with $\partial V_0 = \partial V_1$, then V_0 and V_1 are S^*-equivalent.*

We outline two proofs of this, one 3-dimensional, the other 4-dimensional. Let $l = \partial V_0 = \partial V_1$.

Proof I. First we can use move (C) (and (A)) to ensure that $V_0 \cap V_1$ consists of l together with simple closed curves in $\operatorname{int} V_0 \cap \operatorname{int} V_1$. Using (B), we may also assume that V_0 and V_1 are connected. Let M be the closure of a component of the complement in S^3 of a regular neighbourhood of $V_0 \cup V_1$. Note that M is a manifold, but that if M' is the closure of the corresponding component of $S^3 - (V_0 \cup V_1)$, then M' may fail to be a manifold along some curves of $\operatorname{int} V_0 \cap \operatorname{int} V_1$; we call such a curve *bad*. Regard M as a cobordism between the parts of ∂M coming from V_0, V_1 respectively, and take a handle decomposition of M on one end. We may assume there are only 1- and 2-handles, and that the former precede the latter. Let W be the surface between the 1- and 2-handles. If V_i' is obtained from V_i by deleting the part of V_i corresponding to ∂M and attaching W in the obvious way, then V_i' is obtainable from V_i by a sequence of moves (B) (and (A)), $i = 0, 1$. If ∂M accounts for all of $V_0 \cup V_1$, we are done. If not, there must be some curve $c \subset \operatorname{int} V_0 \cap \operatorname{int} V_1$ which is good for M. (Otherwise, any arc on V_i, $i = 0, 1$, starting at a point coming from ∂M must finish at another such point. Since V_i is connected, all points of $V_0 \cup V_1$ would then come from ∂M.) Pushing one copy of W off the other we get a surface V_1'', ambient isotopic to V_1', such that each curve of $\operatorname{int} V_0 \cap \operatorname{int} V_1$ gives rise to at most one curve of $\operatorname{int} V_0' \cap \operatorname{int} V_1''$. Moreover, by suitably choosing the direction in which to push W, we can arrange that c is eliminated. (We could not necessarily do this if c were bad.) Thus $\operatorname{int} V_0' \cap \operatorname{int} V_1''$ has fewer components than $\operatorname{int} V_0 \cap \operatorname{int} V_1$, and the proof is completed by induction. □

Remark. For the case of orientable surfaces, an outline of the above approach is given by Rice in [16]. However, this does not seem to take into account the possibility of the existence of bad curves (which may occur even in the orientable case).

Proof II. For $i = 0, 1$, let $l^{(i)}$ be a push-off of l missing V_i; by use of move (C) (and (A)) we may assume $Lk(k_j, k_j^{(0)}) = Lk(k_j, k_j^{(1)})$ for each component k_j of l. Hence we can take $l^{(0)} = l^{(1)} = l'$, say. Let V_i' be a section of the normal bundle of V_i in S^3, with $\partial V_i' = l'$. Consider S^4 as

$$D^4 \bigcup_{S^3 \times \{0\}} S^3 \times [0, 1] \bigcup_{S^3 \times \{1\}} -D^4$$

and let $E \subset S^4$ be the closed surface

$$V_0 \times \{0\} \bigcup_{l \times \{0\}} l \times [0, 1] \bigcup_{l \times \{1\}} V_1 \times \{1\}.$$

The sections V_0', V_1' define a section E' of the normal 1-sphere bundle of E, showing that $e(E)=0$.

The following steps will complete the proof.

(1) E bounds a 3-manifold in S^4.

(2) E bounds a 3-manifold M in $S^3\times[0,1]$, with $M\cap S^3\times\{i\}=V_i$ for $i=0,1$.

(3) Take a handle decomposition of M on V_0, with each handle in a level $S^3\times\{t_i\}$, $t_i\in(0,1)$, and with $M\cap S^3\times(t_i,t_{i+1})$ a vertical collar on $\partial(M\cap S^3\times[0,t_i])\cap S^3\times\{t_i\}$. (See [9].)

(4) Eliminate 0- and 3-handles from the decomposition of (3), and order the remaining 1- and 2-handles so that the former precede the latter. Then the intersection of M with a level between the 1- and 2-handles is a spanning surface obtained from V_i by 1-handle moves, for $i=0,1$.

Of these, only (1) needs further explanation.

Actually it is true that if E is any closed surface in S^4 with $e(E)=0$, then E bounds a 3-manifold in S^4. To see this, let T be a tubular neighbourhood of E, and $X=S^4-\operatorname{int}T$. Starting with any section E' of ∂T, it may be modified to produce a section E'' such that $Lk_{\mathbb{Z}_2}(\alpha,E'')=0$ for all $\alpha\in H_1(E)$. By duality and excision this implies that E'' bounds homologically mod 2 in X. Next, we find a map $f\colon \partial T\to RP^\infty$ transverse regular along E'' to a codimension one copy of RP^∞. The fact that E'' bounds mod 2 in X is sufficient to allow the extension of f to $X\to RP^\infty$, which gives a 3-manifold in X with boundary E''. □

Remark. In our situation, if l is a knot, we may take E'' to be the E' defined by V_0' and V_1', but if l has more than one component, this E' may have to be modified.

7. An Application

Let k be a knot, and l a link contained in an unknotted solid torus $T\subset S^3$. Let $f\colon T\to S^3$ be an embedding such that $f(T)$ is a tubular neighbourhood of k. Suppose also that f is faithful, that is, the image of a longitude of T (a simple closed curve on ∂T homologous in T to the core of T and null-homologous in $S^3-\operatorname{int}T$) is a longitude of $f(T)$. Let l^* denote the link $f(l)$, and let $n\in\mathbb{Z}_2$ be the mod 2 homology class $[l]\in H_1(T;\mathbb{Z}_2)$.

Theorem 12. $\xi(l^*)=\begin{cases}\xi(l), & \text{if } n=0\\ \xi(l)+\sigma(k), & \text{if } n=1.\end{cases}$

Remarks. (1) Taking l to be a knot, we recover the result of Shinohara [18, Theorem 9].

(2) Our method can also handle more general situations, in which we start with a link and replace some of its components by links lying in tubular neighbourhoods of these components. But since it does not seem possible to give a concise general statement, we omit the details.

(3) The second author has proved results analogous to those alluded to in (2) above for the Tristram-Levine signatures, using the interpretation of these as eigenspace signatures of higher-order branched covers. For the case of a knot, see [12].

Proof of Theorem 12. First consider the case $n=0$. Then l bounds a surface W, say, in T, and as spanning surface for l^* we may take $f(W)$. Since f is faithful, $Lk(C_1, C_2)=Lk(f(C_1), f(C_2))$ for any two disjoint 1-cycles C_1, C_2 in T. Hence $e(f(W))=e(W)$, and $\mathscr{G}_{f(W)}\cong\mathscr{G}_W$, which implies $\xi(l^*)=\xi(l)$.

Now suppose $n=1$. In this case, there is a surface W_0 in T whose boundary is the union of l and a longitude J of T. Let W be the spanning surface for l obtained from W_0 by capping off J with the obvious 2-disc in $S^3-\operatorname{int} T$. From an orientable spanning surface for k, one obtains an orientable surface V properly embedded in $S^3-\operatorname{int} f(T)$ whose boundary is the longitude $f(J)$ of $f(T)$. Then $V^*=V\cup_{f(J)} f(W_0)$ is a spanning surface for l^*. Since f is faithful, $e(V^*)=e(W)$. From a Mayer-Vietoris exact sequence, we obtain

$$H_1(V^*)\cong H_1(V)\oplus(H_1(f(W_0))/[f(J)])\cong H_1(V)\oplus H_1(W).$$

Moreover, since no 1-cycle in $f(T)$ links any 1-cycle in V, and since f is faithful, $\mathscr{G}_{V^*}$ is isomorphic to the orthogonal direct sum $\mathscr{G}_V\oplus\mathscr{G}_W$. Hence

$$\xi(l^*)=\operatorname{sign}\mathscr{G}_{V^*}+\tfrac{1}{2}e(V^*)=\operatorname{sign}\mathscr{G}_V+\operatorname{sign}\mathscr{G}_W+\tfrac{1}{2}e(W).$$

But $e(V)=0$ since V is orientable. Hence $\xi(l^*)=\xi(l)+\sigma(k)$. □

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Products in *K*-theory and Intersecting Algebraic Cycles

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One of the most elegent theorems of higher algebraic K-theory is Bloch's formula for a non-singular algebraic variety X, proved by Quillen [Q, Sect. 7, Th 5.19]:

$$H^p(X, \mathscr{K}_p) = A^p(X).$$

Here $A^p(X)$ denotes the group of codimension p cycles on X modulo rational equivalence, and $\mathscr{K}_p$ is the sheaf associated to the presheaf $U \mapsto K_p(\Gamma(U, \mathscr{O}_X))$ on X. Summing over p yields:

$$\bigoplus H^p(X, \mathscr{K}_p) = \bigoplus A^p(X).$$

Both sides of this formula have the structure of graded ring. On the left-hand side the product comes from cup-product of cohomology classes and multiplication in higher algebraic K-theory.

On the right-hand side there are two ways to define a product of two cycle classes:

1) When X can be expressed as a subscheme of projective space (i.e. is quasi-projective) the "moving lemma" says there are cycles within the given classes whose intersection has codimension as large as possible. Assigning multiplicities to the components of the intersection yields a cycle and a well-defined cycle class. It is possible to check that this makes $A^*(X)$ into a graded ring.

2) A method due to Baum, Fulton, and MacPherson yields a cycle class directly without moving cycles by using Gysin maps for Chow homology and deformation to the normal bundle of the diagonal, and it agrees with method 1. The discussions of this technique in [BFM] and [V] require X to be quasi-projective, but rumor has it that soon this hypothesis will be discarded. For the time being, we make the blanket assumption that X is quasi-projective.

In this paper we show that the product defined by method 2 agrees (up to $(-1)^{pq}$) with the K-theoretic product in $\bigoplus H^p(X, \mathscr{K}_p)$.

The proof falls into three stages: reduction to the diagonal, compatibility with flat maps, and compatibility with specialization. The hardest part is compatibility

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with specialization and involves proving Gersten's conjecture for coherent sheaves which meet the special fiber "properly" (4.6).

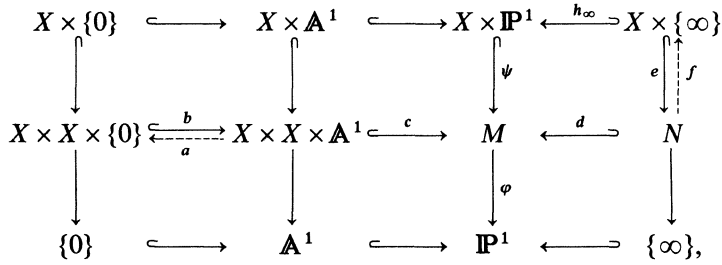
I thank Daniel Quillen for explaining the problem to me, Friedhelm Waldhausen for explaining products in K-theory to me, and finally, Steve Kleiman for a very enlightening discussion and the key to the proof of (4.6).

1. Intersection of Cycles

Suppose C and D are cycles of codimension p and q on a non-singular variety X over a field k. Let $[C]$, $[D]$ denote the corresponding rational equivalence classes. For preliminaries on rational equivalence the reader may refer to [F]. We proceed now to describe method 2 (from the introduction) for intersecting cycles [BFM].

Let $\Delta\colon X \to X \times X$ be the diagonal embedding; it is locally a complete intersection and the normal bundle N is a vector bundle on X – we let N also denote the corresponding variety fibered over X by affine spaces. The zero section $e\colon X \subset N$ is in some respects similar to the embedding Δ; we may obtain it as a special member in a rational family whose generic member is Δ.

There is a diagram:



all of whose squares are cartesian, where φ is flat, ψ is a closed immersion, and the other maps are the usual maps. Thus $N \subset M$ is a principal divisor whose complement is $X \times X \times \mathbb{A}^1$, and $fe = 1_X$, $ab = 1_{X \times X}$. We also let $h_0, h_\infty\colon X \rightrightarrows X \times \mathbb{P}^1$ denote the indicated sections.

By combining reduction to the diagonal [S, V, C 3 d] with Verdier's remark [V, 3.4], we see that $[C] \cdot [D] = (f^*)^{-1} d^* (c^*)^{-1} a^* [C \times D]$. The latter expression is well-defined because:

i) There is an exact sequence

$$A^{p+q-1}(N) \xrightarrow{d_*} A^{p+q}(M) \xrightarrow{c^*} A^{p+q}(X \times X \times \mathbb{A}^1) \longrightarrow 0,$$

ii) $d^* d_* = 0$,

iii) $f^*\colon A^{p+q}(X) \to A^{p+q}(N)$ is an isomorphism.

Let $C^p(X)$ denote the group of codimension p cycles on X. If $s\colon Z \hookrightarrow X$ is a divisor let $C^p(X, Z)$ denote those cycles in $C^p(X)$ which meet Z properly. Intersection with Z defines a map $s^*\colon C^p(X, Z) \to C^p(Z)$. Since $C^p(X) = C^p(X, Z) \oplus C^{p-1}(Z)$ we may define the map $s^*\colon C^p(X) \to C^p(Z)$ to be zero on the second factor.

Let $B^p(X)$ denote $H^p(X, \mathscr{K}_p)$. There is a map $\alpha: C^p(X) \to B^p(X)$, defined in terms of K-theory, which we will explain in more detail later (§2). Quillen proved that there is an isomorphism β which makes the triangle commute:

$$\begin{array}{ccc} C^p(X) & & \\ \downarrow [\ ] & \searrow^{\alpha} & \\ A^p(X) & \xrightarrow[\sim]{\beta} & B^p(X). \end{array}$$

Now $B^p(X)$ is a contravariant functor in X; for if $f: X \to Y$ is a map of schemes, and $U \subset Y$ is an open subset, there is a map $K_p(U) \to K_p(f^{-1}U)$ defined by pulling back vector bundles, thus a map of sheaves $f^{-1}\mathscr{K}_p(Y) \to \mathscr{K}_p(X)$. The map $B^p(Y) \to B^p(X)$ is simply the composite

$$H^p(Y, \mathscr{K}_p(Y)) \to H^p(X, f^{-1}\mathscr{K}_p(Y)) \to H^p(X, \mathscr{K}_p(X)).$$

The product $s \cup t$ of cohomology classes in $B^*(X)$ may be defined as the image of the external products $s \times t$ under the sequence of maps:

$$H^{p+q}(X \times X, \mathscr{K}_p(X) \hat{\otimes} \mathscr{K}_p(X)) \xrightarrow{\Delta^*} H^{p+q}(X, \mathscr{K}_p(X) \otimes \mathscr{K}_p(X)) \longrightarrow H^{p+q}(X, \mathscr{K}_{p+q}(X)).$$

Here $- \hat{\otimes} -$ denotes the external product of sheaves, and $\mathscr{K}_p \otimes \mathscr{K}_q \to \mathscr{K}_{p+q}$ is the product in K-theory which will be explained in more detail later (§2).

We may also define an external product $s \square t$ inside $B^*(X)$; let $s \square t$ be the image of $s \times t$ under the map

$$H^{p+q}(X \times X, \mathscr{K}_p(X) \hat{\otimes} \mathscr{K}_q(X)) \to H^{p+q}(X \times X, \mathscr{K}_{p+q}(X \times X)).$$

The latter map is the one induced by the external product in K-theory: $K_p(X) \otimes K_q(X) \to K_{p+q}(X \times X)$. On the level of vector bundles it is described by $(\mathscr{E}, \mathscr{F}) \mapsto p_1^* \mathscr{E} \otimes p_2^* \mathscr{F}$. (According to [W, 9.2] any bi-exact pairing of exact categories defines a product in K-theory in a natural way.)

(1.1) **Lemma.** $\Delta^*(s \square t) = s \cup t$.

Proof. This formula amounts to the commutativity of a certain diagram of cohomology groups. By pulling the sheaves in question back to X via Δ^*, and taking stalks, we reduce to showing that the diagram:

$$\begin{array}{ccc} K_p(\mathscr{O}_{X,x}) \otimes K_q(\mathscr{O}_{X,x}) & \longrightarrow & K_{p+q}(\mathscr{O}_{X,x}) \\ \downarrow p_1^* \otimes p_2^* & & \downarrow \Delta^* \\ K_p(\mathscr{O}_{X \times X, x \times x}) \otimes K_q(\mathscr{O}_{X \times X, x \times x}) & \longrightarrow & K_{p+q}(\mathscr{O}_{X \times X, x \times x}), \end{array}$$

commutes, but this follows from naturality of the products in K-theory (i.e. apply Δ^* before multiplying). Q.E.D.

(1.2) **Theorem.** The products in $A^*(X)$ and $B^*(X)$ agree, i.e.

$$\beta([C]\cdot[D])=(-1)^{pq}\alpha(C)\cup\alpha(D).$$

Proof. We use the following three lemmas, whose proofs will be postponed:

Lemma A. [Reduction to the diagonal] (2.7). $\alpha(C)\square\alpha(D)=(-1)^{pq}\alpha(C\times D)$.

Lemma B. [Flat maps] (3.1). If $f\colon X\to Y$ is a flat map of non-singular varieties, and $C\in C^p(Y)$, then $f^*(\alpha(C))=\alpha(f^*(C))$.

Lemma C. [Specialization] (4.10). If $s\colon Z\subseteq X$ is a divisor, Z and X are non-singular, and $C\in C^p(X,Z)$, then $\alpha(s^*(C))=s^*(\alpha(C))$.

From Lemma B and surjectivity of $C^p(Y)\to A^p(Y)$, we deduce the formula $\beta f^*=f^*\beta$ for flat maps (e.g. $\beta f^*[c]=\beta[f^*c]=\alpha f^*(c)=f^*\alpha(c)=f^*\beta[c]$).

We claim that $h_0^*=h_\infty^*$ on B^p. Indeed, let g denote the projection $X\times(\mathbb{P}^1-\{1\})\to X$ and i the open immersion $X\times(\mathbb{P}^1-\{1\})\to X\times\mathbb{P}^1$. Let h'_0 and h'_∞ denote the indicated sections of g. On A^p we know that g^* is an isomorphism, so g^* is an isomorphism on B^p by Lemma B. Computing in B^p we see that $h_0^*=h_0'^*i^*=h_0'^*g^*g^{*-1}i^*=h_\infty'^*g^*g^{*-1}i^*=h_\infty^*$.

Let E be the closure of $C\times D\times\mathbb{A}^1$ in M; E meets N properly so $E\in C^{p+q}(M,N)$; we have $c^*(E)=a^*(C\times D)$.

Let $F\in C^{p+q}(X)$ be a cycle such that $f^*[F]=d^*[E]$. We know $[F]=[C]\cdot[D]$. Now the following chain of equalities yields our result

$$\begin{aligned}
(-1)^{pq}\alpha(C)\cup\alpha(D)&=(-1)^{pq}\Delta^*(\alpha(C)\square\alpha(D)) && \text{[Lemma 1.1]}\\
&=\Delta^*(\alpha(C\times D)) && \text{[Lemma A]}\\
&=\Delta^*b^*a^*\alpha(C\times D) && [ab=1]\\
&=\Delta^*b^*\alpha a^*(C\times D) && \text{[Lemma B]}\\
&=\Delta^*b^*\alpha c^*(E) && \text{[def'n of } E]\\
&=\Delta^*b^*c^*\alpha(E) && \text{[Lemma B]}\\
&=h_0^*\psi^*\alpha(E) && [\psi h_0=cb\Delta]\\
&=h_\infty^*\psi^*\alpha(E)\,. && [h_0^*=h_\infty^*]\\
&=e^*d^*\alpha(E) && [de=\psi h_\infty]\\
&=e^*\alpha d^*(E) && \text{[Lemma C]}\\
&=e^*\beta[d^*(E)]\\
&=e^*\beta d^*[E]\\
&=e^*\beta f^*[F] && \text{[def'n of } F]\\
&=e^*f^*\beta[F] && [f \text{ is flat}]\\
&=\beta[F] && [fe=1]\\
&=\beta([C]\cdot[D]).
\end{aligned}$$

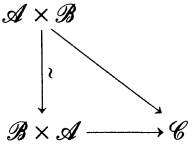
Q.E.D.

2. Reduction to the Diagonal

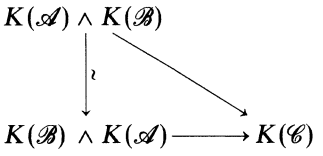
If $\mathscr{A}$, $\mathscr{B}$, and $\mathscr{C}$ are exact categories, a bi-exact functor $\mathscr{A}\times\mathscr{B}\to\mathscr{C}$ is one which is exact in each variable when the other is held fixed (the most common example is the

tensor product of projective modules). In this situation there is a pairing $K_i\mathscr{A} \otimes K_j\mathscr{B} \to K_{i+j}\mathscr{C}$ defined by Waldhausen [W, 9.2]. Let $K(\mathscr{A})$ denote the topological space $\Omega BQ\mathscr{A}$; it has the homotopy type of a CW-complex, and $\pi_i K(\mathscr{A}) \cong K_i\mathscr{A}$. The above pairing results from a map $K(\mathscr{A}) \wedge K(\mathscr{B}) \to K(\mathscr{C})$ which satisfies:

(2.1) Commutativity: If

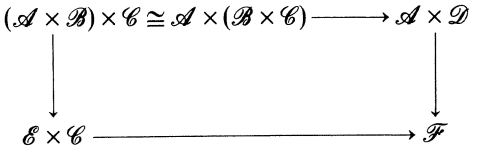


commutes up to natural isomorphism, then

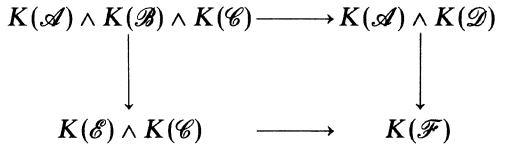


commutes up to homotopy.

(2.2) Associativity: If

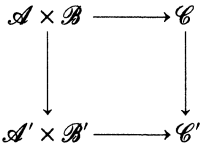


commutes up to natural isomorphism, then



commutes up to homotopy.

(2.3) Naturality: if $\mathscr{A} \to \mathscr{A}'$, $\mathscr{B} \to \mathscr{B}'$, and $\mathscr{C} \to \mathscr{C}'$ are exact functors, and



commutes up to natural isomorphism, then

$$\begin{array}{ccc} K(\mathscr{A}) \wedge K(\mathscr{B}) & \longrightarrow & K(\mathscr{C}) \\ \downarrow & & \downarrow \\ K(\mathscr{A}') \wedge K(\mathscr{B}') & \longrightarrow & K(\mathscr{C}') \end{array}$$

commutes up to homotopy.

(2.4) Agreement in K_0: the map $K_0\mathscr{A} \otimes K_0\mathscr{B} \to K_0\mathscr{C}$ is the usual map given by $[A] \otimes [B] \mapsto [A \otimes B]$.

In the Appendix we show how to define the map $K(\mathscr{A}) \wedge K(\mathscr{B}) \to K(\mathscr{C})$ in terms of Waldhausen' pairing $BQ\mathscr{A} \wedge BQ\mathscr{B} \to BQQ\mathscr{C}$. One must be careful because there are two natural homotopy equivalences $BQ\mathscr{C} \simeq \Omega BQQ\mathscr{C}$ which differ by -1.

Notice that (2.1) leads to anti-commutativity on the level of K-groups: i.e. if $x \in K_i\mathscr{A}$ and $y \in K_j\mathscr{B}$, then $xy = (-1)^{ij} yx$.

From (2.3) we can deduce that boundary maps in K-theory behave as differentials in a certain sense:

(2.5) **Theorem.** Suppose $F_1 \to E_1 \xrightarrow{p_1} B_1$, $F_2 \to E_2 \xrightarrow{p_2} B_2$, and $F \to E \xrightarrow{p} B$ are fibrations up to homotopy of pointed topological spaces, and suppose we are given pairings $E_1 \wedge E_2 \to E$, $B_1 \wedge F_2 \to F$, $F_1 \wedge B_2 \to F$, and $B_1 \wedge B_2 \to B$ such that the following diagrams commute:

$$\begin{array}{ccccc} E_1 \wedge F_2 & \longrightarrow & B_1 \wedge F_2 & \longrightarrow & F \\ & \searrow & & & \downarrow \\ & & E_1 \wedge E_2 & \longrightarrow & E \end{array} \qquad \begin{array}{ccccc} F_1 \wedge E_2 & \longrightarrow & F_1 \wedge B_2 & \longrightarrow & F \\ & \searrow & & & \downarrow \\ & & E_1 \wedge E_2 & \longrightarrow & E \end{array}$$

$$\begin{array}{ccc} E_1 \wedge E_2 & \longrightarrow & E \\ \downarrow & & \downarrow \\ B_1 \wedge B_2 & \longrightarrow & B. \end{array}$$

Then for any $x \in \pi_i B_1$ and $y \in \pi_j B_2$, we have the formula $\partial(xy) = \partial x \cdot y + (-1)^i x \cdot \partial y$, where ∂ denotes the boundary map in the long exact sequences of homotopy groups.

Proof. We may assume the three sequences are actually fibrations (e.g. replace E by $E \times_B B^I$). Let $f: (I^i, \partial I^i) \to (B_1, *)$ represent x, and let $g: (I^j, \partial I^j) \to (B_2, *)$ represent y. Choose $\tilde{f}$: $(I \times I^{i-1},\ I \times \partial I^{i-1} \cup 0 \times I^{i-1}) \to (E_1, *)$ so that $p_1 \circ \tilde{f} = f$; similarly choose $\tilde{g}$. By definition of the boundary map, $\partial x \in \pi_{i-1} F_1$ is represented by the restriction $\bar{f}$ of $\tilde{f}$ to $1 \times I^{i-1} \cong I^{i-1}$; similarly $\bar{g}$ represents ∂y. Consider the product $\tilde{f}\tilde{g}$: $(I \times I^{i-1} \times I \times I^{j-1},\ I \times \partial I^{i-1} \times I \times I^{j-1} \cup 0 \times I^{i-1} \times I \times I^{j-1} \cup I \times I^{i-1} \times I \times \partial I^{j-1} \cup I \times I^{i-1} \times 0 \times I^{j-1}) \to (E, *)$: notice that $p \circ \tilde{f}\tilde{g} = fg$, and that $\tilde{f}\tilde{g}$ sends all but two faces of the $i+j$-cube to the base-point of E, namely, all but $1 \times I^{i-1} \times I \times I^{j-1}$ and $I \times I^{i-1} \times 1 \times I^{j-1}$. The former is the usual face to be omitted in

computing the boundary map, whereas the latter can be brought into standard position via a permutation of the coordinates whose sign is $(-1)^i$. Notice that $\tilde{f}\tilde{g}$ restricted to the former face is just $\bar{f}\tilde{g}=\bar{f}g$, and restricted to the latter is $\tilde{f}\bar{g}=f\bar{g}$. Consequently $\partial(xy)=[\bar{f}g]+(-1)^i[f\bar{g}]=\partial x\cdot y+(-1)^i x\cdot\partial y$. Q.E.D.

(2.6) **Corollary.** Suppose $\mathscr{A}_1\to\mathscr{B}_1\to\mathscr{C}_1$, $\mathscr{A}_2\to\mathscr{B}_2\to\mathscr{C}_2$, and $\mathscr{A}\to\mathscr{B}\to\mathscr{C}$ are exact functors of exact categories such that the composites $\mathscr{A}_1\to\mathscr{C}_1$, $\mathscr{A}_2\to\mathscr{C}_2$, and $\mathscr{A}\to\mathscr{C}$ are all zero, and that $K(\mathscr{A}_1)\to K(\mathscr{B}_1)\to K(\mathscr{C}_1)$, $K(\mathscr{A}_2)\to K(\mathscr{B}_2)\to K(\mathscr{C}_2)$, and $K(\mathscr{A})\to K(\mathscr{B})\to K(\mathscr{C})$ are all fibrations up to homotopy. Suppose there are bi-exact functors $\mathscr{B}_1\times\mathscr{B}_2\to\mathscr{B}$, $\mathscr{C}_1\times\mathscr{A}_2\to\mathscr{A}$, $\mathscr{A}_1\times\mathscr{C}_2\to\mathscr{A}$, and $\mathscr{C}_1\times\mathscr{C}_2\to\mathscr{C}$ which make diagrams analogous to those in the theorem commute. We conclude that for any $x\in K_i\mathscr{C}_1$ and $y\in K_j\mathscr{C}_2$, the formula $\partial(xy)=\partial x\cdot y+(-1)^i x\cdot\partial y$ holds in $K_{i+j-1}\mathscr{A}$.

The above corollary is just what we need. If X is a variety, let $\mathscr{P}(X)$ be the category of vector bundles on X, and let $\mathscr{M}^p(X)$ be the category of coherent sheaves on X of codimension $\geqq p$. Then external tensor product defines bi-exact functors such as:

$$\begin{aligned}
\mathscr{P}(X)\times\mathscr{P}(X) &\longrightarrow \mathscr{P}(X\times X),\\
\mathscr{M}^p(X)\times\mathscr{M}^q(X) &\longrightarrow \mathscr{M}^{p+q}(X\times X),\\
\frac{\mathscr{M}^p(X)}{\mathscr{M}^{p+1}(X)}\times\frac{\mathscr{M}^q(X)}{\mathscr{M}^{q+1}(X)} &\longrightarrow \frac{\mathscr{M}^{p+q}(X\times X)}{\mathscr{M}^{p+q+1}(X\times X)}.
\end{aligned}$$

Adopt the notation $K_i^p(X)=K_i(\mathscr{M}^p(X))$, $K_i(X)=K_i(\mathscr{P}(X))$, and $K_i^{p/p+1}(X)=K_i(\mathscr{M}^p(X)/\mathscr{M}^{p+1}(X))$. There are long exact sequences

$$\cdots\to K_i^p(X)\to K_i^{p/p+1}(X)\to K_{i-1}^{p+1}(X)\to\cdots.$$

We now sketch some details from Quillen's proof of Bloch's formula. Let $\mathscr{K}_i(X)$ denote the sheaf associated to $U\mapsto K_i(U)$; similarly for $\mathscr{K}_i^p(X)$ and $\mathscr{K}_i^{p/p+1}(X)$. The main point is the proof of Gersten's conjecture: $\mathscr{K}_i^{p+1}(X)\to\mathscr{K}_i^p(X)$ is zero if X is non-singular. This splits the long exact sequences of sheaves into short ones which may be spliced together to yield

$$0\to\mathscr{K}_p(X)\to\mathscr{K}_p^{0/1}(X)\to\cdots\to\mathscr{K}_0^{p/p+1}(X)\to 0,$$

a flasque resolution of $\mathscr{K}_p(X)=\mathscr{K}_p^0(X)$. The map $C^p(X)\to H^0(X,\mathscr{K}_0^{p/p+1}(X))$ (which sends a prime cycle Z to the class of its structure sheaf $\mathscr{O}_Z$) is an isomorphism. The map α: $C^p(X)\to H^p(X,\mathscr{K}_p(X))=B^p(X)$ is the evident one defined in terms of this resolution; a computation shows that $B^p(X)$ and $A^p(X)$ are isomorphic as quotients of $C^p(X)$.

(2.7) *Proof* of Lemma A. Let $\mathscr{L}_p^*(X)$ denote the resolution of $\mathscr{K}_p(X)$ obtained above, i.e. $\mathscr{L}_p^i(X)=\mathscr{K}_{p-i}^{i/i+1}(X)$. If $\mathscr{L}_p^*\hat{\otimes}\mathscr{L}_q^*$ denotes the external tensor product of complexes of sheaves, we may define a map of complexes $\mathscr{L}_p^*\hat{\otimes}\mathscr{L}_q^*(X)\to\mathscr{L}_{p+q}^*(X\times X)$ as follows: on the summand $\mathscr{L}_p^i(X)\hat{\otimes}\mathscr{L}_q^j(X)$ take the pairing which is $(-1)^{pj}$ times the pairing into $\mathscr{L}_{p+q}^{i+j}(X\times X)$ defined above; denote it by $*$. Then if $x\in\mathscr{L}_p^i(X)$ and $y\in\mathscr{L}_q^j(X)$ we must check that $\partial(x*y)=\partial x*y+(-1)^i x*\partial y$. We use (2.6) with

the following localization sequences of abelian categories:

$$\frac{\mathscr{M}^{i+1}(X)}{\mathscr{M}^{i+2}(X)} \to \frac{\mathscr{M}^{i}(X)}{\mathscr{M}^{i+2}(X)} \to \frac{\mathscr{M}^{i}(X)}{\mathscr{M}^{i+1}(X)},$$

$$\frac{\mathscr{M}^{j+1}(X)}{\mathscr{M}^{j+2}(X)} \to \frac{\mathscr{M}^{j}(X)}{\mathscr{M}^{j+2}(X)} \to \frac{\mathscr{M}^{j}(X)}{\mathscr{M}^{j+1}(X)},$$

$$\frac{\mathscr{M}^{i+j+1}(X\times X)}{\mathscr{M}^{i+j+2}(X\times X)} \to \frac{\mathscr{M}^{i+j}(X\times X)}{\mathscr{M}^{i+j+2}(X\times X)} \to \frac{\mathscr{M}^{i+j}(X\times X)}{\mathscr{M}^{i+j+1}(X\times X)}.$$

We get

$$\begin{aligned}\partial(x*y) &= (-1)^{pj}\partial(xy) = (-1)^{pj}(\partial x\cdot y + (-1)^{p-i}x\cdot\partial y)\\ &= (-1)^{pj}\partial x\cdot y + (-1)^{i}(-1)^{p(j+1)}x\cdot\partial y = \partial x * y + (-1)^{i}x*\partial y.\end{aligned}$$

Now according to [G, II, Theorem 6.2.1], when we have such a map of complexes, there is a commutative square:

$$\begin{array}{ccc} H^0(X,\mathscr{K}_0^{p/p+1})\otimes H^0(X,\mathscr{K}_0^{q/q+1}) & \xrightarrow{*} & H^0(X\times X,\mathscr{K}_0^{p+q/p+q+1}) \\ \downarrow{\scriptstyle \alpha\otimes\alpha} & & \downarrow{\scriptstyle \alpha} \\ H^p(X,\mathscr{K}_p)\otimes H^q(X,\mathscr{K}_q) & \longrightarrow & H^{p+q}(X\times X,\mathscr{K}_{p+q}). \end{array}$$

By (2.4) and the definition of $*$ we see that $C*D=(-1)^{pq}(C\times D)$, thus $\alpha(C)\square\alpha(D)=(-1)^{pq}\alpha(C\times D)$. Q.E.D.

3. Flat Maps

(3.1) *Proof* of Lemma B. Since $f\colon X\to Y$ is flat, f^* defines an exact functor $\mathscr{M}^p(Y)\to\mathscr{M}^p(X)$ for each p. For each open subset $U\subset Y$, f^* induces an exact functor $\mathscr{M}^p(U)\to\mathscr{M}^p(f^{-1}U)$, and thus a map $f^{-1}\mathscr{K}_q^p(Y)\to\mathscr{K}_q^p(X)$; there is also a map $f^{-1}\mathscr{K}_q^{p/p+1}(Y)\to\mathscr{K}_q^{p/p+1}(X)$. The resulting maps $H^i(Y,\mathscr{K}_q^{p/p+1})\to H^i(X,\mathscr{K}_q^{p/p+1})$ give rise to a commutative diagram

$$\begin{array}{ccccc} C^p(Y)=H^0(Y,\mathscr{K}_0^{p/p+1}) & \xrightarrow{f^*} & H^0(X,\mathscr{K}_0^{p/p+1})=C^p(X) & & \\ \downarrow{\scriptstyle \alpha} & & \downarrow & & \downarrow{\scriptstyle \alpha} \\ B^p(Y)=H^p(Y,\mathscr{K}_p) & \xrightarrow{f^*} & H^p(X,\mathscr{K}_p) & = & B^p(X) \end{array}$$

in light of the naturality of the boundary maps in long exact sequences for cohomology groups. Q.E.D.

4. Specialization

In this section we prove Lemma C, so let $Z\subseteq X$ be a divisor, where Z and X are smooth over k.

(4.1) *Definition.* Let $\mathcal{M}^p(X, Z)$ denote the exact category of coherent sheaves M on X of codimension $\geqq p$ such that $\mathcal{T}or_1^X(M, \mathcal{O}_Z) = 0$. The latter condition is equivalent to the requirement that Z contain no associated points of M, or that any local equation for X is a non-zero-divisor on M. (This definition makes sense even if X and Z are not smooth.)

(4.2) *Definition.* Let $\mathcal{M}^p(X, Z) \to \mathcal{M}^p(Z)$ be the exact functor defined by $M \mapsto M \otimes_X \mathcal{O}_Z$.

(4.3) **Proposition.** If X is affine and $M \in \mathcal{M}^{p+1}(X, Z)$, $p \geqq 0$, then there is a divisor $W \subseteq X$ whose ideal kills M and which meets Z properly. Moreover, $M \in \mathcal{M}^p(W, W \cap Z)$.

Proof. We may assume X is irreducible. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ denote the minimal primes of Z in X, and let $I = \operatorname{ann} M$. We seek a function $g \neq 0$ on X such that $g \in I$ and $g \notin \cup \mathfrak{p}_i$. Since $M \otimes \mathcal{O}_Z \in \mathcal{M}^1(Z)$ no $\mathfrak{p}_i$ is in its support, and thus no $\mathfrak{p}_i$ contains I. It follows that $I \not\subseteq \cup \mathfrak{p}_i$. Q.E.D.

(4.4) **Corollary.** $K_q(\mathcal{M}^{p+1}(X, Z)) = \varinjlim K_q(\mathcal{M}^p(W, Z \cap W))$, where the limit runs over divisors $W \subseteq X$ which meet Z properly.

Proof. Since the scheme-theoretic union of divisors $Z \cup Z'$ is also a divisor, the limit in question is filtering. Since K-theory commutes with filtering direct limits, the result follows from an examination of the categories involved using (4.3). Q.E.D.

(4.5) *Definition.* Let $K_q^p(X, Z) = K_q(\mathcal{M}^p(X, Z))$, and let $\mathcal{K}_q^p(X, Z)$ denote the sheaf for the Zariski topology on X associated to the presheaf $U \mapsto K_q^p(U, U \cap Z)$.

(4.6) **Theorem.** The map of sheaves $\mathcal{K}_q^{p+1}(X, Z) \to \mathcal{K}_q^p(X, Z)$ is the zero map.

Proof. The stalk of this map is $K_q^{p+1}(X_x, Z_x) \to K_q^p(X_x, Z_x)$, at a point $x \in X$, because K_q commutes with direct limits. Here X_x denotes $\operatorname{Spec}(\mathcal{O}_{X,x})$. (Notice that if $x \notin Z$, then the map on this stalk is zero by Quillen's proof of Gersten's conjecture.) Again, $K_q^{p+1}(X_x, Z_x) = \varinjlim K_q^{p+1}(U, Z \cap U)$ where the limit runs over increasingly small open neighborhoods U of x. Thus, it is enough to show that for each U, $K_q^{p+1}(U, Z \cap U) \to K_q^p(X_x, Z_x)$ is zero, and for this purpose we may assume $X = U$ and X is affine. In turn, by Corollary 4.4 it is enough to show that $K_q^p(W, Z \cap W) \to K_q^p(X_x, Z_x)$ is zero, where $W \subset X$ is a divisor that meets Z properly. By Quillen's normalization lemma, [Q, Sect. 7, Lemma 5.12] or [Gr], if $r = \dim X$, there is a map $g\colon X \to \mathbb{A}^{r-1}$ which is smooth at x and whose restriction to $Z \cup W$ is finite.

Let $F\colon \mathcal{M}^p(W, Z \cap W) \to \mathcal{M}^p(X_x)$ be the exact functor $M \mapsto (g^* g_* M)_x$. We claim that $F(M) \in \mathcal{M}^p(X_x, Z_x)$. In fact if N is any coherent sheaf on $\mathbb{A}^{r-1}$, we claim $(g^* N)_x \in \mathcal{M}^0(X_x, Z_x)$. It is enough to show that Z_x contains no associated points of $(g^* N)_x$. By [B, IV, § 2.6, Thm. 2] any associated point of $(g^* N)_x$ is also an associated point in its fiber with respect to the map g. However, each of these fibers is smooth and of dimension 1 near x, and has just one associated point, namely, its generic point; Z contains none of these points because $Z \to \mathbb{A}^{r-1}$ is finite.

If we now proceed just as in [Q, Sect. 7, Thm. 5.11] we obtain an exact sequence of functors:

$$0 \to F(M) \to F(M) \to M_x \to 0,$$

which allows us to conclude that the functor $M \mapsto M_x$ induces the zero map $K_q^p(W, Z \cap W) \to K_q^p(X_x, Z_x)$. Q.E.D.

Recall now that $K_q^p(X, Z)$ is defined to be $\pi_{q+1}(BQ\mathcal{M}^p(X, Z))$, $BQ\mathcal{M}^p(X, Z)$ is an H-space, and contains $BQ\mathcal{M}^{p+1}(X, Z)$ as sub-H-space. Thus the relative homotopy groups are all abelian groups, and we may make:

(4.7) *Definition.* $K_q^{p/p+1}(X, Z) = \pi_{q+1}(BQ\mathcal{M}^p(X, Z),\ BQ\mathcal{M}^{p+1}(X, Z))$. Let $\mathcal{K}_q^{p/p+1}(X, Z)$ be the sheaf associated to the presheaf $U \mapsto K_q^{p/p+1}(U, Z \cap U)$; similarly for $\mathcal{K}_q^p(X, Z)$.

(4.8) **Corollary.** There is a natural exact sequence

$$0 \to \mathcal{K}_p(X) \to \mathcal{K}_p^{0/1}(X, Z) \to \cdots \to \mathcal{K}_0^p(X, Z) \to 0.$$

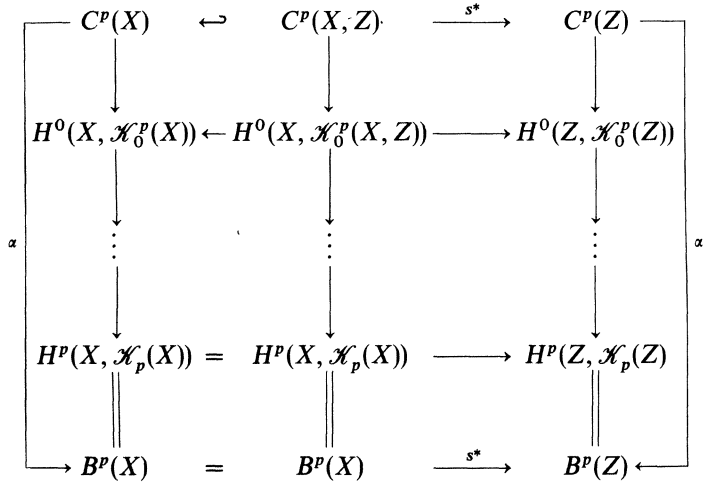
Proof. The long exact sequence of homotopy groups yields, by Theorem 4.6, short exact sequences of sheaves

$$0 \to \mathcal{K}_q^p(X, Z) \to \mathcal{K}_q^{p/p+1}(X, Z) \to \mathcal{K}_{q-1}^{p+1}(X, Z) \to 0.$$

The resolution theorem [Q, Sect. 4] shows that $\mathcal{K}_q(X) \to \mathcal{K}_q^0(X, Z)$ is an isomorphism, so the statement follows from patching together short exact sequences. Q.E.D.

(4.9) *Definition.* Let $C^p(X, Z) \to K_0^p(X, Z)$ be the map which sends a prime cycle Y to the class $[\mathcal{O}_Y]$.

(4.10) *Proof* of Lemma C. The proof hinges on the exact functors $\mathcal{M}^q(X, Z) \to \mathcal{M}^q(Z)$ and $\mathcal{M}^q(X, Z) \subset \mathcal{M}^q(X)$. They induce the horizontal maps in the following diagram:



The commutativity of the diagram is clear, and yields $s^* \alpha = \alpha s^*$. Q.E.D.

Appendix

Here we sketch the proofs of (2.1)–(2.4) using the notation from [W, 9.2].

The map $BIs(\mathscr{A}) \to \Omega BQ\mathscr{A}$ yields, on π_0, a map from the set of isomorphism classes of objects in $\mathscr{A}$ to $K_0\mathscr{A}$ which we use to establish the isomorphism of $K_0\mathscr{A}$ with the Grothendieck group of $\mathscr{A}$.

We define $K(\mathscr{A}) \wedge K(\mathscr{B}) \to K(\mathscr{C})$ by choosing a homotopy inverse for the homotopy equivalence in the following diagram:

$$\begin{array}{ccccc} K(\mathscr{A}) \wedge K(\mathscr{B}) = & \Omega BQ\mathscr{A} \wedge \Omega BQ\mathscr{B} & & \Omega BQ\mathscr{C} & = K(\mathscr{C}) \\ & \downarrow & & \downarrow \wr & \\ & \Omega\Omega(BQ\mathscr{A} \wedge BQ\mathscr{B}) & \longrightarrow & \Omega\Omega BQQ\mathscr{C}. & \end{array}$$

The right-hand vertical map is the one Waldhausen uses, and we may imagine that the second Ω annihilates the second Q, and adopt the following uniform notation:

$$\Omega BQ\mathscr{C} \xrightarrow{\sim} \Omega\overline{\Omega BQ}Q\mathscr{C}.$$

With these definitions (2.3) is clear, and (2.4) follows immediately from [W, Lemma 9.2.6].

To show commutativity (2.1), we consider the following diagram:

$$\begin{array}{ccc} \Omega BQ\mathscr{A} \wedge \Omega BQ\mathscr{B} & \xrightarrow{\sim} & \Omega BQ\mathscr{B} \wedge \Omega BQ\mathscr{A} \\ \downarrow & & \downarrow \\ \Omega\Omega BQQ\mathscr{C} & \xrightarrow{f} & \Omega\Omega BQQ\mathscr{C} \\ \uparrow \wr & & \uparrow \wr \\ \Omega BQ\mathscr{C} & = & \Omega BQ\mathscr{C} \end{array}$$

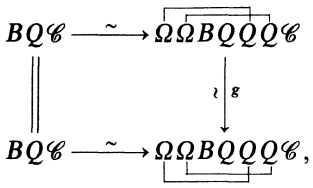
where the horizontal map f interchanges both Ω's and both Q's. The diagram commutes up to homotopy, so it is enough to show that f is homotopic to the identity. Adding another Q merely de-loops, so if g is the map

$$\Omega\Omega BQQQ\mathscr{C} \xrightarrow{\sim} \Omega\Omega BQQQ\mathscr{C}$$

which interchanges both Ω's and interchanges the 2nd and 3rd Q's, then it is enough to show that g is homotopic to the identity. Now the following diagram commutes:

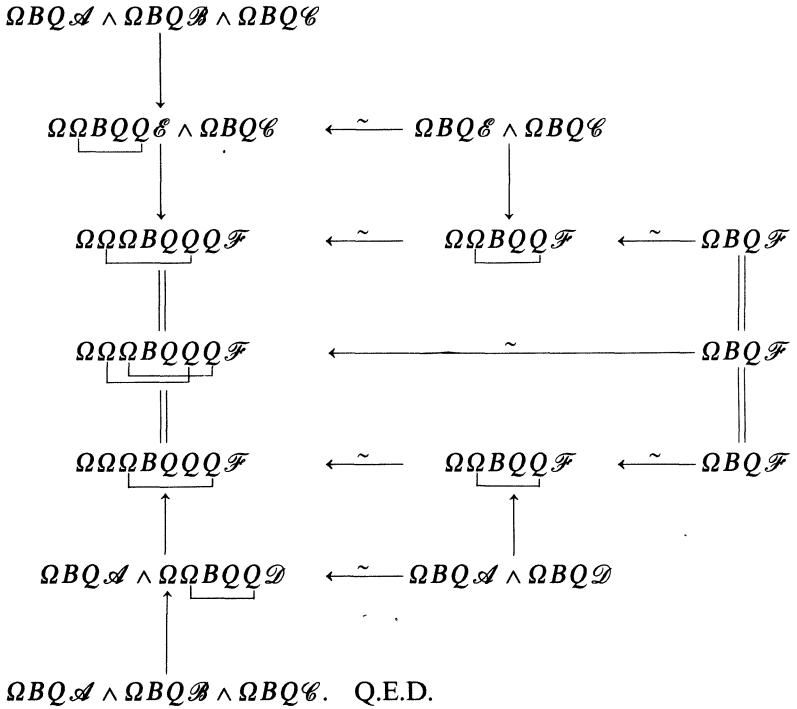
$$\begin{array}{ccc} \Sigma\Sigma BQ\mathscr{C} & \longrightarrow & BQQQ\mathscr{C} \\ \downarrow \wr & & \downarrow \wr \\ \Sigma\Sigma BQ\mathscr{C} & \longrightarrow & BQQQ\mathscr{C} \end{array}$$

where on the left we interchange the Σ's, and on the right we interchange the last two Q's. It follows that the following diagramm commutes:



and since the horizontal maps are equal homotopy equivalences, we have $g \simeq 1$.

Finally, associativity, (2.3), follows from the following diagram which commutes up to homotopy:



$\Omega BQ\mathscr{A} \wedge \Omega BQ\mathscr{B} \wedge \Omega BQ\mathscr{C}$. Q.E.D.

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On the Structure of Certain Galois Groups

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1. Introduction

Let k be a number field and let p be a prime. The field obtained by adjoining to k all p-power roots of unity contains a unique subfield k_∞ such that $\mathrm{Gal}(k_\infty/k) \cong \mathbf{Z}_p$, the additive group of p-adic integers. Let M_∞ denote the maximal abelian pro-p-extension of k_∞ in which only primes of k_∞ lying over p are ramified. One can consider $\mathrm{Gal}(M_\infty/k_\infty)$ as a Λ-module in a natural way, where Λ denotes the complete group ring of $\mathrm{Gal}(k_\infty/k)$ over $\mathbf{Z}_p$. (See Section 2.) In [8], Iwasawa proved the following basic result concerning its structure.

Theorem. *Let r_2 denote the number of complex primes of k. The Λ-module $\mathrm{Gal}(M_\infty/k_\infty)$ has rank r_2. It contains no non-trivial finite Λ-submodule.*

In this paper, we will generalize Iwasawa's theorem, at least in the case when the ground field k is an abelian extension of $\mathbf{Q}$.

We first consider an arbitrary $\mathbf{Z}_p$-extension K of k. Let $G = \mathrm{Gal}(K/k)$ and let Λ_G denote its complete group ring over $\mathbf{Z}_p$. Let M_K be the maximal abelian pro-p-extension of K unramified outside of the set of primes dividing p and let $X_K = \mathrm{Gal}(M_K/K)$. (We will use this notation if K is any field of algebraic numbers.) Then we will prove the following theorem.

Theorem 1. *Assume that k is an abelian extension of $\mathbf{Q}$. Then X_K has rank r_2 as a Λ_G-module and it contains no non-trivial finite Λ_G-submodule.*

In the proof of the above theorem, the only assumption that we actually need about the ground field k is that Leopoldt's conjecture (concerning the p-adic independence of the units of k) is valid. This has been proved by A. Brumer if k is abelian over $\mathbf{Q}$ (see [1]). The first conclusion of Theorem 1 (about the rank) follows very simply. Our approach to proving the second conclusion also depends on knowing something about the p-adic independence of units, but in the field k_n for all sufficiently large n. (Here k_n denotes the subfield of K corresponding to G^{p^n} so that k_n is a cyclic extension of k of degree p^n.) However, the assertion that the rank of X_K equals r_2 turns out to imply that a certain

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weak form of Leopoldt's conjecture is valid for k_n if n is large. This allows us to deduce the second conclusion.

Consider more generally a Galois extension K of k such that $G = \mathrm{Gal}(K/k) \cong \mathbf{Z}_p^d$, where $d \geqq 1$. The complete group ring Λ_G of G over $\mathbf{Z}_p$ is then a noetherian local ring of dimension $d+1$. (It is isomorphic to the ring of formal power series in d variables over $\mathbf{Z}_p$.) Recall that a finitely generated Λ_G-module Z is called pseudo-null if every prime ideal in the support of Z has height $\geqq 2$. (See Bourbaki, Alg. Comm., Chap. 7, §4, no. 4.) Thus $\mathrm{Supp}(Z)$ is of codimension at least 2 in $\mathrm{Spec}(\Lambda_G)$ if Z is pseudo-null. If $G \cong \mathbf{Z}_p$, then a Λ_G-module is pseudo-null if and only if it is finite.

Now X_K can again be viewed as a Λ_G-module and it is not difficult to deduce from Theorem 1 that its rank will also be equal to r_2, if k is abelian over $\mathbf{Q}$. However the following generalization of the second part of Theorem 1 seems rather surprising.

Theorem 2. *Assume that k is abelian over $\mathbf{Q}$. Then X_K contains no non-trivial pseudo-null Λ_G-submodule.*

Letting Y_K denote the torsion Λ_G-submodule of X_K, we could equivalently say that $\mathrm{Supp}(Y_K)$ is purely of codimension 1 in $\mathrm{Spec}(\Lambda_G)$. Also, by using a result proved in [9], there exists a Λ_G-isomorphism φ of Y_K into a module of the form $\bigoplus_{i=1}^{t} \Lambda_G/\mathfrak{p}_i^{e_i}$ such that the cokernel of φ is pseudo-null. Here the $\mathfrak{p}_i$'s are prime ideals of Λ_G of height 1 and are therefore principal ideals since Λ_G is regular. The e_i's are positive integers. Theorem 2 suggests that the principal ideal $\prod_{i=1}^{t} \mathfrak{p}_i^{e_i}$ might be an important invariant of the extension K/k. It would seem interesting to learn more about generators of this ideal. In the rather simple case where k is a totally real abelian extension of $\mathbf{Q}$ and $K = k_\infty$ (which, in this case, is the only extension of k of the kind we are considering), the above ideal has a generator closely related to the p-adic zeta function of k. (See [5] for more details.) Virtually nothing more general than this is known.

It would seem quite interesting to extend the above results further. For example, in [6] M. Harris considers Galois extensions K of k such that $G = \mathrm{Gal}(K/k)$ is a p-adic Lie group. One can again regard X_K as a Λ_G-module. Assuming that G is also pro-p and torsion-free, one can define the rank of a Λ_G-module and, with certain mild restrictions on the extension K/k, Harris proves that $\mathrm{rank}_{\Lambda_G}(X_K) = r_2$. Our arguments allow us to prove that if Leopoldt's conjecture is valid for all finite extensions of k contained in K, then X_K has no non-trivial finite Λ_G-submodule. It should be possible to improve this rather weak analogue of Theorem 2.

In the final section of this paper, we will describe some examples where we can verify that the torsion Λ_G-submodule of X_K is non-trivial. We can construct such examples for any $d \geqq 1$ by suitably choosing the ground field k. However, except when k is totally real, we haven't been able to find such examples where K is the composite of *all* $\mathbf{Z}_p$-extensions of k. It would be interesting to find some examples of this kind.

In conclusion, we would like to thank B. Mazur for several helpful suggestions concerning this paper.

2. Preliminaries

Let K/k be a Galois extension with $G=\mathrm{Gal}(K/k)\cong\mathbf{Z}_p^d$, where $d\geqq 1$. The field M_K defined earlier is easily seen to be Galois over k. We can define a continuous action of G on X_K as follows. If $g\in G$, let $\bar{g}$ be any extension of g to an automorphism of M_K. If $x\in X_K$, we define $g(x)=\bar{g}\,x\,\bar{g}^{-1}$ which is obviously in X_K and well-defined. Since X_K is an abelian pro-p-group, it is a $\mathbf{Z}_p$-module and, by using the above action of G, it is not hard to make X_K into a Λ_G-module. The ring Λ_G is defined as the inverse limit of the group rings of all finite quotient groups of G over $\mathbf{Z}_p$. We will first show that X_K is a finitely generated Λ_G-module. Several parts of our argument here will be useful later. We will need the following lemma (which is proved in [9]).

Lemma 1. *Let X be a compact Λ_G-module. Then X is finitely generated as a Λ_G-module if and only if $X/\mathfrak{m}X$ is finite, where $\mathfrak{m}$ is the maximal ideal of Λ_G.*

In applying the above lemma, it is obviously enough to verify that $X/\mathfrak{a}X$ is finitely generated as a $\Lambda_G/\mathfrak{a}$-module for any proper ideal $\mathfrak{a}$ of Λ_G.

Now assume that $d=1$ so that $G\cong\mathbf{Z}_p$. Choose a topological generator g of G. It is not difficult to show that $(g-1)X_K$ is precisely the commutator subgroup of $\mathrm{Gal}(M_K/k)$ and hence must correspond to the field M_k. (We are using the well-known fact that only primes dividing p can ramify in K/k. See Sect. 2.2 of [8].) Thus $X_K/(g-1)X_K$ is isomorphic to a $\mathbf{Z}_p$-submodule of X_k. But, as we will explain below, it follows from class field theory that X_k is a finitely generated $\mathbf{Z}_p$-module. This together with Lemma 1 implies that X_K is finitely generated as a Λ_G-module.

If $d>1$, let H be a subgroup of G isomorphic to $\mathbf{Z}_p$ and which is a direct summand of G so that $G'=G/H\cong\mathbf{Z}_p^{d-1}$. Then let $\mathfrak{a}_H$ be the kernel of the natural homomorphism $\Lambda_G\to\Lambda_{G'}$. If h is a topological generator of H, then one finds that $\mathfrak{a}_H=(h-1)\Lambda_G$. (It is helpful to think of Λ_G and $\Lambda_{G'}$ as formal power series rings. One can choose one of the variables in Λ_G to be $S=h-1$. The above homomorphism would then essentially be evaluation at $S=0$.) Let K' be the subfield of K corresponding to H so that $\mathrm{Gal}(K'/k)\cong G'$. Then $(h-1)X_K$ is the commutator subgroup of $\mathrm{Gal}(M_K/K')$ and hence corresponds to the field $M_{K'}$. Thus, as $\Lambda_{G'}$-modules, $X_K/\mathfrak{a}_H X_K$ is isomorphic to a submodule of $X_{K'}$. Therefore, if $X_{K'}$ is finitely generated over $\Lambda_{G'}$, X_K will clearly be finitely generated over Λ_G. Hence an induction argument proves what we want.

The $\mathbf{Z}_p$-module $X_k=\mathrm{Gal}(M_k/k)$ can be described by class field theory. For each prime $\mathfrak{p}$ of k lying over p, let $U_\mathfrak{p}$ denote the group of principal units in the completion of k at $\mathfrak{p}$. Let $U=\prod_{\mathfrak{p}\mid p}U_\mathfrak{p}$. Let E_k denote the group of units of k which are $\equiv 1 \pmod{\mathfrak{p}}$ for all $\mathfrak{p}$ dividing p so that E_k is of finite index in the full unit group of k. There is a natural isomorphism of E_k into U and we will identify E_k with its image. Let $\bar{E}_k$ denote the closure of E_k in U. Then by class field theory one can show that $U/\bar{E}_k$ is isomorphic to a subgroup of $\mathrm{Gal}(M_k/k)$ of finite index. Now, as a $\mathbf{Z}_p$-module, $U\cong\mathbf{Z}_p^n\times A$, where $n=[k:Q]$ and A is finite. Therefore, it obviously follows that X_k is finitely generated over $\mathbf{Z}_p$.

Now if r_1 and r_2 denote the number of real and complex primes of k, then $\mathrm{rank}_\mathbf{Z}(E_k)=r_1+r_2-1$ and hence we must have $\bar{E}_k\cong\mathbf{Z}_p^c\times B$, where $c\leqq r_1+r_2-1$

and B is finite. Let $\delta = r_1 + r_2 - 1 - c$. Since $n = r_1 + 2r_2$, we see that $\text{rank}_{\mathbf{Z}_p}(X_k) = n - c = r_2 + 1 + \delta$. The conjecture of Leopoldt that we referred to earlier asserts that $\delta = 0$ (i.e. $\text{rank}_{\mathbf{Z}_p}(\bar{E}_k) = \text{rank}_{\mathbf{Z}}(E_k)$).

We conclude this section with a few general remarks about the structure of Λ_G and its modules. If $G \cong \mathbf{Z}_p^d$, let $g_1, g_2, \ldots, g_d$ be a set of topological generators of G. Then, as mentioned above, Λ_G is isomorphic to the power series ring $\mathbf{Z}_p[[T_1, T_2, \ldots, T_d]]$; the isomorphism can be defined by letting T_i correspond to $g_i - 1$ for $1 \leqq i \leqq d$ and hence depends on the choice of topological generators for G. For $d = 1$, this is proved in [9] but the same proof works for any d.

If G is isomorphic to $\mathbf{Z}_p$, then Λ_G is a regular local ring of dimension 2 and there is a good structure theory for finitely generated Λ_G-modules (see [9]). For any such module X, there is a Λ_G-homomorphism with finite kernel and cokernel from X to a module of the form $\Lambda_G^\rho \oplus Y$, where $\rho = \text{rank}_{\Lambda_G}(X)$ and Y is a torsion Λ_G-module. Let Y_0 be the submodule of Y consisting of elements of finite order. Of course, it is possible for Y_0 to be infinite although it must obviously have bounded exponent. However, if Y_0 is finite then Y must be finitely generated as a $\mathbf{Z}_p$-module. To see this note that Y would have an annihilator $\Phi \in \Lambda_G$ which is not divisible by p (since Y_0 is finite). Hence the ideal (p, Φ) annihilates Y/pY. Since (p, Φ) is not contained in any prime ideal of height 1, it contains some power of the maximal ideal of Λ_G and so Y/pY must be finite which of course implies that Y is a finitely generated $\mathbf{Z}_p$-module.

If $G \cong \mathbf{Z}_p^d$ with $d \geqq 1$, then the above results can be generalized as follows. If X is a finitely generated torsion-free Λ_G-module of rank ρ, then X can obviously be embedded in Λ_G^ρ and the quotient module will be a torsion Λ_G-module. Moreover, if Θ is any non-zero element of Λ_G, then, by a localization argument, one can show that this embedding can be chosen so that the quotient module has an annihilator prime to Θ. Concerning torsion Λ_G-modules, we have the following useful lemma.

Lemma 2. *Let Y be a finitely generated torsion Λ_G-module. Assume that Y has an annihilator $\Phi \in \Lambda_G$ such that $p \nmid \Phi$. Then G contains at least one subgroup F such that $G/F \cong \mathbf{Z}_p$ with the property that Y is finitely generated over Λ_F.*

In this lemma, we consider Λ_F as a subring of Λ_G in the natural way. If $d = 1$, this lemma reduces to our previous remarks. (Take $F = \{\text{id}\}$.) If $d \geqq 2$, our proof will actually show that F could be chosen in infinitely many ways.

Proof. We will construct a subgroup F of G such that $G/F \cong \mathbf{Z}_p$ and such that in the natural ring homomorphism $\Lambda_G \to \Lambda_{G/F}$, the image of Φ is still not divisible by p. Now if $\mathfrak{a}_F$ denotes the kernel of the above ring homomorphism, then $Y/\mathfrak{a}_F Y$ can be viewed as a $\Lambda_{G/F}$-module which is annihilated by $\Phi \bmod \mathfrak{a}_F$. Thus $Y/\mathfrak{a}_F Y$ will be finitely generated as a $\mathbf{Z}_p$-module. Let $f_1, f_2, \ldots, f_{d-1}$ be a set of topological generators for F. Then $\mathfrak{a}_F$ is generated by $\{f_i - 1 | 1 \leqq i \leqq d-1\}$ as an ideal of Λ_G. These elements also generate an ideal I_F of Λ_F (the so-called augmentation ideal). Obviously, $I_F Y = \mathfrak{a}_F Y$. Letting $\mathfrak{m}_F$ denote the maximal ideal of Λ_F, we see that $Y/\mathfrak{m}_F Y$ is finite and hence Lemma 2 follows from Lemma 1.

We will complete the proof by making some simple observations which will also be useful later on.

Let $\tilde{\Lambda}_G$ be the complete group ring for G over $\mathbf{Z}/(p)$; it is isomorphic to $\Lambda_G/p\Lambda_G$ and also to the ring of formal power series in d variables over $\mathbf{Z}/(p)$. Let

$\tilde{\Phi}$ denote the image of Φ in $\tilde{\Lambda}_G$ so that our assumption about Φ is that $\tilde{\Phi} \neq 0$. If H is a direct summand of G isomorphic to $\mathbf{Z}_p$ and if $G' = G/H$, then the kernel $\tilde{\mathfrak{a}}_H$ of the homomorphism $\tilde{\Lambda}_G \to \tilde{\Lambda}_{G'}$ is a prime ideal since $\tilde{\Lambda}_{G'}$ is an integral domain. If h is a topological generator of H, then $h-1$ (viewed as an element of $\tilde{\Lambda}_G$) generates $\tilde{\mathfrak{a}}_H$ and hence is irreducible. Also if $h_1 \in G$ but $\notin H$, then the image of h_1 in G' is not 1 and therefore $h_1 - 1$ is not divisible by $h-1$. If $d \geqq 2$, there are infinitely many possible choices for H but since $\tilde{\Lambda}_G$ is a unique factorization domain, we can see from the above remarks that $h-1$ will divide $\tilde{\Phi}$ for only a finite number of H's. We can therefore choose H so that the image $\tilde{\Phi}'$ of $\tilde{\Phi}$ in $\tilde{\Lambda}_{G'}$ is non-zero. Repeating this process with G' and $\tilde{\Phi}'$, we eventually can construct a subgroup F with the desired property. This completes the proof of the lemma.

The converse of Lemma 2 is also true. Let Y be a Λ_G-module which is finitely generated as a Λ_F-module for some subgroup F of G with $G/F \cong \mathbf{Z}_p$. Let $y_1, y_2, \ldots, y_t$ be a minimal set of generators for Y as a Λ_F-module (so that their images in $Y/\mathfrak{m}_F Y$ form a basis over $\Lambda_F/\mathfrak{m}_F \cong \mathbf{Z}/(p)$). Choose $g \in G$ so that g mod F is a topological generator for G/F and let $T = g - 1$. By a standard linear algebra argument one can construct a monic polynomial $\Phi = \Phi(T) \in \Lambda_F[T]$ of degree t which annihilates Y. Now Λ_G can be considered as the power series ring $\Lambda_F[[T]]$ and so obviously $p \nmid \Phi$.

If Z is a pseudo-null Λ_G-module, then the assumption in Lemma 2 is easily seen to hold. Thus for suitable F, Z will be finitely generated over Λ_F. In fact, it must be a torsion Λ_F-module. To see this, let Φ be the monic polynomial described above. Since Z is pseudo-null, it has an annihilator Ψ relatively prime to Φ. Viewing Ψ as an element of $\Lambda_F[[T]]$, we can assume that Ψ has at least one unit coefficient. (Otherwise just add Φ.) Hence, by the Weierstrass preparation theorem, we may choose Ψ to also be a polynomial over Λ_F. It follows that the ideal generated by Φ and Ψ contains a non-zero element of Λ_F which must annihilate Z. If one assumes that the torsion subgroup of Z is trivial, then Z will clearly have an annihilator in Λ_F not divisible by p. Applying Lemma 2 again, we obtain the following lemma which we will need later.

Lemma 3. *Assume that $d \geqq 2$. Let Z be a pseudo-null Λ_G-module which is torsion-free as a $\mathbf{Z}_p$-module. Then G contains at least one subgroup E such that $G/E \cong \mathbf{Z}_p^2$ with the property that Z is finitely generated as a Λ_E-module.*

It will be useful later to note that if $d \geqq 3$ then E can be chosen in an infinite number of ways.

Conversely, if Z is a Λ_G-module which is finitely generated over Λ_E for some subgroup E of the above type, then Z must be a pseudo-null Λ_G-module. For one can view Λ_G as $\Lambda_E[[S, T]]$ for suitable S and T. One can find monic polynomials $\Phi \in \Lambda_E[S]$ and $\Psi \in \Lambda_E[T]$ which annihilate Z. Since such Φ and Ψ are obviously relatively prime, the ideal generated by Φ and Ψ in Λ_G can't be contained in a prime ideal of height 1 and so Z must be pseudo-null.

3. The Rank of X_K

Let K be an arbitrary $\mathbf{Z}_p$-extension of k. If k_n denotes the subfield of K such that $[k_n : k] = p^n$, then we have

$$k = k_0 \subset k_1 \subset \cdots \subset k_n \subset \cdots.$$

If g is a topological generator of $G=\mathrm{Gal}(K/k)$, then g^{p^n} is also one for $\mathrm{Gal}(K/k_n)$. Let $\omega_n=g^{p^n}-1\in\Lambda_G$ for all $n\geqq 0$. The field M_{k_n} is the maximal abelian extension of k_n contained in M_K and so corresponds to the commutator subgroup $\omega_n X_K$ of $\mathrm{Gal}(M_K/k_n)$. Thus we have

$$\mathrm{rank}_{\mathbf{Z}_p}(X_K/\omega_n X_K)=\mathrm{rank}_{\mathbf{Z}_p}(X_{k_n})-1$$

for all $n\geqq 0$. Now the number of complex primes of k_n is $r_2 p^n$. Therefore, $\mathrm{rank}_{\mathbf{Z}_p}(X_{k_n})=r_2 p^n+1+\delta_n$, where $\delta_n=\mathrm{rank}_{\mathbf{Z}}(E_{k_n})-\mathrm{rank}_{\mathbf{Z}_p}(\bar{E}_{k_n})$. Thus for $n=0, 1, 2, \ldots,$

$$\mathrm{rank}_{\mathbf{Z}_p}(X_K/\omega_n X_K)=r_2 p^n+\delta_n.$$

On the other hand, $\Lambda_G/\omega_n\Lambda_G$ is isomorphic to the group ring $\mathbf{Z}_p[G/G^{p^n}]$ and so has rank p^n as a $\mathbf{Z}_p$-module. Let X be an arbitrary finitely generated Λ_G-module and Y its torsion Λ_G-submodule. It follows easily from the structure theory of Λ_G-modules that for $n\geqq 0$

$$\mathrm{rank}_{\mathbf{Z}_p}(X/\omega_n X)=\rho p^n+\varepsilon_n,$$

where ρ is the rank of X as a Λ_G-module and $\varepsilon_n=\mathrm{rank}_{\mathbf{Z}_p}(Y/\omega_n Y)$. (Thus $0\leqq\varepsilon_n\leqq\lambda(Y)$ for all n.) Applying these facts to $X=X_K$, we immediately find that $\rho\geqq r_2$ and that equality holds if and only if the δ_n's are bounded. (One can give a proof of the first part of Iwasawa's theorem along these lines. This is done in [4] where we show by a Kummer theory argument that the δ_n's are in fact bounded if $K=k_\infty$.) Equality also clearly holds when $\delta_0=0$ (i.e. when Leopoldt's conjecture is valid for the ground field k). In this case, we also have $\varepsilon_0=0$ and hence $Y_K/\omega_0 Y_K$ is finite, where Y_K denotes the torsion Λ_G-submodule of X_K. We summarize some of the above remarks in the following proposition.

Proposition 1. *Let K/k be a $\mathbf{Z}_p$-extension. Then* $\mathrm{rank}_{\Lambda_G}(X_K)\geqq r_2$. *If Leopoldt's conjecture is valid for k, then* $\mathrm{rank}_{\Lambda_G}(X_K)=r_2$.

Assume now that $\mathrm{Gal}(K/k)\cong\mathbf{Z}_p^d$ where $d\geqq 1$. By pursuing the same type of argument which we used to prove that X_K is finitely generated, we will prove the following result.

Proposition 2. *Assume either that Leopoldt's conjecture is valid for k or that the cyclotomic $\mathbf{Z}_p$-extension k_∞ of k is contained in K. Then* $\mathrm{rank}_{\Lambda_G}(X_K)=r_2$.

Proof. We first show inductively that X_K has rank $\geqq r_2$. For $d=1$, this is included in Proposition 1. If $d\geqq 2$, let H be a direct summand of G isomorphic to Z_p and let $G'=G/H$. Let h be a topological generator of H. If X is a finitely generated torsion-free Λ_G-module of rank ρ, then one can show easily that $X/(h-1)X$ has rank ρ as a $\Lambda_{G'}$-module. This is of course obvious if X is a free Λ_G-module, but it follows in general by simply embedding X in Λ_G^ρ so that the quotient module Λ_G^ρ/X has an annihilator prime to $h-1$. If X is an arbitrary finitely generated Λ_G-module of rank ρ and Y is its torsion Λ_G-submodule, then it is not hard to see that

$$\mathrm{rank}_{\Lambda_{G'}}(X/(h-1)X)=\rho+\mathrm{rank}_{\Lambda_{G'}}(Y/(h-1)Y). \tag{1}$$

But if $\Phi \in \Lambda_G$ is a non-zero annihilator of Y, one can choose H so that $h-1 \nmid \Phi$. (See the proof of Lemma 2.) For such H, $Y/(h-1)Y$ is a torsion $\Lambda_{G'}$-module and hence $\operatorname{rank}_{\Lambda_{G'}}(X/(h-1)X)=\rho$. Let K' be the subfield of K such that $\operatorname{Gal}(K/K')=H$. Letting $X=X_K$, we have the following exact sequence of $\Lambda_{G'}$-modules:

$$0 \to X/(h-1)X \to X_{K'} \to H \to 0.$$

Therefore, since H is obviously a torsion $\Lambda_{G'}$-module, we see that

$$\operatorname{rank}_{\Lambda_G}(X_K) \leqq \operatorname{rank}_{\Lambda_{G'}}(X_{K'}) \tag{2}$$

and that equality holds if H is chosen so that $h-1 \nmid \Phi$. It clearly follows that $\operatorname{rank}_{\Lambda_G}(X_K) \geqq r_2$. Also if K contains at least one $\mathbf{Z}_p$-extension L of k such that X_L has rank r_2 as a $\Lambda_{\operatorname{Gal}(L/k)}$-module (e.g. if $k_\infty \subset K$ or if Leopoldt's conjecture is valid for k), then $\operatorname{rank}_{\Lambda_G}(X_K)$ must be r_2 too. To see this, simply construct a tower of fields $L=L_1 \subset L_2 \subset \cdots \subset L_d=K$ such that $\operatorname{Gal}(L_r/k) \cong \mathbf{Z}_p^r$ for $1 \leqq r \leqq d$. Applying (2) successively, one finds that $\operatorname{rank}_{\Lambda_G}(X_K) \leqq r_2$ and so equality definitely holds.

There is one final remark we want to make here. Let $\mathscr{E}$ denote the set of all $\mathbf{Z}_p$-extensions of a number field k and let $\mathscr{E}'$ denote the set of those $\mathbf{Z}_p$-extensions such that X_K has rank r_2. In [3], we put a certain natural topology on $\mathscr{E}$. By using Proposition 2, it is possible to prove that $\mathscr{E}'$ is an open, dense subset of $\mathscr{E}$. Thus, in some sense, equality holds in Proposition 1 for "almost all" $\mathbf{Z}_p$-extensions of k. Although it is not difficult, we will omit the proof.

4. Finite Submodules

In this section, we will prove a result about the non-existence of finite submodules in X_K for a rather general class of fields. Let K be an arbitrary infinite Galois extension such that $G=\operatorname{Gal}(K/k)$ is a pro-p-group. Choose a sequence of subfields $k=k_0 \subset k_1 \subset \cdots \subset k_n \subset \cdots$ of K which are Galois over k and such that $K=\bigcup_{n=0}^{\infty} k_n$. We will assume that Leopoldt's conjecture is valid for k_n for all n. In this context we can still view X_K as a Λ_G-module. We have the following result.

Proposition 3. *Under the above assumptions, X_K contains no non-trivial finite Λ_G-submodule.*

Proof. For brevity we will let $M_n=M_{k_n}$ and $X_n=X_{k_n}$ for all $n \geqq 0$. Now it is not hard to show that $M_K=\bigcup_{n=0}^{\infty} M_n$. (See Lemma 6.1 in [7].) Suppose that X_K did contain a finite non-trivial submodule A. Let A_n denote the image of A in $X_n=\operatorname{Gal}(M_n/k_n)$ under the restriction map $\operatorname{Gal}(M_K/K) \to \operatorname{Gal}(M_n/k_n)$. For n sufficiently large (say, $n \geqq n_0$), A is obviously mapped isomorphically onto A_n.

Let J_n denote the idele group of k_n. If $j \in J_n$, we denote its $\mathfrak{p}$-component by $j_\mathfrak{p}$ for any prime $\mathfrak{p}$ of k_n. Let V_n be the subgroup of J_n consisting of ideles j such that $j_\mathfrak{p}$ is arbitrary if $\mathfrak{p}$ is archimedean, $j_\mathfrak{p}=1$ if $\mathfrak{p}$ divides p, and $j_\mathfrak{p}$ is a $\mathfrak{p}$-adic unit for all other $\mathfrak{p}$. Let $D_n=\overline{k_n^\times V_n} \subseteq J_n$ and let $\tilde{J}_n=J_n/D_n$. By class field theory the Artin

map $\varphi_n: J_n \to X_n$ induces a surjective homomorphism $\tilde{\varphi}_n: \tilde{J}_n \to X_n$ such that $\mathrm{Ker}(\tilde{\varphi}_n)$ is finite and of order prime to p. (The group $\tilde{J}_n$ is isomorphic to the Galois group of the maximal abelian p-ramified extension of k_n.) Let B_n denote the p-primary subgroup of $\tilde{\varphi}_n^{-1}(A_n)$ which is of course mapped isomorphically to A_n by $\tilde{\varphi}_n$. Now φ_n and $\tilde{\varphi}_n$ are actually $\mathrm{Gal}(k_n/k)$-homomorphisms and hence the fact that A is a Λ_G-submodule of X_K implies that $g(B_n)=B_n$ for all $g \in \mathrm{Gal}(k_n/k)$. Furthermore, since A is finite, $\mathrm{Gal}(K/k_{n_1})$ acts trivially on A for some n_1 and so $\mathrm{Gal}(k_n/k_{n_1})$ acts trivially on B_n for all $n \geqq n_1$.

By class field theory, the following diagram is commutative for $m \geqq n$.

$$\begin{array}{ccc} J_m & \xrightarrow{\text{norm}} & J_n \\ \Big\downarrow{\scriptstyle \varphi_m} & & \Big\downarrow{\scriptstyle \varphi_n} \\ X_m & \xrightarrow{\text{rest.}} & X_n \end{array}$$

Let $N_{m,n}: J_m \to J_n$ and $i_{n,m}: J_n \to J_m$ denote the norm and inclusion maps and let $\tilde{N}_{m,n}$ and $\tilde{i}_{n,m}$ denote the corresponding maps on $\tilde{J}_m$ and $\tilde{J}_n$. It follows that $\tilde{N}_{m,n}$ maps B_m onto B_n for $m \geqq n$ and if $n \geqq n_0$, this map is an isomorphism. Let $b_m \in B_m$ and $b_n = \tilde{N}_{m,n}(b_m)$. Then the fact that B_m is invariant under the action of $\mathrm{Gal}(k_m/k_n)$ and the definition of the norm map imply that $\tilde{i}_{n,m}(b_n) \in B_m$. Moreover, if $n \geqq n_1$, then $\tilde{i}_{n,m}(b_n) = b_m^{[k_m:k_n]}$. Therefore, for $m \gg n$, B_n is contained in $\mathrm{Ker}(\tilde{i}_{n,m})$.

To finish the proof of Proposition 3, we will show that $\tilde{i}_{n,m}$ is actually injective if a weak form of Leopoldt's conjecture is valid for k_m. For each $r \geqq 0$, define U_r in the same way that the group U was defined in Section 2, but for k_r. If C is an arbitrary subgroup of E_{k_r}, we will say that Leopoldt's conjecture is valid for C, if $\mathrm{rank}_{\mathbf{Z}_p}(\bar{C}) = \mathrm{rank}_{\mathbf{Z}}(C)$, where $\bar{C}$ denotes the closure of C when C is embedded in U_r in the natural way. This is equivalent to the following statement: For each positive integer s, there exists a positive integer t such that if $\gamma \in C$ is a $\mathfrak{p}$-adic p^t-th power for all primes $\mathfrak{p}$ of k_r lying over p, then $\gamma \in C^{p^s}$. (By a $\mathfrak{p}$-adic p^t-th power, we mean a p^t-th power in the $\mathfrak{p}$-adic completion of k_r.) Now, if $m \geqq n \geqq 0$, consider

$$C_{m,n} = \{\gamma \in E_{k_m} \mid N_{k_m/k_n}(\gamma) = 1\}.$$

What we need is contained in the following lemma.

Lemma 4. *Assume that Leopoldt's conjecture is valid for $C_{m,n}$. Then the homomorphism $\tilde{i}_{n,m}$ is injective.*

Note that the assumption in the lemma is certainly valid if Leopoldt's conjecture is valid for k_m. We also should point out that the proof of the lemma works for an arbitrary finite Galois p-extension k_m/k_n.

Proof. Suppose that $j_n \in J_n$ and that $j_m = i_{n,m}(j_n) \in D_m$. We will show that $j_n \in D_n$ and therefore that $\tilde{i}_{n,m}$ is injective. Since $j_m \in D_m$, $j_m = \lim_{i \to \infty} \alpha_i v_i$, where $\alpha_i \in k_m^\times$ and $v_i \in V_m$

for all $i \geqq 1$. If $g \in \mathrm{Gal}(k_m/k_n)$, then $g(j_m)=j_m$. Let $\gamma_i(g)=g(\alpha_i)/\alpha_i$ and $v_i'(g)=g(v_i)/v_i \in V_m$. Then $\lim_{i\to\infty} \gamma_i(g)\, v_i'(g)=1$. It obviously follows that for i sufficiently large $\gamma_i(g)$ is a unit in k_m (and in fact $\in E_{k_m}$) with norm 1. Thus, we may assume $\gamma_i(g) \in C_{m,n}$ for all $i \geqq 1$. We also see that $\lim_{i\to\infty} \gamma_i(g)=1$ in U_m (because of the definition of V_m). If s is any positive integer, we see that $\gamma_i(g) \in C_{m,n}^{p^s}$ for i sufficiently large (say, $i \geqq i(s)$) and for all g.

Let p^e be the number of p-power roots of unity in $C_{m,n}$ and let p^f be the order of the finite group $H^1(\mathrm{Gal}(k_m/k_n), C_{m,n})$. Let s be a positive integer and, if $i \geqq i(s+e+f)$, write $\gamma_i(g)=\eta_i(g)^{p^{s+e+f}}$, where $\eta_i(g) \in C_{m,n}$. Now $g \to \gamma_i(g)$ is clearly a 1-cocycle for $\mathrm{Gal}(k_m/k_n)$ with values in $C_{m,n}$. It follows that $\eta_i(g)^{p^e}$ is also a 1-cocycle and therefore $\eta_i(g)^{p^{e+f}}$ is a 1-coboundary, say $\eta_i(g)^{p^{e+f}}=g(\rho_i)/\rho_i$ for all g. Thus $\gamma_i(g)=(g(\rho_i)/\rho_i)^{p^s}$. Now when $\rho \in E_{k_m}$, $\rho^{p^s} \to 1$ in the topology of U_m as $s \to \infty$. Therefore, changing the indices if necessary, we can find a sequence ε_i in E_{k_m} such that $\varepsilon_i \to 1$ in U_m as $i \to \infty$ and such that $\gamma_i(g)=g(\varepsilon_i)/\varepsilon_i$ for all $i \geqq 1$ and all $g \in \mathrm{Gal}(k_m/k_n)$.

Let $\beta_i=\alpha_i \varepsilon_i^{-1}$. It follows from the above argument that $\beta_i \in k_n^{\times}$. Also, if $\mathfrak{p}$ is any prime of k_n lying over p, then in the $\mathfrak{p}$-adic completion of k_n, $\lim_{i\to\infty}(\beta_i)=$ the $\mathfrak{p}$-component of j_n. When i is large enough, the fractional ideal determined by j_n will be equal to the principal ideal (β_i) (since these ideals will generate the same ideal in k_m). Hence one can form a sequence of elements $v_i'' \in V_n$ such that, for large i, $j_n \beta_i^{-1}$ and v_i'' have equal $\mathfrak{p}$-components for all primes $\mathfrak{p}$ of k_n not lying over p. We then have $\lim_{i\to\infty}(\beta_i v_i'')=j_n$ and therefore $j_n \in D_n$, concluding the proof of Propositions 3.

Let K be a $\mathbf{Z}_p$-extension of k. Assume that $\mathrm{rank}_{\Lambda_G}(X_K)=r_2$. It follows from our remarks at the beginning of Section 3 that the sequence δ_n defined there is bounded and so achieves a maximum value δ_{n_0}. Hence it is clear that Leopoldt's conjecture must be valid for $C_{m,n}$ provided $m \geqq n \geqq n_0$. Thus we obtain the following consequence of the proof of Proposition 3.

Proposition 4. *If K is a $\mathbf{Z}_p$-extension of k such that* $\mathrm{rank}_{\Lambda_G}(X_K)=r_2$, *then X_K contains no non-trivial finite Λ_G-submodule.*

The above proposition together with Proposition 1 and Brumer's theorem implies Theorem 1. If k is an arbitrary number field then, adding to our comments at the end of Section 3, we could say that Theorem 1 is valid for "almost all" $\mathbf{Z}_p$ -extensions of k.

We should also make the following remark about the theorem of Iwasawa described at the beginning of this paper. If k is an arbitrary number field and if $K=k_\infty$ then, as we mentioned in Section 3, one can prove that $\mathrm{rank}(X_K)=r_2$ and hence one gets a new proof of Iwasawa's theorem. Actually, Iwasawa made the assumption that the ground field contains a primitive p-th root of unity (or a 4-th root of unity if $p=2$). Hence our arguments complete Iwasawa's original result.

Our results can also be extended to the following situation. Let S be a finite set of primes of k containing the primes lying over p. If K is a $\mathbf{Z}_p$-extension of k, we let $M_K(S)$ be the maximal abelian pro-p-extension of K which is unramified outside

of S. By slightly modifying our arguments, one can show that the $\Lambda_{\mathrm{Gal}(K/k)}$-module $\mathrm{Gal}(M_K(S)/K)$ has rank r_2 and contains no non-trivial finite submodule. If S is a subset of the primes dividing p, then one can still sometimes prove results of this type. For example, consider an abelian extension k of an imaginary quadratic field L. Assume that p splits in L and let $\mathfrak{p}$ be one of its prime factors. Let S be the set of primes of k dividing $\mathfrak{p}$. Now L has a unique $\mathbf{Z}_p$-extension in which only the prime $\mathfrak{p}$ is ramified. By translation, one obtains a certain $\mathbf{Z}_p$-extension K of k. Since k is abelian over L, Brumer's theorem in [1] actually implies that the S-adic analogue of Leopoldt's conjecture is valid for k, i.e. the closure of E_k in the product of the unit groups of the completions of k at all primes in S has maximal rank. By imitating our arguments in this case one can show that $\mathrm{Gal}(M_K(S)/K)$ is a torsion module over $\Lambda_{\mathrm{Gal}(K/k)}$ and that it has no finite non-trivial submodule. This resolves a question raised in [2] in connection with elliptic curves with complex multiplication.

5. Multiple $\mathbf{Z}_p$-Extensions

We will prove Theorem 2 by an induction argument, first considering the case when $G=\mathrm{Gal}(K/k)\cong\mathbf{Z}_p^2$. If K' is any $\mathbf{Z}_p$-extension of k contained in K, let $H=\mathrm{Gal}(K/K')$, $G'=\mathrm{Gal}(K'/k)$, and let h be a topological generator for H. Then, as we've mentioned previously, the kernel of the restriction mapping $X_K\to X_{K'}$ is $(h-1)X_K$. Let Y_K denote the torsion Λ_G-submodule of X_K. Then it is known that Y_K is pseudo-isomorphic to a module of the form $\bigoplus_{i=1}^{t}\Lambda_G/\mathfrak{p}_i^{e_i}$, where the $\mathfrak{p}_i$'s are primes ideals of Λ_G of height 1 and the e_i's are positive integers. (See [9] or the reference in Bourbaki given earlier.) Let Z_K be the kernel of such a pseudo-isomorphism so that Z_K is the maximal pseudo-null submodule of Y_K. Now since X_K/Y_K is a torsion-free Λ_G-module, $(h-1)X_K\cap Y_K=(h-1)Y_K$ and so $Y_K/(h-1)Y_K$ is isomorphic to a $\Lambda_{G'}$-submodule of $X_{K'}$. Now assume that Leopoldt's conjecture is valid for k. Then $\mathrm{rank}_{\Lambda_G}(X_K)=\mathrm{rank}_{\Lambda_{G'}}(X_{K'})=r_2$ and hence by (1), $Y_K/(h-1)Y_K$ is a torsion $\Lambda_{G'}$-module. Therefore $h-1\notin\mathfrak{p}_i$ for all i, $1\leqq i\leqq t$. It follows that no non-zero element of Y_K/Z_K is annihilated by $h-1$ and hence $(h-1)Y_K\cap Z_K=(h-1)Z_K$. This implies that $Z_K/(h-1)Z_K$ is isomorphic to a $\Lambda_{G'}$-submodule of $X_{K'}$ (in fact of $Y_{K'}$ since Z_K obviously must have an annihilator prime to $h-1$). Since Leopoldt's conjecture is valid for k, our remarks just before Proposition 1 imply that Z_K/I_GZ_K is finite. Here I_G is the augmentation ideal of Λ_G which is generated by $h-1$ and $g-1$ where g is chosen so that h and g are a set of topological generators for G. Theorem 2 in this case ($d=2$) clearly follows from the following lemma.

Lemma 5. *Let Z be a non-trivial pseudo-null Λ_G-module, where $G\cong\mathbf{Z}_p^2$. Assume that Z/I_GZ is finite. Then there exists at least one subgroup H of G such that $G'=G/H\cong\mathbf{Z}_p$ with the property that $Z/(h-1)Z$ contains a non-trivial finite $\Lambda_{G'}$-submodule, where h is a topological generator of H.*

Proof. We assume at first that Z has the following two properties: Z is torsion-free as a $\mathbf{Z}_p$-module and G contains a subgroup N of finite index which acts

trivially on Z. By Lemma 3, the first property implies that Z is a free $\mathbf{Z}_p$-module of finite rank. Let $V=Z\underset{\mathbf{Z}_p}{\otimes}\Omega_p$, where Ω_p denotes an algebraic closure of the field of p-adic numbers. We regard V as a representation space for the finite group $\tilde{G}=G/N$ over Ω_p. For each 1-dimensional Ω_p-valued character χ of $\tilde{G}$, we let m_χ denote the multiplicity with which the corresponding irreducible representation of $\tilde{G}$ occurs in V. If χ_0 is the trivial character of $\tilde{G}$, then the assumption that $|Z/I_G Z|<\infty$ implies that $m_{\chi_0}=0$. Suppose that the conclusion of the lemma is false for Z. Thus, if $g\in\tilde{G}$ but $\notin\tilde{G}^p$, then $Z/(g-1)Z$ is a torsion-free $\mathbf{Z}_p$-module and so

$$Z/(g-1)Z\cong\mathbf{Z}_p^{d_g}$$

for those g's, where $d_g\geqq 0$. We shall show that $m_\chi=0$ for all χ and therefore $Z=\{0\}$, contradicting the assumption that Z is non-trivial.

Let $g\in\tilde{G}$ but $\notin\tilde{G}^p$. Consider V as a representation space for the cyclic group $\langle g\rangle$. The integer d_g is obviously equal to the multiplicity with which the trivial representation of $\langle g\rangle$ occurs in V. Therefore,

$$d_g=\sum_{\chi(g)=1} m_\chi.$$

On the other hand, it is clear that d_g is also the minimal number of generators for Z as a module over the group ring $\mathbf{Z}_p[\langle g\rangle]$. Thus if ψ is any 1-dimensional character of $\langle g\rangle$, then ψ occurs in V with multiplicity $\leqq d_g$. If $\psi(g)=\zeta$, we obtain the following inequality:

$$\sum_{\chi(g)=\zeta} m_\chi\leqq d_g.$$

If χ is any non-trivial character of $\tilde{G}$, let $\tilde{H}=\tilde{H}_\chi=\mathrm{Ker}(\chi)$ so that $\tilde{G}/\tilde{H}$ is cyclic but $\tilde{H}\neq\tilde{G}$. Then if $\sigma\in\tilde{G}$, it is an easy exercise to compute the number of elements in $\sigma\tilde{H}\cap(\tilde{G}-\tilde{G}^p)$. The result is $|\tilde{H}|$ if $\sigma\tilde{H}$ generates $\tilde{G}/\tilde{H}$ and $\left(1-\frac{1}{p}\right)|\tilde{H}|$ otherwise. Thus, using the fact that $m_{\chi_0}=0$, we find that

$$\sum_g d_g=\sum_\chi\left(1-\frac{1}{p}\right)|H_\chi|\,m_\chi, \tag{3}$$

where $\sum_g$ denotes a sum over all $g\in\tilde{G}-\tilde{G}^p$. Now let p^t be the smallest index $[\tilde{G}:\tilde{H}_\chi]$ that occurs for some χ with $m_\chi\neq 0$ (assuming that m_χ is not always zero). Let ζ be a primitive p^t-th root of unity. Adding the above inequality for all $g\in\tilde{G}-\tilde{G}^p$, we obtain

$$\sum\nolimits_1|H_\chi|\,m_\chi+\sum\nolimits_2\left(1-\frac{1}{p}\right)|H_\chi|\,m_\chi\leqq\sum_g d_g, \tag{4}$$

where $\sum_1$ is a summation over those χ's with $[\tilde{G}:\tilde{H}_\chi]=p^t$ and $\sum_2$ a summation over the remaining χ's. Since $\sum_1$ has at least one non-zero term, (3) and (4) are clearly incompatible and so $m_\chi=0$ for all χ.

Let's assume now that Z is a torsion-free $\mathbf{Z}_p$-module. Let Z_1 be the maximal Λ_G-submodule of Z on which G acts through a finite quotient group. (It's easy to prove the existence of Z_1.) Let $Z_2 = Z/Z_1$. Then Z_2 is again torsion-free as a $\mathbf{Z}_p$-module and we will assume here that it is non-trivial. Consider the representation space $V_2 = Z_2 \underset{\mathbf{Z}_p}{\otimes} \Omega_p$ for G. Since G is commutative, one can triangularize this representation and thus define homomorphisms $\varphi_i\colon G \to \Omega_p^\times$ for $1 \leqq i \leqq \dim_{\Omega_p}(V_2)$ such that for $g \in G$ its eigenvalues are precisely the $\varphi_i(g)$'s. The definition of Z_1 implies that $\varphi_i(G)$ is infinite for each i. Thus $\mathrm{Ker}(\varphi_i)$ contains at most one subgroup H with $G/H \cong \mathbf{Z}_p$. Therefore, except for finitely many H's, a topological generator h of H will act on V_2 with none of its eigenvalues equal to 1. For such h, $Z_2/(h-1)Z_2$ is finite and non-trivial (since Z_2 is). If $Z_1 = \{0\}$, it follows that the lemma is valid for Z. If Z_1 is non-trivial, then we've already proved that $Z_1/(h-1)Z_1$ has a finite non-trivial submodule where h is a topological generator of some suitably chosen H. But since G acts on Z_1 through a finite quotient group, there will clearly be infinitely many such H's and we can therefore assume that h acts on V_2 without 1 as an eigenvalue. Hence $h-1$ doesn't annihilate any non-zero element of Z_2. It follows that $(h-1)Z \cap Z_1 = (h-1)Z_1$ and so $Z_1/(h-1)Z_1$ is isomorphic to a $\Lambda_{G'}$-submodule of $Z/(h-1)Z$. This proves the lemma in the case when Z is a torsion-free $\mathbf{Z}_p$-module. Note that there exists infinitely many distinct H's with the properties described in the lemma.

Finally, let Z be an arbitrary non-trivial pseudo-null Λ_G-module. Let Z_0 denote the torsion $\mathbf{Z}_p$-submodule of Z. Since Z_0 is finitely generated as a Λ_G-module, it clearly has bounded exponent. We will show that with the exception of finitely many H's, $Z_0/(h-1)Z_0$ is finite. Consider $\tilde{Z}_0 = Z_0/pZ_0$ as a module over the ring $\tilde{\Lambda}_G = \Lambda_G/p\Lambda_G$ (which is isomorphic to a power series ring in two variables over $\mathbf{Z}/(p)$). Since Z_0 is pseudo-null, it must have an annihilator $\Phi \in \Lambda_G$ not divisible by p. Then the image $\tilde{\Phi}$ of Φ in $\tilde{\Lambda}_G$ is a non-zero annihilator of $\tilde{Z}_0$. The proof of Lemma 2 shows that, except for finitely many H's, $\tilde{\Phi}$ is not divisible by $h-1$ (considered as an element of $\tilde{\Lambda}_G$). Thus the ideal $(h-1, \tilde{\Phi})$ is of finite index in $\tilde{\Lambda}_G$ and this implies that $\tilde{Z}_0/(h-1)\tilde{Z}_0$ is finite for such H. Since Z_0 has bounded exponent, $Z_0/(h-1)Z_0$ will also be finite (and non-trivial if Z_0 is). Now Z/Z_0 is torsion-free as a $\mathbf{Z}_p$-module and so combining our results, we have proved the lemma.

We now will consider the case when $\mathrm{Gal}(K/k) \cong \mathbf{Z}_p^d$, where $d \geqq 3$. Let K' be a subfield of K such that $G' = \mathrm{Gal}(K'/k) \cong \mathbf{Z}_p^{d-1}$ and let h be a topological generator of $H = \mathrm{Gal}(K/K')$. Denote the maximal pseudo-null Λ_G-submodule of X_K by Z_K. Then if Leopoldt's conjecture is valid for the ground field k, the argument at the beginning of this section shows that $Z_K/(h-1)Z_K$ is isomorphic to a $\Lambda_{G'}$-submodule of $X_{K'}$. We will prove that, for at least one K', $Z_K/(h-1)Z_K$ is a pseudo-null $\Lambda_{G'}$-module. Hence if $X_{K'}$ is known to contain no non-trivial pseudo-null $\Lambda_{G'}$-submodule, then $Z_K/(h-1)Z_K$ and therefore Z_K itself must be trivial.

If Z_K is torsion-free as a $\mathbf{Z}_p$-module, then choose E as in Lemma 3. Thus Z_K is finitely generated over Λ_E. Since $d \geqq 3$, one can choose K' so that $H \subseteq E$. Let E' denote the image of E in G'. Then $G'/E' \cong \mathbf{Z}_p^2$ and $Z_K/(h-1)Z_K$ is finitely generated over $\Lambda_{E'}$. The discussion following Lemma 2 shows that $Z_K/(h-1)Z_K$

must be pseudo-null as a $\Lambda_{G'}$-module. Also because $d \geqq 3$ one could choose E and therefore H in infinitely many distinct ways. On the other hand, if Z_K is a torsion $\mathbf{Z}_p$-module then an argument similar to the one at the end of the proof of Lemma 5 shows that $Z_K/(h-1)Z_K$ must be pseudo-null except for finitely many H's. Combining these arguments, we find that for a suitably chosen K', $Z_K/(h-1)Z_K$ is in fact a pseudo-null $\Lambda_{G'}$-module.

Summing up, we have actually proved the following result.

Proposition 5. *Assume that Leopoldt's conjecture is valid for k. Let K be a Galois extension of k with $G = \mathrm{Gal}(K/k) \cong \mathbf{Z}_p^d$, where $d \geqq 1$. Then X_K contains no non-trivial pseudo-null Λ_G-submodule.*

6. Some Examples

We will now describe a series of examples where the module Y_K is non-trivial. The final example we will construct will be a certain multiple $\mathbf{Z}_p$-extension with a certain specific prime ideal in $\mathrm{Supp}(Y_K)$. We construct such examples for any $d \geqq 1$. Most of our examples were suggested by Theorem 15 in [8].

We first consider certain types of $\mathbf{Z}_p$-extensions. We will denote the n-th layer of a $\mathbf{Z}_p$-extension K/k by k_n. Let $X_n = X_{k_n}$.

I. Let $K = k_\infty$, the cyclotomic $\mathbf{Z}_p$-extension of k. If k is totally real, then Iwasawa's theorem shows that $X_K = Y_K$. If in addition k has a cyclic extension k' of degree p such that $k' \cap K = k$ and such that only primes of k dividing p are ramified in k'/k, then clearly Y_K will be non-trivial (and therefore in fact infinite). One particularly interesting example is the following. Let k be the maximal real subfield of $\mathbf{Q}(\sqrt[p]{1})$. Then it is not hard to show that such a k' exists if and only if p is an irregular prime.

Assume now that k is an arbitrary number field satisfying the following properties: k contains a primitive p-th root of unity (a 4-th root if $p=2$), p has at least two prime factors in k, and the prime factors of p in k are totally ramified in K/k. Theorem 15 of [8] implies that Y_K is non-trivial under these assumptions, but this can also be seen as follows. The field K contains the group W of all p-power roots of unity. We can define a homomorphism $\chi: G \to \mathbf{Z}_p^\times$ by

$$g(\omega) = \omega^{\chi(g)}$$

for all $g \in G = \mathrm{Gal}(K/k)$ and all $\omega \in W$. For each $n \geqq 0$, let $W_n = W \cap k_n$. Following the notation used in Section 4, it is clear that U_n contains a subgroup isomorphic to W_n^t, where t denotes the number of primes of k dividing p. The action of $\mathrm{Gal}(K/k)$ on these subgroups is also described by the homomorphism χ. Now $\bar{E}_n$ contains a subgroup isomorphic to W_n and it is not hard to see that the torsion subgroup of $\bar{E}_n/W_n$ has bounded order as $n \to \infty$, by using the boundedness of the sequence δ_n. Let $T = \varprojlim W_n$, where the inverse limit is defined by means of the norm maps. Thus $T \cong \mathbf{Z}_p$ and G acts on T by $g(\tau) = \chi(g)\tau$ for all $g \in G$ and $\tau \in T$. The above remarks show that X_K contains a Λ_G-submodule isomorphic to T^{t-1} which must obviously be contained in Y_K. Since $t \geqq 2$, Y_K must be non-trivial.

II. Let $\mathfrak{p}$ be a fixed prime of k dividing p. We now consider a $\mathbf{Z}_p$-extension K/k which is "cyclotomic at $\mathfrak{p}$". By this we mean that $K_\mathfrak{p}$ coincides with the cyclotomic $\mathbf{Z}_p$-extension of $k_\mathfrak{p}$, where $k_\mathfrak{p}$ denotes the $\mathfrak{p}$-adic completion of k and $K_\mathfrak{p}$ denotes the union of the completions of the k_n's for $n=1, 2, 3, \ldots$ at some prime of k_n dividing $\mathfrak{p}$. If $[k_\mathfrak{p}: \mathbf{Q}_p] < r_2$, then k will have $\mathbf{Z}_p$-extensions of the above type other than k_∞. To see this, recall that $\mathrm{Gal}(M_k/k)$ has rank at least r_2+1 as a $\mathbf{Z}_p$-module, while the decomposition subgroup for $\mathfrak{p}$ in the extension M_k/k has rank at most $[k_\mathfrak{p}: \mathbf{Q}_p]+1$. Let K_1 denote the composite of all $\mathbf{Z}_p$-extensions of k in which $\mathfrak{p}$ splits completely so that $\mathrm{Gal}(K_1/k) \cong \mathbf{Z}_p^e$, where $e \geqq r_2 - [k_\mathfrak{p}: \mathbf{Q}_p]$. Then any $\mathbf{Z}_p$-extension of k which is contained in $K_1 k_\infty$ but not contained in K_1 will be cyclotomic at $\mathfrak{p}$.

Let $\Gamma = \mathrm{Gal}(K_\mathfrak{p}/k_\mathfrak{p})$. Then Γ can be identified with $\mathrm{Gal}(k_\infty/k)$ and in this way the action of Γ on the p-power roots of unity contained in $K_\mathfrak{p}$ can be described by χ. (We are still assuming that k contains a p-th or 4-th root of unity and that $\mathfrak{p}$ is totally ramified in k_∞/k.) Now Γ can also be identified with the decomposition subgroup of $\mathfrak{p}$ in $\mathrm{Gal}(K/k)$, say $\mathrm{Gal}(K/k_s)$ where $s \geqq 0$. Thus $\mathfrak{p}$ splits into p^s distinct primes in k_s and these primes are totally ramified in K/k_s. Therefore, for $n \geqq 0$, U_{s+n} contains a subgroup isomorphic to $W_n^{p^s}$ and the action of $\mathrm{Gal}(K/k_s)$ on this subgroup is given by χ. If $K \neq k_\infty$ and if Leopoldt's conjecture is valid for k so that the δ_n's are bounded, then it follows that X_K contains a Λ_G-submodule isomorphic to $\mathbf{Z}_p^{p^s}$ as a $\mathbf{Z}_p$-module. The action of $\Gamma = \mathrm{Gal}(K/k_s)$ on this submodule is such that $\gamma - \chi(\gamma)$ (considered as an element of Λ_G) annihilates it. Therefore, Y_K is non-trivial.

III. Let K be an arbitrary $\mathbf{Z}_p$-extension of k contained in the field K_1 defined above, assuming that $e \geqq 1$. Thus $\mathfrak{p}$ splits completely in k_n/k for all $n \geqq 0$ and hence U_n contains a subgroup isomorphic to $(\mathbf{Z}/(p^a))^{p^n}$, where p^a is the number of p-power roots of unity in k. If Leopoldt's conjecture is valid for k so that the torsion subgroup of $\bar{E}_n$ has bounded order, then it follows that Y_K contains a Λ_G-submodule isomorphic to $\Lambda_G/p^a\Lambda_G$ and hence Y_K has a non-zero μ-invariant.

IV. Let $d \geqq 2$. One can easily choose k so that it satisfies all of the assumptions we've made on it so far in this section and so that also $e \geqq d-1$. Then k will have an extension K with the following properties:

$$k_\infty \subseteq K, \quad K \subseteq K_1 k_\infty, \quad \mathrm{Gal}(K/k) \cong \mathbf{Z}_p^d.$$

We will show that Y_K is non-trivial by giving an explicit element in its support.

Let Γ denote the decomposition subgroup for $\mathfrak{p}$ in $G = \mathrm{Gal}(K/k)$. Since $W \subseteq K$, we can define $\chi: G \to \mathbf{Z}_p^\times$ just as before. The restriction map $\mathrm{Gal}(K/k) \to \mathrm{Gal}(k_\infty/k)$ induces an isomorphism $\Gamma \cong \mathrm{Gal}(k_\infty/k)$ and so $\Gamma \cong \mathbf{Z}_p$. Let γ_0 be a topological generator of Γ and let $\chi_0 = \chi(\gamma_0)$. Then it is not hard to see that $\gamma_0 - \chi_0$ is an irreducible element of Λ_G and we will prove that the prime ideal $(\gamma_0 - \chi_0)\Lambda_G$ is contained in $\mathrm{Supp}(Y_K)$.

Let L be an arbitrary $\mathbf{Z}_p$-extension of k contained in K. It is clear from our results in I, II, and III that Y_L contains a non-trivial submodule annihilated by $\gamma_0|_L - \chi_0$. Let $\Phi = \gamma_0 - \chi_0$. Suppose that $(\Phi)\Lambda_G$ is not in fact in $\mathrm{Supp}(Y_K)$. It follows easily from this assumption that one can choose an element Ψ of Λ_G

relatively prime to Φ with the property that X_K contains a free Λ_G-submodule $\mathfrak{X}$ such that $X_K/\mathfrak{X}$ is annihilated by Ψ. As we will explain below, one can find a subfield K' of K with $G'=\mathrm{Gal}(K'/k)\cong \mathbf{Z}_p^{d-1}$ so that the images Φ' and Ψ' of Φ and Ψ in $\Lambda_{G'}$ are still relatively prime. Let h be a topological generator of H. Then $h-1$ and Ψ will be relatively prime and this implies that $X_K/(h-1)X_K$ contains the free $\Lambda_{G'}$-module $\mathfrak{X}/(h-1)\mathfrak{X}$. The corresponding quotient module is annihilated by Ψ'. The exact sequence

$$0\to X_K/(h-1)X_K\to X_{K'}\to H\to 0$$

and the fact that Φ' is not in the augmentation ideal $I_{G'}$ (since $\Phi\notin I_G$) so that Φ' doesn't annihilate H allow one to construct from Ψ' an element Ψ'' in $\Lambda_{G'}$ relatively prime to Φ' with the property that $\Psi'' X_{K'}$ is contained in the free $\Lambda_{G'}$-module $\mathfrak{X}/(h-1)\mathfrak{X}$. Repeating this process we eventually obtain a $\mathbf{Z}_p$-extension L/k such that X_L contains a free $\Lambda_{\mathrm{Gal}(L/k)}$-module such that the corresponding quotient module has an annihilator prime to $\gamma_0|L-\chi_0$. This clearly contradicts the remark made at the beginning of this paragraph.

The existence of the field K' mentioned above can be proved by applying the arguments in Section 5 of this paper to the Λ_G-module $Z=\Lambda_G/(\Phi,\Psi)$. This module is pseudo-null and we must just find a K' so that $Z/(h-1)Z$ is still pseudo-null, where h is as above. Since $Z/(h-1)Z\cong\Lambda_{G'}/(\Phi',\Psi')$, Φ' and Ψ' will then be relatively prime. The case $d=2$ (which will certainly occur when repeating the process described above) can be settled by noticing that the first case considered in the proof of Lemma 5 cannot occur. This can be seen easily from the definition of Φ. If $d\geqq 3$, the argument at the end of Section 5 immediately implies what we want.

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Invariant Measures of Horospherical Flows on Noncompact Homogeneous Spaces

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Introduction

In [18] Veech proves that the "horospherical flow" on G/Γ where G is a semisimple analytic group with no non-trivial compact factor and Γ is a discrete co-compact subgroup, is uniquely ergodic; that is, there is a unique Borel probability measure on G/Γ which is invariant under the flow.

On the other hand there exists a wide class of discrete subgroups Γ in semisimple groups G such that G/Γ admits a G-invariant probability measure, but is noncompact. The discrete subgroup $SL(2, \mathbb{Z})$ of integral unimodular matrices in $SL(2, \mathbb{R})$ the group of all unimodular matrices, constitutes the simplest example of the above phenomenon. It turns out however, that the horospherical flow on a non-compact homogeneous space cannot be uniquely ergodic. This is because there exist proper closed subgroups H containing horospherical subgroups, having closed orbits which admit a finite H-invariant measure. The objective of the present paper is to assert that for a certain class of horospherical flows including the example cited above all the ergodic invariant measures arise in the above manner. We recall that the set of ergodic measures also determines completely the set of all finite invariant measures.

Actually rather than restricting to the class of semisimple groups with no non-trivial compact factor we consider any reductive analytic group T. Apart from being more general this is also convenient in certain arguments. A subgroup U in T is said to be *horospherical* if there exists $t \in T$ such that

$$U = \{u \in T \mid t^j u t^{-j} \to e \text{ as } j \to \infty\}$$

e being the identity in T. Any horospherical subgroup is analytic; i.e. a connected Lie subgroup (cf. § 4, [5]). Further the adjoint action of any element in the Lie subalgebra of U on the Lie algebra of T is nilpotent. A horospherical flow in the sense of [18] is the action of a *maximal* horospherical subgroup.

A discrete subgroup Γ of a locally compact group T is said to be a *lattice* if T/Γ admits a finite T-invariant regular Borel measure.

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A major part of the study of invariant measures of horospherical flows consists of obtaining a characterization of the Haar measure of a semisimple group G without compact factors (cf. Theorem 2.4) in terms of invariance under the right action of a given lattice Γ and the left action of a certain maximal horospherical subgroup N of G, suitably related to Γ (cf. Theorem 2.4). Our approach in obtaining the above characterization is based on the philosophy introduced by Fürstenberg [8], which was also employed in [18].

In Part III we extend the study to reductive Lie groups. Sections 8 and 9 deal with general technical details concerning the extension. Finally in Section 10 we specialize to reductive Lie groups T such that all the non-compact simple factors of the adjoint group of T are of $\mathbb{R}$-rank 1 and obtain a complete classification of (ergodic) invariant measures of the "horospherical flow". We prove the following.

Theorem A. *Let T be a reductive Lie group such that all noncompact simple factors of T are of $\mathbb{R}$-rank 1. Let U be a maximal horospherical subgroup and let Γ be a lattice in T. Let π be a finite U-invariant ergodic measure on T/Γ. Then there exists a closed subgroup L and $t \in T$ such that*

i) *$Lt\Gamma$ is closed and $\pi(T/\Gamma - Lt\Gamma/\Gamma) = 0$ and*

ii) *L contains U, $t\Gamma t^{-1} \cap L$ is a lattice in L and π is the finite L-invariant measure on $Lt\Gamma/\Gamma \simeq L/t\Gamma t^{-1} \cap L$.*

Notice that assertions i) and ii) completely determine π (up to a scalar which is determined by the total measure).

We observe that since every compact minimal subset of any action of an ameanable group is the support of a finite ergodic invariant measure, in particular Theorem A characterizes all compact minimal subsets of the horospherical flow.

Actually our proof incorporates the case of uniform lattices considered in [18]. However for that case there now exist simpler proofs (cf. [3] and [6]). We may note that the proofs in [3] and [6] indeed hold for all reductive groups.

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§ 1. Measures on Homogeneous Spaces

(1.1) For any locally compact, second countable space X, we denote by $\mathcal{M}(X)$ the set of all positive regular Borel measures on X. Let β be a Borel measurable map, or briefly a transformation of a space X onto a space Y. Let $\pi \in \mathcal{M}(X)$. If either β is continuous and proper or π is finite then we have a measure $\beta\pi \in \mathcal{M}(Y)$ defined by $\beta\pi(E) = \pi(\beta^{-1}E)$ for any Borel subset E of Y. If H is a set of transformations of X, $\pi \in \mathcal{M}(X)$ is said to be H-invariant if $h\pi = \pi$ for all $h \in H$.

Let H be a locally compact, second countable group and let $\Phi: H \times X \to X$ be an action of H on a space X. The action is said to be measurable if the map Φ is measurable when $H \times X$ is endowed with product Borel structure. Let $\pi \in \mathcal{M}(X)$

be H-invariant; i.e. invariant under all $\Phi(h, \cdot)$, $h \in H$. Then π is said to be ergodic with respect to the H-action (or that π is H-ergodic or even that the H-action on X is ergodic with respect to π) if for any Borel subset E of X, the assertion $\pi(E \Delta hE)=0$ for all $h \in H$ implies that either $\pi(E)=0$ or $\pi(X-E)=0$. Here and in the sequel hE denotes the set $\Phi(h, \cdot)E$, provided that the action in question is clear from the context.

(1.2) For $\pi \in \mathscr{M}(X)$, $\mathscr{L}^2(X, \pi)$ denotes the Hilbert space of all measurable complex valued functions which are square integrable with respect to π. Let H be a measurable group of transformations of X and let $\pi \in \mathscr{M}(X)$ be H-invariant. Then $h \mapsto V_h$, where V_h is the operator on $\mathscr{L}^2(X, \pi)$ defined by $(V_h f)(x) = f(h^{-1}x)$, is a (strongly continuous) unitary representation of H on $\mathscr{L}^2(X, \pi)$. Following is a simple and useful criterion for ergodicity: a finite H-invariant measure π is ergodic with respect to the H-action if and only if for $f \in \mathscr{L}^2(X, \pi)$, $V_h f = f$ for all $h \in H$ implies that f is a constant function π-a.e.

(1.3) The following proposition asserts that the set of all ergodic H-invariant measures determines the cone of all finite H-invariant measures. The result may be considered folklore. A proof for the case of cyclic and one-parameter group may be found in [17]. The same proof readily generalizes to any locally compact separable group.

(1.4) **Proposition.** *Let (X, π) be a Lebesgue probability space and let Φ: $H \times X \to X$ be a measurable, measure-preserving action of a locally compact, separable group H. Then there exists a measurable partition ξ of X and a family $\{\pi_C\}_{C \in \xi}$ where π_C is a probability measure on X with support contained in C such that if $\bar{\pi}$ denotes the quotient measure of π on X/ξ then we have the following*

i) *For almost all $C \in X/\xi$ (with respect to $\bar{\pi}$), π_C is a H-invariant ergodic measure.*

ii) *Let E be a measurable subset of X. Then for almost all $C \in X/\xi$, $E \cap C$ is measurable, $\pi_C(E \cap C)$ is a measurable function on X/ξ and*

$$\pi(E) = \int_{X/\xi} \pi_C(E \cap C)\, d\bar{\pi}(C).$$

(1.5) We recall some results from [8] regarding measures on homogeneous spaces.

(1.6) **Proposition.** *Let H be a closed unimodular subgroup of a Lie group L. Let dh be a fixed Haar measure on H. Then there is a one-to-one correspondence between $\mathscr{M}(L/H)$ and the subset of $\mathscr{M}(L)$ of measures which are invariant under the right action of H on L. The correspondence is given by $\pi \leftrightarrow \omega$ where*

$$(1.7) \quad \int_L f(x)\, d\omega(x) = \int_{L/H} d\pi(xH) \int_H f(xh)\, dh$$

for all $f \in C_c(L)$, the space of all continuous functions with compact support.

(1.8) **Proposition.** *If H and H' be two closed unimodular subgroups of a Lie group L, then* (1.7) *also sets up a correspondence between measures in $\mathscr{M}(L/H)$ which are invariant under the left action of H' and the subset of $\mathscr{M}(L)$ consisting*

of measures which are invariant under the right action of H and the left action of H'.

(1.9) **Corollary.** *Let ι denote the inversion map of L, i.e. $x \in L \mapsto x^{-1}$. Then the map $\sigma \to \iota\sigma$ of $\mathcal{M}(L)$ induces a one-to-one correspondence between the set of H'-invariant measures on L/H and the set of H-invariant measures on L/H'. Under the correspondence H'-ergodic measures correspond to H-ergodic measures; the L-invariant (resp. absolutely continuous with respect to the L-invariant) measure on L/H corresponds to L-invariant (resp. absolutely continuous with respect to the L-invariant) measure on L/H'.*

Definition. Let G be a Lie group, Γ a lattice in G and let N be a horospherical subgroup of G. A Γ-invariant measure π on G/N is said to be *Γ-finite* if the N-invariant measure ω on G/Γ corresponding to π under the above one-one correspondence is finite.

(1.10) Absolute continuity of invariant measures: Let X be a locally compact, second countable space and let G and H be locally compact groups of homeomorphisms of X. Assume that the actions of G and H commute with each other; i.e. $ghx = hgx$ for all $g \in G$, $h \in H$ and $x \in X$. Assume also that the cone of G-invariant measures on X is one dimensional. Let dh be a fixed Haar measure on H and let $\varphi \in C_c^+(H)$, the cone of positive continuous functions with compact support. For any $\pi \in \mathcal{M}(X)$ and any Borel subset of X define

$$(1.11) \quad \pi_\varphi(E) = \int_H \pi(h^{-1}E)\,\varphi(h)\,dh.$$

Then $\pi_\varphi \in \mathcal{M}(X)$. Then we have the following (cf. Proposition 2.5, [18]).

(1.12) **Proposition.** *Let G, H and X be as above and let μ be a G-invariant measure on X. Let Γ be a subgroup of G such that the action of Γ on X is ergodic with respect to μ. If π is a Γ-invariant measure on X such that $\pi_\varphi \prec \mu$ for all $\varphi \in C_c^+(H)$, then π is a multiple of μ.*

(Here and in the sequel $\prec$ stands for absolute continuity with respect to the latter measure.)

Later in the application of the above proposition we need the following lemma (cf. Theorem 3 and Proposition 6, [15]).

(1.13) **Lemma.** *Let G be a semisimple Lie group with no non-trivial compact factor. Let N be a maximal horospherical subgroup in G and let Γ be a lattice in G. Then the action of Γ on G/N is ergodic with respect to the G-invariant measure on G/N.*

We will also often need the following.

(1.14) **Lemma.** *Let L be a locally compact second countable group and let Γ be a lattice in L. Let H be a (closed) subgroup of L such that H is normalized by Γ and $H\Gamma$ is a closed subgroup. Then $H \cap \Gamma$ is a lattice in H.*

Proof. Observe that since L/Γ admits a finite L-invariant measure $L/H\Gamma$ also admits a finite L-invariant measure. It follows from the criterion for existence of invariant measures on homogeneous spaces (cf. Ch.. II, §9, [20]) that the restrictions of the modular homomorphism of L to Γ and $H\Gamma$ are the modular

homomorphisms of the respective subgroups. Hence in particular we get that $H\Gamma/\Gamma$ admits a $H\Gamma$ invariant measure. But in view of the Fubini-Weil formula (cf. Ch. II, §9, [20]) the $H\Gamma$ invariant measure on $H\Gamma/\Gamma$ must be finite. On the other hand, $H\Gamma/\Gamma$ is canonically isomorphic to $H/H\cap\Gamma$ – under the isomorphism the H-action on $H\Gamma/\Gamma$ corresponds to the H-action on $H/H\cap\Gamma$. Hence we conclude that $H/H\cap\Gamma$ admits a finite H-invariant measure; i.e. $H\cap\Gamma$ is a lattice in H.

Part II. A Characterisation of Haar Measure

Throughout this part G will denote a connected semisimple Lie group with no non-trivial compact factor and with trivial center. Then G is isomorphic to its adjoint group and hence can be considered as the connected component of the identity in the $\mathbb{R}$-algebraic group of all Lie automorphisms of its Lie algebra. The class of maximal horospherical subgroups is then exactly the class of maximal unipotent subgroups of G.

The object of Part II, as the title indicates, is to prove a characterisation of the G-invariant measure on G/N, N being a maximal horospherical subgroup, which will be formulated in §2. The proof follows in several steps through §7.

§2. Lattices and Fundamental Domains

Let G be a Lie group as described above and Γ be a lattice in G. We need a certain special fundamental domain for Γ which we shall describe in the present section.

Definition. A lattice Λ in a semisimple Lie group L is said to be *irreducible* if the only positive dimensional normal subgroup F for which $F\Lambda$ is closed is L itself.

Any lattice Γ in a semisimple Lie group without compact factors can be "decomposed" into irreducible lattices in normal subgroups: More precisely we have the following

(2.1) **Proposition** (cf. [4] *Appendix*). *Let G be a semisimple Lie group without compact factors and with trivial center. Let Γ be a lattice in G. Then there exist normal (semisimple) Lie subgroups G_i, $i\in I$ (a suitable indexing set) such that*

i) $G=\prod_{i\in I} G_i$ *(direct product).*

ii) *For each $i\in I$, $\Gamma_i=\Gamma\cap G_i$ is an irreducible lattice in G_i and*

iii) $\Gamma'=\prod_{i\in I}\Gamma_i$ *is a (normal) subgroup of finite index in Γ.*

The decomposition is unique upto reindexing.

Observe that as a consequence of the above, G/Γ is finitely covered by G/Γ' and the latter is a direct product of homogeneous spaces G_i/Γ_i, $i\in I$. We now classify the components G_i/Γ_i as follows. Put

$$I_0=\{i\in I \mid G_i/\Gamma_i \text{ is compact}\},$$

$$I_1=\{i\in I-I_0 \mid \mathbb{R}\text{-rank } G_i>1\}$$

and

$$I_2 = I - I_0 - I_1.$$

For $i \in I_1$, G_i is a semisimple Lie group without compact factor with trivial center and Γ_i is a non-uniform lattice in G_i. Therefore by Margulis's arithmeticity theorem (cf. [13]) Γ_i is an arithmetic subgroup of G_i. More precisely, G_i admits a structure of the connected component of the identity in an $\mathbb{R}$-algebraic group defined over $\mathbb{Q}$, such that Γ_i is commensurable with group $(G_i)_{\mathbb{Z}}$ of integral elements in G_i (with respect to the $\mathbb{Q}$-structure). We shall henceforth assume to have fixed the $\mathbb{Q}$-structure on each G_i, $i \in I_1$ satisfying the above.

Next let $i \in I_2$. Then G_i is of $\mathbb{R}$-rank 1 and Γ_i is a non-uniform lattice in G_i. For these lattices we plan to use the results of Garland and Raghunathan (cf. [10]). For the present we recall that G_i admits a maximal horospherical subgroup N_i such that $N_i \cap \Gamma_i$ is a lattice in N_i. Let P_i be the normalizer of N_i in G_i. Then P_i is a minimal $\mathbb{R}$-parabolic subgroup of G_i. For future use we note that by conjugating N_i by a suitable element of Γ_i we can find a maximal horospherical subgroup N_i' which is not contained in P_i. If P_i' denotes the normalizer of N_i' then $P_i \neq P_i'$. Since $\mathbb{R}$-rank of G_i is 1, $P_i \cap P_i'$ is a (reductive) Levi subgroup in both P_i and P_i'.

With this information we proceed to introduce some terminology with respect to Γ, generalizing the corresponding notions for arithmetic groups. We must emphasize that this is done only for simplicity in writing the fundamental domain and that the material is essentially well-known.

Definition. A horospherical subgroup U in G is said to be *Γ-rational* if $U \cap \Gamma$ is lattice in U.

A Lie subgroup U is a maximal Γ-rational horospherical subgroup of G if and only if $U = \prod_{i \in I} U_i$ where for $i \in I_0$, $U_i = (e)$, for $i \in I_1$, U_i is a maximal unipotent $\mathbb{R}$-subgroup of G_i defined over $\mathbb{Q}$ and for $i \in I_2$, U_i is a maximal horospherical subgroup such that $U_i \cap \Gamma_i$ is a lattice in U_i. Let U be a maximal Γ-rational horospherical subgroup and let P be the normalizer of U in G. Then again we observe that $P = \prod_{i \in I} P_i$ where for $i \in I_0$, $P_i = G_i$ for $i \in I_1$, P_i is a minimal $\mathbb{R}$-parabolic subgroup defined over $\mathbb{Q}$ and for $i \in I_2$, P_i is a minimal $\mathbb{R}$-parabolic subgroup. The results which we now state can be easily proved by considering separately the components corresponding to indices $i \in I$ and using known results about respective classes of groups and above observations. Hence we shall often omit the details.

Definition. Let U_1 and U_2 be horospherical subgroups in G and let P_1 and P_2 be normalizers of U_1 and U_2 respectively. Then U_1 and U_2 are said to be *opposite* (to each other) if $P_1 \cap P_2$ is a (reductive) Levi subgroup in both P_1 and P_2.

Let U and U^- be two Γ-rational horospherical subgroups opposite to each other, and let P and P^- be the respective normalizers. Then there exists a unique maximal vector subgroup S contained in the center of (the reductive group) $P \cap P^-$ such that $\{\mathrm{Ad}\, s | s \in S\}$ is simultaneously diagonalizable over $\mathbb{R}$. Actually $S = \prod_{i \in I} S_i$ where $S_i = (e)$ for $i \in I_0$, S_i is the connected component of a

maximal $\mathbb{Q}$-split $\mathbb{R}$-torus (algebraic) for $i\in I_1$ and a suitable one parameter group for $i\in I_2$.

Now let $\mathfrak{G}$ be the Lie algebra of G and $\mathscr{S}$ the Lie subalgebra corresponding to S. Then we get a decomposition of $\mathfrak{G}$ as

$$\mathfrak{G}=\mathfrak{Z}(\mathscr{S})+\sum_{\lambda\in\Delta}\mathfrak{G}^{\lambda}$$

where $\Delta\subset\mathscr{S}^*$ (the dual of $\mathscr{S}$) is a root system for the pair $(\mathfrak{G},\mathscr{S})$, $\mathfrak{Z}(\mathscr{S})$ the centralizer of $\mathscr{S}$ in $\mathfrak{G}$, and for each $\lambda\in\Delta$

$$\mathfrak{G}^{\lambda}=\{\xi\in\mathfrak{G}\mid[\alpha,\xi]=\lambda(\alpha)\xi \text{ for all } \alpha\in\mathscr{S}\}$$

the latter being of positive dimension for all $\lambda\in\Delta$. On Δ there exists an ordering such that if Δ^+ is the corresponding set of positive roots then $\sum_{\lambda\in\Delta^+}\mathfrak{G}^{\lambda}$ is the Lie subalgebra corresponding to U and $\sum_{\lambda\in\Delta^+}\mathfrak{G}^{-\lambda}$ is the Lie subalgebra corresponding to U^-. The last assertion may be proved by using Lemma 4.5.4, in [13] for the field $\mathbb{Q}$ in the case of arithmetic components (i.e. $i\in I_1$) and for $\mathbb{R}$ in the case of non-uniform lattices in $\mathbb{R}$-rank 1 groups (i.e. $i\in I_2$).

Now let (U,U^-) be a pair of opposite maximal Γ-rational horospherical subgroups and S and Δ^+ be as chosen above. For $t>0$ we define

$$S_t=\{s\in S\mid\lambda(\log s)\leqq t \text{ for all } \lambda\in\Delta^+\} \tag{2.2}$$

where $\log\colon S\to\mathscr{S}$ is the inverse of the exponential map of $\mathscr{S}$ onto S.

We now state the main proposition of §2.

(2.3) **Proposition.** *Let (U,U^-) be a pair of opposite maximal Γ-rational horospherical subgroups of G and let S and S_t, $t>0$ be as introduced above, corresponding to (U,U^-). Further let H be the connected component of the identity in the normalizer P^- of U^- and let K be a maximal compact subgroup of G. Then there exist a compact subset W of H, a finite subset J of G and $\delta>0$ such that the following properties hold.*

a) $G=\Gamma JWS_{\delta}K$.

b) For all $j\in J$, $j^{-1}\Gamma j\cap U^-$ is a lattice in U^-.

c) Let $G=\prod_{i\in I}G_i$ be the decomposition of G as in Proposition 2.1 and let $i\in I$ be such that $\mathbb{R}$-rank $G_i>1$ and $\Gamma_i=\Gamma\cap G_i$ is a non-uniform lattice in G_i. Then for all $j\in J$, $j^-\Gamma_i j$ is commensurable with Γ_i.

Proof. Let $G=\prod_{i\in I}G_i$ be the decomposition of G as in Proposition 2.1, with respect to Γ. It is obvious that assertions a), b) and c) follow if the corresponding assertions are proved for the lattice Γ_i in G_i for each $i\in I$. In other words in proving the Proposition we may further assume Γ to be an irreducible lattice in G. We now consider three cases separately.

Case i. Γ is a uniform lattice in G.

In this case both U and U^- are the identity subgroup, as there cannot be any Γ-rational horospherical subgroup of positive dimension. In particular $P^-=G$.

But certainly there exists a compact subset W of G such that $G=\Gamma W$. Putting $J=\{e\}$, a) is satisfied. b) is satisfied trivially.

Case ii. Assume that Γ is non-uniform and $\mathbb{R}$-rank $G>1$.

As recalled earlier, in this case Γ is an arithmetic subgroup of G_j (with respect to a $\mathbb{Q}$ structure considered fixed). U and U^- are maximal unipotent algebraic $\mathbb{R}$-subgroups defined over $\mathbb{Q}$, and S is the connected component of the identity in a maximal $\mathbb{Q}$-split (algebraic) $\mathbb{R}$-torus. Applying Borel's construction of fundamental domains for arithmetic groups (cf. Theorem 13.1, [1]) it follows that there exists a compact subset W' of H (actually of a smaller subgroup) a finite subset J' of $G_{\mathbb{Q}}$ and $\delta>0$ such that with

$$S^\delta=\{s\in S\mid \lambda(\log s)\leqq\delta \text{ whenever } -\lambda\in\Delta^+\}$$

(Δ^+ being the set of roots such that the corresponding Lie subalgebra of $\mathfrak{G}$ belongs to U) we have $G=KS^\delta W'J'\Gamma$. Thus $G=G^{-1}=\Gamma JW(S^\delta)^{-1}K$ where $J=(J')^{-1}$ and $W=(W')^{-1}$. Observe that

$$\begin{aligned}(S^\delta)^{-1}&=\{s^{-1}\mid s\in S,\ \lambda(\log s)\leqq\delta \text{ whenever } -\lambda\in\Delta^+\}\\&=\{s\in S\mid\lambda(\log s)\leqq\delta \text{ for all } \lambda\in\Delta^+\}\\&=S_\delta.\end{aligned}$$

Thus we have proved assertion a) in the case at hand. Since $J\subset G_{\mathbb{Q}}$ for each $j\in J$, $j^{-1}\Gamma j$ is commensurable with Γ. This prove c). To prove b) it is enough observe that for any $y\in G_{\mathbb{Q}}$, yUy^{-1} is also a unipotent $\mathbb{R}$-subgroup defined over $\mathbb{Q}$ and hence intersects an arithmetic group in a lattice.

Case iii. Γ is a non-uniform lattice in G and $\mathbb{R}$-rank of G is 1.

In this case we use the construction of fundamental domains by Garland and Raghunathan (cf. Theorem 0.6, [10]). Notice that in this case $G=KSU^-$ (notation as in the statement of the theorem) is an Iwasawa decomposition of G. Then by the theorem of Garland and Raghunathan there exists a compact subset W' of U^-, a finite subset J' of G and $\delta>0$ such that $G=KS^\delta W'J'\Gamma$ where

$$S^\delta=\{s\in S\mid\lambda(\log s)\leqq\delta \text{ whenever } -\lambda\in\Delta^+\}$$

(Δ^+ being the set of roots corresponding to U) and J' is such that for each $j\in J'$, $\Gamma\cap j^{-1}U^-j$ is a lattice in $j^{-1}U^-j$. Then $G=G^{-1}=\Gamma JW(S^\delta)^{-1}K$ where $J=(J')^{-1}$ and $W=(W')^{-1}$. As before we see that $(S^\delta)^{-1}=S_\delta$. Hence $G=\Gamma JWS_\delta K$. Also clearly for each $j\in J$, $j^{-1}\Gamma j\cap U^-$ is a lattice in U^-.

We now formulate a characterisation of the Haar measure whose proof is the subject of Part II. The notation U, U^-, S, K introduced earlier as also the subsets W, J and $\delta>0$ satisfying Proposition 2.3 shall be considered fixed. Also recall that P and P^- denote the normalizers of U and U^- respectively and that H denotes the connected component of the identity in P^-.

Now let A be a maximal vector subgroup containing S, contained in $P\cap P^-$ and such that $\{\mathrm{Ad}\,a\mid a\in A\}$ is diagonalizable over $\mathbb{R}$ and let N be a maximal horospherical subgroup (not necessarily Γ-rational) in G normalized by A and

such that $U \subset N \subset P$. We denote $X = G/N$ and $X_0 = \bigcup_{j \in J} jP^- N/N$. Then $X - X_0$ is a union of lower dimensional manifolds. (This may be proved by using Bruhat decomposition on each component G_i $i \in I$.) Hence if μ is a G-invariant measure on X then $\mu(X - X_0) = 0$. We prove:

(2.4) **Theorem.** *Let π be a Γ-invariant, Γ-finite, ergodic (cf. §1 for definitions) measure on $X = G/N$ such that $\pi(X - X_0) = 0$. Then π is G-invariant.*

We close this section by recalling certain simple properties of the fundamental domains constructed in Proposition 2.3. Put $\Omega = JWS_\delta K/K \subset G/K$. Let μ_K denote a G-invariant measure on G/K. Then it is well-known that $\mu_K(\Omega) < \infty$ (cf. Lemma 12.5, [1]). We also need the following

(2.5) **Lemma.** *Let W_1 be any relatively compact subset of H. Then for any $t > 0$, $\bigcup_{h \in S_t} h^{-1} W_1 h$ is relatively compact.*

Proof. The subgroup H can be expressed as $Z \cdot U^-$ (semidirect product) where Z is the centralizer of S in G. Considering the components it is enough to prove the lemma for W_1 contained U^-. Any element of U^- can be expressed as $(\exp t_1 \xi_1)(\exp t_2 \xi_2) \ldots (\exp t_n \xi_n)$ where $\xi_1, \xi_2, \ldots, \xi_n \in \mathcal{U}^-$, the Lie subalgebra of U^-, are simultaneous eigenvectors of $\{\mathrm{Ad}\, s | s \in S\}$. For any $h \in S$, $t \in \mathbb{R}$ and $i \leqq n$ we have $h^{-1}(\exp t\xi_i)h = \exp v_i(h^{-1})t\xi_i$ where v_i is the character on S such that $(\mathrm{Ad}\, s)\xi_i = v_i(s)\xi_i$ for all $s \in S$. Since $\xi_i \in U^-$ clearly $-\log v_i \in \Delta^+$. Thus if $s \in S_t$, $v_i(h^{-1}) = \exp(-\log v_i(h)) \leqq e^t$. If W_1 is contained in

$$\{(\exp t_1 \xi_1)(\exp t_2 \xi_2) \ldots (\exp t_n \xi_n) | t_1, t_2 \ldots t_n \in [-t', t']\}$$

then $\bigcup h^{-1} W_1 h$ is contained in

$$\{(\exp t_1 \xi_1)(\exp t_2 \xi_2) \ldots (\exp t_n \xi_n) | t_1, t_2 \ldots t_n \in [-e^t t', e^t t']\}.$$

Hence it is relatively compact.

(2.6) *Remark.* $S = JWS_\delta K$ is contained in $JW_0 S^- K$ where $S^- = \{s | \lambda(\log s) \leqq 0$ for all $\lambda \in \Delta^+\}$, and W_0 is a compact subset of SH.

(2.7) *Note.* In the sequel the notations as in Proposition 2.3 and later shall be considered fixed. Also we shall denote by M the group of elements of K which centralize A. Then MAN is a minimal (real) parabolic subgroup of G.

§3. A Measure on the Fürstenberg Boundary

The homogeneous space G/MAN which is canonically isomorphic to K/M is the maximal Fürstenberg boundary of G. (cf. [7, 14].) We denote this space by B. We have a canonical projection of β: $X = G/N \to G/MAN = B$. We now fix a compact subset Q of X such that $\pi(Q) > 0$. Let π_Q be the measure on X defined by $\pi_Q(E) = \pi(E \cap Q)$ for any Borel subset E of X. Since Q is compact, π_Q is a finite measure. Thus we get a (regular, Borel) measure $\beta\pi_Q$ on B defined by

$\beta\pi_Q(Y)=\pi_Q(\beta^{-1}Y)$ for any Borel subset Y of B. Let m denote the K-invariant probability measure on B. The aim of this section is to prove the following:

(3.1) **Proposition.** *If $\beta\pi_Q<m$ then π is G-invariant.*

Proof. Observe that since the subgroup $Z=MA$ normalizes N there exists an action $z\mapsto\psi_z$ of Z on $X=G/N$ defined by $\psi_z(gN)=gzN$, which commutes with the left action of G on X. Therefore in view of Proposition 1.12 and Proposition 1.13 in order to prove the proposition it is enough to prove that for any $\varphi\in C_c^+(Z)$ the measure π_φ defined by

$$(3.2)\quad \pi_\varphi(E)=\int_Z \pi(\psi_z^{-1}E)\,\varphi(z)\,dz$$

for any Borel subset E of X, is absolutely continuous with respect to a G-invariant measure say μ. So let $\varphi\in C_c^+(Z)$ be fixed. Let $Q'=\{\psi_z x \mid z\in\text{support of }\varphi,\ x\in Q\}$ and let $\pi_{\varphi,Q}$ be the measure on X defined by

$$(3.3)\quad \pi_{\varphi,Q}(E)=\pi_\varphi(E\cap Q').$$

(3.4) **Lemma.** *If $\pi_{\varphi,Q}\prec\mu$ then $\pi_\varphi\prec\mu$.*

Proof. Since π is Γ-ergodic and $\pi(Q)>0$ we have $\pi(X-\Gamma Q)=0$. A direct computation now yields $\pi_\varphi(X-\Gamma Q')=0$. Therefore for any Borel subset E of X we have

$$\begin{aligned}(3.5)\quad \pi_\varphi(E)=\pi_\varphi(E\cap\Gamma Q')&\leqq\sum_{\gamma\in\Gamma}\pi_\varphi(E\cap\gamma Q')\\&=\sum_{\gamma\in\Gamma}\pi_\varphi(\gamma^{-1}E\cap Q')\\&=\sum_{\gamma\in\Gamma}\pi_{\varphi,Q}(\gamma^{-1}E).\end{aligned}$$

Now if $\mu(E)=0$ then for any $\gamma\in\Gamma$, $\mu(\gamma^{-1}E)=0$. Therefore if $\pi_{\varphi,Q}\prec\mu$ by (3.5), $\mu(E)=0$ implies $\pi_\varphi(E)=0$.

Proof of Proposition (3.1). It is easy to verify that there exists a character χ: $Z\to\mathbb{R}^+$ such that for any Borel set $E\subset G/N=X$

$$\mu(E)=\int_B dm(xP)\int_Z \chi_E(\psi_z x)\,\chi(z)\,dz$$

(here and in the sequel χ_E denotes the characteristic function of E in the appropriate space). Assume that $\mu(E)=0$ and put

$$B_1=\{b=xP\in B\mid\int_Z \chi_E(\psi_z x)\,\chi(z)\,dz=0\}$$

Then $m(B_1)=1$. Put $E_1=E\cap\beta^{-1}B_1$ and $E_2=E-E_1$. Then

$$(3.6)\quad \pi_\varphi(E_1)=\int_X d\pi(x)\int_Z \chi_{E_1}(\psi_z x)\,\varphi(z)\,dz=0.$$

Put $B_2 = B - B_1$. Then $E \subset \beta^{-1} B_2$ and $m(B_2) = 0$. As in Lemma 3.4, we have

$$\begin{aligned}(3.7)\quad \pi(\beta^{-1}B_2) &\leqq \sum_{\gamma\in\Gamma} \pi_Q(\gamma^{-1}\beta^{-1}B_2)\\ &= \sum_{\gamma\in\Gamma} \pi_Q(\beta^{-1}\gamma^{-1}B_2)\\ &= \sum_{\gamma\in\Gamma} \beta\pi_Q(\gamma^{-1}B_2).\end{aligned}$$

It is easy to see that since $m(B_2)=0$, for all $\gamma\in\Gamma$, $m(\gamma^{-1}B_2)=0$. Therefore if $\beta\pi_Q \prec m$ by (3.7) we have $\pi(\beta^{-1}B_2)=0$. Therefore

$$\begin{aligned}\pi_{\varphi,Q}(E_2) \leqq \pi_{\varphi,Q}(\beta^{-1}B_2) &= \int_X d\pi(x) \int_Z \chi_{\beta^{-1}B_2}(\psi_z x)\, \chi_{Q'}(\psi_z x)\, \varphi(z)\, dz\\ &= \int_X \chi_{\beta^{-1}B_2}(x)\, d\pi(x) \int_Z \chi_{Q'}(\psi_z x)\, \varphi(z)\, dz\\ &= 0.\end{aligned}$$

Combining with (3.6) we conclude that $\pi_{\varphi,Q}(E)=0$. Since E was an arbitrary Borel set with $\mu(E)=0$ we have thus proved that $\pi_{\varphi,Q} \prec \mu$. Hence by Lemma 3.4, $\pi_\varphi \prec \mu$. Again since $\varphi \in C_c^+(Z)$ was arbitrary as remarked in the beginning we get that π is G-invariant.

§ 4. Harmonic Functions on G/K

Recall that we have fixed an Iwasawa decomposition $G = KAN$. For $g\in G$ let $k(g)\in K$ and $a(g)\in A$, be the uniquely determined elements such that $g = k(g)\cdot a(g)\cdot n$ for some $n\in N$. Also for $g\in G$ let $H(g) = \log a(g) \in \mathfrak{A}$. The following useful relation is a simple consequence of the definition. If $g_1, g_2 \in G$ then

$$(4.1)\quad H(g_1 g_2) = H(g_1 k(g_2)) + H(g_2).$$

Let $\rho\in\mathfrak{A}^*$ be the element defined by $2\rho = \sum_{\lambda\in\Delta^+} (\dim \mathfrak{G}^\lambda)\,\lambda$. The Poisson kernel $\mathbb{P}$ on $G/K \times K/M$ is defined by

$$(4.2)\quad \mathbb{P}(gK, kM) = \exp -2\rho H(g^{-1}k).$$

$\mathbb{P}(\cdot,\cdot)$ is clearly well-defined. Using this kernel to each Borel measure σ on $B = K/M$ one associates a non-negative function h_σ on G/K defined by

$$(4.3)\quad h_\sigma(gK) = \int_B \mathbb{P}(gK, kM)\, d\sigma(kM).$$

Though we will not need it explicitly it may be recalled that h_σ defined as above is a harmonic function with respect to the Laplace-Beltrami operator (indeed with respect to any second order, elliptic G-invariant differential operator vanishing on constant functions) on G/K. Conversely any non-negative harmonic function h on G/K admits a representation as in (4.3) for some measure $\sigma = \sigma_h$ on B. Further the

measure σ_h can be 'constructed' from h as follows. Let A^+ be the positive Weyl chamber in A with respect to Δ^+; viz.

$$A^+ = \{a \in A \mid \lambda(\log a) > 0 \text{ for all } \lambda \in \Delta^+\}.$$

For $a \in A^+$ let $\sigma(h, a)$ be the measure on B defined by

$$\sigma(h, a)(E) = \int_E h(k a K)\, dm(kM)$$

for any Borel subset E of B. Since $h(k m a K) = h(k a K)$ the integral is well-defined. Now define '$a \to \infty$' to mean that $\lambda(\log a) \to \infty$ for all $\lambda \in \Delta^+$. Then σ_h is determined by the following result of Knapp and Williamson [12].

(4.4) **Proposition.** *With notations as above*

$$\sigma_h = \lim_{a \to \infty} \sigma(h, a)$$

in the space of all bounded measures which is assumed to be endowed with the weak-topology.*

(4.5) **Corollary** (cf. [18]). *Let σ be a finite Borel measure on B and h_σ the harmonic function defined by* (4.3). *If there exists a sequence $\{a_j\}$ in A^+, $a_j \to \infty$ in the above sense, such that*

$$\limsup_{j \to \infty} \int_K h_\sigma(k a_j K)^2\, dk = l^2 < \infty$$

then $\sigma \prec m$.

Proof. By Proposition 4.4 and Schwartz inequality for any continuous function ψ on B, we get

$$\begin{aligned} \left|\int_B \psi(kM)\, d\sigma(kM)\right| &= \lim_{j \to \infty} \left|\int_B \psi(kM)\, h_\sigma(k a K)\, dm(kM)\right| \\ &\leqq \limsup_{j \to \infty} \left|\int_B \psi(kM)^2\, dm\right|^{1/2} \left|\int_B h_\sigma(k a K)\, dm\right|^{1/2} \\ &= l \left|\int_B \psi(kM)^2\, dm\right|^{1/2} \end{aligned}$$

which shows that $\sigma \prec m$.

On G/K there exists a G-invariant metric defined by

$$D(g_1 K, g_2 K) = \max_{k \in K} |\rho H(g_1^{-1} g_2 k)| \tag{4.6}$$

(cf. [18], §5). Further it is also proved that the distance is given by

$$D(g_1 K, g_2 K) = \rho(\log a)$$

where $a \in \mathrm{cl}\, A^+$ is the unique element such that $g_1^{-1} g_2 = k_1 a k_2$ (Cartan decomposition) where $k_1, k_2 \in K$.

The *disc of radius* τ in G/K with center K is the set

(4.7) $U_\tau = \{gK \mid D(gK, K) \leqq \tau\}$.

Let μ_K be a G-invariant measure on G/K. Proofs of the following lemmas may be found in [18].

(4.8) **Lemma** (cf. Lemma 5.10, [18]). *For $\tau_0 > 0$ there exists a constant $c(\tau_0)$ such that for all $\tau > 0$, $\mu_K(U_{\tau+\tau_0}) \leqq c(\tau_0)\,\mu_K(U_\tau)$.*

(4.9) **Lemma** (cf. Lemma 5.11, [18]). *Let h be a locally integrable, non-negative function on G/K. If there exists a constant c such that for all $\tau > 0$*

$$\int_{U_\tau} h(gK)^2\, d\mu_K \leqq c\,\mu_K(U_\tau)$$

then there exists a sequence $\{a_j\}_1^\infty$ in A^+, $a_j \to \infty$ such that

$$\limsup_{j\to\infty} \int_K h(k a_j K)^2\, dk\, M < \infty.$$

(4.10) *Remark.* Let f be the harmonic function on G/K corresponding to the meaure $\beta\pi_Q$ on B; viz.

$$f(gK) = \int_B \mathbb{P}(gK, kM)\, d\beta\pi_Q(kM).$$

In view of Proposition 3.1, Corollary 4.5 and Lemma 4.9, if there exists a constant c such that for all $\tau > 0$

$$\int_{U_\tau} f(gK)^2\, d\mu_K \leqq c\,\mu(U_\tau)$$

then π is G-invariant.

§5. Estimation of $\mathscr{L}^2$-Norms on Discs

Recall that we have $G/K = \Gamma\Omega$. Now let $\gamma \in \Gamma$, $b \in \Omega$ and $g \in G$ be such that $gK = \gamma b$. We have for $f(gK)$ as in the last section

$$\begin{aligned} f(gK) &= \int_B \exp -2\rho H(g^{-1}k)\, d\beta\pi_Q(kM) \\ &= \int_X \exp -2\rho H(g^{-1}k(x))\, d\pi_Q(x). \end{aligned}$$

(Notice that the functions k and a defined on G are invariant under the right action of N and hence define functions on X which also we denote by the same letters.) Since Q is a compact subset of X there exists a constant c_1 such that for all $x \in Q$

(5.1) $c_1^{-1} \leqq \exp -2\rho H(a(x)) \leqq c_1$.

Since $H(g^{-1}k(x))=H(g^{-1}x)-H(a(x))$ using (5.1) we conclude that

$$f(gK)\leqq c_1\int_X \exp -2\rho H(g^{-1}x)\,d\pi_Q(x)$$
$$=c_1\int_X \chi_Q(x)\exp -2\rho H(b^{-1}\gamma^{-1}x)\,d\pi(x).$$

Here by $H(b^{-1}\gamma^{-1}x)$ we mean $H(b_0^{-1}\gamma^{-1}x)$ for some $b_0\in G$ such that $b=b_0K$, the function being obviously independent of the representative. Now since π is Γ-invariant we get

$$f(gK)\leqq c_1\int_X \chi_Q(\gamma x)\exp -2\rho H(b^{-1}x)\,d\pi(x).$$

For the sake of brevity henceforth let the function $\exp -2\rho H(\cdot)$ on G or X be denoted by $F(\cdot)$ (on G or X respectively). Thus

(5.2) $$f(gK)\leqq c_1\int_X \chi_Q(\gamma x)\,F(b^{-1}x)\,d\pi(x).$$

Now let $\tau>0$ and U_τ be the disc of radius τ and center K. Put $\Gamma_\tau=\{\gamma\in\Gamma\mid \gamma\Omega\cap U_\tau \text{ is nonempty}\}$ and for $\gamma\in\Gamma_\tau$ let $D_\gamma=\Omega\cap\gamma^{-1}U_\tau$. Then $U_\tau=\bigcup_{\gamma\in\Gamma_\tau}\gamma D_\gamma$. Hence

$$\int_{U_\tau} f(gK)^2\,d\mu_K\leqq\sum_{\gamma\in\Gamma_\tau}\int_\Omega f(\gamma bK)^2\,\chi_{D_\gamma}(b)\,d\mu_K(b)$$
$$\leqq c_1\sum_{\gamma\in\Gamma_\tau}\int_\Omega \chi_{D_\gamma}(b)\,d\mu_K(b)\int_{X\times X}\chi_Q(\gamma x)\,\chi_Q(\gamma y)\,F(b^{-1}x)$$
$$\cdot F(b^{-1}y)\,d\pi(x)\,d\pi(y)$$

in view of (5.2). Using Fubini theorem, which henceforth we apply repeatedly without mention, the last expression may be written as

(5.3) $$c_1\int_\Omega d\mu_K(b)\int_{X\times X}F(b^{-1}x)\,F(b^{-1}y)\,E(x,y,b)\,d\pi(x)\,d\pi(y)$$

where for all $x,y\in X$ and $b\in\Omega$

(5.4) $$E(x,y,b)=\sum_{\gamma\in\Gamma_\tau}\chi_Q(\gamma x)\,\chi_Q(\gamma y)\,\chi_{D_\gamma}(b)$$
$$=\sum_{\gamma\in\Gamma}\chi_Q(\gamma x)\,\chi_Q(\gamma y)\,\chi_{U_\tau}(\gamma b).$$

We first obtain an integral which majorizes $E(x,y,b)$. Let V be a compact, symmetric neighborhood of the identity e in G with $V^2\cap\Gamma=\{e\}$. Then there exists $\tau_0>0$ such that $VU_\tau\subset U_{\tau+\tau_0}$. Also put $Q_1=KVQ$. With this notation it is straightforward to verify that there exists a constant c_2 such that for any $x,y\in X$ and $b\in\Omega$

(5.5) $$\chi_Q(x)\,\chi_Q(y)\,\chi_{U_\tau}(b)\leqq c_2\int_V \chi_{Q_1}(gx)\,\chi_{Q_1}(gy)\,\chi_{U_{\tau+\tau_0}}(gb)\,dg$$

where dg is a fixed Haar measure on G. Combining (5.4) and (5.5), we deduce that

$$E(x,y,b) \leqq c_2 \sum_{\gamma\in\Gamma} \int_V \chi_{Q_1}(g\gamma x)\,\chi_{Q_1}(g\gamma y)\,\chi_{U_{\tau+\tau_0}}(g\gamma b)\,dg$$
$$= c_2 \sum_{\gamma\in\Gamma} \int_{V\gamma} \chi_{Q_1}(gx)\,\chi_{Q_1}(gy)\,\chi_{U_{\tau+\tau_0}}(gb)\,dg.$$

Since V is symmetric and $V^2\cap\Gamma=\{e\}$, $\{V\gamma\}_{\gamma\in\Gamma}$ is a family of pairwise disjoint subsets of G. Hence

$$E(x,y,b) \leqq c_2 \int_G \chi_{Q_1}(gx)\,\chi_{Q_1}(gy)\,\chi_{U_{\tau+\tau_0}}(gb)\,dg.$$

Now for any $\tau'>0$ let $G_{\tau'}=\{g\in G \mid gK\in U_{\tau'}\}$. We also fix a Borel cross-section $O:\Omega\to G$ of the canonical projection of G onto G/K, such that $O(\Omega)$ is contained in JWS_t. For any $b\in\Omega$ let b_0 denote $O(b)$. Then for x, $y\in X$ and $b\in\Omega$

$$E(x,y,b) \leqq c_2 \int_G \chi_{Q_1}(gx)\,\chi_{Q_1}(gy)\,\chi_{U_{\tau+\tau_0}}(gb_0K)\,dg \tag{5.6}$$
$$= c_2 \int_{G_{\tau+\tau_0}} \chi_{Q_1}(gb_0^{-1}x)\,\chi_{Q_1}(gb_0^{-1}y)\,dg.$$

Now consider the Cartan decomposition $G=K(\mathrm{cl}\,A^+)K$ where A^+ is the positive Weyl chamber in A. Let $g=k_1\alpha k$ be the expression for the generic element in G. Correspondingly the Haar measure dg has an expression $\nu(\alpha)\,dk_1\,d\alpha\,dk$ where $\nu(\alpha)$ is a function on $\mathrm{cl}\,A^+$, $d\alpha$ is a Haar measure on A and dk_1 and dk are normalized Haar measures on K.

Since $KQ_1=Q_1$ the integral in (5.6) is invariant under the left action of K. Also $G_{\tau+\tau_0}$ is invariant under the left action. Therefore using the above decomposition of dg we get

$$E(x,y,b) \leqq \int_{A^+_{\tau+\tau_0}} \nu(\alpha)\,d\alpha \int_K \chi_{Q_1}(\alpha k b_0^{-1}x)\,\chi_{Q_1}(\alpha k b_0^{-1}y)\,dk \tag{5.7}$$

where $A^+_{\tau+\tau_0}=\{\alpha\in A^+ \mid \alpha K\in U_{\tau+\tau_0}\}$. Here we have also used the fact that the complement of A^+ in $\mathrm{cl}\,A^+$ has zero measure. Now for $\alpha\in A^+$, put $Q_\alpha=K\alpha^{-1}Q_1$. Then the integral over K appearing in (5.7) is nonzero only if x, $y\in b_0Q_\alpha$. Further, using Schwartz lemma we may conclude that

$$\int_K \chi_{Q_1}(\alpha k b_0^{-1}x)\,\chi_{Q_1}(\alpha k b_0^{-1}y)\,dk \tag{5.8}$$
$$\leqq \chi_{Q_\alpha}(b_0^{-1}x)\,\chi_{Q_\alpha}(b_0^{-1}y)\Big\{\int_K \chi_{Q_1}(\alpha k b_0^{-1}x)\,dk\Big\}^{1/2}\Big\{\int_K \chi_{Q_1}(\alpha k b_0^{-1}y)\,dk\Big\}^{1/2}.$$

Fortunately for us an estimate of the integrals appearing in (5.8) which is sufficient for our purposes has indeed been obtained in [18].

(5.9) **Proposition** (Lemma 6.7 in [18]). *For any compact subset Q_1 of X there exist constants $\varepsilon>0$ and c_3 such that for all $\alpha\in A^+$ and $x\in X$*

$$\int_K \chi_{Q_1}(\alpha k x)\,dk \leqq c_3\,F(x)^\varepsilon\,F(\alpha)^\varepsilon.$$

Therefore the right hand side of (5.8) is majorized by

(5.10) $c_3^2 \chi_{Q_\alpha}(b_0^{-1}x)\chi_{Q_\alpha}(b_0^{-1}y)F(b^{-1}x)^{\varepsilon/2}F(b^{-1}y)^{\varepsilon/2}F(\alpha)^{\varepsilon}$.

Putting together (5.3), (5.6) and (5.10) we now conclude that there exists a constant c_4 such that

$$\begin{aligned}(5.11)\quad \int_{U_\tau} f(gK)^2\,d\mu_K &\leqq c_4 \int_\Omega d\mu_K(b) \int_{X\times X} F(b^{-1}x)^{1+\frac{\varepsilon}{2}} F(b^{-1}y)^{1+\frac{\varepsilon}{2}}\,d\pi(x)\,d\pi(y)\\ &\quad\cdot \int_{A_{\tau'+\tau_0}^+} \chi_{Q_\alpha}(b_0^{-1}x)\chi_{Q_\alpha}(b_0^{-1}y)F(\alpha)^{\varepsilon}\nu(\alpha)\,d\alpha\\ &= c_4 \int_\Omega d\mu_K(b) \int_{A_{\tau'+\tau_0}^+} F(\alpha)^{\varepsilon}\nu(\alpha)\,d\alpha\\ &\quad\cdot \{\int_X \chi_{Q_\alpha}(b_0^{-1}x)F(b^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi(x)\}^2.\end{aligned}$$

Now consider the integral

$$I(b,\alpha)=\int_X \chi_{Q_\alpha}(b_0^{-1}x)F(b_0^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi(x).$$

Recall that $b_0 \in JWS_t \subset JS^-W_0$ where W_0 is a compact subset of SH (cf. Corollary 2.6). In view of (4.1) there exists a constant c_5 such that for all $x' \in X$ and $w \in W_0$

$$F(w^{-1}x') \leqq c_5 F(x')$$

Hence if $b_0 = jhw$ where $j \in J$, $h \in S^-$ and $w \in W_0$ then

$$\begin{aligned}(5.12)\quad I(b,\alpha) &= \int_X \chi_{Q_\alpha}(w^{-1}h^{-1}j^{-1}x)F(w^{-1}h^{-1}j^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi(x)\\ &\leqq c_5^{1+\frac{\varepsilon}{2}} \int_X \chi_{W_0Q_\alpha}(h^{-1}j^{-1}x)F(h^{-1}j^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi(x)\\ &= c_5^{1+\frac{\varepsilon}{2}} \int_X \chi_{W_0Q_\alpha}(h^{-1}x)F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi_j(x)\end{aligned}$$

where π_j is the measure on X defined by

$$(5.13)\quad \pi_j(E)=\pi(jE)$$

for any Borel subset E of X. Then π_j is $j^{-1}\Gamma j$-invariant and $\pi_j(X - P^-N/N)=0$.

(5.14) **Proposition.** *There exists a compact subset E of H and a constant c_6 such that for all $j \in J$, $h \in S^-$ and any measurable subset Q of X*

$$\int_Q F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi_j(x) \leqq c_6 \int_{EQ} F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\mu(x).$$

Proof of this proposition is postponed until the next section. Using the Proposition we now complete the proof of Theorem 2.4. Firstly from (5.12) we have

$$I(b,\alpha) \leqq c_5^{1+\frac{\varepsilon}{2}} \int_{hW_0Q_\alpha} F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\, d\pi_j(x)$$

$$\leqq c_7 \int_{EhW_0Q_\alpha} F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\, d\mu(x) \quad \text{(where } c_7 = c_6\, c_5^{1+\frac{\varepsilon}{2}}\text{)}$$

$$= c_7 \int_{h(h^{-1}Eh)W_0Q_\alpha} F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\, d\mu(x)$$

$$\leqq c_7 \int_{hE_0W_0Q_\alpha} F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\, d\mu(x)$$

where $E_0 = \bigcup_{h\in A^-} h^{-1}Eh$. Since E is a compact subset of H by Lemma 2.5, E_0 is also a relatively compact subset of H. How

$$(5.15)\quad I(b,\alpha) \leqq c_7 \int \chi_{E_0W_0Q_\alpha}(h^{-1}x)\, F(h^{-1}x)^{1+\frac{\varepsilon}{2}}\, d\mu(x)$$

$$= c_7 \int \chi_{E_0W_0Q_\alpha}(x)\, F(x)^{1+\frac{\varepsilon}{2}}\, d\mu(x).$$

Using the Iwasawa decomposition $G = KAN$, X may be identified with $K \times A$ (topologically). It is well-known that under this identification $d\mu(x)$ has the expression (up to a scalar multiple) $\exp 2\rho H(a)\, dk\, da$. Since $F(x) = \exp -2\rho H(a(x))$ it follows that

$$(5.16)\quad \int_X \chi_{E_0W_0Q_\alpha}(x)\, F(x)^{1+\frac{\varepsilon}{2}}\, d\mu(x) = \int_{E_0W_0Q_\alpha} F(x)^{\varepsilon/2}\, dk\, da$$

$$= \int_{E_0W_0K\alpha^{-1}Q_1} F(x)^{\varepsilon/2}\, dk\, da.$$

(5.17) **Proposition** (Lemma 6.21 in [18]). *There exists a constant c_8 depending only on $E' = E_0W_0$, Q_1 and ε such that for all $\alpha \in A^+$*

$$\int_{E'K\alpha^{-1}Q_1} F(x)^{\varepsilon/2}\, dk\, da \leqq c_8\, F(\alpha)^{-\varepsilon/2}.$$

Combining (5.11), (5.15), (5.16) and Proposition 5.17, we now have

$$\int_{U_\tau} f(gK)^2\, d\mu_K \leqq c_9 \int_\Omega d\mu_K(b) \int_{A^+_{\tau+\tau_0}} v(\alpha)\, d\alpha$$

where c_9 is a suitable constant. Now recall that $\mu_K(\Omega) < \infty$ and that $\mu(U_{\tau+\tau_0}) \leqq c(\tau_0)\, \mu_K(U_\tau)$. Thus finally we may conclude that there exists a constant c such that for all $\tau > 0$

$$\int_{U_\tau} f(gK)^2\, d\mu_K \leqq c\, \mu(U_\tau).$$

By Remark 4.10 this implies that π is G-invariant, thus proving Theorem 2.4.

§ 6. A Convolution Technique

In this section we develop some techniques to compare certain integrals with respect to a measure which is invariant under a lattice with the corresponding integrals with respect to a G-invariant measure. The ideas have their origin in [8] and [18].

In the sequel we let Γ, U, U^-, P, P^-, S, S^-, N, J etc. be the same as before. Recall that their choice depended on Γ. However we now let Λ denote an arbitrary lattice in G. The results in the sequel will be applicable when we set $\Lambda = j^{-1}\Gamma j$, $j \in J$. In what follows μ' will denote a G-invariant measure on $X' = G/U$.

(6.1) **Lemma.** *Let σ be a Λ-invariant, Λ-finite (cf. § 1 for definition) measure on X'. Let V be a compact neighborhood of the identity e in G. Then there exists a constant d_1 such that for any non-negative measurable function φ on X'*

$$\int_{X'} \int_{G} \varphi(gx)\, \chi_V(g)\, dg\, d\sigma(x) \leqq d_1 \int_{X'} \varphi(x)\, d\mu'(x).$$

χ_V being the characteristic function of V and dg being a Haar measure on G.

Proof. For any $\varphi \in C_c(X')$ put

$$\nu(\varphi) = \int_{X'} \int_{G} \varphi(gx)\, \chi_V(g)\, dg\, d\sigma(x).$$

Then ν defines a (locally finite, Borel) measure on X'. It is easy to see, as in § 1, that ν is absolutely continuous with respect to μ'. To prove the Lemma we only need to show that the Radon-Nikodym derivative is bounded (a.e.).

Let σ' and ν' be the measures on G corresponding to σ and ν respectively under the correspondence introduced in Proposition 1.6. Clearly it is enough to prove that there exists a bounded measurable function ζ which is invariant under the right action of N and is a Radon-Nikodym derivative for ν' with respect to the Haar measure dg on G.

It is straightforward to verify from the definitions of σ' and ν' that for any $\psi \in C_c(G)$

$$(6.2) \quad \int_G \psi(y)\, d\nu'(y) = \int_G \int_G \psi(gy)\, \chi_V(g)\, dg\, d\sigma'(y).$$

Using right invariance of dg we get

$$\begin{aligned} (6.3) \quad \int_G \psi(y)\, d\nu'(y) &= \int_G \int_G \psi(g)\, \chi_V(gy^{-1})\, dg\, d\sigma'(y) \\ &= \int_G \int_G \psi(g)\, \chi_{V^{-1}g}(y)\, dg\, d\sigma'(y) \\ &= \int_G \psi(g)\, \sigma'(V^{-1}g)\, dg. \end{aligned}$$

In other words $\sigma(V^{-1}g)$ is a representative in the a.e. class $\frac{d\nu'}{dg}$. Also $\sigma(V^{-1}g)$ is invariant under the right action of N. We now show that $\zeta(g) = \sigma'(V^{-1}g)$ is a bounded function on G.

Let D be a compact, symmetric neighbourhood of e in G such that for any $\gamma \in \Lambda - \{e\}$ the sets D and $D\gamma$ are disjoint. Since V is compact there exist $g_1, g_2, \ldots, g_k$ in G such that $V \subset \bigcup_1^k g_i D$. Hence $V^{-1} g \subset \bigcup_1^k D g_i^{-1} g$. Hence it is enough to prove that $\sigma'(Dg)$ is a bounded function on G. Observe that for any $\gamma \in \Lambda - \{e\}$ and $g \in G$ the sets Dg and γDg are disjoint. Hence if σ'' is the measure on $\Lambda \backslash G$ corresponding to σ' then $\sigma'(Dg) \leqq \sigma''(\Lambda \backslash G)$. Since σ is Λ-finite $\sigma''(\Lambda \backslash G) < \infty$. Hence $\sigma'(Dg)$ is bounded.

(6.4) **Corollary.** *Let σ be as in Lemma* 6.1. *and let V be any compact neighbourhood of e in G. Then there exists a constant d_2 such that for any measurable subset E of X'*

$$\sigma(E) \leqq d_2 \mu'(VE).$$

Proof. Let E be any measurable subset of X'. Then for all $g \in V$ we have

$$\chi_E(x) \leqq \chi_{VE}(gx)$$

for all $x \in X$. Thus for every $g \in V$ we have

$$\sigma(E) = \int_{X'} \chi_E(x)\, d\sigma(x) \leqq \int_{X'} \chi_{VE}(gx)\, d\sigma(x).$$

Integrating both sides of the inequality over V we get

$$\sigma(E) \leqq d' \int_V \int_{X'} \chi_{VE}(gx)\, d\sigma(x)\, dg$$

where d' is the inverse of the Haar measure of V. By Lemma 6.1 (and Fubini theorem) it follows that

$$\sigma(E) \leqq d'\, d_1 \mu'(VE).$$

We now specialize to Λ-invariant measures on $X' = G/U$ for which the measure of $X' - P^- U/U$ is zero. Denote $Y = P^- U/U$. Y is the unique open orbit of P^- on X' and its complement is a union of lower dimensional manifolds. The restriction of the G-invariant measure μ' on G/U to Y, being P^--invariant, is a constant multiple of the quotient of $d_l h$ on Y. Here $d_l h$ denotes a left Haar measure on P^-. By choosing $d_l h$ appropriately we may assume, as we do, that for any measurable function φ on Y

$$(6.5) \qquad \int_{P^-} \varphi(hU)\, d_l h = \int_Y \varphi(y)\, d\mu'(y)$$

(6.6) **Lemma.** *Let Λ be any lattice in G such that $\Lambda \cap U^-$ is a lattice in U^- and let σ be a Λ-invariant measure on X' such that $\sigma(X' - Y) = 0$. Then there exists a neighbourhood Ω of the identity e in P^- and a constant d_3 such that for any non-negative measurable function φ on Y*

$$\int_Y \int_{P^-} \varphi(hy)\, \chi_\Omega(h)\, d_l h\, d\sigma(y) \leqq d_3 \int_Y \varphi(y)\, d\mu'(y).$$

Proof. Let H be the connected component of e in P^-. Then $H=U^-\cdot Z$ (semidirect product) where Z is the connected component of e in the centralizer of S in G. Let $D\subset U^-$, $F\subset Z$ and $V\subset G$ be compact neighbourhoods of e in the respective subgroups such that DF is contained in V and $V^{-1}V\cap\Lambda=\{e\}$. Put $\Omega=(DF)^{-1}$. Let ν be the measure on Y defined by

$$(6.7)\quad \int_Y \varphi(y)\,d\nu(y)=\int_Y\int_{P^-}\varphi(hy)\,\chi_\Omega(h)\,d_l h\,d\sigma(x).$$

We claim that $\nu\prec\mu'/Y$, the restiction of μ to Y. For if $E\subset Y$ is a Borel set such that $\mu'(E)=0$, then for any $y\in Y$ say $y=h_0U$

$$\begin{aligned}\int_{P^-}\chi_E(hy)\,d_l h&=\int_{P^-}\chi_E(hh_0U)\,d_l h\\&=\delta(h_0)\int_{P^-}\chi_E(hU)\,d_l h\\&=\delta(h_0)\,\mu'(E)\quad\text{(by (6.5))}\\&=0.\end{aligned}$$

Here and in the sequel δ denotes the modular homomorphism of P^-; i.e. for any integrable function θ on P^- and $h_0\in P^-$

$$(6.8)\quad \int_{P^-}\theta(hh_0)\,d_l h=\delta(h_0)\int_{P^-}\theta(h)\,d_l h.$$

To prove the Lemma it is enough to prove that the Radon-Nikodym derivative $d\nu/d\mu'$ (on Y) is bounded. Observe that the map $\tau\colon P^-\to Y$ defined by $h\to hy_0$ where $y_0\in Y$ is fixed (arbitrarily) is bijective. Therefore using τ we now identify Y with P^-. It is easy to see that under this identification the Radon-Nikodym derivative may be written as

$$(6.9)\quad \frac{d\nu}{d\mu'}(h)=\int_Y\chi_\Omega(hy^{-1})\,\delta(y)\,d\sigma(x).$$

Since $\Omega\subset H$ is compact and δ is a continuous homomorphism, there exists a constant d_4 such that whenever $hy^{-1}\in\Omega$, $\delta(y)\leqq d_4\delta(h)$. Therefore

$$\begin{aligned}(6.10)\quad \frac{d\nu}{d\mu'}(h)&\leqq d_4\,\delta(h)\int_Y\chi_\Omega(hy^{-1})\,d\sigma(y)\\&=d_4\delta(h)\,\sigma(\Omega^{-1}h).\end{aligned}$$

Recall that $\Omega^{-1}=DF$ and consider the function $\sigma(\Omega^{-1}h)=\sigma(DFh)$. Let $p_1,p_2,\ldots,p_k$ be a set of representatives for P^-/H. Then any element h of P^- can be uniquely represented as $h=u^-p_jz$ where $u^-\in U^-$, $z\in Z$ and $1\leqq j\leqq k$.

Since $\Lambda\cap U$ is a lattice in U^- there exists a relatively compact Borel subset C of U^- such that i) for $\gamma\in U^-\cap\Lambda$, $\gamma\neq e$ the sets C and γC are disjoint and ii) $U^-=\bigcup_{\gamma\in U^-\cap\Lambda}\gamma C$, that is C is a fundamental domain for $U^-\cap\Lambda$ in U^-. Recall also that any lattice in a nilpotent Lie group is uniform. Hence a relatively compact fundamental domain exists. Now let $h\in P^-$ and $h=u^-p_jz$ be the

representation as above. For each $\zeta \in F$ and $\gamma \in U^- \cap \Lambda$ denote

$$C(\gamma, \zeta) = \{\gamma^{-1} D \zeta u^- \zeta^{-1}\} \cap C$$

and for each $\gamma \in U^- \cap \Lambda$ let

$$C_\gamma = \bigcup_{\zeta \in F} C(\gamma, \zeta)\, \zeta p_j z.$$

It is easy to see that for each $\gamma \in U^- \cap \Lambda$, C_γ is a Borel subset of P^-. Clearly $\{\gamma C_\gamma\}_{\gamma \in U^- \cap \Lambda}$ is a family of pairwise disjoint sets whose union is $DFu^- p_j z$. Therefore using Λ-invariance of σ we now get

$$\begin{aligned} (6.11) \quad \sigma(\Omega^{-1} h) = \sigma(DFu^- p_j z) &= \sum_{\gamma \in U^- \cap \Lambda} \sigma(\gamma C_\gamma) \\ &= \sum_{\gamma \in U^- \cap \Lambda} \sigma(C_\gamma). \end{aligned}$$

We next claim that C_γ, $\gamma \in U^- \cap \Lambda$ are pairwise disjoint. For otherwise let $\gamma \neq \gamma'$ be such that $h_0 = u_0^- p_l z_0 \in C\gamma \cap C\gamma'$ where $u_0^- \in U^-$, $z_0 \in Z$ and $1 \leqq l \leqq k$. In other words there exist $\zeta, \zeta' \in F$, $u_1^- \in C(\gamma, \zeta)$ and $u_2^- \in C(\gamma', \zeta')$ such that

$$(6.12) \quad h_0 = u_0^- p_l z_0 = u_1^- \zeta p_j z = u_2^- \zeta' p_j z.$$

We may write $\zeta p_j = n_1^- p_j z_1$ and $\zeta' p_j = n_2^- p_j z_2$ where $n_1^-, n_2^- \in U^-$ and $z_1, z_2 \in Z$. Substituting in (6.12) and using uniqueness of the representation we get $u_1^- n_1^- = u_2^- n_2^-$ and $z_1 = z_2$. But then $\zeta' \zeta^{-1} = (\zeta' p_j)(\zeta' p_j)^{-1} = n_2^- (n_1^-)^{-1} = u_2^- (u_1^-)^{-1}$. Since $H = U^- \cdot Z$ is a semidirect product it follows that $\zeta = \zeta'$ and $u_1^- = u_2^-$. However this means that $\gamma^{-1} D \cap \gamma'^{-1} D$ is non-empty as it contains $u_1^- (\zeta u^- \zeta^{-1})^{-1}$. But since $D \subset V$ and $\{\gamma V, \gamma \in \Lambda\}$ is a family of pairwise disjoint sets we arrive at a contradiction. Therefore C_γ, $\gamma \in U^- \cap \Lambda$ are pairwise disjoint. Also each C_γ is contained in $CFp_j z$. Therefore by (6.11)

$$\sigma(\Omega^{-1} h) \leqq \sigma(CF_0 z)$$

where $F_0 = \bigcup_1^k F p_j$. Recalling the identifications and using Corollary 6.4 we conclude that

$$\begin{aligned} (6.13) \quad \sigma(\Omega^{-1} hU) &= \sigma(CF_0 zU) \\ &\leqq d_2 \mu'(VCF_0 zU). \end{aligned}$$

Recall that since Z normalizes U there is an action $z \to \psi_z$ of Z on $X' = G/U$ given by $\psi_z(gU) = gzU$. Since this action commutes with the G-action on left there exists a character $\chi\colon Z \to \mathbb{R}^+$ such that for any Borel subset E of X', $\mu'(\psi_z(E)) = \chi(z)\mu'(E)$. We claim that $\chi(z) = \delta(z)^{-1}$. To see this let Q be a compact subset of H with non-empty interior and let $E = QU/U$. Then E is a compact

subset Y with non-empty interior. Then in view of (6.5) we have

$$\chi(z)\mu'(E)=\mu'(\psi_z(E))=\mu'(QzU/U)=\int_H \chi_Q(hz^{-1})\,d_l h$$
$$=\delta(z^{-1})\int_H \chi_Q(h)\,d_l h=\delta(z^{-1})\,\mu'(E).$$

Since $0<\mu'(E)<\infty$ we must have $\chi(z)=\delta(z^{-1})$. Using (6.10) and (6.13) we now conclude that

$$(6.14)\quad \frac{dv}{d\mu'}(h)\leqq d_4 d_2\delta(h)\,\mu'(VCF_0zU/U)$$
$$=d\delta(h)\,\delta(z^{-1})$$

where $d=d_4 d_2\mu'(VCF_0U/U)$. Since VCF_0 is a relatively compact subset, d is finite. It is easy to see that the restriction of the modular homomorphism δ to U^- is identically 1. Hence $\delta(h)=\delta(u^- p_j z)=\delta(p_j)\delta(z)$. Therefore by (6.14) $dv/d\mu'\leqq d\sup_{1\leqq j\leqq k}\delta(p_j)$. This proves Lemma 6.6.

(6.15) **Corollary.** *Let σ be a Λ-invariant measure on $X=G/N$ such that $\sigma(X-P^-N/N)=0$. Then there exists a neighbourhood Ω of e in H and a constant $d_3'>0$ such that for any non-negative measurable function φ on X*

$$\int_X\int_H \varphi(hx)\,\chi_\Omega(h)\,d_l h\leqq d_3'\int_X \varphi(x)\,d\mu(x).$$

Proof. Let Ω be the neighbourhood of e in H as in Lemma 6.6. In view of the one-one correspondence of measures on X and X' with those on G introduced in Proposition 1.6 the corollary is an obvious consequence of Lemma 6.6.

§ 7. Proof of Proposition (5.14)

Recall that the function F on X involved in the integrals in Proposition 5.14 is defined by the (right) N-invariant function on G, (also denoted by F) defined by

$$F(g)=\exp -2\rho H(g)$$

where ρ and $H(g)$ are as introduced in the beginning of § 4. We now need the following alternative formulation of F.

Let $\mathfrak{G}$ be the Lie algebra of G and let $\mathfrak{G}_{\mathbb{C}}$ be its complexification. Let $\mathscr{C}$ be a Cartan subalgebra of $\mathfrak{G}_{\mathbb{C}}$ containing the Lie subalgebra $\mathfrak{A}$ of A. Let $\Delta_{\mathbb{C}}^+$ be a system of positive roots for the pair $(\mathfrak{G}_{\mathbb{C}},\mathscr{C})$ such that the set of restrictions of elements of $\Delta_{\mathbb{C}}^+$ to $\mathfrak{A}$ contains all roots of $\mathfrak{A}$ on $\mathcal{N}$, the Lie subalgebra of N. Let $\Lambda\in\mathscr{C}^*$, the dual of $\mathscr{C}$ be such that 2Λ is the sum of all roots in $\Delta_{\mathbb{C}}^+$. Then there exists a unique finite dimensional irreducible representation $d\theta$ of $\mathfrak{G}_{\mathbb{C}}$ with highest weight Λ, say on a vector space $\mathscr{L}$. Let θ be the corresponding

representation of G on $\mathscr{L}$. Then $\mathscr{L}$ is endowed with a norm $\|\cdot\|$ such that for all $g\in G$

$$\|\theta(g)v_0\| = \exp \rho H(g) \tag{7.1}$$

where v_0 is an element in the weight space corresponding to the highest weight.

Now let as before H be the connected component of e in P^- and let $\mathfrak{H}$ be the Lie subalgebra corresponding to H. Using (7.1) in this section we first prove the following result.

(7.2) **Lemma.** *Let σ be any measure on $X=G/N$. Let $\eta\in\mathfrak{H}$ be a simultaneous eigenvector of $\{\mathrm{Ad}\, a | a\in A\}$ such that $\mathrm{ad}\,\eta$ is a nilpotent transformation of $\mathfrak{G}$. Let q and $t_0>0$ be given. Let $E_\eta=\{\exp t\eta \,||t|\leqq t_0\}$. Then there exists a constant d_5 such that for any Borel subset Q of X and $\xi\in S^-$ (cf. § 2 for notation)*

$$\int_Q F(\xi^{-1}x)^q\, d\sigma(x) \leqq d_5 \int_{E_\eta Q} F(\xi^{-1}x)^q\, d\sigma_\eta(x)$$

where σ_η is the measure on X such that for all $\varphi\in C_c(X)$

$$\int_X \varphi(x)\, d\sigma_\eta(x) = \int_X \int_{\mathbb{R}} \varphi(\exp t\eta\cdot x)\, \chi_{E_\eta}(\exp t\eta)\, dt\, d\sigma(x).$$

Proof. Since η is a nilpotent element of $\mathfrak{G}$, $\tau=d\theta(\eta)$ is a nilpotent linear transformation of $\mathscr{L}$. Consequently there exists $p\in\mathbb{Z}^+$ such that

$$\theta(\exp t\eta) = \sum_{i=0}^{p-1} a_i t^i \tau^i \tag{7.3}$$

where a_i, $0\leqq i<p$ are the first p coefficients in the exponential series. For $v\in\mathscr{L}$ and $t'\in\mathbb{R}$, put

$$\begin{aligned}\mathscr{P}(v,t') &= \frac{d}{dt}\|\theta(\exp t\eta)\cdot\theta(\exp t'\eta)\, v\|\,|_{t=0} \\ &= \frac{d}{dt}\|\theta(\exp t\eta)\, v\|\,|_{t=t'}.\end{aligned}$$

We claim that for any (fixed) $v\in\mathscr{L}$ either $\mathscr{P}(v,t')=0$ for all $t'\in\mathbb{R}$ or the number of solutions of $\mathscr{P}(v,t')=0$ is bounded by $2p$. We first observe that for $v\neq 0$, $\mathscr{P}(v,t')=0$ if and only if

$$\frac{d}{dt}\|\theta(\exp t\eta)\, v\|^2\,|_{t=t'}=0.$$

Now let $\{v_j\}_1^n$ be coordinates of v with respect to an orthonormal basis of $\mathscr{L}$. Then in view of (7.1) the left hand side is a polynomial in t' whose degree is at most $2p$. Hence if $\mathscr{P}(v,t')$ is not identically zero it can have at most $2p$ solutions.

Now for any $t_1>0$ define

$$\mathscr{L}_0(t_1) = \{v\in\mathscr{L} \,|\, \mathscr{P}(v,t')=0 \text{ for some } t'\in[-t_1,t_1]\},$$
$$\mathscr{L}^+(t_1) = \{v\in\mathscr{L} \,|\, \mathscr{P}(v,0)>0\} - \mathscr{L}_0(t_1)$$

and

$$\mathscr{L}^-(t_1)=\{v\in\mathscr{L}\mid\mathscr{P}(v,0)<0\}-\mathscr{L}_0(t_1).$$

Then obviously for any $t_1>0$ the above three sets constitute a partition of $\mathscr{L}$.

(7.5) **Sub-Lemma.** *In above notation if either* $v\in\mathscr{L}^+(t_0)$ *and* $t\in[-t_0,0]$ *or* $v\in\mathscr{L}^-(t_0)$ *and* $t\in[0,t_0]$ *then*

$$\|\theta(\exp t\eta)v\|\leqq\|v\|.$$

Proof. We consider the former case; the latter can be dealt with similarly. Since $v\in\mathscr{L}^+(t_0)$ either the desired inequality holds for all $t\leqq 0$ or there exists $t'>0$ such that

(7.6) $\|\theta(\exp t\eta)v\|<\|v\|$ for $t\in(-t',0)$

and

$$\|\theta(\exp t'\eta)v\|=\|v\|.$$

By Rolle's theorem there exists $t''\in(-t',0)$ such that

$$\mathscr{P}(v,t'')=\frac{d}{dt}\|\theta(\exp t\eta)v\|\,|_{t=t''}=0.$$

Since $v\notin\mathscr{L}_0(t_0)$ we must have $-t''>t_0$. Hence $t'>-t''>t_0$ in view of (7.6) the claim is proved.

(7.7) **Corollary.** *Let* $\xi\in S^-$. *Then if either* $v\in\theta(\xi)\mathscr{L}^-(t_0)$ *and* $t\in[0,t_0]$ *or* $v\in\theta(\xi)\mathscr{L}^+(t_0)$ *and* $t\in[-t_0,0]$ *then*

$$\|\theta(\xi^{-1}\exp t\eta)v\|\leqq\|\theta(\xi^{-1})v\|.$$

Proof. Let λ be the character on A such that $(\mathrm{Ad}\,a)\eta=\lambda(a)\eta$ for all $a\in A$. Since $\eta\in\mathfrak{H}$ it follows that for any $\xi\in S^-$, $0<\lambda(\xi^{-1})\leqq 1$. Thus if $t\in[-t_0,0]$ then $\xi^{-1}(\exp t\eta)\xi=\exp t\lambda(\xi^{-1})\eta=\exp t'\eta$ for some $t'\in[-t_0,0]$. Hence by Sub-Lemma 7.5, in either of the cases we get

$$\begin{aligned}\|\theta(\xi^{-1}\exp t\eta)v\|&=\|\theta(\xi^{-1}(\exp tn)\,\xi)\,\theta(\xi^{-1})v\|\\&\leqq\|\theta(\xi^{-1})v\|.\end{aligned}$$

(7.8) **Sub-Lemma.** *Let* $t_0>0$ *be given. Then there exists a function* ζ *defined on* $\mathscr{L}_0(2t_0)$ *and a constant* $d_6>0$ *such that* i) *for* $v\neq 0$, $\zeta(v)>0$

ii) *for all* $v\in\mathscr{L}_0(t_0)$ *and* $t\in[-t_0,t_0]$, $\zeta(\theta(\exp t\eta)v)=\zeta(v)$ *and*

iii) $v\in\theta(\xi)\mathscr{L}_0(t_0)$ *where* $\xi\in S^-$ *and* $t\in[-t_0,t_0]$,

$$d_6^{-1}\zeta(\theta(\xi^{-1})v)\leqq\|\theta(\xi^{-1}\exp t\eta)v\|\leqq d_6\zeta(\theta(\xi^{-1})v).$$

Proof. On $\mathscr{L}_0(2t_0)$ consider the partition D whose elements are connected components of $\{\theta(\exp t\eta)v\mid t\in\mathbb{R}\}\cap\mathscr{L}_0(2t_0)$, $v\in\mathscr{L}_0(2t_0)$. Let $\sim$ denote the cor-

responding equivalence relation. Now for $v \in \mathscr{L}_0(2t_0)$ define

$$\zeta(v) = \inf\{\|v'\| \,|\, v' \in \mathscr{L}_0(2t_0) \text{ and } v' \sim v\}. \tag{7.9}$$

We first observe that since η is nilpotent either $\|\theta(\exp t\eta)v\| \to \infty$ as $t \to \pm\infty$ or v is a fixed point of $\theta(\exp t\eta)$, $t \in \mathbb{R}$. Hence for any $v \neq 0$, $\zeta(v) > 0$.

Now if $v \in \mathscr{L}_0(t_0)$ and $t \in [-t_0, t_0]$ then $\theta(\exp t\eta)v \in \mathscr{L}_0(2t_0)$ and $\theta(\exp t\eta)v \sim v$. Therefore ζ satisfies ii). Next let $v \in \mathscr{L}_0(t_0)$ and consider the function $\mathscr{P}(v,t)$. First suppose $\mathscr{P}(v,t) = 0$ for all $t \in \mathbb{R}$. Then $\|\theta(\exp t\eta)v\|$ is the constant function $\|v\|$. On the other hand if $\mathscr{P}(v,t)$ is not identically zero then, as observed earlier, the set of solutions of $\mathscr{P}(v,t) = 0$ has at most $2p$ elements. Consequently the "length" of the equivalence class $D(v)$ of v, defined by

$$l(v) = \operatorname{Sup}\left\{|t_2 - t_1| \,\middle|\, \begin{array}{l}\theta(\exp t_1\eta)v, \theta(\exp t_2\eta)v \in \mathscr{L}_0(2t_0) \\ \text{and } \theta(\exp t_1\eta)v \sim \theta(\exp t_2\eta)v\end{array}\right\}$$

satisfies $l(v) \leqq 8pt_0$. Therefore if we put

$$d_6 = \sup\{\|\theta(\exp t\eta)v'\|/\|v'\| \,|\, v' \in \mathscr{L}, |t| \leqq 8pt_0\}$$

then for any $v_1, v_2 \in D(v)$ we have

$$d_6^{-1}\|v_1\| \leqq \|v_2\| \leqq d_6\|v_1\|.$$

Also by compactness of $D(v)$, $\zeta(v) = \|v'\|$ for some $v' \in D(v)$. Thus we have proved iii) in Lemma 7.8 for the case when $\xi = e$. For arbitrary $\xi \in S^-$, iii) can be proved by applying the above to $\theta(\xi^{-1})v$ and observing as in Corollary 7.7 that $\{\xi^{-1}(\exp t\eta)\xi \,|\, |t| \leqq t_0\}$ is contained in $\{(\exp t\eta) \,|\, |t| \leqq t_0\}$.

We now complete the proof of Lemma 7.2. Let Q be a Borel subset of $X = G/N$. Let $v_0 \in \mathscr{L}$ be as in (7.1). Now for $\xi \in S^-$ define

$$Q_0(\xi) = \{gN \in Q \,|\, \theta(g)v_0 \in \theta(\xi)\mathscr{L}_0(t_0)\},$$
$$Q^+(\xi) = \{gN \in Q \,|\, \theta(g)v_0 \in \theta(\xi)\mathscr{L}^+(t_0)\},$$
$$Q^-(\xi) = \{gN \in Q \,|\, \theta(g)v_0 \in \theta(\xi)\mathscr{L}^-(t_0)\}.$$

Then for each ξ the above three sets form a partition of Q. We intend to prove the inequality in Lemma 7.2 by proving it separately on each of these sets.

Firstly consider $Q^+(\xi)$. Since $F(\xi^{-1}x) = \|\theta(\xi^{-1}g)v_0\|^{-2}$ in view of Corollary 7.7 for $x \in Q^+(\xi)$ and $t \in [-t_0, 0]$ we get

$$\begin{aligned} F(\xi^{-1}x) &\leqq \|\theta(\xi^{-1}(\exp t\eta)g)v_0\|^{-2} \\ &= F(\xi^{-1}(\exp t\eta)x). \end{aligned}$$

Let $E_\eta^- = \{\exp t\eta \,|\, t \in [-t_0, 0]\}$. Then for given $q > 0$ and $t \in [-t_0, 0]$, we have

$$\chi_{Q^+(\xi)}(x)F(\xi^{-1}x)^q \leqq \chi_{E_\eta^- Q}((\exp t\eta)x)F(\xi^{-1}(\exp t\eta)x)^q.$$

for all $x \in X$. Therefore

$$(7.10)\quad \int_{Q^+(\xi)} F(\xi^{-1}x)^q\,d\sigma(x)=\frac{1}{t_0}\int_{E_\eta^-} dt\int_X \chi_{Q^+(\xi)}(x)\,F(\xi^{-1}x)^q\,d\sigma(x)$$

$$\leqq\frac{1}{t_0}\int_{E_\eta^-} dt\int_X \chi_{E_\eta^- Q}((\exp t\eta)x)\,F(\xi^{-1}(\exp t\eta)x)^q\,d\sigma(x)$$

$$\leqq\frac{1}{t_0}\int_{E_\eta Q} F(\xi^{-1}x)^q\,d\sigma_\eta(x)$$

where σ_η and E_η are as in the statement of the lemma. Similarly we can prove

$$(7.11)\quad \int_{Q^-(\xi)} F(\xi^{-1}x)^q\,d\sigma(x)\leqq\frac{1}{t_0}\int_{E_\eta Q} F(\xi^{-1}x)^q\,d\sigma_\eta(x).$$

Now consider $\int_{Q_0(\xi)} F(\xi^{-1}x)^q\,d\sigma(x)$. For $x\in Q_0(\xi)$, in view of iii) in Sub-Lemma 7.8 and (7.1), for $t\in[-t_0,t_0]$ we have

$$d_6^{-2}\zeta^{-2}(\xi^{-1}x)\leqq F(\xi^{-1}(\exp t\eta)x)\leqq d_6^2\zeta^{-2}(\xi^{-1}x).$$

Hence in particular for any $t\in[-t_0,t_0]$, $x\in Q_0(\xi)$

$$F(\xi^{-1}x)^q\leqq d_6^{2q}\zeta^{-2}(\xi^{-1}x)^q\leqq d'F(\xi^{-1}(\exp t\eta)x)^q$$

where $d'=d_6^{4q}$. Hence for any $t\in[-t_0,t_0]$ and $x\in X$

$$\chi_{Q_0(\xi)}(x)\,F(\xi^{-1}x)^q\leqq d'\chi_{E_\eta Q}((\exp t\eta)x)\,F(\xi^{-1}(\exp t\eta)x)^q.$$

Integrating both sides with respect to x and $t\in[-t_0,t_0]$ we conclude that

$$(7.12)\quad \int_{Q_0(\xi)} F(\xi^{-1}x)^q\,d\sigma(x)\leqq\frac{d'}{2t_0}\int_{E_\eta Q} F(\xi^{-1}x)^q\,d\sigma_\eta(x).$$

Combining (7.10), (7.11) and (7.12) we conclude that there exists a constant d_5 (independent of ξ) such that

$$\int_Q F(\xi^{-1}x)^q\,d\sigma(x)\leqq d_5\int_{E_\eta Q} F(\xi^{-1}x)^q\,\sigma_\eta(x).$$

(7.13) **Lemma.** *Let $G=KAN$ be the Iwasawa decomposition as fixed in §2. Let M be the subgroup of K consisting of elements which centralize A. Let Σ be a compact neighbourhood of e in MA. Let σ be any measure on $X=G/N$ and let $q>0$ be given. Then there exists a constant d_7 such that for any Borel subset Q of X*

$$\int_Q F(\xi^{-1}x)^q\,d\sigma(x)\leqq d_7\int_{\Sigma Q} F(\xi^{-1}x)^q\,d\sigma_\Sigma(x)$$

where σ_Σ is the measure on X defined by putting for $\varphi\in C_c(X)$

$$\int_X \varphi(x)\,d\sigma_\Sigma(x)=\int_X\int_{MA}\varphi(zx)\,\chi_\Sigma(z)\,dz\,d\sigma(x)$$

dz being a fixed Haar measure on MA.

Proof. In the notation used earlier in the section define

$$d_8 = \mathrm{Sup}\{\|\theta(z)v\|/\|v\| \mid v \in \mathscr{L} - (0), z \in \Sigma\}.$$

Since Σ is compact d_8 is finite. Then using (7.1) for any $z \in Z$, $\xi \in S^-$ and $x = gN \in X$ we have

$$\begin{aligned} F(\xi^{-1}x) &= \|\theta(\xi^{-1}g)v_0\|^{-2} \\ &\leqq d_8^2 \|\theta(z\xi^{-1}g)v_0\|^{-2} \\ &= d_8^2 F(z\xi^{-1}x) \\ &= d_8^2 F(\xi^{-1}zx). \end{aligned}$$

For the last step recall that $S \subset A$. Now as before for the Borel set Q and $z \in \Sigma$ we can write

$$\chi_Q(x) F(\xi^{-1}x)^q \leqq d_8^{2q} \chi_{\Sigma Q}(zx) F(\xi^{-1}zx)^q \ldots \quad \text{for all } x \in X.$$

Let d_9 be the inverse of the Haar measure of Σ in MA. Then

$$\begin{aligned} \int_Q F(\xi^{-1}x)^q \, d\sigma(x) &= d_9 \int_\Sigma dz \int_X \chi_Q(x) F(\xi^{-1}x)^q \, d\sigma(x) \\ &\leqq d_9 d_8^{2q} \int_X \int_\Sigma \chi_{\Sigma Q}(zx) F(\xi^{-1}zx)^q \, dz \, d\sigma(x) \\ &= d_7 \int_{\Sigma Q} F(\xi^{-1}x)^q \, d\sigma_\Sigma(x) \end{aligned}$$

where $d_7 = d_9 d_8^{2q}$.

(7.14) **Proposition.** *Let Λ be a lattice in G such that $\Lambda \cap U^-$ is a lattice in U^-. Let σ be a Λ-invariant measure on $X = G/N$ such that $\sigma(X - Y) = 0$ where $Y = P^- N/N$. Let $q > 0$ be given. Then there exists a compact neighbourhood E of e in $H \subset P^-$ and a constant d such that for any $\xi \in S^-$ and any Borel subset Q of X*

$$\int_Q F(\xi^{-1}x)^q \, d\sigma(x) \leqq d \int_{EQ} F(\xi^{-1}x)^q \, d\mu(x).$$

Proof. Let Ω be a compact neighbourhood of e in H, (the connected component of P^-) for which Corollary 6.15 is satisfied. In the Lie subalgebra $\mathfrak{H}$ of H there exists elements $\eta_1, \eta_2, \ldots, \eta_r$, which are nilpotent elements of $\mathfrak{G}$ and such that the map $\mathscr{M}$: $\mathbb{R}^r \times MA \to H$ defined by $(t_1, t_2, \ldots, t_r, z) \mapsto (\exp t_1\eta_1) \cdot (\exp t_2\eta_2) \ldots (\exp t_r\eta_r) \cdot z$ is regular at $(0, 0, \ldots, 0, e)$. Hence there exists a neighbourhood Σ of e in MA and t_0 such that if $R = \{(t_1, t_2, \ldots, t_r, z) \| t_i| \leqq t_0$ for $1 \leqq i \leqq r$ and $z \in \Sigma\}$ and $E_i = E_{\eta_i} = \{\exp t\eta_i \mid |t_i| \leqq t_0\}$ then $\mathscr{M}/R$ is a diffeomorphism onto $E = E_1 E_2 \cdot \ldots \cdot E_r \Sigma$ and the latter is a neighbourhood of e in H. By choosing smaller Σ and t_0 we may also assume that $E \subset \Omega$.

Applying Lemma 7.13 and then Lemma 7.2 repeatedly we conclude the following: For any $q > 0$ there exists a constant d' such that for any $\xi \in S^-$ and any Borel subset Q of X

$$(7.15) \quad \int_Q F(\xi^{-1}x)^q \, d\sigma(x) \leqq d' \int_{EQ} F(\xi^{-1}x)^q \, d((\ldots(\sigma_\Sigma)_{\eta_r})_{\eta_{r-1}})_{\eta_1}.$$

Since $\mathscr{M}/R$ is a diffeomorphism onto E there exists a differentiable function ψ on E such that for any $\varphi \in C_c(H)$

$$\int_\Sigma dz \int_{E_r} dt_r \int_{E_{r-1}} \dots \int_{E_1} dt_1 \{\varphi((\exp t_1\eta_1)(\exp t_2\eta_2)\dots(\exp t_r\eta_r)z\} = \int_E \varphi(h)\psi(h)\,d_l h.$$

Thus if φ is a non-negative function on X

$$\int_X \varphi(x)\,d((\dots(\sigma_\Sigma)_{\eta_r})\dots)_{\eta_1}(x) = \int_X \int_H \varphi(hx)\,\chi_E(h)\,\psi(h)\,d_l h\,d\sigma(x)$$

$$\leqq d'' \int_X \int_H \varphi(hx)\,\chi_\Omega(h)\,d_l h\,d\sigma(x)$$

where $d'' = \sup_{h\in E} \psi(h) < \infty$. By Lemma 6.6 we further have

$$\int_X \int_H \varphi(hx)\,\chi_\Omega(h)\,d_l h\,d\sigma(x) \leqq d_3 \int_X \varphi(x)\,d\mu(x).$$

In particular for $\varphi(x) = F(\xi^{-1}x)^q \chi_{EQ}(x)$ we get

$$(7.16)\quad \int_X F(\xi^{-1}x)^q \chi_{EQ}(x)\,d((\dots(\sigma_\Sigma)_{\eta_r}))\dots)_{\eta_1} \leqq d''\,d_3 \int_X F(\xi^{-1}x)^q \chi_{EQ}(x)\,d\mu.$$

Combining (7.15) and (7.16) we conclude that for any $\xi \in S^-$

$$\int_Q F(\xi^{-1}x)^q\,d\sigma(x) \leqq d \int_{EQ} F(\xi^{-1}x)^q\,d\mu(x)$$

where $d = d'd''d_3$. This proves Proposition.

Proof of Proposition (5.14). Recall that for each $j \in J$, $j^{-1}\Gamma j \cap U^-$ is a lattice in U^-. Further the measure π_j defined by (5.13) is $j^{-1}\Gamma j$-invariant and $\pi_j(X - Y) = 0$. Thus for each $j \in J$ there exists a compact neighbourhood E_j of e in H, and a constant $d_j > 0$ such that for any $\xi \in S^-$ and any Borel subset Q of X

$$\int_Q F(\xi^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\pi_j(x) \leqq d_j \int_{E_jQ} F(\xi^{-1}x)^{1+\frac{\varepsilon}{2}}\,d\mu(x).$$

Since J is finite, putting $E = \bigcup_{j\in J} E_j$ and $c_6 = \max_{j\in J} d_j$ we get the Proposition.

Part III. Classification of Invariant Measures

We now consider the general case. In the sequel T denotes a (connected) reductive Lie group. This means that the adjoint representation is completely reducible. The adjoint group T^* admits a direct product decomposition $T^* = C \cdot G$ wher C and G are normal (and hence semisimple) subgroups of T^*, C is compact and G is a product of noncompact simple Lie groups. We denote by ρ: $T \to G$ the canonical projection homomorphism of T onto G. There exists in T a unique maximum normal semisimple analytic (connected) subgroup G_0

without compact factors. The restriction of ρ to G_0 is a covering of G. Any horospherical subgroup U of T (see Introduction for definition) is contained in G_0.

Let U be a maximal horospherical subgroup of T, and let Γ be a lattice in T. In § 8 we first show that any ergodic U-invariant measure on T/Γ is concentrated on a translate of a closed orbit of a reductive subgroup L of T containing G_0 such that the L-invariant measure on the orbit is U-ergodic. This reduces the task of proving Theorem A only in the special case when the action of U on T/Γ is ergodic with respect to the Haar measure. Then using Proposition 1.12 and the duality principle (Corollary 1.9) in § 9 we reduce the problem to the special case of semisimple groups with trivial center and without compact factors. In § 10 we show that if the measure (or more precisely the dual of it) fails to satisfy the condition of Theorem 2.4 then it is concentrated on a closed orbit of a proper subgroup. We are then able to produce an inductive argument to complete the proof of Theorem A.

§ 8. Ergodicity of Haar Measure

(8.1) **Lemma.** *Let H be any Lie group (not necessarily reducitve) and Λ be a lattice in H. Let V be a normal analytic subgroup of H. Then there exists an analytic subgroup L of H such that*

(i) *L is normalized by Λ, $L\Lambda$ is a closed subgroup and $L\cap\Lambda$ is a lattice in L.*

ii) *L contains V and the action of V on $L/L\cap\Lambda$ is ergodic with respect to the L-invariant measure on $L/L\cap\Lambda$.*

Proof. We construct inductively, a decreasing sequence $\{L_i\}_0^\infty$ of analytic subgroups of H such that for each i, L_i contains V, L_i is normalized by Λ and $L_i\Lambda$ is closed. Put $L_0=H$ and suppose $L_0, L_1, \ldots, L_{s-1}$ are constructed satisfying the above conditions. Let L_s be the connected component of the identity in $cl(V\Lambda_s)$ where $\Lambda_s=L_{s-1}\cap\Lambda$. Then L_s contains V and is normalized by Λ. We show that $L_s\Lambda$ is closed. Let $x_\alpha\gamma_\alpha$ be a net in $L_s\Lambda$ converging to $y\in H$. Since $L_s\subset L_{s-1}$ and $L_{s-1}\Lambda$ is closed (by induction hypothesis) y has the form $x\gamma$ where $x\in L_{s-1}$ and $\gamma\in\Lambda$. Hence $x_\alpha(\gamma_\alpha\gamma^{-1})\to x\in L_{s-1}$. But $L_{s-1}\Lambda$ being closed, is a discrete union of left cosets of certain elements of Λ. Therefore $\gamma_\alpha\gamma^{-1}$ must eventually belong to L_{s-1}. Choosing a subnet we may assume that for all α, $\gamma_\alpha\gamma^{-1}\in\Lambda\cap L_{s-1}=\Lambda_s$. But clearly $L_s\Lambda_s$ is closed. Hence $x_\alpha(\gamma_\alpha\gamma^{-1})$ converges in $L_s\Lambda_s$. Thus $x_\alpha\gamma_\alpha$ converges in $L_s\Lambda$. Hence $L_s\Lambda$ is closed.

Now put $L=\bigcap_{i=0}^{\infty} L_i$. Since $\{L_i\}_0^\infty$ is a decreasing sequence of analytic subgroups there exists $j\geqq 0$ such that $L=L_j$. It follows that L is normalized by Λ and that $L\Lambda$ is closed. By Lemma 1.14 $L\cap\Lambda$ is a lattice in L.

Clearly L contains V. Further since $L=L_j=L_{j+1}$ we conclude that $V(L\cap\Lambda)$ is dense in L. The ergodicity of the action of V on $L/L\cap\Lambda$ is now a consequence of the following Lemma (due to Mostow) which we separate out because of its usefulness.

(8.2) **Lemma** (Mostow). *Let H be a locally compact second countable group and let Λ be a closed subgroup such that H/Λ admits a finite H-invariant measure, say σ. Let V be a normal subgroup H such that $V\Lambda$ is dense in H. Then the action of V on H/Λ is ergodic with respect to σ.*

Proof. Let $f\in\mathscr{L}^2(H/\Lambda,\sigma)$ such that $f(gx\Lambda)=f(x\Lambda)$ a.e. σ for all $g\in V$. Let $\tilde{f}$ be the function on H defined by $\tilde{f}(x)=f(x\Lambda)$. For any $g\in V$ and $\gamma\in\Lambda$ we have

$$\tilde{f}(xg\gamma)=\tilde{f}((xgx^{-1}x\gamma)=f((xgx^{-1}x\Lambda)=f(x\Lambda)=\tilde{f}(x) \tag{8.3}$$

a.e. on $V\times H$.

But observe that $\tilde{f}$ is a locally integrable function on H. Also the action of H (on right) on the space of locally integrable functions on H is continuous when the latter space is endowed with the topology induced by local seminorms. Since $V\Lambda$ is dense (8.3) now implies that $\tilde{f}$ is a constant function. Hence so is f. Therefore the action of V on $H/H\cap\Lambda$ is ergodic with respect to σ.

We now return to the earlier notation – as introduced in the introduction to Part III.

(8.3) **Proposition.** *There exists a reductive analytic subgroup L of T containing G_0 such that*

i) *L is normalized by Γ, $L\Gamma$ is closed and $L\cap\Gamma$ is a lattice in L.*

ii) *The action of U on $L/L\cap\Gamma$ is ergodic with respect to the L-invariant measure.*

iii) *If π is any U-invariant ergodic measure on T/Γ then there exists $t\in T$ such that $\pi(T/\Gamma-tL\Gamma/\Gamma)=0$.*

Proof. By Lemma 8.1 there exists an alaytic subgroup L containing G_0 for which i) is satisfied, and the action of G_0 on $L/L\cap\Gamma$ is ergodic. We also observe that any analytic subgroup containing G_0 is reductive.

The assertion ii) is a consequence of Calvin Moore's ergodicity theorem. Observe that if p_i is a canonical projection of G_0 onto any simple factor of the adjoint group then $\mathrm{cl}(p_i(U))$ is noncompact. The ergodicity theorem asserts that any subgroup satisfying the above has the following property (cf. Theorem 1, [15] and also [4]). Let τ be a unitary representation of G_0 on a separable Hilbert space $\mathscr{H}$. If $\psi\in\mathscr{H}$ is such that $\tau(u)\psi=\psi$ for all $u\in U$ then $\tau(g)\psi=\psi$ for all $g\in G_0$. Applying the result to the unitary representation corresponding to the action of G_0 on $L/L\cap\Gamma$, we get that if $f\in\mathscr{L}^2(L/L\cap\Gamma)$ such that $f(ux)=f(x)$ a.e. for all $u\in U$ then $f(gx)=f(x)$ a.e. for all $g\in G_0$. Since the action of G_0 is ergodic such an f must be a constant function; i.e. the action of U on $L/L\cap\Gamma$ is ergodic.

We now prove iii). Since $L\Gamma$ is a closed subgroup of T the partition of T/Γ into $\{xL\Gamma/\Gamma|x\in T\}$ is countably separated; i.e. there exists a countable family $\{E_j\}$ of Borel subsets such that each E_j is a union of certain elements of the partition and given any two distinct elements of the partition there exists j such that E_j contains one of them but not the other. Since L contains G_0 which is a normal subgroup containing U any element of the above partition is U-invariant. Therefore if π is a U-invariant ergodic measure on T/Γ there exists $t\in T$ such that $\pi(T/\Gamma-tL\Gamma/\Gamma)=0$.

(8.4) *Remark.* In view of Proposition 8.3, Theorem A needs to be proved only under the additional hypothesis that the action of U on T/Γ is ergodic with respect to the T-invariant measure. For suppose this is done. Let π be a U-invariant ergodic measure on T/Γ. Let L be an analytic subgroup and $t\in T$ be as obtained in Proposition 8.3. Let π' be the measure on $L/L\cap\Gamma\simeq L\Gamma/\Gamma$ defined by $\pi'(E)=\pi(tE)$ for any Borel subset E of $L\Gamma/\Gamma$. Then π' is a $t^{-1}Ut$-invariant measure on $L/L\cap\Gamma$. Thus we get a subgroup H of L and $x\in L$ such that $H(L\cap\Gamma)$ is closed and π' is the xHx^{-1}-invariant measure supported on $xH(L\cap\Gamma)/L\cap\Gamma\simeq xH\Gamma/\Gamma$. It follows that π is the $txHx^{-1}t^{-1}$ invariant measure supported on $txH\Gamma/\Gamma$.

§ 9. Reduction to Semisimple Groups

In this section we reduce proving Theorem A to the special case when T is a semisimple group without compact factors. Recall that G is the product of all noncompact simple normal subgroups of the adjoint group of T and $\rho: T\to G$ is the canonical projection of T onto G. Let $\bar{\rho}: T/\Gamma\to G/\rho(\Gamma)$ be the map defined by $\bar{\rho}(t\Gamma)=\rho(t)\,\rho(\Gamma)$ for all $t\in T$.

(9.1) **Lemma.** *$\rho(\Gamma)$ is a lattice in G and $\bar{\rho}$ is proper.*

Proof. Let L denote the connected component of the identity in $\ker\rho$. Also let R be the radical of T. Then R is also the radical of L and L/R is compact. Now by a theorem of L. Auslander (cf. Theorem 8.24, [16]) the connected component A of the identity in $\mathrm{cl}\,R\Gamma$ is solvable. Since L/R is compact AL is a closed subgroup. Further AL is normalized by Γ and $AL\Gamma$ is a closed subgroup. Since Γ is a lattice in T it follows easily that the subgroup $A\Gamma L/L$ has the Selberg property (A subgroup H of T is said to have Selberg property if for any neighborhood Ω of the identity in T and any $g\in T$ there exists $\gamma\in\Gamma$ and $n\in\mathbb{Z}^+$ such that $g^n\in\Omega\gamma\Omega^{-1}$). Therefore by Borel's density theorem (cf. Theorem 5.5, [16], see also [9]) $A\Gamma L/L$ is a Zariski dense subgroup of T/L. In particular it follows that the connected component AL/L is normal in T/L. But since A is solvable and T/L is semisimple this means that $AL=L$ or that A is contained in L. Consequently $L\Gamma/L$ is a discrete subgroup. Further by Lemma 1.14 it follows that $L\Gamma/L$ is a lattice in T/L. Next notice that $\ker\rho/L$ is a central subgroup of T/L. Therefore again because of Borel's density theorem $(\ker\rho)\,\Gamma/L$ is discrete (cf. Corollary 5.17, [16]). Thus $\rho(\Gamma)$ is discrete and indeed a lattice.

To prove that $\bar{\rho}$ is proper we proceed as follow: Since A is a closed subgroup (notation as above) normalized by Γ and $A\Gamma$ is closed it follows that $A\cap\Gamma$ is a lattice in A. Since A is solvable it follows from a result of Mostow (cf. Corollary 3.5 and Theorem 2.1, [16]) that $A/A\cap\Gamma$ is compact. Since L/A is compact it follows that $L/L\cap\Gamma$ is compact. Now the same argument as above shows that $(\ker\rho)\cap\Gamma$ is a lattice in $\ker\rho$. Since each orbit of L on $\ker\rho/(\ker\rho)\cap\Gamma$ is open it follows that there are only finitely many orbits. Also each orbit, being topologically isomorphic to $L/L\cap\Gamma$ is compact. Hence $\ker\rho/(\ker\rho)\cap\Gamma$ is compact. Hence $\bar{\rho}$ is proper.

(9.2) **Lemma.** *Let Q be an analytic group. Let U' be a normal analytic subgroup of Q such that Q/U' is reductive. Let U be an analytic subgroup containing U' such that U/U' is a maximal horospherical subgroup of Q/U'. Let V be the smallest normal analytic subgroup of Q containing U. Then the restriction of any ergodic measure-preserving action of V to U is ergodic.*

Proof. Since U' is a normal subgroup contained in U it is enough to show that any ergodic action of V/U' restricts to an ergodic action of U/U'. But V/U' is clearly the maximum normal semisimple subgroup without compact factors in Q/U'. Also U/U' is a maximal horospherical subgroup in V/U'. The desired result now follows from C.C. Moore's ergodicity in the same way as argued in Proposition 8.3, ii).

(9.3) **Proposition.** *Let π be a U-invariant ergodic measure on T/Γ. Suppose that there exists an analytic subgroup H of G and $g_0 \in G$ such that*

a) *H contains $N = \rho(U)$. There exists an analytic subgroup N' of N which is normal in H and H/N' is reductive.*

b) *$H \cap g_0 \rho(\Gamma) g_0^{-1}$ is a lattice in H, $Hg_0\rho(\Gamma)$ is closed and $\operatorname{supp} \bar{\rho}\pi = Hg_0\rho(\Gamma)/\rho(\Gamma)$.*

c) *$\bar{\rho}\pi$ is H-invariant.*

Then there exists an analytic subgroup L of $\rho^{-1}(H)$ and $t_0 \in T$ such that

i) *$L \cap t_0 \Gamma t_0^{-1}$ is a lattice in L, $Lt_0\Gamma$ is closed and $\operatorname{supp} \pi = Lt_0\Gamma/\Gamma$ and*

ii) *L contains U and π is L-invariant.*

Proof. Let $t_1 \in \rho^{-1}(g_0)$. Put $Q = \rho^{-1}(H)$ and $\Lambda = Q \cap t_1 \Gamma t_1^{-1}$. Since $g_0\rho(\Gamma)g_0^{-1} \cap H$ is a lattice in H and $\bar{\rho}$ is proper it follows that Λ is a lattice in Q. Also Λ is the isotropy subgroup for the action of Q at the point $t_1\Gamma$. Since $\bar{\rho}\pi$ is supported on $Hg_0\rho(\Gamma)/\rho(\Gamma)$ it follows that the support of π is contained in $Qt_0\Gamma/\Gamma \simeq Q/\Lambda$. Therefore π may be visualized as a U-invariant, ergodic measure on Q/Λ.

Recall that the restriction of ρ to G_0 is a covering of G. Hence there exists a unique analytic subgroup U' of U such that the restriction of ρ to U' is an isomorphism of U' onto N'. Further it is clear from Lie algebra considerations that U' is normal in Q and that Q/U' is reductive. Also U/U' is a maximal horospherical subgroup in Q/U'.

Now let V be the smallest normal analytic subgroup of Q containing U. By Lemma 8.1 there exists an analytic subgroup L' of Q such that L' is normalized by Λ, $L'\Lambda$ is a closed subgroup, $L' \cap \Lambda$ is a lattice in L', L' contains V and the action of V on $L'/L' \cap \Lambda$ is ergodic. By Lemma 9.2 it follows that the action of U on $L'/L' \cap \Lambda$ is ergodic. Also since $\{tL'\Lambda \mid t \in Q\}$ is a countably separated partition as in the proof of Proposition 8.3 we conclude that there exists $t_2 \in Q$ such that the U-invariant ergodic measure is supported on $t_2 L'\Lambda/\Lambda$. Under the canonical isomorphism of Q/Λ with $Qt_1\Gamma/\Gamma$ the supporting set corresponds to $t_2 L' t_1 \Gamma/\Gamma$. Now put $L = t_2 L' t_2^{-1}$ and $t_0 = t_2 t_1$. Then π is supported on $Lt_0\Gamma/\Gamma \simeq L/\Delta$ where $\Delta = L \cap t_0 \Gamma t_0^{-1}$ is a lattice in L. Observe that U is contained in L and the action of U on L/Δ is ergodic. This is because ergodicity of the action of a subgroup on a homogeneous space depends only on the conjugacy class of the subgroup.

We also observe that the restriction of ρ to L maps L onto H. For, since $\bar{\rho}$ is proper from considerations of supports of π and $\bar{\rho}\pi$ we can deduce that $\rho(Lt_0\Gamma) = Hg_0\rho(\Gamma)$. In other words $\rho(L)$ acts transitively on the H-orbit $Hg_0\rho(\Gamma)/\rho(\Gamma) \simeq H/H \cap g_0\rho(\Gamma)g_0^{-1}$. However since $\rho(\Gamma)$ is a discrete subgroup of G and $\rho(L)$ is analytic this is possible only if $\rho(L)=H$.

Let $\rho' : L/\Delta \to H/\rho(\Delta)$ be the map induced by the restriction of ρ to L. Clearly ρ' is a proper map. Now we may visualize π as a U-invariant ergodic measure on L/Δ, such that $\rho'\pi$ is a H-invariant measure and we need to prove that π is L-invariant. Since ρ' is proper we can decompose π along the fibers of ρ': There exists a family $\{\pi_p\}_{p\in H/\rho(\Delta)}$ where π_p is a probability measure on $\rho'^{-1}(p)$ such that for any $\varphi \in C_c(L/\Delta)$

$$\int_{L/\Delta} \varphi(x)\, d\pi(x) = \int_{H/\rho(\Delta)} d\rho'\pi(p) \int_{\rho'^{-1}(p)} \varphi(x)\, d\pi_p(x).$$

Now let $R = (\ker \rho) \cap L$. Let dr be a Haar measure on R. For any $\theta \in C_c^+(R)$ let π_θ be the measure defined by

$$\int_{L/\Delta} \varphi(x)\, d\pi_\theta(x) = \int_{L/\Delta} \int_R \varphi(rx)\, \theta(r)\, dr\, d\pi(x)$$

for all $\varphi \in C_c(L/\Delta)$. Since the actions of the subgroups R and U commute it follows that each π_θ is U-invariant. We claim that for all $\theta \in C_c(R)$, π_θ is absolutely continuous with respect to the L-invariant measure on L/Δ. Note that R acts transitively on each $\rho'^{-1}(p)$. Since $\rho'\pi$ is the H-invariant measure, in view of Fubini-Weil formula for invariant measures on homogeneous spaces it is enough to assert that for each $p \in H/\rho(\Delta)$ and $\theta \in C_c^+(R)$ the measure $(\pi_p)_\theta$ defined by

$$\int_{\rho'^{-1}(p)} \varphi(x)\, d(\pi_p)_\theta = \int_{\rho'^{-1}(p)} \int_R \varphi(rx)\, \theta(r)\, dr\, d\pi_p(x)$$

where $\varphi \in C_c(\rho'^{-1}(p))$ is absolutely continuous with respect to the R-invariant measure on $\rho'^{-1}(p)$. But this obviously follows in the same way as in the proof of Lemma 6.6.

Now recall the one-to-one correspondence between the U-invariant measures on L/Δ and Δ-invariant measures on L/U established in Corollary 1.9. (Notice that since L contains a lattice it is necessarily unimodular.) Let ω and ω_θ, $\theta \in C_c^+(R)$ denote the Δ invariant measures on L/U corresponding to π and π_θ, $\theta \in C_c^+(R)$ respectively. It is straightforward to check that for any $\theta \in C_c^+(R)$ and any Borel subset E of L/U

$$\omega_\theta(E) = \int_R \varphi(\psi_r^{-1} E)\, \theta(r)\, dr$$

where $r \to \psi_r$ is the action of R on L/U on right defined by $\psi_r(xU) = xrU$ (it is well-defined since R normalizes U). Since the L-invariant measure on L/Δ is ergodic for the action of U it follows that the L-invariant measure on L/U is ergodic for the action of Δ. Since for each $\theta \in C_c^+(R)$, π_θ is absolutely continuous

with respect to the L-invariant measure, each ω_θ is absolutely continuous with respect to the L-invariant measure on L/U. Therefore it now follows from Proposition 1.12 that ω is a L-invariant measure on L/U. Hence π is a L-invariant measure on L/Δ.

§ 10. Applications

Using the characterization of Haar measure obtained in earlier sections we now determine all ergodic invariant measures of the 'horospherical flow' on G/Γ where G is a semisimple analytic group such that $\mathbb{R}$-rank of each factor of G is 1. A generalization of the result – without the condition on rank of the factors – involves study of orbits of the horospherical flow and will be considered later.

(10.1) **Theorem.** *Let G be a semisimple analytic group with trivial center and such that $\mathbb{R}$-rank of each factor of G is 1. (In particular G has no compact factors.) Let Γ be a lattice in G. Let U be a maximal horospherical subgroup of G. Let π be an ergodic U-invariant finite measure on G/Γ. Then there exists an analytic subgroup L of G and $g_0 \in G$ such that*

i) *$Lg_0\Gamma$ is closed, $L \cap g_0 \Gamma g_0^{-1}$ is a lattice in L and* $\operatorname{supp} \pi = Lg_0\Gamma/\Gamma$, *and*

ii) *L contains U and π is L-invariant. Also*

iii) *There exists an analytic subgroup U' of U which is normal in L and such that L/U' is reductive.*

(10.2) *Remark.* If there exists a maximal horospherical subgroup of G for which the result holds then it holds for all maximal horospherical subgroups.

Proof of Remark. Recall that all maximal horospherical subgroups are conjugate. Assume the theorem to be true for U and let $U' = gUg^{-1}$ where $g \in G$. Let π' be a U'-invariant ergodic measure on G/Γ. Let π be the measure defined by $\pi(E) = \pi'(gE)$ for any Borel set E. Then π is a N-invariant ergodic measure on G/Γ. If L is an analytic subgroup and $g_0 \in G$ be such that i)–iii) are satisfied for N and π then $L' = gLg^{-1}$ and $g_0' = gg_0$ clearly have the desired properties with respect to N' and π'.

(10.3) **Lemma.** *Let G be as in Theorem* 10.1 *and let Λ be any lattice in G. Let $G = \prod_{i \in I} G_i$ be the decomposition such that $\Lambda_i = \Lambda \cap G_i$ is an irreducible lattice in G_i (cf. Proposition* 2.1). *Let $i \in I$ be such that G_i/Λ_i is non-compact. Then G_i is a simple Lie group and in particular $\mathbb{R}$-rank of G_i is* 1.

Proof. Let $G_i = \prod_{j=1}^{l} H_j$ be the decomposition of G_i into simple factors and if possible let $l > 1$. Let $U \neq (e)$ be any Λ_i-rational horospherical subgroup of G_i. Also U can be expressed as $\prod_{j=1}^{l} U_j$ where U_j, $1 \leqq j \leqq l$ is a horospherical subgroup H_j. Since $\mathbb{R}$-rank of each H_j is 1, U_j is either the identity subgroup (e) or a maximal horospherical subgroup in H_j. If there exists j such that $U_j = (e)$ then we get a normal subgroup F of G_i containing U and such that $(e) \subset F \subset G_i$. However

since Λ_i is an irreducible lattice in G_i no proper subgroup contains a non-trivial element of Λ_i (cf. Lemma 1.5 [13]). Thus we conclude that any non-trivial Λ_i-rational horospherical subgroup U is a maximal horospherical subgroup of G_i.

However recall that by arithmeticity theorem (cf. [13]) of Margulis, G_i admits a $\mathbb{Q}$-structure such that Λ_i is an arithmetic lattice. For that $\mathbb{Q}$-structure the conclusion of last paragraph implies that $\mathbb{Q}$-rank G_i is same as the $\mathbb{R}$-rank of G_i. Since $\mathbb{Q}$-rank (being equal to the maximum length of an increasing chain of non-trivial Λ_i-rational *non-trivial* horospherical subgroups) is clearly 1 so is the $\mathbb{R}$-rank. Since each H_j is non-compact $\mathbb{R}$-rank of each H_j is positive. This gives a contradiction unless $l=1$.

(10.4) *Remark.* Let G and Γ be as in the statement of Theorem 10.1. Let G be decomposed as $G=G_0\cdot G_1$ where G_0 is the maximum normal Lie subgroup of G such that $G_0\cap\Gamma$ is a uniform lattice in G (product of all G_i, $i\in I_0$ (in the notation of §2) and G_1 is such that $G_1\cap\Gamma$ is a (non-uniform) lattice in G_1. Let U, U^-, P, P^-, S, A, N etc. be as in §2. Then in view of Lemma 10.3 we have the following:

i) $U=G_1\cap N$, and U is a maximal horospherical subgroup in G_1

ii) $P=G_0\cdot(P\cap G_1)$; $P^-=G_0\cdot(P^-\cap G_1)$. $P\cap G_1$ and $P^-\cap G_1$ are minimal $\mathbb{R}$-parabolic subgroups of G_1 and we have Langlands decompositions (cf. [19], pp. 75) as

$$P\cap G_1=M\cdot(A\cap G_1)\cdot U$$

and

$$P^-\cap G_1=M\cdot(A\cap G_1)\cdot U^-$$

where M is the centralizer of $A\cap G_1$ in a maximal compact subgroup G_1.

iii) $$P^-U=G_0\cdot(P^-\cap G_1)U=(G_0\cdot M\cdot(A\cap G_1)\cdot U^-)U$$
$$=U^-(G_0\cdot M(A\cap G_1)\cdot U)=U^-P.$$

iv) Using the Bruhat decomposition of G_1 with respect to $A\cap G_1$ we have a Bruhat decomposition of G as

$$(10.4)\quad G=G_0\cdot G_1=G_0\bigcup_{\omega\in W}U\omega(P\cap G_1)\quad\text{(disjoint union)}$$
$$=\bigcup_{\omega\in W}U\omega P$$

where W is the Weyl group of G_1 with respect to $A\cap G_1$.

v) There exists a unique $\omega_0\in W$ such that $\omega_0^{-1}U\omega_0=U^-$. Also since $\mathbb{R}$-rank of each factor of G is 1 it follows that for $\omega\neq\omega_0$, $(\omega^{-1}U\omega)P$ is contained in a proper parabolic subgroup say $Q(\omega)$ of G. Thus by (10.4) we have

$$(10.5)\quad G=\bigcup_{\omega\in W}\omega(\omega^{-1}U\omega)P$$
$$=\omega_0U^-P\cup\bigcup_{\omega\neq\omega_0}\omega Q(\omega).$$

We may also note that each $Q(\omega)$, $\omega\neq\omega_0$ is of the form $L\cdot P'$ where L is a normal (semisimple) Lie subgroup of G and P' is a minimal parabolic subgroup

in a semisimple normal Lie subgroup L' of G such that $G = L \cdot L'$. Indeed $P' = P \cap L'$.

Proof of Theorem (10.1). We proceed by induction on the dimension of G (satisfying the hypothesis of the theorem).

In view of Remark 10.2 it is also enough to prove the theorem for the maximal horospherical subgroup N as chosen in §2. Let π be a finite, N-invariant ergodic measure on G/Γ and σ be the Γ-invariant measure on G/N corresponding to π (cf. Corollary 1.3). Let $X = G/N$ and $X_0 = \bigcup_{j \in J} jP^- U/N$ with notations as in §2. If $\sigma(X - X_0) = 0$ then by Theorem 2.4, σ is G-invariant. Hence so is π. Then clearly G itself has the desired properties.

Now consider the complementary possibility viz. there exists $j \in J$ such that $\sigma(X - jP^- U/N) > 0$. In view of (10.5) it follows that there exists $\omega \neq \omega_0$ such that $\sigma(j\omega_0^{-1} \omega Q(\omega)/N) > 0$. Since σ is ergodic we conclude that

$$\sigma(X - \Gamma j \omega_0^{-1} \omega Q(\omega)/N) = 0.$$

Put $p = j\omega_0^{-1} \omega$ and $\Gamma' = p^{-1} \Gamma p$. Then we have

$$\sigma(X - p\Gamma' Q(\omega)/N) = 0. \tag{10.6}$$

Recall from Remark 10.4, v) that $Q(\omega) = L \cdot (P \cap L')$ where L and L' are normal subgroup of G such that $G = L \cdot L'$ (direct product). Let $P \cap L' = M' A' N'$ be a Langlands decomposition of $P \cap L'$ in L' where $A' = A \cap L'$ and $N' = N \cap L'$. Put $Q' = LM'N'$. We claim that a) $\Gamma' \cap Q(\omega)$ is contained and is a lattice in Q' and b) $\Gamma' Q'$ is closed.

Firstly consider $\Gamma' \cap N'$. It is clear from the definition of L' that ω centralizes L'. Hence $\Gamma' \cap N' = (j\omega_0^{-1} \omega)^{-1} \Gamma (j\omega_0^{-1} \omega) \cap N' = \omega_0 j^{-1} \Gamma j \omega_0^{-1} \cap N'$. By choice of J for each $j \in J$, $j^{-1} \Gamma j \cap U^-$ is a lattice in U^- (cf. Proposition 2.3). Hence $\omega_0 j^{-1} \Gamma j \omega_0^{-1} \cap U$ is a lattice in U. Since by Lemma 10.3 for any lattice Λ, $(\Lambda \cap L) \cdot (\Lambda \cap L')$ is a subgroup of finite index in Λ and N' is the L'-component of U, the above implies that $\omega_0 j^{-1} \Gamma j \omega_0^{-1} \cap N'$ is a lattice in N'. Since M' is compact we may further conclude that $\Gamma' \cap M'N'$ is a lattice in $M'N'$.

By Lemma 10.3 in particular $L\Gamma'$ is closed. Since $\Gamma' \cap M'N'$ is a (necessarily uniform) lattice in $M'N'$, straightforward verification now shows that $\Gamma' Q' = \Gamma' LM'N'$ is closed. Further clearly $\Gamma' \cap Q'$ is lattice in Q'. To complete the claim we only need to show that $\Gamma' \cap Q(\omega) = \Gamma' \cap Q'$. Suppose this is not true. Then again in view of Lemma 10.3 there exists a simple normal subgroup V of L' such that $V \cap \Gamma'$ is a lattice in V, and $V \cap Q(\omega) \cap \Gamma' \supsetneq V \cap Q' \cap \Gamma'$. Repeating the earlier argument we see that $V \cap Q' \cap \Gamma'$ is a lattice $V \cap Q'$. Since $\mathbb{R}$-rank of each simple factor of G is 1, $V \cap Q'$ is of codimension one in $V \cap Q(\omega)$. Hence if $V \cap Q(\omega) \cap \Gamma' \supsetneq V \cap Q' \cap \Gamma'$ it follows that $V \cap Q(\omega) \cap \Gamma'$ is a lattice in $V \cap Q(\omega)$. However this is a contradiction since $V \cap Q(\omega)$ is clearly not unimodular. Hence $\Gamma' \cap Q(\omega) = \Gamma' \cap Q'$.

We now return to the measure σ. Observe that $\Gamma p Q(\omega) = p\Gamma' Q' A'$. Since A' normalizes N we have

$$\Gamma p Q(\omega)/N = \{\Gamma p \Gamma' Q'/N\} A'.$$

Since $p\Gamma' Q'$ is closed it follows that $\{p\Gamma' Q' aN/N \mid a \in A'\}$ is a countably separated partition of $p\Gamma' Q(\omega)/N$, each element of which is invariant under the Γ-action (on left). Since σ is a Γ-invariant ergodic measure it now follows that there exists $a \in A'$ such that

$$\sigma(X - p\Gamma' Q' aN/N) = 0$$

i.e. $\sigma(X - \Gamma q\, Q'/N) = 0$ where $q = pa$. Under the one-to-one correspondence of Γ-invariant measures on G/N and N-invariant measures on G/Γ the last assertion corresponds to the following: $\pi(G/\Gamma - Q'\, q\Gamma/\Gamma) = 0$.

Recall that $q^{-1}\Gamma q = a^{-1}\Gamma' a$ intersects Q' as well as N' in lattices in respective subgroups. Therefore $\Lambda = \{q^{-1}\Gamma q \cap Q'\}\, N'/N'$ is a lattice $Q'/N' = T$ (say). Let π' be the measure on T/Λ defined by $\pi'(E) = \pi(\bar{\eta}^{-1}E)$ where $\bar{\eta}$: $Q' q^{-1}\Gamma/\Gamma \simeq Q/q^{-1}\Gamma q \cap Q' \to T/\Lambda$ is induced by the projection homomorphism $\eta: Q' \to Q'/N'$. Since $N' \cap \Gamma'$ is a lattice in N', η is a proper map. Hence π is a regular measure on T/Λ. Clearly π' is invariant under the action of $U' = \eta(N)$ and is ergodic. It is easy to verify that U' is a maximal horospherical subgroup in T. Therefore by induction hypothesis and Proposition 9.3 there exists an analytic subgroup H of T and $t_0 \in T$ such that assertions i)–iii) are satisfied (for appropriate subgroups). Put $L = \eta^{-1}(H)$ and $g_0 = tq^{-1}$ where $t \in \eta^{-1}(t_0)$. Then it is evident that assertions i)–iii) of Theorem 10.1 are satisfied.

Combining Theorem 10.1 and Proposition 9.3 we conclude Theorem A.

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Singularites rationnelles et deformations

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Le but de cet article est d'étudier le comportement, par déformations, des singularités rationnelles et d'invariants numériques de singularités isolées.

Les schémas considérés seront toujours supposés de type fini sur un corps k de caractéristique zéro.

On établit les résultats suivants:

– Toute déformation d'une singularité rationnelle est une singularité rationnelle (Théorème IV).

– Soit $X \xrightarrow{f} S$ un morphisme plat dont la base et les fibres sont à singularités rationnelles. Alors X est à singularités rationnelles (Théorème V).

– Soit $X \xrightarrow{f} S$ un morphisme plat de dimension relative d, à lieu singulier fini et à fibres normales. Pour tout point s de S soit $X'_s \xrightarrow{\varphi_s} X_s$ une désingularisation de la fibre X_s. On définit les caractéristiques d'Euler-Poincaré partielles suivantes:

$$\chi_i(X_s) = \sum_{j=0}^{d-i-1} (-1)^j \lg_{k(s)} R^{d-j-i} \varphi_{s*} O_{X'_s} \qquad i \in [1, d-1].$$

En particulier on a:

$$\chi_1(X_s) = \lg R^{d-1} \varphi_{s*} O_{X'_s} - \lg R^{d-2} \varphi_{s*} O_{X'_s} + \cdots + (-1)^{d-2} \lg R^1 \varphi_{s*} O_{X'_s},$$

$$\chi_{d-1}(X_s) = \lg R^1 \varphi_{s*} O_{X'_s}.$$

Ces nombres sont indépendants de la résolution $X'_s \xrightarrow{\varphi_s} X_s$ choisie et on démontre que les fonctions $s \to \chi_i(X_s)$ sont semi continues supérieurement sur S (Théorème I).

I. Notations – Quelques rappels

Si S est un schéma on désigne par O_S son faisceau structural, par $O_{S,s}$, l'anneau local d'un point s de S, par $k(s)$ le corps résiduel en ce point. Si $\mathscr{F}$ est un faisceau de O_S modules on note $\mathscr{F}_s$ le $O_{S,s}$ module $\mathscr{F} \underset{O_s}{\otimes} O_{S,s}$ et X_s la fibre $X \underset{S}{\times} \operatorname{Spec} k(s)$ d'un

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S-schéma X. On identifie quand cela est utile un faisceau $\mathscr{F}$ à un complexe dont l'unique terme non nul est $\mathscr{F}$ placé en degré 0 et si $K^\cdot$ est un complexe on désigne par $K^\cdot[n]$, $n\in\mathbb{Z}$, le complexe obtenu à partir de $K^\cdot$ par translation de n vers la gauche. On écrira souvent K^i pour $H^i(K^\cdot)$. Si $X\xrightarrow{f}Y$ est un morphisme de schémas, on note $Lf^*: D^+(Y)\to D^+(X)$ le morphisme dérivé de f^*, ([H]).

Complexe dualisant

Soit $X\xrightarrow{f}S$ un morphisme de k-schémas de type fini. On peut définir dans $D^+(X)$ un complexe dualisant relatif noté $\omega^\cdot_{X/S}$ ($f^!(O_S)$ avec les notations de [H]), dont on utilisera la propriété suivante:

Soit $X'\xrightarrow{\varphi}X$ un S-morphisme projectif. Pour tout faisceau quasi-cohérent $\mathscr{F}$ sur X' on a dans $D^+(X)$

$$R\varphi_*\,\underline{R\mathrm{Hom}}_{X'}(\mathscr{F},\omega^\cdot_{X'/S})=\underline{R\mathrm{Hom}}_X(R\varphi_*\mathscr{F},\omega^\cdot_{X/S})$$

([H], chap. III). Si X est plongé dans un schéma Z lisse sur S, de dimension relative n, et si $\omega_{Z/S}$ désigne la puissance extérieure maximale du module des différentielles relatives de Z sur S on a: $\omega^\cdot_{X/S}=\underline{R\mathrm{Hom}}_Z(O_X,\omega_{Z/S}[n])$.

Supposons maintenant X et S quasi projectifs et f plat équidimensionnel de dimension d. On peut alors représenter $\omega^\cdot_{X/S}$ par un complexe borné de O_X modules cohérents et S plats concentré en degré $[-d,0]$ (i.e. $\omega^\cdot_{X/S}$ est de Tor dimension finie sur S, d'amplitude $[-d,0]$ [SGA VI]). Son homologie est concentrée en degré $-d$ si et seulement si f est de Cohen-Macaulay. On désigne alors par $\omega_{X/S}$ cet unique faisceau d'homologie c'est-à-dire qu'on a:

$$\omega^\cdot_{X/S}=\omega_{X/S}[d].$$

Pour tout morphisme $T\xrightarrow{h}S$ le complexe dualisant relatif de $X\underset{S}{\times}T$ sur T est égal à $Lh^*\omega^\cdot_{X/S}$. Lorsque S est le spectre d'un corps, on écrit $\omega^\cdot_X$ pour $\omega^\cdot_{X/S}$. Dans ce cas et plus généralement lorsque S est Gorenstein, $\omega^\cdot_{X/S}$ est un objet dualisant de $D^+(X)$.

II. Désingularisation et complexe Dualisant

Soit X un k-schéma de type fini. On dit que $X'\xrightarrow{\varphi}X$ est une désingularisation de X si φ est un morphisme projectif birationnel et si X' est lisse sur k. (On suppose toujours qu'un ouvert dense de X est réduit.) Supposons X équidimensionnel de dimension d. Le théorème de dualité ([H. chap. III]) permet d'écrire dans $D^+(X)$:

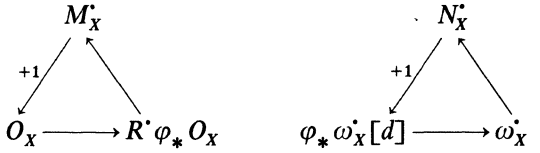
$$R^\cdot\varphi_*\,\omega_{X'}[d]=\underline{R\mathrm{Hom}}_X(R^\cdot\varphi_*O_{X'},\omega^\cdot_X).$$

Les complexes $R^\cdot\varphi_*O_{X'}$, et $R^\cdot\varphi_X\omega_X$, ne dépendent que de X et non de la résolution choisie $X'\xrightarrow{\varphi}X$ [Hi]. De plus Grauert et Riemenschneider ont montré [G.R.] que $R^i\varphi_*\omega_{X'}=0$, $\forall i>0$.

On a donc: $\varphi_* \omega_{X'}[d] = \underline{R\mathrm{Hom}}_X(R^\cdot \varphi_* O_{X'}, \omega_X^\cdot)$.

Soit $j: \varphi_* \omega_{X'}[d] \to \omega_X^\cdot$, la flèche duale de l'application naturelle $i: O_X \to R^\cdot \varphi_* O_{X'}$ et soit $M_X^\cdot$ (resp. $N_X^\cdot$) le troisième sommet du triangle construit sur i (resp. j).

On a:



On déduit du Théorème de Dualité $N_X^\cdot = \underline{R\mathrm{Hom}}_X(M_X^\cdot, \omega_X^\cdot)$ et, puisque $\omega_X^\cdot$ est dualisant, $M_X^\cdot = \underline{R\mathrm{Hom}}_X(N_X^\cdot, \omega_X^\cdot)$. L'homologie de ces complexes est à support dans le lieu singulier de X. On a $N_X^i = 0$ pour $i < -d$ ou $i > -\mathrm{prof}\, X$, $N_X^i = \omega_X^i$ $i \in [-d+1, -\mathrm{prof}\, X]$ et $N_X^{-d} = \mathrm{coker}\, \varphi_* \omega_{X'} \to \omega_X^{-d}$. $M_X^i = 0$ pour $i < -1$ ou $i > d-1$, $M_X^{-1} = \ker O_X \to O_{X_{\mathrm{red}}}$, $M_X^0 = \tilde{O}_{X/O_X}$ où $O_{\tilde{X}}$ est la normalisation de O_X, $M_X^i = R^i \varphi_* O_{X'}$ pour $i > 0$ [1].

Quelques exemples

1) Singularités rationnelles

La définition suivante est due à Kempf [T.E., chap. I § 3].

Définition. On dit que X a une singularité rationnelle en un point x (ou que x est une singularité rationnelle de X) si l'une des deux propriétés équivalentes suivantes est vérifiée:

a) $M_{X,x}^\cdot = 0$,

b) $N_{X,x}^\cdot = 0$.

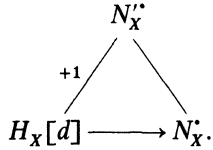
Ceci implique en particulier que l'anneau local $O_{X,x}$ est normal et Cohen-Macaulay.

L'ensemble des singularités rationnelles de X est un ouvert de X. On dira que X est à singularités rationnelles s'il en est ainsi en tout point.

2) Supposons X de Cohen-Macaulay

L'homologie de $N_X^\cdot$ est alors concentrée en degré $-d$ et est égale à $H_X = \mathrm{coker}\, (\varphi_* \omega_{X'} \to \omega_X)$. On a alors $M_X^i = \mathrm{Ext}^{i+1}(H_X, \omega_X)$, $\forall i \geqq 0$.

De manière générale, on désignera par H_X le conoyau $\varphi_* \omega_{X'} \to \omega_X^{-d}$ c'est-à-dire N_X^{-d} et par $N_X'^\cdot$ le sommet du triangle:



Ce dernier complexe est connu sans références a une désingularisation de X.

On en déduit par application de $\underline{R\mathrm{Hom}}_X(.,\omega_X^\cdot)$ une longue suite exacte:

$$0\to \mathrm{Ext}^0(N'^\cdot,\omega_X^\cdot)\to M_X^{-1}\to \mathrm{Ext}^0(H,\omega_X^\cdot)\to \mathrm{Ext}^1(N'^\cdot,\omega_X^\cdot)\to M_X^0,$$
$$\mathrm{Ext}^1(H_X,\omega_X^\cdot)\to\cdots\to M_X^{d-1}\to \mathrm{Ext}^d(H_X,\omega_X)\to 0$$

($\mathrm{Ext}^0(M,\omega_X^\cdot)$ est toujours nul).

Dans le cas particulier où X présente une singularité isolée x tous les complexes considérés ont une homologie à support dans ce point. Si I est une enveloppe injective du corps résiduel de $O_{X,x}$ on a pour tout faisceau $\mathscr{F}$ à support x:

$$\underline{R\mathrm{Hom}}_X(\mathscr{F},\omega_X^\cdot)=\underline{\mathrm{Ext}}^d(\mathscr{F},\omega_X^\cdot)=\underline{\mathrm{Hom}}_{O_X}(\mathscr{F},I).$$

Et la suite exacte précédente se réduit à:

$$M_X^i=\mathrm{Hom}(N'^{-i-1},I)\qquad i\in[-1,d-2]$$
$$M_X^{d-1}=\mathrm{Hom}(H_X,I).$$

3) Lorsque X est Cohen-Macaulay en dehors d'un fermé F fini sur k, les faisceaux $\omega_X^i=N_X'^i$ sont, pour $i>-d$, de longueur finie sur k et on peut définir les caractéristiques d'Euler-Poincaré partielles suivantes:

$$\chi_i(X)=\sum_{j\geqq 0}(-1)^j \lg_k N_X'^{i+j-d}\qquad i\in[1,d].$$

Si de plus les singularités de X sont à support dans F on définit

$$\psi_i(X)=\sum_{j=0}^{d-i}(-1)^j \lg_k M_X^{d-j-i-1}\qquad i\in[0,d]$$

et

$$\chi_0(X)=\lg_X H_X-\chi_1(X).$$

Des dualités énoncées au paragraphe précédent il résulte que

$$\lg_k N_X^{-i-1}=\lg_k M_X^i$$

et donc

$$\chi_i(X)=\psi_i(X)\qquad \forall i\in[0,d].$$

On démontrera dans la suite (Théorème I) que les différentes fonctions introduites ici varient semi-continûment supérieurement si X varie dans une famille plate.

III. Déformations de singularités isolées. Semi-Continuité des caractéristiques d'Euler-Poincaré

Lemme 1. *Soient R un anneau de valuation discrète essentiellement de type fini sur k, s le point fermé de* Spec R *et $X\xrightarrow{f}S=$* Spec R *un morphisme plat equidimensionnel de dimension d. On désigne par j: $X_s\hookrightarrow X$ l'inclusion naturelle. Soient $X'\xrightarrow{\varphi}X$ une désingularisation de X et $Z\xrightarrow{\psi}X_s$ une désingularisation de X_s. Alors*

i) *il existe une flèche injective naturelle:* $\psi_* \omega_Z \to \varphi_* \omega_{X'_s}$ *telle que le diagramme*

$$\begin{array}{ccc} \psi_* \omega_Z[d] & \longrightarrow & \varphi_* \omega_{X'_s}[d] \\ & \searrow \quad \swarrow & \\ & \omega^{\cdot}_{X_s} & \end{array}$$

soit commutatif;

ii) *il existe dans* $D^+(X_s)$ *un morphisme naturel* $\gamma: N^{\cdot}_{X_s} \to Lj^* N'^{\cdot}_X$ *tel que le sommet du triangle construit sur* γ *ait un unique terme d'homologie en degré* $-d-1$.

Puisque S est local régulier $\omega^{\cdot}_X$ (resp. $\omega^{\cdot}_{X'}$) est égal à $\omega^{\cdot}_{X/S}$ (resp. $\omega^{\cdot}_{X'/S}$) et $\omega^{\cdot}_{X_s} = Lj^* \omega^{\cdot}_X$. Si t est une uniformisante de R on a une suite exacte $0 \to \omega_{X'} \xrightarrow{t} \omega_{X'} \to \omega_{X'_s} \to 0$ et par [G.R] $R^{\cdot}\varphi_* \omega_{X'_s} = \varphi_* \omega_{X'_s} = \varphi_* \omega_{X'} | t\varphi_* \omega_{X'}$.

Ce module est indépendant de la résolution X' de X choisie. Quitte à modifier X' on peut supposer que le transformé strict de X_s dans X' est une désingularisation de X_s $[H_i]$, et donc se limiter au cas où Z est une réunion de composantes irréductibles de X'_s.

ω_Z est alors un sous-faisceau de $\omega_{X'_s}$ défini par $\omega_Z = \underline{\mathrm{Hom}}_{X'_s}(O_Z, \omega_{X'_s})$ et le lemme résulte de cette inclusion $\omega_Z \hookrightarrow \omega_{X'_s}$.

Soit $X \xrightarrow{f} S$ un morphisme plat équidimensionnel de dimension relative d de k-schémas quasi-projectifs. On dit que $X' \xrightarrow{\varphi} X$ est une désingularisation simultanée de X sur S si φ est projectif si $f \circ \varphi$ est plat et si pour tout s dans S, X'_s est une désingularisation de X_s. On étend à cette situation les définitions du paragraphe précédent:

On a $R^i \varphi_* (\omega_{X'/S} \underset{O_S}{\otimes} k(s)) = 0 \ \forall i > 0, \forall s \in S$ [G.R] et donc $R^i \varphi_* \omega_{X'/S} = 0 \ \forall i > 0$ et $\varphi_* (\omega_{X'/S} \underset{O_S}{\otimes} k(s)) = \varphi_* \omega_{X'/S} \underset{O_S}{\otimes} k(s)$. Le Théorème de Dualité appliqué à φ s'écrit donc:

$$\varphi_* \omega_{X'/S}[d] = \underline{R\mathrm{Hom}}_X(R^{\cdot}\varphi_* O_{X'}, \omega^{\cdot}_{X/S}).$$

On désigne par $M^{\cdot}_{X/S}$ le troisième sommet du triangle construit sur $i: O_X \to R^{\cdot}\varphi_* O_{X'}$ et on en déduit par application de $\underline{R\mathrm{Hom}}_X(., \omega^{\cdot}_{X/S})$ le triangle

$$\begin{array}{ccc} & N^{\cdot}_{X/S} & \\ & \overset{+1}{\swarrow} \quad \nwarrow & \\ \varphi_* \omega_{X'/S}[d] & \longrightarrow & \omega^{\cdot}_{X/S} \end{array}$$

où $N^{\cdot}_{X/S} = \underline{R\mathrm{Hom}}_X(M^{\cdot}_{X/S}, \omega^{\cdot}_{X/S})$.

Si j_s est l'inclusion $X_s \hookrightarrow X$ on a

$$L^*_{j_s} M^{\cdot}_{X/S} = M^{\cdot}_{X_s}, \qquad L^*_{j_s} N^{\cdot}_{X/S} = N^{\cdot}_{X_s} \quad [1].$$

Lemme 2. *Sous les hypothèses précédentes, il existe sur* X *un complexe borné de* O_X *modules cohérents et* S *plats* $\mathscr{M}^{\cdot}$ *(resp.* $\mathscr{N}^{\cdot}$*) concentré en degré* $[-1, d-1]$ *(resp.* $[-d, 0]$*) tel que pour tout* s *dans* S *le complexe* $\mathscr{M}^{\cdot} \underset{O_S}{\otimes} k(s)$ *(resp.* $\mathscr{N}^{\cdot} \underset{O_S}{\otimes} k(s)$*) soit quasi-isomorphe à* $M^{\cdot}_{X_s}$ *(resp.* $N^{\cdot}_{X_s}$*).*

En effet les objets $M^{\cdot}_{X/S}$ et $N^{\cdot}_{X/S}$ sont comme O_X, $\omega^{\cdot}_{X/S}$, $\varphi_* \omega_{X'/S}$ et $R^{\cdot}\varphi_* O_{X'}$ de Tor-dimension finie sur S et il est clair que l'amplitude est celle indiquée dans l'énoncé.

Lemme 3. *Sous les hypothèses précédentes supposons de plus f à lieu singulier relatif fini. Les fonctions $s \to \lg_{k(s)} N^i_{X_s}$ $i \in [-d, 0]$ et $s \to \chi_i(X_s)$ $i \in [0, d]$ sont semi continues supérieurement sur S (c'est-à-dire constructible et ne pouvant que croitre par spécialisation). La fonction $\chi_0(X_s)$ est constante.*

Les groupes d'homologie et caractéristiques d'Euler-Poincaré considérés sont égaux à ceux de $Rf_* N^{\cdot}_{X/S}$ qui est quasi isomorphe à un complexe borné de O_S-modules localement libres de type fini.

Théorème 1. *Soit $f: X \to S$ un morphisme plat equidimensionnel de dimension relative d de k-schémas quasi-projectifs.*

1) *Supposons f de Cohen-Macaulay en dehors d'un fermé F fini sur S alors les fonctions $s \to \lg_{k(s)} \omega^i_{X_s}$ $i \in [-d+1, 0]$ et $s \to \chi_i(X_s)$ $i \in [1, d]$ sont semi-continues supérieurement sur S.*

2) *Supposons X lisse sur S en dehors de F alors les fonctions*

$s \to \lg_{k(s)} N^i_{X_s}$ $\quad i \in [-d, 0]$,

$s \to \chi_i(X_s)$ $\quad i \in [0, d]$

sont semi-continues supérieurement sur S.

1) Quitte à localiser S pour la topologie étale et X au voisinage de F on peut supposer f affine, de sorte que le foncteur Rf_* conserve l'homologie des complexes. Le complexe $Rf_* \omega^{\cdot}_{X/S}$ est de Tor-dimension finie d'amplitude $[-d, 0]$ et son homologie en degré $\geqq -d+1$ est cohérente sur S. Il existe donc un complexe $L^{\cdot}$: $0 \to L^{-d} \to L^{-d+1} \to \cdots L^0 \to 0$ de O_S-modules plats, quasi isomorphe à $Rf_* \omega^{\cdot}_{X/S}$, tel que L^i soit cohérent pour tout $i > -d+1$ et dont l'homologie est universellement ω^i_X [SGA VI, chap. I, 5]. L'assertion 1 en résulte.

2) Supposons S réduit et soit X' une désingularisation de l'espace total X. Il existe un ouvert lisse de S au-dessus duquel X' est une résolution simultanée de X et la semi continuité des fonctions considérées sur cet ouvert est établie dans le lemme précédent. Il suffit donc pour compléter la démonstration du théorème de montrer que les χ_i sont des fonctions croissantes par spécialisation. Le théorème est donc conséquence du lemme suivant:

Lemme 4. *Soit $S = \operatorname{Spec} R$ le spectre d'un anneau de valuation discrète essentiellement de type fini sur k de point fermé s et de point générique η. Soit $X \xrightarrow{f} S$ un morphisme plat de type fini, de dimension d, et à lieu singulier relatif fini. On a alors:*

$$\lg_{k(s)} N^i_{X_s} \geqq \lg_{k(\eta)} N^i_{X_\eta} \quad \forall i \geqq -d,$$

$$\chi_i(X_s) \geqq \chi_i(X_\eta) \quad \forall i \geqq 0.$$

C'est une conséquence facile du lemme 1:

Désignons par j_s (resp. j_η) l'inclusion $X_s \to X$ (resp. $X_\eta \to X$). Les modules d'homologie et caractéristiques d'Euler-Poincaré considérées sont égales à celles des complexes $R^{\cdot} f_{\eta *} N^{\cdot}_{X_\eta} = (R^{\cdot} f_* N^{\cdot}_X)_\eta$ et $R^{\cdot} f_* N^{\cdot}_{X_s}$. $R^{\cdot} f_* N^{\cdot}_X$ est quasi-isomorphe à un complexe parfait de S modules et d'après le lemme 1 il existe un morphisme

canonique $Rf_* N_{X_s}^{\cdot} \to Rf_* Lj_s^* N_X^{\cdot}$ qui induit un isomorphisme sur l'homologie en degré $i > -d$ et une surjection en degré $-d$.

Corollaire 1. *Soit A une k algèbre locale essentiellement de type fini équidimensionnelle de dimension d. Pour que A admette une déformation lisse il faut que l'on ait $\chi_i(\operatorname{Spec} A) \geqq 0$, $i \in [0, d]$.*

Pour qu'il existe une déformation lisse de A admettant une résolution simultanée il faut que l'on ait $\chi_0(A) = 0$. En particulier si A est de plus Cohen-Macaulay il faut que A soit à singularités rationnelles.

C'est clair.

Exemples. Soient X une variété projective lisse, $\mathscr{L}$ un faisceau très ample sur X et $C = \operatorname{Spec} \bigoplus_{n \geqq 0} H^0(\mathscr{L}^{\otimes n})$. Soit $V = \operatorname{Spec} \operatorname{Sym} \mathscr{L}$ le fibré vectoriel sur X associé à $\mathscr{L}$ le morphisme naturel $\varphi\colon V \to C$ est une désingularisation de C. On a: $R^i \varphi_* O_v = \bigoplus_{n \geqq 0} H^i(\mathscr{L}^{\otimes n})$. Le lemme précédent donne des conditions nécessaires sur la dimension de ces H^i pour que C admette une déformation lisse. Soit par exemple E une courbe lisse de genre $g \geqq 1$, $\mathscr{E}$ un faisceau localement libre de rang 2 sur E et prenons $X = \mathbb{P}(\mathscr{E}) \xrightarrow{\pi} E$, le fibré projectif sur X associé à $\mathscr{E}$. On a $H^1(O_X) = H^1(O_E) \neq 0$. Soit $O_X(1)$ le faisceau inversible canonique sur X et $\mathscr{H}$ un faisceau suffisamment ample sur E. Soit $\mathscr{L}$ un multiple suffisamment grand de $f^* \mathscr{H} \otimes O_X(1)$ pour que $\mathscr{L}$ soit très ample sur X. Le cône $C = \operatorname{Spec} \bigoplus_{n \geqq 0} H^0(\mathscr{L}^{\otimes n})$ n'est pas lissifiable. En effet on a $H^2(X, \mathscr{L}^{\otimes n}) = H^2(E, \pi_* \mathscr{L}^{\otimes n}) = 0 \; \forall n$ et $\bigoplus_{n \geqq 0} H^1(X, \mathscr{L}^{\otimes n}) \neq 0$ puisque $H^1(O_X)$ est lui-même non nul. On a donc $\lg_k R^2 \varphi_* O_V - \lg_k R^1 \varphi_* O_V < 0$.

IV. Déformations de singularités rationnelles

Théorème 2. *Soit $X \xrightarrow{f} S$ un morphisme plat de k-schémas de type fini. Soit s un point lisse de S et x une singularité rationnelle de X_s. Alors X a une singularité rationnelle en x.*

L'idée de la démonstration est voisine de celle du lemme 1.

Quitte à se restreindre à un voisinage de x on peut supposer X (normal et) Cohen-Macaulay. Il reste à montrer que $H_{X,x}$ est nul c'est-à-dire que si $X' \xrightarrow{\varphi} X$ est une désingularisation de X l'homomorphisme $\varphi_* \omega_{X'} \to \omega_X$ est surjectif.

On peut supposer que S est le spectre d'un anneau R local régulier de dimension n et raisonner par récurrence sur n. Soit donc t un paramètre régulier de R et supposons que $X_t = X \times_S \operatorname{Spec} R/tR$ a une singularité rationnelle en x. Soit $X' \xrightarrow{\varphi} X$ une désingularisation de X, $X'_t = X \times_S \operatorname{Spec} R/tR$ est un diviseur de X'. Quitte à modifier X' on peut supposer que la composante irréductible de X'_t qui s'envoie birationnellement sur X_t est une désingularisation de X_t [Hi]. Soit X'_1 cette composante. On a sur X (resp. X') les suites exactes:

$$0 \to \omega_X \xrightarrow{t} \omega_X \to \omega_{X_t} \to 0$$

(resp. $0 \to \omega_{X'} \xrightarrow{t} \omega_{X'} \to \omega_{X'_t} \to 0$).

D'autre part ω_{X_i} est un sous-faisceau de ω_{X_t} défini par

$\omega_{X_i} = \mathrm{Hom}_{OX_t}(O_{X_i}, \omega_{X_t})$.

Par application de φ_* on obtient le diagramme commutatif suivant

$$\begin{array}{ccccccccc} & & & & & & \varphi_*\omega_{X_i} & & \\ & & & & & \swarrow & & & \\ 0 & \longrightarrow & \varphi_*\omega_{X'} & \xrightarrow{t} & \varphi_*\omega_{X'} & \longrightarrow & \varphi_*\omega_{X_t} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \omega_X & \xrightarrow{t} & \omega_X & \longrightarrow & \omega_{X_t} & \longrightarrow & 0 \end{array}$$

dans lequel la 1ère suite horizontale est exacte par [G.R.].

Par hypothèse $\varphi_*\omega_{X_i} \to \omega_{X_t}$ est un isomorphisme au voisinage de x. Les applications $\varphi_*\omega_{X_t} \to \omega_{X_t}$ et $\varphi_*\omega_{X'} \to \omega_X$ sont donc surjectives au voisinage de x.

Théorème 3. *Soit $f: X \to S$ un morphisme plat de k-schémas de type fini, s un point lisse de S et x un point de X_s. Supposons que X admette une résolution simultanée sur S au voisinage de x. Alors x est une singularité rationnelle de X si et seulement si c'est une singularité rationnelle de X_S.*

Compte-tenu du théorème 2 il suffit de démontrer le »seulement si«.

Soit $X' \xrightarrow{\varphi} X$ une résolution simultanée. La lissité de S en s implique qu'on a au voisinage de s $\omega_{X'} = \omega_{X'/S}$ et $\omega_X = \omega_{X/S}$.

Le diagramme suivant est commutatif

$$\begin{array}{ccccccc} \varphi_*\omega_{X'} & \longrightarrow & \varphi_*\omega_{X_S} = \varphi_*\omega_{X'} \underset{O_S}{\otimes} k(s) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \\ \omega_X & \longrightarrow & \omega_{X_S} = \omega_X \underset{O_S}{\otimes} k(s) & \longrightarrow & 0. \end{array}$$

Il est clair que les deux flèches verticales sont simultanément des isomorphismes au voisinage de x.

Théorème 4. *Soit $X \xrightarrow{f} S$ un morphisme plat de k-schémas de type fini. L'ensemble des points x de X où $X_{f(x)}$ est à singularité rationnelle est ouvert dans X.*

Soit $\mathcal{U}$ l'ensemble de points considéré. Il résulte du théorème 2 que $\mathcal{U}$ est stable par générisation. En effet soit x un point de $\mathcal{U}$ d'image s et x' une générisation de x d'image s' dans S. On peut supposer s' différent de s. Soit R un anneau de valuation discrète essentiellement de type fini de $k(s')$ dominant l'anneau local de s dans l'adhérence de s'. Il résulte du théorème 2 après le changement de base $\mathrm{Spec}\, R \to S$ que x' est une singularité rationnelle de $X_{S'}$.

De plus $\mathcal{U}$ est constructible. En effet il existe un ouvert V non vide de S lisse sur k (quitte à réduire S) au-dessus duquel X admet une résolution simultanée.

$\mathcal{U} \cap f^{-1}(V)$ est l'ensemble des singularités rationnelles de $f^{-1}(V)$ d'après le théorème 3, et est donc ouvert.

Théorème 5. *Avec les notations précédentes supposons que s soit une singularité rationnelle de S et soit x une singularité rationnelle de X_s. Alors x est une singularité rationnelle de X.*

Quitte à se restreindre à un voisinage de s et de x on peut supposer grâce au théorème précédent que S et les fibres de f sont à singularités rationnelles.

On montre qu'il en est de même de l'espace total X.

Soit $S' \xrightarrow{\varphi} S$ une désingularisation de S, $X' = X \underset{S}{\times} S'$ et $\psi\colon Y \to X'$ une désingularisation de X' (donc de X). Soit:

$$\begin{array}{ccccc} X & \xleftarrow{\varphi'} & X' & \xleftarrow{\psi} & Y \\ \Big\downarrow f & & \Big\downarrow f' & & \\ S & \xleftarrow{\varphi} & S' & & \end{array}$$

le diagramme obtenu.

D'après le théorème 2, X' est à singularités rationnelles, on a donc $R^{\cdot}\psi_{*}\mathcal{O}_{Y} = \mathcal{O}_{X'}$.

Par hypothèse sur S, $R^{\cdot}\varphi_{*}\mathcal{O}_{S'} = \mathcal{O}_{S}$ et on en déduit par changement de base plat $R^{\cdot}\varphi'_{*}\mathcal{O}_{X'} = \mathcal{O}_{X}$ et donc $R^{\cdot}(\varphi' \circ \psi)_{*}\mathcal{O}_{Y} = \mathcal{O}_{X}$.

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Hybrid Bounds for Dirichlet *L*-Functions

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1. Introduction

Let χ be a character (mod q) and let $L(s,\chi)$ be the corresponding Dirichlet L-function. In this paper we consider the order of magnitude of $L(s,\chi)$ along the critical line $\mathrm{Re}(s)=\frac{1}{2}$. The trivial bound in this context is

$$L(\tfrac{1}{2}+it,\chi)\ll(qT)^{1/4}, \tag{1}$$

where, as later, $T=|t|+1$. The estimate (1) follows, for example, from Lemma 1. Burgess [2] has given bounds for $L(s,\chi)$ that are sharper than (1) with respect to q; although he does not give the dependence on T explicitly, it is clear that his method yields

$$L(\tfrac{1}{2}+it,\chi)\ll q^{3/16+\varepsilon}T, \tag{2}$$

for any $\varepsilon>0$. This estimate has had many applications, for example to sharpenings of the Brun-Titchmarsh theorem on primes in arithmetic progressions.

Burgess' bound (2) is weaker than the trivial bound (1) for $q\leqq T^{12}$. However there is an alternative method which improves upon (1) for sufficiently large T; one treats the q-dependence trivially and applies van der Corput's method to sums of the type

$$\sum_n (nq+r)^{-s}.$$

This leads to the bound

$$L(\tfrac{1}{2}+it,\chi)\ll q^{1/2}T^{1/6}(\log qT). \tag{3}$$

The estimate (3) is weaker than (1) for $q\geqq T^{1/3}$; thus neither (2) nor (3) improves upon (1) for the intermediate range $T^{1/3}<q<T^{12}$. The object of this paper is to prove the following estimates which, together, supersede (1), (2) and (3) for all values of q and T.

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Theorem 1.

$$L(\tfrac{1}{2}+it,\chi) \ll d(q)\, q^{3/16}\, T^{1/4} (\log q)^2.$$

Theorem 2. *Let $q_0 | q$. Then*

$$L(\tfrac{1}{2}+it,\chi) \ll (d(q))^3 \,(q_0^{1/2} + (q/q_0)^{1/4} (\log qT)^{1/2} + (qT)^{1/6})(\log qT).$$

Here as elsewhere, $d(q)$ denotes the number of divisors of q. We note that (2) is superseded by Theorem 1, and (3) by Theorem 2 with $q_0 = 1$. The corollary below improves upon (1) for all values of q and T.

Corollary 1.

$$L(\tfrac{1}{2}+it,\chi) \ll (qT)^{1/5} (\log q)^2.$$

From Theorem 2 we may deduce an estimate without the factor $(d(q))^3$.

Corollary 2.

$$L(\tfrac{1}{2}+it,\chi) \ll (q^{1/4} (\log q)^{1/2} + (d(q))^3\, (qT)^{1/6})(\log qT).$$

We also note that Theorem 2 trivially contains the following result.

Corollary 3. *Let $\mathscr{S}$ be a set with the property that for all $q \in \mathscr{S}$ there exists $f_q | q$ such that*

$$(\log f_q)/(\log q) \to \tfrac{1}{3}$$

as $q \to \infty$ through members of $\mathscr{S}$. Then for any $\varepsilon > 0$ we have

$$L(\tfrac{1}{2}+it,\chi) \ll (qT)^{1/6+\varepsilon}$$

for all $q \in \mathscr{S}$, where the implied constant depends on $\mathscr{S}$ and ε only.

Corollary 3 improves upon Theorem 4 of Fujii, Gallagher and Montgomery [3], in which the set $\mathscr{S}$ is restricted to the products of powers of a given finite set of primes. However in our result one may take $\mathscr{S}$ to be the set of cubes, 1^3, 2^3, $3^3, \ldots$, or the set of factorials, $1!, 2!, 3!, \ldots$, for example.

Theorem 1 is proved by a simple modification of Burgess' method. The proof of Theorem 2, which constitutes the major part of this paper, is motivated by van der Corput's method; indeed the use of the factor q_0 of q may be looked on as a q-analogue of the technique. In the course of the work the sums

$$S(q;\chi, h, n) = \sum_{m=0}^{q-1} \chi(m+h)\, \bar{\chi}(m)\, e(mn/q)$$

arise; here we employ the usual notation $e(x) = \exp(2\pi i x)$. The estimation of this sum is the main difficulty in the proof. Although the treatment is largely elementary we have to apply estimates, due to Weil [5], for the Kloostermann sum and for $S(p;\chi,h,n)$; both these bounds depend on the Riemann hypothesis for curves over finite fields.

2. Preliminaries

In this section we show how $L(\frac{1}{2}+it,\chi)$ may be estimated in terms of Dirichlet polynomials. We remark at the outset that it suffices to consider the case of primitive χ. For if χ is induced by a primitive character $\chi_1 (\mathrm{mod}\, q_1)$, then $q_1 | q$ and

$$L(s,\chi)=L(s,\chi_1)\prod_p (1-\chi_1(p)p^{-s}),$$

where the product is over certain prime factors p of q/q_1. Since, for any $\varepsilon>0$,

$$\prod_p (1-\chi_1(p)p^{-s}) \ll (q/q_1)^{\varepsilon}, \quad (\mathrm{Re}(s)=\tfrac{1}{2}),$$

we see that the estimates required for $L(\frac{1}{2}+it,\chi)$ by Theorems 1 and 2 follow from the corresponding bounds for $L(\frac{1}{2}+it,\chi_1)$. Moreover, we may assume $q\neq 1$; for if $q=1$, $L(s,\chi)=\zeta(s)$, and the required bounds follow from Theorem 5.12 of Titchmarsh [4].

We now write $\xi(s,\chi)=(q/\pi)^{(s+a)/2}\,\Gamma((s+a)/2)\,L(s,\chi)$, where $\chi(-1)=(-1)^a$, $a=0$ or 1. Then, by the functional equation for $L(s,\chi)$,

$$\xi(1-s,\bar{\chi})=i^a q^{1/2}(\tau(\chi))^{-1}\,\xi(s,\chi),$$

where $|\tau(\chi)|=q^{1/2}$. We write $s=\frac{1}{2}+it$, $t\geqq 0$, and consider the integral

$$I(s,\chi)=\frac{1}{2\pi i}\int_{1-i\infty}^{1+i\infty}\xi(s+w,\chi)\,e^{w^2-i\pi w/4}\,dw/w.$$

We move the line of integration to $\mathrm{Re}(w)=-1$ and use the functional equation, whence

$$I(s,\chi)=\xi(s,\chi)-i^{-a}q^{-1/2}\tau(\chi)\overline{I(s,\chi)}.$$

Thus

$$|L(s,\chi)|\leqq \pi^{-1}\left|\sum_{n=1}^{\infty}\chi(n)n^{-s}J(n\pi^{1/2}q^{-1/2})\right|, \tag{4}$$

where

$$J(x)=\int_{1-i\infty}^{1+i\infty}x^{-w}f(w)\,e^{w^2-i\pi w/4}\,dw/w, \tag{5}$$

$$f(w)=\Gamma((w+s+a)/2)\,(\Gamma((s+a)/2))^{-1}.$$

We wish to remove the factor $J(n\pi^{1/2}q^{-1/2})$ from (4) by partial summation. This will require certain estimates for $J(x)$, which we proceed to derive. We move the line of integration in (5) to $\mathrm{Re}(w)=-\frac{1}{4}$. By Stirling's approximation we have, on the line of integration,

$$f(w)\ll \exp((\pi/4)(t-|t+v|)-\tfrac{1}{8}\log T+O(\log(2+|v|))),$$

where $v = \mathrm{Im}(w)$. Hence, allowing for a pole of residue $2\pi i$ at $w = 0$,

$$J(x) \ll 1 + (x/T^{1/2})^{1/4} \int_{-\infty}^{\infty} e^{-v^2 + (\pi/4)(t+v-|t+v|)} (2+|v|)^A \, dv,$$

for some absolute constant A. Since

$$t + v - |t+v| \leqq 0$$

for all values of v, we conclude that

$$J(x) \ll 1 + (x/T^{1/2})^{1/4} \int_{-\infty}^{\infty} e^{-v^2} (2+|v|)^A \, dv \ll 1 + (x/vT)^{1/4}. \tag{6}$$

Similarly, after differentiating under the sign and moving the line of integation to $\mathrm{Re}(w) = 0$, we have

$$J'(x) \ll x^{-1}. \tag{7}$$

We next replace part of the line of integration in (5) by a circular arc, of radius $K \geqq 1$ about the origin, in the right-hand half plane. On the remaining semi-infinite line segments we find that

$$x^{-w} f(w) e^{w^2 - i\pi w/4} w^{-1} \ll e^{-v^2/2},$$

if $x \geqq 1$, so that the contribution to $J(x)$ is $\ll \exp(-K^2/2)$. On the circular arc, Stirling's approximation yields

$$\mathrm{Re}(-w \log x + \log f(w) + w^2 - i\pi w/4) = -u \log(x/T^{1/2}) + \pi/4(t+v-|t+v|) + u^2 - v^2 + O(u \log(2+K)),$$

where $w = u + iv$, $|w| = K$. We choose $K = \frac{1}{2}\log(x/T^{1/2})$ and assume $x \geqq CT^{1/2}$ where C is a suitably large constant. Then the expression above is

$$\leqq -2uK + 2u^2 - K^2 + Bu \log(2+K) \leqq -K^2 + BK \log(2+K) \leqq -K^2/2,$$

where B is some absolute constant. It now follows that the integral along the circular arc contributes $\ll \exp(-K^2/2)$ to $J(x)$, whence

$$J(x) \ll \exp(-\tfrac{1}{2}(\log(x/T^{1/2}))^2), \quad (x \geqq CT^{1/2}). \tag{8}$$

In a precisely similar way we find

$$J'(x) \ll x^{-1} \exp(-\tfrac{1}{2}(\log(x/T^{1/2}))^2), \quad (x \geqq CT^{1/2}). \tag{9}$$

We write, temporarily, $g(n) = n^{-1/2} J(n\pi^{1/2} q^{-1/2})$. Then the sum on the righthand side of (4) may be written

$$\sum_{\substack{k=0, \\ N=2^{k-1}}}^{\infty} \Big(\sum_{N < n \leqq 2N} \chi(n) n^{-it} g(n) \Big).$$

Put

$$S_N(x)=\sum_{N<n\leqq x}\chi(n)n^{-it};$$

then on summing by parts we have

$$\sum_{N<n\leqq 2N}\chi(n)n^{-it}g(n)=S_N(2N)g(2N)-\int_N^{2N}S_N(x)g'(x)\,dx$$

By (6) and (8)

$$g(2N)\ll N^{-1/2}\exp(-(\tfrac{1}{3})(\log(1+N/(q_T)^{1/2}))^2),$$

while by (7) and (9)

$$g'(x)\ll x^{-3/2}\exp(-(\tfrac{1}{3})(\log(1+N/(q_T)^{1/2}))^2).$$

Thus if we write

$$M(N)=\operatorname*{Max}_{M\leqq N}\Big|\sum_{N<n\leqq N+M}\chi(n)n^{-it}\Big|,$$

we have the following estimate for $L(\frac{1}{2}+it,\chi)$.

Lemma 1. *In the above notation*

$$L(\tfrac{1}{2}+it,\chi)\ll 1+\sum_{\substack{k=0,\\ N=2^k}}^{\infty}M(N)N^{-1/2}\exp(-(\tfrac{1}{3})(\log(1+N/(q_T)^{1/2}))^2).$$

Since $M(N)\leqq N$ we see that the terms with $N\geqq(qT)^{1/2+\varepsilon}$ contribute $O(1)$, so we may restrict our attention to the estimation of $M(N)$ for $N\leqq(qT)^{1/2+\varepsilon}$. We also note that the bound $M(N)\leqq N$ yields (1), as commented in §1; here we use the bound

$$\sum_{\substack{k=0,\\ N=2^k}}^{\infty}N^{\alpha}\exp(-\tfrac{1}{3}(\log(1+N/(qT)^{1/2}))^2)\ll(qT)^{\alpha/2}(\log qT)^{\delta} \tag{10}$$

in which $\alpha\geqq 0$ and $\delta=0$ or 1 according as $\alpha>0$ or $\alpha=0$.

3. Burgess' Method and the Proof of Theorem 1

We begin this section by giving a minor sharpening to Burgess' work [1, 2]. We prove

Lemma 2.

$$\sum_{n=1}^{q}\Big|\sum_{m=1}^{h}\chi(n+m)\Big|^4\ll qh^2+q^{1/2}h^4(d(q))^6.$$

For the proof we follow Burgess' demonstration of Lemma 8 in [1]; however we give a slightly sharper estimate for

$$\sum_{m_1,\dots,m_4} (q, A_j),$$

where

$$A_j = \prod_{i \neq j} (m_i - m_j),$$

and the summation is over values for which $1 \leqq m_i \leqq h$ and $A_j \neq 0$. Without loss of generality we take $j=1$. Let $K=(q, A_1)$, $K_i=(q, m_i-m_1)$, $(i \neq 1)$, so that $K=(q, \prod K_i)$. Then

$$\sum_{m_i} (q, A_1) \leqq \sum_{K_i | q} K \sum_{\substack{m_1,\dots,m_4; \\ K_i | m_i - m_1}} 1.$$

Since $A_1 \neq 0$, $m_i \neq m_1$; hence, for fixed m_1, the conditions $1 \leqq m_i \leqq h$ and $K_i | m_1 - m_i$ give $\leqq h/K_i$ possible values for m_i. Thus

$$\sum_{m_i} (q, A_1) \leqq \sum_{K_i | q} K h \prod_{2}^{4} (h/K_i) \leqq h^4 \sum_{K_i | q} 1 \leqq h^4 (d(q))^3.$$

The proof of Lemma 2 may now be completed as in [1].

We now follow Burgess [2] as far as Lemma 5. Note that in our notation q is the modules of χ, (Burgess uses p); conversely we shall use p where Burgess writes q. As in (18) of [2] we find that

$$\eta = \Big| \sum_{N < n \leqq N+M} \chi(n) n^{-it} \Big| \leqq \sum_{s=0}^{p-1} \Big| \sum_{z \in I(p,s)} \chi(z) (zp - qs)^{-it} \Big|,$$

where

$$I(p,s) = \left(\frac{N+qs}{p}, \frac{N+M+qs}{p} \right], \qquad (0 \leqq s < p),$$

and $(\alpha, \beta]$ denotes the set of real numbers x for which $\alpha < x \leqq \beta$. We let p run over the primes in $[q_1, 2q_1]$ which do not divide q; then, just as in Lemma 5 of [2], we deduce the following result.

Lemma 3. *Let*

$$(\log q)^2 < q_1 < q/(4M).$$

Then there exists a set $\mathscr{S}$ of disjoint intervals $I = I(p,s)$ such that

$$|\mathscr{S}| \leqq q_1^2/(w \log q)^2$$

for some $w \geqq 1$ and

$$\sum_{I \in \mathscr{S}} \Big| \sum_{z \in I} \chi(z) (zp - qs)^{-it} \Big| \gg q_1 \eta / (w (\log q)^2). \tag{11}$$

We now split $I=I(p,s)$ into g equal subintervals $(\alpha,\beta]$. Then, writing

$$S(x)=\sum_{\alpha<z\leqq x}\chi(z),$$

we have, on summing by parts

$$\begin{aligned}&\sum_{\alpha<z\leqq\beta}\chi(z)(zp-qs)^{-it}\\&\quad=S(\beta)(\beta p-qs)^{-it}+ipt\int_{\alpha}^{\beta}S(x)(xp-qs)^{-it-1}\,dx\\&\quad\ll(1+p|t|(\beta-\alpha)/N)\max_{\alpha<x\leqq\beta}|S(x)|.\end{aligned}$$

Since $\beta-\alpha=M/pg$, the choice $g=[T]$ yields

$$\sum_{z\in I}\chi(z)(zp-qs)^{-it}\ll\sum_{J}|\sum_{z\in J}\chi(z)|,$$

where there are g disjoint intervals J. We use this bound in (11), whence

$$\sum_{J}|\sum_{z\in J}\chi(z)|\gg q_1\eta/(w(\log q)^2),\tag{12}$$

where there are $K\leqq q_1^2T(w\log q)^{-2}$ disjoint intervals J, of length $\leqq M/(q_1T)$. We number the intervals $J_1,\ldots,J_K$ so that $J_i=(\alpha_i,\beta_i]$ with α_i increasing. Then

$$\sum_{i}|\sum_{z\in J_i}\chi(z)|=\sum_{i}|\sum_{z\in J_{2i}}\chi(z)|+\sum_{i}|\sum_{z\in J_{2i+1}}\chi(z)|.$$

We suppose for example, that the sum over J_{2i} makes the larger contribution. Then (12) holds with $J=J_{2i}$, and therefore with the additional hypothesis that

$$|\alpha-\alpha'|,\ |\beta-\beta'|\geqq M/(2q_1T)$$

for distinct intervals $J=(\alpha,\beta]$, $J'=(\alpha',\beta']$. We now follow the proof of Lemma 6 of [2], with $q_1=M/(2^vT\log q)$, to obtain the estimate below.

Lemma 4. *Let*

$$4M^2/(qT\log q)<2^v<M/(T(\log q)^3).\tag{13}$$

Then, either

$$\eta\ll M/(2^v\log q),$$

or

$$\sum_{n=1}^{q}\left|\sum_{m=1}^{h}\chi(n+m)\right|^4\gg\eta^4M^{-2}T^{-1}4^vh(\log q)^{-4},$$

for some integer h satisfying $1\leqq h\leqq 2^v$.

Lemmas 2 and 4 now yield either

$$\eta^4 \ll qM^2T2^{-\nu}(\log q)^4 + (d(q))^6 q^{1/2} M^2 T 2^{\nu}(\log q)^4, \tag{14}$$

or

$$\eta \ll M/(2^{\nu}\log q). \tag{15}$$

We choose ν so that

$$2^{\nu-1} \leqq (d(q))^{-3} q^{1/4} < 2^{\nu};$$

then (13) holds, under the condition $A < M < B$, where

$$A = 2(d(q))^{-3} q^{1/4} T(\log q)^3,$$
$$B = \tfrac{1}{2}(d(q))^{-3/2} q^{5/8} T^{1/2} (\log q)^{1/2}.$$

Moreover (14) immediately yields

$$\eta \ll d(q) q^{3/16} T^{1/4} M^{1/2} \log q$$

and from (15) and $M < B$ we have

$$\begin{aligned}\eta &\ll (d(q))^3 q^{-1/4} M \ll (d(q))^3 q^{1/16} T^{1/4} M^{1/2} (\log q)^{1/2} \\ &\ll d(q) q^{3/16} T^{1/4} M^{1/2} \log q.\end{aligned}$$

Since $\eta \leqq M$ trivially, we deduce that

$$M(N) = \operatorname*{Max}_{M \leqq N} \eta \ll d(q) q^{3/16} T^{1/4} M^{1/2} \log q + \operatorname{Min}(A, N),$$

for $N < B$. Thus, if

$$T \leqq (d(q))^{10} q^{1/4} (\log q)^2, \tag{16}$$

Lemma 1 yields

$$\begin{aligned}L(\tfrac{1}{2} + it, \chi) &\ll d(q) q^{3/16} T^{1/4} (\log q)^2 + A^{1/2} \\ &\ll d(q) q^{3/16} T^{1/4} (\log q)^2.\end{aligned}$$

Note that $B \geqq (qT)^{9/16}$ by (16). This proves Theorem 1 except when (16) is false, in which case the required estimate will follow from Theorem 2 with $q_0 = 1$. We shall see in the sections that follow that the proof of Theorem 2 is completely independent of Theorem 1.

4. A 'qT-Analogue' of van der Corput's Method

In this section we commence the proof of Theorem 2 by applying an analogue of van der Corput's method to the estimation of $M(N)$.

We define temporarily $f(n) = \chi(n) n^{-it}$ if $N < n \leqq N + M$, $f(n) = 0$ otherwise, and

$$S = \sum_n f(n);$$

thus

$$M(N) \ll \operatorname*{Max}_{M \leqq N} |S|.$$

Let M, H be positive integers such that $M \leqq N$ and

$$H \leqq N/q_0, \tag{17}$$

where $q_0 | q$ as in Theorem 2. Then

$$HS = \sum_{h=1}^{H} \sum_{n} f(n+4hq_0) = \sum_{n} \sum_{h=1}^{H} f(n+4hq_0),$$

whence, by Cauchy's inequality,

$$H^2|S|^2 \leqq \Big(\sum_{N-4Hq_0 < n \leqq N+M} 1\Big)\left(\sum_{n}\left|\sum_{h=1}^{H} f(n+4hq_0)\right|^2\right),$$

and

$$\begin{aligned}|S|^2 &\ll NH^{-2}\Big(\sum_{1 \leqq h_1, h_2 \leqq H} \Big|\sum_{n} f(n+4h_1q_0)\,\overline{f(n+4h_2q_0)}\Big|\Big)\\ &\ll N^2H^{-1} + NH^{-1}\sum_{h=1}^{H}\Big|\sum_{n} f(n+4hq_0)\overline{f(n)}\Big|.\end{aligned}$$

For brevity we write

$$S(h) = \operatorname*{Max}_{M \leqq N} \Big|\sum_{N < n \leqq N+M} \chi(n+4hq_0)\,\bar{\chi}(n)\,(1+4hq_0/n)^{-it}\Big|,$$

so that

$$M(N) \ll NH^{-1/2} + N^{1/2}H^{-1/2}\left(\sum_{h=1}^{H} S(h)\right)^{1/2}. \tag{18}$$

We have now to estimate $S(h)$. We write

$$\begin{aligned}&\sum_{N < n \leqq N+M} \chi(n+4hq_0)\,\bar{\chi}(n)\,(1+4hq_0/n)^{-it}\\ &\quad = \sum_{r \,(\mathrm{mod}\, q)} \chi(r+4hq_0)\,\bar{\chi}(r) \sum_{A_r < n \leqq B_r} (1+4hq_0/(qn+r))^{-it},\end{aligned}$$

where

$$A_r = (N-r)/q, \qquad B_r = (N+M-r)/q.$$

To calculate the inner sum we take

$$g(x) = (1+4hq_0/(qx+r))^{-it}$$

in the formula

$$\sum_{A < n \leqq B} g(n) = \int_{A}^{B} g(x)\,dx + \int_{A}^{B} g'(x)\,\{x\}\,dx - [g(x)\,\{x\}]_{A}^{B}, \tag{19}$$

where $\{x\}=x-[x]-\frac{1}{2}$. We first consider the contribution from the second integral on the right of (19). Here we substitute

$$\{x\}=\sum_{n\neq 0}(-2\pi i n)^{-1}e(nx);$$

we may then integrate termwise, as the sum is convergent in mean square. On putting $y=qx+r$ this yields

$$\sum_{n\neq 0}(-2\pi i n)^{-1}\int_N^{N+M} e(ny/q-nr/q)\frac{d}{dy}(e(h(y)))\,dy,$$

where we have written

$$h(y)=-t(2\pi)^{-1}\log(1+4hq_0/y)$$

for convenience. Now, on summing over r, each term contributes

$$S(q;\chi,4hq_0,-n)(-2\pi i n)^{-1}\int_N^{N+M} e(ny/q)\frac{d}{dy}(e(h(y)))\,dy.$$

To estimate the above integral we apply Lemma 4.3 of Titchmarsh [4] which states that

$$\int_a^b G(x)\,e^{iF(x)}\,dx \ll \mathrm{Sup}(|G(x)/F'(x)|)$$

if $F'(x)/G(x)$ is monotonic. Hence, if $|n|\geqq qTq_0hN^{-2}$,

$$\int_N^{N+M} e(ny/q)\frac{d}{dy}(e(h(y)))\,dy \ll qTq_0hN^{-2}|n|^{-1},$$

and the resulting contribution to $S(h)$ is

$$\ll \sum_{|n|\geqq qTq_0hN^{-2}} qTq_0hN^{-2}n^{-2}|S(q;\chi,4hq_0,n)|. \tag{20}$$

In case $|n|<qTq_0hN^{-2}$ we integrate by parts, to give

$$(-2\pi i n)^{-1}\int_N^{N+M} e(ny/q)\frac{d}{dy}(e(h(y)))\,dy$$
$$=[(-2\pi i n)^{-1}e(ny/q+h(y))]_N^{N+M}+q^{-1}\int_N^{N+M} e(ny/q+h(y))\,dy.$$

We estimate the integral on the right, and also the first integral on the right of (19), by means of Lemma 4.4 of Titchmarsh [4], which states that

$$\int_a^b e(F(x))\,dx \ll \mathrm{Sup}(|F''(x)|^{-1/2}).$$

Thus

$$q^{-1}\int_N^{N+M} e(n\,y/q+h(y))\,dy \ll q^{-1}T^{-1/2}q_0^{-1/2}h^{-1/2}N^{3/2},$$

so that the total contribution to $S(h)$ is

$$\ll \sum_{|n|<qTq_0hN^{-2}} q^{-1}T^{-1/2}q_0^{-1/2}h^{-1/2}N^{3/2}|S(q;\chi,4hq_0,n)|. \tag{21}$$

The remaining terms are

$$-\sum_{r(\mathrm{mod}\,q)} \chi(r+4hq_0)\,\bar{\chi}(r)\,[e(h(y))\,\{(y-r)/q\}]_N^{N+M}$$
$$+\sum_{0<|n|<qTq_0hN^{-2}} (-2\pi i n)^{-1}\,S(q;\chi,4hq_0,-n)\,[e(h(y)+n\,y/q)]_N^{N+M}. \tag{22}$$

To evaluate the first sum we note that

$$\sum_{r=0}^{q-1} \chi(r+4hq_0)\,\bar{\chi}(r)\,\{(y-r)/q\}$$
$$=q^{-1}\sum_{n=0}^{q-1}\sum_{s=0}^{q-1} \chi(s+4hq_0)\,\bar{\chi}(s)\,e(-sn/q)\sum_{r=0}^{q-1} e(nr/q)\,\{(y-r)/q\}$$
$$=q^{-1}\sum_{n=0}^{q-1} S(q;\chi,4hq_0,-n)\sum_{r=0}^{q-1} e(nr/q)\,\{(y-r)/q\}.$$

The final sum over r is $\{y\}$ if $q|n$, or $e(n[y]/q)/(e(-n/q)-1)$, if $q\nmid n$. If we replace the range of the n-summation by $-q/2<n\leqq q/2$ we obtain

$$q^{-1}\{y\}\,S(q;\chi,4hq_0,0)$$
$$+\sum_{\substack{-q/2<n\leqq q/2,\\ n\neq 0}} S(q;\chi,4hq_0,-n)\,((-2\pi i n)^{-1}\,e(n\,y/q)+O(q^{-1}))$$
$$=\sum_{0<|n|<q/2} (-2\pi i n)^{-1}\,S(q;\chi,4hq_0,-n)\,e(n\,y/q)$$
$$+O\left(q^{-1}\sum_{n=0}^{q-1}|S(q;\chi,4hq_0,n)|\right).$$

On comparing this with (22) we see that the total contribution to $S(h)$ is

$$\ll q^{-1}\sum_{n=0}^{q-1}|S(q;\chi,4hq_0,n)|$$
$$+\sum_{\substack{\mathrm{Min}(q/2,qTq_0hN^{-2})\\ \leqq|n|<\mathrm{Max}(q/2,qTq_0hN^{-2})}} |n|^{-1}\,|S(q;\chi,4hq_0,n)|. \tag{23}$$

Since $S(q;\chi,4hq_0,n)$ has period q in n, the expressions (20) and (23) are

$$\ll \sum_{n\in\mathscr{S}_h} |n|^{-1}\,|S(q;\chi,4hq_0,n)|,$$

where

$$\mathscr{S}_h = \{n: \mathrm{Min}(qTq_0hN^{-2}, q) \leqq |n| \leqq qTq_0hN^{-2} + q\},$$

and so, when we include the contribution (21), we have

$$S(h) \ll q^{-1}T^{-1/2}q_0^{-1/2}h^{-1/2}N^{3/2} \sum_{|n| < qTq_0hN^{-2}} |S(q;\chi,4hq_0,n)| + \sum_{n \in \mathscr{S}_h} |n|^{-1} |S(q;\chi,4hq_0,n)|. \tag{24}$$

5. The Sum $S(q;\chi,4hq_0,n)$

We begin this section by proving the following multiplicative property of $S(q;\chi,h,n)$.

Lemma 5. *Let $q = uv$, where $(u,v)=1$, and let*

$$v\bar{v} \equiv 1 \pmod{u}, \quad u\bar{u} \equiv 1 \pmod{v}. \tag{25}$$

Then there are primitive characters $\chi_u \pmod{u}$ and $\chi_v \pmod{v}$ such that

$$S(q;\chi,h,n) = S(u;\chi_u, h\bar{v}, n)\, S(v;\chi_v, h\bar{u}, n).$$

We note that the conguences (25) are solvable since $(u,v)=1$. Define

$$\chi_u(n) = \chi(n + u\bar{u}(1-n)), \qquad \chi_v(n) = \chi(n + v\bar{v}(1-n));$$

then χ_u, χ_v are primitive characters modulo u and v respectively, and $\chi(n) = \chi_u(n)\chi_v(n)$. Thus

$$\begin{aligned} S(q;\chi,h,n) &= \sum_{m=1}^{q} \chi(m+h)\bar{\chi}(m)\, e(nm/q) \\ &= \sum_{k=1}^{u}\sum_{l=1}^{v} \chi(vk+ul+h)\bar{\chi}(vk+ul)\, e(n(vk+ul)/q) \\ &= \sum_{k,l} \chi_u(vk+h)\chi_v(ul+h)\bar{\chi}_u(vk)\bar{\chi}_v(ul)\, e(nvk/q)\, e(nul/q) \\ &= \Big(\sum_k \chi_u(k+h\bar{v})\bar{\chi}_u(k)\, e(nk/u)\Big)\Big(\sum_l \chi_v(l+h\bar{u})\bar{\chi}_v(l)\, e(nl/v)\Big) \\ &= S(u;\chi_u, h\bar{v}, n)\, S(v;\chi_v, h\bar{u}, n) \end{aligned}$$

as required.

We have now to estimate $S(p^r;\chi,k,n)$ where $k \equiv h n_p \pmod{p^r}$, n_p depends only on p^r and q, and $p \nmid n_p$. For convenience we write S for $S(p^r;\chi,k,n)$. We also use the standard notation $p^s \| m$, to mean $p^s | m$ and $p^{s+1} \nmid m$. We first consider the case $p | k$ for which we prove the following result.

Lemma 6. *If $p^r | k$, then*

$$|S| \leqq p^r, \ (p^r | n); \qquad |S| \leqq p^{r-1}, \ (p^{r-1} \| n); \qquad |S| = 0, \ (p^{r-1} \nmid n).$$

If $p^s \| k$ *with* $1 \leqq s \leqq r-1$, *and* $s \geqq 2$ *if* $p=2$, *then*

$$|S| \leqq p^{r-1}, \ (s=r-1, p^r | n); \quad |S| \leqq 2p^{(r+s)/2}, \ \text{or} \ \leqq 6p^{(r+s)/2} \ \text{if} \ p=2, \ (p^s \| n);$$

$$|S|=0, \quad (p^s \nmid n, s \leqq r-2, \text{or } p^s \nmid n, \ s=r-1).$$

Let $p^r | k$; then

$$S = \sum_{\substack{m=1, \\ p \nmid m}}^{p^r} e(n\,m/p^r) = \sum_{m=1}^{p^r} e(n\,m/p^r) - \sum_{u=1}^{p^{r-1}} e(n\,u/p^{r-1}).$$

The first sum in the above expression is zero unless $p^r | n$, when it is p^r; the second sum is zero unless $p^{r-1} | n$, when it is p^{r-1}. The first part of Lemma 6 then follows.

Henceforth let $\bar{x}$ denote the solution of $x\,\bar{x} \equiv 1 \pmod{p^r}$, with $0 < \bar{x} \leqq p^r$, when $p \nmid x$. All summations involving $\bar{x}$ will be understood to exclude values for which $p | x$. Let $p^s \| k$, $1 \leqq s \leqq r-1$. Then

$$S = \sum_{1}^{p^r} \chi(1+k\,\bar{m})\, e(n\,m/p^r) = \sum_{u=1}^{p^{r-s}} \sum_{v=1}^{p^s} \chi(1+k\overline{(u+v\,p^{r-s})})\, e(n(u+v\,p^{r-s})/p^r)$$

$$= \Big(\sum_{u} \chi(1+k\,\bar{u})\, e(n\,u/p^r)\Big)\Big(\sum_{v} e(n\,v/p^s)\Big).$$

The sum with respect to v, and hence also S, will be zero unless $p^s | n$, as we now assume. In case $s=r-1$ we then have

$$S = p^s \sum_{u=1}^{p} \chi(1+k\,\bar{u})\, e(n'\,u/p),$$

where $n'\,p^s = n$. However, since χ is primitive (mod p^r),

$$\chi(1+k\,\bar{u}) = e(c\,\bar{u}/p),$$

for some c prime to p. Then if $p | n'$

$$S = p^s \sum_{u=1}^{p-1} e(c\,\bar{u}/p) = -p^s,$$

and if $p \nmid n'$

$$|S| = p^s \left| \sum_{u=1}^{p-1} e((c\,\bar{u}+n'\,u)/p) \right| \leqq 2p^{s+1/2},$$

by Weil's estimate for the Kloostermann sum. This deals with the case $s=r-1$.

We can now suppose $s \leqq r-2$. We write $t=r-s$, $a=[t/2]$, $b=t-a$, $c=t-2a$. Then

$$S = \sum_{u=1}^{p^a} \sum_{v=1}^{p^{r-a}} \chi(1+k(u+p^a\,v))\, e(n\overline{(u+p^a\,v)}/p^r).$$

Since $2(s+a) \geqq r$ we have

$$1+k(u+p^a v) \equiv (1+k u)(1+k p^a)^{v\overline{(1+ku)}} \pmod{p^r},$$

and since χ is primitive $(\bmod p^r)$

$$\chi(1+k p^a) = e(K/p^b), \qquad p \nmid K.$$

Moreover

$$\overline{u+p^a v} \equiv \bar{u} - p^a \bar{u}^2 v + p^{2a} \bar{u}^3 v^2 \pmod{p^t},$$

since $t \geqq 2$. Thus

$$S = \sum_u \chi(1+ku)\, e(n\bar{u}/p^r) \sum_v e((\alpha v^2 + \beta v)/p^b), \tag{26}$$

where $\alpha = p^a n' \bar{u}^3$, $\beta = K\overline{(1+ku)} - n'\bar{u}^2$ and $n' = n p^{-s}$. On writing $v = x + p^c y$ the inner sum becomes

$$\sum_{x=1}^{p^c} \sum_{y=1}^{p^{a+s}} e((\alpha x^2 + \beta x)/p^b)\, e(\beta y/p^a).$$

The summation with respect to y is zero unless $p^a \mid \beta$, in which case $p \nmid n'$; thus $S = 0$ unless $p^s \| n$. In case $p^a \mid \beta$ the summation with respect to y gives p^{a+s}, so that the sum over v in (26) yields

$$p^{a+s} \sum_{x=1}^{p^c} e((\alpha' x^2 + \beta' x)/p^c),$$

where $\alpha' = \alpha p^{-a}$, $\beta' = \beta p^{-a}$. Note that $c = 0$ or 1 always and, since $p^a \mid \beta$ and $p \nmid n'$, that $p \nmid \alpha'$. Thus

$$\left| \sum_{x=1}^{p^c} e((\alpha' x^2 + \beta' x)/p^c) \right| \leqq \delta p^{c/2},$$

where $\delta = 1$ if $p \geqq 3$, or $\delta = 2^{1/2}$, $(p = 2)$, by the usual estimate for Gauss' sum. We now have from (26)

$$|S| \leqq \delta p^{a+s+c/2} \sum_{1 \leqq u \leqq p^a} 1,$$

where u is such that $p^a \mid \beta$, i.e. u ranges over solutions of $K\overline{(1+ku)} \equiv n'\bar{u}^2 \pmod{p^a}$. However this condition is equivalent to $K u^2 \equiv n'(1+ku) \pmod{p^a}$, which has at most 2 solutions $(\bmod p^a)$ if $p > 2$, or 4 solutions $(\bmod p^a)$ if $p = 2$. Here we use the facts that $p \nmid K n'$, and that, if $p = 2$, $4 \mid k$; this is the reason for the introduction of the factor 4 in $S(h)$. Since $a + s + c/2 = (r+s)/2$ the proof of Lemma 6 is complete.

We turn now to the case $r \geqq 2$, $p \nmid k$, $p \geqq 3$. On writing $s = [r/2]$ we have

$$S = \sum_{u=1}^{p^s} \sum_{v=1}^{p^{r-s}} \chi(1+k(u+v p^s))\, e(n\overline{(u+v p^s)}/p^r).$$

Here

$$\overline{u+p^s v} \equiv \bar{u} - p^s \bar{u}^2 v + p^{2s} \bar{u}^3 v^2 \pmod{p^r}$$

and

$$1 + k(u + v p^s) \equiv (1 + k u)(1 + p^s)^w \pmod{p^r},$$

where

$$w = k v \overline{(1+ku)} + \bar{2} k v \overline{(1+ku)}\, p^s - \bar{2} k^2 v^2 \overline{(1+ku)^2}\, p^s.$$

We write $\chi(1+p^s) = e(c/p^{r-s})$ where $c = c(p, \chi, r)$ and, as χ is primitive $(\bmod p^r)$, $p \nmid c$. Then

$$S = \sum_u \chi(1+ku)\, e(n\bar{u}/p^r) \sum_v e((\alpha v^2 + \beta v)/p^{r-s}),$$

with

$$\alpha = p^s(n\bar{u}^3 - c\bar{2}k^2\overline{(1+ku)^2}), \qquad \beta = -n\bar{u}^2 + ck\overline{(1+ku)} + c\bar{2}k\overline{(1+ku)}\, p^s.$$

For the v-summation we set $v = x + p^{r-2s} y$, yielding

$$\sum_{x=1}^{p^{r-2s}} e((\alpha x^2 + \beta x)/p^{r-s}) \sum_{y=1}^{p^s} e(\beta y/p^s);$$

this is zero if $p^s \nmid \beta$, and otherwise has modulus

$$p^s \Big| \sum_x e((\alpha' x^2 + \beta' x)/p^{r-2s}) \Big| \leqq p^{r/2} (p^{r-2s}, \alpha', \beta')^{1/2},$$

where $\alpha' p^s = \alpha$, $\beta' p^s = \beta$. Note that $r - 2s = 0$ or 1 always, and that $p \mid \beta$ requires $p \nmid n$. Let N_1 denote the number of solutions of $p^s \mid \beta$, $(1 \leqq u \leqq p^s)$, and N_2 the number of solutions of $p^{s+1} \mid \alpha, \beta, (1 \leqq u \leqq p^s)$ if $r - 2s = 1$, or $N_2 = 0$ if $r - 2s = 0$. Then

$$|S| \leqq p^{r/2} N_1 + p^{(r+1)/2} N_2.$$

We write the condition $p^t \mid \beta$, where $t = s$ or $s+1$, as

$$X^2 \equiv m \pmod{p^t},$$

$$X = ck(1 - \bar{2}p^s)u - \bar{2}nk, \qquad m = nck(1 - \bar{2}p^s) + \bar{4}n^2 k^2.$$

If $(p^t, m) = p^l$ we require $l = t$ or $l = 2j < t$. In the former case $p^{[(t+1)/2]} \mid X$ and, as $p \nmid ck(1 - \bar{2}p^s)$, u has $p^{s-[(t+1)/2]}$ possible values in the range $0 < u \leqq p^s$. When $l = 2j < t$ we require

$$(X p^{-j})^2 \equiv m p^{-2j} \pmod{p^{t-2j}},$$

yielding at most $2p^{s+j-t}$ solutions u with $0 < u \leqq p^s$, except when $j = 0$, $t = s+1$. These estimates show that $N_1 \leqq 2(m, p^s)^{1/2}$, and that, if $j \neq 0$, $N_2 \leqq 2(m, p^s)^{1/2} p^{-1/2}$. However if $p^{s+1} \mid \alpha, \beta$ has a solution u, then

$$p \mid n\bar{u}^3 - c\bar{2}k^2\overline{(1+ku)^2}, \qquad p \mid n\bar{u}^2 - ck\overline{(1+ku)}.$$

On eliminating $\overline{1+ku}$ we find that

$$p|2cn\bar{u}^3-n^2\bar{u}^4,$$

whence, since $p\nmid n$,

$$p|ck u-\bar{2}nk, \quad p|X, \quad p|m.$$

Thus there is no solution u for $j=0, t=s+1$. Since $(p^s, m)=(p^s, 4c+nk)$ we have now proved the following estimate.

Lemma 7. *If $r\geqq 2$, $p\nmid k$ and $p\geqq 3$ then $S=0$ if $p|n$, and*

$$|S|\leqq(r+1)p^{r/2}(p^s, 4c+nk)^{1/2},$$

where $s=[r/2]$, $(c,p)=1$ and $c=c(p,\chi,r)$, if $p\nmid n$.

Finally we require an estimate for the case $r=1$, $p\nmid k$, for which

$$S=\sum_{m=1}^{p-1}\chi(1+k\bar{m})e(mn/p).$$

For $p|n$ we have $S=-1$ trivially. If $p\nmid n$ we use the following estimate, due to Weil [5].

$$|\sum_m \chi(R(m))\psi(m)|\leqq dp^{1/2},$$

(this is the unnumbered displayed formula at the bottom of page 206 of [5]). Here $R(m)$ is any rational function, ψ any non-trivial additive character (mod p), d the total number of zeros and poles of R, and m runs over all values (mod p) for which $R(m)$ is defined. In our case $R(m)=(m+k)/m$ with $m=-k$ as a zero and $m=0$ as a pole, so that $d=2$. We thus have the following bound.

Lemma 8. *If $r=1$ and $p\nmid k$ then*

$$|S|\leqq 2p^{1/2}, \quad \text{or } \leqq 2 \text{ if } p|n.$$

6. The Average Value of $S(q;\chi,4hq_0,n)$

In this section we apply Lemmas 5–8 to estimate the double sum

$$\sum=\sum_{1\leqq h\leqq A}\sum_{1\leqq n\leqq B}|S(q;\chi,4hq_0,n)|.$$

Given n and $m=4hq_0$ we decompose q as a product

$$q=q_1q_2q_3q_4$$

as follows. Each q_i is a product $\prod p^\alpha$ of certain primes p, where $p^\alpha\|q$. For q_1 we take the primes p such that either $p^\alpha|m,n$, or $p^\beta\|m,n$ with $1\leqq\beta<\alpha$. For q_2 we take the primes p for which $p^\alpha|m$, $p^{\alpha-1}\|n$. For q_3 we use the p such that $p^{\alpha-1}\|m$, $p^\alpha|n$. The remaining primes p make up q_4. By Lemmas 5 and 6, $S=0$ unless $p\nmid m$ for each

$p|q_4$. Moreover, if $2|q$ and $S\neq 0$, then, since $2|m, 2|q_1 q_2 q_3$; thus q_4 may be supposed to be odd. We now write $u_2=\prod p$ for $p|q_2, u_3=\prod p$ for $p|q_3$, and $u_4=\prod p^{[\alpha/2]}$ for $p^\alpha \| q_4$. Then according to Lemmas 5–8,

$$|S|\leqq 3d(q)\, q^{1/2}\, \delta_1^{1/2}\delta_2^{1/2} u_2^{-1}\, \delta_3^{1/2} u_3^{-1/2}\delta_4^{1/2},$$

where $\delta_i=(m, q_i)$, $1\leqq i\leqq 3$, and $\delta_4=(u_4, e+mn)$ for some e, prime to u_4, and depending only on q, χ and q_4. Moreover by construction

$$f_2=\delta_1\delta_2 u_2^{-1}\delta_3 u_3 | n.$$

We now sum for all values of h and n which give rise to a particular set $\mathscr{S}$ of values of q_i, δ_i and u_i, and for which $S\neq 0$. This requires $\delta_4|e+4hq_0 n, f_2|n$ and $\delta_1\delta_2\delta_3|4hq_0$. By construction $q_0|\delta_1\delta_2\delta_3$ and so, if

$$f_1=q_0^{-1}\delta_1\delta_2\delta_3/(4, q_0^{-1}\delta_1\delta_2\delta_3),$$

then $f_1|h$. Putting $h=xf_1$, $n=yf_2$, we have $1\leqq x\leqq A/f_1$, $1\leqq y\leqq B/f_2$ and, since $(\delta_4, 4q_0 f_1 f_2)=1$, $\delta_4|e'+xy$. Hence

$$\sum \ll d(q)\, q^{1/2}\sum_{\mathscr{S}}(\delta_1\delta_2\delta_3\delta_4)^{1/2} u_2^{-1} u_3^{-1/2} S(A/f_1, B/f_2; \delta_4, e'),$$

where

$$S(X, Y; k, l)=\sum_{1\leqq x\leqq X}\ \sum_{\substack{1\leqq y\leqq Y,\\ k|xy+l}} 1.$$

We shall prove below that

$$S(X, Y; k, l)\ll (XY/k+1)\,\mathrm{Min}(k^\varepsilon, (XY)^\varepsilon), \tag{27}$$

for $(l, k)=1$ and $\varepsilon>0$. We will then have

$$\sum \ll d(q)\, q^{1/2}\sum_{\mathscr{S}}(q_0(\delta_1\delta_2\delta_3)^{-3/2}\delta_4^{\varepsilon-1/2} AB + \\ +(\delta_1\delta_2\delta_3\delta_4)^{1/2} u_2^{-1}\,\mathrm{Min}(\delta_4^\varepsilon, (ABq_0 u_2(\delta_1\delta_2\delta_3)^{-2})^\varepsilon)). \tag{28}$$

We now suppose $\varepsilon<\frac{1}{8}$. Then, since $q_0|\delta_1\delta_2\delta_3$

$$\sum_{\delta_i} q_0(\delta_1\delta_2\delta_3)^{-3/2}\delta_4^{\varepsilon-1/2}\ll q_0^{-1/2}\sigma_{-1/4}(q),$$

where

$$\sigma_\alpha(n)=\sum_{d|n} d^\alpha.$$

As q_1, q_2, q_3, q_4 take at most $(d(q))^3$ sets of values, we have

$$d(q)\, q^{1/2}\sum_{\mathscr{S}} q_0(\delta_1\delta_2\delta_3)^{-3/2}\delta_4^{\varepsilon-1/2} AB\ll (d(q))^4\sigma_{-1/4}(q)\, q^{1/2} AB q_0^{-1/2}.$$

For the second part of the sum in (28) we note that f_1 must be $\leqq A$, whence

$$(\delta_1\delta_2\delta_3\delta_4)^{1/2}u_2^{-1}(ABq_0u_2(\delta_1\delta_2\delta_3)^{-2})^{\varepsilon}$$
$$\ll(\delta_1\delta_2\delta_3\delta_4^2q^{-1})^{1/4}(Aqq_0)^{1/4}(AB/q_0)^{\varepsilon}.$$

Then, since

$$\sum_{\delta_i}(\delta_1\delta_2\delta_3\delta_4^2)^{1/4}\leqq(q_1q_2q_3u_4^2)^{1/4}(\sigma_{-1/4}(q))^4\leqq q^{1/4}(\sigma_{-1/4}(q))^4,$$

we see that the total contribution to (28) is

$$\ll(d(q))^4(\sigma_{-1/4}(q))^4q^{1/2}(Aqq_0)^{1/4}(AB/q_0)^{\varepsilon}.$$

We have now proved the first part of Lemma 9 below.

Lemma 9. *If $q_0|q$ and $\varepsilon>0$, then*

$$\sum_{1\leqq h\leqq A}\sum_{1\leqq n\leqq B}|S(q;\chi,4hq_0,n)|\ll(d(q)\sigma_{-1/4}(q))^4q^{1/2}\{ABq_0^{-1/2}$$
$$+(qAq_0)^{1/4}(AB/q_0)^{\varepsilon}\},\tag{29}$$

and

$$\sum_{1\leqq h\leqq A}|S(q;\chi,4hq_0,0)|\ll(d(q))^2q_0A.$$

To prove the second part of Lemma 9 we note that

$$|S(q;\chi,4hq_0,0)|\ll d(q)(4hq_0,q),$$

by Lemmas 5–8, and that

$$\sum_{1\leqq h\leqq A}(4hq_0,q)\ll d(q)q_0A.$$

It remains to prove (27). Suppose firstly that $X,Y\leqq k$. Then

$$S(XY;k,l)\leqq\sum_{\substack{n\leqq XY,\\ k|n+l}}d(n)\leqq(XY/k+1)\operatorname*{Max}_{n\leqq XY}d(n)$$
$$\ll(XY/k+1)(XY)^{\varepsilon/2}\ll(XY/k+1)\operatorname{Min}(k^{\varepsilon},(XY)^{\varepsilon}).$$

Alternatively let X, say, be greater than k. Then $(l,k)=1$ implies that $(y,k)=1$. Thus

$$\sum_{\substack{1\leqq x\leqq X,\\ k|xy+l}}1\leqq(X/k+1)\leqq2X/k.$$

Hence

$$S(X,Y;k,l)\leqq2XY/k\ll(XY/k+1)\operatorname{Min}(k^{\varepsilon},(XY)^{\varepsilon})$$

as required.

We may now see the reason for the introduction of q_0. The first term of (29) has been reduced by a factor $q_0^{-1/2}$; this is because $S=0$ unless $f_2|n$, and f_2 is essentially divisible by q_0. The improvement relative to the case $q_0=1$ is directly reflected in Theorem 2.

7. Proof of Theorem 2

We first use (24) and Lemma 9 to estimate

$$\sum_{G<h\leqq 2G} S(h),$$

for $G\leqq N$. By (24) there exists B,

$$\mathrm{Min}(qTq_0GN^{-2},q)\leqq B\leqq 2qTq_0GN^{-2}+q,$$

such that

$$S(h)\ll q^{-1}T^{-1/2}q_0^{-1/2}G^{-1/2}N^{3/2}\sum_{|n|<2qTq_0GN^{-2}}|S(q;\chi,4hq_0,n)|$$
$$+B^{-1}(\log qT)\sum_{0<|n|\leqq 2B}|S(q;\chi,4h_0,n)|,$$

for $G<h\leqq 2G$. By an application of Lemma 9 we obtain

$$\sum_{G<h\leqq 2G} S(h)\ll(d(q)\sigma_{-1/4}(q))^4(q^{-1}T^{-1/2}q_0^{1/2}G^{1/2}N^{3/2}+q^{1/2}T^{1/2}G^{3/2}N^{-1/2}$$
$$+q^{1/2}q_0^{-1/2}G\log(qT)+q^{-1/4}T^{-1/2}q_0^{-1/4}G^{-1/8}N^{3/2}(qTN^{-2})^{\varepsilon}$$
$$+B^{-1}q^{3/4}q_0^{1/4}G^{3/8}(qTN^{-2})^{\varepsilon}\log(qT)).$$

We note that $B^{-1}\ll q^{-1}+q^{-1}T^{-1}q_0^{-1}G^{-1}N^2$; then on putting $G=H2^{-k}$ and summing over k, we obtain

$$\sum_{1\leqq h\leqq H} S(h)\ll(d(q)\sigma_{-1/4}(q))^4(q^{-1}T^{-1/2}q_0^{1/2}H^{1/2}N^{3/2}+q^{1/2}T^{1/2}H^{3/2}N^{-1/2}$$
$$+q^{1/2}q_0^{-1/2}H\log(qT)$$
$$+q^{-1/4}T^{-1/2}q_0^{-1/4}N^{3/2}(qTN^{-2})^{\varepsilon}+q^{-1/4}q_0^{1/4}H^{3/8}(qT)^{2\varepsilon}$$
$$+q^{-1/4}T^{-1}q_0^{-3/4}N^2(qTN^{-2})^{\varepsilon}\log(qT)),$$

whence, by (18)

$$M(N)\ll NH^{-1/2}+(d(q)\sigma_{-1/4}(q))^2(q^{-1/2}T^{-1/4}q_0^{1/4}H^{-1/4}N^{5/4}$$
$$+q^{1/4}T^{1/4}H^{1/4}N^{1/4}+q^{-1/8}T^{-1/4}q_0^{-1/8}H^{-1/2}N^{5/4}(qTN^{-2})^{\varepsilon/2}$$
$$+q^{-1/8}q_0^{1/8}H^{-5/16}N^{1/2}(qT)^{\varepsilon}+q^{1/4}q_0^{-1/4}N^{1/2}(\log qT)^{1/2}$$
$$+q^{-1/8}T^{-1/2}q_0^{-3/8}H^{-1/2}N^{3/2}(qTN^{-2})^{\varepsilon/2}(\log qT)^{1/2}).$$

We choose

$$H=\left[\frac{1}{2}\left(\frac{N}{q_0+V^{8/3}(qT)^{1/3}}+\frac{N}{q_0T}\right)\right],$$

where $V=d(q)\sigma_{-1/4}(q)$. Then H is an integer $\leqq N/q_0$, as required by (17). If $H=0$ then

$$M(N)\ll N^{1/2}\cdot N^{1/2}\ll N^{1/2}(q_0^{1/2}+V^{4/3}(qT)^{1/6}).$$

Otherwise, writing K for $N/(qT)^{1/2}$,

$$NH^{-1/2}\ll N(N/(q_0+V^{8/3}(qT)^{1/3}))^{-1/2}\ll N^{1/2}(q_0^{1/2}+V^{4/3}(qT)^{1/6}),$$

$$V^2q^{-1/2}T^{-1/4}q_0^{1/4}H^{-1/4}N^{5/4}\ll V^2q^{-1/4}q_0^{1/4}(q_0+V^{8/3}(qT)^{1/3})^{1/4}N^{1/2}K^{1/2}$$
$$\ll N^{1/2}K^{1/2}q^{\varepsilon-1/4}q_0^{1/4}(q_0^{1/4}+(qT)^{1/12})\ll N^{1/2}K^{1/2}(q_0^{1/2}+(qT)^{1/6}),$$

$$V^2q^{1/4}T^{1/4}H^{1/4}N^{1/4}\ll V^2N^{1/2}q^{1/4}T^{1/4}(V^{-8/3}(qT)^{-1/3}+(q_0T)^{-1})^{1/4}$$
$$\ll N^{1/2}(V^{4/3}(qT)^{1/6}+V^2(q/q_0)^{1/4}),$$

$$V^2q^{-1/8}T^{-1/4}q_0^{-1/8}H^{-1/2}N^{5/4}(qTN^{-2})^{\varepsilon/2}$$
$$\ll N^{1/2}K^{1/4-\varepsilon}V^2T^{-1/8}q_0^{-1/8}(q_0+V^{8/3}(qT)^{1/3})^{1/2}$$
$$\ll N^{1/2}K^{1/4-\varepsilon}(q_0^{1/2}+(q/q_0)^{1/4}+(qT)^{1/6}),$$

$$V^2q^{-1/8}q_0^{1/8}H^{-5/16}N^{1/2}(qT)^{\varepsilon}\ll N^{1/2}q_0^{1/8}T^{1/12}\ll N^{1/2}(q_0^{1/2}+(qT)^{1/6}),$$

$$V^2q^{-1/8}T^{-1/2}q_0^{-3/8}H^{-1/2}N^{3/2}(qTN^{-2})^{\varepsilon/2}$$
$$\ll N^{1/2}K^{1/2-\varepsilon}V^2q^{1/8}T^{-1/4}q_0^{-3/8}\operatorname{Min}(q_0^{1/2}+V^{4/3}(qT)^{1/6},(q_0T)^{1/2})$$
$$\ll N^{1/2}K^{1/2-\varepsilon}V^{8/3}((q/q_0)^{1/4}+(qT)^{1/6}+q_0^{1/2}).$$

The above estimates yield

$$M(N)\ll V^{8/3}N^{1/2}(q_0^{1/2}+(q/q_0)^{1/4}(\log qT)^{1/2}+(qT)^{1/6})$$
$$\cdot(1+K^{1/2}+K^{7/16}(\log qT)^{1/2});$$

here we have chosen $\varepsilon=1/16$, whence the constant implied by the notation $\ll$ is absolute. Theorem 2 now follows at once from Lemma 1 and (10), since $\sigma_{-1/4}(q)\ll(d(q))^{1/10}$.

8. Proof of the Corollaries

The deduction of Corollary 3 from Theorem 2 is self-evident. In this section we first derive Corollary 2 and then Corollary 1. We begin by showing that, for any constant A, there exists q_0 such that

$$q_0 | q, \quad (d(q))^4 \leqq q_0 \leqq q^{1/4}, \tag{30}$$

providing that $d(q)$, and hence q, is sufficiently large. Corollary 2 will follow immediately.

For convenience we write $D=(d(q))^4$. We may suppose that q is so large that $D^8 \leqq q$. Define

$$q_1 = \prod_{\substack{p^\alpha \| q, \\ p \leqq D}} p^\alpha.$$

Suppose firstly that $q_1 > D$, and let q_2 be the smallest factor of q_1 such that $q_2 > D$. Then q_2 cannot be prime, by definition of q_1, so we may write $q_2 = u\,v$, where $u, v < q_2$, and hence $u, v \leqq D$. Then $D < q_2 \leqq D^2 \leqq q^{1/4}$, and q_2 is an admissable value for q_0. Now suppose that $q_1 \leqq D$. Then if p is any prime factor of q/q_1, $p > D$. Some such p provides a possible value for q_0 unless all prime factors of q/q_1 are greater than $q^{1/4}$. But then $d(q/q_1) \leqq 8$,

$$d(q) \leqq d(q/q_1)\, d(q_1) \ll D^\varepsilon,$$

and, taking $\varepsilon = \frac{1}{2}A$, we have

$$d(q) \ll (d(q))^{1/2}$$

which is false if $d(q)$ is large enough. This shows that (30) is possible, and on choosing $A=12$ Theorem 2 yields

$$\begin{aligned} L(\tfrac{1}{2}+i\,t, \chi) &\ll (q^{1/4}(\log q\,T)^{1/2} + (d(q))^3\,(q\,T)^{1/6})\,(\log q\,T) \\ &\ll (q^{1/4}(\log q)^{1/2} + (d(q))^3\,(q\,T)^{1/6})\,(\log q\,T), \end{aligned}$$

as required.

Corollary 1 is an immediate consequence of Theorem 1 if

$$T \leqq (d(q))^{-20}\, q^{1/4}, \tag{31}$$

and of Corollary 2 if

$$T \geqq q^{1/4}(\log q)^{-5/2}. \tag{32}$$

For T in the intermediate range, the required estimate follows from Theorem 2, if there exists q_0 such that

$$(d(q))^3 (q_0^{1/2} + (q/q_0)^{1/4})(\log q)^{3/2} \leqq (q\,T)^{1/5} (\log q)^2.$$

Since we may now suppose that (31) is false it suffices that

$$(d(q))^{28} \leqq q_0 \leqq q^{1/4},$$

and we have shown that this is always possible, if $d(q)$ is large enough. Of course, if $d(q)$ is bounded, Corollary 1 follows at once from Theorem 1 and Corollary 2.

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Almost-Primes Represented by Quadratic Polynomials

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1. Introduction and Statement of Results

The only polynomials that are known to represent infinitely many primes are of degree one (Dirichlet's theorem on primes in arithmetic progressions). Some heuristic arguments suggest that any irreducible polynomial in $Z[x]$ with positive leading coefficient and without any fixed prime divisor assumes infinitely many prime values. Schinzel [19] formulated a more general conjecture and Bateman and Horn [1] have given a quantitative form of his conjecture

$$|\{n \leqq x;\ G(n)=p\}| \sim \Gamma_G \frac{x}{\log x} \quad \text{as } x \to \infty,$$

where $\Gamma_G = \frac{1}{g}\prod \left(1-\frac{\rho(p)}{p}\right)\left(1-\frac{1}{p}\right)^{-1}$, $g=\deg G$, and $\rho(p)$ is the number of incongruent solutions of the congruence

$$G(n) \equiv 0 \pmod{p}.$$

The case of quadratic polynomial is the content of the Hardy-Littlewood Conjecture F [6]. In the present state of knowledge the conjecture seems to be extremely difficult. A very nice approximation to the conjecture was offered by Brun's sieve, the powerful tool for studying almost-primes.

Let P_r denote an almost-prime of order r, i.e., a number having at most r prime factors counted with multiplicity. The problem was first investigated by Rademacher [16] who proved that there exist infinitely many n such that $G(n) = P_{4g-1}$, and Ricci [17] reduced r from $4g-1$ to $3g-1$. Great progress has been made since the work of Kuhn [12], who first noticed that if weights of a certain kind are attached to the elements of the sequence to be sifted, the sieve results concerning almost primes become much stronger. By using Selberg's sieve with Kuhn's weights, Kuhn [13] and Wang [21] obtained $r = g + c\log g$. For small values of g a further improvement was obtained by Levin [14].

The problem of finding the optimal weights is rather complicated. Buchstab [2] transfered the problem to linear programming and was led to a very

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complicated set of constant weights. In 1968 Richert [18] discovered his simple weighted sum with logarithmic weights of the type

$$w(a)=1-\lambda\sum_{p|a}\frac{\log^+(y/p)}{\log y}$$

where $\log^+ x=\max(0,\log x)$. Either weight permits one to prove that there exist infinitely many n such that

$$G(n)=P_{g+1}.$$

We should mention that the same result was proved earlier by Kuhn [13] if $g=2$, by Wang [21] if $g=3,4,5$ and by Levin [14] if $g=6,7$ (in his definition of almost-primes of order r the prime factors are not counted with multiplicity).

Our aim is to prove the following result.

Theorem. *Let $G(n)=an^2+bn+c$ be an irreducible polynomial with $a>0$ and $c\equiv 1 \pmod 2$. Then there exist infinitely many integers n such that $G(n)$ has at most two prime factors. Moreover, if x is sufficiently large the following estimate holds*

$$|\{n\leqq x;\ G(n)=P_2\}|>\frac{1}{77}\Gamma_G\frac{x}{\log x}. \tag{1}$$

Actually, in order to avoid some technical complications, we shall restrict our proof to the polynomial $G(n)=n^2+1$, the general case being similar.

The method involves an injection into a new Richert weighted sum [5] of a combination of two ideas due to Hooley [8, 10] with a new form of the error term in the linear sieve [11]. These four innovations are presented in sections 2, 3 and 4.

Acknowledgement. This is due to several people with whom I have discussed the problem since two years ago. I deeply appreciate Professor Friedlander's assistence in studying Hooley's methods [7, 8] (during our sabbaticals at the Scuola Normale Superiore di Pisa in 1976/77). Actually, a slightly weaker estimate of the type (5) was already obtained in our joint paper [4]. I should say that it was the suggestion of Professor Halberstam that the ideas of Chen [3] and Hooley [8] might be used to attack the problem. I am very grateful to Professor Richert for showing me his new unpublished results and for many valuable comments. Finally I would like to express my thanks to Professor Hooley for his critical remarks and some corrections in the first version of the paper. For the financial support I wish to thank the Mittag-Leffler Institut of the Royal Swedish Academy of Sciences.

2. A New Weighted Sum of Richert

Let p_n and $\omega(n)$ denote the least prime factor and the number of prime divisors of the integer n respectively. For given $x>2$ and for all $p<x$, $p|n$ define

$$\omega_p(n)=\begin{cases}1-\dfrac{\log p}{\log x} & \text{if } p=p_n\\[2ex] \dfrac{\log p_n}{\log x} & \text{if } p>p_n \text{ and } p<x^{\frac{1}{2}}\\[2ex] 1-\dfrac{\log p}{\log x} & \text{if } p>p_n \text{ and } p\geqq x^{\frac{1}{2}}.\end{cases}$$

Let $2 \leqq \lambda < 3$ and

$$w(n) = 1 - \frac{1}{3-\lambda} \sum_{p \mid n,\ p < x} \omega_p(n).$$

Lemma 1. *If $n \leqq x^{\lambda}$ and $w(n) > 0$ then n has at most two distinct prime factors.*

Proof. If n had two or more different prime factors less then $x^{\frac{1}{2}}$ its weight would be

$$w(n) \leqq 1 - \frac{1}{3-\lambda}\left\{1 - \frac{\log p_n}{\log x} + \frac{\log p_n}{\log x}\right\} = \frac{2-\lambda}{3-\lambda} \leqq 0.$$

Therefore one can assume that n has at most one prime factor less than $x^{\frac{1}{2}}$. In this case we have

$$w(n) = 1 - \frac{1}{3-\lambda} \sum_{p \mid n} \frac{\log^+(x/p)}{\log x} \leqq 1 - \frac{1}{3-\lambda}\left\{\omega(n) - \frac{\log n}{\log x}\right\} \leqq \frac{3-\omega(n)}{3-\lambda}$$

whence $\omega(n) \leqq 2$.

Let $\mathscr{A}$ be a finite sequence of positive integers $a \leqq x^{\lambda}$. For any $z \leqq x^{\frac{1}{2}}$, by Lemma 1, we have

$$|\{a \in \mathscr{A};\ a = P_2\}| \geqq \sum_{a \in \mathscr{A},\ (a,\ P(z)) = 1}^{*} w(a) = W^*(\mathscr{A}, z), \quad \text{say}$$

where the summation is over all square-free numbers from $\mathscr{A}$ which are coprime with

$$P(z) = \prod_{p < z} p.$$

We shall be concerned with the sequence

$$\mathscr{A} = \{n^2 + 1;\ n \leqq x\}$$

in which case we choose $\lambda = 2$. The non-square-free numbers are easy to handle. We have

$$\sum_{\substack{a \in \mathscr{A},\ (a,\ P(z)) = 1 \\ a - \text{non-square-free}}} 1 \leqq \sum_{z \leqq p \leqq x z^{-\frac{1}{2}}} \sum_{n \leqq x,\ n^2 + 1 \equiv 0 \,(\mathrm{mod}\, p^2)} 1$$
$$\leqq \sum_{z \leqq p \leqq x z^{-\frac{1}{2}}} 2(x p^{-2} + 1) \leqq 4 x z^{-\frac{1}{2}}.$$

Therefore we are led to consider the expression

$$W(\mathscr{A}, z) = \sum_{a \in \mathscr{A},\ (a,\ P(z)) = 1} w(a)$$

with the aim of showing that

$$W(\mathscr{A}, z) > \frac{1}{77} \Gamma_G \frac{x}{\log x} \tag{2}$$

for all sufficiently large x and for some suitable value of z. Letting

$$\mathscr{A}_q=\{a\in\mathscr{A};\ a\equiv 0(\mathrm{mod}\ q)\}$$

the sum $W(\mathscr{A},z)$ may be interpreted in terms of the sifting functions

$$S(\mathscr{A}_q,u)=|\{a\in\mathscr{A}_q;\ (a,P(u))=1\}|.$$

We have

$$W(\mathscr{A},z)=S(\mathscr{A},z)-\sum_{z\leqq p<x}\ \sum_{a\in\mathscr{A}_p,\ (a,\,P(z))=1}\omega_p(a)$$

and the double sum is equal to

$$\sum_{z\leqq p<x^{\frac12}}\left(1-\frac{\log p}{\log x}\right)S(\mathscr{A}_p,p)+\sum_{z\leqq p_1<p<x^{\frac12}}\frac{\log p_1}{\log x}S(\mathscr{A}_{pp_1},p_1)$$
$$+\sum_{x^{\frac12}\leqq p<x}\left(1-\frac{\log p}{\log x}\right)S(\mathscr{A}_p,z).$$

Applying to the middle sum the Buchstab formula

$$\sum_{z\leqq p_1<p}S(\mathscr{A}_{pp_1},p_1)=S(\mathscr{A}_p,z)-S(\mathscr{A}_p,p)$$

we arrive at

$$W(\mathscr{A},z)=S(\mathscr{A},z)+\sum_{z\leqq p<x^{\frac12}}\ \sum_{z\leqq p_1<p}\frac{\log(p_1/p)}{\log x}S(\mathscr{A}_{pp_1},p_1)$$
$$-\sum_{z\leqq p<x^{\frac12}}\left\{\left(1-2\frac{\log p}{\log x}\right)S(\mathscr{A}_p,p)+\frac{\log p}{\log x}S(\mathscr{A}_p,z)\right\}$$
$$-\sum_{x^{\frac12}\leqq p<x}\left(1-\frac{\log p}{\log x}\right)S(\mathscr{A}_p,z).\tag{3}$$

By (3), the proof of (2) is transformed to the linear sieve problem of estimating the sifting function $S(\mathscr{A}_q,u)$. The usual estimates, being not sufficient, will be bettered in two respects, for individual $S(\mathscr{A},z)$ and for $S(\mathscr{A}_q,u)$ on average. We utilize a variant of the Linnik dispersion method (the idea of Hooley used in [10] for the arithmetic progressions). The method when applied directly would yield the improvement for almost all q in the range $x^{14/15}<q<x$ only. Our purpose will be effected by a combination of the Hooley's method with a new very flexible form of the error term in the linear sieve [11].

3. A New Form of the Error Term in the Linear Sieve

Let $\mathscr{B}$ be a finite sequence of integers (not necessary distinct) and suppose that the quantity

$$|\mathscr{B}_d|=|\{b\in\mathscr{B};\ b\equiv 0(\mathrm{mod}\ d)\}|$$

can be approximated by an expression of the form $\frac{\rho(d)}{d}X$ with some parameter $X>0$ independent of d and with some multiplicative function $\rho(d)$ such that $0\leqq\rho(p)<p$ and

$$\prod_{w\leqq p<z}\left(1-\frac{\rho(p)}{p}\right)^{-1}\leqq\frac{\log z}{\log w}\left(1+\frac{K}{\log w}\right)$$

for any $2\leqq w<z$ with some constant $K\geqq 1$. By [11] we have

Lemma 2. *Let $z\geqq 2$, $M\geqq 2$ and $N\geqq 2$. For any $\eta>0$ we have*

$$S(\mathscr{B},z)\leqq V(z)X\{F(s)+E\}+2^{\eta^{-7}}R(\mathscr{B};M,N),$$
$$S(\mathscr{B},z)\geqq V(z)X\{f(s)-E\}-2^{\eta^{-7}}R(\mathscr{B};M,N),$$

where $s=\log MN/\log z$, $E\ll\eta s^2e^K+\eta^{-8}e^{K-s}(\log MN)^{-1/3}$ and

$$V(z)=\prod_{p<z}\left(1-\frac{\rho(p)}{p}\right).$$

The functions $F(s)$ and $f(s)$ are the continuous solutions of the system of differential-difference equations

$$\begin{aligned}
sf(s) &= 0 && \text{if } 0<s\leqq 2,\\
sF(s) &= 2e^C && \text{if } 0<s\leqq 3,\\
(sf(s))' &= F(s-1) && \text{if } s>2,\\
(sF(s))' &= f(s-1) && \text{if } s>3,
\end{aligned}$$

where $C=0.577\ldots$ is the Euler constant. The second error term $R(\mathscr{B};M,N)$ has the form

$$R(\mathscr{B};M,N)=\sum_{m<M,\ n<N,\ mn\mid P(z)}a_mb_nr(\mathscr{B};mn) \tag{4}$$

where $r(\mathscr{B};d)=|\mathscr{B}_d|-\frac{\rho(d)}{d}X$ and the coefficients a_m,b_n are bounded by 1 in absolute value and depend at most on M,N,z and η.

Since $V(z)X$ should be the order of magnitude of $S(\mathscr{B},z)$ the error term $R(\mathscr{B};M,N)$ is required to be $o(V(z)X)$. There are several treatments of the double sum (4) that lead to $R(\mathscr{B};M,N)=o(V(z)X)$ with values of MN larger than those that are possible for the conventional error term

$$R(\mathscr{B};MN)=\sum_{d<MN,\ (d,\ P(z))=1}|r(\mathscr{B},d)|.$$

4. Estimation of the Error Term $R(\mathscr{A};M,N)$

Here we use a method based on the concept of dispersion. For a given sequence of real numbers b_n such that $|b_n|\leqq 1$ and $b_n=0$ unless n is square-free, let us denote

$$B(x;m,N)=\sum_{n<N,\ (n,\ m)=1}b_nr(\mathscr{A};mn).$$

The bulk of this section will be devoted to the proof of the following result.

Proposition 1. *Let $M<x$ and $\varepsilon>0$. We have*

$$\sum_{M<m<2M} B^2(x;m,N) \ll (1+N^{7/2}M^{-5/4}x)\,x^{1+\varepsilon},$$

the implied constant depending on ε.

Corollary. *For any $\varepsilon>0$ we have*

$$\sum_{m<x^{1-4\varepsilon}} \Big| \sum_{n<x^{1/15-\varepsilon},\,(n,m)=1} b_n r(\mathscr{A};mn)\Big| \ll x^{1-\varepsilon}$$

the implied constant depending on ε.

Proof of Corollary. Letting $N=x^{1/15-\varepsilon}$ it is sufficient to proof that for all $M<x^{1-4\varepsilon}$

$$\sum_{M<m<2M} |B(x;m,N)| \ll x^{1-3\varepsilon/2}.$$

If $M<x^{14/15-\varepsilon}$ the result follows from the trivial estimate

$$|B(x;m,N)| \leqq \rho(m) \sum_{n<N} \rho(n) \ll \rho(m)\,N.$$

If $x^{14/15-\varepsilon} \leqq M < x^{1-4\varepsilon}$, by Proposition 1 and the Cauchy-Schwarz inequality we get

$$\sum_{M<m<2M} |B(x;m,N)| \ll \{(Mx)^{1/2}+N^{7/4}M^{-1/8}x\}\,x^{\varepsilon/8} \ll x^{1-3\varepsilon/2}.$$

Remark. Our former treatment of $R(\mathscr{A};M,N)$ (jointly with Friedlander, see [4]) was based on another idea of Hooley [8]. The error terms $r(\mathscr{A};mn)$ were expressed by some exponential sums being estimated through the use of Lemmata 5 and 6. Analysing this method with greater detail one can prove the following estimate

$$R(\mathscr{A};M,N) \ll \{N^{-1/2}+(N^5M^{-1})^{1/8}+(x^4NM^{-5})^{1/12}\}\,MN\,x^{\varepsilon}. \tag{5}$$

We should mention that the method of Motohashi [16] can also give some useful results.

Turning our attention to the proof of Proposition 1 we write

$$B(x;m,N) = \sum_{n<N,\,(n,m)=1} b_n |\mathscr{A}_{mn}| - \frac{\rho(m)}{m}\,x \sum_{n<N,\,(n,m)=1} b_n \frac{\rho(n)}{n} \tag{6}$$

and

$$\sum_{n<N,\,(n,m)=1} b_n |\mathscr{A}_{mn}| = \sum_{\substack{0<v<m \\ v^2+1\equiv 0 \,(\mathrm{mod}\, m)}} \sum_{\substack{n<N \\ (n,m)=1}} b_n \sum_{\substack{k<x,\ k\equiv v\,(\mathrm{mod}\, m) \\ k^2+1\equiv 0\,(\mathrm{mod}\, n)}} 1.$$

Since $(m, n)=1$ one may expect that the sum $\sum_n b_n \sum_k 1$ is approximately

$$\frac{x}{m} \sum_{n<N,\,(n,m)=1} b_n \frac{\rho(n)}{n} = \frac{x}{m} Y(m), \quad \text{say.} \tag{7}$$

We are thus led to consider the expression

$$\begin{aligned}
\mathscr{M}(x; M, N) &= \sum_{M<m<2M} \sum_{\substack{0<v<m \\ v^2+1\equiv 0 \,(\mathrm{mod}\, m)}} \left\{ \sum_{\substack{n<N \\ (n,m)=1}} b_n \sum_{\substack{k<x \\ k\equiv v \,(\mathrm{mod}\, m) \\ k^2+1\equiv 0 \,(\mathrm{mod}\, n)}} 1 - \frac{x}{m} Y(m) \right\}^2 \\
&= \sum_{M<m<2M} \sum_{\substack{0<v<m \\ v^2+1\equiv 0 \,(\mathrm{mod}\, m)}} \left\{ \sum_{\substack{n_1, n_2<N \\ (n_1 n_2, m)=1}} b_{n_1} b_{n_2} \sum_{\substack{k_1, k_2<x \\ k_1\equiv k_2\equiv v \,(\mathrm{mod}\, m) \\ k_1^2+1\equiv 0 \,(\mathrm{mod}\, n_1) \\ k_2^2+1\equiv 0 \,(\mathrm{mod}\, n_2)}} 1 \right. \\
&\quad \left. -2\frac{x}{m} Y(m) \sum_{\substack{n<N \\ (n,m)=1}} b_n \sum_{\substack{k<x,\, k\equiv v \,(\mathrm{mod}\, m) \\ k^2+1\equiv 0 \,(\mathrm{mod}\, n)}} 1 + \left(\frac{x}{m} Y(m)\right)^2 \right\} \\
&= W(x; M, N) - 2x\, V(x; M, N) + x^2\, U(M, N), \quad \text{say.}
\end{aligned}$$

For the estimation of V and W we shall require the following result about the uniform distribution of the solutions of the congruence $\Omega^2+1\equiv 0 \pmod{mq}$.

Lemma 4. *Let q be a square-free number, $d \mid q$, $2 \nmid d$, $(\mu, d)=1$ and $\omega^2+1\equiv 0 \pmod{d}$. Let $M<M_1<2M$ and $0\leqq\alpha<\beta<1$. Denote by $P(M_1, M; q, d, \mu, \omega, \alpha, \beta)$ the number of all pairs of integers m, Ω satisfying simultaneous conditions*

$$M<m<M_1, \quad (m, q)=1, \quad m\equiv\mu \pmod{d},$$
$$\alpha mq\leqq\Omega<\beta mq, \quad \Omega^2+1\equiv 0 \pmod{mq}, \quad \Omega\equiv\omega \pmod{d}.$$

Then for any $\varepsilon>0$ we have

$$P(M_1, M; q, d, \mu, \omega, \alpha, \beta) = \frac{\pi}{4}(\beta-\alpha)(M_1-M)\,\rho\left(\frac{q}{d}\right)\frac{A(q)}{\varphi(d)} + O((qM)^{3/4+\varepsilon}),$$

where $A(q)=\dfrac{1}{(2, q)}(\varphi(q)/q)^2$.

Corollary 1. *Let qq_1 be a square-free number with $\rho(qq_1)\neq 0$. Let $M<M_1<2M$ and $0\leqq\alpha<\beta<1$. Then for any $\varepsilon>0$ we have*

$$\sum_{\substack{M<m<M_1,\,(m, qq_1)=1 \\ \alpha mq\leqq\Omega<\beta mq,\ \Omega^2+1\equiv 0 \,(\mathrm{mod}\, mq)}} 1 = \frac{\pi}{4}(\beta-\alpha)(M_1-M)\,\rho(q)\,A(qq_1) + O((qq_1 M)^{3/4+\varepsilon}).$$

Proof. The sum in question is equal to $\dfrac{1}{\rho(q_1)} P(M_1, M; 1, 1, 1, \alpha, \beta)$.

Corollary 2. *Let q be a square-free number with $\rho(q)\neq 0$. Let $M<M_1<2M$. Then for any $\varepsilon>0$ we have*

$$\sum_{M<m<M_1,\,(m,q)=1}\rho(m)=\frac{\pi}{4}(M_1-M)\,A(q)+O((qM)^{3/4+\varepsilon}).$$

Proof. The sum in question is equal to $\dfrac{1}{\rho(q)}\,P(M_1, M; q, 1, 1, 1, 0, 1)$.

For the proof of Lemma 4 we shall require three lemmata.

Lemma 5. *There exists a one to one correspondence between the solutions $\Omega \pmod D$ of the congruence $\Omega^2+1\equiv 0 \pmod D$ and the pairs of numbers (r, s) such that*

$$D=r^2+s^2, \quad (r,s)=1, \quad |r|<s.$$

The correspondence is given through the formula

$$\Omega=\frac{\bar r}{s}(r^2+s^2)-\frac{r}{s}.$$

The notation $\bar r$, used either in $\dfrac{\bar r}{s}$ or in a congruence $(\bmod s)$ means that $r\bar r\equiv 1 \pmod s$. This result, due of course to Lagrange is to be found in Article 86 of [20]. The second lemma, due to Hooley, (Lemma 3 of [8]) is a consequence of Weil's estimate for Kloosterman sums.

Lemma 6. *If h and s are integers and if $0<r_2-r_1<2s$ then*

$$\sum_{\substack{r_1<r<r_2,\,(r,s)=1\\ r\equiv\lambda \pmod\Delta}} e\left(h\frac{\bar r}{s}\right)\ll s^{\frac12+\varepsilon}(h,s)^{\frac12}$$

the implied constant depending on ε.

We shall also require an approximation of the characteristic function $\psi(t)$ of the interval $[\alpha, \beta) \pmod 1$ by two absolutely convergent Fourier series.

Lemma 7. *Let $2\Delta<\beta-\alpha<1-2\Delta$. There exist two functions $A(t)$ and $B(t)$ such that*

$$|\psi(t)-A(t)|=B(t)$$

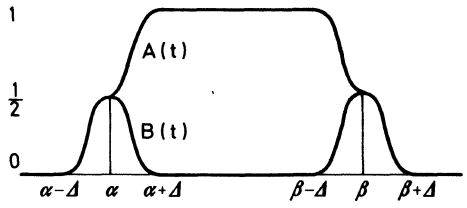
and

$$A(t)=\beta-\alpha+\sum_{h\neq 0}A_h\,e(ht)$$

$$B(t)=\Delta+\sum_{h\neq 0}B_h\,e(ht)$$

with Fourier coefficients A_h, B_h satisfying $|A_h|, |B_h|\leqq\min\left(\dfrac{1}{|h|}, \dfrac{\Delta^{-2}}{|h|^3}\right)=C_h$.

Proof. Take the functions drawn below



Now we are ready to prove Lemma 4. By Lemma 7 we have

$$P(M_1, M; q, d, \mu, \omega, \alpha, \beta) = (\beta-\alpha) \sum_{\substack{M<m<M_1,\ (m,q)=1,\ m\equiv\mu \pmod d \\ 0\leqq\Omega<qm,\ \Omega^2+1\equiv 0 \pmod{mq},\ \Omega\equiv\omega \pmod d}} 1$$
$$+O\{\rho(q)\,\Delta M+\sum_{h\neq 0} C_h \Big| \sum_{\substack{M<m<M_1,\ (m,q)=1,\ m\equiv\mu \pmod d \\ 0\leqq\Omega<qm,\ \Omega^2+1\equiv 0 \pmod{mq},\ \Omega\equiv\omega \pmod d}} e(h\Omega/mq)\Big|\}.$$

The main term is easy to handle. Since $d \mid q$, $(d, mq/d)=1$ and $(\mu, d)=1$ it is equal to

$$\rho(q/d) \sum_{\substack{M<m<M_1,\ (m,q)=1 \\ m\equiv\mu \pmod d}} \rho(m) = \rho(q/d) \sum_{\substack{a\leqq M^{\frac12} \\ (a,q)=1}} \chi_4(a) \sum_{\substack{\frac{M}{a}<b<\frac{M_1}{a},\ (b,q/d)=1 \\ b\equiv\mu\bar a \pmod d}} 1$$
$$+\rho(q/d) \sum_{\substack{b<2M^{\frac12} \\ (b,q)=1}} \sum_{\substack{\max\left(\frac{M}{b}, M^{\frac12}\right)<a<\frac{M_1}{b} \\ a\equiv\mu\bar b \pmod d,\ (a,q/d)=1}} \chi_4(a)$$

$$=\rho(q/d) \sum_{a\leqq M^{\frac12},\ (a,q)=1} \chi_4(a) \left\{\varphi\left(\frac{q}{d}\right)\frac{M_1-M}{aq}+O(d(q))\right\}+O(\rho(q)\,M^{\frac12})$$
$$=\rho\left(\frac{q}{a}\right)\varphi\left(\frac{q}{d}\right)\frac{M_1-M}{q}\,L(1,\chi_4)\prod_{p\mid q}\left(1-\frac{\chi_4(p)}{p}\right)+O(\rho(q)\,M^{\frac12})$$

and this is the required main term. Proceeding to the exponential sum $\sum_{m,\Omega} e(h\Omega/mq)$ we first eliminate the condition $(m, q)=1$ as follows

$$\sum_{m,\Omega} e(h\Omega/mq) = \sum_{l\mid\frac{q}{d}} \mu(l) \sum_{\substack{qM<D<qM_1,\ D\equiv 0 \pmod{lq},\ D\equiv\mu q \pmod{dq} \\ 0\leqq\Omega<D,\ \Omega^2+1\equiv 0 \pmod D,\ \Omega\equiv\omega \pmod d}} e(h\Omega/D).$$

Now we apply Lemma 5. Writing $D=r^2+s^2$, $(r, s)=1$, $|r|<s$, the congruence conditions $D\equiv 0 \pmod{lq}$, $D\equiv\mu q \pmod{dq}$ and $\Omega\equiv\omega \pmod d$ are equivalent to

$$r^2+s^2\equiv 0 \pmod{lq}, \quad r^2+s^2\equiv\mu q \pmod{dq}, \text{ and } r+\omega s\equiv 0 \pmod d.$$

Therefore

$$\sum_{D,\Omega} e(h\Omega/D) \leqq d(q) \sum_{(\frac12 qM)^{\frac12}<s<(2qM)^{\frac12}} \sup_{\substack{r_1, r_2 \\ \lambda, \Lambda}} \left| \sum_{\substack{(r,s)=1 \\ r_1<r<r_2 \\ r\equiv\lambda \pmod\Lambda}} e\left(h\frac{\bar r}{s}-\frac{hr}{s(r^2+s^2)}\right)\right|$$

where the supremum is taken over all integers r_1, r_2, λ, Λ such that $0<r_2-r_1<2s$. By Lemma 6 and partial summation we obtain

$$\sum_{D,\Omega} e(h\Omega/D) \ll \left(1+\frac{h}{qM}\right)(qM)^{1/4+\varepsilon} \sum_{s<(2qM)^{\frac{1}{2}}} (h,s)^{\frac{1}{2}}$$

$$\ll \left(1+\frac{h}{qM}\right)(qM)^{3/4+\varepsilon} d(h),$$

the implied constant depending on ε. Obviously the same holds for the sum $\sum_{m,\Omega} e(h\Omega/qm)$. Since $C_h \leqq \min\left(\frac{1}{|h|}, \Delta^{-2}|h|^{-3}\right)$ we have

$$\sum_{h\neq 0} C_h \left(1+\frac{h}{qM}\right) d(h) \ll \left(1+\frac{1}{\Delta qM}\right)(\log \Delta)^2.$$

On choosing $\Delta = 1/qM$ we complete the proof of Lemma 4.

Estimation of $U(M,N)$

By the definition of $U(M,N)$ given by (7) and (8), we have

$$\begin{aligned} U(M,N) &= \sum_{M<m<2M} \sum_{\substack{0\leqq v<m \\ v^2+1\equiv 0 \,(\mathrm{mod}\, m)}} \frac{1}{m^2} \sum_{\substack{n_1,n_2<N \\ (n_1n_2,m)=1}} b_{n_1} b_{n_2} \frac{\rho(n_1)\rho(n_2)}{n_1 n_2} \\ &= \sum_{n_1,n_2<N} b_{n_1} b_{n_2} \frac{\rho(n_1)\rho(n_2)}{n_1 n_2} \sum_{\substack{M<m<2M \\ (m,n_1n_2)=1}} \frac{\rho(m)}{m^2} \\ &= \sum_{n_1,n_2<N} b_{n_1} b_{n_2} \frac{\rho(n_1)\rho(n_2)}{n_1 n_2} \\ &\quad \cdot \left\{\frac{\pi}{8} A(n_1n_2) M^{-1} + O((n_1n_2M)^{3/4+\varepsilon} M^{-2})\right\} \\ &= \frac{\pi}{8} M^{-1} \sum_{n_1,n_2<N} b_{n_1} b_{n_2} \frac{\rho(n_1)\rho(n_2)}{n_1 n_2} A(n_1n_2) + O(M^{-2}(N^2M)^{3/4+\varepsilon}) \end{aligned} \tag{9}$$

the penultimate line having been formed through the use of Corollary 2 of Lemma 4 and partial summation. This is the required estimate for $U(M,N)$.

Estimation of $V(x;M,N)$

Expressing $V(x;M,N)$ as a multiple sum through (7) and (8), we have

$$\begin{aligned} V(x;M,N) = \sum_{\substack{M<m<2M \\ 0\leqq v<m,\, v^2+1\equiv 0 \,(\mathrm{mod}\, m)}} \frac{1}{m} \Bigg(\sum_{\substack{n_1<N \\ (m,n_1)=1}} b_{n_1} \sum_{\substack{k<x \\ k\equiv v \,(\mathrm{mod}\, m) \\ k^2+1\equiv 0 \,(\mathrm{mod}\, n_1)}} 1 \Bigg) \\ \cdot \left(\sum_{\substack{n_2<N \\ (n_2,m)=1}} b_{n_2} \frac{\rho(n_2)}{n_2} \right). \end{aligned}$$

Changing the order of summation, we obtain

$$V(x; M, N) = \sum_{n_1, n_2 < N} b_{n_1} b_{n_2} \frac{\rho(n_2)}{n_2} \sum_{\Theta_{n_1, n_2}} \frac{1}{m}$$
$$= \sum_{n_1, n_2 < N} b_{n_1} b_{n_2} \frac{\rho(n_2)}{n_2} S(n_1, n_2; x, M)$$

where, for given n_1, n_2, the symbol Θ_{n_1, n_2} indicates summation over all pairs of integers m, v and k satisfying simultaneous conditions

$$M < m < 2M, \quad (m, n_1 n_2) = 1,$$
$$0 \leqq v < m, \quad v^2 + 1 \equiv 0 \pmod{m},$$
$$k < x, \quad k \equiv v \pmod{m}, \quad k^2 + 1 \equiv 0 \pmod{n_1}.$$

To evaluate $S(n_1, n_2; x, M)$ it is necessary to reinterpret the conditions implied by Θ_{n_1, n_2}. First the condition $k \equiv v \pmod{m}$ taken with the inequalities satisfied by m, v and k, means that there is a non-negative integer $l < x/M$ such that

$$k = v + ml.$$

Therefore $S(n_1, n_2; x, M)$ can be transformed into a sum over variables of summation m, v and l, where the latter are subject to the conditions

$$M < m < \min\left(2M, \frac{x - v}{l}\right), \quad (m, n_1 n_2) = 1,$$
$$0 \leqq v < m, \quad v^2 + 1 \equiv 0 \pmod{m}, \quad (ml + v)^2 + 1 \equiv 0 \pmod{n_1}. \tag{10}$$

Since $0 \leqq v < m$, if one replaces $\frac{x - v}{l}$ by $\frac{x}{l}$ the total error will be less than

$$\sum_{n_2 < N} \frac{\rho(n_2)}{n_2} \sum_{1 \leqq l < \frac{x}{M}} \sum_{\frac{x - 2M}{l} < m < \frac{x}{l}} m^{-1} \sum_{\substack{0 \leqq v < m \\ v^2 + 1 \equiv 0 \pmod{m}}} \sum_{n_1 \mid (ml + v)^2 + 1} 1 \ll x^{\varepsilon} \tag{11}$$

and this is sufficient for our needs. Let $S^*(n_1, n_2; x, M)$ stand for the modified sum. We shall reinterpret the congruence conditions (10) once more. Letting c to be the solution of

$$c \equiv l \pmod{n_1} \quad \text{and} \quad 0 \leqq c < n_1$$

and defining $\Omega = cm + v$ we obtain

$$cm \leqq \Omega < (c + 1)m \quad \text{and} \quad \Omega^2 + 1 \equiv 0 \pmod{mn_1}.$$

We deduce from the above that

$$S^*(n_1, n_2; x, M) = \sum_{l < \frac{x}{M}} \sum_{\substack{M < m < \min\left(2M, \frac{x}{l}\right), (m, n_1 n_2) = 1 \\ cm \leqq \Omega < (c+1)m,\ \Omega^2 + 1 \equiv 0 \pmod{mn_1}}} m^{-1}.$$

We infer from Corollary 1 of Lemma 4 and partial summation that

$$S^*(n_1, n_2; x, M) = \sum_{1 < x/M} \frac{\pi}{4} \frac{p(n_1)}{n_1} A(n_1 n_2) \cdot \left\{ \log \left(\min \left(2, \frac{x}{lM} \right) \right) + O(M^{-1}(n_1 n_2 M)^{3/4+\varepsilon}) \right\}$$
$$= \frac{\pi}{8} \frac{x}{M} \frac{\rho(n_1)}{n_1} A(n_1 n_2) + O\left(\frac{\rho(n_1)}{n_1} \left(\log x + \frac{x}{M^2} (n_1 n_2 M)^{3/4+\varepsilon} \right) \right) \quad (12)$$

The required estimate for $V(x; M, N)$ is now available. We obtain from (11) and (12) that

$$V(x; M, N) = \frac{\pi}{8} \frac{x}{M} \sum_{n_1, n_2 < N} b_{n_1} b_{n_2} \frac{\rho(n_1)\rho(n_2)}{n_1 n_2} A(n_1 n_2) + O(x^{\varepsilon}) + O(xM^{-2}(N^2 M)^{3/4+\varepsilon}). \quad (13)$$

Estimation of $W(x; M, N)$

By the definition of $W(x; M, N)$ given by (8), after changing the order of summations, we obtain

$$W(x; M, N) = \sum_{n_1, n_2 < N} b_{n_1} b_{n_2} \sum_{\Phi_{n_1, n_2}} 1 = \sum_{n_1, n_2 < N} b_{n_1} b_{n_2} T(n_1, n_2; x, M) \quad (14)$$

where, for given n_1, n_2, the symbol Φ_{n_1, n_2} indicates that the summation is over all sets of integers m, v, k_1 and k_2 satisfying the simultaneous conditions

$$M < m < 2M, \quad (m, n_1 n_2) = 1,$$
$$0 \leqq v < m, \quad v^2 + 1 \equiv 0 \pmod{m},$$
$$k_1 < x, \quad k_2 < x, \quad k_1 \equiv k_2 \equiv v \pmod{m},$$
$$k_1^2 + 1 \equiv 0 \pmod{n_1}, \quad k_2^2 + 1 \equiv 0 \pmod{n_2}.$$

We scrutinize the significance of Φ_{n_1, n_2} much as we did that of Θ_{n_1, n_2}. Writing

$$k_1 = ml_1 + v, \quad k_2 = ml_2 + v,$$

where l_1, $l_2 < x/M$ we see that $T(n_1, n_2; x, M)$ can be transformed into a sum over variables of summation m, v, l_1 and l_2 that are constrained by the conditions

$$M < m < \min \left(2M, \frac{x - v}{l_1}, \frac{x - v}{l_2} \right), \quad (m, n_1 n_2) = 1,$$
$$0 \leqq v < m, \quad v^2 + 1 \equiv 0 \pmod{m}, \quad (ml_1 + v)^2 + 1 \equiv 0 \pmod{n_1},$$
$$(ml_2 + v)^2 + 1 \equiv 0 \pmod{n_2}.$$

Since $0 \leqq v < m$, if one replaces $x - v$ by x the total error will be less than

$$2 \sum_{l_1 \leqq l_2 < \frac{x}{M}} \sum_{\frac{x-2M}{l_2} < m < \frac{x}{l_2}} \sum_{\substack{0 \leqq v < m \\ v^2+1 \equiv 0 \pmod m}} \sum_{\substack{n_1 \mid (ml_1+v)^2+1 \\ n_2 \mid (ml_2+v)^2+1}} 1 \ll x^{1+\varepsilon}$$

and this is just what we need. Let $T^*(n_1, n_2; x, M)$ stand for the modified sum. For computing $T^*(n_1, n_2; x, M)$ we shall further reinterpret the congruence conditions (15), as we did that of (10). Letting c to be the solution of

$$c \equiv l_1 \left(\mathrm{mod}\ \frac{n_1}{(n_1, n_2)}\right), \quad c \equiv l_2 \left(\mathrm{mod}\ \frac{n_2}{(n_1, n_2)}\right),$$
$$c \equiv l_1 \ (\mathrm{mod}\ (n_1, n_2)), \quad 0 \leqq c < [n_1, n_2]$$

and defining $\Omega = cm + v$ we obtain

$$cm \leqq \Omega < (c+1)m \quad \text{and} \quad \Omega^2 + 1 \equiv 0 \ (\mathrm{mod}\ m[n_1, n_2]).$$

If the congruence conditions (15) are soluble then

$$l_1 \equiv l_2 \ (\mathrm{mod}\ (2, n_1, n_2)),$$

so $d = (n_1, n_2)/(n_1, n_2, l_1 - l_2)$ is odd number. From the congruences $(\mathrm{mod}\ n_1)$ and $(\mathrm{mod}\ n_2)$ we derive

$$v \equiv -m\frac{l_1+l_2}{2} \ (\mathrm{mod}\ d) \quad \text{and} \quad \left(m\frac{l_1-l_2}{2}\right)^2 + 1 \equiv 0 \ (\mathrm{mod}\ d).$$

Thus, if μ stands for the class of $m \ (\mathrm{mod}\ d)$ then

$$\left(\mu\frac{l_1-l_2}{2}\right)^2 + 1 \equiv 0 \ (\mathrm{mod}\ d) \quad \text{and} \quad \Omega \equiv \omega \ (\mathrm{mod}\ d)$$

where $\omega = \left(c - \frac{l_1+l_2}{2}\right)\mu = \frac{l_1-l_2}{2}\mu$. We deduce from the above discussion that

$$T^*(n_1, n_2; x, M) = \sum_{\substack{l_1, l_2 < x/M \\ l_1 \equiv l_2 \ (\mathrm{mod}\ (2, n_1, n_2))}} \sum_{\mu} \sum_{\substack{M < m < \min\left(2M, \frac{x}{l_1}, \frac{x}{l_2}\right), \ (m, n_1 n_2) = 1 \\ m \equiv \mu \ (\mathrm{mod}\ d), \ \Omega \equiv \omega \ (\mathrm{mod}\ d) \\ cm \leqq \Omega < (c+1)m, \ \Omega^2 + 1 \equiv 0 \ (\mathrm{mod}\ m[n_1, n_2])}} 1.$$

The inner sum is just $P(M_1, M; q, d, \mu, \omega, \alpha, \beta)$ with $M_1 = \min\left(2M, \frac{x}{l_1}, \frac{x}{l_2}\right)$, $q = [n_1, n_2]$, $\alpha = c/q$ and $\beta = (c+1)/q$, thus by Lemma 4 we obtain

$$T^*(n_1, n_2; x, M)$$

$$= \sum_{\substack{l_1, l_2 < x/M \\ l_1 \equiv l_2 \,(\mathrm{mod}\,(2, n_1, n_2))}} \rho(d) \left\{ \frac{\pi}{4q} \rho\left(\frac{q}{d}\right) \frac{A(q)}{\varphi(d)} \left[\min\left(2M, \frac{x}{l_1}, \frac{x}{l_2}\right) - M \right] + O((qM)^{3/4+\varepsilon}) \right\}$$

$$= \frac{\pi}{2} \frac{\rho([n_1, n_2])}{n_1 n_2} \frac{A(n_1 n_2)}{\varphi(n_1, n_2)} \cdot \sum_{\substack{l_1 < l_2 < x/M \\ l_1 \equiv l_2 \,(\mathrm{mod}\,(2, n_1, n_2))}} \varphi((n_1, n_2, l_1 - l_2)) \left[\min\left(2M, \frac{x}{l_2}\right) - M \right] + O\left(\frac{(n_1, n_2)}{n_1 n_2} \rho(n_1 n_2)\, x\right) + O\left(\left(\frac{x}{M}\right)^2 (n_1 n_2 M)^{3/4+\varepsilon}\right). \tag{16}$$

Writing $\varphi = \psi * 1$ we find $\psi(p) = p - 2$ and hence

$$\sum_{\substack{0 < l_1 < l_2 \\ l_1 \equiv l_2 \,(\mathrm{mod}\,(2, n_1, n_2))}} \varphi((n_1, n_2, l_1 - l_2)) = \sum_{t \mid (n_1, n_2)} \psi(t) \sum_{\substack{0 < l_1 < l_2 \\ l_1 \equiv l_2 \,(\mathrm{mod}\,(2, n_1, n_2)\, t)}} 1$$

$$= \sum_{t \mid (n_1, n_2)} \psi(t) \left\{ \frac{l_2}{(2, n_1, n_2)\, t} + O(1) \right\}$$

$$= \frac{d((n_1, n_2))}{(2, n_1, n_2)} \frac{\varphi((n_1, n_2))}{(n_1, n_2)} l_2 + O(\varphi((n_1, n_2)). \tag{17}$$

We have also

$$\sum_{l_2 < x/M} l_2 \left[\min\left(2M, \frac{x}{l_2}\right) - M \right] = \frac{x^2}{4M} + O(x). \tag{18}$$

Collecting together (16), (17) and (18), by

$$\rho(n_1, n_2) \frac{d((n_1, n_2))}{(2, n_1, n_2)} = \rho(n_1)\, \rho(n_2).$$

we obtain

$$T^*(n_1, n_2; x, M) = \frac{\pi x^2}{8M} \frac{\rho(n_1)\, \rho(n_2)}{n_1 n_2} A(n_1 n_2) + O\left(\frac{\rho(n_1 n_2)}{n_1 n_2} (n_1, n_2)\, x\right) + O\left(\left(\frac{x}{M}\right)^2 (n_1 n_2 M)^{3/4+\varepsilon}\right).$$

Therefore

$$W(x; M, N) = \frac{\pi}{8} \frac{x^2}{M} \sum_{n_1, n_2 < N} b_{n_1} b_{n_2} \frac{\rho(n_1)\, \rho(n_2)}{n_1 n_2} A(n_1 n_2) + O(x^{1+\varepsilon}) + O\left(\left(\frac{x}{M}\right)^2 N^2 (N^2 M)^{3/4+\varepsilon}\right). \tag{19}$$

This is the required estimate for $W(x; M, N)$.

If we introduce (9), (13) and (19) into (8) we find that the leading terms disappear throughout and we are left with

$$\mathscr{M}(x; M, N) \ll (x + N^{7/2} M^{-5/4} x^2) x^{\varepsilon}, \tag{20}$$

the implied constant depending on ε. The proof of Proposition 1 follows from (6), (8), (20), the Cauchy-Schwarz inequality and the inequality $\rho(m) \ll m^{\varepsilon}$.

Concluding this section we remark that assuming Hooley's Hypothesis R about incomplete Kloosterman-Ramanujan's sums (see [9]) the exponent 1/15 in Corollary of Proposition 1 may be increased to 1/9.

5. Estimation of the Sifting Functions

The following result is almost immediate consequence of Lemma 2 and Corollary of Proposition 1.

Proposition 2. *Let* $y = x^{16/15}$, $0 < \gamma < \frac{1}{2}$, $z = x^{\gamma}$, $z \leqq z_q < x^{\frac{1}{2}}$ *and* $0 \leqq c_q \leqq 1$. *Then for any* $\varepsilon > 0$ *we have*

$$\sum_{\substack{q < x^{1-\varepsilon} \\ (q, P(z_q)) = 1}} c_q S(\mathscr{A}_q; z_q)$$

$$< V(z) x \left\{ \sum_{\substack{q < x^{1-\varepsilon} \\ (q, P(z_q)) = 1}} c_q \frac{\rho(q)}{q} F\left(\frac{\log y/q}{\log z_q}\right) \frac{\log z}{\log z_q} + O_{\gamma}(\varepsilon) \right\},$$

provided $x > x_0(\varepsilon, \gamma)$. *A similar result holds for lower bound, the function* $F(s)$ *being replaced by* $f(s)$.

Proof. For any pair of real numbers Q, Z such that $1 \leqq Q < x^{1-\varepsilon}$ and $z \leqq Z < x^{\frac{1}{2}}$ define $H(Q, Z)$ to be the set of that integers q which satisfy simultaneous conditions

$$Q \leqq q < 2Q, \quad Z \leqq z_q < 2Z, \quad (q, P(z_q)) = 1.$$

Now, for each q from $H(Q, Z)$ we apply Lemma 2 to the sequence $\mathscr{B} = \mathscr{A}_q$ with the parameter $X = \dfrac{\rho(q)}{q} x$ as to have $r(\mathscr{B}, d) = r(\mathscr{A}_q, d) = r(\mathscr{A}, qd)$ for all $d \mid P(Z)$. We obtain

$$S(\mathscr{A}_q, z_q) \leqq S(\mathscr{A}_q, Z) < V(Z) \frac{\rho(q)}{q} x \left\{ F\left(\frac{\log MN}{\log Z}\right) + O(\eta) \right\}$$

$$+ 2^{\eta^{-10}} \sum_{m < M,\, m \mid P(Z)} \Big| \sum_{n < N,\, (m, n) = 1} b_n r(\mathscr{A}, qmn) \Big| \tag{21}$$

with arbitrary $\eta > 0$ and $M > 1$, $N > 1$ subject to $MN > c(\gamma, \eta)$. The implied constant in the symbol $O(\eta)$ depends at most on γ. The best choice for M and N is

$$M = x^{1-\varepsilon}/Q \quad \text{and} \quad N = x^{1/15-\varepsilon}.$$

If we multiply (21) throughout by c_q and sum with respect to q from $H(Q,Z)$ we get

$$\sum_{q\in H(Q,\,Z)} c_q S(\mathscr{A}_q, z_q)$$
$$< \left(\sum_{q\in H(Q,\,Z)} c_q \frac{\rho(q)}{q}\right) V(Z)\, x\left\{F\left(\frac{\log(y/Qx^{2\varepsilon})}{\log Z}\right)+O(\eta)\right\}$$
$$+O_\varepsilon(2^{\eta^{-10}}x^{1-\varepsilon}),$$

the latter error term being estimated by means of Corollary of Proposition 1. But we have

$$V(Z)=V(z)\frac{\log z}{\log z_q}\left(1+O\left(\frac{1}{\log z}\right)\right)$$

and

$$F\left(\frac{\log(y/Qx^{2\varepsilon})}{\log Z}\right)=F\left(\frac{\log(y/q)}{\log z_q}\right)+O_\gamma(\varepsilon).$$

Therefore, since the number of disjoint classes $H(Q,Z)$ covering all q can be less than $4(\log x)^2$, the summation over the classes completes the proof.

6. Estimation of $W(\mathscr{A}, z)$

Apart from the short sum

$$\sum_{x^{1-\varepsilon}\le p<x}\left(1-\frac{\log p}{\log x}\right)S(\mathscr{A}_p, z) \tag{22}$$

in (3) the weighted sum $W(\mathscr{A}, z)$ can be estimated by means of Proposition 2. For (22) the crude estimate

$$S(\mathscr{A}_p, z)\ll\frac{x}{p\log(x/p)}$$

is sufficient giving the bound $O(\varepsilon x/\log x)$. If we replace sums by integrals, as we can do by partial summation, we obtain

$$W(\mathscr{A}, z) > V(z)\, x\left\{f\left(\frac{\alpha}{\gamma}\right)+\int_\gamma^{\frac12}\int_\gamma^{u}(u-t)\frac{\gamma}{t}f\left(\frac{\alpha-u-t}{t}\right)\frac{du}{u}\frac{dt}{t}\right.$$
$$-\int_\gamma^{\frac12}\left[(1-2u)\frac{\gamma}{u}F\left(\frac{\alpha-u}{u}\right)+uF\left(\frac{\alpha-u}{\gamma}\right)\right]\frac{du}{u}$$
$$\left.-\int_{\frac12}^{1}(1-u)F\left(\frac{\alpha-u}{\gamma}\right)\frac{du}{u}-\varepsilon\right\}$$
$$=V(z)\,x\{W-\varepsilon\},\quad \text{say}$$

where $\alpha = 16/15$ and ε is any positive number, provided $x > x_0(\varepsilon, \gamma)$. Note that from Mertens prime number theorem we have

$$V(z) \sim \Gamma_G e^{-C} (\log z)^{-1} = \Gamma_G e^{-C} (\gamma \log x)^{-1}.$$

Now our aim is to prepare the integrals for estimation by manual calculations. We succeed partly in reducing the problem to the estimation of few logarithms of rational numbers. The functions $F(s)$ and $f(s)$ are elementary in the intervals $0 < s \leqq 3$ and $0 < s \leqq 4$ only, while our choice of $\gamma = 1/5$ necessitaties our considering the arguments of F and f beyond the elementary ranges.

From the differential-difference equations for $F(s)$ and $f(s)$ it is easy to derive the formulae

$$sF(s) = 2e^C \left\{1 + \int_2^{s-1} \log(u-1) \frac{du}{u}\right\} \quad \text{if } 3 \leqq s \leqq 5,$$

$$sf(s) = 2e^C \left\{\log(s-1) + \int_3^{s-1} \int_2^{t-1} \log(u-1) \frac{du}{u} \frac{dt}{t}\right\} \quad \text{if } 4 \leqq s \leqq 6.$$

Hence one can find that

$$W = 2e^C \gamma \left\{\log(\alpha - \gamma) - \frac{\alpha - 1}{\alpha} \log(\alpha - 1) - \int_2^{\frac{\alpha}{\gamma} - 2} \left[t \log \frac{\alpha(t+1)}{(\alpha-\gamma)(t+2)} + \log\left(1 - \frac{\gamma t}{\alpha - \gamma}\right)(t+1)\right] \frac{\log(t-1)}{t(t+1)} dt\right\}$$

and applying the inequality $\log(1+a) \leqq a$ we obtain

$$W \geqq 2e^C \gamma \left\{\log(\alpha - \gamma) - \frac{\alpha - 1}{\alpha} \log(\alpha - 1) - \frac{\alpha}{\alpha - \gamma} \int_4^{\alpha/\gamma} \left(\frac{1}{t} - \frac{\gamma}{\alpha}\right) \log(t-3)\, dt\right\}$$

$$\geqq 2e^C \gamma \left\{\log(\alpha - \gamma) - \frac{\alpha - 1}{\alpha} \log(\alpha - 1) - \frac{\alpha}{\alpha - \gamma} \int_4^{\alpha/\gamma} \left(\frac{1}{t} - \frac{\gamma}{\alpha}\right)(t-4)\, dt\right\}$$

$$= 2e^C \gamma \left\{\log(\alpha - \gamma) - \frac{\alpha - 1}{\alpha} \log(\alpha - 1) - \frac{\alpha}{\alpha - \gamma} \left(\frac{\alpha}{2\gamma} - 8\frac{\gamma}{\alpha} - 4 \log \frac{\alpha}{4\gamma}\right)\right\}.$$

On substituting $\alpha = 16/15$ and $\gamma = 1/5$ we finally get

$$W > 2e^C \gamma / 154 \quad \text{and} \quad W(\mathcal{A}, z) > \frac{\Gamma_G}{77} \frac{x}{\log x}.$$

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Intersection Numbers for Quotients of the Complex 2-Ball and Hilbert Modular Forms

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§1. Introduction

Let $D=\{z=(z_1,z_2)\in\mathbf{C}^2$ with $|z_1|^2+|z_2|^2<1\}$ be the complex 2-ball. Take a totally real number field $\mathfrak{k}$ with $|\mathfrak{k}:\mathbf{Q}|=g$, and a totally imaginary quadratic extension $K/\mathfrak{k}$. Let V be a 3-dimensional vector space over K, and H a non-degenerate σ-Hermitian form on V, where σ is the Galois automorphism of $K/\mathfrak{k}$. Assume that the signature of H is (2, 1) at the first r archimedian completions of $\mathfrak{k}$ and (3, 0) at the remaining $g-r$ such completions. Then the unit group $\Gamma(L)\subset U(V,H)$ of an $\mathcal{O}_K$-lattice $L\subset V$ contains a congruence subgroup Γ which acts freely on D^r. If we assume that $r<g$, then H is anisotropic and the quotient $U=\Gamma\backslash D^r$ is a compact complex manifold with $\dim_{\mathbf{C}} U=2r$. In this paper we will show there is a relation between a certain subspace of $H^{2r}(U,\mathbf{C})$ and certain space of Hilbert modular forms for $\mathfrak{k}$.

To be more precise, for each $X\in V$ such that $H[X]\gg 0$ we will construct an r-cycle $C_X\subset U$, and to each $\beta\in\mathcal{O}_{\mathfrak{k}}$, $\beta\gg 0$, we will associate a reducible cycle $C_\beta=\bigcup_j C_{Y_j}$; constructed out of the Γ-non-associate representations Y_j of β by H. Let $I(C_X, C_Y)$ denote the intersection number of the (co)-homology classes $[C_X]$ and $[C_Y]\in H^{2r}(U,\mathbf{Z})$. Then the function

$$G(\tau,X)=2^{-r}E(\Gamma_X\backslash D_X)+|U_K^1|\sum_{\substack{\beta\in\mathcal{O}_{\mathfrak{k}}\\ \beta\gg 0}} I(C_X,C_\beta)\,\mathbf{e}[tr_{\mathfrak{k}}(\tau\beta)]$$

with $\tau\in\mathfrak{H}^g$, $\mathfrak{H}=$ upper half plane, is a Hilbert modular form of Nebentypus and character $\left(\frac{*}{\Delta}\right)$, where Δ is the discriminant of $K/\mathfrak{k}$. See Sections 1, 2 and 6 for details. Extending by linearity, we get a mapping from the subspace of $H^{2r}(U,\mathbf{C})$ spanned by the $[C_X]$'s to Hilbert modular forms.

A mapping of this sort for quotients of $\mathfrak{H}\times\mathfrak{H}$ was first observed and proved by Hirzebruch and Zagier [7], [6], who observed that it is the adjoint of the

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Doi-Naganuma lifting with respect to the Petersson inner product [18]. On the other hand, in a much more general setting, Millson [11] has shown that the cycles produced from certain subsymmetric spaces, in arithmetic quotients of Riemannian symmetric spaces represent non-trivial cohomology classes, and that in some cases these cycles are not dual to any invariant form. The results of this paper and recent results of Oda [12], raise the hope that a connection between intersection numbers and modular forms may actually extend to this general setting. Such a connection should probably be viewed as a geometric embodiment of the correspondence between automorphic forms which arise from the Weil representation, although the connection is not so clearly apparent in the proofs given here. The analogue of the Doi-Naganuma lifting in our case has recently been investigated by Howe [9] for finite fields.

If $r=g$, so that $U=\Gamma\backslash D^r$ is not compact, then our main result, namely that $G(\tau, X)$ is a Hilbert modular form of weight 3 and Nebentypus, remains true if the Euler number $E(\Gamma_X\backslash D_X)$ in the definition of $I(C_X, C_\beta)$ (cf. Propositions 5.1 and 5.2) is replaced by the volume integral $V(\Gamma_X\backslash D_X)=\int_{\Gamma_X\backslash D_X}\omega$ (ω=Gauss-Bonnet form).

At present I do not have enough information about the cohomology of U and about the analogue of the Doi-Naganuma map to make any precise conjectures about the dimensions of the subspace of $H^{2r}(U)$ spanned by the $[C_X]$'s and about the image of the map $[C_X]\mapsto G(\tau, X)$. More detailed information might be available in the case $k=\mathbf{Q}$ provided the contributions of the additional intersections occurring in the cusps were taken into account.

In fact, if $r=g=1$ so that $k=\mathbf{Q}$ and the quotient $\Gamma\backslash D=U$ is non-compact, our main theorem should still hold with the following modification. A smooth compactification $\tilde{U}$ of U can be obtained by adding a torus S_σ to each cusp [4]. Then the cycles C_X, compactified when necessary, define cohomology classes $[C_X]\in H^2(\tilde{U})$. Now $H^2(\tilde{U})=\mathrm{Im}(H_c^2(U)\to H^2(\tilde{U}))\oplus H'$ where H' is the subspace generated by the S_σ's. Motivated by the results of [7], let $[C_X]^c$ denote the first component of $[C_X]$ in this decomposition. Then it *should* again be true that the intersection number $[C_X]^c\cup[C_Y]^c=I(C_X, C_Y)$, where I is defined as above, using the volume integral. I hope to return to this case in the future.

Finally, the methods of this paper carry over with only slight modification to the compact analogues of the Hilbert modular varieties defined by quaternion algebras.

Sections 2 and 3 contain the construction of U and the C_X. Sections 4 and 5 describe the intersection of such cycles. Section 6 contains the definition of the C_β's and the statement of the main theorem, whose proof occupies sections 7 and 8.

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§2. Construction of U

We will use the following notation:

$\mathfrak{k}$ = totally real number field, $|\mathfrak{k}:\mathbf{Q}|=g$.
$\mathfrak{k}_\lambda$, $\lambda=1, \ldots, g$, the archimedian completions of $\mathfrak{k}$.
$\mathcal{O}_\mathfrak{k}$ = integers of $\mathfrak{k}$.
$\mathfrak{D}_\mathfrak{k}$ = different of $\mathfrak{k}/\mathbf{Q}$.
For $\alpha\in\mathfrak{k}$, $\alpha\gg 0$ means α is totally positive.
K = totally imaginary quadratic extension of $\mathfrak{k}$.
σ = Galois automorphism of $K/\mathfrak{k}$.
K_λ, $\lambda=1, \ldots, g$, the archimedian completions of K.

Let V be a vector space over K, $\dim_K V=3$, and let H be a non-degenerate σ-Hermitian form on V. Suppose that

$$\text{signature } (V_\lambda, H)=\begin{cases}(2,1) & \text{if } 1\leqq\lambda\leqq r\\ (3,0) & \text{if } r<\lambda\leqq g.\end{cases}$$

where we:

Assume I: $1\leqq r<g$.

In particular H must then be indefinite but anisotropic. Let $G=U(V,H)$ be the unitary group of (V,H) which we will view as the $\mathfrak{k}$-rational points of an algebraic group defined over $\mathfrak{k}$. Let $G_\lambda=U(V_\lambda,H)$, and, let $G_\infty=\prod_{\lambda=1}^{g} G_\lambda$. Then $G_\infty\simeq U(2,1)^r\times U(3)^{g-r}$.

Fix some identification of V with K^3, and let H also denote the matrix for the σ-Hermitian form H on V. Then $H[X]={}^tX^\sigma HX$, for $X\in V=K^3$, and $(X,Y)={}^tX^\sigma HY$ is the $\mathfrak{k}$-bilinear form associated to H.

For each λ, $1\leqq\lambda\leqq r$, let

$$D_\lambda=\{\ell\subset V_\lambda,\ \ell=\text{complex line in } V_\lambda \text{ such that } H|_\ell<0\}$$

(this is isomorphic to the complex 2-ball), and let

$$D=\prod_{\lambda=1}^{r} D_\lambda.$$

Then $\dim_{\mathbf{C}} D=2r$ and G_∞ acts transitively on D via its non-compact factor. If $Z=$ center of G, and $Z_\lambda=$ center of G_λ, then $G/Z\hookrightarrow\prod_{\lambda=1}^{r} G_\lambda/Z_\lambda$ acts faithfully on D. Note that

$$Z\simeq K^1=\{\alpha\in K \text{ such that } N_{K/\mathfrak{k}}(\alpha)=1\}.$$

Let $L\subset V$ be an $\mathcal{O}_K$-lattice, maximal in the sense of Shimura [16].

$\mu(L)=\mathcal{O}_\mathfrak{k}$-ideal generated by $\{H[X], X\in L\}$
$\Gamma(L)=\{g\in G \text{ such that } gL=L\}$.

Take a congruence subgroup $\Gamma \subset \Gamma(L)$. We will suppose, for convenience, that $\Gamma \supset Z(\Gamma(L))$. Then Γ can be viewed as a discrete subgroup of $\prod_{\lambda=1}^{r} G_\lambda$, and $\Gamma/Z(\Gamma)$ acts faithfully on D where $Z(\Gamma) = \Gamma \cap Z = Z(\Gamma(L))$. Note that $Z(\Gamma(L)) \simeq U_K^1$.

Assume II. $\Gamma/Z(\Gamma)$ acts freely on D.

Such Γ is exist by [2].

Let $U = \Gamma \backslash D$ and let $\pi: D \to U$ be the natural projection. By assumptions I and II, U is a compact complex manifold with $\dim_{\mathbf{C}} U = 2r$. In fact by Bailey-Borel [1], U can be embedded as a non-singular complex projective variety.

§3. Construction of Cycles in U

Let $X \in V$ such that $H[X] \gg 0$. Let $\langle X \rangle$ be the subspace of V spanned by X, and let V_X be the orthogonal complement of $\langle X \rangle$ with respect to H. Similarly, let X_λ be the image of X in V_λ, $\langle X_\lambda \rangle = \langle X \rangle_\lambda =$ span of X in V_λ, and $V_{X_\lambda} =$ the orthogonal complement of $\langle X \rangle_\lambda$ in V_λ with respect to H. Then

$$\text{signature } (V_{X_\lambda}, H) = \begin{cases} (1,1) & \text{if } 1 \leqq \lambda \leqq r \\ (2,0) & \text{if } r < \lambda \leqq g. \end{cases}$$

Let

$$G_X = \{g \in G \text{ such that } gX = X\}$$

$$G_{X_\lambda} = \{g_\lambda \in G_\lambda \text{ such that } g_\lambda X_\lambda = X_\lambda\}.$$

For $1 \leqq \lambda \leqq r$, let

$$D_{X_\lambda} = \{\ell \subset V_{X_\lambda},\ \ell = \text{complex line in } V_{X_\lambda} \text{ such that } H|_\ell < 0\},$$

and put

$$D_X = \prod_{\lambda=1}^{r} D_{X_\lambda}.$$

Then $D_X \subset D$ with $\dim_{\mathbf{C}} D_X = r$, and $\prod_{\lambda=1}^{r} G_{X_\lambda}$ acts transitively on D_X. Note that $G_{X_\lambda} \simeq U(1,1)$, if $1 \leqq \lambda \leqq r$ and $G_{X_\lambda} \simeq U(2)$ if $\lambda > r$.

Let $\Gamma_X = \Gamma \cap G_X$, and $\Gamma_X^* = \{g \in \Gamma \text{ such that } gD_X = D_X\}$. Then $\Gamma_X^* = \Gamma_X \cdot Z(\Gamma)$. Note that $Z(\Gamma(L)) \cap \Gamma_X = 1$, so that, by assumption II, Γ_X acts freely on $D_X \subset D$. We then have

$$\begin{array}{ccc} D_X & \longrightarrow & D \\ \downarrow{\scriptstyle \pi_X} & & \downarrow \\ \Gamma_X \backslash D_X & \xrightarrow[\iota_X]{} & U \end{array}$$

By assumptions I and II, $\Gamma_X \backslash D_X = \Gamma_X^* \backslash D_X$ is a compact complex manifold with $\dim_{\mathbf{C}} \Gamma_X \backslash D_X = r$. Let $C_X = \pi(D_X)$. Note that the mapping ι_X is generically $1-1$, but that the r-cycle $C_X \subset U$ may have singularities coming from $\gamma \in \Gamma - \Gamma_X^*$ which identify points of D_X. We will see that these singularities are isolated points of transverse self intersection.

If, X, $Y \in V$ with $H[X] \gg 0$, and $H[Y] \gg 0$, then the corresponding cycles C_X and C_Y in U define cohomology classes, $[C_X]$, $[C_Y] \in H^{2r}(U, \mathbf{Z})$. We want to compute the intersection number

$$I(C_X, C_Y) = [C_X] \cup [C_Y] \in H^{4r}(U, \mathbf{Z}) \simeq \mathbf{Z}.$$

§4. Intersections in D

Suppose X, $Y \in V$ such that $H[X] = A \gg 0$, $H[Y] = B \gg 0$. The following lemma describes the intersections of D_X and D_Y in D.

Lemma 4.1. *Let* $C = {}^t X^\sigma H Y$. *Then*

i) $B - A^{-1}[C] \gg 0 \Leftrightarrow D_X \cap D_Y = \text{point}$,
ii) $B - A^{-1}[C] = 0 \Leftrightarrow D_X = D_Y$,
iii) $(B - A^{-1}[C])_\lambda < 0$ *for some* λ, $1 \leqq \lambda \leqq r \Leftrightarrow D_X \cap D_Y = \emptyset$.

Proof. Let $\langle X, Y \rangle$ denote the subspace of V generated by X and Y.

$$H[X, Y] = \begin{pmatrix} {}^t X^\sigma \\ {}^t Y^\sigma \end{pmatrix} H(X, Y) = \begin{pmatrix} A & C \\ {}^t C^\sigma & B \end{pmatrix} = \begin{pmatrix} A & \\ & B - A^{-1}[C] \end{pmatrix} \left[\begin{pmatrix} 1 & A^{-1}C \\ & 1 \end{pmatrix} \right].$$

Then

$$B - A^{-1}[C] \gg 0 \Leftrightarrow \langle X, Y \rangle_\lambda > 0 \quad \forall \lambda = 1, \ldots, g \text{ and } \langle X, Y \rangle \text{ has } \dim = 2.$$

Then

$$D_X \cap D_Y = \{(\ell_1, \ldots, \ell_r), \ell_\lambda \in \langle X, Y \rangle_\lambda^\perp \ \forall \lambda = 1, \ldots, r\}.$$

So

$$D_X \cap D_Y = \{\ell\} = \{(\ell_1, \ldots, \ell_r)\} \Leftrightarrow \langle X, Y \rangle_\lambda > 0 \quad \forall \lambda = 1, \ldots, r \text{ with } \dim = 2.$$

If $(B - A^{-1}[C])_\lambda = 0$ for some λ, then

$$B - A^{-1}[C] = H[Y - XA^{-1}C] = 0 \Leftrightarrow Y - XA^{-1}C = 0$$

by assumption I. Thus $\dim_K \langle X, Y \rangle = 1$, and $\langle X, Y \rangle = \langle X \rangle = \langle Y \rangle$ so $D_X = D_Y$. Conversely, if $D_X = D_Y$, then $\langle X \rangle = \langle Y \rangle$ so that ${}^t X^\sigma H(Y - XA^{-1}C) = C - AA^{-1}C = 0 \Rightarrow Y - XA^{-1}C = 0$. Finally if $\exists \lambda$, $1 \leqq \lambda \leqq r$ such that $(B - A^{-1}[C])_\lambda < 0$, then H has signature $(1,1)$ on $\langle X, Y \rangle_\lambda$ and so $D_{X_\lambda} \cap D_{Y_\lambda} = \emptyset$.

Conversely, $D_X \cap D_Y = \emptyset \Leftrightarrow D_{X_\lambda} \cap D_{Y_\lambda} = \emptyset$ for some λ and so $\langle X_\lambda, Y_\lambda \rangle$ has signature $(1,1)$ for H, so that $(B - A^{-1}[C])_\lambda < 0$.

Thus, via the lemma and the fact that D_X and D_Y are totally geodesic submanifolds of D, we see that the only possibilities are i) D_X and D_Y intersect transversely in a point, ii) D_X and D_Y coincide, iii) D_X and D_Y don't intersect. Since $\pi: D \to U$ is a covering by assumption II, we conclude that the intersections of C_X and C_Y are transverse except when $C_X = C_Y$.

§5. Intersections of C_X and C_Y

In this section we consider intersections of the cycles $C_X = \pi(D_X)$ and $C_Y = \pi(D_Y)$ with X and Y as in the previous section.

Let

$$\Delta_{X,Y} = \{\gamma \in \Gamma \text{ such that } D_X \cap \gamma D_Y \neq \emptyset\}$$

and

$$\Delta^0_{X,Y} = \{\gamma \in \Gamma \text{ such that } D_X \cap \gamma D_Y = \text{point}\}$$

Proposition 5.1. *Suppose that* $C_X \neq C_Y$. *Then* $I(C_X, C_Y) = |\Gamma_X U_K^1 \setminus \Delta^0_{X,Y}/\Gamma_Y|$.

Proof. By the lemma of the previous section, $C_X \cap C_Y$ consists of a finite number of points in which C_X and C_Y intersect transversely. Thus

$$I(C_X, C_Y) = \sum_{p \in C_X \cap C_Y} |\iota_X^{-1}(p)|\,|\iota_Y^{-1}(p)|.$$

Let

$$\Delta^0_{X,Y}(p) = \{\gamma \in \Gamma \text{ such that } \pi(D_X \cap \gamma D_Y) = p\}.$$

Then it is sufficient to prove that, for $p \in C_X \cap C_Y$,

$$|\iota_X^{-1}(p)|\,|\iota_Y^{-1}(p)| = |\Gamma_X U_K^1 \setminus \Delta^0_{X,Y}(p)/\Gamma_Y|.$$

For $\eta \in \Delta^0_{X,Y}(p)$ let $\ell(\eta) = \ell = D_X \cap \eta D_Y$ and

$$\ell'(\eta) = \ell' = \eta^{-1} D_X \cap D_Y = \eta^{-1}\ell.$$

Then define a map

$$\varphi: \Gamma_X U_K^1 \setminus \Delta^0_{X,Y}(p)/\Gamma_Y \to \iota_X^{-1}(p) \times \iota_Y^{-1}(p)$$

by

$$\varphi(\eta) = (\pi_X(\ell), \pi_Y(\ell')).$$

Given $(q, q') \in \iota_X^{-1}(p) \times \iota_Y^{-1}(p)$, $q = \pi_X(\ell)$ for some $\ell \in D_X$ and $q' = \pi_Y(\ell')$, $\ell' \in D_Y$. Since $\pi(\ell) = \pi(\ell') = p$, $\exists \eta \in \Gamma$ such that $\eta^{-1}\ell = \ell'$. Then $D_X \cap \eta D_Y = \ell = \eta\ell'$, so that $\eta \in \Delta_{X,Y}(p)$ and $\varphi(\eta) = (q, q')$. So φ is surjective. Also φ is injective, for suppose $\varphi(\eta_1) = \varphi(\eta_2)$. Set

$$\ell_1 = D_X \cap \eta_1 D_Y, \quad \ell_1' = \eta_1^{-1}\ell_1 \quad \text{and} \quad \ell_2 = D_X \cap \eta_2 D_Y, \quad \ell_2' = \eta_2^{-1}\ell_2.$$

Now $\pi_X(\ell_1)=\pi_X(\ell_2)$ and $\pi_Y(\ell_1')=\pi_Y(\ell_2') \Rightarrow \exists \gamma\in\Gamma_X U_K^1$ and $\gamma'\in\Gamma_Y U_K^1$ such that $\gamma\ell_1=\ell_2, \gamma'\ell_2'=\ell_1'$. But then

$$\eta_2\ell_2'=\gamma\eta_1\ell_1'=\gamma\eta_1\gamma\ell_2' \Rightarrow \eta_2^{-1}\gamma\eta_1\gamma'\ell_2'=\ell_2' \Rightarrow \eta_2\in U_K^1\Gamma_X\eta_1\Gamma_Y$$

since $\Gamma/Z(\Gamma)$ acts freely on D.

Proposition 5.2. *$I(C_X, C_X)=|\Gamma_X U_K^1\setminus\Delta_{X,X}^0/\Gamma_X|+2^{-r}E(\Gamma_X\setminus D_X)$ where $E(\Gamma_X\setminus D_X)$ is the Euler number of $\Gamma_X\setminus D_X$.*

Remark. An argument similar to the proof of Proposition 5.1 shows that

$$|\Gamma_X U_K^1\Delta_{X,X}^0/\Gamma_X|=\sum_{p\in C_X}|\iota_X^{-1}(p)|(|\iota_X^{-1}(p)|-1).$$

Then if $r=1$, so that C_X is a curve on a surface, Proposition 5.2 follows from the familiar adjunction formula [5], [7]. As noted in [5], $2^{-r}E(\Gamma_X\setminus D_X)$ is the arithmetic genus of $\Gamma_X\setminus D_X$.

Proof. First suppose that C_X is non-singular that $\Delta_{X,X}^0=\emptyset$.

One can then apply (18) of [3], p. 72 to get: $I(C_X, C_X)=c_r(\nu)[C_X]$ where $c_r(\nu)$ is the r^{th} Chern class of the normal bundle of C_X in U. Then a simple computation shows that $c_r(\nu)$ is represented by the (r,r) form $\omega=2^{-r}\omega_1\wedge\cdots\wedge\omega_r$ where ω_λ is the Gauss-Bonnet form on D_{X_λ}. Hence

$$c_r(\nu)[C_X]=\int_{\Gamma_X\setminus D_X}\omega=2^{-r}E(\Gamma_X\setminus D_X)$$

as claimed.

Next suppose that C_X has singularities. These will occur at the points $\pi(D_X\cap\gamma D_X)$ for $\gamma\in\Delta_{X,X}^0$. To compute the self intersection number in this case, we first pass to a normal covering desingularize C_X. Let

$$T^0(X)=\{X'\in T(X), \text{ such that } D_X\cap D_{X'}=\text{point}\}.$$
$$T(X)=\{\gamma X, \gamma\in\Gamma\}.$$

Since $D_X=D_{\alpha X}$, for $\alpha\in K^X$, we may as well assume $X\in L$. Now let $\{X_k\}$ be a set of orbit representatives for $\Gamma_X\setminus T^0(X)$, and choose an ideal $\mathfrak{a}$ of $\mathcal{O}_K$ such that $X_k\not\equiv X \bmod \mathfrak{a}L\ \forall k$. Note that this condition implies $X'\not\equiv X \bmod \mathfrak{a}L, \forall X'\in T^0(X)$. Take

$$\Gamma'=\Gamma(\mathfrak{a})=\{\gamma\in\Gamma \text{ such that } \gamma=1 \bmod \mathfrak{a}\}.$$

Then the singularities of $C_X'=\iota_X'(\Gamma_X'\setminus D_X)\subset\Gamma'\setminus D$ must occur at the points $\pi'(D_X\cap\gamma D_X)$ for $\gamma\in\Delta_{X,X}^0(\Gamma' U_K^1)$. But if $\gamma u\in\Delta_{X,X}^0(\Gamma' U_K^1)$, $u\gamma X$ and $\gamma X\in T^0(X)$ with $\gamma X\equiv X \bmod \mathfrak{a}L$ which contradicts the choice of $\mathfrak{a}$. Hence $\Delta_{X,X}^0(\Gamma' U_K^1)=\emptyset$ and C_X' is non-singular as claimed.

Let $\Phi=\Gamma/\Gamma' U_K^1$, $\Phi_X=\Gamma_X/\Gamma_X'$ and let $C_{X_i}', i=1,\ldots,|\Phi/\Phi_X|$ be the distinct cycles in the Φ orbit of C_X'. Then

$$I(C_X, C_X)=|\Phi|^{-1}\sum_{i,j}I(C_{X_i}', C_{X_j}')$$
$$=|\Phi|^{-1}\sum_i I(C_{X_i}', C_{X_i}')+|\Phi|^{-1}\sum_{i\neq j}I(C_{X_i}', C_{X_j}').$$

The first term contributes:

$$|\Phi|^{-1}\sum_i I(C'_{X_i}, C'_{X_i}) = |\Phi|^{-1}|\Phi/\Phi_X|\, I(C'_X, C'_X)$$
$$= |\Phi_X|^{-1}\, 2^{-r} E(\Gamma'_X \setminus D_X)$$
$$= 2^{-r} E(\Gamma_X \setminus D_X).$$

The second term gives

$$|\Phi|^{-1}\sum_{i\neq j} I(C'_{X_i}, C'_{X_j}) = |\Phi_X|^{-1}\sum_{j\neq 1} I(C'_X, C'_{X_j})$$
$$= |\Phi_X|^{-1}\sum_{j\neq 1} |\Gamma'_X U^1_X \setminus \Delta^0_{X,X_j}(\Gamma' U^1_K)/\Gamma'_{X_j}|$$

where we may assume $X_1 = X$. Now

$$T^0(X) = \Delta^0_{X,X}(\Gamma)\, X = \coprod_{j\neq 1} \Delta^0_{X,X_j}(\Gamma' U^1_K)\, X_j$$

and so

$$\Gamma'_X U^1_K \setminus \Delta^0_{X,X}(\Gamma)/\Gamma_X = \coprod_{j\neq 1} \Gamma'_X U^1_K \setminus \Delta^0_{X,X_j}(\Gamma' U^1_K)/\Gamma'_{X_j}.$$

Finally we observe that $\Phi_X = \Gamma_X/\Gamma'_X$ acts freely on the left hand side of this equation. In fact, if $\gamma\in\Gamma_X$, $\eta\in\Delta^0_{X,X}(\Gamma)$,

$$\gamma\Gamma'_X U^1_K \eta\Gamma_X = \Gamma'_X U^1_K \eta\Gamma_X \Rightarrow \gamma\eta X = \gamma' u \eta X$$
$$\Rightarrow \quad \eta X = \gamma^{-1}\gamma' u \eta X.$$

But then $\gamma^{-1}\gamma'$ preserves the span of X and ηX and hence fixes the totally negative line $\langle X, \eta X\rangle^\perp$. Thus $\gamma^{-1}\gamma'$ has a fixed point on D, and so $\gamma^{-1}\gamma' = 1$, and $\gamma\in\Gamma'_X$. Therefore the contribution of the second term is just

$$|\Phi_X|^{-1}|\Gamma'_X U^1_K \setminus \Delta^0_{X,X}(\Gamma)/\Gamma_X| = |\Phi_X|^{-1}|\Phi_X|\,|\Gamma_X U^1_K \setminus \Delta^0_{X,X}(\Gamma)/\Gamma_X|$$

which completes the formula for $I(C_X, C_X)$.

§6. Arithmetic Cycles

It will be convenient to consider certain reducible cycles which are, in some sense, analogues of the graphs of Hecke correspondences. Such cycles were first introduced by Hirzebruch [5, 6, 7] in his study of the Hilbert modular surfaces. For convenience we will assume that $\mu(L) = \mathcal{O}_t$.

For $\beta\in\mathcal{O}_t$ with $\beta \gg 0$, let

$$T_\beta = T_{\beta,L} = \{Y\in L \text{ such that } H[Y] = \beta\}.$$

Take a set $\{Y_j\}$ of orbit representatives for the action of Γ in T_β, and define:

$$C_\beta = \bigcup_j C_{Y_j}.$$

Note, for example, that if $\varepsilon \in N U_K$, then $C_{\varepsilon\beta} = C_\beta$ and that $C_\beta = \emptyset$ if (L, H) does not represent β.

Let $X \in V$ with $H[X] = \alpha \gg 0$ and let $C_X \subset U$ be the corresponding cycle. The main result of this paper is the following theorem which says that the generating function for the intersection numbers $I(C_X, C_\beta)$ is a Hilbert modular form.

Main Theorem. *Assume I and II. Let* $\tau \in \mathfrak{H}^g$, $\mathfrak{H}$ = *upper half-plane, and* $X \in V$ *such that* $H[X] \gg 0$. *Let*

$$G(\tau, X) = 2^{-r} E(\Gamma_X \backslash D_X) + |U_K^1| \sum_{\substack{\beta \in \mathcal{O}_k \\ \beta \gg 0}} I(C_X, C_\beta)\, \mathbf{e}[\mathrm{tr}_k(\tau\beta)].$$

Then $G(\tau, X) \in M_3\left(\Gamma_0(\mathfrak{a}), \left(\frac{*}{\Delta}\right)\right)$, that is

$$G\left(\frac{a\tau + b}{c\tau + d}, X\right) = \left(\frac{d}{\Delta}\right) N_k(c\tau + d)^3\, G(\tau, X)$$

$\forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathcal{O}_k)$ with $c \in \mathfrak{a}$. Here $\Delta = \Delta_{K/k}$ = discriminant of K/k, $\left(\frac{d}{\Delta}\right) = \prod_{p \nmid \Delta} (d, \Delta)_p (d, \Delta)_\infty$ where $(a, b)_v$ is the Hilbert symbol for k_v, [14]; and $\mathfrak{a}$ is the $\mathcal{O}_k$ ideal defined by

$$\mathfrak{a} = \{\alpha \in \mathcal{O}_k \text{ such that } \alpha L^* \subset 2L\}$$

where

$$L^* = \{Y \in V \text{ such that } {}^t Y^\sigma H X \in \mathfrak{D}_K^{-1},\ \forall X \in L\}.$$

Remark: It should be noted that the level of $G(\tau, X)$ does not depend on $X \in V$ or on $\Gamma \subset \Gamma(L)$. Moreover, it can be shown from the structure of maximal lattices that

$$\mathfrak{a} = 2\mathfrak{D}_k \prod_p p^{r_p}$$

where

$$r_p = \begin{cases} 0 & \text{if } K_p \simeq k_p \times k_p \\ 0 & \text{if } K_p/k_p \text{ is unramified and } \partial_p(H) \equiv 0(2) \\ 1 & \text{if } K_p/k_p \text{ is unramified and } \partial_p(H) \equiv 1(2) \\ \delta_p/2 & \text{if } \mathfrak{D}_{K_p}/k_p = \beta^{\delta_p},\ \delta_p \text{ even} \\ \frac{1}{2}(\delta_p + 1) & \text{if } \mathfrak{D}_{K_p}/k_p = \beta^{\delta_p},\ \delta_p \text{ odd.} \end{cases}$$

Here

$$\partial_p(H)=\begin{cases}0 & \text{if } \det(H)\in NK_p^\times\\ 1 & \text{if } \det(H)\notin NK_p^\times\end{cases}$$

and β is the prime ideal of K_p if $K_p/\mathfrak{k}_p$ is ramified.

The proof of the main theorem occupies Sections 7 and 8. There are two steps. First, the intersection numbers $I(C_X, C_\beta)$ are expressed as "convolutions" of certain representation numbers the Hermitian spaces $\langle X\rangle$ and $\langle X\rangle^\perp$, Proposition 7.5. This expression implies that $G(\tau, X)$ can be written as a sum of products of theta-type series attached to these Hermitian forms. Then the standard transformation laws of such functions give the claimed behavior of $G(\tau, X)$.

Remark: R. Howe has pointed out that τ ought to be viewed as a variable for $U(1,1)^g$ rather than for $SL_2(\mathbf{R})^g$, but we will stick to a more classical point of view here.

§7. Another Formula for $I(C_X, C_\beta)$

Let C_β and C_X be as in the previous section. Let

$$T_\beta(X)=\{Y\in T_\beta \text{ such that } D_X\cap D_Y\neq\varnothing\}$$
$$T_\beta^0(X)=\{Y\in T \text{ such that } D_X\cap D_Y=\text{point}\}$$
$$T_\beta^r(X)=\{Y\in T \text{ such that } D_X=D_Y\}$$
$$=T_\beta(X)-T_\beta^0(X).$$

Note that $\Gamma_X\cdot U_K^1\cdot T_\beta(X)=T_\beta(X)$. The superscript 0 or r is the complex dimension of the intersection of D_X and D_Y. Write

$$I(C_X, C_\beta)=I^{\mathrm{tr}}(C_X, C_\beta)+I^0(C_X, C_\beta)$$

where $I^{\mathrm{tr}}(C_X, C_\beta)$ is the contribution of the points of transverse intersection and $I^0(C_X, C_\beta)$ is the contribution from the common components of C_X and C_β. Then:

Proposition 7.1.

$$I^{\mathrm{tr}}(C_X, C_\beta)=|\Gamma_X\cdot U_K^1\backslash T_\beta^0(X)|$$
$$I^0(C_X, C_\beta)=|\Gamma_X\cdot U_K^1\backslash T_\beta^r(X)|\,2^{-r}E(\Gamma_X\backslash D_X).$$

Proof. By the definition of the Y_j's, we have disjoint unions,

$$T_\beta=\coprod_j \Gamma Y_j \rightleftarrows \coprod_j \Gamma/\Gamma_{Y_j}$$

and

$$T_\beta(X)=\coprod_j \Delta_{X,Y_j}\cdot Y_j \rightleftarrows \coprod_j \Delta_{X,Y_j}/\Gamma_{Y_j}.$$

Then

$$T_\beta^0(X) \rightleftarrows \coprod_j \Delta^0_{X,Y_j}/\Gamma_{Y_j}$$

and

$$T_\beta^r(X) \rightleftarrows \coprod_j (\Delta_{X,Y_j} - \Delta^0_{X,Y_j})/\Gamma_{Y_j}.$$

By Proposition 5.1,

$$\begin{aligned} |\Gamma_X \cdot U_K^1 \backslash T_\beta^0(X)| &= \sum_j |\Gamma_X \cdot U_K^1 \backslash \Delta^0_{X,Y_j}/\Gamma_{Y_j}| \\ &= \sum_j I^{\mathrm{tr}}(C_X, C_{Y_j}) \\ &= I(C_X, C_\beta). \end{aligned}$$

Note that $\Delta_{X,Y_j} - \Delta^0_{X,Y_j} = \{\gamma \in \Gamma \text{ such that } \gamma D_{Y_j} = D_X\}$. If γ_1 and γ_2 are both in this set, then $\gamma_2 \gamma_1^{-1} \in U_K^1 \Gamma_X$ so that

$$\Delta_{X,Y_j} - \Delta^0_{X,Y_j} = \begin{cases} \Gamma_X \cdot U_K^1 \gamma & \text{if } \exists \gamma \in \Gamma \text{ with } D_X = \gamma D_{Y_j} \\ \emptyset & \text{otherwise} \end{cases}.$$

Therefore $|\Gamma_X U_K^1 \backslash T_\beta^r(X)| = \#$ of j for which $C_{Y_j} = C_X$. This proves the second formula of the proposition.

The next step will be to reparameterize $T_\beta(X)$.

For $X \in V$, $H[X] = \alpha \gg 0$ as before, and $Y \in T_\beta(X)$.

$$H[X,Y] = \begin{pmatrix} \alpha & \lambda \\ \lambda^\sigma & \beta \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \beta - \alpha^{-1}[\lambda] \end{pmatrix} \left[\begin{pmatrix} 1 & \alpha^{-1}\lambda \\ & 1 \end{pmatrix} \right].$$

where $\lambda = {}^tX^\sigma HY$. Set $\mu = \lambda\lambda^\sigma$ and let

$$T_{\beta,\mu}(X) = \{Y \in T_\beta \text{ such that } \lambda\lambda^\sigma = \mu, \text{ with } \lambda = {}^tX^\sigma HY\}.$$

Note that $T_{\beta,\mu}(X)$ is left $\Gamma_X \cdot \mathrm{U}_K^1$ invariant. Moreover, by the intersection lemma, $Y \in T_\beta(X) \Leftrightarrow \beta - \alpha^{-1}\mu \gtrsim 0$. Thus

$$T_\beta(X) = \coprod_{\substack{\mu \\ \alpha\beta - \mu \gtrsim 0}} T_{\beta,\mu}(X)$$

$$T_\beta^0(X) = \coprod_{\substack{\mu \\ \alpha\beta - \mu \gg 0}} T_{\beta,\mu}(X)$$

and

$$T_\beta^r(X) = T_{\beta,\alpha\beta}(X).$$

Write $V=\langle X\rangle+\langle X\rangle^{\perp}$, $V_X=\langle X\rangle^{\perp}$ so that any $Y\in L$ can be written $Y=Y_0+Y_1$, $Y_0\in\langle X\rangle$, $Y_1\in V_X$ with $H[Y]=H[Y_0]+H[Y_1]$. Also $H[Y_0]=\alpha^{-1}\lambda\lambda^{\sigma}$, if $\lambda={}^tX^{\sigma}HY$. Then

$$T_{\beta,\mu}(X)=\{Y=Y_0+Y_1\in L \text{ such that } H[Y_0]=\alpha^{-1}\mu \text{ and } H[Y_1]=\beta-\alpha^{-1}\mu\}.$$

If we write $L_0=L\cap\langle X\rangle$, $L_1=L\cap V_X$, $L'=L_0+L_1$ and

$$L=\coprod_i Y^{(i)}+L'$$

for some choice of coset representatives, $i=0,\ldots,|L:L'|-1$, $Y^{(0)}=0$, then

$$T_{\beta,\mu}(X)=\coprod_i\{Y=Y_0+Y_1,\ Y_0\equiv Y_0^{(i)} \bmod L_0,\ H[Y_0]=\alpha^{-1}\mu,$$
$$Y_1\equiv Y_1^{(i)} \bmod L_1,\ H[Y_1]=\beta-\alpha^{-1}\mu\}.$$

We now want to construct a set of coset representatives for the action of $U_K^1\Gamma_X$ in $T_{\beta,\mu}(X)$.

Lemma 7.2.

i) *If* $\beta-\alpha^{-1}\mu\gneqq 0$, *then* U_K^1 *acts faithfully in* $\Gamma_X\backslash T_{\beta,\mu}(X)$.
ii) $Y_0^{(i)}\equiv Y_0^{(j)} \bmod L_0 \Rightarrow Y^{(i)}=Y^{(j)}$.
iii) Γ_X *acts trivially on* L/L'.

Proof. Suppose $Y\in T_{\beta,\mu}(X)$, $\gamma\in\Gamma_X$ and $u\in U_K^1$ such that $\gamma Y=uY$. Write $Y=Y_0+Y_1$ with $H[Y_1]=\beta-\alpha^{-1}\mu$. First suppose $\mu=\alpha\beta$. Then $H[Y_1]=0$, hence $Y=Y_0$. But then $\gamma Y=\gamma Y_0=Y_0$ as $Y_0\in\langle X\rangle$ so that $\gamma Y=uY\Rightarrow Y_0=uY_0$, $Y_0\neq 0$ so that $u=1$. Next suppose that $\beta-\alpha^{-1}\mu\gg 0$. Then $\gamma Y=uY\Rightarrow\gamma$ preserves the span $\langle X,Y\rangle$ of X and Y in V, which, since $\langle X,Y\rangle=\langle X,Y_1\rangle$ is totally positive for H. But then $\gamma l=l$ for $l=\langle X,Y\rangle^{\perp}\in D$. Hence $\gamma\in Z\cap\Gamma_X=1$, and $\gamma Y=Y=uY\Rightarrow u=1$. To show (ii) observe that

$$Y_0^{(i)}\equiv Y_0^{(j)} \bmod L_0 \Rightarrow Y_1^{(i)}-Y_1^{(j)}=Y^{(i)}-Y^{(j)}-(Y_0^{(i)}-Y_0^{(j)})\in L\cap V_X=L_1,$$

so that $Y^{(i)}-Y^{(j)}\in L'$, i.e., $Y^{(i)}=Y^{(j)}$. Finally, by definition, $\gamma\in\Gamma_X\Rightarrow\gamma Y_0^{(i)}=Y_0^{(j)}$, so by (ii), $\gamma Y^{(i)}\equiv Y^{(j)} \bmod L'$.

Corollary 7.3. *If* $\beta-\alpha^{-1}\mu\gg 0$, *then*

$$|U_K^1\Gamma_X\backslash T_{\beta,\mu}(X)|=|U_K^1|^{-1}\,|\Gamma_X\backslash T_{\beta,\mu}(X)|.$$

Thus we need only consider the action of Γ_X.

Corollary 7.4.

$$\Gamma_X\backslash T_{\beta,\mu}(X)\rightleftarrows\coprod_j\{(Y_0,Y_1),\ Y_0\equiv Y_0^{(i)} \bmod L_0,\ H[Y_0]=\alpha^{-1}\mu$$
$$Y_1\equiv Y_1^{(i)} \bmod L_1,\ H[Y_1]=\beta-\alpha^{-1}\mu\ \ Y_1 \bmod \Gamma_X\}.$$

Combining this result with Proposition 7.1 and the description of $T_\beta^0(X)$ given above we have:

Proposition 7.5.

$$I^{\mathrm{tr}}(C_X, C_\beta) = |U_K^1|^{-1} \sum_i \sum_{\substack{\mu \\ \alpha\beta - \mu \gg 0}} |\{Y_0, Y_0 \equiv Y_0^{(i)} \bmod L_0,\ H[Y_0] = \alpha^{-1}\mu\}|$$
$$\times |\Gamma_X \backslash \{Y_1 \in V_X,\ Y_1 \equiv Y_1^{(i)} \bmod L_1,\ H[Y_1] = \beta - \alpha^{-1}\mu\}|$$
$$I^0(C_X, C_\beta) = |U_K^1|^{-1} |\{Y_0 \in L_0 \text{ such that } H[Y_0] = \beta\}| 2^{-r} E(\Gamma_X \backslash D_X).$$

§8. Proof of the Main Theorem

The proposition at the end of §7 says that the intersection numbers $I(C_X, C_\beta)$ are expressible in terms of representation numbers for the Hermitian spaces $\langle X \rangle$ and V_X. This allows us to express $G(\tau, X)$ as a sum of products of ϑ-type series; the behavior of $G(\tau, X)$ with respect to $SL_2(\mathcal{O}_\ell)$ follows.

As in the previous section, let $L_0 = \langle X \rangle \cap L$, $L_1 = V_X \cap L$ and set $L' = L_0 + L_1$. Let

$$L^* = \{Y \in V \text{ such that } {}^t Y^\sigma H X \in \mathfrak{D}_K^{-1}\ \forall X \in L\}$$
$$L'^* = \{Y \in V \text{ such that } {}^t Y^\sigma H X \in \mathfrak{D}_K^{-1}\ \forall X \in L'\}.$$

Recall that $\mu(L) = \mathcal{O}_\ell$.

Lemma 8.1. $L'^* = L_0^* + L_1^* \supset L^* \supset L \supset L' = L_0 + L_1$.

Proof. Set $\mathfrak{m} = \{{}^t X^\sigma H Y, X, Y \in L\}$. Then $\mu(L)\mathcal{O}_\ell \subset \mathfrak{m} \subset \mu(L)\mathfrak{D}_K^{-1}$ and $L^* = \mathfrak{D}_K^{-1}\mathfrak{m}^{-1}M$ where $M = \{Y \in V$ such that ${}^t X^\sigma H Y \in \mathfrak{m}, \forall X \in L\}$. So $M \supseteq L$ and, since $\mu(L) = \mathcal{O}_\ell$,

$$\mathfrak{m} \subset \mu(L)\mathfrak{D}_K^{-1} = \mathfrak{D}_K^{-1} \Rightarrow \mathfrak{m} \subset \mathfrak{D}_K^{-1},$$

hence

$$L^* = \mathfrak{m}^{-1}\mathfrak{D}_K^{-1} M \supseteq M \supseteq L.$$

We now introduce ϑ-type series.
For $c \in \mathcal{O}_\ell$, $c \gg 0$, let

$$\Gamma(L', c) = \{\gamma \in \Gamma(L') \text{ such that } \gamma \text{ acts trivially in } L'^*/cL'\}$$

and let $\Gamma_X^{(c)} = \Gamma_X \cap \Gamma(L', c)$. Observe that because $\Gamma_X \subset \Gamma(L')$ and $\Gamma(L', c)$ is normal in $\Gamma(L')$, $\Gamma_X^{(c)}$ is normal in Γ_X. Moreover, since Γ_X acts freely in $\{Y_1 \in V_X$ such that $H[Y_1] \gg 0\}$, $\Gamma_X/\Gamma_X^{(c)}$ acts faithfully in

$$\Gamma_X^{(c)} \backslash \{Y_1 \in V_X \text{ such that } H[Y_1] \gg 0\}.$$

For $Y^{(k)} \in L'^*/cL'$, define

$$g(\tau, c, Y^{(k)}) = g_0(\tau, c, Y_0^{(k)})\, g_1(\tau, c, Y_1^{(k)})$$

where

$$g_0(\tau, c, Y_0^{(k)}) = \sum_{Y_0 \equiv Y_0^{(k)} \bmod cL_0} \mathbf{e}[\mathrm{tr}_{\mathfrak{k}}(\tau c^{-1} H[Y_0])]$$

and

$$g_1(\tau, c, Y_1^{(k)}) = 2^{-r} E(\Gamma_X \backslash D_X)\delta_{k,0} + |\Gamma_X : \Gamma_X^{(c)}|^{-1} \sum_{\substack{Y_1 \equiv Y_1^{(k)} \bmod cL_1 \\ H[Y_1] \gg 0 \\ \bmod \Gamma_X^{(c)}}} \mathbf{e}[\mathrm{tr}_{\mathfrak{k}}(\tau c^{-1} H[Y_1])].$$

If $c=1$, and $Y^{(k)} \in L$, $\Gamma_X/\Gamma_X^{(1)}$ acts faithfully in

$$\Gamma_X^{(1)} \backslash \{Y_1 \equiv Y_1^{(k)} \bmod L_1,\ H[Y_1] \gg 0\}$$

because, by Lemma 7.2, Γ_X preserves $Y_1^{(k)} \bmod L_1$. Thus

$$g_1(\tau, 1, Y_1^{(k)}) = 2^{-r} E(\Gamma_X \backslash D_X)\delta_{k,0} + \sum_{\substack{Y_1 \equiv Y_1^{(k)} \bmod L_1 \\ H[Y_1] \gg 0 \\ \bmod \Gamma_X}} \mathbf{e}[\mathrm{tr}_{\mathfrak{k}}(\tau H[Y_1])].$$

Proposition 7.5 then yields

Proposition 8.2.

$$G(\tau, X) = \sum_{Y^{(k)} \in L/L'} g(\tau, 1, Y^{(k)}).$$

Proof. The coefficient of β in $g(\tau, 1, Y^{(k)})$ is

$$\sum_{\substack{\eta, \nu \\ \eta+\nu=\beta \\ \eta \geq 0, \nu \gg 0}} |\{Y_0 \equiv Y_0^{(k)} \bmod L_0,\ H[Y_0] = \eta\}|\,|\Gamma_X \backslash \{Y_1 \equiv Y_1^{(k)} \bmod L_1,\ H[Y_1] = \nu\}|$$
$$+ 2^{-r} E(\Gamma_X \backslash D_X)\,\delta_{k,0}\,|\{Y_0 \equiv Y_0^{(k)} \bmod L_0,\ H[Y_0] = \beta\}|.$$

Setting $\eta = \alpha^{-1}\mu$ and $\nu = \beta - \alpha^{-1}\mu$, and summing over $Y^{(k)}$ yields

$$|U_K^1|(I^{\mathrm{tr}}(C_X, C_\beta) + I^0(C_X, C_\beta)) = |U_K^1| I(C_X, C_\beta)$$

by Proposition 7.5. The constant term is $2^{-r} E(\Gamma_X \backslash D_X)$ as required.

The basic transformation properties of the $g(\tau, c, Y^{(k)})$'s are as follows:

Proposition 8.3.

i) *For* $b \in \mathcal{O}_{\mathfrak{k}}$,

$$g(\tau + b, c, Y^{(k)}) = \mathbf{e}[\mathrm{tr}_{\mathfrak{k}}(bc^{-1} H[Y^{(k)}])]\, g(\tau, c, Y^{(k)}),$$

ii) $$g\left(-\frac{1}{\tau}, c, Y^{(k)}\right) = (-1)^r i^g N_{\mathfrak{k}}(\tau)^3 |L^*/cL'|^{-1/2}$$
$$\times \sum_{Y^{(j)} \in L'^*/cL'} e[\mathrm{tr}_{\mathfrak{k}}(c^{-1} \mathrm{tr}({}^t Y^{(k)\sigma} H Y^{(j)}))]\, g(\tau, c, Y^{(j)}),$$

iii) *If* $c' \in \mathcal{O}_{\mathfrak{k}}$, $c' \gg 0$, *then*

$$g(\tau, c, Y^{(k)}) = \sum_{\substack{Y' \equiv Y^{(k)} \bmod cL' \\ \bmod cc'L}} g(c'\tau, cc', Y').$$

Proof (sketch). The factors $g_0(\tau, c, Y_0^{(k)})$ and $g_1(\tau, c, Y_1^{(k)})$ can be identified with the analogues over $\mathfrak{k}$ of the classical ϑ-series attached to an ideal in an imaginary quadratic field by Hecke [3], and to an ideal in an indefinite quaternion algebra by Schoenberg [13] and Siegel [15]. The well known transformation laws of functions then yield those of $g(\tau, c, Y^{(k)})$.

Remark. It should be possible to derive the behavior of $g(\tau, c, Y^{(k)})$ from the Weil representation as is done for indefinite quadratic forms over $\mathbf{Q}$ by Shintani [17].

Let $\psi_0(\alpha) = \mathbf{e}[\mathrm{tr}_{\mathfrak{k}}(\alpha)]$ for $\alpha \in \mathfrak{k}$.

Proof of the main theorem. Proceed in the standard way: take $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathcal{O}_{\mathfrak{k}})$ and write $\dfrac{a\tau+b}{c\tau+d} = \dfrac{a}{c} - \dfrac{1}{c(c\tau+d)}$. Assume $c \gg 0$, then

$$
\begin{aligned}
& g\left(\frac{a\tau+b}{c\tau+d}, 1, Y^{(k)}\right) \\
&\quad = g\left(\frac{a}{c} - \frac{1}{c(c\tau+d)}, 1, Y^{(k)}\right) \\
&\quad = \sum_{\substack{Y' \equiv Y^{(k)} \bmod L' \\ \bmod cL'}} g\left(a - \frac{1}{c\tau+d}, c, Y'\right) \\
&\quad = \sum_{Y'} \psi_0(c^{-1} a H[Y'])\, g\left(-\frac{1}{c\tau+d}, c, Y'\right) \\
&\quad = (-1)^r i^g N(c\tau+d)^3 \, |L^*/cL'|^{-1/2} \\
&\qquad \times \sum_{Y'} \sum_{Y^{(j)} \in L'^*/cL'} \psi_0(ac^{-1} H[Y'] + c^{-1} \mathrm{tr}({}^t Y'^{\sigma} H Y^{(j)}))\, g(c\tau+d, c, Y^{(j)}) \\
&\quad = (-1)^r i^g N(c\tau+d)^3 \, |L^*/cL'|^{-1/2} \\
&\qquad \times \sum_{Y^{(j)} \in L'^*/cL'} \varphi(Y^{(k)}, Y^{(j)})\, g(c\tau, c, Y^{(j)})
\end{aligned}
$$

where

$$
\begin{aligned}
& \varphi(Y^{(k)}, Y^{(j)}) \\
&\quad = \sum_{\substack{Y' \equiv Y^{(k)} \bmod L' \\ \bmod cL'}} \psi_0(ac^{-1} H[Y'] + c^{-1} \mathrm{tr}({}^t Y'^{\sigma} H Y^{(j)}) + dc^{-1} H[Y^{(j)}])
\end{aligned}
$$

with

$$Y^{(k)} \in L'^*/L' \quad \text{and} \quad Y^{(j)} \in L'^*/cL'.$$

In fact, by the standard argument, $\varphi(Y^{(k)}, Y^{(j)})$ depends only on $Y^{(j)} \bmod L'$, not $\bmod cL'$, so that the sum becomes

$$\sum_{Y^{(j)} \in L'^*/L'} \varphi(Y^{(k)}, Y^{(j)})\, g(\tau, 1, Y^{(j)}).$$

Next, observe that if we sum over $Y^{(k)} \in L$ we get

$$\sum_{Y^{(k)} \in L/L'} g(\tau, 1, Y^{(k)}) = G(\tau, X).$$

Now let

$$\Phi(Y^{(j)}) = \sum_{Y^{(k)} \in L/L'} \varphi(Y^{(k)}, Y^{(j)})$$

$$= \sum_{Y \in L/cL'} \psi_0(ac^{-1} H[Y] + c^{-1} \operatorname{tr}({}^t Y^\sigma H Y^{(j)}) + dc^{-1} H[Y^{(j)}]).$$

If $Y_1 \in L$,

$$H[Y + cY_1] = H[Y] + c \operatorname{tr}({}^t Y^\sigma H Y_1) + c^2 H[Y_1]$$

so that, since $\mu(L) = \mathcal{O}_\ell$,

$$\psi_0(ac^{-1} H[Y + cY_1]) = \psi_0(ac^{-1} H[Y]).$$

Then

$$\Phi(Y^{(j)}) = \sum_{Y \in L/cL} \psi_0(ac^{-1} H[Y] + c^{-1} \operatorname{tr}({}^t Y^\sigma H Y^{(j)}) + dc^{-1} H[Y^{(j)}])$$

$$\times \sum_{Y_1 \in L/L'} \psi_0(c^{-1} \operatorname{tr}(c\, {}^t Y_1^\sigma H Y^{(j)}))$$

and the inner sum is

$$\begin{cases} 0 & \text{if } Y^{(j)} \notin L^* \\ |L/L'| & \text{if } Y^{(j)} \in L^*. \end{cases}$$

Thus

$$\Phi(Y^{(j)}) = \sum_{Y \in L/cL} \psi_0(ac^{-1} H[Y] + c^{-1} \operatorname{tr}({}^t Y^\sigma H Y^{(j)}) + dc^{-1} H[Y^{(j)}])\, |L/L'|$$

$$\text{if } Y^{(j)} \in L^*$$

$$\Phi(Y^{(j)}) = 0 \quad \text{otherwise.}$$

Now suppose:

$$dL^* \subset 2L,$$

so that $dH[Y] = \frac{1}{2} \operatorname{tr}({}^t Y^\sigma H dY) \in \mathfrak{O}_\ell^{-1}\ \forall\, Y \in L^*$. Then adding $-dY^{(j)}$ to Y in $\Phi(Y^{(j)})$ leaves the whole sum unchanged while the first factor of any term becomes

$$\psi_0(ac^{-1} H[Y] + ac^{-1} \operatorname{tr}(-{}^t Y^\sigma H dY^{(j)}) + ac^{-1} H[dY^{(j)}])$$

$$= \psi_0(ac^{-1} H[Y] - c^{-1} \operatorname{tr}({}^t Y^\sigma H Y^{(j)}) + b\, dH[Y^{(j)}] + dc^{-1} H[Y^{(j)}])$$

via $ad=bc+1$. The third factor of this last expression is 1. The second factor of any term becomes:

$$\psi_0(-2c^{-1}\,dH[Y^{(j)}]+c^{-1}\,\mathrm{tr}({}^tY^{\sigma}HY^{(j)})).$$

Therefore, the whole term becomes simply

$$\psi_0(ac^{-1}H[Y]).$$

Thus:

$$\Phi(Y^{(j)})=|L/L'|\sum_{Y\in L/cL}\psi_0(ac^{-1}H[Y])=|L/L'|F(c).$$

Therefore we conclude that

$$G\left(\frac{a\tau+b}{c\tau+d},X\right)=(-1)^r i^g N(c\tau+d)^3\,|L'^*/cL'|^{-1/2}\,|L/L'|F(c)$$

$$\cdot\sum_{Y^{(j)}\in L^*/L'} g(\tau,1,Y^{(j)})$$

provided $dL^*\subset 2L$. Applying $\tau\mapsto -\dfrac{1}{\tau}$ again yields:

$$G\left(\frac{b\tau-a}{d\tau-c},X\right)=(-1)^r i^g N\left(-\frac{c}{\tau}+d\right)^3|L'^*/cL'|^{-1/2}$$

$$\cdot|L/L'|F(c)(-1)^r i^g N(\tau)^3|L'^*/L'|^{-1/2}$$

$$\cdot\sum_{\substack{Y^{(j)}\in L^*\\ \mathrm{mod}\ L'}}\sum_{Y^{(k)}\in L'^*/L'}\psi_0\left(\mathrm{tr}({}^tY^{(j)\sigma}HY^{(k)})\right)g(\tau,1,Y^{(k)}).$$

Observe that

$$\sum_{Y\in L^*/L'}\psi_0\left(\mathrm{tr}({}^tY^{\sigma}HY^{(k)})\right)=\begin{cases}0 & \text{if } Y^{(k)}\notin(L^*)^*=L\\ |L^*/L'| & \text{if } Y^{(k)}\in(L^*)^*=L.\end{cases}$$

Also

$$|L'^*/cL'|^{-1/2}|L'^*/L'|^{-1/2}|L/L'|\,|L^*/L'|=N(c)^{-3}|L'^*/L'|^{-1}|L/L'|^2|L^*/L|$$
$$=N(c)^{-3}.$$

Finally,

Lemma 8.4. *Since L is a maximal lattice in V,*

$$Nc^{-3}F(c)=\left(\frac{K/k}{c\mathcal{O}_k}\right),$$

where $\left(\frac{K/k}{\cdot}\right)$ *is the Artin symbol. Note in particular that the condition* $dL^* \subset 2L \Rightarrow c\mathcal{O}_k$ *is relatively prime to* $\Delta_{K/k}$, *the discriminant of* K/k.

Proof. In the usual way one can write

$$F(c) = \sum_{Y \in L/cL} \psi_0(ac^{-1}H[Y])$$

$$= \prod_{p|c\mathcal{O}_k} \sum_{Y \in L_p/cL_p} \psi_p(ac^{-1}H[Y]) = \prod_{p|c\mathcal{O}_k} F_p(c)$$

where ψ_p is the standard additive character of k_p. Then, again by the usual technique, write $c = \pi^m u$, $\pi =$ prime of $\mathcal{O}_{k_p}$, $u \in U_{k_p}$,

$$F_p(c) = \begin{cases} Np^{3m} & \text{if } m \text{ even} \\ Np^{3(m-1)} \sum_{Y \in L_p/\pi L_p} \psi_p(au^{-1}\pi^{-1}H[Y]) & \text{if } m \text{ is odd} \end{cases}$$

Now since $ad - bc = 1$ and $dL^* \subset 2L$, $p | c\mathcal{O}_k \Rightarrow L_p^* = L_p$, and, in particular $p \nmid 2$. By the results of Shimura [16] on maximal lattices and the condition $L_p^* = L_p$ can only be satisfied in the following cases:

1) $K_p = k_p \times k_p$

$$L_p = \sum_{i=1}^{3} (\mathcal{O}_{k_p} e + \mathcal{O}_{k_p} e^{\sigma}) z_i,$$

2) $K_p/k_p =$ unramified quadratic extension and

$$L_p = \mathcal{O}_{K_p} x + \mathcal{O}_{K_p} y + \mathcal{O}_{K_p} z$$

for a Witt basis x, y, z with $H[z] = 1$.

It is then easy to check that

$$\sum_{Y \in L_p/\pi L_p} \psi_p(au^{-1}\pi^{-1}H[Y]) = \begin{cases} Np^3 & \text{in case 1)} \\ -Np^3 & \text{in case 2).} \end{cases}$$

Thus

$$F_p(c) = \left(\frac{K_p/k_p}{c\mathcal{O}_{k_p}}\right) Np^m, \quad \text{if } c = \pi^m u.$$

Combining these observations gives:

$$G\left(\frac{a\tau + b}{c\tau + d}, X\right) = (-1)^g N(c\tau + d)^3 \left(\frac{K/k}{d\mathcal{O}_k}\right) G(\tau, X)$$

where $d \ll 0$ and $cL^* \subset 2L$.

To finish the theorem, take $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathcal{O}_k)$ with $cL^* \subset 2L$, and choose $b' \in \mathcal{O}_k$

such that $cb' \ll 0$ and $cb'+d \ll 0$. Write $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & ab'+b \\ c & cb'+d \end{pmatrix} \begin{pmatrix} 1 & -b' \\ & 1 \end{pmatrix}$. Then since $G(\tau+b', X) = G(\tau, X)\ \forall\, b' \in \mathcal{O}_k$.

$$G\left(\frac{a\tau+b}{c\tau+d}, X\right) = (-1)^g N(c\tau+d)^3 \left(\frac{K/k}{(cb'+d)\,\mathcal{O}_k}\right) G(\tau, X).$$

Now $\left(\frac{K/k}{(cb'+d)\,\mathcal{O}_k}\right) = \prod_{p \nmid \Delta_{K/k}} (cb'+d, \Delta_{K/k})_p$ where $\Delta_{K/k}$ is the discriminant of K/k, and, $(\ ,\ )_p$ is the Hilbert symbol [14]. By the results of Shimura [16], on the structure of maximal lattices, we find that the condition $cL_p^* \subset 2L_p$ implies that $c \equiv 0 \bmod p$, $\forall\, p | \Delta_{K/k}$. Hence, since $ad-bc=1$, d is a unit for $p | \Delta_{K/k}$. Therefore, for $p | \Delta_{K/k}$,

$$\begin{aligned}(cb'+d, \Delta_{K/k})_p &= ((cb'd^{-1}+1)\,d, \Delta_{K/k})_p \\ &= (cb'd^{-1}+1, \Delta_{K/k})_p (d, \Delta_{K/k})_p.\end{aligned}$$

Multiplying b' by a suitable power of 2 if necessary, one gets $(cb'd^{-1}+1, \Delta_{K/k})_p = 1$ since $cb'd^{-1}+1 \in N U_{K_p}$. Thus:

$$\begin{aligned}(-1)^g \left(\frac{K/k}{(cb'+d)\,\mathcal{O}_k}\right) &= \prod_{\lambda=1}^{g} (cb'+d, \Delta_{K/k})_\lambda \prod_{p \nmid \Delta_{K/k}} (cb'+d, \Delta_{K/k})_p \\ &= \prod_{p | \Delta_{K/k}} (cb'+d, \Delta_{K/k})_p \\ &= \prod_{p | \Delta_{K/k}} (d, \Delta_{K/k})_p \\ &= \prod_{\lambda=1}^{g} (d, \Delta_{K/k})_\lambda \prod_{p \nmid \Delta_{K/k}} (d, \Delta_{K/k})_p \\ &= \left(\frac{d}{\Delta_{K/k}}\right)\end{aligned}$$

by our definition of this symbol. This gives the transformation laws for $G(\tau, X)$ asserted in the main theorem.

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Algebraic Surfaces of General Type with Small c_1^2. III

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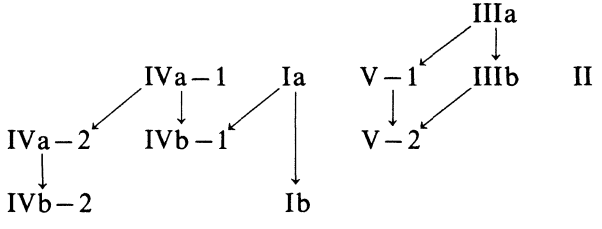
Introduction

In this Part III, we shall study minimal algebraic surfaces defined over $\mathbb{C}$ with $p_g=4$, $q=0$, and $c_1^2=6$. Our purpose is to determine the structures and deformations of these surfaces. We shall use the terminology and notation of our preceding papers [10, 11, I and II], which are respectively referred to as [Q], Part I, and Part II.

Minimal surfaces with $p_g=4$, $q=0$, and $c_1^2=6$ are classified into five types: I to V. These are further divided into eleven types: Ia, Ib, II, IIIa, IIIb, IVa−1, IVa−2, IVb−2, V−1, and V−2. A surface of type Ia is birationally equivalent to a sextic surface in $\mathbb{P}^3$ which has a double curve along a plane cubic curve, while a surface of type Ib is birationally equivalent to a double covering of a cubic surface in $\mathbb{P}^3$. A surface of type II is birationally equivalent to a triple covering of a quadratic cone in $\mathbb{P}^3$. A surface of type IIIa or IIIb has a pencil of curves of genus 3 with a base point, and a surface of type IV or V has a pencil of curves of genus 2. We have the following hierarchy among them:



Here A→B means that B is a specialization of A, i.e. there exists a 1-parameter family with a special fibre being of type B and a generic fibre being of type A. Several points are left unsettled. In particular, we can not prove nor disprove the following specializations: II→IIIb or V, Ia→V, and IV→V−2[1]. The number of moduli for the generic surfaces are 38 for Ia, II, IVa−1, and 39

[1] This is not the case as we shall prove in Part IV, §6

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for IIIa. These surfaces except for those of type IV and V were already known to Enriques (see [6], pp. 271–273).

Here we mention two points about deformations. First, the generic surfaces S of type II have non-reduced Kuranishi spaces of deformations. This phenomenon is connected with the existence of a non-singular rational curve G with $G^2=-2$ on such S, which is stable under small deformations. In other words, for any sufficiently small deformation S_t of S, the canonical bundle of S_t is not ample. Secondly, the surfaces of type Ia and those of type IVa-1 both specialize to surfaces of type IVb-1. This fact as well as its proof is quite similar to that in the case of numerical quintic surfaces [Q].

In addition to the three papers cited above, we shall often use the result of our paper on pencils of curves of genus 2 [12], which will be referred to as [P]. Other surfaces with $c_1^2=2p_g-2$ will be studied in a subsequent paper.

Among others we shall use the following notation. For any non-negative integer d, Σ_d denotes the $\mathbb{P}^1$-bundle $\mathbb{P}(\mathcal{O}\oplus\mathcal{O}(d))$ over $\mathbb{P}^1$, and we call it the Hirzebruch surface of degree d. We let Γ denote a fibre of Σ_d and Δ_0 a section with $\Delta_0^2=-d$. The latter will be called the 0-section of Σ_d. For any sheaf $\mathcal{F}$ on a compact complex manifold X, we set $h^i(\mathcal{F})=\dim H^i(X,\mathcal{F})$, and $\chi(\mathcal{F})=\sum(-1)^i h^i(\mathcal{F})$. If $\mathcal{F}=\mathcal{O}(D)$ with a divisor or a line bundle D, we set $h^i(D)=h^i(\mathcal{O}(D))$ and $\chi(D)=\chi(\mathcal{O}(D))$. Finally Θ_X denotes the sheaf of germs of holomorphic vector fields on X.

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§1. Canonical Maps of Algebraic Surfaces

Let S be a minimal algebraic surface of general type defined over $\mathbb{C}$ and let K be the canonical bundle of S. As usual, we denote by p_g the geometric genus of S. Then the complete linear system $|K|$ defines a rational map $\Phi_K: S\to\mathbb{P}^n$ with $n=p_g-1$. We shall say that $|K|$ is composite with a linear pencil if the image of Φ_K is a rational curve.

Theorem 1.1. *Let c_1 be the first Chern class of S. If $c_1^2 \leqq 3p_g - 4$ with $p_g \geqq 3$, then $|K|$ is not composite with a linear pencil.*

Proof. Suppose that $|K|$ is composite with a linear pencil $|D|$. Then we have $|K| = |nD| + F$, where F is the fixed part of $|K|$. Since S is of general type, we have $KF \geqq 0$. Hence

$$nKD \leqq K^2 < 3n.$$

It follows that $KD = 2$, because $KD = 1$ would imply $K^2 = 1$ and $p_g \leqq 2$ (see [16] or [3]). Therefore, we have the equality

$$2 = nD^2 + DF.$$

Since D^2 is even and since $n \geqq 2$, we conclude $D^2 = 0$ and $DF = 2$. Hence $|D|$ defines a holomorphic map $g: S \to \mathbb{P}^1$ whose general fibre is of genus 2.

We now recall the result of [P]. By the construction there, we obtain a "canonical" rational map f of degree 2 of S onto some Hirzebruch surface Σ_d. We let B denote the branch locus of f. Its singularities are classified in [P], Lemma 6 into six types: (0), (I_k), (II_k), (III_k), (IV_k), and (V). Let $\nu(T)$ denote the number of singular fibres of type (T). Then we have, by [P], Theorem 3,

$$c_1^2 - (2p_a - 4)$$
$$= \sum_k (2k-1)\{\nu(\mathrm{I}_k) + \nu(\mathrm{III}_k)\} + \sum_k 2k\{\nu(\mathrm{II}_k) + \nu(\mathrm{IV}_k)\} + \nu(\mathrm{V}), \tag{1.1}$$

where $p_a = p_g - q$ is the arithmetic genus of S.

On the other hand, assume that B is linearly equivalent to $6\Delta_0 + 2(m+d+2)\Gamma$. Then we have

$$p_g = \dim H^0(\Sigma_d, \mathcal{O}(\Delta_0 + (m - \deg \mathfrak{c})\Gamma)),$$
$$q = \dim H^1(\Sigma_d, \mathcal{O}(\Delta_0 + (m - \deg \mathfrak{c})\Gamma)),$$

where $\mathfrak{c}$ is a divisor on $\mathbb{P}^1$ determined in terms of singular fibres (see [P], p. 87). Since $|K|$ is composite with a pencil, we have the equalities $m - \deg \mathfrak{c} = p_g - 1$, and $d = p_g + q$.

Since B has no multiple component, we have

$$-d \leqq \Delta_0 B = 2\deg \mathfrak{c} - 2d + 2 - 2q.$$

From Theorem 3 in [P], we obtain

$$2\deg \mathfrak{c} - (c_1^2 - 2p_a + 4) = \sum_k (\nu(\mathrm{I}_k) + \nu(\mathrm{III}_k)) + \nu(\mathrm{V}).$$

We call the right hand side r. Then B contains r fibres $\Gamma_1, \Gamma_2, \ldots, \Gamma_r$. So let us write $B = B_0 + \Gamma_1 + \cdots + \Gamma_r$. Then we have

$$\Delta_0 B_0 = c_1^2 - (4p_g - 6) - 2q,$$

and, by assumption, $\Delta_0 B_0$ is negative. Therefore B_0 contains Δ_0 as a component. If we set $C = B_0 - \Delta_0$, then we have

$$\Delta_0 C = c_1^2 - 3p_g + 6 - q. \tag{1.2}$$

Next we prove that $\nu(\mathrm{III}_k) = \nu(\mathrm{IV}_k) = 0$. In fact, suppose that Γ is a fibre of type (III_k) or (IV_k). Then the curve B_0 intersects Γ at a unique point s, and B_0 has infinitely near triple points at s. Since Δ_0 is a component of B_0, s is on Δ_0. Let $\pi: W_1 \to \Sigma_d$ be the quadratic transformation with center s, $E = \pi^{-1}(s)$, and let $\tilde{\Gamma}$ be the proper transform of Γ. Then, by definition, the proper transform $\tilde{B}_0$ of B_0 has a triple point at $E \cap \tilde{\Gamma}$. On the other hand, $\tilde{B}_0$ contains the proper transform of Δ_0. These facts imply that $\tilde{B}_0 E \geqq 4$, which contadicts that s is a triple point.

Suppose that Γ is a singular fibre of type (I_k). Then, by definition, $B_0 = C + \Delta_0$ has at least $(2k-1)$-fold triple point at $s = \Gamma \cap \Delta_0$. This implies that C has a contact of order $\geqq 4k-2$ with Δ_0 at s, i.e. $(\Delta_0 C)_s \geqq 4k-2$. In a similar way, if Γ is of type (III_k), we have $(\Delta_0 C)_s \geqq 4k$, and, if Γ is of type (V), we have $(\Delta_0 C)_s \geqq 3$. From these facts we obtain

$$\Delta_0 C \geqq \sum_k (4k-2)\,\nu(\mathrm{I}_k) + \sum_k 4k\,\nu(\mathrm{II}_k) + 3\nu(\mathrm{V}).$$

Combining with (1.1) and (1.2), we obtain $c_1^2 \leqq p_g - 2 - 5q$. This contradicts the inequality $c_1^2 \geqq 2p_g - 4$ (see [Q], Lemma 2). Q.E.D.

Theorem 1.2. *Assume that $p_g \geqq 5$ and $c_1^2 \leqq 4p_g - 7$. Then $|K|$ is not composite with a linear pencil.*

Proof. We suppose that $|K|$ is composite with a linear pencil, and write $|K| = |nD| + F$ as before. Then we have

$$nKD \leqq K^2 \leqq 4n - 3.$$

This implies $KD = 2$ or 3. But, if $KD = 3$, we have $3 = nD^2 + DF$, and D^2 is odd. It follows that $n \leqq 3$, which contradicts the assumption $p_g \geqq 5$. Thus we have $KD = 2$ and we can repeat the above proof. Q.E.D.

Theorem 1.3. *Assume that $c_1^2 = 2p_g - 2$ or $2p_g - 1$ and $p_g \geqq 3$. Then $|K|$ is not composite with a pencil.*

Proof. If $|K|$ is composite with a pencil $\{D\}$, then K is algebraically equivalent to $\mu D + F$ for some integer $\mu \geqq n$, where F denotes the fixed part of $|K|$. From the inequalities $\mu KD \leqq K^2 \leqq 2n+1$, we infer that $\mu = n$. This implies that $\{D\}$ is a linear pencil. But this is impossible by Theorem 1.1. Q.E.D.

§2. Holomorphic Maps of Degree 2

The purpose of this section is to supplement the result of [Q], §2. Let $f: S \to W$ be a surjective holomorphic map of degree 2 of compact complex surfaces S and

W, both being non-singular. We assume that no exceptional curve (of the first kind) on S is mapped into a point by f. We define the ramification divisor R on S, and the branch locus B on W as in [Q], §2. The following lemma proves that the assumptions in [Q], Lemma 4 are always satisfied.

Lemma 2.1. *There exists a line bundle F on W which satisfies the following conditions:*

(i) $[B]=2F$;

(ii) *S is the minimal resolution of the singularities of the double covering S' of W in F with branch locus B (see* [P], §2);

(iii) *there exists an effective divisor Z (possibly 0) in $|f^*F-R|$ such that $f^*B-2R=2Z$.*

Proof. Let $S\to S'\to W$ be the Stein decomposition of f, and let $\pi: S\to S'$, $f': S'\to W$ be the induced maps. Then S' is normal and $\pi_*\mathcal{O}_S=\mathcal{O}_{S'}$. Hence $f_*\mathcal{O}_S=f'_*\mathcal{O}_{S'}$ is locally free of rank 2, and we have an exact sequence

$$0\to\mathcal{O}_W\to f_*\mathcal{O}_S\to\mathcal{O}(-F)\to 0$$

for some line bundle F on W (see [P], Lemma 4). This F satisfies the condition (i) and S' is nothing but the double covering of W in F with branch locus B.

Let w_i be a fibre coordinate on F over an open set $U_i\subset W$, and let $b_i=0$ be an equation of B on U_i. Then S' is defined by $w_i^2=b_i$ over U_i (see [Q], p. 48). Moreover, the equations $w_i=0$ define a Cartier divisor on S', and it induces one on S. We call this R'. Then we have $R'\in|f^*F|$ and $2R'=f^*B$. If we set $Z=R'-R$, then $2Z$ is equal to f^*B-2R, and is effective by [Q], Lemma 3. Therefore Z itself is effective. The second assertion is proved in [Q], Lemma 4. Q.E.D.

§3. Surfaces of Type I

From now on, S denotes a minimal algebraic surface with $p_g=4$, $q=0$, and $c_1^2=6$. Let $\pi:\tilde{S}\to S$ be a composition of quadratic transformations such that the variable part $|L|$ of $|\pi^*K|$ has no base point. We assume that π is the shortest one among such compositions. We let F denote the fixed part of $|\pi^*K|$, and we write the canonical bundle $\tilde{K}$ of $\tilde{S}$ in the form $\pi^*K+[E]$ with an effective divisor E.

Lemma 3.1. *We have $L^2=6$ or 4.*

Proof. By Theorem 1.3, $|L|$ is not composite with a pencil. Hence, by [Q], Lemma 2, we have $4\leqq L^2\leqq 6$. It remains to exclude the possibility $L^2=5$.

Suppose $L^2=5$. Then $|L|$ defines a birational map of S onto a quintic surface in $\mathbb{P}^3$. Let C be a general member of $|L|$, which we assume to be non-singular. Then, by the adjunction formula, C is of genus $6+L(E+F)/2$. On the other hand, the image of C is a plane quintic curve, which is at most of genus 6. Therefore we have $LF=0$. But this implies $F^2=1$, which contradicts Hodge's index theorem. Q.E.D.

In the rest of this section, we assume $L^2=6$, i.e., by [Q], Lemma 2, the canonical system $|K|$ has no base point. The following theorem is obvious.

Theorem 3.1. *Suppose that the canonical system $|K|$ has no base point. Such surfaces are classified into the following three types:*

(Ia) *the canonical map Φ_K induces a birational map of S onto a sextic surface in $\mathbb{P}^3$;*

(Ib) *Φ_K induces a map of degree 2 onto a cubic surface in $\mathbb{P}^3$;*

(II) *Φ_K induces a map of degree 3 onto a quadric in $\mathbb{P}^3$.*

We shall study surfaces of type I in this section, and those of type II in the next section.

Theorem 3.2. *Let S be a surface of type* Ia, *and let $S'=\Phi_K(S)$ be its canonical image. Then S' is defined in $\mathbb{P}^3$ by the equation of the form*

$$g^2+Agh+Bh^2=0,$$

where g, h, A, and B are homogeneous forms of degree 3, 1, 2, *and* 4, *respectively. Moreover, S is the minimal resolution of the singularities of S'.*

Proof. Let $\{\varphi_0,\varphi_1,\varphi_2,\varphi_3\}$ be a basis of $H^0(S,\mathcal{O}(K))$. Then the products $\varphi_i\varphi_j$ generate a 10-dimensional subspace of $H^0(S,\mathcal{O}(2K))$. Since we have $h^0(2K)=11$ (see [16, 3]), there exists $\psi\in H^0(S,\mathcal{O}(2K))$ which is independent from the $\varphi_i\varphi_j$.

The products $\varphi_i\varphi_j\varphi_k$ and $\varphi_i\psi$ determine 24 elements of $H^0(S,\mathcal{O}(3K))$. Because of the equality $h^0(3K)=23$, there exists a relation of the form

$$h\psi=g, \tag{3.1}$$

where h and g are linear and cubic forms in $(\varphi_0,\varphi_1,\varphi_2,\varphi_3)$, respectively. We may assume $h=\varphi_0$. Then, by the choice of ψ, g is not divisible by φ_0.

Suppose that we have another relation $h'\psi=g'$ of the same form. Then we have $hg'=h'g$. Since g is not divisible by φ_0, we conclude that $h'=\alpha h$ and $g'=g/\alpha$ for some constant $\alpha\neq 0$. This implies that the collection

$$\varphi_i\varphi_j\varphi_k \quad (0\leqq i\leqq j\leqq k\leqq 3), \qquad \varphi_i\psi \quad (1\leqq i\leqq 3)$$

forms a basis of $H^0(S,\mathcal{O}(3K))$.

Next we prove that the following 41 elements

$$\begin{aligned} &\varphi_i\varphi_j\varphi_k\varphi_l \quad (0\leqq i\leqq j\leqq k\leqq l\leqq 3),\\ &\varphi_i\varphi_j\psi \quad (1\leqq i\leqq j\leqq 3) \end{aligned} \tag{3.2}$$

are linearly independent in $H^0(S,\mathcal{O}(4K))$. Suppose we have a relation $p\psi=q$, where p and q are respectively of degree 2 and 4 in the φ_i's. Then we have $hq=pg$. This is a relation of degree 5 in the φ_i's. Hence p is divisible by φ_0. This proves that there is no relation among the elements in (3.2).

Comparing the dimensions, we see that (3.2) forms a basis. Therefore, we can write ψ^2 in the form

$$\psi^2=-A\psi-B, \tag{3.3}$$

where A and B are homogeneous forms in the φ_i's of degree 2 and 4, respectively. Combining with (3.1), we obtain

$$g^2+Agh+Bh^2=0.$$

This is a non-trivial relation of degree 6 among the φ_i's. Hence S' is defined by this equation.

Next we consider the line bundle $\varpi: V\to\mathbb{P}^3$ of degree 2, and let θ be a fibre coordinate on V. By this, we mean that θ is a section of $\varpi^*\mathcal{O}(2)$ which defines the 0-section of ϖ. Then, setting $\theta=\psi$, we obtain a well-defined holomorphic map $\Psi: S\to V$ which satisfies $\varpi\circ\Psi=\Phi_K$. Since (3.2) is a basis of $H^0(S,\mathcal{O}(4K))$, the image $\Psi(S)$ has only rational double points (see [3], Main Theorem). From (3.1) and (3.3), it follows that $\Psi(S)$ is defined by

$$\begin{aligned}&\theta^2+A\theta+B=0,\\ &\quad h\theta-g=0.\end{aligned}\tag{3.4}$$

Since the projection $\Psi(S)\to S'$ is finite, $\Psi(S)$ is the normalization of S'. Hence S is the minimal resolution of the singularities of S'. Q.E.D.

Conversely, if S'' is a subvariety of V of the form (3.4), and if S'' has only rational double points, then its minimal resolution of singularities is a minimal surface with $p_g=4$, $q=0$, and $c_1^2=6$ which is of type Ia. These surfaces were already known to M. Noether [21] in the generic case. However, the double curve $g=h=0$ may be singular.

Corollary. *The image of the bicanonical map Φ_{2K} is normal and has at most rational double points.*

Next we shall study the structure of surfaces of type Ib.

Theorem 3.3. *Let S be a surface of type* Ib. *Then*

(i) *the canonical map is of degree 2 onto a cubic surface $g=0$ in $\mathbb{P}^3$,*

(ii) *the bicanonical map Φ_{2K} is birational and its image is defined by*

$$\theta^2+A\theta+B=0,\quad g=0\tag{3.5}$$

in the line bundle $\varpi: V\to\mathbb{P}^3$ of degree 2, where θ denotes a fibre coordinate on V, and A and B are forms of degree 2 and 4 on $\mathbb{P}^3$, respectively.

(i) is obvious by definition. To prove (ii), we first prove the following

Lemma 3.2. (i) *The canonical image $W=\Phi_K(S)$ has only isolated singularities.*

(ii) *A general member C of $|K|$ is a non-hyperelliptic curve of genus 7.*

(iii) *The bicanonical map Φ_{2K} is birational.*

Proof. (i) Suppose that W has a singular locus of dimension 1. Let D be a hyperplane section of W and let C be its inverse image on S. We assume that C is irreducible and non-singular. Let $\tilde{D}\to D$ be the normalization of D. Then the induced map $C\to D$ factors through $\tilde{D}$. If $K_{|C}$ denotes the restriction of K to C, then we have $\dim H^0(C,\mathcal{O}(K_{|C}))>3$, because D is a rational curve of degree 3.

On the other hand, from the exact sequence

$$0 \to \mathcal{O} \to \mathcal{O}(K) \to \mathcal{O}(K_{|C}) \to 0,$$

and $H^1(S, \mathcal{O}) = 0$, it follows that $\dim H^0(C, \mathcal{O}(K_{|C})) = 3$. This is a contradiction.

(ii) Let D be a generic hyperplane section of W, and let C be as above. Then the induced map $C \to D$ is 2-sheeted and the composition $C \to D \to \mathbb{P}^2$ is defined by the linear system $|K_{|C}|$. Since $2K_{|C}$ is the canonical bundle of C, the canonical image of C dominates D. Since D is an elliptic curve, this implies that C is not hyperelliptic.

(iii) We consider the exact sequence

$$0 \to \mathcal{O}(K) \to \mathcal{O}(2K) \to \mathcal{O}(2K_{|C}) \to 0.$$

From the assumption $q = 0$, it follows that the restriction map

$$H^0(S, \mathcal{O}(2K)) \to H^0(C, \mathcal{O}(2K_{|C}))$$

is surjective. Combining with (ii), we conclude that Φ_{2K} induces a birational map of C onto its image. Hence Φ_{2K} induces a birational map of S onto its image. Q.E.D.

Proof of Theorem 3.3. We take a basis $\{\varphi_0, \varphi_1, \varphi_2, \varphi_3\}$ of $H^0(S, \mathcal{O}(K))$ and a section $\psi \in H^0(S, \mathcal{O}(2K))$ in the same way as in Theorem 3.2. We claim that the products

$$\varphi_i \varphi_j \varphi_k \quad (0 \leqq i \leqq j \leqq k \leqq 3), \qquad \varphi_i \psi \quad (0 \leqq i \leqq 3)$$

generate $H^0(S, \mathcal{O}(3K))$. In fact, the products $\varphi_i \varphi_j \varphi_k$ generate a 19-dimensional subspace. Hence, if our claim is false, there is a non-trivial relation $p\psi = q$, where p and q are linear and cubic forms in φ_i's. But this contradicts that Φ_{2K} is birational.

In the same manner, we can prove that

$$\varphi_i \varphi_j \varphi_k \varphi_l \quad (0 \leqq i \leqq j \leqq k \leqq l \leqq 3), \qquad \varphi_i \varphi_j \psi \quad (0 \leqq i \leqq j \leqq 3) \tag{3.6}$$

generate $H^0(S, \mathcal{O}(4K))$. It follows that there exist homogeneous polynomials A and B in the φ_i's of degree 2 and 4, respectively, such that

$$\psi^2 + A\psi + B = 0.$$

Moreover, ψ defines a holomorphic map $\psi: S \to V$ whose image is contained in (3.5). Since the collection (3.6) generates $H^0(S, \mathcal{O}(4K))$, the image $\Psi(S)$ is birationally equivalent to S (see [3], Main Theorem). Hence the projection $\Psi(S) \to W$ is two-sheeted and $\Psi(S)$ is defined by (3.5).

Conversely, if (3.5) defines a normal surface with at most rational double points, then its minimal resolution is a minimal surface with $p_g = 4$, $q = 0$, and $c_1^2 = 6$ which is of type Ib. By Bertini's theorem, these surfaces actually exist.

§4. Surfaces of Type II

Let S be a surface of type II. That is, $\Phi_K: S \to \mathbb{P}^3$ induces a holomorphic map of degree 3 onto a quadric W.

Lemma 4.1. *W is singular.*

Proof. Suppose that W is non-singular. Then W is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$. Hence the canonical bundle is of the form $[C+D]$ with $C^2=D^2=0$, $CD=3$. This contradicts that $KC+C^2$ is even. Q.E.D.

Let Σ_2 be the Hirzebruch surface of degree 2. Then Σ_2 is the minimal resolution of the singularity of W. We let Γ and Δ_0 denote a fibre and the 0-section of Σ_2, respectively.

Theorem 4.1. *Let $\varpi: V \to \Sigma_2$ be the line bundle associated with $2\Delta_0+3\Gamma$, and let w be a fibre coordinate on V. Then S is birationally equivalent to a surface S' in V defined by the equation*

$$w^3+\alpha\zeta w^2+\beta\zeta w+\gamma\zeta^2=0, \tag{4.1}$$

where

$$\zeta\in H^0(\Sigma_2, \mathcal{O}([\Delta_0])), \qquad \alpha\in H^0(\Sigma_2, \mathcal{O}([\Delta_0+3\Gamma])),$$
$$\beta\in H^0(\Sigma_2, \mathcal{O}([3\Delta_0+6\Gamma])), \qquad \gamma\in H^0(\Sigma_2, \mathcal{O}([4\Delta_0+9\Gamma])).$$

More precisely, let $\tilde{S}$ be the minimal resolution of S'. Then $\tilde{S}$ contains an exceptional curve E over Δ_0 and S is obtained from $\tilde{S}$ by contracting E to a point.

Proof. Since $\Phi_K(S)$ is a quadratic cone, we can find a pencil $|D|$ and an effective divisor G such that $|K|=|2D+G|$ and $KG=0$ (see [Q], p. 46). From $K^2=6$, it follows that $KD=3$, which in turn implies $D^2=DG=1$ and $G^2=-2$. In particular, the pencil $|D|$ has a unique base point b. We let $\pi: \tilde{S} \to S$ be the quadratic transformation with center b, and set $E=\pi^{-1}(b)$. The variable part $|\tilde{D}|$ of $|\pi^* D|$ defines a holomorphic map $\tilde{S} \to \mathbb{P}^1$. Since $|K|$ has no base point, there exists a section $\eta\in H^0(\tilde{S}, \mathcal{O}(\pi^* K))$ which does not vanish on $\pi^* G \cup E$. Furthermore, we take a non-zero section ω of $[\pi^* G+2E]$ over $\tilde{S}$, which is unique up to a constant. Then the pair (ω, η) defines a holomorphic map $h: \tilde{S} \to \Sigma_2$ such that $h^* \Delta_0=\pi^* G+2E$. We note that the canonical bundle $\tilde{K}$ of $\tilde{S}$ is given by $h^*[\Delta_0+2\Gamma]+[E]$.

Next we shall lift the map h to $\tilde{h}: \tilde{S} \to V$. For this purpose, we consider the line bundle

$$h^*[2\Delta_0+3\Gamma]=\tilde{K}+[\tilde{D}+\pi^* G+E]. \tag{4.2}$$

By the Riemann-Roch theorem and the vanishing of $H^1(\tilde{S}, \mathcal{O}(-\tilde{D}))$ (see [3], Theorem A), we have

$$\dim H^0(\tilde{S}, \mathcal{O}(\tilde{K}+\tilde{D}))=\tfrac{1}{2}\tilde{D}(\tilde{K}+\tilde{D})+5=7.$$

If ξ is a non-zero section of $[\pi^*G+E]$, then, for any $\varphi \in H^0(\tilde{S}, \mathcal{O}(\tilde{K}+[\tilde{D}]))$, the product $\varphi\xi$ determines a section of $h^*[2\Delta_0+3\Gamma]$ over $\tilde{S}$. Since we have $\dim H^0(\Sigma_2, \mathcal{O}(2\Delta_0+3\Gamma))=6$, there exists φ such that $\varphi\xi$ is not induced by a section of $[2\Delta_0+3\Gamma]$ over Σ_2. Setting $w=\varphi\xi$, we obtain a holomorphic map $\tilde{h}: \tilde{S} \to V$.

Lemma 4.2. *The holomorphic map $\tilde{h}$ induces a birational map of $\tilde{S}$ onto its image.*

Proof. Let S' be the image of $\tilde{h}$. Since h is of degree 3, if S' is not birationally equivalent to $\tilde{S}$, then the projection $\varpi: S' \to \Sigma_2$ is birational. Hence we have an isomorphism

$$H^0(\Sigma_2, \mathcal{O}([2\Delta_0+3\Gamma])) \xrightarrow{\sim} H^0(S', \mathcal{O}(\varpi^*[2\Delta_0+3\Gamma])).$$

But this contradicts the choice of φ. Q.E.D.

Lemma 4.3. *The image of $\tilde{h}$ is defined by the equation of the form* (4.1).

Proof. We consider the subspace of $H^0(\tilde{S}, \mathcal{O}(h^*[6\Delta_0+9\Gamma]))$ which consists of those sections vanishing on $2\pi^*G+3E$. It contains

$$\begin{array}{ll} \varphi^3\xi^3 & \\ \varphi^2\xi^2\alpha\zeta & \text{with } \alpha \in H^0(\Sigma_2, \mathcal{O}([\Delta_0+3\Gamma])), \\ \varphi\xi\beta\zeta & \text{with } \beta \in H^0(\Sigma_2, \mathcal{O}([3\Delta_0+6\Gamma])), \\ \gamma\zeta^2 & \text{with } \gamma \in H^0(\Sigma_2, \mathcal{O}([4\Delta_0+9\Gamma])). \end{array} \tag{4.3}$$

When α, β, and γ move in bases of the corresponding spaces, the above list represents 53 sections. On the other hand, we have

$$[h^*(6\Delta_0+9\Gamma)-(2\pi^*G+3E)] = \tilde{K}+h^*[3\Delta_0+7\Gamma].$$

Hence, by the Riemann-Roch theorem and the Kodaira vanishing theorem (see [16, 3]), we have

$$h^0(h^*(6\Delta_0+9\Gamma)-(2\pi^*G+3E))=52.$$

Therefore we can find a non-trivial relation of the form

$$\delta(\varphi\xi)^3+\alpha\zeta(\varphi\xi)^2+\beta\zeta(\varphi\xi)+\gamma\zeta^2=0,$$

where $\delta \in \mathbb{C}$ and α, β, γ, ζ are as above. From Lemma 4.2, we infer that $\delta \neq 0$ and that this is the only relation among the sections in (4.3). Q.E.D.

Lemma 4.4. *Let Λ be the curve defined by $w=\zeta=0$. Then $S'=\tilde{h}(\tilde{S})$ has a double curve along Λ, but other singular points of S' are rational double points.*

Proof of Lemma 4.4 consists of several lemmas. Let $\mu: V^* \to V$ be the monoidal transformation with center Λ and let S^* be the proper transform of S' by μ.

Lemma 4.5. *The holomorphic map $\tilde{h}: \tilde{S} \to S'$ factors through S^*.*

Proof. Let $\mathscr{I}$ be the ideal sheaf of Δ on S' and let $\mathscr{J}$ be the ideal of $\mathcal{O}_{\tilde{S}}$ generated by $\tilde{h}^* \mathscr{I}$. We shall show that $\mathscr{J}$ is invertible. In fact, we have

$$\tilde{h}^* w = \varphi \xi, \quad \tilde{h}^* \zeta = e \xi,$$

where e is a non-zero section in $H^0(\tilde{S}, \mathcal{O}([E]))$. It suffices to prove that φ and e have no common zero. To see this, we first note that, since $(\tilde{K}+\tilde{D})E=0$, φ is constant on E. Secondly, we have the exact sequence

$$0 \to \mathcal{O}(\tilde{K}) \to \mathcal{O}(\tilde{K}+[\tilde{D}]) \to \mathcal{O}(K_{\tilde{D}}) \to 0,$$

where $K_{\tilde{D}}$ denotes the canonical bundle of $\tilde{D}$, and the restriction map

$$H^0(\tilde{S}, \mathcal{O}(\tilde{K}+[\tilde{D}])) \to H^0(\tilde{D}, \mathcal{O}(K_{\tilde{D}}))$$

is surjective. Since $|K_{\tilde{D}}|$ has no base point, it follows that $|\tilde{K}+\tilde{D}|$ has no base point on $\tilde{D}$.

Thirdly, by virtue of the choice of φ, we can find a basis $\{\varphi, \psi_1, \ldots, \psi_6\}$ of $H^0(\tilde{S}, \mathcal{O}(\tilde{K}+[\tilde{D}]))$ such that the $\psi_i \xi$ are induced by the elements of $H^0(\Sigma_2, \mathcal{O}([2\Delta_0+3\Gamma]))$. In view of the fact that Δ_0 is a fixed component of $|2\Delta_0 + 3\Gamma|$ and the equality $h^*\Delta_0 = \pi^* G + 2E$, the $\psi_i \xi$ vanish twice along E. Hence the ψ_i vanish along E. Therefore, if φ is 0 on E, the linear system $|\tilde{K}+\tilde{D}|$ has base points along E. This contradicts what we have proved above. Q.E.D.

Lemma 4.6. *The singular locus of S' consists of the double curve Λ and a finite number of points.*

Proof. Let Δ be a general member of $|\Delta_0+2\Gamma|$ on Σ_2 and let C be its inverse image on $\tilde{S}$. We may assume that C is irreducible and non-singular. Since C is a member of $|\pi^* K|$, it is of genus 7. Let C' be the inverse image of Δ on S'. Then we have the sequence of maps $C \to C' \to \Delta$. We consider the restriction $V_{|\Delta} \to \Delta$ of V to Δ, which is a line bundle of degree 3. From the form of (4.1), we readily infer that C' is of arithmetic genus 7. Hence C' is non-singular. This implies that there is no multiple curve other than Λ. Q.E.D.

Lemma 4.7. *The proper transform S^* by μ has only isolated singularities.*

Proof. In view of Lemma 4.6, it suffices to consider $\mu^{-1}(\Lambda)$. Let U be a coordinate neighborhood on Σ_2 which meets Δ_0, and let (z, ζ) be a system of coordinates on U such that Δ_0 is defined by $\zeta=0$. For a while, we let w denote a fibre coordinate on $\tilde{U}=\varpi^{-1}(U)$. Then $S' \cap \tilde{U}$ is defined by the equation of the form (4.1), where α, β, γ are regarded as functions of (z, ζ). By replacing w by $w + \alpha\zeta/3$, we may assume $\alpha=0$. The inverse image $\mu^{-1}(\tilde{U})$ is covered by two open sets $V_i, i=1,2$, and the V_i are covered by systems of coordinates (z_i, u_i, v_i), which satisfy

$$z = z_1 = z_2, \quad \zeta = u_1 v_1 = u_2, \quad w = u_1 = u_2 v_2.$$

Suppose first that at least one of β and γ does not vanish identically on Δ_0. Then the proper transform S^* of S' is defined by

$$\begin{aligned} &u_1 + \beta v_1 + \gamma v_1^2 = 0 \quad \text{on } V_1, \\ &u_2 v_2^3 + \beta v_2 + \gamma = 0 \quad \text{on } V_2. \end{aligned} \tag{4.4}$$

It is clear that $S^* \cap V_1$ is non-singular. On the other hand, the intersection $\mu^{-1}(\Delta) \cap S^* \cap V_2$ is defined by $u_2 = \beta v_2 + \gamma = 0$, and is non-singular. Hence $S^* \cap V_2$ is non-singular in a neighborhood of $\mu^{-1}(\Delta) \cap S^* \cap V_2$.

Next suppose that β and γ both vanish on Δ_0. Then Δ is a triple curve of S'. We shall prove that this is impossible. For this purpose, we take a non-singular rational curve $\bar{C} \in |\Delta_0 + 3\Gamma|$ on Σ_2, and let C be its preimage on $\tilde{S}$. We may assume C is non-singular. Then, being a member of $|\pi^* K + \tilde{D}|$, C is of genus 12. Let C' be the image of C on S'. Then C' is defined by the Equation (4.1) in the restriction of V over $\bar{C}$. This restriction is a line bundle of degree 5 over $\bar{C}$. From these facts, it follows that C' is of arithmetic genus 13. This implies that C' has only one double point. Hence Δ is not a triple curve. Q.E.D.

Lemma 4.8. *S^* has only rational double points as singularities.*

Proof. In this proof we let V denote the $\mathbb{P}^1$-bundle associated with $[2\Delta_0 + 3\Gamma]$. Since the normal sheaf $\mathcal{N} = \mathcal{N}_{\Delta/V}$ of Δ in V is isomorphic to $\mathcal{O}(-1) \oplus \mathcal{O}(-2)$, the exceptional divisor $\mathscr{E} = \mu^{-1}(\Delta)$ is isomorphic to Σ_1. Furthermore, the normal bundle of $\mathscr{E}$ in V is given by $[-b-2f]$, where b and f denote the 0-section and a fibre of Σ_1, respectively. This latter fact can be seen, for example, as follows. We note that $\bigwedge^2 \mathcal{N} = \mathcal{O}(-3)$. Hence the canonical bundle K_V induces a line bundle of degree 1 on Δ. The canonical bundle of $\mathscr{E}$ is induced by $\mu^* K_V + [2\mathscr{E}]$, and it coincides with $[-2b-3f]$. Since $\mathscr{E}$ has no torsion line bundle, we conclude $[\mathscr{E}]_{|\mathscr{E}} = [-b-2f]$.

From (4.4), we infer that $[S^*]_{|\mathscr{E}} = [2b+f]$. Hence, combined with the above observation, it follows that $[\mu^* S']$ induces $[-3f]$ on $\mathscr{E}$. Hence we obtain the exact sequence

$$0 \to \mathcal{O}(-\mu^* S' + (k-1)\mathscr{E}) \to \mathcal{O}(-\mu^* S' + k\mathscr{E}) \to \mathcal{O}_{\mathscr{E}}(-kb - (2k-3)f) \to 0 \qquad (4.5)$$

for any k. First, by a standard calculation, we prove $\chi(\mathcal{O}_V(-S')) = -3$. Then using (4.5), we obtain $\chi(\mathcal{O}(-\mu^* S' + 2\mathscr{E})) = -4$. Finally we obtain $\chi(\mathcal{O}_{S^*}) = 5$.

Thus we have proved $\chi(\mathcal{O}_{\bar{S}}) = \chi(\mathcal{O}_{S^*})$. Since S^* is normal by Lemma 4.7, it follows that S^* has only rational singularities. Since S^* is embedded in a non-singular threefold, it has only rational double points (see [1], Corollary 6). This completes the proof of Lemma 4.8, and is the end of the proof of Lemma 4.4.

As we have seen in the proof of Lemma 4.7, S^* is non-singular in a neighborhood of $\mu^{-1}(\Delta) \cap S^*$. Now we shall prove that S^* contains an exceptional curve in $\mu^{-1}(\Delta) \cap S^*$. We divide the case according to whether β vanishes on Δ_0 or not.

If β does not vanish on Δ_0, $\mu^{-1}(\Delta) \cap S^*$ consists of the two disjoint curves E and $\tilde{G}$, where E is defined by $v_1 = 0$ and $\tilde{G}$ is defined by $\beta + \gamma v_1 = 0$ on V_1 and $\beta v_2 + \gamma = 0$ on V_2, in the previous notation. We claim $E^2 = -1$. In fact, let U' be another coordinate neighborhood on Σ_2 and let ζ', w', v_1' etc. be the corresponding coordinates as above. Then we have

$$\zeta/\zeta' = (w/w')(v_1/v_1').$$

We identify E with Λ by the map μ, and note that ζ/ζ' and w/w' are respectively the transition functions of $\varpi^*[\Delta_0]$ and the line bundle associated with the 0-section of V. On Λ, the former is of degree -2, and the latter -1. This implies that $E^2=-1$. Moreover, from the equality $\zeta=-v_1^2(\beta+\gamma v_1)$, it follows that the pull-back of Δ_0 on S^* is $2E+\tilde{G}$. Hence we have $\tilde{G}^2=-2$.

Next we suppose that β vanishes identically on Δ_0. Then $\mathscr{E}_{|S^*}$ is of the form $2E+G'$, where E is defined by $v_1=0$ and G' by $\gamma=0$. In the same way as above, we can prove $E^2=-1$. On the other hand, since γ has a simple zero on Δ_0, G' is a fibre of the projection $\mu^{-1}(\Lambda)\to\Lambda$. Hence we have $EG'=1$. Furthermore, the inverse image of Δ_0 is $3E+G'$. This proves that $G'^2=-3$. We set $\tilde{G}=G'+E$ in accordance with the preseding case.

In the first case, S^* is the normalization of S', while, in the second case, S^* is obtained by blowing up a point on the normalization of S'. In either case $\tilde{S}$ is the minimal resolution of the singularities of S'.

Conversely, we start with S' defined by (4.1) and assume it has only rational double points except for the double curve Λ. Let $\mu\colon S^*\to S'$ be the monodial transformation with center Λ and let $\tilde{S}$ be the minimal resolution of S^*. Then $\tilde{S}$ contains an exceptional curve E over Δ_0. We let $\pi\colon \tilde{S}\to S$ be the contraction of E. We now prove S is a minimal surface with $p_g=4$, $q=0$, and $c_1^2=6$. First, by a standard calculation using (4.5), we obtain $p_g=4$ and $q=0$. Secondly, the canonical bundle of $\tilde{S}$ is induced by $h^*(2\Delta_0+2\Gamma)-\mu^{-1}(\Lambda)$, where $h\colon \tilde{S}\to\Sigma_2$ denotes the natural map. As we have seen above, $h^*\Delta_0=2E+\tilde{G}$ and $\mu^{-1}(\Lambda)=E+\tilde{G}$. Hence, $\pi^*K=h^*[\Delta_0+2\Gamma]$. This implies $K^2=6$, and that $|K|$ has no base point. In particular, S is minimal. This completes the proof of Theorem 4.1.

It remains to prove the existence of the Equations (4.1) which define S' with the properties required above. For this purpose, we regard (4.1) as a linear system of divisors on V. It can be easily checked that it has no base point outside of Λ. Furhtermore we have already shown that (4.4) is non-singular provided one of β and γ does not vanish identically on Δ_0. Therefore, for a generic choice of (4.1), S^* is non-singular. This proves the existence of surfaces of type II.

§5. Surfaces of Type III

In this and the next sections we shall study the case in which $|K|$ has base points. We let $\pi\colon \tilde{S}\to S$, L, E, and F be the same as in §3.

Lemma 5.1. *Assume that $|K|$ has base points. Then we have $L^2=4$, $LF=2$, $F^2=-2$, and $LE=0$ or 2.*

Proof. By Lemma 3.1, we have $L^2=4$ and $(L+\pi^*K)F=K^2-L^2=2$. The second equality implies that $LF=1$ or 2. On the other hand, by the Riemann-Roch theorem and the Kodaira vanishing theorem, we have

$$h^0(\tilde{K}+L)=L^2+\tfrac{1}{2}L(E+F)+5, \tag{5.1}$$

which is at most equal to $h^0(2\tilde{K})=11$. We also note that $F-E$ is effective, and hence $LF\geqq LE$. This is obvious if π is a single quadratic transformation at a base point of $|K|$. The general case immediately follows from this special case.

If $LF=1$, we have $LE=1$, and hence E is irreducible (see [Q], p. 45). From the equalities $LE=LF=-EF=1$, we infer that $F=E+F'$ with $LF'=EF'=0$ (see [Q], p. 46). From $F^2=K^2-L^2-2LF=0$, it follows that $F'^2=1$, which contradicts Hodge's index theorem. Therefore, we have $LF=2$ and $F^2=-2$. Since $L(E+F)$ is even by (5.1), we have $LE=0$ or 2. Q.E.D.

We shall call S of *type* III if $|K|$ has base points and if S does not admit a pencil $|D|$ whose general member is a non-singular curve of genus 2. We refer to such a pencil as a pencil of curves of genus 2. Surfaces with such pencil will be studied in the next section. Furthermore, we call S of *type* IIIa or IIIb according to whether $LE=2$ or 0.

Theorem 5.1. *Let S be a surface of type* IIIa. *Then $|L|$ defines a holomorphic map of degree 2 onto a non-singular quadric W in $\mathbb{P}^3$. If we identify W with Σ_0, the branch locus B is in $|8\Delta_0+8\Gamma|$, and, in the generic case, it consists of a fibre Γ, a section Δ_0, and a curve B_0 which has a quadruple point at $x=\Gamma\cap\Delta_0$ and two triple points, one each on Γ and Δ_0.*

We first prove the following lemma.

Lemma 5.2. *The equality $LE=2$ implies that $F=E$ and $E^2=-2$.*

Proof. By (5.1) we have $h^0(\tilde{K}+L)=h^0(2K)$. Since $F-E$ is effective, we have

$$|2\pi^*K|=|\tilde{K}+L|+(F-E).$$

Since $|2K|$ has no base point (see [3], Theorem 2), we obtain $F=E$. Q.E.D.

Corollary. *π is the composition of two quadratic transformations.*

Let $f\colon \tilde{S}\to\mathbb{P}^3$ be the holomorphic map defined by $|L|$. We first suppose that W is singular and derive a contradiction. In the same manner as in [Q], p. 46, we can find a pencil $|D|$ and an effective divisor G such that $2D+G\in|L|$ and $LG=0$. From $L^2=4$, we get

$$2=LD=2D^2+DG. \tag{5.2}$$

Hence $\tilde{K}D$ and D^2 are even integers. By (5.2), we obtain $D^2=0$. $DG=2$, and $G^2=-4$. Using the inequality

$$6=(\pi^*K)L\geqq 2(\pi^*K)D=4+2DE,$$

we obtain $DE=0$ or 1. But $DE=0$ would imply that $|D|$ is a pencil of genus 2. Hence we have $DE=1$, and hence $GE=0$. Let E_0 be the unique component of E such that $DE_0=1$, and set $E_1=E-E_0$. Then, by the above Corollary, $E_1\neq 0$, and

$$2\geqq LE_0=2+GE_0, \qquad LE_1=GE_1.$$

Suppose first that $GE_0=0$. Then, combining with $GE=0$, we get $GE_1=LE_1=0$. This implies that π is the blowing up of two infinitely near points, for otherwise, we would have $LE_1>0$. But, then E is of the form $2E_0'+E_1'$, and we necessarily have $E_0=E_0'$. This contradicts the equality $DE=1$.

Next suppose $GE_0<0$. Then G contains E_0. Hence, from $LG=0$, we obtain $LE_0=0$. This again implies that π is the blowing up of two infinitely near points. This time we have $E_0^2=-2$ and $E_1=2E_1'$ with $E_0E_1'=-E_1'^2=1$. We also have

$$DE_1'=0, \quad LE_1'=1. \tag{5.3}$$

By the same method as in [Q], p. 46, f induces a holomorphic map $h: \tilde{S}\to\Sigma_2$ of degree 2 such that $h^*\Delta_0=G$, and the ramification divisor R of h is linearly equivalent to $6D+3G+2E$. Hence $DR=8$, $GR=0$. It follows that the branch locus B is linearly equivalent to $8\Delta_0+16\Gamma$. We consider an involution ϕ of $\tilde{S}$ which commutes with h. Such ϕ exists because $\tilde{S}$ is the minimal resolution of the double covering of Σ_2 with branch locus B (see Lemma 2.1 and [Q], Lemma 4). Since ϕ induces an involution on the minimal model S, and since it fixes the base points of $|K|$, it follows that $\phi(E_0)=E_0$ and $\phi(E_1')=E_1'$. Moreover, the equalities $DE_0=LE_1'=1$ imply that h is generically one-to-one on E_0 and E_1'. Therefore $h(E_0)$ and $h(E_1')$ are contained in the branch locus B. By construction, E_0 is mapped onto Δ_0, and, by (5.3), E_1' is mapped onto a fibre Γ_0. Since B is in $|8\Delta_0+16\Gamma|$, the divisor $B-\Delta_0-\Gamma_0$ intersects Δ_0 transversally at one point. This implies that B has no infinitely near triple points in a neighborhood of Δ_0. Hence the minimal resolution coincides with the canonical resolution in a neighborhood of the preimage of Δ_0 (see [Q], §2, Lemma 5). But, by the canonical resolution, we have to blow up the intersection of Δ_0 and Γ_0, and it turns out that E_0 and E_1' cannot intersect. This contradicts our previous observation.

Thus we have proved that $f(S)$ is a non-singular quadric W. We identify W with Σ_0, and use the same symbol f to denote the induced map $\tilde{S}\to\Sigma_0$. From the equality $L=f^*[\Delta_0+\Gamma]$, we obtain

$$\tilde{K}f^*\Delta_0+\tilde{K}f^*\Gamma=8.$$

Since S has no pencil of genus 2, it follows that $\tilde{K}f^*\Delta_0=\tilde{K}f^*\Gamma=4$, and hence $Ef^*\Delta_0=Ef^*\Gamma=1$. We take an irreducible component E_0 of E such that $E_0f^*\Gamma=1$ and set $E_1=E-E_0$. Note that E_0 is a simple component of E. Hence, by Corollary to Lemma 5.2, E_1 is either an exceptional curve or is double in case the two base points are infinitely near. Anyway, E_1 is not mapped into a point by f. Therefore we have $LE_1>0$. Since $E_1f^*\Gamma=0$, it follows that $E_1f^*\Delta_0=1$ and $E_0f^*\Delta_0=0$. Eventually, E_1 is reduced and irreducible.

By a straightforward calculation we see that the ramification divisor of f is linearly equivalent to $3L+2E$ and the branch locus B is linearly equivalent to $8\Delta_0+8\Gamma$. By Lemma 2.1, there exists a divisor $Z\in|f^*(\Delta_0+\Gamma)-2E|$. This implies the existence of

$$Z_0\in|f^*\Delta_0-2E_0|, \quad \text{and} \quad Z_1\in|f^*\Gamma-2E_1|.$$

Therefore B contains $f(E_0)$ and $f(E_1)$ as components.

Let S' be the double covering of Σ_0 with branch locus B, and let S^* be the canonical resolution of S'. Then S^* has the following numerical characters:

$$p_a(S^*)=9-\tfrac{1}{2}\sum[m_i/2]([m_i/2]-1),$$
$$c_1^2(S^*)=16-2\sum([m_i/2]-1)^2,$$

where the m_i denote the multiplicities appearing in the process of the canonical resolution, and the brackets denote the integral part (see [Q], §2). We recall $p_a(S^*)=4$ and $c_1^2(S^*)\leqq c_1^2(\tilde{S})=4$. Therefore, we have $[m_i/2]=3$ for one i and $[m_i/2]=2$ for two i's. Hence, we have $c_1^2(S^*)=c_1^2(\tilde{S})$, and $\tilde{S}$ is the canonical resolution of the singularities of S'.

Let $\tilde{x}$ be one of the intersection $E_0\cap Z_0$ and set $x=f(\tilde{x})$. Let q: $W_1\to\Sigma_0$ be the quadratic transformation with center x. Then, by the above observation, f can be lifted to f_1: $\tilde{S}\to W_1$. Let $E_x=q^{-1}(x)$ and let R_1 be the ramification divisor of f_1. Then $(f^*E_x)R_1$ is 6 or 4 according to whether x is on $f(E_1)$ or not. Applying a similar consideration to E_1, we see that B has a sextuple point at $f(E_0)\cap f(E_1)$. Now we take x to be $f(E_0)\cap f(E_1)$. Then the branch locus B_1 of f_1 has two quadruple points, one each on the proper transforms of $f(E_0)$ and $f(E_1)$.

Conversely, we take a section Δ_0 and a fibre Γ on Σ_0, and we let x be the intersection $\Delta_0\cap\Gamma$. Let q: $W_1\to\Sigma_0$ be the blowing up of x, and let q: $\tilde{W}\to W_1$ be the blowing up of two points y and z, which are on the proper transforms of Δ_0 and Γ, respectively. Here y and z may be over x. We consider the composition $\mu=q_1\circ q$, and a divisor

$$\tilde{B}_0\in|\mu^*(7\Delta_0+7\Gamma)-4E_x-3E_y-3E_z|,$$

where $E_x=q_1^*(q^{-1}(x))$, $E_y=q_1^{-1}(y)$, and $E_z=q_1^{-1}(z)$. We assume that $\tilde{B}_0$ has no multiple component nor infinitely near triple points, and does not contain the proper transforms $\tilde{\Delta}_0$ and $\tilde{\Gamma}$ of Δ_0 and Γ. Let $\tilde{S}$ be the double covering of $\tilde{W}$ with branch locus $\tilde{B}_0+\tilde{\Delta}_0+\tilde{\Gamma}$. Then $\tilde{S}$ contains two exceptional curves E_0 and E_1 over $\tilde{\Delta}_0$ and $\tilde{\Gamma}$. Contracting E_0 and E_1, we obtain a minimal surface S with $p_g=4$, $q=0$, and $c_1^2=6$.

We prove that S is of type IIIa. For this purpose, we first note that the canonical bundle $\tilde{K}$ of $\tilde{S}$ is defined by $f^*(\Delta_0+\Gamma)+2E_0+2E_1$ (see [Q], §2). Hence the canonical system $|K|$ of S has two base points. Next we shall prove that there exists no pencil $|D|$ of genus 2 on S. In fact, if such $|D|$ exists, it has no base point (see [P], Theorem 5). Hence we have

$$2=(\pi^*D)\tilde{K}=(\pi^*D)(f^*\Delta_0)+(\pi^*D)(f^*\Gamma),$$

where π denotes the contraction map $\tilde{S}\to S$. If $(\pi^*D)(f^*\Gamma)=1$, then D is mapped birationally onto $\mathbb{P}^1$, which is impossible. Hence, one of the above summands vanishes. Suppose, for instance, $(\pi^*D)(f^*\Gamma)=0$. Then, since $(\pi^*D)^2=0$, π^*D is numerically equivalent to a rational multiple of $f^*\Gamma$ (see, e.g. [23], p. 92). But this is impossible. This completes the proof of Theorem 5.1.

Remarks 1. The existence of $\tilde{B}_0$ can be checked by a method similar to [Q], pp. 52–53. Here we note that y and z may be both infinitely near to x, in which case B_0 has a quintuple point at x.

2. We can contract on W_1 the proper transforms of Δ_0 and Γ. This transforms W_1 into $\mathbb{P}^2$, and B into a curve of degree 10 which has two 2-fold triple points (cf. Enriques [6], p. 272). We shall use this fact in §11.

Theorem 5.2. *Let S be a surface of type* IIIb. *Then S is birationally equivalent to a double covering of Σ_2 whose branch locus B consists of the 0-section Δ_0 and $B_0 \in |7\Delta_0 + 14\Gamma|$ which has a quadruple point at $x \in \Gamma$ and a 2-fold triple point at $y \in \Gamma$ on a fibre Γ, with x and y being possibly infinitely near.*

Proof. Using the previous notation, we have $E=0$. Let $f: S \to \mathbb{P}^3$ be the holomorphic map defined by the variable part $|L|$ of $|K|$. We first prove that the image W of f is singular. If not, then $W=\Sigma_0$, and $L=f^*[\Delta_0+\Gamma]$. From $KL=6$, we obtain $K(f^*\Delta_0+f^*\Gamma)=6$. Since $Kf^*\Delta_0$ and $Kf^*\Gamma$ are both even, one of them must be equal to 2. This contradicts that S has no pencil of genus 2.

Thus W is a quadratic cone, and hence $|L|$ is of the form $|2D+G|$ with a pencil $|D|$ and $LG=0$ (see [Q], p. 46). It follows that

$$2D^2+DG=DL=2, \qquad 2DF+GF=2, \tag{5.4}$$

where F denotes the fixed part of $|K|$. Hence we have $D^2=0$ or 1. But, if $D^2=0$, then KD is even, and hence so is DF. Since $GF=KF$ is non-negative, from the second formula in (5.4), we obtain $DF=0$. This contradicts that S has no pencil of genus 2. Thus we have proved $D^2=1$, $DG=0$, and $G^2=0$. By Hodge's index theorem, we obtain $G=0$, and hence $DF=1$. In particular, $|D|$ has a base point b.

Let $\pi: \tilde{S} \to S$ be the quadratic transformation with center b, and set $E=\pi^{-1}(b)$, $\tilde{D}=\pi^*D-E$. Then we obtain a holomorphic map $f: \tilde{S} \to \Sigma_2$ of degree 2 such that $f^*\Delta_0=2E$. The ramification divisor R and the branch locus B are linearly equivalent to $\pi^*(3L+F)+E$ and $8\Delta_0+14\Gamma$, respectively. In particular, B is a disjoint sum Δ_0+B_0 with $B_0 \in |7\Delta_0+14\Gamma|$. By Lemma 2.1, there exists an effective divisor $Z \in |\tilde{D}+E-\pi^*F|$. Since $f(Z)$ is a finite set, Z does not contain E. On the other hand, E is a fixed component of $|\tilde{D}+E|$. Hence π^*F is of the form $E+F'$ with $F' \geqq 0$.

Let S^* be the canonical resolution of the double covering S' of Σ_2 with branch locus B. Then S^* has the following numerical characters:

$$p_a = 6 - \tfrac{1}{2}\sum [m_i/2]([m_i/2]-1),$$
$$c_1^2 = 8 - 2\sum([m_i/2]-1)^2$$

(see [Q], §2). It follows that there are exactly two i's with $[m_i/2]=2$. Hence S^* is a quadratic transform of $\tilde{S}$. Moreover, since Z is linearly equivalent to $\tilde{D}-F'$, the essential singularities of B are on a single fibre Γ (cf. [Q], Lemma 5).

We have the following four possibilities:

1) B has two quadruple points.

2) B has a quadruple point and a 2-fold triple point.

3) B has a quadruple point x which, after a quadratic transformation at x, gives a 2-fold triple point.

4) B has a 4-fold triple point.

Among these, only 2) and 3) are possible. In the case 1), S^* would contain no exceptional curve which is mapped to a point on S'. In the case 4), S^* would contain two of such exceptional curves. Neither of them is the case.

First we consider the case 2). Let x and y be two points on a fibre Γ and not on Δ_0, and let q: $W_1 \to \Sigma_2$ be the blowing up of x and y. We set $E_x = q^{-1}(x)$, $E_y = q^{-1}(y)$. Then, take a point y' on E_y which is not on the proper transform of Γ, and let q_1: $\tilde{W} \to W_1$ be the blowing up of y'. We set $\mu = q_1 \circ q$, $E'_y = q_1^{-1}(y')$, and $\tilde{E}_y = q_1^* E_y - E'_y$, and for simplicity we write E_x instead of $q_1^* E_x$. We consider divisors $\tilde{B}_0$ on $\tilde{W}$ linearly equivalent to

$$\mu^*(7\Delta_0 + 14\Gamma) - 4E_x - 3\tilde{E}_y - 6E'_y.$$

Let $\Delta_x \in |\Delta_0 + 2\Gamma|$ be an irreducible curve through x, and let $\Delta_y \in |\Delta_0 + 2\Gamma|$ be through y at the direction of y'. We let $\tilde{\Delta}_x$ and $\tilde{\Delta}_y$ be their proper transforms by μ. Finally we denote by $\tilde{\Gamma}$ the proper transform of Γ by μ. Then $|\tilde{B}_0|$ contains

$$4\tilde{\Delta}_x + 3\tilde{\Delta}_y, \quad 6\tilde{\Gamma} + 2E_x + 3\tilde{E}_y + 7\mu^*\Delta_0 + |8\mu^*\Gamma|, \quad \text{and}$$
$$4\tilde{\Gamma} + 3\tilde{\Delta}_y + 4\tilde{E}_y + 4E'_y + |\mu^*(4\Delta_0 + 4\Gamma)|.$$

It follows that $|\tilde{B}_0|$ has no base point on $\tilde{W}$. Using the Riemann-Roch theorem, we obtain $\dim |\tilde{B}_0| = 41$.

Remark 3. The above argument can be applied with a slight modification to the case where x and y are not on a single fibre, and $\dim |\tilde{B}_0|$ remains constant. But in this case, we obtain a surface of type IIIa instead of type IIIb. This fact will be used in constructing a family of deformations of a surface of type IIIb (see §9).

Next we shall prove that the singularities of 3) actually occur. We distinguish the case according to whether the 2-fold triple point is on the proper transform of the fibre Γ or not. Let z and η be inhomogeneous coordinates on the base curve and on Γ, respectively. We may assume that x is at $z = \eta = 0$. Then any section of $[7\Delta_0 + 14\Gamma]$ over Σ_2 is represented by a linear combination of

$$\eta^i z^j \quad \text{with} \quad j \leqq 14 - 2i. \tag{5.5}$$

Let Λ be the subsystem spanned by $\eta^i z^j$ with $3i + j \geqq 10$. Then a generic member B of Λ has the singularity of type 3) at x. In fact, after a quadratic transformation at x, the proper transform of B is defined by a linear combination of $(\eta/z)^i z^{i+j-4}$ with $2i + (i + j - 4) \geqq 6$. This has a 2-fold triple point at $z = \eta/z = 0$ (which is not on the proper transform of Γ). We can also check that Λ has no base point other than x. Hence B is non-singular at other points.

Next we consider the linear subsystem Λ' of (5.5) spanned by

$$\eta^i z^j \quad \text{with} \quad i + 2j \geqq 10$$
$$\eta(\eta^2 + z)^3, \quad z^2(\eta^2 + z)^2, \quad \eta^3 z^2(\eta^2 + z).$$

Then a member of Λ' has, after a quadratic transformation at $z = \eta = 0$, a 2-fold triple point at $\eta = z/\eta = 0$, with the tangent being $\eta + (z/\eta) = 0$. Moreover, Λ' has no base point other than x. Hence we can also find a seeked branch locus in this case.

Conversely, if we start with a branch locus B of the above form, then we obtain a minimal surface with $p_g = 4$, $q = 0$, and $c_1^2 = 6$. We can easily prove that

S has no pencil of genus 2 (cf. the final part of the proof of Theorem 5.1). Hence S is of type III. In the case 2), S is of type IIIb (but, see Remark 3). However, in the case 3), S is of type IIIb or IIIa according to whether the 2-fold triple point is on the proper transform of Γ or not.

Remark 4. A surface of type IIIb is birationally equivalent to a double covering of $\mathbb{P}^2$ with branch locus of degree 10 which has a 4-fold triple point. To obtain this, we apply the elementary transformation to Σ_2 at the quadruple point x. Then we obtain Σ_1 and this can be transformed into $\mathbb{P}^2$ by contracting an exceptional curve.

§6. Surfaces of Types IV and V

In this section we shall study the case in which S has a pencil $|D|$ of curves of genus 2. We apply the result of [P] to such surfaces. By [P], Theorem 5, $|D|$ defines a surjective holomorphic map $g: S \to \mathbb{P}^1$. For each integer m, we set

$$b(m) = h^0(K+mD) - h^0(K+(m-1)D).$$

Then we have $b(m) \leqq b(m+1) \leqq 2$. Since $h^1(-D)$ vanishes (see [3], Theorem A), we have $h^0(K+D) = p_g + 2$. Hence it follows that $b(1) = 2$. We set

$$m_j = \mathrm{Min}\,\{m \mid b(m) \geqq j\}, \quad j = 1, 2,$$

and $d = m_2 - m_1$. We consider the rational map Φ defined by $|K+mD|$ for sufficiently large m. By [P], Theorems 1 and 2, the image of Φ is isomorphic to Σ_d with $d \equiv p_g \bmod 2$. Since $p_g = 4$, and since $|K|$ is not composite with a pencil (Theorem 1.3), we have $-2 \leqq m_1 \leqq m_2 \leqq 1$. This implies $d = 0$ or 2. Therefore, Φ induces a rational map $f: S \to \Sigma_d$ of degree 2, and $d = 0$ or 2. If $d = 0$, S is called of *type* IVa. If $d = 2$, S is called of *type* V or of *type* IVb according to whether the 0-section Δ_0 is contained in the branch locus or not. The singularities of the branch locus B are classified in [P], Lemma 6. In the present case, by [P], Theorem 3, one of the following holds:

(i) S has two singular fibres each of which is of type (I_1), (III_1) or (V).
(ii) S has one singular fibre of type (II_1) or (IV_1).

We shall add -1 or -2, e.g. IVa-1 or IVa-2, to indicate that S has singular fibers of the above type (i) or (ii).

Let us first suppose that S has singular fibres of (i). Then, employing the notation of [P], Lemmas 9 and 10, we have $\deg \mathfrak{c} = 2$, and, by [P], Theorem 2, $\deg \mathfrak{f} = 3 + d/2$. Hence B is linearly equivalent to $6\Delta_0 + (10+3d)\Gamma$.

Theorem 6.1. *Let S be a surface of type* IVa-1, IVb-1, *or* V-1. *Set $d = 0$ in the first case, and $d = 2$ in the second and third cases. Then S is birationally equivalent to a double covering of Σ_d whose branch locus B is linearly equivalent to $6\Delta_0 + (10 + 3d)\Gamma$ and is of the following form: B contains two fibers Γ_1 and Γ_2, and $B_0 = B - \Gamma_1 - \Gamma_2$ has two triple points on each of Γ_1 and Γ_2, which may be infinitely near. In this case $|K|$ has two simple base points.*

Next suppose that B has a singular fibre of type (ii). In this case, we have $\deg \mathfrak{c}=1$ and $\deg \mathfrak{f}=2+d/2$. Hence B is linearly equivalent to $6\Delta_0+(8+3d)\Gamma$.

Theorem 6.2. *Let S be a surface of type* IVa−2, IVb−2, *or* V−2. *Set $d=0$ in the first case, and $d=2$ in the second and third cases. Then S is birationally equivalent to a double covering of Σ_d whose branch locus B is linearly equivalent to $6\Delta_0+(8+3d)\Gamma$ and has two 2-fold triple points on a fibre Γ, which may be infinitely near. In this case, $|K|$ has a unique fixed component F which is a non-singular rational curve with $F^2=-2$.*

These are all in [P], except that we regarded a singular fibre of type (V) as a degenerate case of a singular fibre of type (III_1) to simplify the statement. Especially, the last statement about the fixed component follows from Lemma 10 in [P].

Remark. The pencil of genus 2 is unique on each surface of type IV or V, because it is induced from the canonical map in a natural way.

§7. Deformations of Surfaces of Type I

Theorem 7.1. *Let S be a surface of type* Ia, *and assume that K is ample. Then the number of moduli $m(S)$ is defined and is equal to* $\dim H^1(S, \Theta_S)=38$, *where Θ_S denotes the sheaf of germs of holomorphic vector fields on S.*

Proof. See [9].

Theorem 7.2. *The surfaces of type* I *have one and the same deformation type.*

Proof. In view of the simultaneous resolution of the rational double points [4], it follows from the constructions in §3 that each type Ia or Ib has one and the same deformation type. Therefore, it remains to show that a surface of type Ib is a deformation of a surface of type Ia.

We let t be a parameter which moves in a neighborhood of the origin in $\mathbb{C}$. Using the notation in §3, we define S_t in V by the equations

$$\begin{aligned} &\theta^2+A\theta+B=0,\\ &\quad th\theta-g=0. \end{aligned} \tag{7.1}$$

Then S_0 is of type Ib. If $t\neq 0$, (7.1) reduces to $g^2+tAgh+t^2Bh^2=0$. Hence the $S_t, t\neq 0$, are of type Ia (cf. [6], p. 280). Q.E.D.

§8. Deformations of Surfaces of Type II

The main result of this section is the following.

Theorem 8.1. (i) *Let S be a surface of type* II. *Then any sufficiently small deformation of S is of type* II.

(ii) *If S is of type* II *and generic, then the number of moduli $m(S)$, in the sense of Kodaira-Spencer* [17], §11, *is defined and is equal to* 38.

(iii) *We have* $\dim H^1(S, \Theta_S)=39$ *and the Kuranishi family of deformations of* S *is parametrized by a non-reduced space* M *for which* M_{red} *is non-singular of dimension* 38.

Remark. The assertion (i), combined with the result of §3, implies that S cannot be directly deformed to a surface with ample canonical bundle. This seems to be a new example to the author. We shall find a similar example in [14], with $p_g=6$, $q=0$, and $c_1^2=11$.

Proof of (i). We let $\mathbf{D}$ denote a small disc at the origin of $\mathbb{C}$ with coordinate t, and let $p: \mathscr{S}\to\mathbf{D}$ be a family of surfaces such that $S=p^{-1}(0)$ is of type II. We take a generic member C of $|K|$ on S. Then, since $H^1(S, \mathscr{O}(C))$ vanishes, C can be extended to a family $q: \mathscr{C}\to\mathbf{D}$ of curves of genus 7. This family carries a line bundle $\mathscr{L}$ such that $2\mathscr{L}$ is the canonical bundle $\mathscr{K}$ of $\mathscr{C}$. The following lemma corresponds to our classification of surfaces with $p_g=4$, $q=0$, and $c_1^2=6$.

Lemma 8.1. *Let* C *be a non-singular curve of genus* 7, *and suppose that there exists a line bundle* L *on* C *such that* $2L$ *is the canonical bundle* K_C *and such that* $\dim|L|=2$. *Then one of the following is true.*

(Ia) C *is birationally equivalent to a plane sextic curve with three double points on a line.*

(Ib) C *is a double covering of an elliptic curve.*

(II) C *is a triple covering of* $\mathbb{P}^1$ *and* L *is induced by* $\mathscr{O}(2)$.

(III) C *is a hyperelliptic curve and* L *is induced by* $5Q_1+Q_2$ *with two Weierstrass points* Q_1 *and* Q_2.

Proof is parallel to §§3–5, and is much simpler.

If C is a canonical curve on a surface of type II, then C is of type II in Lemma 8.1. We need some more precise informations. We first prove a general lemma.

Lemma 8.2. *Let* C *be a non-singular curve which is a triple covering of another non-singular curve* D. *Then* C *can be embedded in a* $\mathbb{P}^1$*-bundle over* D.

Proof. Obviously, we can find a possibly singular model C' of C in the product $D\times\mathbb{P}^1$, which intersects a general fibre Γ at three points. If C' has a singular point s, we apply the elementary transformation at s. Since $C'\Gamma=3$, and since s is a singular point, it does not produce any new singularity, and it has the same effect on C' as the blowing up of s. Hence, after a finite number of such elementary transformations, C' is transformed into a non-singular curve in a $\mathbb{P}^1$-bundle over D. Q.E.D.

Lemma 8.3. *Let* C *be a non-singular curve of genus* 7 *which is a triple covering of* $\mathbb{P}^1$. *Then* C *can be embedded either in* Σ_1 *or in* Σ_3. *Moreover,* C *carries a line bundle* L *as in Lemma* 8.1, *if and only if* C *can be embedded in* Σ_3.

Proof. By Lemma 8.2, C can be embedded in Σ_d for some d. If C is linearly equivalent to $3\Delta_0+m\Gamma$, then K_C is induced by $\Delta_0+(m-d-2)\Gamma$. Hence, we have the equality $m=(3d+9)/2$. Since C is irreducible, we have $m\geqq 3d$. Hence d is 1 or 3. Conversely, such C certainly exists for $d=1$ and 3.

If $d=3$, then C is disjoint from Δ_0. Therefore K_C is induced by 4Γ. Hence $L=[2\Gamma]_{|C}$ satisfies the condition of Lemma 8.1. Conversely, if C is of type II in

Lemma 8.1, then L can be written as $2L_0$, and, by the Riemann-Roch theorem, we have $h^0(3L_0)=5$. If $\{\varphi_0, \varphi_1\}$ is a basis of $H^0(\mathcal{O}(L_0))$, then $H^0(\mathcal{O}(3L_0))$ is generated by the products $\varphi_0^i \varphi_1^{3-i}, 0 \leqq i \leqq 3$, together with an element ψ. Hence we can define a holomorphic map Ψ: $C \to \Sigma_3$ such that ψ corresponds to the ∞-section of Σ_3. To prove that Ψ is an embedding, it is enough to note that

$$\varphi_0^i \varphi_1^{4-i} \quad (0 \leqq i \leqq 4), \qquad \varphi_0 \psi, \quad \varphi_1 \psi$$

form a basis of $H^0(\mathcal{O}(4L_0)) = H^0(\mathcal{O}(K_C))$, and that C is not hyperelliptic. Q.E.D.

Let $\mathcal{M}_0 \to M_0$ be the family of the triple coverings of $\mathbb{P}^1$ in Σ_3 which are close to C, and let $T_0(M_0)$ be the tangent space of M_0 at the point 0 corresponding to C (see [15]). This determines an infinitesimal deformation map

$$\rho_0\colon T_0(M_0) \to H^1(C, \Theta_C).$$

In a similar way, let $\mathcal{M}_1 \to M_1$ be the family of all triple coverings of $\mathbb{P}^1$ (not necessarily in Σ_3) which are close to $g\colon C \to \mathbb{P}^1$ (see [8], I). We denote its infinitesimal deformation map by

$$\rho_1\colon T_0(M_1) \to H^1(C, \Theta_C).$$

Lemma 8.4. *We have* $\dim \operatorname{Im} \rho_0 = 13$ *and* $\dim \operatorname{Im} \rho_1 = 15$.

Proof. The tangent space $T_0(M_0)$ can be identified with $H^0(C, \mathcal{N})$, where $\mathcal{N}$ is the normal sheaf of C in $W = \Sigma_3$. Since C is linearly equivalent to $3\Delta_0 + 9\Gamma$, we have $\dim H^0(\mathcal{N}) = 21$. In view of the exact sequence

$$0 \to H^0(\Theta_{W|C}) \to H^0(\mathcal{N}) \to H^1(\Theta_C),$$

we need to calculate $\dim H^0(\Theta_{W|C})$. For this purpose, we consider the exact sequence

$$0 \to \Theta_{W/\mathbb{P}^1} \to \Theta_W \to p^* \Theta_{\mathbb{P}^1} \to 0, \tag{8.1}$$

where $p\colon W \to \mathbb{P}^1$ is the projection and $\Theta_{W/\mathbb{P}^1}$ is the sheaf of vector fields along the fibres. As is well known, we have the isomorphisms $\Theta_{W/\mathbb{P}^1} \cong \mathcal{O}(2\Delta_0 + 3\Gamma)$ and $p^* \Theta_{\mathbb{P}^1} \cong \mathcal{O}(2\Gamma)$. The sequence (8.1) induces the exact sequence

$$0 \to \mathcal{O}(3L_0) \to \Theta_{W|C} \to \mathcal{O}(2L_0) \to 0$$

on C, where L_0 denotes the restriction of $[\Gamma]$ to C. We note that the two maps

$$H^0(W, \Theta_W) \to H^0(W, \mathcal{O}(2\Gamma)), \qquad H^0(W, \mathcal{O}(2\Gamma)) \to H^0(C, \mathcal{O}(2L_0))$$

are surjective. Hence, it follows that

$$\dim H^0(C, \Theta_{W|C}) = h^0(3L_0) + h^0(2L_0) = 8.$$

This proves the first equality $\dim \operatorname{Im} \rho_0 = 13$.

To prove the second equality we use the theory of deformations of holomorphic maps [8], and consider the exact sequence

$$0 \to \Theta_C \to g^* \Theta_{\mathbb{P}^1} \to \mathcal{T}_{C/\mathbb{P}^1} \to 0. \tag{8.2}$$

By [8], I, Theorem 3.1, we have

$$\dim \operatorname{Im} \rho_1 = \dim H^0(\mathcal{T}_{C/\mathbb{P}^1}) - \dim H^0(g^* \Theta_{\mathbb{P}^1}).$$

Using (8.2) and the fact that Supp $\mathcal{T}_{C/\mathbb{P}^1}$ is discrete, we obtain $h^0(\mathcal{T}_{C/\mathbb{P}^1}) = 18$. Then, it immediately follows $\dim \operatorname{Im} \rho_1 = 15$. Q.E.D.

The embedding of $\mathbb{P}^1$ into $\mathbb{P}^2$ as a conic depends on 5 parameters. Adding these parameters to M_1, we obtain a family $\mathcal{M}_2 \to M_2$ and a holomorphic map $\Psi\colon \mathcal{M}_2 \to \mathbb{P}^2 \times M_2$. Let $f\colon C \to \mathbb{P}^2$ be the holomorphic map defined by $|2L_0|$. Then $(\mathcal{M}_2, \Psi)$ contains those small deformations of f which factor through conics. We now change the meaning of M_0 and let it denote the set of those points of M_2 which correspond to the triple coverings of $\mathbb{P}^1$ contained in Σ_3.

Let $q\colon \mathcal{C} \to \mathbf{D}$ be the family of curves of genus 7 which was obtained form the family of surfaces at the beginning of the proof. Then $f\colon C \to \mathbb{P}^2$ extends to a holomorphic map $\Phi\colon \mathcal{C} \to \mathbb{P}^2 \times \mathbf{D}$ over $\mathbf{D}$ provided $\mathbf{D}$ is sufficiently small.

Lemma 8.5. *There exists a holomorphic map* $s\colon \mathbf{D} \to M_0$ *with* $s(0) = 0$ *such that* $\mathcal{C}$ *and* Φ *are respectively induced from* $\mathcal{M}_2$ *and* Ψ *by* s.

Proof. We set $\mathbf{D}_\mu = \operatorname{Spec}(\mathbb{C}[t]/t^{\mu+1})$ and prove the existence of holomorphic maps $s_\mu\colon \mathbf{D}_\mu \to M_0$, $\mu = 1, 2, \ldots$ such that $(\mathcal{C}, \Phi)$ modulo $t^{\mu+1}$ is induced from $(\mathcal{M}_2, \Psi)$ by s_μ.

For $\mu = 1$, $\mathcal{C} \times \mathbf{D}_1$ (the product being taken over $\mathbf{D}$) corresponds to an element ρ of $H^1(C, \Theta_C)$. The existence of Φ implies that ρ is in the image of the coboundary map

$$\delta\colon H^0(C, \mathcal{T}_{C/\mathbb{P}^2}) \to H^1(C, \Theta_C)$$

of the exact sequence

$$0 \to \Theta_C \to f^* \Theta_{\mathbb{P}^2} \to \mathcal{T}_{C/\mathbb{P}^2} \to 0. \tag{8.3}$$

In order to calculate $h^i(f^* \Theta_{\mathbb{P}^2})$, we use the exact sequence

$$0 \to \mathcal{O}_C \to \mathcal{O}(2L_0)^3 \to f^* \Theta_{\mathbb{P}^2} \to 0. \tag{8.4}$$

The second arrow induces the map $H^1(\mathcal{O}_C) \to H^1(\mathcal{O}(2L_0))^3$ which is dual to the map $H^0(\mathcal{O}(2L_0))^3 \to H^0(\mathcal{O}(4L_0))$ defined by

$$(a, b, c) \to a\varphi_0^2 + b\varphi_0\varphi_1 + c\varphi_1^2,$$

where $\{\varphi_0, \varphi_1\}$ is a basis of $H^0(\mathcal{O}(L_0))$. Since the image of f is a conic, this map has a 5-dimensional image. Therefore, we have

$$\dim \operatorname{Ker}(H^1(\mathcal{O}_C) \to H^1(\mathcal{O}(2L_0))^3) = 2.$$

Hence, from (8.4), we obtain $h^0(f^* \Theta_{\mathbb{P}^2}) = 10$, $h^1(f^* \Theta_{\mathbb{P}^2}) = 4$.

On the other hand, we have the exact sequence

$$0 \to \mathcal{T}_{C/\mathbb{P}^1} \to \mathcal{T}_{C/\mathbb{P}^2} \to g^* \mathcal{N}_{\mathbb{P}^1} \to 0,$$

where $\mathcal{N}_{\mathbb{P}^1}$ denotes the normal sheaf of $\mathbb{P}^1$ in $\mathbb{P}^2$ as a conic. Using $h^0(\mathcal{T}_{C/\mathbb{P}^1})=18$ and $h^0(g^*\mathcal{N}_{\mathbb{P}^1})=h^0(4L_0)=7$, we obtain $h^0(\mathcal{T}_{C/\mathbb{P}^2})=25$ and $h^1(\mathcal{T}_{C/\mathbb{P}^2})=1$. Then, using (8.3), we conclude $\dim \operatorname{Im}\delta=15$. Since $\operatorname{Im}\delta$ contains $\operatorname{Im}\rho_1$, these two spaces coincide (Lemma 8.4). This implies that there exists a holomorphic map $s_1: \mathbf{D}_1 \to M_2$ such that $\mathscr{C}\times\mathbf{D}_1$ is isomorphic to $s_1^*\mathscr{M}_2$.

Here we insert the following lemma.

Lemma 8.6. *Let $s: N \to M_2$ be a holomorphic map with N being possibly non-reduced and $s(0)=0$, and let $\mathscr{C}_1 \to N$ be the family of deformations of C induced by s from $\mathscr{M}_2$. Suppose that $\mathscr{C}$ carries a line bundle $\mathscr{L}_1$ inducing L on C such that $2\mathscr{L}_1$ is the relative canonical bundle of $\mathscr{C}_1 \to N$ and such that any section ψ of L over C extends to a section of $\mathscr{L}_1$ over $\mathscr{C}_1$. Then s factors through M_0.*

Proof. We define a line bundle $\mathscr{L}$ on $\mathscr{M}_2$ by the conditions $\mathscr{L}_{|C}=L$ and that $2\mathscr{L}$ is the canonical bundle of $\mathscr{M}_2$. We note that such $\mathscr{L}$ is uniquely determined. Recall that the obstruction for extending $\psi\in H^0(C,\mathcal{O}(L))$ to a section of $\mathscr{L}$ lies in $H^1(C,\mathcal{O}(L))$. We define M_0^* to be the maximal subspace of M_2 over which any ψ extends (cf. [20, 20a]). Then, by assumption, s factors through M_0^*. We remark that M_0^* is possibly non-reduced.

Sublemma 1. $(M_0^*)_{\mathrm{red}}=M_0$.

Proof. Let C_t denote the curve over $t\in M_0^*$, and let L_t be the restriction to C_t of the line bundle $\mathscr{L}$. Then, for any $t\in M_0^*$, we have $\dim H^0(C_t,\mathcal{O}(L_t))=3$, and $2L_t$ is the canonical bundle of C_t. Hence, by Lemma 8.3, t belongs to M_0. Q.E.D.

Sublemma 2. *M_0^* is reduced.*

Proof. By Lemma 8.4, M_0 is a non-singular submanifold of M_2 of codimension 2. Hence, it suffices to find two linearly independent tangent vectors on M_2 at 0 for which some $\psi\in H^0(C,\mathcal{O}(L))$ does not extend.

Recall that C is a triple covering $g: C\to\mathbb{P}^1$ defined by the linear system $|L_0|$ such that $2L_0=L$. We take a point P_1 at which g ramifies. We assume that g has ramification index 2 at P_1. The case where the index equals 3 will be discussed later. We further take two points P_2 and P_3 with $g(P_1)=g(P_2)\neq g(P_3)$ such that g is unramified at P_2 and P_3. We set $P=P_1+P_2+P_3$, and consider the elements in the kernel of the natural map

$$i_P: H^1(C,\Theta)\to H^1(C,\Theta(P)), \tag{8.5}$$

where $\Theta=\Theta_C$ denotes the tangent sheaf to C. To investigate the kernel, we consider the exact commutative diagram

$$\begin{array}{ccccccc} 0\to\Theta & \longrightarrow & \mathcal{O}(2L_0) & \longrightarrow & \mathcal{T}_{C/\mathbb{P}^1} & \to 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0\to\Theta(P) & \longrightarrow & \mathcal{O}(2L_0+P) & \longrightarrow & \mathcal{T}_{C/\mathbb{P}^1}(P) & \to 0. \end{array}$$

It induces the exact commutative diagram

$$\begin{array}{ccccc} 0 \to H^0(\mathscr{O}(2L_0)) & \longrightarrow & H^0(\mathscr{T}_{C/\mathbb{P}^1}) & \longrightarrow & H^1(\Theta) \\ \downarrow & & \downarrow & & \downarrow \\ 0 \to H^0(\mathscr{O}(2L_0+P)) & \longrightarrow & H^0(\mathscr{T}_{C/\mathbb{P}^1}(P)) & \longrightarrow & H^1(\Theta(P)). \end{array} \tag{8.6}$$

First, we note that $h^0(\mathscr{T}_{C/\mathbb{P}^1}) = h^0(\mathscr{T}_{C/\mathbb{P}^1}(P)) = 18$, and, in fact, this is true for any divisor P. Secondly, by the choice of P, we have $h^1(2L_0+P)=1$, and hence, by the Riemann-Roch theorem, $h^0(2L_0+P)=4$. Since we have $h^0(2L_0)=3$, this implies that there exists a non-zero element $\rho \in H^1(\Theta)$ which is the image of an element $\tau \in H^0(\mathscr{T}_{C/\mathbb{P}^1})$ and which is annihilated by i_P. Such ρ is unique up to a constant factor. By specifying $\tau \in H^0(\mathscr{T}_{C/\mathbb{P}^1})$ over ρ, we may consider that ρ corresponds to a tangent vector on M_2.

Let $S_\rho \to \mathbf{D}_1$ be the infinitesimal deformation corresponding to ρ. We shall prove that some $\psi \in H^0(C, \mathscr{O}(L))$ does not extend over S_ρ.

Let $\{U_i\}$ be an open covering of C by sufficiently small discs U_i. We assume that each P_i belongs only to U_i, $1 \leqq i \leqq 3$, and that U_j, $j \geqq 4$ intersects at most one of the U_i, $1 \leqq i \leqq 3$. We let z_i be a coordinate on U_i, and assume that it is centered at P_i for $1 \leqq i \leqq 3$. Then, the above cohomology class ρ is represented by the 1-cocycle $\{\rho_{ij}\}$ on the nerve of the covering $\{U_i\}$ with

$$\rho_{ij} = \frac{\alpha_j}{z_j}\frac{\partial}{\partial z_j} - \frac{\alpha_i}{z_i}\frac{\partial}{\partial z_i},$$

where $\alpha_j = 0$ for $j \geqq 4$. Let $\{l_{ij}\}$ *and* $\{\kappa_{ij}\} = \{\partial z_j/\partial z_i\}$ be systems of transition functions for L and K_C, respectively. Then, we have a 0-cochain $\{u_i\}$ of non-vanishing holomorphic functions such that

$$l_{ij}^2 = u_j \kappa_{ij} u_i^{-1} \quad \text{on } U_i \cap U_j.$$

Let K_ρ be the canonical bundle of S_ρ. Then, it is defined by the transition functions

$$\kappa_{ij} + \kappa_{ij}(-\operatorname{div} \rho_{ij})\, t,$$

where $\operatorname{div} \rho_{ij} = \partial \beta_{ij}/\partial z_i$ if $\rho_{ij} = \beta_{ij}(\partial/\partial z_i)$ (see [Q], p. 68). Let L_ρ be the extension of L over S_ρ such that $2L_\rho = K_\rho$. Then L_ρ is defined by the transition functions

$$v_{ij} = l_{ij} + \tfrac{1}{2} l_{ij}(-\operatorname{div} \rho_{ij} - \rho_{ij} \cdot \log u_i)\, t.$$

In fact, it can be easily verified that

$$u_j \kappa_{ij}(1 - \operatorname{div} \rho_{ij}\, t) = v_{ij}^2 u_i (z_j + \beta_{ij}\, t)$$

over $\mathbf{D}_1$, where the last factor means the u_i evaluated at $z_i = z_j + \beta_{ij} t$.

Let ψ be a section of L which vanishes at the P_i's. We represent ψ by a 0-cochain $\{\psi_i\}$ with $\psi_i = l_{ij}\psi_j$. If ψ extends to a section of L_ρ, then we can find a 0-cochain $\{h_i\}$ such that

$$\rho_{ij} \cdot \psi_i + \tfrac{1}{2}\psi_i(-\operatorname{div} \rho_{ij} - \rho_{ij} \cdot \log u_i) = l_{ij} h_j - h_i \tag{8.7}$$

(see [Q], §5, or [13]). We take $i \leqq 3$ and $j \geqq 4$, with $U_i \cap U_j \neq \emptyset$. Then, we have

$$\rho_{ij} = -\frac{\alpha_i}{z_i}\frac{\partial}{\partial z_i}, \qquad \operatorname{div} \rho_{ij} = \frac{\alpha_i}{z_i^2}.$$

Hence, (8.7) reduces to

$$-\frac{\alpha_i}{z_i}\frac{\partial \psi_i}{\partial z_i} - \frac{1}{2}\psi_i\left(\frac{\alpha_i}{z_i^2} - \frac{\alpha_i}{z_i}\frac{\partial}{\partial z_i}\log u_i\right) = l_{ij}h_j - h_i. \tag{8.8}$$

The expression on the left side does not depend on j. So we call it γ_i, and set $\gamma_j = 0$ for $j \geqq 4$. Then (8.8) implies

$$h_i + \gamma_i = l_{ij}(h_j + \gamma_j) \qquad \text{on } U_i \cap U_j.$$

This obviously holds for any pair (i, j). Since ψ vanishes to second order at P_1, it follows that γ_1 is holomorphic. On the other hand, γ_2 and γ_3 may have poles of order 1 at P_2 and P_3, respectively. Hence, the collection $\{h_i + \gamma_i\}$ defines an element of $H^0(C, \mathcal{O}(L + P_2 + P_3))$. But, by the Riemann-Roch theorem, we have $h^0(L) = h^0(L + P_2 + P_3) = 3$. Therefore, γ_2 and γ_3 actually have no poles. This implies $\alpha_2 = \alpha_3 = 0$. Recalling the choice of ρ, we conclude that ρ is annihilated by (8.5) with P being replaced by P_1. But, in the diagram (8.6) with $P = P_1$, we have $h^0(2L_0) = h^0(2L_0 + P_1) = 3$. Therefore, there is no such ρ except 0. This proves that ψ does not extend over S_ρ.

Next suppose that g has ramification index 3 at P_1. In this case, we consider i_P with $P = 2P_1$ and $P = 3P_1$. From the equalities $h^0(2L_0 + 2P_1) = 4, h^0(2L_0 + 3P_1) = 5$, it follows that there exists an element $\rho \in H^1(\Theta)$ which is the image of an element $\tau \in H^0(\mathcal{T}_{C/\mathbb{P}^1})$ such that $i_{2P_1}(\rho) \neq 0$ and $i_{3P_1}(\rho) = 0$. We take as ψ a section of L which vanishes exactly to order 3 at P_1. We repeat the above argument. In this case, we have

$$\rho_{1j} = -\left(\frac{\alpha_1}{z_1^3} + \frac{\alpha_2}{z_1^2} + \frac{\alpha_3}{z_1}\right)\frac{\partial}{\partial z_1}.$$

Hence, (8.8) becomes

$$-\left(\frac{\alpha_1}{z_1^3} + \frac{\alpha_2}{z_1^2} + \frac{\alpha_3}{z_1}\right)\frac{\partial \psi_1}{\partial z_1} - \frac{1}{2}\psi_1\left[\frac{3\alpha_1}{z_1^4} + \frac{2\alpha_2}{z_1^3} + \frac{\alpha_3}{z_1^2} - \left(\frac{\alpha_1}{z_1^3} + \frac{\alpha_2}{z_1^2} + \frac{\alpha_3}{z_1}\right)\frac{\partial}{\partial z_1}\log u_1\right]$$
$$= l_{1j}h_j - h_1.$$

We call the left side γ_1, and set $\gamma_j = 0$ for $j \neq 1$. Then we obtain an element $\{h_i + \gamma_i\}$ of $H^0(C, \mathcal{O}(L + P_1))$. Since it actually has no pole at P_1, it follows that α_1 vanishes. This contradicts $i_{2P_1}(\rho) \neq 0$.

In any case, we can find a divisor P of degree 3 and $\rho \in H^1(\Theta)$ with $i_P(\rho) = 0$ such that some $\psi \in H^0(C, \mathcal{O}(L))$ does not extend over S_ρ. We choose a similar divisor Q of degree 3 which is disjoint from P, and let ρ' be the corresponding element of $H^1(\Theta)$.

Sublemma 3. *The two elements ρ and ρ' are linearly independent.*

Proof. Using the natural injections $\mathcal{O} \to \mathcal{O}(P)$ and $\mathcal{O} \to \mathcal{O}(Q)$ we obtain the exact sequence

$$0 \to \mathcal{O} \to \mathcal{O}(P) \oplus \mathcal{O}(Q) \to \mathcal{O}(P+Q) \to 0.$$

This induces

$$0 \to \Theta \to \Theta(P) \oplus \Theta(Q) \to \Theta(P+Q) \to 0.$$

If ρ and ρ' are linearly dependent, we have $i_P(\rho) = i_Q(\rho) = 0$. But, since $H^0(\Theta(P+Q))$ vanishes, this is impossible.

This completes the proof of Lemma 8.6.

We continue the proof of Lemma 8.5. By Lemma 8.6, the map $s_1 : \mathbf{D}_1 \to M_2$ factors through M_0. Then, the two holomorphic maps induced by Φ and $s_1^* \Psi$ are both defined by the same line bundle. Hence they differ by an automorphism of $\mathbb{P}^2 \times \mathbf{D}_1$. Therefore, by modifying s_1, we may assume these two maps coincide.

Now suppose that $s_\mu : \mathbf{D}_\mu \to M_0$ is constructed in such a manner $(\mathscr{C}, \Phi)$ over $\mathbf{D}_\mu$ is induced by s_μ from $(\mathscr{M}_2, \Psi)$. We take an arbitrary extension $s_{\mu+1} : \mathbf{D}_{\mu+1} \to M_2$. Then the two families $\mathscr{C} \times \mathbf{D}_{\mu+1}$ and $s_{\mu+1}^* \mathscr{M}_2$ differ by an element of $\delta H^0(C, \mathscr{T}_{C/\mathbb{P}^2})$. Since $\operatorname{Im} \rho_1$ coincides with $\operatorname{Im} \delta$, we can modify $s_{\mu+1}$ so that these two families should be isomorphic over $\mathbf{D}_{\mu+1}$ (cf. proof of the theorem of completeness [18]). By Lemma 8.6, $s_{\mu+1}$ factors through M_0. Then, we can modify $s_{\mu+1}$ so that Φ and $s_{\mu+1}^* \Psi$ should coincide over $\mathbf{D}_{\mu+1}$. (This part may be replaced by a general theory [24].)

The existence of a formal solution implies that of an analytic solution (see [2], or proof of Theorem 2.1 in [8], I). Thus, we obtain a holomorphic map $s : \mathbf{D} \to M_0$ such that $q : \mathscr{C} \to \mathbf{D}$ is induced by s from $\mathscr{M}_2$. It follows that any member of $\mathscr{C}$ is of type II. This completes the proof of the assertion (i).

Proof of (ii) *and* (iii). We assume that S is generic, and consider the family $\tilde{\mathscr{S}}$ of deformations of $\tilde{S}$ defined by (4.1), in which we regard α, β, and γ as parameters. We further assume that β does not vanish on Δ_0. Using the notation of §4, we have $\dim H^0(\tilde{S}, \mathscr{N}_{\tilde{S}/V^*}) = 52$, where $\mathscr{N}_{\tilde{S}/V^*}$ denotes the normal sheaf of $\tilde{S}$ in V^*.

Next we use the exact sequences

$$0 \to \mathscr{N}_{\tilde{S}/V^*} \to \mathscr{T}_{\tilde{S}/V} \to \mathscr{T}_{V^*/V|\tilde{S}} \to 0,$$
$$0 \to \mathscr{N}_{\mathscr{E}/V^*} \to \mu^* \mathscr{N}_{\Delta/V} \to \mathscr{T}_{V^*/V} \to 0.$$

The second one results from the exact commutative diagram

$$\begin{array}{ccccccc}
0 \to & \Theta_{\mathscr{E}} & \longrightarrow & \Theta_{V^*|\mathscr{E}} & \longrightarrow & \mathscr{N}_{\mathscr{E}/V^*} \to 0 \\
 & \downarrow & & \downarrow & & \downarrow & \\
0 \to & \mu^* \Theta_\Delta & \longrightarrow & \mu^* \Theta_{V|\mathscr{E}} & \longrightarrow & \mu^* \mathscr{N}_{\Delta/V} & \to 0 \\
 & & & \downarrow & & \downarrow & \\
 & & & \mathscr{T}_{V^*/V} \otimes \mathcal{O}_{\mathscr{E}} & = & \mathscr{T}_{V^*/V} \otimes \mathcal{O}_{\mathscr{E}} & \\
 & & & \downarrow & & \downarrow & \\
 & & & 0 & & 0 &
\end{array}$$

Here, using the standard coordinates on $\mathscr{E}$, we can easily verify $\mathscr{T}_{V^*/V} \otimes \mathscr{O}_{\mathscr{E}} = \mathscr{T}_{V^*/V}$ and that $\mathscr{N}_{\mathscr{E}/V^*} \to \mu^* \mathscr{N}_{\Delta/V}$ is injective.

From these exact sequences we infer that $h^0(\mathscr{T}_{V^*/V|S}) = 0$ and $h^0(\mathscr{T}_{\tilde{S}/V}) = 52$. We let $\tilde{h}: \tilde{S} \to V$ be the natural map and calculate $h^0(\tilde{h}^* \Theta_V)$. This sheaf fits in the exact sequence

$$0 \to \mathscr{O}(h^*(2\Delta_0 + 3\Gamma)) \to \tilde{h}^* \Theta_V \to h^* \Theta_{\Sigma_2} \to 0,$$

where h is the projection $\tilde{S} \to \Sigma_2$ (cf. (8.1)). We also use the exact sequence

$$0 \to \mathscr{O}(h^*(2\Delta_0 + 2\Gamma)) \to h^* \Theta_{\Sigma_2} \to \mathscr{O}(h^*(2\Gamma)) \to 0.$$

From the second one, we get $h^0(h^* \Theta_{\Sigma_2}) = 7$. To calculate $h^i(h^*(2\Delta_0 + 3\Gamma))$, we use the equality (4.2). Since $\tilde{D} + \pi^* G + E$ is a reduced connected curve, we have $h^1(h^*(2\Delta_0 + 3\Gamma)) = 0$ (see [3], Theorem A). Hence, by the Riemann-Roch theorem, we get $h^0(h^*(2\Delta_0 + 3\Gamma)) = 7$. Using the first exact sequence, we obtain $h^0(\tilde{h}^* \Theta_V) = 14$. Therefore, from the exact sequence

$$0 \to \Theta_{\tilde{S}} \to \tilde{h}^* \Theta_V \to \mathscr{T}_{\tilde{S}/V} \to 0,$$

it follows that our family $\tilde{\mathscr{S}}$ contains 38-dimensional infinitesimal deformations. Let $\mathscr{S}$ be the family of the minimal models of the members of $\tilde{\mathscr{S}}$ (see [8], III). Then $\mathscr{S}$ contains the same number of infinitesimal deformations (cf. [Q], Lemma 26)[2].

We now want to prove $h^1(\Theta_S) = 39$. First recall that S contains a rational curve G with $G^2 = -2$ and that the natural map $\zeta_*: H^1(S, \Theta_S) \to H^1(G, \mathscr{N}_{G/S}) \cong \mathbb{C}$ is surjective (see [5]). Since G extends to a family of curves on $\mathscr{S}$, the infinitesimal deformations in $\mathscr{S}$ are annihilated by ζ_*. Conversely, suppose that $\rho \in H^1(S, \Theta_S)$ is annihilated by ζ_*. Then the divisor G extends over the corresponding deformation $S_\rho \to \mathbf{D}_1$. Since the canonical bundle extends automatically, it follows that the line bundle $[2D]$, and hence $[D]$ extend over S_ρ. Moreover, we can easily prove $h^1(D) = 0$, and $h^1(\tilde{K} + \tilde{D} + \pi^* G + E) = 0$ as remarked above. Therefore, the construction in Theorem 4.1 can be carried out over S_ρ. This implies that ρ is contained among the infinitesimal deformations in $\mathscr{S}$. Thus we conclude $h^1(\Theta_S) = 39$ and $h^2(\Theta_S) = 1$.

To complete the proof of (ii) and (iii), we take the Kuranishi family $\mathscr{S}_0 \to M_0$ of deformations of S. Then M_0 is defined at the origin of $\mathbb{C}^{39}$ by a single equation (see [19]). It contains a non-singular component M_1 of dimension 38 which corresponds to our family $\mathscr{S}$. We shall prove $(M_0)_{\text{red}} = M_1$. In fact, let $\mathscr{S}' \to M'$ be any family of deformations of S with reduced base space M'. Then, by virtue of (i), any member of $\mathscr{S}'$ is of type II.

Here we use the following lemma.

Lemma 8.7. *Suppose that the canonical image of each member of $\mathscr{S}' \to M'$ is a quadric. Then the images form a flat family.*

Proof. We take a basis $\{\varphi_0, \varphi_1, \varphi_2, \varphi_3\}$ of $H^0(S, \mathscr{O}(K))$ and assume that $\varphi_0^2 = Q(\varphi_1, \varphi_2, \varphi_3)$, where Q is a quadratic form. We extend each φ_i to a section $\tilde{\varphi}_i$ of the canonical bundle of $\mathscr{S}'$. Let $\{\psi_0, \psi_1, \ldots, \psi_{10}\}$ be a basis of $H^0(S, \mathscr{O}(2K))$ and suppose that these sections are already extended to those over $\mathscr{S}'$. We may further

[2] See Part IV, §5 for a detailed proof

assume that the nine of them, for instance, $\psi_2, \psi_3, \ldots, \psi_{10}$ are quadratic forms in $\tilde{\varphi}_0, \ldots, \tilde{\varphi}_3$. Then we can write

$$\tilde{\varphi}_0^2 = \sum_{i=0}^{10} \alpha_i(t)\, \psi_i,$$

where the $\alpha_i(t)$ are holomorphic functions on M'. By assumption, the rank of the space generated by the products $\tilde{\varphi}_i\, \tilde{\varphi}_j$ is 9 at each point t of M'. Therefore, we have $\alpha_0(t) = \alpha_1(t) = 0$. Hence $\tilde{\varphi}_0, \ldots, \tilde{\varphi}_3$ satisfy a quadratic relation which depends holomorphically on t. Q.E.D.

Changing the basis of $H^0(S, \mathcal{O}(K))$, if necessary, we may assume $\tilde{\varphi}_0^2 = \tilde{\varphi}_1\, \tilde{\varphi}_2$ on $\mathcal{S}'$. Then, as is proved in §4, the locus $\tilde{\varphi}_0 = \tilde{\varphi}_1 = \tilde{\varphi}_2 = 0$ has two connected components, because we assumed that β does not vanish on Δ_0. One of them induces G on S, and the other is the locus of the base point of $|D|$. This implies that G extends to a family of divisors on $\mathcal{S}'$. Therefore, by the same argument as before, we can prove that $\mathcal{S}'$ is induced from $\mathcal{S}$. This completes the proof of the equality $(M_0)_{\mathrm{red}} = M_1$. Since the equation of M_0 is at least of order 2, it follows that M_0 is non-reduced (see [19]).

Remark. Enriques' proof of (i) is as follows (see [6], p. 273). Let $\{C_t\}$ be a 1-parameter family of plane sextic curves with three double points on a line for $t \neq 0$, and suppose that C_t tends to a triple conic $3Q$ as $t \to 0$. Then the line passing through the three double points tends to a tangent line to Q. Hence these three points come together to a point P. If t is sufficiently close to 0, C_t is close to a union of three conics in a neighborhood of P. If two of them intersect like as in Figure 1, then the third component must be reducible, which is a contradiction.

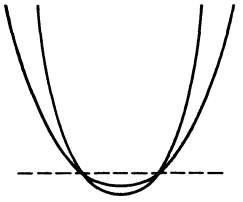

Fig. 1

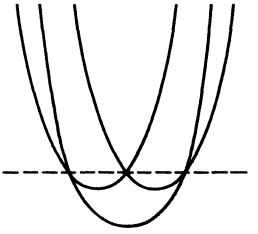

Fig. 2

Enriques did not mention the possibility of Figure 2. Some examples indicate that, in this case, the limit of C_t is a curve of arithmetic genus 7 with one double point. But the author could not produce a rigorous proof of (i) on this line.

§9. Deformations of Surfaces of Type III

We first assume that S is of type III a. Let $\pi: \tilde{S} \to S$ be the blowing up of the two base points of $|K|$. Then $|K|$ defines a holomorphic map $f: \tilde{S} \to \Sigma_0$ of degree 2. It follows that $|K|$ is composed of two pencils $|C|$ and $|D|$ of genus 3 with $C^2 = D^2 = 1$.

Lemma 9.1. *Let $g: S \to \mathbb{P}^1$ be the rational map associated with $|C|$. Then, for any family $p: \mathscr{S} \to M$ of deformations of S, g extends to a family $\Psi: \mathscr{S} \to \mathbb{P}^1 \times M$ of deformations of g, provided M is sufficiently small. More precisely, the line bundle $[C]$ and a basis $\{g_0, g_1\}$ of $H^0(S, \mathcal{O}([C]))$ extend to a line bundle on $\mathscr{S}$ and to a pair $\{\Psi_0, \Psi_1\}$ of its sections*

Proof. We apply [13], Theorem 5. Recall that the sheaf $\mathscr{K}_g$ is defined by the exact sequence

$$0 \to \mathcal{O} \to \mathcal{O}([C])^2 \to \mathscr{K}_g \to 0,$$

where the first map is defined by $\alpha \to (\alpha g_0, \alpha g_1)$. By the theorem cited above, it suffices to prove $H^1(S, \mathscr{K}_g) = 0$, or equivalently, $H^1(\tilde{S}, \pi^* \mathscr{K}_g) = 0$. For this purpose, we write $\pi^* C = \tilde{C} + E_1$ on $\tilde{S}$, where $\tilde{C}$ is the proper transform of C and E_1 is an exceptional curve. From the commutative diagram

$$\begin{array}{ccccccc} 0 \to & \mathcal{O} & \longrightarrow & \mathcal{O}(\tilde{C}+E_1)^2 & \longrightarrow & \pi^*\mathscr{K}_g & \to 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 \to & \mathcal{O}(E_1) & \longrightarrow & \mathcal{O}(\tilde{C}+E_1)^2 & \longrightarrow & \mathcal{O}(2\tilde{C}+E_1) & \to 0, \end{array}$$

we obtain the exact sequence

$$0 \to \mathcal{O}_{E_1}(-1) \to \pi^* \mathscr{K}_g \to \mathcal{O}(2\tilde{C} + E_1) \to 0.$$

Hence, it suffices to prove $h^1(2\tilde{C} + E_1) = 0$. To prove this, we use the exact sequence

$$0 \to \mathcal{O}(\tilde{C} + E_1) \to \mathcal{O}(2\tilde{C} + E_1) \to \mathcal{O}_{\tilde{C}}(E_1) \to 0.$$

From $\tilde{C}E_1 = 1$, it follows that $\dim H^0(\tilde{C}, \mathcal{O}_{\tilde{C}}(E_1)) = 1$. On the other hand, we have $h^0(\tilde{C} + E_1) = 2$. Therefore, $h^0(2\tilde{C} + E_1) \leqq 3$. Since the converse inequality is obvious, we have $h^0(2\tilde{C} + E_1) = 3$. Then by using the Riemann-Roch theorem, we obtain $h^1(2\tilde{C} + E_1) = 0$. Q.E.D.

Theorem 9.1. *Let S be a surface of type* IIIa. *Then any sufficiently small deformation of S is of type* IIIa.

Proof. Let $p: \mathscr{S} \to M$ be a family of deformations of S. Then, by Lemma 9.1, the rational map $g: S \to \mathbb{P}^1$ extends to a rational map $\Psi: \mathscr{S} \to \mathbb{P}^1 \times M$. Similarly, the rational map $h: S \to \mathbb{P}^1$ defined by $|D|$ extends. Therefore, we have two pencils $|C_t|$ and $|D_t|$ on each $S_t = p^{-1}(t)$. If we take the bases $\{g_{1t}, g_{2t}\}$ of $H^0(S_t, \mathcal{O}([C_t]))$ and $\{h_{1t}, h_{2t}\}$ of $H^0(S_t, \mathcal{O}([D_t]))$, then the products $g_{it} h_{jt}$, $i, j = 1, 2$, form a basis of $H^0(S_t, \mathcal{O}(K_t))$, where K_t denotes the canonical bundle of S_t. Hence, $|K_t|$ is composed of two pencils of genus 3. This proves that S_t is of type IIIa. Q.E.D.

Theorem 9.2. *Let S be a generic surface of type* IIIa. *Then the number of moduli $m(S)$ of S is defined and is equal to* $\dim H^1(S, \Theta_S) = 39$.

Proof. Let $\tilde{W} \to W = \Sigma_0$ be the blowing up of the three points as in §5. Then we can prove

$$h^0(\mathscr{T}_{\tilde{S}/\tilde{W}}) = 41, \quad h^1(\mathscr{T}_{\tilde{S}/\tilde{W}}) = 2, \quad \text{and} \quad h^0(h^* \mathscr{T}_{\tilde{W}/W}) = 6,$$

where $h: \tilde{S} \to \tilde{W}$ is the induced map (cf. [Q], §4). Next we consider the exact sequence

$$0 \to \mathcal{T}_{\tilde{S}/\tilde{W}} \to \mathcal{T}_{\tilde{S}/W} \to h^* \mathcal{T}_{\tilde{W}/W} \to 0.$$

In the same way as in [Q], Lemma 15, we can prove that the coboundary map

$$H^0(\tilde{S}, h^* \mathcal{T}_{\tilde{W}/W}) \to H^1(\tilde{S}, \mathcal{T}_{\tilde{S}/\tilde{W}})$$

is surjective. Therefore, we have $h^0(\mathcal{T}_{\tilde{S}/W}) = 45$ and $h^1(\mathcal{T}_{\tilde{S}/W}) = 0$.

Finally, we use the exact sequence

$$0 \to \Theta_{\tilde{S}} \to f^* \Theta_W \to \mathcal{T}_{\tilde{S}/W} \to 0.$$

We note that $h^0(f^* \Theta_W) = 6$, $h^1(f^* \Theta_W) = 4$, and $h^2(f^* \Theta_W) = 0$. Hence we obtain $h^1(\Theta_{\tilde{S}}) = 43$ and $h^1(\Theta_S) = 39$.

We take a complete family $\tilde{\mathcal{S}} \to \Sigma_0 \times M$ of deformations of $\tilde{S} \to \Sigma_0$ such that the characteristic map $\tau: T_0(M) \to H^0(\tilde{S}, \mathcal{T}_{\tilde{S}/W})$ is surjective (see [8], I, Theorem 3.1). Then $\tilde{\mathcal{S}}$ contains 39-dimensional infinitesimal deformations of $\tilde{S}$. We let $\mathcal{S}$ be the family obtained from $\tilde{\mathcal{S}}$ by contracting exceptional curves (see [8], III). Then, using [Q], Lemma 26, it is not difficult to prove that $\mathcal{S}$ contains 39-dimensional infinitesimal deformations[3]. Hence the map $\rho: T_0(M) \to H^1(S, \Theta_S)$ is surjective. Since $H^0(S, \Theta_S)$ vanishes, there exists a 39-dimensional family of deformations of S, which is complete and effectively parametrized. Q.E.D.

As to the deformations of surfaces of type IIIb, we only prove the following

Theorem 9.3. *A generic surface of type* IIIb *is a specialization of surfaces of type* IIIa.

Proof. Let S be a generic surface of type IIIb. Then, it is birationally equivalent to a double covering of Σ_2 whose branch locus B has a quadruple point x and a 2-fold triple point y on a fibre Γ (see Theorem 5.2). Let x_t be a point depending holomorphically on a parameter t with $x_0 = x$, and consider the curves in $|8\Delta_0 + 14\Gamma|$ on Σ_2 having the above singularities at x_t and y. Then, by Remark 3 in §5, and by Grauert's theorem (see [7], Satz 5), we can extend B to a family $\{B_t\}$ of such curves. Let $\{\tilde{S}_t\}$ be the family of the canonical resolutions of the double coverings of Σ_2 with branch loci B_t. Contracting the exceptional curves on $\tilde{S}_t$, we obtain a family $\{S_t\}$ of deformations of S (see [8], III). Suppose that the x_t, $t \neq 0$, are not on the same fibre as y. Then, by Remark 3, in §5, the S_t, $t \neq 0$, are of type IIIa. Q.E.D.

With some more efforts, we can prove that any surface of type IIIb is not a specialization of surfaces of type I, IV, nor V. But we do not go into details. On the other hand, we do not know whether a surface of type IIIb is a specialization of surfaces of type II. This is not the case for the generic surfaces of type IIIb, as follows from the fact that the generic surfaces of type IIIb depend on 38 parameters which is equal to the number of moduli of the generic surfaces of type II.

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[3] See Part IV, §5 for a detailed proof

§10. Deformations of Surfaces of Type IV

We begin with the following two theorems.

Theorem 10.1. *Let S be a surface of type* IVa. *Then any sufficiently small deformation of S is of type* IVa.

Proof. Let $|D|$ be a pencil of curves of genus 2 on S. Then we have $h^1(2D)=0$. This implies that the pencil $|D|$ is stable under small deformations (cf. [Q], proof of Proposition 1). Hence, if $\{S_t\}$ is a family of deformations of $S=S_0$, then each S_t is of type IV or V. By Lemma 8.7, the canonical images form a flat family. Hence they are non-singular quadrics. This proves that S_t is of type IVa. Q.E.D.

Theorem 10.2. *A generic surface of type* IVa-2 *is a specialization of surfaces of type* IVa-1.

Proof. Let S be a generic surface of type IVa-2. For simplicity, we assume that S has one singular fibre of type (III_1). Then the corresponding branch locus B has two 2-fold triple points x, y on a fibre Γ. As in the proof of Theorem 9.3, we consider x_t with $x_0=x$. By the same argument as there, B extends to a family $\{B_t\}$ of curves having 2-fold triple points at x_t and y. This gives rise to a family of deformations of S. If x_t and y are not on a single fibre, then the corresponding member S_t is of type IVa-1. This can be seen by applying the elementary transformations at x_t and y. The proof in the case where S has a singular fibre of type (IV_1) is similar. Q.E.D.

Next we study deformations of surfaces of type IVb.

Theorem 10.3. *Let S be a generic surface of type* IVb-1, *and let $p:\mathscr{S}\to M$ be the Kuranishi family of deformations of S* (see [19]). *Then M has two 38-dimensional non-singular components M_0 and M_1, which intersect transversally along a 37-dimensional manifold N. The points of M_0-N and M_1-N respectively correspond to surfaces of type* Ia *and* IVa-1, *and the points of N to surfaces of type* IVb-1.

Proof is similar to [Q], §§4–6. So we shall give only an outline. Let $g:S\to\mathbb{P}^1$ be the unique pencil of genus 2 on S. The following two propositions can be proved in the same way as in [Q], §4.

Proposition 10.1. *Let $p:\mathscr{S}\to M$ be a family of deformations of S which is of type* IVb -1. *Suppose that each fibre $S_t=p^{-1}(t)$ is of type* III, IV, *or* V. *Then g extends to a holomorphic map $\Psi:\mathscr{S}\to\mathbb{P}^1\times M$ over M (shrinking M if necessary).*

Proposition 10.2. *Let S be a generic surface of type* IVb-1. *Then, there exists a family $p_1:\mathscr{S}_1\to M_1$ of deformations of $S=p^{-1}(0)$ with $0\in M_1$ such that*

i) *M_1 is a 38-dimensional manifold,*

ii) *the infinitesimal deformation map $\rho:T_0(M_1)\to H^1(S,\Theta_S)$ induces an isomorphism of $T_0(M_1)$ onto the kernel of the canonical homomorphism*

$$G:H^1(S,\Theta_S)\to H^1(S,g^*\Theta_{\mathbb{P}^1}),$$

iii) *each fibre S_t is of type* IVa-1 *or* IVb-1, *and the generic members are of type* IVa-1.

The calculation in [Q], §5 can be carried out with the following modifications. First, $|K|$ is of the form $|2D+\bar{G}|$ with $\bar{G}^2=-2$ and $K\bar{G}=2$. Hence $\bar{G}$ is an elliptic curve. With minor changes in Lemmas 21 and 22, we obtain $h^1(\Theta_S)=39$ and $h^2(\Theta_S)=1$. The analogue of Lemma 24 can be proved in a similar way. That is, the canonical homomorphism

$$\zeta_*: H^1(S,\Theta_S)\to H^1(\bar{G},\mathcal{N}_{\bar{G}})$$

is surjective, where $\mathcal{N}_{\bar{G}}$ denotes the normal sheaf of $\bar{G}$ in S. By the same way as in [Q], pp. 71–72, we define a linear map

$$\gamma: H^1(S,\Theta_S)\to H^0(\bar{G},\mathcal{O}(K_{|\bar{G}})),$$

which has the following property.

Lemma 10.1. *Let b_1, $b_2\in\bar{G}$ be the base points of $|K|$. Then*

1) *$G\rho=0$ if and only if $\gamma(\rho)$ vanishes at b_1 and b_2,*
2) *$\zeta_*\rho=0$ if and only if $\gamma(\rho)$ vanishes identically on $\bar{G}$.*

We now take a basis $\{\rho_1,\rho_2,\dots,\rho_{39}\}$ of $H^1(S,\Theta_S)$ such that

$$G\rho_1\neq 0,\quad G\rho_2=0,\quad \zeta_*\rho_2\neq 0,\quad \zeta_*\rho_\lambda=0\quad(\lambda\geqq 3),$$

and let $(t_1,t_2,\dots,t_{39})$ be the corresponding coordinates on $\mathbb{C}^{39}$.

Lemma 10.2. *We fix a hermitian metric on S and let $\varphi(t)$ be a $(0,1)$-form with coefficients in Θ_S which defines the Kuranishi family. Then we have*

$$\mathbf{H}[\varphi(t),\varphi(t)]=\sum_{\lambda=1}^{39} a_\lambda t_1 t_\lambda+(\text{higher terms}),$$

with $a_2\neq 0$, where $\mathbf{H}$ denotes the projection to the harmonic part with respect to the given hermitian metric on S.

Proof. Lemmas 30 and 31 can be applied without change. Hence, it suffices to prove that the product $\gamma(\rho_1)\gamma(\rho_2)\in H^0(\bar{G},\mathcal{O}(2K_{|\bar{G}}))$ is not a restriction of an element of $H^0(S,\mathcal{O}(2K))$. For this purpose, let $f:\tilde{S}\to\Sigma_2$ be the map of degree 2 defined in §6. Then the branch locus B has four quadruple points x_i, $1\leqq i\leqq 4$. From the construction in §6, we infer that $|2\pi^*K|$ contains the linear system

$$\left|2\Delta_0+6\Gamma-\sum_{i=1}^{4}x_i\right|. \tag{10.1}$$

Comparing the dimension of these two linear systems, we see that they coincide. Let Δ_1 and $\Delta_2\in|\Delta_0+2\Gamma|$ be irreducible curves through x_1,x_2, and x_3,x_4, respectively. Then, $|2\Gamma|+\Delta_1+\Delta_2$ is a subsystem of (10.1), and induces a 2-dimensional linear system on $\bar{G}$. On the other hand, since $h^0(K+2D)=8$, the image of the restriction map

$$\text{res}: H^0(S,\mathcal{O}(2K))\to H^0(\bar{G},\mathcal{O}(2K_{|\bar{G}}))$$

is 3-dimensional. Hence it coincides with the image of the pull-back

$$H^0(\mathbb{P}^1, \mathcal{O}(2)) \to H^0(\bar{G}, \mathcal{O}([2b_1+2b_2]))$$

by the restriction $\bar{f}: \bar{G} \to \mathbb{P}^1$ of f. We note that $\bar{f}$ is a double covering which is ramified at 4 points including b_1 and b_2. Therefore, if $\gamma(\rho_1)\gamma(\rho_2)$ is in the image of res, its divisor is $2b_1+2b_2$, for, it vanishes at b_1+b_2. But this is impossible by Lemma 10.1.

We shall prove that S is not a specialization of surfaces of type II in Theorem 10.4 below, and the remaining assertions of Theorem 10.3 can be proved in the same way as in [Q], Theorem 3. Q.E.D.

In order to complete the proof of the hierarchy in Introduction, we need to prove the following theorems.

Theorem 10.4. *Let S be a surface of type* IV. *Then, any sufficiently small deformation of S is either of type* I *or* IV.

Theorem 10.5. *Generic surfaces of type* IVb-1 *and* IVb-2 *are respectively the specializations of surfaces of type* IVa-1 *and* IVa-2.

We first prepare with the following lemma.

Lemma 10.3. *Let S be a surface of type* IVb, *and let $|D|$ be a pencil of genus 2 on S. Then there exists a unique divisor $G \in |K-2D|$, and G is one of the following.*

If S is of type IVb-1, *then*

(i) *G is a reduced irreducible curve, which is an elliptic curve or a rational curve with one node, and $G^2=-2$, or*

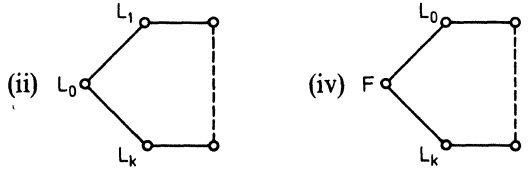
(ii) *G is a sum $\sum_{i=0}^{k} L_k$ of non-singular rational curves L_i, with $k \geqq 1$, $L_0^2=-4$, $DL_0=2$, and $L_i^2=-2$ for $1 \leqq i \leqq k$.*

If S is of type IVb-2, *then $|K|$ has a unique fixed component F which is a non-singular rational curve with $F^2=-2$. In this case,*

(iii) *$G=F+L_0$ where L_0 is a non-singular rational curve with $L_0^2=-4$, and $FL_0=DL_0=2$, or*

(iv) *$G=F+\sum_{i=0}^{k} L_i$ with $k \geqq 1$, where the L_i are non-singular rational curves with $L_0^2=L_k^2=-3$, $DL_0=DL_k=1$, and $L_i^2=-2$, $DL_i=0$ for $1 \leqq i \leqq k-1$.*

In the cases (ii) *and* (iv), *the curves form the configurations whose dual graphs are as follows:*



These can be proved by examining the possible singularities of the branch locus given in §6.

Proof of Theorem 10.4. Let $p: \mathcal{S} \to \mathbf{D}$ be a 1-parameter family of deformations of $S = p^{-1}(0)$, where $\mathbf{D}$ is a small disc centered at $0 \in \mathbb{C}$. In view of Theorem 10.1, we may

assume that S is of type IVb. We suppose that the general fibres $S_t = p^{-1}(t)$ are not of type I. Then, the canonical images of S_t form a flat family of quadrics in $\mathbb{P}^3$ (see Lemma 8.7). Therefore, we can find a basis $\{\varphi_0(t), \ldots, \varphi_3(t)\}$ of $H^0(S_t, \mathcal{O}(K_t))$[4] such that

$$\varphi_2(t)^2 = \varphi_0(t)\,\varphi_1(t) + \alpha(t)\,\varphi_3(t)^2,$$

where $\alpha(t)$ is a holomorphic function in t with $\alpha(0) = 0$.

First suppose that $\alpha(t)$ is identically 0. In this case, $g = \varphi_0/\varphi_2 = \varphi_2/\varphi_1$ defines a meromorphic function on $\mathcal{S}$. We write the divisor (g) in the form $\mathcal{D} - \mathcal{D}_1$, where $\mathcal{D}$ and $\mathcal{D}_1$ are effective divisors without common component on $\mathcal{S}$. Furthermore, there exists an effective divisor $\mathcal{G}$ on $\mathcal{S}$ such that

$$(\varphi_0) = 2\mathcal{D} + \mathcal{G}, \quad (\varphi_1) = 2\mathcal{D}_1 + \mathcal{G}, \quad (\varphi_2) = \mathcal{D} + \mathcal{D}_1 + \mathcal{G} \tag{10.2}$$

(cf. [Q], p. 46). The restrictions $\mathcal{D}_{|S}$ and $\mathcal{D}_{1|S}$ may have some common components. So we express them as

$$\mathcal{D}_{|S} = D + C, \quad \mathcal{D}_{1|S} = D_1 + C,$$

where D and D_1 have no common component. By construction, $|D|$ is the unique pencil of genus 2 on S. Moreover, it follows that $2C + \mathcal{G}_{|S}$ coincides with the unique divisor G in $|K - 2D|$. But, by Lemma 10.3, G has no multiple component. Hence $|D|$ extends to pencils on S_t. Since we have assumed $\alpha(t) \equiv 0$, this implies that each S_t is of type IVb or V. To exclude the second possibility, we consider the family $\{G_t\} = \{\mathcal{G}_{|S_t}\}$ of divisors on S_t. As remarked above, G_0 has no multiple component. Hence neither G_t has one, provided $|t|$ is sufficiently small. But, if S_t is of type V, then Δ_0 is in the branch locus. Hence, G_t contains a double component. This proves that S_t is not of type V, and settles the case when $\alpha(t) \equiv 0$.

Next suppose that $\alpha(t)$ is not identically 0. In this case, taking a square root of t, we may assume that $\alpha(t)$ is of the form $\beta(t)^2$. Then we have two meromorphic functions

$$g = \frac{\varphi_0}{\varphi_2 + \beta\varphi_3} = \frac{\varphi_2 - \beta\varphi_3}{\varphi_1}, \qquad h = \frac{\varphi_0}{\varphi_2 - \beta\varphi_3} = \frac{\varphi_2 + \beta\varphi_3}{\varphi_1} \tag{10.3}$$

on $\mathcal{S}$. We write $(g) = \mathcal{D} - \mathcal{D}_1$ and $(h) = \mathcal{D}' - \mathcal{D}_1'$ as above. Then we can write

$$(\varphi_0) = \mathcal{D} + \mathcal{G}_1, \qquad (\varphi_2 + \beta\varphi_3) = \mathcal{D}_1 + \mathcal{G}_1,$$
$$(\varphi_2 - \beta\varphi_3) = \mathcal{D} + \mathcal{G}_2, \quad (\varphi_1) = \mathcal{D}_1 + \mathcal{G}_2.$$

By (10.3), we have

$$(g) + (h) = (\varphi_0) - (\varphi_1) = (\mathcal{D} - \mathcal{D}_1) + (\mathcal{G}_1 - \mathcal{G}_2).$$

Hence, we have $\mathcal{G}_1 - \mathcal{G}_2 = \mathcal{D}' - \mathcal{D}_1'$, and we can write

$$\mathcal{G}_1 = \mathcal{D}' + \mathcal{G}_0, \quad \mathcal{G}_2 = \mathcal{D}_1' + \mathcal{G}_0$$

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[4] K_t denotes the canonical bundle of S_t

with an effective divisor $\mathscr{G}_0$. Finally, we get

$$\begin{aligned} &(\varphi_0)=\mathscr{D}+\mathscr{D}'+\mathscr{G}_0, && (\varphi_2+\beta\varphi_3)=\mathscr{D}_1+\mathscr{D}'+\mathscr{G}_0,\\ &(\varphi_2-\beta\varphi_3)=\mathscr{D}+\mathscr{D}'_1+\mathscr{G}_0, && (\varphi_1)=\mathscr{D}_1+\mathscr{D}'_1+\mathscr{G}_0. \end{aligned} \tag{10.4}$$

We write as before

$$\begin{aligned} &\mathscr{D}_{|S}=D+C, && \mathscr{D}_{1|S}=D_1+C,\\ &\mathscr{D}'_{|S}=D'+C', && \mathscr{D}'_{1|S}=D'_1+C'. \end{aligned}$$

Then $|D|$ and $|D'|$ both coincide with the unique pencil of genus 2 on S. Hence, it follows that $C+C'+\mathscr{G}_{0|S}$ coincides with $G\in|K-2D|$.

We now suppose that a general member S_t is not of type I nor IV. Then, since we have assumed $\alpha(t)\not\equiv 0$, S_t is necessarily of type IIIa. This implies that $|D|+C$ and $|D|+C'$ both extend to pencils of genus 3 with one base point on S_t. Therefore, we have

$$\begin{aligned} &K(D+C)=K(D+C')=3,\\ &(D+C)^2=(D+C')^2=1. \end{aligned}$$

The first line implies $KC=KC'=1$. By Lemma 10.3, this is possible only in the case (iv). But then we have $DC=1$ and $C^2\leqq -3$, which contradicts the above second line. Q.E.D.

Proof of Theorem 10.5. The assertions can be proved by the same method as in the proof of [Q], Proposition 2.

Remark. If we apply the construction of Theorem 10.2 to a surface of type IVb−2, we obtain a family of surfaces of type IVa−1 which specialize to a surface of type IVb−2.

§11. Deformations of Surfaces of Type V

In this section we shall prove the following three theorems concerning the deformations of surfaces of type V.

Theorem 11.1. *Surfaces of type* V−1 *and* V−2 *are specializations of surfaces of type* IIIa *and* IIIb, *respectively.*

Theorem 11.2. *A surface of type* V−1 *is not a specialization of surfaces of type* IV.

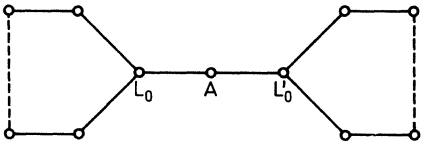
Theorem 11.3. *A surface of type* V−2 *is a specialization of surfaces of type* V−1.

Proof of Theorem 11.1. We recall that a surface of type III is birationally equivalent to a double covering of $\mathbb{P}^2$ with branch locus of degree 10 which has two 2-fold triple points or a 4-fold triple point (see §5, Remarks 2 and 4). On the other hand, a surface of type V is birationally equivalent to a double covering of Σ_4 with branch locus B in $|6\Delta_0+20\Gamma|$ which has the same singularities as above. This can be seen as follows. First, it is birationally equivalent to a double covering of Σ_2 as in §6. Then, we apply the elementary transformations at the two singular points on Δ_0 of the branch locus.

Using these facts, we can construct, by the same method as in Lemma 4.4 in Part I, a family in which surfaces of type IIIa (or IIIb) specialize to a surface of type V−1 (or V−2). Q.E.D.

Before starting the proof of Theorem 11.2, we prepare with the following lemma.

Lemma 11.1. *Let S be a surface of type* V−1, *$|D|$ the unique pencil of genus 2 on S, and let G be the unique divisor in $|K-2D|$. Then, G contains a unique double component A, which is a non-singular rational curve with $A^2=-2$, and two components L_0 and L_0' which satisfy $KL_0=KL_0'=1$. Other components, if any, are non-singular rational curves with self-intersection number −2. They form the configuration whose dual graph is as follows.*



Moreover, A is the unique component of G which intersects D.

This lemma, like Lemma 10.3, can be proved by examining the singularities of the branch locus. Here A is the unique component over Δ_0, and, in the generic case, L_0 and L_0' are elliptic curves with $L_0^2=L_0'^2=-1$, and there are no other components. The above notation for the curves on S will be used in the following.

Proof of Theorem 11.2. Let $p\colon \mathcal{S}\to\mathbf{D}$ be a 1-parameter family of deformations of $S=p^{-1}(0)$ of type V−1, and suppose that a general member $S_t=p^{-1}(t)$ is of type IV. Since $|K|$ has no fixed component, S_t is of type IVa−1 or IVb−1. Hence, the canonical system $|K_t|$ of S_t has two base points. Therefore, we can consider the family $\tilde{\mathcal{S}}$ obtained from $\mathcal{S}$ by blowing up the locus of base points, and we have a holomorphic map $\Phi\colon \tilde{\mathcal{S}}\to\mathbb{P}^3\times\mathbf{D}$, which induces the canonical map on each fibre. By Lemma 8.7, we can find a basis $\{\varphi_0(t),\ldots,\varphi_3(t)\}$ of $H^0(S_t,\mathcal{O}(K_t))$ such that

$$\varphi_2(t)^2=\varphi_0(t)\,\varphi_1(t)+\alpha(t)\,\varphi_3(t)^2,\qquad \alpha(0)=0.$$

First suppose that $\alpha(t)\equiv 0$. Then, a pencil of the form $|D|+C$ extends to a pencil $|D_t|$ on S_t (cf. proof of Theorem 10.4). If $C\neq 0$, C is the unique double component A of G in Lemma 11.1. Since S_t is of type IVb−1, $|D_t|$ is a pencil of genus 2. Hence, in (10.2), the divisor $\mathcal{G}_{|S_t}$ is connected for $t\neq 0$, and is disconnected for $t=0$. This contradiction proves that $C=0$, i.e. the pencil $|D|$ extends to $|D_t|$.

From this fact, we readily infer that Φ can be lifted to $\tilde{\Phi}\colon \tilde{\mathcal{S}}\to\Sigma_2\times\mathbf{D}$. Now let $\tilde{W}\to\Sigma_2$ be the blowing up of four quadruple points of the branch locus of $\tilde{S}\to\Sigma_2$ (Theorem 6.1). Then, we can prove that $\tilde{\Phi}$ can be lifted to $\tilde{\mathcal{S}}\to\tilde{\mathcal{W}}$ extending the natural map $h\colon \tilde{S}\to\tilde{W}$, where $q\colon \tilde{\mathcal{W}}\to\mathbf{D}$ is a family of deformations of $\tilde{W}$. In fact, by [8], II, Proposition 7.1, it suffices to prove that the natural map

$$H^0(\tilde{W},\mathcal{T}_{\tilde{W}/\Sigma_2})\to H^0(\tilde{S},h^*\mathcal{T}_{\tilde{W}/\Sigma_2})$$

is surjective. This can be proved to be bijective by a method similar to [Q], Lemma 13.

In this way, we get a family of maps $h_t: \tilde{S}_t \to \tilde{W}_t = q^{-1}(t)$ of degree 2. We note that the branch locus of h has no infinitely near triple points. Hence, it follows that the branch loci B_t of the h_t form a flat family (see Part I, p. 369). But, while B_0 has 4 connected components, B_t, $t \neq 0$ has three, which is a contradiction.

Next suppose that $\alpha(t)$ is not identically 0. Then the general fibres are of type IVa -1. By the second half of the proof of Theorem 10.4, we may assume that two pencils $|D|+C$ and $|D|+C'$ extend to those on S_t. Since $|K_t|$ has no fixed component, we have $C+C'=G$ (cf. (10.4)). Moreover, since a generic member S_t is of type IVa, we may assume that $|D|+C$ extends to a pencil $|D_t|$ of genus 2. This implies that $KC=0$ and $KC'=2$. Suppose $C \neq 0$. Then, C consists of non-singular rational curves with self-intersection number -2. If $DC=0$, then from $(D+C)^2=0$, we have $C^2=0$, and hence $C=0$, by Hodge's index theorem. So we have $DC \geqq 1$. Similarly, $DC' \geqq 1$. Combined with $DG=2$, these imply $DC=DC'=1$. Then from $(D+C)^2=0$, we get $C^2=-2$. By Lemma 11.1,

$$C = A + (\text{rational curves}),$$

and the equality $KC=0$ implies that this is a disjoint sum. Hence, using $C^2=A^2=-2$, we conclude $C=A$.

We now want to apply an elementary operation to A (see Part II, Appendix B). For this purpose, we need to show that A does not extend to a family $\{A_t\}$ of divisors on S_t. Suppose A extends. Then, A_t is the rational curve on S_t with $K_t A_t = 0$, and is away from the base points of $|K_t|$. Hence A_t is mapped into a point by the canonical map Φ_{K_t} on S_t. This contradicts the equality $D_t A_t = 1$.

By the elementary operation, we may assume $|D|$ itself extends to $|D_t|$. Then, we have

$$(\varphi_0) = \mathscr{D} + \mathscr{D}', \quad (\varphi_2 + \beta\varphi_3) = \mathscr{D}_1 + \mathscr{D}', \tag{11.1}$$

where $\mathscr{D}$ and $\mathscr{D}_1$ are disjoint (cf. (10.4)). Recall that the image of $\Phi: \tilde{\mathscr{S}} \to \mathbb{P}^3$ is defined by the equation

$$(Z_2 + \beta Z_3)(Z_2 - \beta Z_3) = Z_0 Z_1,$$

for a system of homogeneous coordinates (Z_0, Z_1, Z_2, Z_3) on $\mathbb{P}^3$. By blowing up the curve $Z_0 = Z_2 + \beta Z_3 = 0$, we obtain a family $\mathscr{W}$ of deformations of Σ_2. From (11.1), it follows that the ideal sheaf generated by φ_0 and $\varphi_2 + \beta\varphi_3$ on $\tilde{\mathscr{S}}$ is invertible. Therefore, the map Φ factors through $\mathscr{W}$. This leads us to a contradiction as before. Q.E.D.

Proof of Theorem 11.3. A surface of type V -2 is birationally equivalent to a double covering of Σ_2 such that its branch locus has two 2-fold triple points on a fibre, one of them being on Δ_0. By displacing the other triple point out of the fibre, we obtain a family of double coverings of Σ_2 (cf. Theorem 10.2). Applying elementary transformations, we see that the general members are of type V -1. Q.E.D.

§12. Simply-Connectedness

Theorem 12.1. *Any minimal algebraic surface S with $p_g=4$, $q=0$, and $c_1^2=6$ is simply connected.*

Proof. It suffices to prove the theorem, for instance, for surfaces of type Ib, II, and IIIa. First, it can be easily proved that S has no finite non-trivial unramified covering (see [3], p. 212). Hence, it is enough to show that the fundamental group $\pi_1(S)$ is abelian. For surfaces of type Ib and IIIa, this last fact follows from [22], Proposition 3 (cf. the proof of Theorem 3.4 in Part I). The proof for surfaces of type II is as follows. Since they have one and the same deformation type, we may assume, in (4.1), that $\alpha=\beta=0$ and that the divisor $\gamma=0$ is non-singular. Then the branch locus B of $\tilde{S}\to\Sigma_2$ is set-theoretically defined by $\gamma\zeta=0$. By the result cited above, the fundamental group of Σ_2-B is abelian. This implies that $\pi_1(S)$ is abelian. Q.E.D.

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The Campbell-Hausdorff Formula and Invariant Hyperfunctions

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Introduction

Let $\mathbf{G}$ be a Lie group and $\mathfrak{g}$ its Lie algebra. We denote by V the underlying vector space of $\mathfrak{g}$.

There is a canonical isomorphism between the ring $\mathbf{Z}(\mathfrak{g})$ of the biinvariant differential operators on $\mathbf{G}$ and the ring $\mathbf{I}(\mathfrak{g})$ of the constant coefficient operators on V which are invariant by the adjoint action of $\mathbf{G}$. When $\mathfrak{g}$ is semi-simple, this is the "Harish-Chandra isomorphism"; for a general Lie algebra, this was established by Duflo [4].

We shall prove here, that when $\mathbf{G}$ is solvable the Duflo isomorphism extends to an isomorphism Φ of the algebra of "local" invariant hyperfunctions under the group convolution and the algebra of invariant hyperfunctions on V under additive convolution (the exact result will be stated below). This gives a partial answer to a conjecture of Rais [12].

The existence of such an isomorphism Φ is of importance for the harmonic analysis on $\mathbf{G}$, and for the study of the solvability of biinvariant operators on $\mathbf{G}$ (see [7]). It reflects and explains the "orbit method" ([8, 9]), i.e. the correspondence between orbits of $\mathbf{G}$ in V^*, the dual vector space of V, and unitary irreducible representations of $\mathbf{G}$: let T be an irreducible representation of $\mathbf{G}$, then the infinitesimal character of T is a character of the ring $\mathbf{Z}(\mathfrak{g})$. Let $\mathcal{O}$ be an orbit in V^*, the map $\rho_{\mathcal{O}}(P)=P(f)$ $(f\in\mathcal{O})$ is a character of the ring $\mathbf{I}(\mathfrak{g})$ ($\mathbf{I}(\mathfrak{g})$ being identified with the ring of invariant polynomials on V^*). The principle of the orbit method is to assign to a (good) orbit $\mathcal{O}$ a representation $T_{\mathcal{O}}$ of $\mathbf{G}$ (or $\mathfrak{g}$), whose infinitesimal character corresponds to $\rho_{\mathcal{O}}$ via the isomorphism Φ. This is the technique used by M. Duflo to construct the ring isomorphism Φ.

Furthermore let $t_{\mathcal{O}}$ be (when defined) the distribution on V which is the Fourier transform of the canonical measure on the orbit $\mathcal{O}$, then $t_{\mathcal{O}}$ is clearly an invariant positive eigendistribution of every operator P in $\mathbf{I}(\mathfrak{g})$ of eigenvalue $\rho_{\mathcal{O}}(P)$. Kirillov conjectured that the global character of the representation $T_{\mathcal{O}}$ (when

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defined) should be intimately connected with the "orbit distribution" $\Phi^{-1}(t_{\mathcal{O}})$, as proven in numerous cases. It is an essential result of Duflo [4] that these "orbit distributions" are indeed eigenfunctions for every biinvariant operator P in $\mathbf{Z}(\mathfrak{g})$; as in Rais [11], this implies the local solvability of P [4].

We will here derived the existence of Φ from a property of the Campbell-Hausdorff formula, that we conjecture and can prove in the solvable case. It is then a natural corollary of our conjecture, that biinvariant operators are locally solvable and that "orbit distributions" are eigendistributions for $\mathbf{Z}(\mathfrak{g})$. Hence the correspondence between orbits and representations is already engraved in the structure of the multiplication law.

Let us describe with some details our technique and results: We denote by $\mathfrak{g}_t$ the Lie algebra whose underlying vector space is V itself and in which the bracket $[*,*]_t$ is given by $[X,Y]_t = t[X,Y]$. Then $\mathfrak{g}_t$ gives a deformation between $\mathfrak{g}$ and the abelian Lie algebra, in which the fact is trivial.

In the course of the proof we encounter the following problem: Let $\mathbf{L}$ be the free Lie algebra generated by two indeterminates x and y and $\hat{\mathbf{L}}$ its completion. Since $x+y-\log e^y e^x$ belongs to $[\hat{\mathbf{L}},\hat{\mathbf{L}}]$, by Campbell-Hausdorff formula, we can write it in $x+y-\log e^y e^x = (1-e^{-\operatorname{ad} x})F + (e^{\operatorname{ad} y}-1)G$ for F and G in $\hat{\mathbf{L}}$. F and G are not uniquely determined by this property.

Conjecture. *For any Lie algebra* $\mathfrak{g}$ *of finite dimension, we can find* F *and* G *such that they satisfy*

a) $x+y-\log e^y e^x = (1-e^{-\operatorname{ad} x})F + (e^{\operatorname{ad} y}-1)G$.

b) F *and* G *give* $\mathfrak{g}$*-valued convergent power series on* $(x,y)\in\mathfrak{g}\times\mathfrak{g}$.

c) $\operatorname{tr}((\operatorname{ad} x)(\partial_x F);\mathfrak{g}) + \operatorname{tr}((\operatorname{ad} y)(\partial_y G);\mathfrak{g})$

$$= \tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{\operatorname{ad} x}-1} + \frac{\operatorname{ad} y}{e^{\operatorname{ad} y}-1} - \frac{\operatorname{ad} z}{e^{\operatorname{ad} z}-1} - 1;\mathfrak{g}\right).$$

Here $z = \log e^x e^y$ *and* $\partial_x F$ (resp. $\partial_y G$) *is the* $\mathbf{End}(\mathfrak{g})$*-valued real analytic function defined by*

$$\mathfrak{g}\ni a \mapsto \frac{d}{dt}F(x+ta,y)|_{t=0} \quad \left(\text{resp. } \mathfrak{g}\ni a\mapsto \frac{d}{dt}G(x,y+ta)|_{t=0}\right),$$

and tr *denotes the trace of an endomorphism of* $\mathfrak{g}$.

When $\mathfrak{g}$ is nilpotent, this conjecture is easily verified because $(\operatorname{ad} x)(\partial_x F)$, $1-(\operatorname{ad} x/(e^{\operatorname{ad} x}-1))$ etc. are nilpotent endomorphisms of $\mathfrak{g}$ so that their traces vanish. However, we get the following fact.

Proposition 0. *If* $\mathfrak{g}$ *is solvable, then* Conjecture *is true.*

Let K be a non-empty closed cone in $\mathfrak{g}$. Let $\mathscr{I}(K)$ (resp. $\tilde{\mathscr{I}}(K)$) be the vector space of the germs at the unit element $e\in\mathbf{G}$ (resp. the origon $0\in\mathfrak{g}$) of the functions (i.e. either distributions, or hyperfunctions or micro-functions) $u(g)$ (resp. $\tilde{u}(x)$) such that $\operatorname{supp} u \subset \exp K$ (resp. $\operatorname{supp}\tilde{u}\subset K$) infinitesimally (see § 2) and that $u(ghg^{-1}) = |\det(\operatorname{Ad}(g);\mathfrak{g})|^{-1}u(h)$ (resp. $\tilde{u}(\operatorname{Ad}(g)x) = |\det(\operatorname{Ad}(g);\mathfrak{g})|^{-1}\tilde{u}(x)$). We shall set $j(x) = \det((1-e^{-\operatorname{ad} x})/\operatorname{ad} x;\mathfrak{g})$ for $x\in\mathfrak{g}$ sufficiently near the origin. We

define the isomorphism Φ: $\mathscr{I}(K)\to\tilde{\mathscr{I}}(K)$ by $(\Phi u)(x)=j(x)^{1/2}u(e^x)$ for $u\in\mathscr{I}(K)$. If two closed cones K_1 and K_2 satisfy $K_1\cap(-K_2)=\{0\}$, then we can define the product $\mathscr{I}(K_1)\times\mathscr{I}(K_2)\to\mathscr{I}(K_1+K_2)$ (resp. $\tilde{\mathscr{I}}(K_1)\times\tilde{\mathscr{I}}(K_2)\to\tilde{\mathscr{I}}(K_1+K_2)$) by the convolution $*$, i.e.

$$(u*v)(g)=\int_{\mathbf{G}}u(h)\,v(h^{-1}g)\,dh \quad\text{and}\quad (\tilde{u}*\tilde{v})(x)=\int_{\mathfrak{g}}\tilde{u}(y)\,\tilde{v}(-y+x)\,dy.$$

The exact statement which we shall prove is the following:

Theorem. *If* Conjecture *is true for the group* **G**, *then we have*

$$(\Phi u)*(\Phi v)=\Phi(u*v)$$

for $u\in\mathscr{I}(K_1)$ *and* $v\in\mathscr{I}(K_2)$.

If we apply this theorem when v is supported at the origin, then we obtain the following corollary:

Corollary 0. *Suppose that* Conjecture *is true for* **G**, *then with any biinvariant differential operator* P *on* **G** *we can associate a constant coefficient differential operator* $\tilde{P}$ *on* $\mathfrak{g}$ *so that* $\tilde{P}\Phi(u)=\Phi(Pu)$ *holds for any* $u\in\mathscr{I}(\mathfrak{g})$.

In paragraph 4, we will prove directly this particular case of our theorem. In fact, applying the same technique, we can prove a more precise result, giving a partial answer to a conjecture of Dixmier.

Let $\gamma(P)=\beta(D(j^{1/2})P)$ the Duflo isomorphism from $\mathbf{I}(\mathfrak{g})$ to $\mathbf{Z}(\mathfrak{g})$, where β is the symmetrization map and $D(j^{1/2})$ the "differential" operator (of infinite order) defined by $j^{1/2}$, let us look at the operator $\gamma(P)$ as a biinvariant differential operator on **G**; we denote by $(\exp)^*(\gamma(P))$ the differential operator on $\mathfrak{g}$ with analytic coefficients, which is the inverse image of $\gamma(P)$ by the exponential mapping. Let D be the ring of the germs at 0 of differential operators with analytic coefficients. We consider the left ideal $\mathscr{L}$ of D generated by the elements $\langle[A,x],\partial_x\rangle+\operatorname{tr}(\operatorname{ad}A;\mathfrak{g})$, $A\in\mathfrak{g}$ (here $\langle[A,x],\partial_x\rangle$ is the adjoint vector field given by $\dfrac{d}{d\varepsilon}\varphi(\exp\varepsilon A\cdot x)|_{\varepsilon=0}$). Every invariant distribution on $\mathfrak{g}$ is annihilated by $\mathscr{L}$. So Corollary 0 is implied by:

Corollary 1. *Suppose that* Conjecture *is true for* **G**, *then*

$$(\exp)^*(\gamma(P))-j(x)^{-\frac{1}{2}}Pj(x)^{\frac{1}{2}}\in\mathscr{L}.$$

Since Conjecture is solved in the solvable case the above theorem and its corollaries are true for a solvable group **G**. Recall that the result stated in Corollary 0 holds for $\mathfrak{g}$ semi-simple as proved by Harish-Chandra [6]. Howe [16] says that he proved Theorem for a nilpotent group **G** and a restricted class of functions u, v.

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§1

For The Theory of Microfunctions, we Refer to [1, 10, 15]. Let $\mathbf{G}$ be a Lie group, $\mathfrak{g}$ its Lie algebra and exp: $\mathfrak{g}\to G$ the exponential map. Let $\mathbf{M}$ be a real analytic manifold on which $\mathbf{G}$ acts real analytically. A hyperfunction $u(x)$ on $\mathbf{M}$ is called a *relative invariant* with respect to a character χ of G if $u(gx)=\chi(g)u(x)$ holds on $G\times\mathbf{M}$. Here $u(gx)$ is the pull-back of u by the map r: $\mathbf{G}\times\mathbf{M}\to\mathbf{M}$ defined by $(g,x)\mapsto gx$, and $\chi(g)u(x)$ is the product of a real analytic function $\chi(g)$ on $\mathbf{G}\times\mathbf{M}$ and the pull-back of u by the projection from $\mathbf{G}\times\mathbf{M}$ onto $\mathbf{M}$. More generally, let A be a subset of M, $\mathbf{G}_A=\{g\in\mathbf{G};\ gA=A\}$. A hyperfunction $u(x)$ defined in a neighborhood U of A is called *relative invariant* locally at A if there is a neighborhood W of $\mathbf{G}_A\times A$ such that $r(W)\subset U$ and that $u(gx)=\chi(g)u(x)$ on W.

For any $X\in\mathfrak{g}$, we denote by D_χ the vector field defined by $(D_\chi u)(x)=\frac{d}{dt}u(\exp(-tX)x)|_{t=0}$, and by $\delta\chi$ the derivative of χ $\left(\text{i.e. } \delta\chi(X)=\frac{d}{dt}\chi(\exp tX)|_{t=0}\right)$.

Lemma 1.1. *If u is a relative invariant locally on A hyperfunction then $(D_\chi+\delta\chi(X))u=0$ in a neighborhood of A for any $X\in\mathfrak{g}$.*

Proof. We define the map φ: $\mathbb{R}\times\mathbf{M}\to\mathbf{G}\times\mathbf{M}$ by $(t,x)\mapsto(\exp(-tX),x)$. Then the pull-back of $u(gx)$ is the pull-back $(r\varphi)^*u$ of u by the map $r\circ\varphi$, and the pull-back of $\chi(g)u(x)$ is $\chi(e^{-tX})u(x)$. Since $r\circ\varphi$ has maximal rank, this is justified. Thus $(r\circ\varphi)^*u=\chi(e^{-tX})u(x)$. If we differentiate the both-sides with respect to t, and restrict them at the variety $t=0$ in $\mathbb{R}\times\mathbf{M}$, we obtain $D_\chi u$ from the left hand side and $-\delta\chi(X)u$ from the right hand side. Q.E.D.

§2

Let $\mathbf{G}$ be a Lie group, $\mathfrak{g}$ its Lie algebra and exp: $\mathfrak{g}\to\mathbf{G}$ the exponential map. We denote by $\mathbf{dg}$ the left invariant Haar measure and by $\mathbf{dx}$ the Euclidean measure on $\mathfrak{g}$. After the normalization, $\mathbf{dg}$ and $\mathbf{dx}$ are related under the exponential map by the formula: $d(e^x)=j(x)\,dx$ where $j(x)=\det((1-e^{-\mathrm{ad}x})/\mathrm{ad}x;\mathfrak{g})$ in a neighborhood of $x=0$, because the derivative of $\exp x$ at x is given by $(1-e^{-\mathrm{ad}x})/\mathrm{ad}x$ when we identify $\mathbf{TG}$ with $\mathfrak{g}\times\mathbf{G}$ by the left translation. We define the character $\chi_0(g)$ of G by $|\det(\mathrm{Ad}(g);\mathfrak{g})|$, we denote by $d\chi_0$ the corresponding character of $\mathfrak{g}$, i.e. $d\chi_0(x)=\mathrm{tr}(\mathrm{ad}x;\mathfrak{g})$.

Let A and B be subsets of a C^1-manifold $\mathbf{M}$, x a point in $\mathbf{M}$. Take a local coordinate system $(x_1,\dots,x_l)$ of $\mathbf{M}$. The set of limits of the sequence $a_n(y_n-z_n)$ where $a_n>0$, $y_n\in A$, $z_n\in B$ and y_n, z_n converge to x when $n\to\infty$, is denoted by $\mathbf{C}_x(A;B)$ regarded as a closed subset of the tangent space $\mathbf{T}_x\mathbf{M}$ of $\mathbf{M}$ at x. $\mathbf{C}_x(A;\{x\})$ is simply denoted by $\mathbf{C}_x(A)$. If f is a differential map from $\mathbf{M}$ to a C^1 manifold N, then we have $(df)_x(\mathbf{C}_x(A;B))\subset\mathbf{C}_{f(x)}(fA;fB)$. If $\mathbf{C}_x(A;B)\cap\mathrm{Ker}\,df(x)\subset\{0\}$, then there is a neighborhood U of x such that

$$(df)_x\,\mathbf{C}_x(A;B)=\mathbf{C}_{f(x)}(f(A\cap U);f(B\cap U)).$$

If $\mathbf{C}_x(A;B)=\{0\}$, then x is an isolated point of $\bar{A}$ and $\bar{B}$. $\mathbf{C}_x(A;B)=\emptyset$ if and only if $\bar{A}\cap\bar{B}\neq x$.

Let K be a closed cone of $\mathfrak{g}$. We shall denote by $\mathscr{F}(K)$ (resp. $\tilde{\mathscr{F}}(K)$) the space of the germs of function $u(g)$ (resp. $\tilde{u}(x)$) on $\mathbf{G}$ (resp. on $\mathfrak{g}$) at $e\in\mathbf{G}$ (resp. $0\in\mathfrak{g}$) satisfying

(2.1) $\mathbf{C}_e(\operatorname{supp} u)\subset K\subset\mathfrak{g}=\mathbf{T}_e\mathbf{G}$ (resp. $C_0(\operatorname{supp}\tilde{u})\subset K\subset\mathfrak{g}=\mathbf{T}_0\mathfrak{g}$)

and

(2.2) u is a relative invariant locally at e with respect to the character $\chi_0(g)^{-1}$.

Let K_1 and K_2 be two closed cones in $\mathfrak{g}$ such that $K_1\cap(-K_2)=\{0\}$. If $u\in\mathscr{F}(K_1)$ and $v\in\mathscr{F}(K_2)$, then $(\operatorname{supp} u)\cap(\operatorname{supp} v)^{-1}$ is contained in $\{e\}$ locally. Suppose that u and v are defined on a neighborhood U_0 of e. For any open neighborhood $U\subset U_0$ of e, we can find neighborhoods W and V of e such that $W\subset U$, $W^{-1}\subset U$, $\{h\in W; h\in\operatorname{supp} u, h^{-1}\in\operatorname{supp} v\}\subset\{e\}$ and that the map $(g,h)\mapsto g$ from $\{(g,h)\in V\times W; h^{-1}g\in\operatorname{supp} v, h\in\operatorname{supp} u\}$ to V is a proper map. Hence we can define $(u*v)(g)$ by

$$\int_W u(h)\,v(h^{-1}g)\,dh \quad \text{on } g\in V.$$

This gives the bilinear homomorphism $\mathscr{F}(K_1)\times\mathscr{F}(K_2)\to\mathscr{F}(K_1+K_2)$ because $\mathbf{C}_e((\operatorname{supp} u)\cdot(\operatorname{supp} v))\subset K_1+K_2$. In the same way, we can define the convolution

$$(\tilde{u}*\tilde{v})(x)=\int_{\mathfrak{g}}\tilde{u}(y)\,\tilde{v}(-y+x)\,dy$$

which gives the homomorphism $\tilde{\mathscr{F}}(K_1)\times\tilde{\mathscr{F}}(K_2)\to\tilde{\mathscr{F}}(K_1+K_2)$.

Note that if u belongs to $\mathscr{F}(\mathfrak{g})$, then we have $\chi_0(g)u(g)=u(g)$. In fact, if we restrict the identity $u(g_1gg_1^{-1})=\chi_0(g_1)^{-1}u(g)$ on the submanifold $\{(g_1,g)\in\mathbf{G}\times\mathbf{G}; g_1=g^{-1}\}$, then we obtain the above identity. Hence we have $u(g)=\chi_0(g)^{\lambda}u(g)$ for any $\lambda\in\mathbb{C}$. We shall define the isomorphism Φ: $\mathscr{F}(K)\to\tilde{\mathscr{F}}(K)$ by $(\Phi u)(x)=j(x)^{\frac{1}{2}}u(e^x)$. The above remark shows us $(\Phi u)(x)=\chi_0(e^x)^{\lambda}j(x)^{\frac{1}{2}}u(e^x)$ for any λ.

For any $\tilde{u}(x)$ in $\tilde{\mathscr{F}}(\mathfrak{g})$, we have $d\chi_0(x)\tilde{u}(x)=0$. In fact, by Lemma 1.1, we have $\langle[A,x],\partial_x\rangle\tilde{u}(x)=-d\chi_0(A)\tilde{u}(x)$ for any $A\in\mathfrak{g}$. Here, for any $\mathfrak{g}$-valued real analytic function $E(x)$ on $\mathfrak{g}$, $\langle E(x),\partial_x\rangle$ is the vector field defined by $\langle E(x),\partial_x\rangle u(x)=\dfrac{d}{dt}u(x+tE(x))|_{t=0}$. Thus, we have the identity $\langle[A,x],\partial_x\rangle\tilde{u}(x)=-d\chi_0(A)\tilde{u}(x)$ on $(x,A)\in\mathfrak{g}\times\mathfrak{g}$. If we restrict this on the submanifold $A=x$, we obtain $d\chi_0(x)\tilde{u}(x)=0$. These observations also show the following:

Let us denote by $\mathbf{G}_0$ the kernel of χ_0 and $\mathfrak{g}_0$ its Lie algebra. Then, $\mathbf{G}_0$ is a unimodular group. For any $u\in\mathscr{F}(\mathfrak{g})$, we can find an absolute invariant v on $\mathbf{G}_0$ such that $u=v\,\delta(\chi_0)$. Similarly, for any $\tilde{u}\in\tilde{\mathscr{F}}(\mathfrak{g})$, we can find an absolute invariant $\tilde{v}$ on $\mathfrak{g}_0$ such that $\tilde{u}=\tilde{v}\,\delta(d\chi_0)$. Thus we can reduce the study of $\mathscr{F}(\mathfrak{g})$ and $\tilde{\mathscr{F}}(\mathfrak{g})$ into the case where the group is unimodular, although we will not employ this fact.

§3. We Shall Prove Theorem

Take two closed cones K_1 and K_2 of $\mathfrak{g}$ such that $K_1 \cap (-K_2) = \{0\}$ and two functions u in $\mathscr{I}(K_1)$ and v in $\mathscr{I}(K_2)$. Set $w(g) = \int_G u(h)\, v(h^{-1} g)\, dh$, and $\tilde{u} = \Phi u$, $\tilde{v} = \Phi v$, $\tilde{w} = \Phi w$.

In order to prove Theorem we shall compute $\tilde{w}$.

$$\begin{aligned}\tilde{w}(z) &= j(z)^{\frac{1}{2}} \int_G u(h)\, v(h^{-1} e^z)\, dh \\ &= j(z)^{\frac{1}{2}} \int_{\mathfrak{g}} u(e^x)\, v(e^{-x} e^z)\, j(x)\, dx \\ &= j(z)^{\frac{1}{2}} \int_{\mathfrak{g}} dx \int_{\mathfrak{g}} dy\, u(e^x)\, v(e^y)\, j(x)\, \delta(y - \log e^{-x} e^z).\end{aligned}$$

Lemma 3.1. $\delta(y - \log e^{-x} e^z) = j(y) j(z)^{-1} \delta(z - \log e^x e^y)$.

Proof. We have $\delta(y - f(z)) = |Jf|^{-1} \delta(z - f^{-1}(y))$ where Jf is the Jacobian of f. Setting $f(z) = \log e^{-x} e^z$, we shall apply this. We have, for $a \in \mathfrak{g}$

$$f(z + \varepsilon a) = \log e^{-x} e^{z + \varepsilon a}$$

which equals $\log e^{-x} e^z \exp(\varepsilon(1 - e^{-\operatorname{ad} z})/\operatorname{ad} z)\ a$ modulo ε^2. As we can set $y = \log e^{-x} e^z$, this is equal to

$$\log e^y \exp(\varepsilon(1 - e^{-\operatorname{ad} z})/\operatorname{ad} z)\, a = y + \varepsilon \frac{\operatorname{ad} y}{1 - e^{-\operatorname{ad} y}} \frac{1 - e^{-\operatorname{ad} z}}{\operatorname{ad} z} a \quad \text{modulo } \varepsilon^2.$$

Thus we obtain $Jf = \det \dfrac{\operatorname{ad} y}{1 - e^{-\operatorname{ad} y}} \dfrac{1 - e^{-\operatorname{ad} z}}{\operatorname{ad} z}$, which implies the desired result. Q.E.D.

By this lemma, we have

$$\begin{aligned}(3.1)\quad \tilde{w}(z) &= \iint u(e^x)\, v(e^y)\, j(x) j(y) j(z)^{-\frac{1}{2}}\, \delta(z - \log e^x e^y)\, dx\, dy \\ &= \iint \left(\frac{j(x) j(y)}{j(z)}\right)^{\frac{1}{2}} \tilde{u}(x)\, \tilde{v}(y)\, \delta(z - \log e^x e^y)\, dx\, dy.\end{aligned}$$

We want to prove that this integral equals

$$(\tilde{u} * \tilde{v})(z) = \int \tilde{u}(x)\, \tilde{v}(y)\, \delta(z - x - y)\, dx\, dy.$$

Given a vector space V and two functions $\tilde{u}$ and $\tilde{v}$ on V, given a structure μ of Lie algebra on V, we want to prove for the Lie algebra $\mathfrak{g} = (V, \mu)$ the equality:

$$\int \left(\frac{j(x) j(y)}{j(z)}\right)^{\frac{1}{2}} \tilde{u}(x)\, \tilde{v}(y)\, \delta(z - \log e^x e^y)\, dx\, dy = \int \tilde{u}(x)\, \tilde{v}(y)\, \delta(z - x - y)\, dx\, dy.$$

If we consider the Lie algebra $\mathfrak{g}_t = (V, t\mu)$ i.e. $[x, y]_t = t[x, y]$, the first member of the equality becomes

$$(3.2)\quad \varphi_t(z) = \int \left(\frac{j(tx) j(ty)}{j(tz)}\right)^{\frac{1}{2}} \tilde{u}(x)\, \tilde{v}(y)\, \delta\left(z - \frac{1}{t} \log e^{tx} e^{ty}\right) dx\, dy,$$

and this must be equal to the second member which is the value of φ_t for $t=0$. Therefore it is enough to show that φ_t does not depend on t, or equivalently $\frac{\partial}{\partial t}\varphi_t=0$. Let us calculate this derivative.

Lemma 3.2. *Let $F(x,y)$ and $G(x,y)$ be two $\mathfrak{g}$-valued real analytic functions on $(x,y)\in\mathfrak{g}\times\mathfrak{g}$ defined in a neighborhood of the origin. Suppose that $F(0,0)=G(0,0)=0$ and that*

$$x+y-\log e^y e^x=(1-e^{-\operatorname{ad}x})F(x,y)+(e^{\operatorname{ad}y}-1)G(x,y).$$

Then, we have

$$\frac{\partial}{\partial t}\frac{1}{t}\log e^{tx}e^{ty}=\left(\left\langle\left[x,\frac{1}{t}F(tx,ty)\right],\partial_x\right\rangle+\left\langle\left[y,\frac{1}{t}G(tx,ty)\right],\partial_y\right\rangle\right)\frac{1}{t}\log e^{tx}e^{ty}. \tag{3.3}$$

Here $\langle A(x),\partial_x\rangle$ is the derivation defined by

$$(\langle A(x),\partial_x\rangle u)(x)=\frac{d}{d\varepsilon}u(x+\varepsilon A(x))|_{\varepsilon=0}.$$

Proof. Set $F_t=t^{-1}F(tx,ty)$ and $G_t=t^{-1}G(tx,ty)$. Then, the right hand side of (3.3) is the value of

$$t^{-1}\frac{d}{d\varepsilon}\log\exp(tx+\varepsilon[tx,F_t])\exp(ty+\varepsilon[ty,G_t])$$

at $\varepsilon=0$. We shall calculate

$$A=\exp(tx+\varepsilon[tx,F_t])\exp(ty+\varepsilon[ty,G_t])$$

modulo ε^2. We have

$$\begin{aligned}\exp(tx+\varepsilon[tx,F_t])&=e^{tx}\exp\varepsilon\frac{1-e^{-\operatorname{ad}tx}}{\operatorname{ad}(tx)}[tx,F_t]\\&=e^{tx}\exp\varepsilon(1-e^{-\operatorname{ad}tx})F_t\quad\text{modulo }\varepsilon^2,\end{aligned}$$

and similarly $\exp(ty+\varepsilon[ty,G_t])=\exp\varepsilon(e^{\operatorname{ad}ty}-1)G_t\exp ty$ modulo ε^2. Thus, we have

$$\begin{aligned}A&=e^{tx}\exp\varepsilon((1-e^{-\operatorname{ad}tx})F_t+(e^{\operatorname{ad}ty}-1)G_t)e^{ty}\\&=e^{tx}\exp\varepsilon\left(x+y-\frac{1}{t}\log e^{ty}e^{tx}\right)e^{ty}\\&=e^{(t+\varepsilon)x}\exp\varepsilon\left(y-\frac{1}{t}\log e^{ty}e^{tx}\right)e^{ty}\\&=e^{(t+\varepsilon)x}e^{ty}\exp\varepsilon\left(y-\frac{1}{t}\log e^{tx}e^{ty}\right)\\&=e^{(t+\varepsilon)x}e^{(t+\varepsilon)y}\exp-\varepsilon\left(\frac{1}{t}\log e^{tx}e^{ty}\right)\quad\text{modulo }\varepsilon^2.\end{aligned}$$

We have therefore

$$\log A = \log e^{(t+\varepsilon)x} e^{(t+\varepsilon)y} - \frac{\varepsilon}{t} \log e^{tx} e^{ty}$$

$$= \frac{t}{t+\varepsilon} \log e^{(t+\varepsilon)x} e^{(t+\varepsilon)y}.$$

This implies Lemma 3.2. Q.E.D.

This lemma shows in particular

$$(3.4) \quad \frac{\partial}{\partial t} \delta\left(z - \frac{1}{t} \log e^{tx} e^{ty}\right) = (\langle [x, F_t], \partial_x \rangle + \langle [y, G_t], \partial_y \rangle)\, \delta\left(z - \frac{1}{t} \log e^{tx} e^{ty}\right).$$

Therefore, integrating by parts, we have the equality

$$(3.5) \quad p_1 = \int \left(\frac{j(tx)j(ty)}{j(tz)}\right)^{\frac{1}{2}} \tilde{u}(x)\tilde{v}(y) \frac{\partial}{\partial t} \delta\left(z - \frac{1}{t} \log e^{tx} e^{ty}\right) dx\, dy$$

$$= -\int \left\{ (\langle [x, F_t], \partial_x \rangle + \langle [y, G_t], \partial_y \rangle + \mathrm{div}_x [x, F_t] \right.$$

$$\left. + \mathrm{div}_y [y, G_t]) \left(\frac{j(tx)j(ty)}{j(tz)}\right)^{\frac{1}{2}} \tilde{u}(x)\tilde{v}(y) \right\} \delta\left(z - \frac{1}{t} \log e^{tx} e^{ty}\right) dx\, dy.$$

Here div_x (resp. div_y) signifies the divergent with respect to the variable x (resp. y), i.e. the function $\mathrm{div}_x E(x)$ is the sum of the vector field $\langle E(x), \partial_x \rangle$ and its formal adjoint.

If a function $\varphi(x)$ satisfies $\varphi(\mathrm{Ad}(g)x) = \chi(g)\varphi(x)$ with a character $\chi(g)$, then we have

$$\langle [A, x], \partial_x \rangle \varphi = (\delta\chi)(A)\varphi(x) \quad \text{for } A \in \mathfrak{g}.$$

Here, $\delta\chi$ is the derivative of χ. Hence, if φ is an absolute invariant, φ and $\langle [A, x], \partial_x \rangle$ commute. Since $(j(x)j(y)/j(z))^{\frac{1}{2}}$ is an absolute invariant

$$\langle [x, F_t], \partial_x \rangle + \langle [y, G_t], \partial_y \rangle + \mathrm{div}_x [x, F_t] + \mathrm{div}_y [y, G_t]$$

commutes with this function. Since $\tilde{u}(x)$ is a relative invariant with respect to the character $|\det(\mathrm{Ad}(g); \mathfrak{g})^{-1}|$, we have

$$\langle [A, x], \partial_x \rangle \tilde{u}(x) = -\mathrm{tr}(\mathrm{ad} A)\tilde{u}(x).$$

Thus, we obtain

$$(3.6) \quad p_1 = -\int (\mathrm{tr}(\mathrm{ad}(F_t + G_t), \mathfrak{g}) + \mathrm{div}_x [x, F_t]$$

$$+ \mathrm{div}_y [y, G_t])(j(tx)j(ty)/j(tz))^{\frac{1}{2}} \tilde{u}(x)\tilde{v}(y)\, \delta\left(z - \frac{1}{t} \log e^{tx} e^{ty}\right) dx\, dy.$$

Lemma 3.3. $\dfrac{\partial}{\partial t} \log j(tx) = \mathrm{tr}\left(\dfrac{\mathrm{ad}\, x}{e^{t\,\mathrm{ad}\, x} - 1} - \dfrac{1}{t}\right).$

Proof.

$$\frac{\partial}{\partial t}\log\det\frac{1-e^{-\operatorname{ad}tx}}{\operatorname{ad}(tx)}=\operatorname{tr}\frac{\operatorname{ad}tx}{1-e^{-\operatorname{ad}tx}}\frac{\partial}{\partial t}\frac{1-e^{-\operatorname{ad}tx}}{\operatorname{ad}tx}$$
$$=\operatorname{tr}\left(\frac{\operatorname{ad}x}{e^{t\operatorname{ad}x}-1}-\frac{1}{t}\right).$$

By this lemma we have

$$\frac{\partial}{\partial t}\left(\frac{j(tx)j(ty)}{j(tz)}\right)^{\frac{1}{2}}=\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{e^{t\operatorname{ad}x}-1}+\frac{\operatorname{ad}y}{e^{t\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{t\operatorname{ad}z}-1}-\frac{1}{t}\right)\left(\frac{j(tx)j(ty)}{j(tz)}\right)^{\frac{1}{2}}.$$

We obtain finally

$$\frac{\partial}{\partial t}\varphi_t=-\int\left\{\operatorname{div}_x[x,F_t]+\operatorname{div}_y[y,G_t]+\operatorname{tr}\operatorname{ad}(F_t+G_t)\right.$$
$$\left.-\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{e^{t\operatorname{ad}x}-1}+\frac{\operatorname{ad}y}{e^{t\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{t\operatorname{ad}z}-1}-\frac{1}{t}\right)\right\}$$
$$\cdot\left(\frac{j(tx)j(ty)}{j(tz)}\right)^{\frac{1}{2}}\tilde{u}(x)\tilde{v}(y)\,\delta\left(z-\frac{1}{t}\log e^{tx}e^{ty}\right)dx\,dy.$$

In order to see that $\partial\varphi_t/\partial t$ vanishes, it is enough to show

$$\operatorname{div}_x[x,F_t]+\operatorname{div}_y[y,G_t]+\operatorname{tr}\operatorname{ad}(F_t+G_t)$$
$$-\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{e^{t\operatorname{ad}x}-1}+\frac{\operatorname{ad}y}{e^{t\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{t\operatorname{ad}z}-1}-\frac{1}{t}\right)=0 \tag{3.7}$$

when $z=\frac{1}{t}\log e^{tx}e^{ty}$. Since the left hand side of this formula is homogeneous of degree 1 when we assign degree -1 to t and degree 1 to x and y, it is enough to show (3.7) when $t=1$.

For a $\mathfrak{g}$-valued function $A(x)$, let us denote by $\partial_x A$ the endomorphism of $\mathfrak{g}$ defined by $\mathfrak{g}\ni a\mapsto\frac{d}{dt}A(x+ta)|_{t=0}$. Then $\operatorname{div}_x A(x)=\operatorname{tr}\partial_x A(x)$.

Since $\partial_x[x,A(x)]=(\operatorname{ad}x)\partial_x A-\operatorname{ad}A$, the formula (3.7) is equivalent to

$$\operatorname{tr}(\operatorname{ad}x)(\partial_x F)+\operatorname{tr}(\operatorname{ad}y)(\partial_y G)=\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{e^{\operatorname{ad}x}-1}+\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}-1\right) \tag{3.8}$$

with $z=\log e^x e^y$. This completes the proof of Theorem.

§4. Biinvariant Differential Operators

We consider the algebra $\mathbf{I}(\mathfrak{g})$ of the $\mathbf{G}$-invariant elements of $\mathbf{S}(\mathfrak{g})$. We identify $\mathbf{S}(\mathfrak{g})$ with the algebra of constant coefficient differential operators on $\mathfrak{g}$, hence $\mathbf{I}(\mathfrak{g})$ is identified with the ring of constant coefficient differential operators on $\mathfrak{g}$

invariant by the action of $\mathbf{G}$. We consider the universal enveloping algebra $\mathbf{U}(\mathfrak{g})$ of $\mathfrak{g}$ and its center $\mathbf{Z}(\mathfrak{g})$. We identify $\mathbf{U}(\mathfrak{g})$ with the algebra of the left invariant differential operators, hence $\mathbf{Z}(\mathfrak{g})$ will be identified with the ring of biinvariant differential operators on $\mathbf{G}$.

We denote by δ the Dirac function on $\mathbf{G}$ supported at the unit e, then $u*\delta=\delta*u=u$. On the other hand, we have $P(u*v)=u*Pv$ for $P\in\mathbf{U}(\mathfrak{g})$. This shows that $Pu=u*P\delta$. We shall denote by the same letter δ the Dirac function on $\mathfrak{g}$ supported at the origin. Similarly if $P\in\mathbf{S}(\mathfrak{g})$, $Pu=u*P\delta=P\delta*u$. We shall denote by $(\exp)^*$ (resp. $(\exp)_*$) the pull-back of functions or differential operators on $\mathbf{G}$ to those on $\mathfrak{g}$ (resp. the inverse of $(\exp)^*$), by the exponential map.

We shall denote by β the linear mapping from $\mathbf{S}(\mathfrak{g})$ onto $\mathbf{U}(\mathfrak{g})$ obtained by symmetrization. We have $(\beta(P)\varphi)(e)=(P\tilde{\varphi})(0)$ with $\tilde{\varphi}(x)=\varphi(e^x)$, hence $(\beta(P)\delta)(e^x)=j(-x)^{-1}(P\delta)(x)$.

For a real analytic function $f(x)$ on $\mathfrak{g}$ defined on a neighborhood of the origin, and $P\in\mathbf{S}(\mathfrak{g})$, we define

$$D(f)P\in\mathbf{S}(\mathfrak{g}) \quad \text{by} \quad ((D(f)P)\delta)(x)=f(-x)P\delta(x),$$

or

$$((D(f)P)\varphi)(0)=P(x\mapsto f(x)\varphi(x))(0).$$

We shall denote by γ the map from $\mathbf{I}(\mathfrak{g})$ onto $\mathbf{Z}(\mathfrak{g})$ defined by $P\mapsto\beta(D(j^{\frac{1}{2}})P)$.

Duflo [4] has proved that for any Lie algebra $\mathfrak{g}$, γ is an isomorphism of the rings $\mathbf{I}(\mathfrak{g})$ and $\mathbf{Z}(\mathfrak{g})$.

We have seen that for any $P\in\mathbf{I}(\mathfrak{g})$,

$$\chi_0(e^x)(P\delta)(x)=P\delta(x),$$

and hence $\chi_0(e^x)$ and P commute. In fact,

$$\chi_0(e^{x-y})(P\delta)(x-y)=(P\delta)(x-y)$$

and this implies

$$\chi_0(e^x)(P\delta)(x-y)=\chi_0(e^y)(P\delta)(x-y).$$

Let us denote by $\mathfrak{g}_0=\{A\in\mathfrak{g};\operatorname{tr}\operatorname{ad}A=0\}$, this implies that $P\in\mathbf{S}(\mathfrak{g}_0)$ (see also [3, 13]). In particular, we have $j(x)^{\frac{1}{2}}(P\delta)\ x)=j(-x)^{\frac{1}{2}}(P\delta)_{,}x)$, as $j(x)=(\det e^{-\operatorname{ad}x})j(-x)$. So we have $\Phi(\gamma(P)\delta)=P\delta$. If we take $v=\gamma(P)\delta$ then we can get from Theorem the following proposition.

Proposition 4.1. *If* Conjecture *is true for* $\mathfrak{g}$, *then for every* $\tilde{u}\in\tilde{\mathcal{J}}(\mathfrak{g})$ *and* $P\in\mathbf{I}(\mathfrak{g})$

$$((\exp)^*\gamma(P))\tilde{u}=(j(x)^{-\frac{1}{2}}Pj(x)^{\frac{1}{2}})\tilde{u}.$$

(*In particular* γ *is an isomorphism of the ring* $\mathbf{I}(\mathfrak{g})$ *and* $\mathbf{Z}(\mathfrak{g})$.)

However, we can get a more precise result applying the same method as in the preceeding paragraphs. Let us denote by $\mathbf{D}$ the ring of the germs of the differential operators at the origin.

Proposition 4.2. *Suppose that* Conjecture *is true for* $\mathfrak{g}$, *then for any* $P \in \mathbf{I}(\mathfrak{g})$

$$j(x)^{\frac{1}{2}}((\exp)^* \gamma(P))\, j(x)^{-\frac{1}{2}} - P$$

is contained in the left ideal of $\mathbf{D}$ *generated by the* $(\langle [A,x], \partial_x \rangle + \operatorname{tr} \operatorname{ad} A)$*'s* $(A \in \mathfrak{g})$.

(As we have $(\langle [A,x], \partial_x \rangle + \operatorname{tr} \operatorname{ad} A)\tilde{u}(x) = 0$ for every $\tilde{u} \in \tilde{\mathcal{F}}(\mathfrak{g})$, this implies Proposition 4.1.)

Proof. Remark that for $P \in \mathbf{S}(\mathfrak{g})$, $\exp^*(\beta(P))$ is the differential operator defined by

$$((\exp)^* \beta(P) u)(x) = P_y(u(\log e^x e^y))|_{y=0},$$

where P_y means that P operates on the y variable. Hence

$$Q = j(x)^{\frac{1}{2}}((\exp)^* \gamma(P))\, j(x)^{-\frac{1}{2}}$$

is the operator:

$$(Qu)(x) = P_y\left(\frac{j(x)^{\frac{1}{2}} j(y)^{\frac{1}{2}}}{j(\log e^x e^y)^{\frac{1}{2}}} u(\log e^x e^y)\right)\Bigg|_{y=0}.$$

As before we introduce the Lie algebra $\mathfrak{g}_t$ and the corresponding operator Q_t, then

$$(Q_t u)(x) = P_y\left(\frac{j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}}}{j(\log e^{tx} e^{ty})^{\frac{1}{2}}} u\left(\frac{1}{t} \log e^{tx} e^{ty}\right)\right)\Bigg|_{y=0}.$$

Let us remark that if we define the left ideal $\mathscr{L}_t$ of $\mathbf{D}$ generated by the element $\langle [x,A]_t, \partial_x \rangle + \operatorname{tr}(\operatorname{ad}_t A; \mathfrak{g}_t)$ then for $t \neq 0$ $\mathscr{L}_t = \mathscr{L}$. Hence we have to prove that: $Q_t - P \in \mathscr{L}$. As $Q_0 = P$, it is sufficient to prove that $\dfrac{\partial}{\partial t} Q_t \in \mathscr{L}$, where

$$\left(\left(\frac{\partial}{\partial t} Q_t\right) u\right)(x) = \frac{\partial}{\partial t}(Q_t u)(x)$$

$$= P_y\left(\frac{\partial}{\partial t} \frac{j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}}}{j(\log e^{tx} e^{ty})^{\frac{1}{2}}} u\left(\frac{1}{t} \log e^{tx} e^{ty}\right)\right)\Bigg|_{y=0}.$$

Let F and G be as in Lemma 3.2,

$$F_t(x,y) = \frac{F(tx,ty)}{t}, \qquad G_t(x,y) = \frac{G(tx,ty)}{t},$$

and

$$d(x,y,t) = \tfrac{1}{2} \operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{t \operatorname{ad} x} - 1} + \frac{\operatorname{ad} y}{e^{t \operatorname{ad} y} - 1} - \frac{\operatorname{ad} z}{e^{t \operatorname{ad} z} - 1} - \frac{1}{t}\right)$$

$$- \operatorname{tr}((\operatorname{ad} x)\, \partial_x F_t + (\operatorname{ad} y)\, \partial_y G_t)$$

where $z = \dfrac{\log e^{tx} e^{ty}}{t}$. Then we prove:

$$(4.1)\quad \frac{\partial}{\partial t}\left(j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j(\log e^{tx}e^{ty})^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx}e^{ty}\right)\right)$$

$$= d(x,y,t)\, j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j(\log e^{tx}e^{ty})^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx}e^{ty}\right)$$

$$+ \sum_{i=1}^{n} \alpha_i(x,y,t)((\langle [e_i,z],\partial_z\rangle + \operatorname{tr}\operatorname{ad} e_i)\cdot u)\left(\frac{1}{t}\log e^{tx}e^{ty}\right)$$

$$+ \sum_{i=1}^{n} \langle [y,e_j],\partial_y\rangle \cdot \beta_i(x,y,t)\, u\left(\frac{1}{t}\log e^{tx}e^{ty}\right).$$

Here, $e_i (i=1,2,\dots,n)$ is a basis of the Lie algebra $\mathfrak{g}$, $\langle [e_i,z],\partial_z\rangle$ denotes the adjoint field corresponding to e_i, and $\alpha_i(x,y,t)$, $\beta_i(x,y,t)$ are analytic functions defined near the origin.

To prove (4.1), we compute as in Lemma 3.2

$$\frac{1}{t+\varepsilon}\log e^{(t+\varepsilon)x}e^{(t+\varepsilon)y} \qquad \text{modulo } \varepsilon^2$$

$$= \frac{1}{t}\log e^{tx}e^{\varepsilon x}e^{ty}e^{\varepsilon y}e^{-\varepsilon\frac{\log e^{tx}e^{ty}}{t}}$$

$$= \frac{1}{t}\log e^{tx}e^{ty}e^{\varepsilon\left(e^{-t\operatorname{ad} y}\left(x+y-\frac{\log e^{ty}e^{tx}}{t}\right)\right)}$$

$$= \frac{1}{t}\log e^{tx}e^{ty}e^{\varepsilon(e^{-t\operatorname{ad} y}((1-e^{-t\operatorname{ad} x})F_t+(e^{t\operatorname{ad} y}-1)G_t))}.$$

We write

$$e^{-t\operatorname{ad} y}((1-e^{-t\operatorname{ad} x})F_t+(e^{t\operatorname{ad} y}-1)G_t)$$
$$= (1-e^{-\operatorname{ad}(\log e^{tx}e^{ty})})F_t + (1-e^{-t\operatorname{ad} y})(G_t-F_t).$$

So we have

$$\frac{d}{dt}\left(\frac{1}{t}\log e^{tx}e^{ty}\right) = \langle [z,F_t],\partial_z\rangle\left(\frac{1}{t}\log e^{tx}e^{ty}\right) + \langle [y,G_t-F_t],\partial_y\rangle\left(\frac{1}{t}\log e^{tx}e^{ty}\right)$$

(if $F_t(x,y)=\sum f_i(x,y,t)\,e_i$, and $\mathbf{I}(x)=x$

$$\langle [F_t,z],\partial_z\rangle\left(\frac{1}{t}\log e^{tx}e^{ty}\right) = \sum f_i(x,y)(\langle [e_i,z],\partial_z\rangle\cdot\mathbf{I})\left(\frac{1}{t}\log e^{tx}e^{ty}\right)\Big).$$

We write

$$\frac{\partial}{\partial t} j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j(\log e^{tx}e^{ty})^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx}e^{ty}\right)\Big|_{t=t_0}$$

$$= \frac{\partial}{\partial t} j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j\left(t\frac{\log e^{tx}e^{ty}}{t}\right)^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx}e^{ty}\right)\Big|_{t=t_0}$$

$$= \tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{t\operatorname{ad} x}-1}+\frac{\operatorname{ad} y}{e^{t\operatorname{ad} y}-1}-\frac{\operatorname{ad} z}{e^{t\operatorname{ad} z}-1}-\frac{1}{t}\right)$$

$$\cdot j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j(\log e^{tx} e^{ty})^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx} e^{ty}\right)$$

$$+ j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} \frac{\partial}{\partial t}\left(j\left(t_0 \frac{\log e^{tx} e^{ty}}{t}\right)^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx} e^{ty}\right)\right)\Big|_{t=t_0}$$

by Lemma 3.3.

Now if $(G_t - F_t)(x, y) = \sum \lambda_i(x, y, t)\, e_i$ we have

$$\langle [y, G_t - F_t], \partial_y \rangle = \sum_{i=1}^{n} \langle [y, e_i], \partial_y \rangle\, \lambda_i(x, y, t) - \operatorname{tr}\operatorname{ad} y\, \partial_y (G_t - F_t).$$

As j is an absolute invariant, j commutes with the adjoint fields.

Hence from the preceeding calculation, we obtain that the left hand side of (4.1) is equal to

$$\left(\frac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{t\operatorname{ad} x}-1}+\frac{\operatorname{ad} y}{e^{t\operatorname{ad} y}-1}-\frac{\operatorname{ad} z}{e^{t\operatorname{ad} z}-1}-\frac{1}{t}\right) - \operatorname{tr}\operatorname{ad} y\, \partial_y (G_t - F_t)\right)$$

$$\cdot j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j(\log e^{tx} e^{ty})^{-\frac{1}{2}} u\left(\frac{1}{t}\log e^{tx} e^{ty}\right)$$

$$+ j(tx)^{\frac{1}{2}} j(ty)^{\frac{1}{2}} j(\log e^{tx} e^{ty})^{-\frac{1}{2}} (\langle [z, F_t], \partial_z \rangle \cdot u)\left(\frac{1}{t}\log e^{tx} e^{ty}\right)$$

$$+ \sum_{i=1}^{n} \langle [y, e_i], \partial_y \rangle \cdot \left(\beta_i(x, y, t)\, u\left(\frac{1}{t}\log(e^{tx} e^{ty})\right)\right).$$

But, we have

$$\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{t\operatorname{ad} x}-1}+\frac{\operatorname{ad} y}{e^{t\operatorname{ad} y}-1}-\frac{\operatorname{ad} z}{e^{t\operatorname{ad} z}-1}-\frac{1}{t}\right) - \operatorname{tr}\operatorname{ad} y\, \partial_y (G_t - F_t)$$

$$= d(x, y, t) + \operatorname{tr}(\operatorname{ad} y\, \partial_y F_t + \operatorname{ad} x\, \partial_x F_t).$$

Let us remark here that if E is in $\hat{\mathbf{L}}$, we have $g \cdot E(x, y) = E(gx, gy)$ for every $g \in \mathbf{G}$. The operator $(\partial_x E)\operatorname{ad} x + (\partial_y E)\operatorname{ad} y$ is the linear operator

$$c \mapsto \frac{d}{d\varepsilon} E(x + \varepsilon[x, c], y + \varepsilon[y, c])|_{\varepsilon=0}$$

$$= \frac{d}{d\varepsilon} E(\exp \varepsilon c \cdot x, \exp \varepsilon c \cdot y)|_{\varepsilon=0}$$

$$= \frac{d}{d\varepsilon} \exp \varepsilon c \cdot E(x, y)|_{\varepsilon=0}$$

$$= -[E(x, y), c]$$

hence is the operator $-\operatorname{ad} E$.

We then obtain that the left side of (4.1) is equal to

$$d(x,y,t)\,j(tx)^{\frac{1}{2}}j(ty)^{\frac{1}{2}}j(\log e^{tx}e^{ty})^{-\frac{1}{2}}\,u\left(\frac{1}{t}\log e^{tx}e^{ty}\right)$$

$$-j(tx)^{\frac{1}{2}}j(ty)^{\frac{1}{2}}j(\log e^{tx}e^{ty})^{-\frac{1}{2}}((\langle[F_t,z],\partial_z\rangle+\operatorname{tr}\operatorname{ad}F_t)\cdot u)\left(\frac{1}{t}\log e^{tx}e^{ty}\right)$$

$$+\sum_{i=1}^{n}\langle[y,e_i],\partial_y\rangle\,\beta_i(x,y,t)\,u\left(\frac{1}{t}\log(e^{tx}e^{ty})\right),$$

which is of the required form.

Now if our conjecture is true for $\mathfrak{g}$, then we can find F and G such that $d(x,y,t)=0$. Now we remark that if $P\in\mathbf{I}(\mathfrak{g})$,

$$P_y\langle[y,e_i],\partial_y\rangle=\langle[y,e_i],\partial_y\rangle P_y$$

hence $(P_y\langle[y,e_i],\partial_y\rangle\,\psi(y))|_{y=0}=0$. Let $R_i(t)$ denote the differential operator

$$(R_i(t)\varphi)(x)=P_y\left(\alpha_i(x,y,t)\,\varphi\left(\frac{1}{t}\log e^{tx}e^{ty}\right)\right)\Big|_{y=0}.$$

We obtain from (4.1)

$$\frac{\partial}{\partial t}Q_t=\sum_{i=1}^{n}R_i(t)(\langle[e_i,x],\partial_x\rangle+\operatorname{tr}\operatorname{ad}e_i),$$

i.e. $\dfrac{\partial}{\partial t}Q_t\in\mathscr{L}$. Q.E.D.

Remark. The same proof shows the corresponding fact for biinvariant integral operators.

Remark. We will see in the next section that our conjecture is true for **G** solvable; we can easily deduce from Proposition 4.1, the fact that every biinvariant operator on **G** is locally solvable, which was already obtained by Rouvière [14] and Duflo-Raïs [5]. In fact P being invariant by the action of **G** we can find a fundamental solution for P, which is invariant by **G**. It follows that $(\exp)^*\gamma(P)$ has a local fundamental solution. If **G** is exponential solvable, the maps F and G can be constructed in the whole space $\mathfrak{g}$ hence the Propositions 4.1 and 4.2 hold on the whole space $\mathfrak{g}$. So $\exp^*(j(P))$ has a fundamental solution on the space G, (Weita Chang [2] has proven recently that every biinvariant operator on an simply connected solvable group is globally solvable). We recall that M. Duflo has shown that every biinvariant differential operator on a Lie group **G** is locally solvable [4].

§ 5. Proof of Proposition 0

First we shall translate our conjecture into another form. Let us write for an $A\in\hat{\mathbf{L}}$

$$2(x+y-\log e^y e^x)=((x+y-\log e^y e^x)+A)+(x+y-\log e^y e^x)-A.$$

Hence we will consider $A \in \hat{\mathbf{L}}$ such that $(x+y-\log e^y e^x)+A$ is divisible by x (i.e. in $[x, \hat{\mathbf{L}}]$) and $(x+y-\log e^y e^x)-A$ is divisible by y (i.e. in $[y, \hat{\mathbf{L}}]$). As $x+y-\log e^y e^x \equiv \frac{1}{2}[x, y]$ mod $[[\hat{\mathbf{L}}, \hat{\mathbf{L}}], \hat{\mathbf{L}}]$ and $[x, y]$ is divisible by x and y, we may take A in $[[\hat{\mathbf{L}}, \hat{\mathbf{L}}], \hat{\mathbf{L}}]$. We will write $x+y-\log e^y e^x+A=[x, P]$, $A-(x+y-\log e^y e^x)=[y, Q]$, choose $F=\frac{1}{2}\frac{\operatorname{ad} x}{1-e^{-\operatorname{ad} x}}P$, $G=-\frac{1}{2}\frac{\operatorname{ad} y}{e^{\operatorname{ad} y}-1}Q$ and translate our conjecture in terms of A.

We shall first give two preliminary lemmata.

Lemma 5.1.

i) $$\partial_x \log e^x e^y = \frac{\operatorname{ad} z}{e^{\operatorname{ad} z}-1}\frac{e^{\operatorname{ad} x}-1}{\operatorname{ad} x}$$

and

ii) $$\partial_y \log e^x e^y = \frac{\operatorname{ad} z}{1-e^{-\operatorname{ad} z}}\frac{1-e^{-\operatorname{ad} y}}{\operatorname{ad} y}.$$

Here $z=\log e^x e^y$.

Proof. We have, modulo ε^2,

$$\log e^{(x+\varepsilon a)} e^y = \log e^{\varepsilon \frac{e^{\operatorname{ad} x}-1}{\operatorname{ad} x} a} e^x e^y = \log e^{\varepsilon \frac{e^{\operatorname{ad} x}-1}{\operatorname{ad} x} a} e^z$$
$$= z+\varepsilon \frac{\operatorname{ad} z}{e^{\operatorname{ad} z}-1}\frac{e^{\operatorname{ad} x}-1}{\operatorname{ad} x} a.$$

The formula ii) is shown in the same way. Q.E.D.

Lemma 5.2. *Let* $a \in \mathfrak{g}$, $f(\lambda)$ *and* $g(\lambda)$ *two power series on* λ. *Then*

$$\operatorname{tr}(f(\operatorname{ad} x)\partial_x(g(\operatorname{ad} x)a)) = \operatorname{tr}\left(f(\operatorname{ad} x)\frac{g(0)-g(\operatorname{ad} x)}{\operatorname{ad} x}\operatorname{ad} a; \mathfrak{g}\right).$$

Proof. By linearity, we may assume $g(\lambda)=\lambda^n$. If $n=0$, the lemma is evident. Suppose $n \geqq 1$. Then we have

$$\operatorname{ad}(x+\varepsilon c)^n a - (\operatorname{ad} x)^n a = \varepsilon \sum_{k=0}^{n-1} (\operatorname{ad} x)^{n-1-k}(\operatorname{ad} c)(\operatorname{ad} x)^k a$$
$$= -\varepsilon \sum_{k=0}^{n-1} (\operatorname{ad} x)^{n-1-k} \operatorname{ad}((\operatorname{ad} x)^k a)c \quad \text{modulo } \varepsilon^2.$$

Thus we have

$$\partial_x(g(\operatorname{ad} x)a) = -\sum_{k=0}^{n-1} (\operatorname{ad} x)^{n-1-k} \operatorname{ad}((\operatorname{ad} x)^k a).$$

If $k>0$, $\operatorname{tr} f(\operatorname{ad} x)(\operatorname{ad} x)^{n-1-k} \operatorname{ad}((\operatorname{ad} x)^k a)$ vanishes. In fact, if we set $b=(\operatorname{ad} x)^{k-1} a$ and $\varphi(\lambda)=\lambda^{n-1-k} f(\lambda)$, then

$$\operatorname{tr} \varphi(\operatorname{ad} x) \operatorname{ad}((\operatorname{ad} x)b) = \operatorname{tr} \varphi(\operatorname{ad} x)(\operatorname{ad} x \operatorname{ad} b - \operatorname{ad} b \operatorname{ad} x) = 0.$$

Therefore, we obtain

$$\operatorname{tr} f(\operatorname{ad} x)\partial_x g(\operatorname{ad} x)a = -\operatorname{tr} f(\operatorname{ad} x)(\operatorname{ad} x)^{n-1}(\operatorname{ad} a). \quad \text{Q.E.D.}$$

Proposition 5.3. Conjecture *is implied from the following: For any Lie algebra* $\mathfrak{g}$, *we can find* A *in* $[[\hat{\mathfrak{L}}, \hat{\mathfrak{L}}], \hat{\mathfrak{L}}]$ *satisfying the conditions* i), ii) *and* iii):

i) *There is* P *in* $\hat{\mathfrak{L}}$ *such that* $A+x+y-\log e^y e^x=[x,P]$ *and that* P *gives a convergent power series on* $(x,y)\in\mathfrak{g}\times\mathfrak{g}$.

ii) *There is* Q *in* $\hat{\mathfrak{L}}$ *such that* $A-(x+y-\log e^y e^x)=[y,Q]$ *and that* Q *gives a convergent power series on* $(x,y)\in\mathfrak{g}\times\mathfrak{g}$.

iii) $$\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x A-\operatorname{tr}\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}\partial_y A=\operatorname{tr}\left(\frac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}-1+\tfrac{1}{2}\operatorname{ad}z\right),$$

where $z=\log e^x e^y$.

Proof. We have $x+y-\log e^y e^x=\frac{1}{2}[x,P]-\frac{1}{2}[y,Q]$. Let $F=\frac{1}{2}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}P$ and $G=-\frac{1}{2}\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}Q$. Then we have

$$x+y-\log e^y e^x=(1-e^{-\operatorname{ad}x})F+(e^{\operatorname{ad}y}-1)G.$$

We have $[x,P]=2(1-e^{-\operatorname{ad}x})F$. Therefore, by Lemma 5.2, we have

$$\begin{aligned}\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x[x,P]&=2\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\frac{0-(1-e^{-\operatorname{ad}x})}{\operatorname{ad}x}\operatorname{ad}F+2\operatorname{tr}(\operatorname{ad}x)\partial_x F\\&=2\operatorname{tr}(\operatorname{ad}x)\partial_x F-2\operatorname{tr}\operatorname{ad}F.\end{aligned}$$

Similarly, we have $-\operatorname{tr}\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}\partial_y[y,Q]=2\operatorname{tr}(\operatorname{ad}y)\partial_y G-2\operatorname{tr}\operatorname{ad}G$. Set $\tilde{z}=\log e^y e^x$, we have, by Lemma 5.1

$$\partial_x[x,P]=\partial_x(x+y-\log e^y e^x+A)=1-\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\frac{1-e^{-\operatorname{ad}x}}{\operatorname{ad}x}+\partial_x A.$$

Hence, we obtain

$$\begin{aligned}\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x[x,P]&=\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x A+\operatorname{tr}\left(\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}-\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\right)\\&=\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x A+\operatorname{tr}\left(\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}-\frac{\operatorname{ad}z}{1-e^{-\operatorname{ad}z}}\right).\end{aligned}$$

In the same way, we have

$$-\operatorname{tr}\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}\partial_y[y,Q]=-\operatorname{tr}\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}\partial_y A+\operatorname{tr}\left(\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}\right).$$

Thus, we obtained

$$\begin{aligned}&\operatorname{tr}(\operatorname{ad}x)(\partial_x F)+\operatorname{tr}(\operatorname{ad}y)(\partial_y G)\\&\quad=\operatorname{tr}(\operatorname{ad}F)+\operatorname{tr}(\operatorname{ad}G)+\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x A-\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}\partial_y A\right)\\&\qquad+\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}+\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}-\frac{\operatorname{ad}z}{1-e^{-\operatorname{ad}z}}\right)\end{aligned}$$

$$=\operatorname{tr}(\operatorname{ad}F)+\operatorname{tr}(\operatorname{ad}G)+\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}+\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{1-e^{-\operatorname{ad}z}}-1+\tfrac{1}{2}\operatorname{ad}z\right).$$

Since $\lambda/(1-e^{-\lambda})=\lambda/(e^{\lambda}-1)+\lambda$ and $\operatorname{tr}\operatorname{ad}z=\operatorname{tr}(\operatorname{ad}x+\operatorname{ad}y)$, this equals

$$\operatorname{tr}(\operatorname{ad}F)+\operatorname{tr}(\operatorname{ad}G)+\tfrac{1}{4}\operatorname{tr}(\operatorname{ad}x-\operatorname{ad}y)+\tfrac{1}{2}\operatorname{tr}\left(\frac{\operatorname{ad}x}{e^{\operatorname{ad}x}-1}+\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}-\frac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}-1\right).$$

Hence, it is enough to show that

(5.1) $\operatorname{tr}(\operatorname{ad}F)+\operatorname{tr}(\operatorname{ad}G)=\tfrac{1}{4}\operatorname{tr}(\operatorname{ad}y-\operatorname{ad}x)$.

However, adding a constant multiple of x (resp. y) to P (resp. Q), we may assume that P (resp. Q) is equal to αy (resp. βx) modulo $[\hat{\mathbf{L}},\hat{\mathbf{L}}]$. However $x+y-\log e^{y}e^{x}\equiv-\frac{1}{2}[x,y]$ modulo $[[\hat{\mathbf{L}},\hat{\mathbf{L}}],\hat{\mathbf{L}}]$ and hence $P\equiv\frac{1}{2}y$ (resp. $Q\equiv\frac{1}{2}x$). Thus, we have $F\equiv\frac{1}{4}y$ (resp. $G=-\frac{1}{4}x$) modulo $[\hat{\mathbf{L}},\hat{\mathbf{L}}]$. Since $\operatorname{tr}\operatorname{ad}[\hat{\mathbf{L}},\hat{\mathbf{L}}]=0$, (5.1) is satisfied. Q.E.D.

Let A satisfy i), ii), iii), of the Proposition 4.3. We may remark that $A'(x,y)=\frac{1}{4}(A(x,y)-A(y,x)-A(-x,-y)+A(-y,-x))$ satisfies also 1), 2), and 3). This follows from the following observations:

a) if $m(x,y)=x+y-\log e^{y}e^{x}$, then $m(x,y)=-m(-y,-x)$;

$$\begin{aligned}m(x,y)-m(y,x)&=\log e^{x}e^{y}-\log e^{y}e^{x}\\&=(e^{\operatorname{ad}x}-1)\log e^{y}e^{x}\\&=(1-e^{\operatorname{ad}y})\log e^{x}e^{y}\end{aligned}$$

hence is divisible by x and y.

b) if $t(x,y)=\operatorname{tr}\left(\dfrac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}-1+\dfrac{\operatorname{ad}z}{2}\right)$ then $t(x,y)=t(y,x)=t(-x,-y)$.

c) for any $E\in[\hat{\mathbf{L}},\hat{\mathbf{L}}]$,

$$\operatorname{tr}\frac{\operatorname{ad}x}{1-e^{-\operatorname{ad}x}}\partial_x E-\operatorname{tr}\frac{\operatorname{ad}y}{e^{\operatorname{ad}y}-1}\partial_y E=\operatorname{tr}\frac{\operatorname{ad}x}{e^{\operatorname{ad}x}-1}\partial_x E-\operatorname{tr}\frac{\operatorname{ad}y}{1-e^{-\operatorname{ad}y}}\partial_y E.$$

In fact the difference is

$$\begin{aligned}\operatorname{tr}(\operatorname{ad}x\,\partial_x E+\operatorname{ad}y\,\partial_y E)&=\operatorname{tr}(\partial_x E\operatorname{ad}x+\partial_y E\operatorname{ad}y)\\&=-\operatorname{tr}\operatorname{ad}E(x,y)\quad\text{(see 4.2)}\\&=0\quad\text{as } E\in[\hat{\mathbf{L}},\hat{\mathbf{L}}].\end{aligned}$$

We will now construct A in $[[\hat{\mathbf{L}},\hat{\mathbf{L}}],\hat{\mathbf{L}}]$ such that

$$A(x,y)=-A(y,x)=-A(-x,-y)$$

and i) $x+y-\log e^{y}e^{x}+A(x,y)=[x,P]$ and P gives a convergent power series on $(x,y)\in\mathfrak{g}\times\mathfrak{g}$. ((ii) follows then). If $\mathfrak{g}$ is solvable we will be able to prove that A satisfies also the condition iii).

We consider now the condition i):

$$x+y-\log e^y e^x+A(x,y)=[x,P(x,y)].$$

Then for every t, we will have

$$tx+ty-\log e^{ty}e^{tx}+A(tx,ty)=t[x,P(tx,ty)].$$

Hence $\frac{\partial}{\partial t}(tx+ty-\log e^{ty}e^{tx})+\frac{\partial}{\partial t}A(tx,ty)\in[x,\hat{\mathbf{L}}]$ and $\frac{\partial}{\partial t}A(tx,ty)$ satisfies the same antisymmetry relation as A.

Let θ be the vector field $\langle x,\partial_x\rangle+\langle y,\partial_y\rangle$ (or the derivation of $\hat{\mathbf{L}}$ defined by $\theta|\mathbf{L}_n=n\,\mathrm{id}\,\mathbf{L}_n$ where $\mathbf{L}_n$ is the space of elements of $\mathbf{L}$ of degree n) then $t\frac{\partial}{\partial t}B(tx,ty)|_{t=1}=\theta B$, for $B\in\hat{\mathbf{L}}$. We compute

$$\theta(x+y-\log e^y e^x)=x+y-\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot y-\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\cdot x$$

with $\tilde{z}=\log e^y e^x$ and we will write $\theta(x+y-\log e^y e^x)$ as an antisymmetric element mod $[x,\hat{\mathbf{L}}]$.

For any real analytic function $g(\lambda)$, we have $g(\operatorname{ad}z)=e^{\operatorname{ad}x}g(\operatorname{ad}\tilde{z})e^{-\operatorname{ad}x}$, in particular $g(\operatorname{ad}\tilde{z})\cdot x\equiv g(\operatorname{ad}z)\cdot x$ modulo $[x,\hat{L}]$ and

$$g(\operatorname{ad}z)\cdot y\equiv g(\operatorname{ad}\tilde{z})e^{-\operatorname{ad}x}y\equiv g(\operatorname{ad}\tilde{z})e^{-\operatorname{ad}\tilde{z}}\cdot y \quad \text{modulo } [x,\hat{L}].$$

Hence we write modulo $[x,\hat{L}]$

$$\begin{aligned}\theta(x+y-\log e^y e^x)&=\left(1-\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\right)\cdot x+y-\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}e^{-\operatorname{ad}\tilde{z}}\cdot y\\&\equiv f(\operatorname{ad}\tilde{z})\cdot x+f(\operatorname{ad}z)\cdot y,\quad \text{where}\quad f(\lambda)=\left(1-\frac{\lambda}{1-e^{-\lambda}}\right)\\&\equiv f(\operatorname{ad}\tilde{z})\cdot x-f(\operatorname{ad}z)\cdot y+2f(\operatorname{ad}z)\cdot y.\end{aligned}$$

We write, as $f(0)=0$,

$$\begin{aligned}f(\operatorname{ad}z)\cdot y&=\frac{f(\operatorname{ad}z)}{e^{\operatorname{ad}z}-1}(e^{\operatorname{ad}z}-1)y\\&=\frac{f(\operatorname{ad}z)}{e^{\operatorname{ad}z}-1}(e^{\operatorname{ad}x}-1)y,\end{aligned}$$

therefore $f(\operatorname{ad}z)\cdot y\equiv\left(\frac{f(\operatorname{ad}\tilde{z})}{e^{\operatorname{ad}\tilde{z}}-1}-\frac{f(\operatorname{ad}z)}{e^{\operatorname{ad}z}-1}\right)\cdot y.$

As $\left(\frac{f(\operatorname{ad}\tilde{z})}{e^{\operatorname{ad}\tilde{z}}-1}-\frac{f(\operatorname{ad}z)}{e^{\operatorname{ad}z}-1}\right)\cdot x\equiv 0$ we obtain that

$$\theta(x+y-\log e^y e^x)\equiv f(\operatorname{ad}z)\cdot x-f(\operatorname{ad}z)\cdot y+2\left(\frac{f(\operatorname{ad}\tilde{z})}{e^{\operatorname{ad}\tilde{z}}-1}-\frac{f(\operatorname{ad}z)}{e^{\operatorname{ad}z}-1}\right)\cdot(x+y).$$

Let us denote by $\alpha(x, y)$ the second member of this equality. We have obviously $\alpha(x, y) = -\alpha(y, x)$, hence if we define $\beta(x, y) = \frac{1}{2}(\alpha(x, y) + \alpha(-y, -x))$, β will satisfy the relation $\beta(x, y) = -\beta(y, x) = -\beta(-x, -y)$ and $\theta(x+y-\log e^y e^x) \equiv \beta(x, y)$ mod $[x, \hat{L}]$. We remark that the function $h(\lambda) = \left(1 - \frac{\lambda}{1-e^{-\lambda}}\right) \frac{1}{e^\lambda - 1}$ verifies $h(\lambda) = -h(-\lambda) - 1$ as $\frac{1}{1-e^{-\lambda}} = \frac{1}{e^\lambda - 1} + 1$, hence

$$\beta(x, y) = 2\left(\frac{f(\operatorname{ad}\tilde{z})}{e^{\operatorname{ad}\tilde{z}} - 1} - \frac{f(\operatorname{ad} z)}{e^{\operatorname{ad} z} - 1}\right) \cdot (x+y) + \tfrac{1}{2}(f(\operatorname{ad}\tilde{z}) + f(-\operatorname{ad} z)) \cdot x$$
$$- \tfrac{1}{2}(f(\operatorname{ad} z) + f(-\operatorname{ad}\tilde{z})) \cdot y.$$

We can therefore define $A(x, y)$ by the differential equation:

$$\theta A = 2\left(1 - \frac{\operatorname{ad} z}{1-e^{-\operatorname{ad} z}}\right) \frac{1}{e^{\operatorname{ad} z} - 1}(x+y) - 2\left(1 - \frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\right) \frac{1}{e^{\operatorname{ad}\tilde{z}} - 1} \cdot (x+y)$$
$$+ \tfrac{1}{2}\left(\frac{\operatorname{ad} z}{e^{\operatorname{ad} z} - 1} - 1\right) \cdot x + \tfrac{1}{2}\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}} - 1\right) \cdot x - \tfrac{1}{2}\left(\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}} - 1} - 1\right) \cdot y$$
$$- \tfrac{1}{2}\left(\frac{\operatorname{ad} z}{1-e^{-\operatorname{ad} z}} - 1\right) \cdot y, \tag{5.2}$$

with the initial condition $A(0, 0) = 0$ ($\tilde{z} = \log e^y e^x$, $z = \log e^x e^y$). As the second member is a convergent power series at the origin, so is $A(x, y)$.

The preceding calculation implies now 1) and 2) of the:

Lemma 5.4.

1) $A(x, y) = -A(y, x) = -A(-x, -y)$,
2) $x + y - \log e^y e^x + A \in [x, \hat{L}]$,
3) $A \in [\hat{L}, [\hat{L}, \hat{L}]]$.

For 3) we remark that $A \in [\hat{L}, \hat{L}]$, and the properties $A(x, y) = -A(-x, -y)$ implies that $A \in [\hat{L}, [\hat{L}, \hat{L}]]$. The lemma is proven. Q.E.D.

Let q be a power series of the two non commutative variables x and y, i.e. q is in the completion of the tensor algebra $\hat{T}(x, y)$ of the vector space $\mathbb{C}x + \mathbb{C}y$. We denote by $c(q)$ the image of q under the map $\hat{T}(x, y) \to \hat{S}(x, y) = C[[x, y]]$, i.e. $c(q)$ is a power series in the commutative variables x and y.

Lemma 5.5. *If* $\mathfrak{g}$ *is solvable,* $\operatorname{tr}(q(\operatorname{ad} x, \operatorname{ad} y))$ *depends only on* $c(q)$.

Proof. There is a basis of $\mathfrak{g}^{\mathbb{C}}$ where the operators $\operatorname{ad} x, \operatorname{ad} y$ are lower triangular, then $\operatorname{ad}[x, y] = \operatorname{ad} x \operatorname{ad} y - \operatorname{ad} y \operatorname{ad} x$ have zeros on the diagonal, and the lemma follows.

Let us write $A = p(\operatorname{ad} x, \operatorname{ad} y) \cdot [x, y]$, where p is a convergent power series in the non commutative variables x and y.

Lemma 5.6. *Let* $\mathfrak{g}$ *be solvable, then*

$$\operatorname{tr} \frac{\operatorname{ad} x}{1-e^{-\operatorname{ad} x}} \partial_x A - \operatorname{tr} \frac{\operatorname{ad} y}{e^{\operatorname{ad} y} - 1} \partial_y A$$

$$= -\operatorname{tr}\left((e^{\operatorname{ad} z}-1)\left(\frac{\operatorname{ad} x}{e^{\operatorname{ad} x}-1}\right)\left(\frac{\operatorname{ad} y}{e^{\operatorname{ad} y}-1}\right)p(\operatorname{ad} x, \operatorname{ad} y)\right).$$

Proof. Let us consider the endomorphism

$$\mathfrak{g} \ni c \mapsto \frac{d}{d\varepsilon} p(\operatorname{ad} x + \varepsilon \operatorname{ad} c, \operatorname{ad} y)\cdot[x, y]|_{\varepsilon=0};$$

this is a sum of terms of the form

$$p_1(\operatorname{ad} x, \operatorname{ad} y)\operatorname{ad} c\, p_2(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y]$$
$$= -p_1(\operatorname{ad} x, \operatorname{ad} y)\operatorname{ad}(p_2(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y])\cdot c.$$

The trace of the endomorphism $\frac{\operatorname{ad} x}{1-e^{-\operatorname{ad} x}} p_1(\operatorname{ad} x, \operatorname{ad} y)\operatorname{ad}(p_2(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y])$ vanishes by the preceding lemma. So the only term appearing in $\operatorname{tr}\frac{\operatorname{ad} x}{1-e^{-\operatorname{ad} x}}\partial_x A$ will come from the trace of the endomorphism

$$c \mapsto \frac{d}{d\varepsilon}\frac{\operatorname{ad} x}{1-e^{-\operatorname{ad} x}} p(\operatorname{ad} x, \operatorname{ad} y)[x+\varepsilon c, y]|_{\varepsilon=0}.$$

We obtain that the left side of the equality is:

$$-\operatorname{tr}\left(\frac{\operatorname{ad} x}{1-e^{-\operatorname{ad} x}}\operatorname{ad} y + \frac{\operatorname{ad} y}{e^{\operatorname{ad} y}-1}\operatorname{ad} x\right) p(\operatorname{ad} x, \operatorname{ad} y)$$
$$= -\operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{\operatorname{ad} x}-1}\right)\left(\frac{\operatorname{ad} y}{e^{\operatorname{ad} y}-1}\right)(e^{\operatorname{ad} z}-1)\, p(\operatorname{ad} x, \operatorname{ad} y).$$

If we restrict our attention when $\mathfrak{g}$ is solvable, we have to prove:

$$-\operatorname{tr}\left(\frac{\operatorname{ad} x}{e^{\operatorname{ad} x}-1}\frac{\operatorname{ad} y}{e^{\operatorname{ad} y}-1}(e^{\operatorname{ad} z}-1)\, p(\operatorname{ad} x, \operatorname{ad} y)\right) = \operatorname{tr}\left(\frac{\operatorname{ad} z}{e^{\operatorname{ad} z}-1} - 1 + \tfrac{1}{2}\operatorname{ad} z\right).$$

Hence, considering the commutative ring $\mathbb{C}[[x, y]]$ we need only to prove:

$$\mathbf{c}(p)(x, y) = \left(1 - \frac{x+y}{2} - \frac{x+y}{e^{x+y}-1}\right)\frac{1}{e^{x+y}-1}\frac{e^x-1}{x}\frac{e^y-1}{y}.$$

We denote by $q(x, y)$ the right hand side.

Let us consider the homomorphism $\mathrm{h}: [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}] \to [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}]/[[\hat{\mathfrak{L}}, \hat{\mathfrak{L}}], [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}]]$ and let us write for $m \in [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}]$, $m = \varphi(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y]$ then clearly $\mathbf{h}(m)$ depends only on $\mathbf{c}(\varphi)$. Therefore, for $f(x, y) \in \mathbb{C}[[x, y]]$, we shall write $f(\operatorname{ad} x, \operatorname{ad} y)[x, y]$ for the element $\varphi(\operatorname{ad} x, \operatorname{ad} y)[x, y]$ modulo $[[\hat{\mathfrak{L}}, \hat{\mathfrak{L}}], [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}]]$ with $f = \mathbf{c}(\varphi)$.

Remark 5.7. If $f(x, y) \in \mathbb{C}[[x, y]]$ is such that $f(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y] \equiv 0$ modulo $[[\hat{\mathfrak{L}}, \hat{\mathfrak{L}}], [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}]]$, then $f(x, y) = 0$. In fact if $\varphi(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y] \in [[\hat{\mathfrak{L}}, \hat{\mathfrak{L}}], [\hat{\mathfrak{L}}, \hat{\mathfrak{L}}]]$, with $f = c(\varphi)$ then $\operatorname{tr}(\partial_x \varphi(\operatorname{ad} x, \operatorname{ad} y)\cdot[x, y]; \mathfrak{g}) = 0$ for any solvable Lie algebra $\mathfrak{g}$.

On the other hand the same calculation as in Lemma 5.6 shows that

$$\mathrm{tr}\,(\partial_x(\varphi(\mathrm{ad}\,x, \mathrm{ad}\,y)\cdot[x, y]); \mathfrak{g}) = -\mathrm{tr}\,(\varphi(\mathrm{ad}\,x, \mathrm{ad}\,y)\,\mathrm{ad}\,y; \mathfrak{g}).$$

Considering the 2 dimension Lie algebra $\mathfrak{g}$ with basis H, A and relation $[H, A] = A$, we have for $x = x_1 H + x_2 A$, $y = y_1 H + y_2 A$,

$$\mathrm{tr}\,(\varphi(\mathrm{ad}\,x, \mathrm{ad}\,y)\,\mathrm{ad}\,y; \mathfrak{g}) = f(x_1, y_1)\,y_1,$$

hence $f(x_1, y_1)\,y_1 = 0$, and so is f.

Proposition 0 will result from the following lemma.

Lemma 5.8. *Let*

$$\alpha = \left(1 - \frac{\mathrm{ad}\,\tilde{z}}{e^{\mathrm{ad}\,\tilde{z}} - 1}\right)\frac{1}{\mathrm{ad}\,\tilde{z}}\cdot(x + y - \tilde{z}) + \tfrac{1}{2}\tilde{z} - \left(1 - \frac{\mathrm{ad}\,z}{e^{\mathrm{ad}\,z} - 1}\right)\frac{1}{\mathrm{ad}\,z}\cdot(x + y - z) - \tfrac{1}{2}z$$

then

1) $\mathbf{h}(\alpha) = q(\mathrm{ad}\,x, \mathrm{ad}\,y)\cdot[x, y]$,
2) $\mathbf{h}(\alpha) = \mathbf{h}(A)$.

Proof. 1) We have as $(x + y - \tilde{z}) \in [\hat{\mathbf{L}}, \hat{\mathbf{L}}]$,

$$\begin{aligned}\alpha &\equiv \left(1 - \frac{\mathrm{ad}\,z}{e^{\mathrm{ad}\,z} - 1}\right)\frac{1}{\mathrm{ad}\,z}\cdot(x + y - \tilde{z}) + \tfrac{1}{2}\tilde{z} \\ &\quad - \left(1 - \frac{\mathrm{ad}\,z}{e^{\mathrm{ad}\,z} - 1}\right)\frac{1}{\mathrm{ad}\,z}(x + y - z) - \tfrac{1}{2}z \quad \text{modulo } [[\hat{\mathbf{L}}, \hat{\mathbf{L}}], [\hat{\mathbf{L}}, \hat{\mathbf{L}}]] \\ &\equiv \left(1 - \frac{\mathrm{ad}\,z}{e^{\mathrm{ad}\,z} - 1}\right)\frac{1}{\mathrm{ad}\,z}(z - \tilde{z}) - \tfrac{1}{2}(z - \tilde{z})\end{aligned}$$

and 1) will result from the following formula:

$$(5.3)\quad (z - \tilde{z}) \equiv \frac{\mathrm{ad}\,z}{e^{\mathrm{ad}\,z} - 1}\,\frac{e^{\mathrm{ad}\,x} - 1}{\mathrm{ad}\,x}\,\frac{e^{\mathrm{ad}\,y} - 1}{\mathrm{ad}\,y}\cdot[x, y] \quad \text{modulo } [[\hat{\mathbf{L}}, \hat{\mathbf{L}}], [\hat{\mathbf{L}}, \hat{\mathbf{L}}]].$$

Proof of (5.3). Let

$$\varphi_1(x, y) = (e^x - e^{-y})^{-1}\left(\frac{e^x - 1}{x} - \frac{1 - e^{-y}}{y}\right)$$

$$\varphi_2(x, y) = (e^y - e^{-x})^{-1}\left(\frac{1 - e^{-x}}{x} - \frac{e^y - 1}{y}\right).$$

then φ_1 and φ_2 are analytic functions at the origin. We have

a) $x + y - \tilde{z} \equiv \varphi_1(\mathrm{ad}\,x, \mathrm{ad}\,y)\cdot[x, y]$,
b) $(x + y - z) \equiv \varphi_2(\mathrm{ad}\,x, \mathrm{ad}\,y)\cdot[x, y] \mod [[\hat{\mathbf{L}}, \hat{\mathbf{L}}], [\hat{\mathbf{L}}, \hat{\mathbf{L}}]]$.

For a) we consider

$$e^{\mathrm{ad}\,x} - e^{-\mathrm{ad}\,y})(x + y - \tilde{z}) = (e^{\mathrm{ad}\,x} - e^{-\mathrm{ad}\,y})(x + y)$$

(as $(e^{\operatorname{ad} x}-e^{-\operatorname{ad} y})(\tilde{z})=e^{-\operatorname{ad} y}(e^{\operatorname{ad}\tilde{z}}-1)\cdot\tilde{z}=0$) so

$$(e^{\operatorname{ad} x}-e^{-\operatorname{ad} y})(x+y-\tilde{z})=(e^{\operatorname{ad} x}-1)+(1-e^{-\operatorname{ad} y})\cdot(x+y)$$
$$=\left(\frac{e^{\operatorname{ad} x}-1}{\operatorname{ad} x}-\frac{1-e^{-\operatorname{ad} y}}{\operatorname{ad} y}\right)\cdot[x,y]$$

and we obtain the equality a) by Remark 5.7. Now

$$z-\tilde{z}\equiv(\varphi_1-\varphi_2)(\operatorname{ad} x,\operatorname{ad} y)\cdot[x,y]$$

but

$$\varphi_1=(e^z-1)^{-1}\left(\frac{e^z-e^y}{x}-\frac{e^y-1}{y}\right),\quad \varphi_2=(e^z-1)^{-1}\left(\frac{e^x-1}{x}-\frac{e^z-e^x}{y}\right),$$

with $z=x+y$, and

$$\varphi_1-\varphi_2=(e^z-1)^{-1}\left(\frac{(e^x-1)(e^y-1)}{x}+\frac{(e^x-1)(e^y-1)}{y}\right)$$
$$=(e^z-1)^{-1}\frac{z}{xy}(e^x-1)(e^y-1)$$

and this proves Formula (5.3).

Let us prove 2) in Lemma 5.8. We let

$$\zeta(x,y)=\left(1-\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\right)\frac{1}{\operatorname{ad}\tilde{z}}(x+y-\tilde{z})+\tfrac{1}{2}\tilde{z},$$

then $\alpha(x,y)=\zeta(x,y)-\zeta(y,x)$. As $x+y-\tilde{z}\in[\hat{\mathbf{L}},\hat{\mathbf{L}}]$, we have

$$\zeta(tx,ty)\equiv\left(1-\frac{t\operatorname{ad}\tilde{z}}{e^{t\operatorname{ad}\tilde{z}}-1}\right)\frac{1}{t\operatorname{ad}\tilde{z}}(tx+ty-\tilde{z}(tx,ty))+\tfrac{1}{2}\tilde{z}(tx,ty)$$
$$\equiv\left(1-\frac{t\operatorname{ad}\tilde{z}}{e^{t\operatorname{ad}\tilde{z}}-1}\right)\frac{1}{\operatorname{ad}\tilde{z}}\left(x+y-\frac{1}{t}\tilde{z}(tx,ty)\right)+\tfrac{1}{2}\tilde{z}(tx,ty)$$
$$\text{modulo } [[\hat{\mathbf{L}},\hat{\mathbf{L}}],[\hat{\mathbf{L}},\hat{\mathbf{L}}]].$$

Here $\tilde{z}$ still denotes $\log e^y e^x$ and $\tilde{z}(tx,ty)=\log e^{ty}e^{tx}$. We have

$$\frac{\partial}{\partial t}\left(\left(1-\frac{tz}{e^{tz}-1}\right)\frac{1}{z}\right)_{t=1}=\frac{1}{e^z-1}\left(\frac{z}{1-e^{-z}}-1\right).$$

So

$$(\theta\zeta)(x,y)\equiv\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y-\tilde{z})+\left(1-\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\right)\frac{1}{\operatorname{ad}\tilde{z}}(\tilde{z}-\theta\tilde{z})+\tfrac{1}{2}\theta\tilde{z}$$
$$\equiv\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)-\left(1-\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\right)\frac{1}{\operatorname{ad}\tilde{z}}\cdot\theta\tilde{z}+\tfrac{1}{2}\theta\tilde{z}$$

as

$$\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot\tilde{z}=\tfrac{1}{2}\tilde{z}=\left(1-\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\right)\frac{1}{\operatorname{ad}\tilde{z}}\cdot\tilde{z}.$$

Recalling that

$$\theta\tilde{z}=\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot y+\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\cdot x=\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)+\operatorname{ad}\tilde{z}\cdot x,$$

we obtain:

$$\begin{aligned}\theta\zeta(x,y)\equiv&\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)+\left(\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)\\&+\left(\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}-1\right)\cdot x+\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot y+\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\cdot x\\\equiv&2\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)-\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)\\&+\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}x-x+\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot y+\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\cdot x\\\equiv&2\left(\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)+\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\cdot x-x-\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}\cdot y.\end{aligned}$$

After antisymmetrization, we obtain

$$\begin{aligned}\theta\alpha(x,y)\equiv&2\left(1-\frac{\operatorname{ad}z}{1-e^{-\operatorname{ad}z}}\right)\frac{1}{e^{\operatorname{ad}z}-1}\cdot(x+y)-2\left(1-\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}\right)\frac{1}{e^{\operatorname{ad}\tilde{z}}-1}\cdot(x+y)\\&+\left(\tfrac{1}{2}\frac{\operatorname{ad}z}{e^{\operatorname{ad}z}-1}+\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{1-e^{-\operatorname{ad}\tilde{z}}}-1\right)\cdot x-\left(\tfrac{1}{2}\frac{\operatorname{ad}\tilde{z}}{e^{\operatorname{ad}\tilde{z}}-1}+\tfrac{1}{2}\frac{\operatorname{ad}z}{1-e^{-\operatorname{ad}z}}-1\right)\cdot y\\\equiv&\theta A.\quad\text{c.q.f.d.}\end{aligned}$$

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Geometric Modules over the Burnside Ring

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0. Introduction and Results

This paper deals with the homotopy theory of group representations and its relation to transformation groups on manifolds.

Let G be a compact Lie group and let E and F be unitary or orthogonal representations of G, so that the unit spheres $S(E)$ and $S(F)$ are preserved by the G-action. Then a basic problem in the theory of transformation groups is: Determine the set (or group) of G-homotopy classes of maps $S(E) \to S(F)$. Since this problem in general involves the homotopy groups of spheres we restrict our investigation to the simplest situation: We study phenomena which can be analyzed using the concept of mapping degree.

Question 1. *When are $S(E)$ and $S(F)$ G-homotopy-equivalent?*

A G-map $f\colon S(E) \to S(F)$ is a G-homotopy-equivalence if and only if for each subgroup H of G (always closed, notation $H < G$) the H-fixed point mapping $f^H\colon S(E)^H \to S(F)^H$ is an ordinary homotopy equivalence (i.e. has degree ± 1). In particular, when $S(E)$ and $S(F)$ are G-homotopy-equivalent, then $\dim E^H = \dim F^H$ for all $H < G$. We use the dimension function

$$\mathrm{Dim}(E-F)\colon\ H \mapsto \dim E^H - \dim F^H, \qquad H < G \tag{1}$$

which only depends on the class of $E-F$ in the real (resp. complex) representation ring $RO(G)$ (resp. $R(G)$) of G. We are thus led to consider the additive subgroups of the representation rings

$$\begin{aligned} RO_0(G) &= \{x \in RO(G) \mid \mathrm{Dim}(x) = 0\} \\ R_0(G) &= \{x \in R(G) \mid \mathrm{Dim}(x) = 0\}. \end{aligned} \tag{2}$$

The G-equivariant stable homotopy groups $\{SE), S(F)\}$, the direct limit of the equivariant homotopy sets $[S(E \oplus U), S(F \oplus U)]_G$ taken over complex G-modules U and suspension maps, shall be denoted

$$\omega_\alpha = \omega_\alpha^G = \{S(E), S(F)\}, \qquad \alpha = E - F. \tag{3}$$

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As usual in homotopy theory the ω_α should better be defined using base points. Smashed product of representatives then induces a bilinear pairing

$$\omega_\alpha \times \omega_\beta \to \omega_{\alpha+\beta}. \tag{4}$$

In particular ω_α is a module over the ring ω_0, the stable equivariant homotopy ring of spheres in dimension zero. The pairing (4) induces a homomorphism

$$m_{\alpha,\beta}\colon\ \omega_\alpha \otimes_{\omega_0} \omega_\beta \to \omega_{\alpha+\beta}. \tag{5}$$

The modules ω_α are useful for dealing with question 1. This may be seen from our first result.

Theorem 1. *Let $\alpha = E - F$ be in $RO_0(G)$ (see (2)). Then the following holds:*

i) *The module ω_α is a projective ω_0-module of rank one.*

ii) *For each $\beta \in RO(G)$ the pairing* (5)

$$\omega_\alpha \otimes_{\omega_0} \omega_\beta \to \omega_{\alpha+\beta}$$

is an isomorphism.

iii) *The ω_0-module ω_α is free if and only if the spheres $S(E)$ and $S(F)$ are stably G-homotopy-equivalent* (i.e. *$S(E \oplus U)$ and $S(F \oplus U)$ are G-homotopy-equivalent for suitable U).*

The isomorphism classes of projective ω_0-modules of rank one with tensor product as composition law form an abelian group ([4], II. §5), the Picard group $\mathrm{Pic}(\omega_0)$. Using the additive subgroup of $RO(G)$

$$RO_h(G) = \{E - F \mid S(E), S(F) \text{ stably } G\text{-homotopy equiv.}\} \tag{6}$$

we may reformulate part of Theorem 1:

The assignment $\alpha \mapsto \omega_\alpha$ induces an injective homomorphism

$$tO(G)\colon\ RO_0(G)/RO_h(G) \to \mathrm{Pic}(\omega_0^G). \tag{7}$$

In order to use this result we have to describe the ring ω_0 and its projective modules. For simplicity we only deal with complex representations from now on. This has the advantage that all the fixed point sets E^H are complex vector spaces and therefore the spaces $S(E^H)$ may be coherently oriented. We use coherent orientations to define the degree of a map. Given $x \in \omega_\alpha$ represented by

$$f\colon\ S(E \oplus U) \to S(F \oplus U) \quad \text{with } U^G \neq \{0\}$$

we define the degree maps

$$d_{\alpha,H}(x) = \begin{cases} \text{degree } f^H & \text{if } \dim E^H = \dim F^H \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

This does not depend on the chosen representative. Then

$$d_{\alpha,H}\colon\ \omega_\alpha \to \mathbb{Z} \tag{9}$$

is an additive homomorphism and for $\alpha=0$ even a ring homomorphism. We list some elementary properties of the $d_{\alpha,H}$.

a) If H is conjugate to K then $d_{\alpha,H}=d_{\alpha,K}$.

b) If H is a normal subgroup of K and K/H is a torus then $d_{\alpha,H}=d_{\alpha,K}$. Therefore it is only necessary to consider $H<G$ which have finite index in their normalizer NH (see [6], proof of Prop. 12).

c) For each $x\in\omega_\alpha$ the function $H\mapsto d_{\alpha,H}(x)$ is continuous in the following sense: If the space S of closed subgroups of G is endowed with the Hausdorff topology and $\mathbb{Z}$ with the discrete topology then $S\to\mathbb{Z}$: $H\mapsto d_{\alpha,H}(x)$ is continuous.

It is convenient to assemble the degree maps into a single homomorphism, taking into account a), b), and c) above. Let therefore $\Phi=\Phi(G)$ be the space of conjugacy classes of subgroups H of G which have finite index in their normalizer. Then Φ is a quotient of a closed subspace of S, and is a countable (hence totally disconnected) compact Hausdorff space. Let ($=:$ means "defined" or "abbreviated")

$$C(\Phi(G),\mathbb{Z})=:C(G) \tag{10}$$

be the ring of continuous (= locally constant) functions $\Phi(G)\to\mathbb{Z}$. For finite G this is of course just a finite direct product of rings $\mathbb{Z}$. The degree homomorphism

$$d_\alpha\colon\ \omega_\alpha\to C(\Phi,\mathbb{Z}) \tag{11}$$

is given by $(d_\alpha(x))(H)=d_{\alpha,H}(x)$. This is an additive homomorphism, and a ring homomorphism for $\alpha=0$. Moreover it is compatible with the pairing (5)

$$d_{\alpha+\beta}\,m_{\alpha,\beta}(x\otimes y)=d_\alpha(x)\,d_\beta(y). \tag{12}$$

(Serious sign problems would arise at this point if we were not using complex representations.)

In certain favourable cases the degree homomorphism characterizes elements of ω_α. Using elementary obstruction theory as in [13], §5 one shows

Proposition 1. *For* $\alpha\in R(G)$ *with* $\operatorname{Dim}\alpha\leqq 0$ (i.e. $\alpha=E-F, H<G\Rightarrow\dim E^H\leqq\dim F^H$) *the map*

$$d_\alpha\colon\ \omega_\alpha\to C(\Phi,\mathbb{Z})$$

is injective.

Using our work Hauschild [12] has generalized this to unstable homotopy groups.

Using Proposition 1 we identify ω_α, for $\alpha\in R_0(G)$, with a subgroup of $C(\Phi,\mathbb{Z})$. Here is another simple device for constructing ω_0-modules in $C(\Phi,\mathbb{Z})$: For $e\in C(\Phi,\mathbb{Z})$ put

$$M(e)=\{x\in C(\Phi,\mathbb{Z})\mid x\,e\in\omega_0\}. \tag{13}$$

This is certainly an ω_0-module if we define (in accordance with (12)!) the action of $y\in\omega_0\subset C(\Phi,\mathbb{Z})$ as multiplication with y. If M and N are ω_0-submodules of $C(\Phi,\mathbb{Z})$ we can define their product

$$MN\in C(\Phi,\mathbb{Z}) \tag{14}$$

as the module generated by all products mn, $m \in M$, $n \in N$. We call M invertible if there exists N such that $MN = \omega_0$. The invertible submodules with product (14) form a commutative group $\mathrm{Inv}(\omega_0^G)$.

If G is finite let c be a multiple of the group order $|G|$. In general let c be a multiple of all the $|NH/H|$ for $(H) \in \Phi(G)$. It was shown in [7] that such integers exist. Let

$$r\colon\ C(\Phi, \mathbb{Z}) \to C(\Phi, \mathbb{Z}/c\mathbb{Z})$$

be the canonical map and put

$$C_c := C(\Phi, \mathbb{Z})_c := r^{-1}\, C(\Phi, \mathbb{Z}/c\mathbb{Z})^* \tag{15}$$

where a star denotes units of the ring. So C_c consists of functions whose values are invertible mod c.

We collect the necessary information about projective ω_0-modules in a single theorem. We need one further piece of notation: Call two complex G-modules E and F stably oriented homotopy equivalent ($E \sim_{oh} F$) if there exists a G-homotopy equivalence $f\colon S(E \oplus U) \to S(F \oplus U)$ such that f^H has degree one for each $H < G$. Let $R_h(G)$ be the additive subgroup of $R(G)$ consisting of all $E - F$, where $E \sim_{oh} F$.

Theorem 2. i) *Invertible modules are projective of rank one. Each projective module of rank one is isomorphic to an invertible module. The canonical map*

$$\mathrm{Inv}(\omega_0^G) \to \mathrm{Pic}(\omega_0^G)\colon\ M \mapsto [M]$$

is a surjective homomorphism.

ii) *If $e \in C_c$ then $M(e) \in \mathrm{Inv}(\omega_0^G)$. Each invertible module has the form $M(e)$ for a suitable $e \in C_c$. If $e, f \in C_c$ then $M(e)\,M(f) = M(ef)$. If $r(e) = r(f)$ then $M(e) = M(f)$.*

iii) *If $f \in M$, $r(ef) = 1$, and M is invertible then $M = M(e)$.*

iv) *Let α be in $R_0(G)$. Then $\omega_\alpha \subset C(\Phi, \mathbb{Z})$ is invertible. Moreover $\omega_\alpha = \omega_0$ if and only if $\alpha \in R_h(G)$.*

As a consequence of Theorem 2 we see that the assignment $\alpha \mapsto \omega_\alpha$ induces an injective homomorphism

$$t(G)\colon\ R_0(G)/R_h(G) \to \mathrm{Inv}(\omega_G^0). \tag{16}$$

In view of Theorem 2 we would like to determine an $e_\alpha \in C_c$ such that $M(e_\alpha) = \omega_\alpha$. This is done using part iii) of this Theorem. In order to achieve this we need a method which enables us to decide when an element $e \in C(\Phi, \mathbb{Z})$ is actually contained in a particular module ω_α. This leads to the second main topic of this paper.

Question 2. *What are the relations between the mapping degrees $d_{\alpha, H}$?*

Up to now we have not mentioned the fact that the ring ω_0^G is canonically isomorphic to the Burnside ring. For finite G this was announced by Segal [15]. For G a compact Lie group the Burnside ring $A(G)$ was defined in [6] as follows: On the

set of isomorphism classes of compact differentiable G-manifolds introduce the equivalence relation

$$M \sim N \Leftrightarrow \text{ for all } H < G \text{ the Euler characteristics } \chi(M^H) \text{ and } \chi(N^H) \text{ are equal.} \tag{16}$$

Disjoint union (addition) and cartesian product (multiplication) make the set of equivalence classes into a commutative ring $A(G)$. From the definition of $A(G)$ we see that $M \mapsto \chi(M^H)$ induces a ring homomorphism

$$d'_H\colon A(G) \to \mathbb{Z}. \tag{17}$$

Since we want to use the results of [6] we have to include a proof of

Proposition 2. *There exists a unique isomorphism* $t\colon A(G) \to \omega_0^G$ *such that for all* $H < G$ $d_{0,H}\, t = d'_H$.

In view of this Proposition an answer to Question 2 will include an answer to

Question 3. *What are the relations between the Euler characteristics* $\chi(M^H)$ *for compact differentiable G-manifolds M?*

Note that classical relations are $\chi(M) \equiv \chi(M^G) \bmod p$ if G is a p-group.

Since the answer to Question 2 is somewhat involved we first explain, how Dress (unpublished) determined the relations between the d'_H for finite G using the definition of $A(G)$ as Burnside ring of finite G-sets. One basic idea of his argument will then be extended to the general case.

For finite G it is easy to see that the cokernel of the homomorphism

$$d\colon A(G) \to C(\Phi, \mathbb{Z}); \quad d(S)(H) = |S^H|,$$

is a finite group of order $\prod |NH/H|$ where H runs through a complete set of non-conjugate subgroups of G. If $E(S)$ is the permutation representation of a finite G-set S then its character (denoted by the same symbol) satisfies

$$\sum_{g \in G} E(S)(g) = |G| \dim E(S)^G \equiv 0 \bmod |G|. \tag{18}$$

But $E(S)(g)$ is equal to $|S^g|$, hence (18) can be rewritten as

$$\sum_{g \in G} d_{\langle g \rangle}(x) \equiv 0 \bmod |G| \tag{19}$$

for any $x \in A(G)$, where $\langle g \rangle$ denotes the cyclic group generated by g and $d_{\langle g \rangle}\colon A(G) \to \mathbb{Z}\colon S \mapsto |S^{\langle g \rangle}|$. If we apply this argument to $E(S^H)$ considered as NH/H-module and collect conjugate subgroups we obtain

$$\sum_{(K)} |NK/NK \cap NH|\, \varphi(|K/H|)\, d_K(x) \equiv 0 \bmod |NH/H| \tag{20}$$

where the summation is taken over the NH-conjugacy classes (K) of subgroups K such that H is normal in K and K/H is cyclic; and where φ is the Euler function. The congruences (20) are independent (triangular matrix!). The above remark on the

order of the cokernel of d then shows that they must be a complete set of congruences.

In order to generalize this argument to ω_α and compact Lie groups we only need a substitute for the congruences (20) in the more general case. This is provided using the K-theory mapping degree. We recall that the equivariant K-theory $\tilde{K}_G(S^V)$ of the one-point-compactification $S^V = V \cup \{\infty\}$ of the complex G-module V is a free $R(G)$-module on the generator $\lambda(V)$, the Bott class of V (see [1]). The K-theory degree $k(f) \in R(G)$ of a pointed G-map $f\colon S^E \to S^F$ is then defined by the equation

$$f^* \lambda(F) = k(f) \cdot \lambda(E). \tag{21}$$

This degree only depends on the class of f in ω_{E-F}. We shall compute the character $k(f)$ as

$$k(f)(g) = \lambda_{-1}(F_g - E_g)(g)\,\text{degree}(f^C) \tag{22}$$

where C is the closure of the subgroup generated by g, where E_g is a complement of $E^g = E^C$ in E, where λ_{-1} denotes the alternating sum of the exterior powers, and where degree (f^C) is the ordinary degree of f^C (which, by definition, is zero if $\dim E^C \neq \dim F^C$). (The formula (22) is related to the localization theorem of K_G-theory.) Since $k(f)$ is a character we obtain for finite G from $\sum_{g \in G} k(f)(g) \equiv 0 \bmod |G|$ congruences which are analogous to (19). We apply the same argument to the NH/H-modules E^H and F^H.

The following Theorem answers Question 2 and should now be plausible as a generalization of the result of Dress.

Theorem 3. *Let $\alpha \in R(G)$ be an element with $\mathrm{Dim}(\alpha) \leqq 0$ (see (1)). There exists a collection of integers $n_{H,K}(\alpha)$, depending on α, $(H) \in \Phi$, and (K) with H normal in K and K/H cyclic, such that $n_{H,H}(\alpha) = 1$ and such that the following holds: $x \in C(\Phi, \mathbb{Z})$ is contained in the image of d_α if and only if:*

i) $x(H) = 0$ *whenever* $\dim \alpha^H \neq 0$.

and

ii) *For each H with finite NH/H*

$$\sum_{(K)} n_{H,K}(\alpha)\, x(K) \equiv 0 \bmod NH/H.$$

The sum is taken over the conjugacy classes (K) such that H is normal in K and K/H is cyclic.

Theorem 3 may be used to give a simple proof of the following result of Tornehave which is relevant for the applications of Theorem 2 (see passage before Question 2). Let Ψ^k denote the Adams operation on the ring $R(G)$. For finite groups G and k prime to $|G|$ the element $x - \Psi^k x$ is always contained in $R_0(G)$. (Actually $R_0(G)$ is generated by such elements, see e.g. [14], 3.17.)

Theorem 4 (Tornehave). *Let G be a finite group and let k be prime to $|G|$. For a complex G-module E put $\alpha = E - \Psi^k E$. Then ω_α contains the function*

$$k^{\mathrm{Dim}\, E}\colon (H) \mapsto k^{\dim E^H}.$$

For $(l, |G|)=1$ we have $k^{\mathrm{Dim}E}=k^{\mathrm{Dim}\Psi^l E}$. We therefore conclude from Theorem 1 and 2 that for any $x\in R(G)$

$$\omega_\alpha=\omega_\beta \quad \text{for } \beta=\Psi^l\alpha, \quad \alpha=x-\Psi^k x \tag{23}$$

(equality of subgroups of $C(\Phi,\mathbb{Z})$).

Since the action of the Ψ^k on $R(G)$ may be obtained by an action of an appropriate Galois group on $R(G)$ (see [2], II. §3), we like to interpret (23) as Galois invariance of the modules ω_α, $\alpha\in R_0(G)$. One can also show $\omega_\alpha\cong\omega_\beta$ for $\alpha\in RO_0(G)$ and $\beta=\Psi^l\alpha$. This result and Theorem 1 and 2 have our final Theorem as a Corollary. We put

$$\begin{aligned} RO_1(G) &:= \{\Psi^l x - x \mid (l,|G|)=1, x\in RO_0(G)\} \\ R_1(G) &:= \{\Psi^l x - x \mid (l,|G|)=1, x\in R_0(G)\}. \end{aligned} \tag{24}$$

Theorem 5. $R_1(G)\subset R_h(G)$, $RO_1(G)\subset RO_h(G)$.

One can show that these inclusions are equalities if G is a p-group (compare [9]).

We prove everything that has been stated as Proposition or Theorem so far: Proposition 2 in 1.; Theorem 1 in 2.; Theorem 2 in 3.; Theorem 3 and 4 in 4. Section 5 contains various examples.

In the meantime, using methods of this paper, Theorem 3 and 4 have been generalized to real representations by Tornehave.

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1. The Burnside Ring

We prove Proposition 2.

If M is a compact differentiable G-manifold then the Lefschetz index of the identity id: $M\to M$ is by the equivariant analogue of [10] an element $I(M)$ in ω_0^G. The Lefschetz index is compatible with taking H-fixed points, i.e.

$$d_H I(M)=I(M^H)\in\omega_0^{\{1\}}\cong\mathbb{Z}.$$

The non-equivariant Lefschetz index $I(M^H)$ is of course the Euler characteristic, hence

$$d_H I(M)=\chi(M^H). \tag{25}$$

It was show in [5], Satz 2 and Satz 3, that ω_0^G is the free abelian group on the elements $I(G/H)$, where H runs over a complete set of non-conjugate subgroups $H<G$ with finite NH/H. This implies together with (25) that the map (11) d_0 is injective and that the t of Proposition 2 is well-defined, bijective and satisfies $d_{0,H}t=d'_H$.

2. Projective Modules

This section contains the proof of Theorem 1.

We write $\omega = \omega_0$ and $\omega_\alpha = \omega^{-\alpha}$ (in order to have space for lower indices). Let P be an ω-module. Then P is a projective ω-module of rank one if and only if P is finitely generated and for each maximal ideal q of ω the localization P_q of P at q is a free ω_q-module of rank one ([4], §5 Théorème 2).

Let q be a maximal ideal of ω. According to [6], Theorem 4, there exists a group $H < G$ (unique up to conjugation) such that NH/H is finite, the order of NH/H is prime to the characteristic $p \neq 0$ of ω/q and q is the kernel of $d_H \bmod p\colon \omega \to \mathbb{Z}/(p)$. We denote this ideal $q = q(H, p)$.

Let $\alpha = E - F \in RO_0(G)$. We show that ω_q^α is a free ω_q-module. Take $x \in \omega^\alpha$, $y \in \omega^{-\alpha}$. Then multiplication with x resp. y, using the pairing (5), induces ω-linear maps

$$y_*\colon \omega^\alpha \to \omega \quad \text{resp.} \quad x_*\colon \omega \to \omega^\alpha.$$

The composition $y_* x_*$ is multiplication with $yx \in \omega$. By definition of $q(H, p)$ this element becomes a unit in ω_q if

$$d_H(yx) = \pm d_H(y)\, d_H(x) \not\equiv 0 \bmod p.$$

A similar argument applies to $x_* y_*$. If $x\, y$ is a unit in ω_q then x_{*q} is an isomorphism, thus proving our assertion. So we need only find an element $x \in \omega^\alpha$ (depending on H) such that $d_H x \not\equiv 0 \bmod p$. (Note that d_H is only defined up to sign for elements α in the real representation ring. This causes no problem here.) To find such an x we first construct an H-equivariant map $f\colon S^E \to S^F$ with degree $f^H = 1$. This is done using the obstruction theory of [13], §5; see the proof of Proposition 3 for more details. The assumption $\alpha \in RO_0(G)$ guarantees that all obstructions vanish. Then we use the transfer homomorphism

$$\mathrm{Top}_H^G\colon \omega_{\alpha|H}^H \to \omega_\alpha^G \tag{26}$$

which satisfies

$$d_{\alpha,K}(\mathrm{Top}_H^G y) = \chi(G/H^K)\, d_{\alpha|H,K}(y) \tag{27}$$

(compare [13], 2.2). Then $x = \mathrm{Top}_H^G[f]$ has the desired property.

Now let $\alpha, \beta \in RO_0(G)$. We show that the pairing (5) is an isomorphism in this case by showing that all localizations $(m_{\alpha,\beta})_q$ are isomorphisms. But then we are dealing with a map

$$(m_{-\alpha,-\beta})_q\colon \omega_q^\alpha \otimes_{\omega_q} \omega_q^\beta \to \omega_q^{\alpha+\beta} \tag{28}$$

between free ω_q-modules of rank one. If x resp. y are generators of ω_q^α resp. ω_q^β then $z = (m_{-\alpha,-\beta})_q(x \otimes y)$ satisfies $d_H z \not\equiv 0 \bmod p$ (if $q = q(H, p)$) and is therefore a generator of $\omega_q^{\alpha+\beta}$. Hence (28) is an isomorphism.

To finish the proof of Theorem 1 i) it remains to be shown that ω_α is finitely generated. But if under an isomorphism $\omega_\alpha \otimes_\omega \omega_{-\alpha} \cong \omega$ the element 1 corresponds to $\sum m_i \otimes n_i$ then ω_α is generated as ω-module by the m_i.

For the proof of Theorem 1 ii) we note that an inverse to $m_{\alpha,\beta}$ is given by

$$\omega_{\alpha+\beta} \cong \omega_\alpha \otimes (\omega_{-\alpha} \otimes \omega_{\alpha+\beta}) \to \omega_\alpha \otimes \omega_\beta.$$

Finally we show iii). If $S(E)$ and $S(F)$ are stably G-homotopy-equivalent then a stable equivalence induces an isomorphism $\omega_\alpha \cong \omega_0$ by composition. Conversely assume that ω_α is free, with generator x say. Then $\omega_{-\alpha}$, representing the inverse of ω_α in the Picard group, is free too, say with generator y. The product $xy \in \omega$ therefore has to be a generator of this module, so has to be a unit in the ring ω. This implies $d_H(xy) = \pm 1$ for all $H < G$. Hence representatives for x must satisfy $d_H(x) = \pm 1$ thus being G-homotopy equivalences.

3. Picard Group and Invertible Modules

This section contains the proof of Theorem 2. We should point out that the algebraic results are essentially well known, but there does not seem to exist a reference which contains exactly the technical details we need. We abbreviate $C = C(\Phi, \mathbb{Z})$.

From the definition of the integer c before the statement of Theorem 2 we conclude that

$$\begin{array}{ccc} \omega & \longrightarrow & C \\ \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle r} \\ \omega/c\,C & \longrightarrow & C/c\,C \end{array} \tag{29}$$

is a pull-back of rings. Here p and r are the canonical quotient maps. The horizontal maps are injective. The groups $\mathrm{Pic}(C)$ and $\mathrm{Pic}(\omega/c\,C)$ are zero: This is clear for finite G (see [4], 5.3. Prop. 5) and follows by a limit argument for general G. Therefore the Mayer-Vietoris sequence for Pic([3], p. 482) yields the exact sequence

$$0 \to \omega^* \to (\omega/c\,C)^* \oplus C^* \to (C/c\,C)^* \xrightarrow{d} \mathrm{Pic}\,\omega_0 \to 0. \tag{30}$$

Here S^* denotes the group of units in a ring S. The map d is given as follows: If $y \in (C/c\,C)^*$ is represented by $e \in C$ then

$$d(y) = M(e). \tag{31}$$

We now enter the proof of Theorem 2. This is done in several steps which are numbered so that we can refer to them.

3.1. $r(e) = r(f) \Rightarrow M(e) = M(f)$. Proof: Let $e = f + ch$. Then: $x \in M(e) \Rightarrow ex \in \omega_0 \Rightarrow x(f + ch) \in \omega_0$. Since $cC \subset \omega$ we conclude that $xch \in \omega$ and therefore $xf \in \omega$. We have shown $M(e) \subset M(f)$.

3.2. If M is invertible then M is projective of rank one [4]. By (31) if $e \in C_c$ then $M(e)$ is projective of rank one hence invertible. Therefore, by (31) again, the natural map $\mathrm{Inv}\,\omega \to \mathrm{Pic}\,\omega$ is surjective.

3.3. Let $M \subset C$ be invertible. An element $x \in M$ is a generator of the localized module M_q, $q = q(H,p)$, if and only if $d_H x \not\equiv 0 \bmod p$. Therefore $M/q\,M \cong M_q/q\,M_q \cong \omega_q/q\,\omega_q \cong \omega/q\,\omega$.

3.4. Let $M \subset N$ be invertible modules. Then $M/qM \cong N/qN$ for every maximal ideal q of ω by 3.3. By [4], §3.3 Proposition 11 we have $M = N$.

3.5. Let M be invertible. Then $c\,C \subset M$. Proof: Choose N such that $MN = \omega$. Choose $m_1, \ldots, m_k \in M, n_1, \ldots, n_k \in N$ such that $\sum m_i n_i = 1$. For $x \in C$ we have $c\,x \in \omega$ and therefore $c\,x = \sum m_i (c\,n_i\,x) \in M$.

3.6. If $e, f \in C_c$ then $M(e)\,M(f) = M(ef)$. Proof: The inclusion $M(e)\,M(f) \subset M(ef)$ follows from the definitions. Using 3.2 and 3.4 we see that the inclusion must be an equality.

3.7. Let M be invertible, $f \in M$, $r(ef) = 1$. If $x \in M(e)$ then $e\,x \in \omega$ and therefore $e\,x f \in M$. By 3.5 therefore $x \in M$. By 3.2 and 3.4 we have $M = M(e)$.

3.8. Let M be invertible. Then $M \cap C_c \neq \emptyset$. Proof: Let $M \cong M(f), f \in C_c$. Such f exist by 3.2. If $M(e)\,M(f) = \omega$, then $M(e)\,M \cong \omega$. Hence $M(e)\,M$ is free with generator x say. Then x has to lie in C^*. Choose $h \in C_c \cap M(f)$. Then $h\,x$ is contained in $M(f)\,M(e)\,M = M$ and C_c.

3.9. By Theorem 1 and (12) the module $\omega_\alpha \subset C, \alpha \in R_0(G)$ is invertible. Now assume that $\omega_\alpha = \omega$. Then $1 \in \omega_\alpha$, hence this module contains a stable map which has degree one on each fixed point set; so $\alpha \in R_h(G)$. If $\alpha \in R_h(G)$ then $1 \in \omega_\alpha$ and therefore $\omega_\alpha = \omega$ by 3.7.

The reader may check that by now we have proved Theorem 2.

We mention as Corollary that the following two sequences are exact (with $D(r\,e) = M(e)$)

$$0 \to (\omega_0/c\,C)^* \to (C/c\,C)^* \xrightarrow{D} \mathrm{Inv}(\omega_0) \to 0 \tag{32}$$

$$0 \to \omega_0^* \to C^* \to \mathrm{Inv}(\omega_0) \to \mathrm{Pic}(\omega_0) \to 0. \tag{33}$$

4. Congruences for Fixed Point Degrees

This section contains a proof of Theorem 3 and 4.

We first show that any set of congruence relations of the type considered in Theorem 3 suffices to describe the modules ω_α. Later we derive specific congruences as indicated, using K-theory.

Suppose we are given for each $(H) \in \Phi = \Phi(G)$ a map $r_H\colon C(\Phi, \mathbb{Z}) \to \mathbb{Z}/(w_H), w_H = |NH/H|$, of the form

$$r_H(z) = z(H) + \sum n_{H,K}\, z(K) \bmod w_H, \tag{34}$$

where the $n_{H,K}$ are integers and the sum is taken over the conjugacy classes (K) such that H is normal in K and K/H is cyclic and non-trivial. Suppose that for a certain $\alpha = E - F$ with $\dim \alpha \leqq 0$ the image of d_α: $\omega_\alpha \to C(\Phi, \mathbb{Z}) =: C$ is contained in

$$C_\alpha = \{z \in C \mid (H) \in \Phi \Rightarrow r_H(z) = 0; \dim \alpha^H < 0 \Rightarrow z(H) = 0\}. \tag{35}$$

Then we claim

Proposition 3. $d_\alpha \omega_\alpha = C_\alpha$.

Proof. Given $z \in C_\alpha$. We have to show that for a suitable U there exists a map f: $S(E \oplus U) \to S(F \oplus U)$ such that for each $(H) \in \Phi$ degree $f^H = z(H)$. To begin with we choose U large enough so as to satisfy the following conditions (36) and (37). (The reader is warned: These technical considerations are only necessary for non-finite G.) Let $\mathrm{Iso}(X)$ denote the set of conjugacy classes of isotropy groups appearing on X.

$$\begin{aligned} &\mathrm{Iso}(E \oplus U) = \mathrm{Iso}(F \oplus U) \\ &(1), (G) \in \mathrm{Iso}(E \oplus U) \\ &(K), (L) \in \mathrm{Iso}(E \oplus U) \Rightarrow (K \cap L) \in \mathrm{Iso}(E \oplus U). \end{aligned} \tag{36}$$

Choose an integer $n \neq 0$ such that $x = n z$ is contained in C_0 (this exists by [7]). Then there shall exist a representative $S(E \oplus U) \to S(E \oplus U)$ for x in ω_0. (37)

Once (37) is satisfied for U it is also satisfied for any U' that contains U as a direct summand. So without loss of generality we assume (37) is true for $U = \{0\}$ to simplify the notation. We can also assume (1), $(G) \in \mathrm{Iso}(E) \cap \mathrm{Iso}(F)$. Since by the descending chain condition for subgroups of G the finite intersections of isotropy groups of $E \oplus F$ contain only finitely many conjugacy classes there exists an integer k such that $\mathrm{Iso}(k(E \oplus F))$ is closed under intersections. Choose k with this property and put $U = k(E \oplus F)$. Then (36) holds.

We set $X = S(E \oplus U)$ and $Y = S(F \oplus U)$. Let $\mathrm{Iso}(X) = \{(H_1), \ldots, (H_t)\}$ where the indexing is chosen so that $(H_i) > (H_j)$ implies $i < j$. $((H) < (K)$ means: H subconjugate to $K)$. If $X_i := \{x \in X \mid (G_x) = (H_j)$ for some $j \leqq i\}$ with G_x the isotropy group at x, then we construct inductively G-maps f_r: $X_r \to Y$ such that

$$\begin{aligned} &\text{degree } f_r^L = z(L) \qquad \text{if } (L) \in \Phi, (L) \geqq (H_i), \quad i \leqq r \\ &\qquad\qquad\qquad \text{or if } (L) > (H_{r+1}), (L) \in \Phi. \end{aligned} \tag{38}$$

(Note that $X_i^L = X^L$ for such L.) Put $H = H_{r+1}$. The G-extensions f_{r+1}: $X_{r+1} \to Y$ of f_r correspond via restriction bijectively to the NH/H-extensions h: $X^H \to Y^H$ of $f_r' := f_r | X_H$: $X_H \to Y^H$, where $X_H = X^H \cap X_r$. The obstructions to the existence of h lie in $H^*(X^H/NH, X_H/NH; \pi_{*-1}(Y^H))$, see [13]. These groups are zero because $\dim X^H \leqq \dim Y^H$. Let f_{r+1}': $X^H \to Y^H$ be an NH/H-extension of f_r'. Actually (using this obstruction argument) we see that there exists f_1: $X \to Y$ with $f_1^H = f_{r+1}'$. Therefore, if $(H) \in \Phi$, the fixed point degrees satisfy

$$d(f_1^H) + \sum n_{H,K} d(f_1^K) \equiv 0 \bmod |NH/H|.$$

By induction $d(f_1^K)=z(K)$ so that in this case $d(f'_{r+1})\equiv z(H) \bmod |NH/H|$. Since NH/H acts freely on $X^H \setminus X_H$ we can alter f'_{r+1} rel X_H to an NH-map f''_{r+1} so that $d(f''_{r+1})=z(H)$. Let f_{r+1} be the map with $f_{r+1}|X^H=f''_{r+1}$ if $(H)\in\Phi$ and $f_{r+1}|X^H=f'_{r+1}$ if $(H)\notin\Phi$. Then $d(f^L_{r+1})=z(L)$ whenever $(L)\geqq(H_i)$, $i\leqq r+1$. Suppose $(L)>(H_{r+1})$, $(L)\in\Phi$. Since $\mathrm{Iso}(X)=\mathrm{Iso}(Y)$ is closed under intersections there exists a unique minimal isotropy group $(P)=(H_s)$ such that $(P)\geqq(L)$ and then $(P)\in\Phi$, $X^L=X^P$, $Y^L=Y^P$, degree $f^L_{r+1}=$ degree $f^P_{r+1}=z(P)$. We have to show $z(L)=z(P)$. But since by the assumption (37) nz is represented by a map $g\colon X\to X$ the equality $g^P=g^L$ implies $nz(L)=nz(P)$. This finishes the proof of Proposition 3.

We now derive specific functions of the type (34). Let $f\colon E\to F$ be a proper G-map between complex G-modules. Let $C<G$ be a topologically cyclic group with generator h. Put $E=E^C\oplus E_C$, $j_E\colon E_C\subset E$. We apply equivariant K-theory with compact support [16] and obtain the diagram

$$\begin{array}{ccc} K_C(F) & \xrightarrow{f^*} & K_C(E) \\ \downarrow{\scriptstyle j_F^*} & & \downarrow{\scriptstyle j_E^*} \\ K_C(F^C) & \xrightarrow{(f^C)^*} & K_C(E^C). \end{array} \tag{39}$$

Using Bott classes as in (21) to define $a\in R(G)$ by $f^*\lambda(F)=a\,\lambda(E)$ we see from (39) that the restriction $a|C\in R(C)$ satisfies

$$(a|C)\,\lambda_{-1}(E_C)=\lambda_{-1}(F_C)\,\mathrm{degree}\,f^C \tag{40}$$

and if we evaluate characters at h

$$a(h)\,\lambda_{-1}(E_C)(h)=\lambda_{-1}(F_C)(h)\,\mathrm{degree}\,f^C. \tag{41}$$

We note that $\lambda_{-1}(E_C)(h)\neq 0$ and that

$$\lambda_{-1}(F_C)(h)/\lambda_{-1}(E_C)(h)=\lambda_{-1}(F_C-E_C)(h). \tag{42}$$

If G is finite then $|G|^{-1}\sum_{h\in G} a(h)$ is the dimension of a^G for any $a\in R(G)$ and therefore

$$\sum_{h\in G} a(h)\equiv 0 \bmod |G|. \tag{43}$$

If $C<G$ is cyclic and C^* the set of its generators we put $a^*(C)=\sum_{h\in C^*} a(h)$ and rewrite (43) as

$$\sum_{h\in G} a(h)=\sum_{(C)} |G|\,|NC|^{-1}\,a^*(C) \tag{44}$$

where the right hand sum is taken over the conjugacy classes of cyclic subgroups of G. From (41) we obtain

$$a^*(C)=n(E-F,C)\ \mathrm{degree}\,f^C \tag{45}$$

with

$$n(E-F, C)=\sum_{h\in C^*}\lambda_{-1}(F_C-E_C)(h) \tag{46}$$

By Galois theory $n(E-F, C)$ is a rational number. It is an integer if $\dim F_C=\dim E_C$ because in this case $\lambda_{-1}(F_C-E_C)(h)$ is a cyclotomic unit.

If we apply these considerations to f^H considered as NH/H-map we obtain from (41)–(46) for each $(H)\in\Phi$

$$\sum_K |NH/NH\cap NK|\, n(E^H-F^H, K/H)\, d(f^K)\equiv 0 \bmod |NH/H| \tag{47}$$

where the sum is taken over the NH-conjugacy classes (K) with H normal in K and K/H cyclic. Writing this in terms of G-conjugacy classes we have found functions of the type used in (34) and (35). This finishes the proof of Theorem 3.

Proposition 1 is proved easily with the help of the obstruction theory used in the proof of Proposition 3. For an elaboration see [12].

Remark. We should point out that for arbitrary E and F and finite G $a(h)$ in (41) is an algebraic integer whereas (42) is not an integer in general. This gives certain integrality relations which degree f^C has to satisfy. These are powerful conditions as shall be shown at another occasion.

Proof of Theorem 4. We have to show that the function $k^{\mathrm{Dim}\,E}$ satisfies the congruences (47). We use the Bott cannibalistic class $\Theta_k(E)$, see [2]. It has the character

$$\Theta_k(E)(g)=k^{\dim E^g}\lambda_{-1}(\Psi^k E_C-E_C)(g)$$

if g generates C, and similarly for $\Theta_k(E^H)$ considered as virtual NH/H-module. These characters are precisely of the form (41) so that $k^{\mathrm{Dim}\,E}$ has to satisfy (47).

5. Examples

Example 1. $G=S^1$ and $N=t^p+t^q$, $M=t^1+t^{pq}$ are the complex 2-dimensional G-modules on which $t\in S^1$ acts via $t(z_1, z_2)=(t^p z_1, t^q z_2)$ respectively $t(z_1, z_2)=(t z_1, t^{pq} z_2)$. Here p and q are relatively prime integers. Set $\beta=M-N$. The only subgroup of G of finite index in its normalizer is G. Since $\mathrm{Dim}\,\beta\geqq 0$ Theorem 3 shows $\omega_G^\beta\cong\mathbb{Z}$. As a matter of fact, ω_G^β is generated by an element f of degree one. On the other hand $\omega_G^\beta=0$ by the proof of Theorem 3 and the fact that $\lambda_{-1}(N-M)\notin R(S^1)$. A generator of ω_G^β can be constructed explicitly. Choose positive integers a and b satisfying $-ap+bq=1$. Define $f=h(\cdot)\cdot\|h(\cdot)\|^{-1}$ where $h(z_1, z_2)=(z_1^a z_2^b, z_1^q+z_2^p)$. There is no $f\in\omega_G^{-\beta}$ with degree one. This shows the necessity of the hypothesis $\mathrm{Dim}\,\beta\geqq 0$ in Theorem 3.

Example 2. $G=\mathbb{Z}/(p\,q)\subset S^1$ where for simplicity p and q are distinct primes. Let M and N be the restrictions of the S^1-modules of Example 1 to $\mathbb{Z}/(p\,q)$. Set $\beta=M-N$.

In this example $\Phi(G)=\{1, \mathbb{Z}/p, \mathbb{Z}/q, \mathbb{Z}/pq\}$ and $C(\Phi, \mathbb{Z}) \cong \prod_{\Phi(G)} \mathbb{Z}$. For $\alpha = \pm\beta$ $\dim \alpha^G = +1$ so far any $z \in C_\alpha$, $z(G)=0$. In fact C_β is the subgroup of C defined by

$$z \in C_\beta \Leftrightarrow z(G)=0, \quad z(\mathbb{Z}/p) \equiv 0(q),$$
$$z(\mathbb{Z}/q) \equiv 0(p),$$
$$z(1) \equiv (u\, z\mathbb{Z}/p) + v\, z(\mathbb{Z}/q)) \bmod pq$$

for integers u, v with $uq \equiv 1(p)$, $vp \equiv 1(q)$. On the other hand

$$z \in C_{-\beta} \Leftrightarrow z(G)=0, \quad z(\mathbb{Z}/p) \equiv 0(q), \quad z(\mathbb{Z}/q) \equiv 0(p)$$
$$z(1) \equiv (q\, z(\mathbb{Z}/p) + p\, z(\mathbb{Z}/q)) \bmod pq.$$

The degree map $\omega_G^{-\beta} \to C_{-\beta}$ is not injective. One can show $\omega_G^{-\beta} \cong \mathbb{Z}/2 \times C_{-\beta}$ where the $\mathbb{Z}/2$-factor appears from the two-stem of the stable homotopy of spheres and the fact that $\dim_{\mathbb{R}} \beta^G = 2$. This is expressed by a surjective homomorphism

$$\omega_G^{-\beta} \to \lim \pi_{n+2}(S^n) = \mathbb{Z}/2 : f \mapsto f^G.$$

Example 3. $G = A_5$ (alternating group on 5 letters). G contains one conjugacy class of each of the subgroups $A_4, D_5, D_3, D_2, \mathbb{Z}/(5), \mathbb{Z}/(3), \mathbb{Z}/(2), 1$ (D_n dihedral group of order $2n$). Note A_5 is simple and perfect. The groups A_5, D_3, A_4, and D_5 are their own normalizers while $N(\mathbb{Z}/(n)) = D_n$, $N(D_2) = A_4$. Here $\omega \subset C$ is the set of $z \in C$ satisfying

(i) zH) arbitrary for $H = A_5, D_3, A_4, D_5$.
(ii) $z(\mathbb{Z}/(n)) \equiv -z(D_n) \bmod 2$, $n = 3, 5$
(iii) $z(D_2) \equiv z(A_4) \bmod 3$
(iv) $z(1) \equiv -(20\, z(\mathbb{Z}/(3)) + 15\, z(\mathbb{Z}/(2)) + 24\, z(\mathbb{Z}/(5))) \bmod 60$.

From this description, we see that ω contains the idempotent e defined by $e(A_5) = 1$, $e(H) = 0$ for $H = A_5$. This illustrates a theorem of Dress [11] (G finite) and tom Dieck [8] (G infinite) that any perfect group G has an idempotent $e_G \in \omega_G^0$ with $e_G(G) = 1$, $e_G(H) = 0$ for $(H) \in \Phi(G)$, $H \neq G$.

Example 4. $G = SO(3)$. The conjugacy classes of G which are their own normalizers are: G, A_5, S_4 (symmetric group), and N (normalizer of a maximal torus). The remaining conjugacy classes which have finite index in their normalizers are: D_n with $N(D_n) = D_{2n}$ for $n \neq 2$, $N(D_2) = S_4$, A_4 with $N(A_4) = S_4$, T with $N(T) = N$. With this information we find $\omega \subset C$ is the subring of functions z satisfying

(i) $z(H)$ arbitrary for $H = G, A_5, S_4, N$
(ii) $z(D_n) \equiv z(D_{2n}) \bmod 2, n \neq 2$
(iii) $z(D_2) \equiv -(2\, z(A_4) + 3\, z(D_4)) \bmod 6$
(iv) $z(A_4) \equiv z(S_4) \bmod 2$
(v) $z(T) \equiv z(N) \bmod 2$.

Note that ω_G^0 contains the idempotent e_G represented by $e_G(G) = 1$ and $e_G(H) = 0$, $H \neq G$.

Example 5. Let G be finite abelian. Then $\mathrm{Inv}(\omega_G^0)$ is isomorphic to $\prod_{(H)}(\mathbb{Z}/(|G/H|))^*$ and under this isomorphism the image of the map $R_0(G)/R_1(G) \to \mathrm{Inv}(\omega_G^0)$ corresponds to the factors with cyclic G/H.

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The Behaviour of the Plurigenera of Surfaces Under Algebraic Smooth Deformations

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Introduction

In this paper we study the behaviour of the plurigenera of projective surfaces under algebraic smooth deformations. By a *surface* we shall mean an irreducible variety of dimension 2 defined over the complex numbers. An *algebraic smooth deformation* of a surface S_0 is given by a flat proper morphism of varieties, $\rho: \mathscr{S} \to Y$ say, with fibre $\rho^{-1}(y_0) = S_0$ for some $y_0 \in Y$ and with the general fibre $\rho^{-1}(y) = S$ being a smooth surface. Alternatively we may consider this as an algebraic family of surfaces whose general member is smooth and which degenerates into the surface S_0. Recall that when we talk about the plurigenera of a singular surface S_0, we mean the plurigenera of a desingularization of S_0 (see [19]).

The question we ask is therefore about the relation between the plurigenera $P_n(S_0)$ of the (possibly singular) surface S_0 and the plurigenera $P_n(S)$ of a general (smooth) fibre S. From the results of Iitaka [1] we know that if S_0 is smooth then the plurigenera are the same. Hence we note that the particular choice of the smooth fibre S above is irrelevant to our considerations.

Smooth deformations of singular surfaces are studied in [14] where the author considers the more general case of the special fibre being reducible. However since the methods employed are mainly Hodge theoretic in nature, it does not seem likely that they may be applied to the behaviour of the plurigenera. Indeed, without any restriction on the surface S_0, it does not seem possible to say much about the behaviour of the plurigenera P_n. However if we impose the condition that S_0 has only isolated Gorenstein singularities then a rather beautiful theory appears, and it is this theory that will be dealt with here.

Recall here that a variety X is said to be *Gorenstein* if all its local rings are Gorenstein in the sense of [7]. Thus we know that a projective variety X is Gorenstein if and only if it is Cohen-Macaulay and its Grothendieck dualizing sheaf ω_X is invertible (see [1] for an elementary introduction to duality theory). Note that from Serre's criterion ([1] page 135) a surface S_0 having only isolated Gorenstein singularities is equivalent to its being normal and Gorenstein.

The theory developed in this paper will enable us to prove the following result:

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Main Theorem. *If S_0 is a normal Gorenstein surface and $\rho: \mathscr{S} \to Y$ is an algebraic smooth deformation of S_0 with general surface S, then $P_n(S_0) \leqq P_n(S)$ for all positive integers n.*

More details of the work contained in this paper may be found in [21]. I should like to thank Dr. Barry Tennison and Dr. Miles Reid for their invaluable advice and encouragement during the writing of this paper. I should also like to thank the Science Research Council and Jesus College, Cambridge for their financial support.

1. Preliminaries

Given the deformation $\rho: \mathscr{S} \to Y$ as in the Introduction, we may join the points y and y_0 by a curve C in Y. Hence we may assume that Y is in fact a curve. We may normalize C and take the pullback of ρ, and in this way we may also assume that C is smooth. Lastly we may clearly assume for our purposes that $\rho^{-1}(y_0)$ is the only singular fibre.

We therefore need only consider deformations of the form $\rho: \mathscr{S} \to C$ where C is a smooth (not necessarily complete) curve and ρ has just one singular fibre $S_0 = \rho^{-1}(y_0)$. As before we denote the general fibre by $S = \rho^{-1}(y)$.

Without any restriction on the singularities of the surface S_0, we may deduce using generalized Hodge theory that $P_1(S_0) \leqq P_1(S)$. This follows either as a trivial corollary to Theorem 2.7.5 in [14] or as an easy consequence of the paper [5].

In order to obtain results on the higher plurigenera we now impose the condition that S_0 is normal and Gorenstein. Given any normal Gorenstein surface V, we have an invertible dualizing sheaf on V which we denote by ω_V (see [1]). Using the analogy of the arithmetic and geometric genus in the case of a curve, we make the following definition.

Definition 1.1. We define the *arithmetic plurigenera* of the surface V, denoted $\bar{P}_m(V)$ for each integer m, by $\bar{P}_m(V) = h^0(V, \omega_V^{\otimes m})$. These of course coincide with the usual (geometric) plurigenera when V is smooth.

Returning now to the given deformation $\rho: \mathscr{S} \to C$ with C smooth, we know that ρ is flat with Gorenstein fibres. Therefore $\mathscr{S}$ itself is Gorenstein by Proposition 9.6 of [9]. Hence we have a canonical line bundle $\omega_{\mathscr{S}/C}$ on $\mathscr{S}$ ($\omega_{\mathscr{S}/C} = \omega_{\mathscr{S}} \otimes (\rho^* \Omega^1_C)^{-1}$) which coincides with $\Omega^2_{\mathscr{S}/C}$ outside the singular locus of $\rho^{-1}(y_0)$, and such that $\omega_{\mathscr{S}_y} = \omega_{\mathscr{S}/C} \otimes_{\mathscr{O}_{\mathscr{S}}} \mathscr{O}_{\mathscr{S}_y}$ for all $y \in C$. In fact one can show using Proposition 2.5 of [20] that $\omega_{\mathscr{S}/C} = (\Omega^2_{\mathscr{S}/C})^{\vee\vee}$, the double dual of the sheaf $\Omega^2_{\mathscr{S}/C}$. We call $\omega_{\mathscr{S}/C}$ the relative dualizing sheaf.

Since ρ is flat and proper, we may now use the base change theorem ([13] page 50) to deduce:

Proposition 1.2. *With the notation as above, for all integers m we have $\bar{P}_m(S_0) \geqq \bar{P}_m(S)$ and $\chi(S, \omega_S^{\otimes m}) = \chi(S_0, \omega_{S_0}^{\otimes m})$.* □

2. Normal Gorenstein Surfaces

Let X be any normal Gorenstein surface with a singularity at $P \in X$. Let $f: \tilde{X} \to X$ denote a minimal resolution of the singularity P (so that $f^{-1}(P)$ contains no exceptional curves of the first kind).

Since X is Gorensten we have $K_{\tilde{X}} = f^* K_X - Z$ where Z is a divisor supported on $f^{-1}(P)$. Throughout this paper we shall use the convention that K represents a divisor corresponding to the invertible sheaf ω. The following Lemma is due to Miles Reid – the proof will be left to the reader although details may be found in [21].

Lemma 2.1. *The divisor Z is effective and either*

(i) *P is a rational double point and $Z=0$, or*

(ii) *P is a non-rational singularity and the support of Z is the whole of $f^{-1}(P)$.* □

Hence if S_0 is a normal Gorenstein surface, $g\colon \tilde{S}_0 \to S_0$ a minimal desingularization, then $K_{\tilde{S}_0} = g^* K_{S_0} - Z$ where $Z \geqq 0$ is supported on g^{-1}(singular locus) and $Z=0$ if and only if all the singularities of S_0 are rational.

If $\mathscr{L}$ is any line bundle on S_0, then $g_* g^* \mathscr{L} = \mathscr{L} \otimes_{\mathscr{O}_{S_0}} g_* \mathscr{O}_{\tilde{S}_0}$. But $g_* \mathscr{O}_{\tilde{S}_0} = \mathscr{O}_{S_0}$ ([8] § 4.3) and so $g_* g^* \mathscr{L}$ is naturally isomorphic to $\mathscr{L}$. Hence we deduce:

Lemma 2.2. $\bar{P}_m(S_0) = h^0(\tilde{S}_0, mg^* K_{S_0})$ *for all integers m.* □

Now since $mK_{\tilde{S}_0} = mg^* K_{S_0} - mZ$, we deduce that

$$h^0(\tilde{S}_0, mK_{\tilde{S}_0}) \leqq h^0(\tilde{S}_0, mg^* K_{S_0})$$

and hence $P_m(S_0) \leqq \bar{P}_m(S_0)$ for all positive integers m. Recall (1.2) that in the smooth deformation of S_0 given by $\rho\colon \mathscr{S} \to C$ with general surface S, we have the inequality $P_m(S) \leqq \bar{P}_m(S_0)$. This of course is the wrong way round for our purposes! However we also have (1.2) that $\chi(S, \omega_S^{\otimes m}) = \chi(S_0, \omega_{S_0}^{\otimes m})$ and so if we could obtain some bound on the behaviour of the h^1 term, we would know more about the behaviour of the h^0 term. This we shall not be able to do immediately, the reason for this being that the surfaces S and S_0 are not minimal in the sense defined below.

Definition 2.3. Let X_0 be any normal Gorenstein surface and let $f\colon \tilde{X}_0 \to X_0$ be its minimal desingularization. We say that X_0 is *minimal* if for all exceptional curves ℓ of the first kind on $\tilde{X}_0$, ℓ meets $f^{-1}(P)$ for some non-rational singularity P. By (2.1) we have that $K_{\tilde{X}_0} = f^* K_{X_0} - Z$ where $Z \geqq 0$ and that the above condition is equivalent to saying that ℓ meets Z.

We say that a curve ℓ on X_0 is an *exceptional curve of the first kind* if ℓ does not meet the singular locus of X_0 and $f^{-1}\ell$ is an exceptional curve of the first kind on $\tilde{X}_0$. Thus for instance if X_0 contains no such curves and also no rational singularities then it is clear that X_0 is minimal.

Proposition 2.4. *If S is a smooth surface with Kodaira dimension $\kappa(S) \geqq 0$ and if r is the number of blow ups of points needed to obtain S from its minimal model $\bar{S}$, then $h^1(S, \omega_S^{\otimes n})/n^2 \to \frac{1}{2}r$ as $n \to \infty$. In particular therefore S is minimal if and only if $h^1(S, \omega_S^{\otimes n})/n^2 \to 0$ as $n \to \infty$.*

Proof. It is clear that $K_S^2 = K_{\bar{S}}^2 - r$ and so the Riemann-Roch theorem yields $\chi(S, \omega_S^{\otimes n}) - \chi(\bar{S}, \omega_{\bar{S}}^{\otimes n}) = -\frac{1}{2}n(n-1)r$ for $n \geqq 1$. Since the plurigenera are birational invariants, $h^0(S, \omega_S^{\otimes n}) = h^0(\bar{S}, \omega_{\bar{S}}^{\otimes n})$ for $n \geqq 1$. Since $\kappa(S) \geqq 0$, it is clear that $h^2(S, \omega_S^{\otimes n}) \leqq 1$ and $h^2(\bar{S}, \omega_{\bar{S}}^{\otimes n}) \leqq 1$ for $n \geqq 1$.

We now comment that $h^1(\bar{S}, \omega_{\bar{S}}^{\otimes n})/n^2 \to 0$ as $n \to \infty$ (since $\bar{S}$ is minimal and $\kappa(\bar{S}) \geqq 0$) and so the result follows. □

Definition 2.5. Given a divisor D on a smooth surface S, we say that D is *arithmetically effective* (denoted a.e.) if $D \cdot L \geqq 0$ for every effective divisor L on S (cf. [22]).

Remark 2.6. A very important example of an a.e. divisor occurs when we have S a smooth minimal surface of Kodaira dimension $\kappa(S) \geqq 0$. In this case we know that K_S is an a.e. divisor on S (see [3]).

Given a minimal normal Gorenstein surface as in (2.3) we have the following vital result.

Proposition 2.7. With the notation as in (2.3), if X_0 is minimal and $\kappa(\tilde{X}_0) \neq -1$, then $f^* K_{X_0}$ is a.e. on $\tilde{X}_0$.

Proof. Since $\kappa(\tilde{X}_0) \geqq 0$, we have a unique minimal model $\bar{X}_0$ of $\tilde{X}_0$ with $g: \tilde{X}_0 \to \bar{X}_0$ say. We therefore have a collection $\Delta_1, \ldots, \Delta_s$ of connected cycles on $\tilde{X}_0$ which collapse down under g to points $Q_1, \ldots, Q_s$ on $\bar{X}_0$.

We need to show that $f^* K_{X_0} \cdot E \geqq 0$ for all irreducible curves E on $\tilde{X}_0$.

(i) First suppose that E is not an exceptional curve, neither over one of the points Q_i nor over the singular locus of X_0'. We denote by $\bar{E}$ the irreducible curve on $\bar{X}_0$ whose strict transform under g is E.

The reader may then verify that $K_{\tilde{X}_0} \cdot E \geqq K_{\bar{X}_0} \cdot \bar{E}$. However by (2.6) $K_{\bar{X}_0} \cdot \bar{E} \geqq 0$ and hence $K_{\tilde{X}_0} \cdot E \geqq 0$.

But $K_{\tilde{X}_0} = f^* K_{X_0} - Z$ and so $f^* K_{X_0} \cdot E = K_{\tilde{X}_0} \cdot E + Z \cdot E$. Since E is not an exceptional curve $Z \cdot E \geqq 0$ and hence $f^* K_{X_0} \cdot E \geqq 0$.

(ii) If E is now an exceptional curve over a singular point of X_0, then $f^* K_{X_0} \cdot E = 0$ and there is nothing further to prove.

(iii) Finally therefore we consider the case when E is in one of the cycles Δ_i, and hence E is rational with $E^2 \leqq -1$. We may assume that E is not also an exceptional curve over the singular locus of X_0, for this eventuality is covered by (ii). Thus we know that $Z \cdot E \geqq 0$.

Now $K_{\tilde{X}_0} \cdot E = -2 - E^2$ and so $f^* K_{X_0} \cdot E = -2 - E^2 + Z \cdot E$. If $E^2 < -1$ then the result is clear. If however $E^2 = -1$, then E is an exeptional curve of the first kind on $\tilde{X}_0$. Since X_0 is minimal we must therefore have $Z \cdot E \geqq 1$ (2.3), and hence $f^* K_{X_0} \cdot E \geqq 0$ follows in this case also. □

Corollary 2.8. *With X_0 satisfying the conditions of* (2.7), *for $\alpha = 12$ we have $h^1(X_0, \omega_{X_0}^{\otimes \alpha n})/n^2 \to 0$ as $n \to \infty$* (cf. *Proposition* 2.4).

Proof. We noted before that $f_* f^* \omega_{X_0}^{\otimes m} = \omega_{X_0}^{\otimes m}$. Hence using the Leray spectral sequence we deduce that $h^1(X_0, \omega_{X_0}^{\otimes m}) \leqq h^1(\tilde{X}_0, f^* \omega_{X_0}^{\otimes m})$ for all positive integers m. Thus $h^1(X_0, \omega_{X_0}^{\otimes \alpha n}) \leqq h^1(\tilde{X}_0, \mathcal{O}_{\tilde{X}_0}(n \alpha f^* K_{X_0}))$.

But from Enriques' classification [3] we know that $\alpha K_{\tilde{X}_0} \geqq 0$ and hence we deduce that $\alpha f^* K_{X_0} \geqq 0$. Thus $\alpha f^* K_{X_0}$ is both effective and a.e., and so we deduce the required result from [22] page 611, results (1), (3a) and (5). □

This result cannot however be applied immediately to the smooth deformation $\rho: \mathcal{S} \to C$ since the surface S_0 is not necessarily minimal. However we shall see in the next section that by using some standard theory this difficulty can be circumvented.

3. The Analytic Smooth Deformation $\pi: X \to N$

Motivated by the results of § 2, our plan now is to replace our original deformation $\rho: \mathscr{S} \to C$ by an analytic smooth deformation $\pi: X \to N$ with just the one singular fibre X_0, which fibre will be normal, Gorenstein and minimal. Here X_0 will be birationally equivalent to the surface $S_0 = \rho^{-1}(y_0)$, and the general surface X_1 will be birationally equivalent to some smooth surface S of the original deformation.

Our first step towards attaining this plan is to resolve the rational singularities of S_0 in the family. By (2.1) the rational singularities of S_0 must be rational double points. Therefore using the results of [18], we can find a neighbourhood U (in the complex topology) of y_0 in C and a commutative diagram of complex space morphisms:

$$\begin{array}{ccc} \mathscr{T} & \xrightarrow{\psi} & \mathscr{S}|U \\ {\scriptstyle\sigma}\downarrow & & \downarrow{\scriptstyle\rho} \\ V & \xrightarrow{\phi} & U \end{array}$$

where σ is flat and proper, ϕ is finite and surjective, ψ is proper and surjective and $\psi_t: \mathscr{T}_t \to \mathscr{S}_{\phi(t)}$ is a resolution (in the sense of [10]) of the rational singularities of $\mathscr{S}_{\phi(t)}$ for all $t \in V$.

In particular this means that $\mathscr{T}_t$ is an algebraic surface for all $t \in V$, and that if W_t denotes the locus of rational double points on $\mathscr{S}_{\phi(t)}$, then the restriction

$$\operatorname{res}(\psi_t): \mathscr{T}_t - \psi_t^{-1}(W_t) \to \mathscr{S}_{\phi(t)} - W_t$$

is an isomorphism.

Thus we see that if $\phi(t) \neq y_0$ then ψ_t is an isomorphism. We also note that by the Corollary to Theorem 1 in [18], since V is of complex dimension one we may assume that V is in fact smooth.

By shrinking V we may assume now that $\sigma: \mathscr{T} \to V$ has only singular fibre, say $\mathscr{T}_0 = \sigma^{-1}(t_0)$. The surface $\mathscr{T}_0$ is birationally equivalent to S_0, whilst for $t \neq t_0$ the surface $\mathscr{T}_t$ is isomorphic to a smooth surface in the original deformation.

Remark 3.1. The surface $\mathscr{T}_0$ now contains no rational singularities. Hence if in our original family S_0 contained only rational singularities, then the above resolution and the results of [11] enable us to deduce immediately that $P_n(S) = P_n(S_0)$ for all $n \geqq 1$.

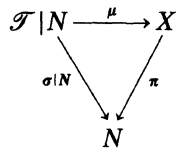
Returning now to the general case, $\mathscr{T}_0$ is not yet minimal since it may well contain exceptional curves of the first kind.

Proposition 3.2. *In our analytic family $\sigma: \mathscr{T} \to V$, if $\mathscr{T}_0$ contains an exceptional curve of the first kind E_0, then there exists a neighbourhood B of t_0 in V and a unique complex fibre submanifold E (using the terminology of* [12]) *of $\sigma^{-1}(B)$ whose restriction to $\mathscr{T}_0$ is the curve E_0. Moreover, for any point $t \in B$, the curve $E_t = E \cap \mathscr{T}_t$ is an exceptional curve of the first kind on $\mathscr{T}_t$.*

Proof. The result follows easily from Theorem 5 of [12]. □

We now assume that the situation of (3.2) obtains.

Theorem 3.3. *With the notation as above, there exists a neighbourhood N of t_0 in B such that we can construct an analytic space X and a commutative diagram:*



Here μ: $\mathscr{T}|N \to X$ is proper and surjective, μ_t: $\mathscr{T}_t \to X_t$ blows down the curve E_t to a smooth point of X_t for all $t \in N$, and the restriction of μ, $\mathrm{res}(\mu)$: $(\mathscr{T}-E)|N \to X-\mu(E|N)$, is an analytic isomorphism. Since σ is proper, so too is π, and from $\dim(B)=1$ we have the fact that π is also flat.

Proof. This is a fairly standard procedure whose details may be reconstructed from [2] and [16]. Details may also be found in [21]. □

In fact one can easily show above that for N sufficiently small the contraction map μ is locally trivial – that is there exists a neighbourhood $\tilde{V}_0$ of E_0 in $\mathscr{T}_0$ with image V_0 und μ_0 such that μ is locally (up to analytic isomorphism) of the form $\mu_0 \times 1$: $\tilde{V}_0 \times N \to V_0 \times N$.

By repeating the procedure described in (3.3) we may obtain an analytic family of algebraic surfaces denoted by π: $X \to N$, where the singular fibre $X_0 = \pi^{-1}(t_0)$ is now minimal, normal and Gorenstein. The surface X_0 is birationally equivalent to our original singular surface S_0 whilst the general surface X_1 of the family is birationally equivalent to a smooth surface S of our original deformation.

Proposition 3.4. *There is a holomorphic line bundle $\omega_{X/N}$ on X which coincides with the sheaf of relative holomorphic 2-forms $\Omega^2_{X/N}$ outside the singular locus of X_0 and such that $\omega^h_{X_t} = \omega_{X/N}|X_t$ for all $t \in N$ (the notation here being that of* [17] *in that $\omega^h_{X_t}$ is the holomorphic sheaf corresponding to ω_{X_t}).*

Proof. Note that there is such a sheaf on $\mathscr{T}$ given by $\psi^*(\omega^h_{\mathscr{T}/C}|U)$. If we then contract E as in (3.3), we see that the sheaf $\omega_{X/N} = \mu_*(\omega_{\mathscr{T}/N} \otimes \mathscr{O}_{\mathscr{T}}(-E))$ is invertible and has the required properties.

Details of the above are left to the reader or may be found in [21]. □

4. First Part of the Proof of the Main Result

We deal with the (flat and proper) analytic smooth deformation π: $X \to N$ obtained in § 3, where the surface $X_0 = \pi^{-1}(t_0)$ is minimal. Letting X_1 denote the general (smooth) surface of the deformation, we shall prove that $P_n(X_0) \leqq P_n(X_1)$ for all $n > 1$ and thus complete the proof of the main result. We shall denote by f: $\tilde{X}_0 \to X_0$ a minimal desingularization of X_0.

From the existence of the line bundle $\omega_{X/N}$, the analytic base change theorem [6] and the results of [17], we deduce for all integers m that $\chi(\omega^{\otimes m}_{X_1}) = \chi(\omega^{\otimes m}_{X_0})$ and $h^i(\omega^{\otimes m}_{X_1}) \leqq h^i(\omega^{\otimes m}_{X_0})$ for $i \leqq 2$ (cf. Proposition 1.2).

Proposition 4.1. *If $\kappa(X_t) \geqq 0$ for all $t \in N$, then since X_0 is minimal, the general surface X_1 is also minimal (here of course $\kappa(X_0)$ is defined to be $\kappa(\tilde{X}_0)$).*

Proof. Since X_0 is minimal and $\kappa(\tilde{X}_0) \geqq 0$, then for $a = 12$ we know from (2.8) that $h^1(X_0, \omega_{X_0}^{\otimes an})/n^2 \to 0$ as $n \to \infty$. But from the analytic base change theorem we deduce as above that $h^1(X_1, \omega_{X_1}^{\otimes m}) \leqq h^1(X_0, \omega_{X_0}^{\otimes m})$ for all positive integers m. Thus $h^1(X_1, \omega_{X_1}^{\otimes an})/n^2 \to 0$ as $n \to \infty$, and the result now follows by (2.4). □

If $\kappa(\tilde{X}_0) = -1$, then $P_n(X_0) = 0$ for all positive n and there is nothing to prove. If $\kappa(X_1) = -1$, then Theorem 3.1.3 of [14] yields $\kappa(\tilde{X}_0) = -1$ (alternatively this may be easily shown using the above theory).

Proposition 4.2. *Unless X_0 is smooth (and so all the singularities of S_0 were rational), if $\kappa(X_1) = 0$ then $\kappa(\tilde{X}_0) = -1$.*

Proof. Suppose that $\kappa(X_0) \geqq 0$. Then by (4.1) X_1 is minimal and so $aK_{X_1} = 0$ for $a = 12$. Hence $P_a(X_1) = 1 = P_{-a}(X_1)$. Using the analytic base change theorem as above therefore $\bar{P}_a(X_0) \geqq 1$ and $\bar{P}_{-a}(X_0) \geqq 1$. Thus both $\omega_{X_0}^{\otimes a}$ and its dual have a non-zero section, so $\omega_{X_0}^{\otimes a}$ must be trivial, i.e. $aK_{X_0} = 0$.

Under the assumption that X_0 is not smooth, $K_{\tilde{X}_0} = f^* K_{X_0} - Z$ with $Z > 0$. We deduce that $aK_{\tilde{X}_0} = -aZ < 0$. Hence $\kappa(\tilde{X}_0) = -1$ and we obtain the required contradiction. □

Proposition 4.3. *If $\kappa(X_1) = 2$, then $P_m(X_0) \leqq P_m(X_1)$ for all $m > 1$.*

Proof. We may assume that $\kappa(\tilde{X}_0) \neq -1$. We deduce from (4.1) that X_1 is minimal. Hence from Enriques classification [3], $K_{X_1}^2 > 0$. But clearly $K_{X_0}^2 = K_{X_1}^2$ and so we deduce that $K_{X_0}^2 > 0$.

Now $h^1(mK_{X_0}) = h^1(-(m-1)K_{X_0}) \leqq h^1(-(m-1)f^* K_{X_0})$ from the Leray spectral sequence.

But for $m > 1$, the divisor $D = (m-1)f^* K_{X_0}$ is arithmetically effective (2.7) with $D^2 > 0$. Hence the Kodaira vanishing theorem [15] yields $h^1(-(m-1)f^* K_{X_0}) = 0$. Thus for $m > 1$, $h^1(\omega_{X_0}^{\otimes m}) = 0$.

It is clear that $h^2(\omega_{X_0}^{\otimes m}) = 0$ for $m > 1$. Hence, since $\chi(\omega_{X_1}^{\otimes m}) = \chi(\omega_{X_0}^{\otimes m})$, this yields $\bar{P}_m(X_0) = \bar{P}_m(X_1)$ for all $m > 1$. Thus the required result follows from (2.2). □

In § 1 we saw that the main result held in the case of the geometric genus. The above results now imply that the main theorem is true when $\kappa(X_1) = -1, 0$ or 2. We now study the case when $\kappa(X_1) = 1$.

Proposition 4.4. *If $\kappa(X_1) = 1$, then $\kappa(\tilde{X}_0) \leqq 1$.*

Proof. This follows easily from (4.1) and (2.8) or alternatively from Theorem 3.2.1 of [14]. □

From the Leray spectral sequence we know that $\chi(\mathcal{O}_{X_0}) = \chi(\mathcal{O}_{\tilde{X}_0}) + r^1 f_* \mathcal{O}_{\tilde{X}_0}$, where $r^1 f_* \mathcal{O}_{\tilde{X}_0} = h^0(X_0, R^1 f_* \mathcal{O}_{\tilde{X}_0})$ is the geometric genus of the singularities of X_0. Thus if $\chi(\mathcal{O}_{X_1}) = 0$ (and hence $\chi(\mathcal{O}_{X_0}) = 0$) it is clear that either X_0 is smooth or $\kappa(X_0) = -1$, in both of which cases the main theorem is true.

Thus we may restrict ourselves to studying the case when $\kappa(X_1) = 1$ and $\chi(\mathcal{O}_{X_1}) \geqq 1$. By the classification theory of surfaces (see [4]) we may then deduce that $P_m(X_1) \geqq 1$ for all $m > 1$. Thus if $\chi(\mathcal{O}_{X_1}) \geqq 1$ and $\kappa(\tilde{X}_0) \leqq 0$ the main theorem is also true. Therefore by (4.4) we need now only consider when $\kappa(X_1) = 1 = \kappa(\tilde{X}_0)$.

5. The Remaining Special Case

We saw in §4 that we are now reduced to the case when both X_0 and X_1 are elliptic surfaces of general type. Both X_0 and X_1 are then minimal (4.1). Let $\bar{X}_0$ denote the minimal model of $\tilde{X}_0$ with morphism $g\colon \tilde{X}_0 \to \bar{X}_0$. So in the notation of §4, we have

$$K_{\tilde{X}_0} = f^* K_{X_0} - Z = g^* K_{\bar{X}_0} + L \tag{1}$$

where L is a collection of curves over the blown up points of $\bar{X}_0$.

We know that $K^2_{X_0} = K^2_{\bar{X}_0} = 0$ and so squaring both sides of equation we obtain $Z^2 = L^2$.

Now for $a = 12$ we have $aK_{X_1} \geqq 0$ and hence also $aK_{X_0} \geqq 0$. Thus $af^* K_{X_0} \geqq 0$. Since $\kappa(X_1) = 1$, we know using the analytic base change theorem that $h^1(naK_{X_0})$ is unbounded as $n \to \infty$. By the Leray spectral sequence $h^1(naK_{X_0}) \leqq h^1(naf^* K_{X_0})$ for $n \geqq 1$, and so $h^1(naf^* K_{X_0})$ is unbounded as $n \to \infty$. Furthermore we know that the effective divisor $af^* K_{X_0}$ is also a.e. on $\tilde{X}_0$.

We now note that $f^* K_{X_0} = g^* K_{\bar{X}_0} + L + Z$ where $ag^* K_{\bar{X}_0} \geqq 0$. Therefore using [22] page 611 result (3a) we deduce from the above facts that $(L+Z)^2 \leqq 0$. Since $L^2 = Z^2$, this may be rewritten as $L^2 + Z \cdot L \leqq 0$.

But $f^* K_{X_0} \cdot L = L^2 + Z \cdot L$, and as $f^* K_{X_0}$ is a.e., we deduce that $L^2 + Z \cdot L \geqq 0$. Therefore

$$L^2 = Z^2 = -L \cdot Z. \tag{2}$$

Denote by E the divisor on $\bar{X}_0$ corresponding to Z on $\tilde{X}_0$. We express Z as the disjoint sum of connected cycles, $Z = Z_1 + \cdots + Z_r$, corresponding to the singular points $P_1, \ldots, P_r$ on X_0 (where by assumption P_i is non-rational for all i).

Now for any irreducible curve ℓ in L, we have ℓ rational and $\ell^2 = -b$ for some $b \geqq 1$. Therefore $K_{\bar{X}_0} \cdot \ell = b - 2$. But $f^* K_{X_0}$ is a.e. (2.7) and from Equations (1) and (2) we have $f^* K_{X_0} \cdot L = 0$. Hence $f^* K_{X_0} \cdot \ell = 0$. Therefore from Equation (1) $-Z \cdot \ell = b - 2$. Thus if ℓ is not contained in Z we must have either $b = 1$ or $b = 2$.

If $b = 1$ (i.e. ℓ is an exceptional curve of the first kind) then $Z \cdot \ell = 1$ and so ℓ meets one and only one of the cycles Z_i for $1 \leqq i \leqq r$. If however $b = 2$, then $Z \cdot \ell = 0$ – that is ℓ does not meet Z at all. But since $\ell^2 = -2$ and $\kappa(\bar{X}_0) \neq -1$, ℓ meets one and only one exceptional curve of the first kind.

Hence we deduce that E is the disjoint sum of connected cycles, $E = E_1 + \cdots + E_r$, where E_i corresponds to Z_i for $1 \leqq i \leqq r$.

Proposition 5.1. *For all $1 \leqq i \leqq r$, E_i is contained in a fibre of $\bar{X}_0$ with respect to the canonical elliptic fibration.*

Proof. From Equations (1) and (2) we deduce that

$$f^* K_{X_0} \cdot L = g^* K_{\bar{X}_0} \cdot Z = 0. \tag{3}$$

Now let e be an irreducible curve in E_i, z the corresponding curve in Z_i, then

$$K_{\bar{X}_0} \cdot e = g^* K_{\bar{X}_0} \cdot g^* e = g^* K_{\bar{X}_0} \cdot z. \tag{4}$$

By the same method of proof as (2.7) we see that $g^* K_{\bar{X}_0}$ is an a.e. divisor on $\tilde{X}_0$. Hence from this and Equations (3) and (4) we deduce that $K_{\bar{X}_0} \cdot e = 0$.

Therefore e is contained in a fibre of the canonical fibration. Since this obtains for all e in E_i and E_i is connected, E_i is itself contained in such a fibre. □

The above Proposition tells us that in passing from $\bar{X}_0$ to X_0 we are altering at most r fibres of $\bar{X}_0$ with respect to the canonical fibration.

Now recall that for any elliptic surface V of general type we have the formula:

$$P_m(V) = m(2g(V) - 2 + \chi(\mathcal{O}_V)) + \sum_{\lambda=1}^{s} [m(1 - 1/m_\lambda)] + 1 - g(V) \qquad (\dagger)$$

for all $m > 1$, where $g(V)$ denotes the geometric genus of the base curve of V with respect to the canonical fibration and the m_λ are the multiplicities of the multiple fibres (see [11]).

We put $\Phi_m(V) = \Sigma[m(1 - 1/m_\lambda)]$. Furthermore for $\bar{X}_0$ we put $\Phi'_m(\bar{X}_0) = \Sigma'[m(1 - 1/m_\lambda)]$ where the sum Σ' is taken over those multiple fibres which are unchanged by f and g.

Since the singularities $P_1, \ldots, P_r$ of X_0 are all non-rational, we know that $\chi(\mathcal{O}_{X_1}) = \chi(\mathcal{O}_{X_0}) + r_1$ where $r_1 \geqq r$. Moreover from the above definitions it is clear that $\Phi_m(\bar{X}_0) - \Phi'_m(\bar{X}_0) \leqq m r_2$ where r_2 is the number of multiple fibres of $\bar{X}_0$ that are changed, and so $r \geqq r_2$.

Consider the following inequalities.

(A) $g(X_1) \geqq g(\bar{X}_0)$

(B) $\Phi_m(X_1) \geqq \Phi'_m(\bar{X}_0)$ for all $m > 1$.

Using the Equation (†), it is clear that (A) and (B) together now imply that $P_m(X_1) \geqq P_m(\bar{X}_0)$ for all $m > 1$. Hence in order to complete the proof of the main theorem we need only demonstrate the inequalities (A) and (B). These proofs will follow almost exactly proofs contained in [11], which I shall refer to extensively.

Proposition 5.2. *In our special case,* (A) *holds.*

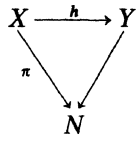
Proof. The proof of this is virtually identical with the first part of the proof of Propositions 9 and 10 on pages 253–4 of [11].

Given the analytic deformation $\pi: X \to N$, we denote by $\mathscr{F}$ the invertible sheaf on X given by $\mathscr{F} = \omega_{X/N}^{\otimes \alpha}$ (where $\alpha = 86$). The construction of Iitaka then provides a holomorphic map

$$h: X - \{\text{singular locus of } X_0\} \to \mathbb{P}(\pi_* \mathscr{F})$$

where h_t for $t \neq t_0$ is merely the morphism given by the linear system $|\alpha K_{X_t}|$, and h_{t_0} is given (outside the singular locus of X_0) by the linear subsystem of $|\alpha K_{X_0}|$ defined by the sections of $\pi_*(\mathscr{F}) \otimes \kappa(t_0)$ on X_0. This linear subsystem however also defines a morphism at the singular points of X_0. This is because any section cuts out an α-canonical divisor on X_0, and thus if the linear subsystem has a fixed point then it must also have a fixed component.

Thus we see that we can extend h to a morphism $h: X \to \mathbb{P}(\pi_* \mathscr{F})$ where h_{t_0} is given by the appropriate linear subsystem $|\alpha K_{X_0}|$. We denote by Y the reduced subcomplex space of $\mathbb{P}(\pi_* \mathscr{F})$ with $h(X)$ as the underlying topological space, so that we have a fibre space $h: X \to Y$ and a commutative diagram:



Now since π is flat and proper, so too is the morphism $Y \to N$ defined above. It is clear that $g(\bar{X}_0)$ is the geometric genus of Y_{t_0}, and so we deduce from the analytic base change theorem that the inequality (A) is true. □

We now consider the multiple fibres of X_0 over $Y_0 = Y_{t_0}$ which do not pass through the singular points of X_0. These clearly correspond to the multiple fibres of $\bar{X}_0$ with respect to the canonical fibration which are unchanged by f and g. As in [11], we are led to the following result.

Proposition 5.3. *For every multiple fibre $h_{t_0}^*(y_0) = mL_{t_0}$ of multiplicity m (whose reduced component is L_{t_0} not containing any singular points of X_0), we have that $(m-1)L_{t_0}$ extends locally to an effective divisor $\mathscr{L}$ on $\pi^{-1}(U)$ where U is a small neighbourhood of t_0. For $t \in U$ we then know that $\mathscr{L} \mid X_t$ decomposes into the sum of multiple fibres,*

$$\mathscr{L} \mid X_t = \sum_{i=1}^{u} (m^{(i)} - 1) L_t^{(i)}$$

where $m^{(i)} \geqq 2$ and $L_t^{(i)}$ is the reduced component of a multiple fibre $h_t^*(y_t^{(i)})$ of h_t where $y_t^{(i)} \in Y_t$. Moreover the multiplicity $m^{*(i)}$ of $h_t^*(y_t^{(i)})$ is greater than or equal to $m^{(i)}$ for each $1 \leqq i \leqq u$, and if $u = 1$ then $m^{(1)} = m$.

Proof. Since the multiple fibre under consideration does not pass through any of the singular points of X_0, we find that the proof of the corresponding result on pages 255–7 of [11] may be taken over virtually unchanged. One uses the sheaf $\omega_{X/N}$ when $\Omega^2_{X/N}$ would be used in Iitaka's proof and in a couple of places one deals with $\bar{P}_1(X_0)$ rather than $p_g(X_0)$. However the required modifications are fairly clear and so will be left to the reader. More details may be found in [21]. □

It is now trivial to see that (5.3) gives the required inequality (B). Thus the main result has been proved in the case of $\kappa(X_1) = 1$ and is therefore now proved in general.

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